



WORK ORDER NO. 1606884

NAVFAC CONTRACT NO. N6247019D5014

**Replace Fuel Pipeline from Pier 3 to
Bulk Storage**

At the
NAVSTA Rota, Spain

PREPARED BY:
Stanley Consultants, Inc.
225 Iowa Avenue
Muscatine, IA 52761

Revised Corrected Final Design

Architectural: N/A
Civil: Cathy Weikel, P.E.
Structural: Daniel Dreyer, P.E.
Geotechnical: N/A

Mechanical: Timothy Schmitt, P.E.
Electrical: Stanley Worcester, P.E.
Fire Protection: N/A
Safety: N/A

Submitted By: Jerome
Schoffler

Digitally signed by
Jerome Schoffler
Date: 2024.07.15
22:25:50 -05'00'

Date: 07/15/2024

APPROVED BY:

Specifications:

For Commander, NAVFAC:
Date:

THRUN.KEMPER.E
LISE.1297275972
Digitally signed by
THRUN.KEMPER.E
LISE.1297275972
Date: 2024.07.25 08:22:08
+02'00'

PROJECT TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

00 01 15	02/11, CHG 1: 08/14	LIST OF DRAWINGS
00 73 04.00 41	08/08	SUPPLEMENTARY CONDITIONS FOR PROJECTS IN MEDITERRANEAN AREA

DIVISION 01 - GENERAL REQUIREMENTS

01 11 00	08/15, CHG 2: 08/21	SUMMARY OF WORK
01 14 00	11/11, CHG 14: 02/22	WORK RESTRICTIONS
01 20 00	11/20, CHG 2: 08/21	PRICE AND PAYMENT PROCEDURES
01 30 00	11/20, CHG 2: 05/22	ADMINISTRATIVE REQUIREMENTS
01 32 17.00 20	05/18, CHG 3: 08/20	COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS)
01 33 00	08/18, CHG 4: 02/21	SUBMITTAL PROCEDURES
01 35 26	11/20, CHG 3: 02/22	GOVERNMENTAL SAFETY REQUIREMENTS
01 42 00	02/19	SOURCES FOR REFERENCE PUBLICATIONS
01 42 15	02/14	METRIC MEASUREMENTS
01 45 00.00 20	11/11, CHG 8: 02/21	QUALITY CONTROL
01 50 00	11/20, CHG 2: 08/22	TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS
01 57 19.01 21		ENVIRONMENTAL CONTROLS
01 58 00	08/19, CHG 5: 08/22	PROJECT IDENTIFICATION
01 74 19	02/19, CHG 3: 11/21	CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL
01 78 00	05/19, CHG 1: 08/21	CLOSEOUT SUBMITTALS
01 78 23	08/15, CHG 2: 08/21	OPERATION AND MAINTENANCE DATA
01 78 24.00 10	05/18, CHG 1: 11/21	FACILITY DATA REQUIREMENTS

DIVISION 02 - EXISTING CONDITIONS

02 41 00	08/22	DEMOLITION
02 61 13	02/10, CHG 1: 02/21	EXCAVATION AND HANDLING OF CONTAMINATED MATERIAL

DIVISION 03 - CONCRETE

03 30 00	02/19, CHG 4: 08/22	CAST-IN-PLACE CONCRETE
----------	---------------------	------------------------

DIVISION 05 - METALS

05 50 13	05/17, CHG 1: 08/18	MISCELLANEOUS METAL FABRICATIONS
----------	---------------------	----------------------------------

DIVISION 09 - FINISHES

09 97 13.27	02/21, CHG 1: 08/22	HIGH PERFORMANCE COATING FOR STEEL STRUCTURES
-------------	---------------------	--

DIVISION 25 - INTEGRATED AUTOMATION

25 05 11	05/21	CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS
----------	-------	---

DIVISION 26 - ELECTRICAL

26 08 00	11/21	APPARATUS INSPECTION AND TESTING
26 20 00	08/19, CHG 3: 11/21	INTERIOR DISTRIBUTION SYSTEM

26	28	01.00	10	08/21	COORDINATED POWER SYSTEM PROTECTION
26	42	17		05/21	IMPRESSED CURRENT CATHODIC PROTECTION (ICCP) SYSTEM

DIVISION 31 - EARTHWORK

31	23	00.00	20	02/11, CHG 2: 08/15	EXCAVATION AND FILL
----	----	-------	----	---------------------	---------------------

DIVISION 32 - EXTERIOR IMPROVEMENTS

32	11	23		05/22	AGGREGATE BASE COURSE
32	12	13		05/17	BITUMINOUS TACK AND PRIME COATS
32	12	16.16		11/20	ROAD-MIX ASPHALT PAVING
32	15	00		05/17	AGGREGATE SURFACING
32	16	19		05/18	CONCRETE CURBS AND SIDEWALKS
32	17	23		08/16, CHG 5: 11/18	PAVEMENT MARKINGS
32	31	13.53		11/21	HIGH-SECURITY FENCES AND GATES
32	92	19		08/17, CHG 1: 08/21	SEEDING

DIVISION 33 - UTILITIES

33	08	55		08/19	FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT)
33	52	23.15		11/18	POL SERVICE PIPING WELDING
33	52	40		11/18, CHG 2: 11/20	FUEL SYSTEMS PIPING (NON-HYDRANT)
33	52	80		08/22	LIQUID FUELS PIPELINE COATING SYSTEMS
33	71	02		08/21	UNDERGROUND ELECTRICAL DISTRIBUTION

-- End of Project Table of Contents --

SECTION TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

DOCUMENT 00 01 15

LIST OF DRAWINGS

02/11, CHG 1: 08/14

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 CONTRACT DRAWINGS
- 1.3 SUPPLEMENTARY DRAWINGS

-- End of Section Table of Contents --

DOCUMENT 00 01 15

LIST OF DRAWINGS
02/11, CHG 1: 08/14

PART 1 GENERAL

1.1 SUMMARY

This section lists the drawings for the project pursuant to contract clause "DFARS 252.236-7001, Contract Drawings, Maps and Specifications."

1.2 CONTRACT DRAWINGS

Contract drawings are as follows:

DRAWING INDEX / ÍNDICE DE PLANOS			
SHEET NUMBER / NÚMERO DE PLANO	NAVFAC NO.	SHEET TITLE	TÍTULO DE PLANO
GENERAL / GENERAL			
G-001	14142702	COVER SHEET	PORTADA
G-002	14142703	INDEX - SHEET 1	ÍNDICE - PLANO 1
G-003	14142704	INDEX - SHEET 2	ÍNDICE - PLANO 2
G-004	14142705	GENERAL ABBREVIATION SHEET	PLANO DE ABREVIATURAS GENERALES
CIVIL: CIVIL-GENERAL / URBANIZACIÓN: URBANIZACIÓN - GENERAL			
C-001	14142706	CIVIL LEGEND - SHEET 1	LEYENDA DE URBANIZACIÓN - PLANO 1
C-002	14142707	CIVIL LEGEND - SHEET 2	LEYENDA DE URBANIZACIÓN - PLANO 2

CIVIL: CIVIL DEMO / URBANIZACIÓN: URBANIZACIÓN - DEMOLICIONES			
CD101	14142708	SITE DEMOLITION KEY PLAN	PLANTA DE REFERENCIA DE DEMOLICIÓN DE EXTERIORES
CD106	14142709	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD107	14142710	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD108	14142711	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD109	14142712	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD110	14142713	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD111	14142714	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD112	14142715	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD113	14142716	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD114	14142717	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD115	14142718	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD116	14142719	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD117	14142720	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD118	14142721	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD119	14142722	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD120	14142723	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CIVIL: CIVIL-SITE / URBANIZACIÓN: URBANIZACIÓN - EXTERIORES			

C-101	14142724	OVERALL SITE KEY PLAN	PLANTA DE REFERENCIA DE EXTERIORES GENERAL
C-106	14142725	SITE PLAN	PLANTA DE EXTERIORES
C-107	14142726	SITE PLAN	PLANTA DE EXTERIORES
C-108	14142727	SITE PLAN	PLANTA DE EXTERIORES
C-109	14142728	SITE PLAN	PLANTA DE EXTERIORES
C-110	14142729	SITE PLAN	PLANTA DE EXTERIORES
C-111	14142730	SITE PLAN	PLANTA DE EXTERIORES
C-112	14142731	SITE PLAN	PLANTA DE EXTERIORES
C-113	14142732	SITE PLAN	PLANTA DE EXTERIORES
C-114	14142733	SITE PLAN	PLANTA DE EXTERIORES
C-115	14142734	SITE PLAN	PLANTA DE EXTERIORES
C-116	14142735	SITE PLAN	PLANTA DE EXTERIORES
C-117	14142736	SITE PLAN	PLANTA DE EXTERIORES
C-118	14142737	SITE PLAN	PLANTA DE EXTERIORES
C-119	14142738	SITE PLAN	PLANTA DE EXTERIORES
C-120	14142739	SITE PLAN	PLANTA DE EXTERIORES
C-401	14142740	ENLARGED SITE PLAN - SHEET 1	PLANTA AMPLIADA DE EXTERIORES - PLANO 1

C-402	14142741	ENLARGED SITE PLAN - SHEET 2	PLANTA AMPLIADA DE EXTERIORES - PLANO 2
C-403	14142742	ENLARGED SITE PLAN - SHEET 3	PLANTA AMPLIADA DE EXTERIORES - PLANO 3
C-404	14142743	ENLARGED SITE PLAN - SHEET 4	PLANTA AMPLIADA DE EXTERIORES - PLANO 4
C-501	14142744	CIVIL DETAILS	DETALLES DE URBANIZACIÓN
C-502	14142745	CIVIL DETAILS	DETALLES DE URBANIZACIÓN
C-503	14142746	CIVIL DETAILS	DETALLES DE URBANIZACIÓN
C-504	14142747	DETAILS - FENCING	DETALLES - VALLA
C-505	14142748	DETAILS - FENCING	DETALLES - VALLA
CIVIL: CIVIL GRADING / URBANIZACIÓN: NIVELACIÓN DE URBANIZACIÓN			
CG401	14142749	ENLARGED GRADING PLAN - SHEET 1	PLANTA AMPLIADA DE NIVELACIÓN - PLANO 1
CG402	14142750	ENLARGED GRADING PLAN - SHEET 2	PLANTA AMPLIADA DE NIVELACIÓN - PLANO 2
STRUCTURAL / ESTRUCTURA			
S-021	14142751	STRUCTURAL NOTES - SHEET 1	NOTAS DE ESTRUCTURA - PLANO 1
S-022	14142752	STRUCTURAL NOTES - SHEET 2	NOTAS DE ESTRUCTURA - PLANO 2
S-521	14142753	TYPICAL REINFORCING DETAILS	DETALLES TÍPICOS DE ARMADO
S-522	14142754	TYPICAL FOUNDATION DETAILS	DETALLES TÍPICOS DE CIMENTACIÓN

S-523	14142755	MISCELLANEOUS DETAILS	DETALLES VARIOS
S-524	14142756	MISCELLANEOUS DETAILS	DETALLES VARIOS
S-530	14142757	TYPICAL PIPE SUPPORT DETAILS	DETALLES TÍPICOS DE SOPORTES DE TUBERÍAS
MECHANICAL: MECHANICAL-GENERAL / MECÁNICA: MECÁNICA - GENERAL			
M-001	14142758	MECHANICAL LEGEND	LEYENDA DE MECÁNICA
MECHANICAL: MECHANICAL - DEMOLITION / MECÁNICA: MECÁNICA - DEMOLICIONES			
MD101	14142759	MECHANICAL DEMOLITION KEY PLAN	PLANTA DE REFERENCIA DE DEMOLICIÓN
MD102	14142760	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD103	14142761	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD104	14142762	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD105	14142763	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD106	14142764	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD107	14142765	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD108	14142766	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA

MD109	14142767	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD110	14142768	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD111	14142769	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD119	14142770	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD120	14142771	MECHANICAL DEMOLITION PLAN ML-1 TO ML-2	PLANTA DE DEMOLICIONES DE MECÁNICA DE ML1 A ML-2
MD401	14142772	DEMOLITION PLAN VP-1 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO VP-1
MD402	14142773	DEMOLITION PLAN VP-1.5 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO VP-1.5
MD403	14142774	DEMOLITION PLAN BPH-VP-1 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO BPH-VP-1
MD404	14142775	DEMOLITION PLAN VP-3 & 3.5 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO VP-3 Y 3.5
MD410	14142776	DEMOLITION PLAN ML-1 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO ML-1
MD411	14142777	DEMOLITION PLAN ML-2 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO ML-2
MECHANICAL: MECHANICAL - SITE / MECÁNICA: MECÁNICA - EXTERIORES			
M-101	14142778	OVERALL MECHANICAL SITE & KEY PLAN	PLANTA GENERAL EXTERIOR Y DE REFERENCIA
M-102	14142779	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA

M-103	14142780	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-104	14142781	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-105	14142782	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-106	14142783	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-107	14142784	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-108	14142785	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-109	14142786	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-110	14142787	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-111	14142788	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-112	14142789	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-113	14142790	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-114	14142791	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-115	14142792	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-116	14142793	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-117	14142794	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-118	14142795	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-119	14142796	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-120	14142797	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA

M-401	14142798	ENLARGED SITE PLAN VP-1 MANHOLE	PLANTA EXTERIOR AMPLIADA DEL POZO VP-1
M-402	14142799	ENLARGED SITE PLAN BPH-VP-1.5 MANHOLE	PLANTA EXTERIOR AMPLIADA DEL POZO BPH-VP-1.5
M-403	14142800	ENLARGED SITE PLAN - BPH-VP-1 MANHOLE	PLANTA EXTERIOR AMPLIADA DEL POZO BPH-VP-1
M-404	14142801	ENLARGED SITE PLAN BPH-VP- 3 & 3.5 MANHOLE	PLANTA EXTERIOR AMPLIADA DEL POZO BPH-VP-3 Y 3.5
M-410	14142802	ENLARGED SITE PLAN - ML1	PLANTA EXTERIOR AMPLIADA - ML1
M-411	14142803	ENLARGED SITE PLAN - ML2	PLANTA EXTERIOR AMPLIADA - ML2
M-501	14142804	MECHANICAL DETAILS - SHEET 1	DETALLES DE MECÁNICA - PLANO 1
M-502	14142805	MECHANICAL DETAILS - SHEET 2	DETALLES DE MECÁNICA - PLANO 2
M-503	14142806	MECHANICAL FUELING DETAILS	DETALLES DE COMBUSTIBLE MECÁNICO
M-504	14142807	MECHANICAL FUELING DETAILS	DETALLES DE COMBUSTIBLE MECÁNICO
M-505	14142808	TYPICAL PIPING PLAN	PLANTA TÍPICA DE TUBERÍAS
M-506	14142809	MECHANICAL SECTION	SECCIÓN MECÁNICA
M-601	14142810	ALIGNMENT & PIPE SCHEDULES - SHEET 1	CUADROS DE ALINEACIONES Y TUBERÍAS - PLANO 1
M-602	14142811	ALIGNMENT & PIPE SCHEDULES - SHEET 2	CUADROS DE ALINEACIONES Y TUBERÍAS - PLANO 2
M-611	14142812	PROCESS FLOW DIAGRAM - SHEET 1	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 1

M-612	14142813	PROCESS FLOW DIAGRAM - SHEET 2	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 2
M-613	14142814	PROCESS FLOW DIAGRAM - SHEET 3	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 3
M-614	14142815	PROCESS FLOW DIAGRAM - SHEET 4	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 4
M-615	14142816	PROCESS FLOW DIAGRAM - SHEET 5	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 5
M-616	14142817	PROCESS FLOW DIAGRAM - SHEET 6	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 6
ELECTRIC: ELECTRIC GENERAL / ELECTRICIDAD: ELECTRICIDAD GENERAL			
E-001	14142818	ELECTRICAL SYMBOLS & LEGEND SHEET	PLANO DE SÍMBOLOS ELÉCTRICOS Y LEYENDA
E-002	14142819	ELECTRICAL SYMBOLS & LEGEND SHEET	PLANO DE SÍMBOLOS ELÉCTRICOS Y LEYENDA
E-003	14142820	ELECTRICAL SYMBOLS & LEGEND SHEET	PLANO DE SÍMBOLOS ELÉCTRICOS Y LEYENDA
E-004	14142821	ELECTRICAL SYMBOLS & LEGEND SHEET	PLANO DE SÍMBOLOS ELÉCTRICOS Y LEYENDA
E-005	14142822	GENERAL NOTES SHEET	PLANO DE NOTAS GENERALES
E-006	14142823	GENERAL NOTES SHEET	PLANO DE NOTAS GENERALES
ELECTRIC: ELECTRIC DEMOLITION / ELECTRICIDAD: DEMOLICIONES ELÉCTRICAS			
ED401	14142824	ELECTRIC DEMOLITION PLAN - VP-1 MANHOLE	PLANTA DE DEMOLICIÓN ELÉCTRICA DE POZO VP-1

ED404	14142825	ELECTRIC DEMOLITION PLAN - VP-3 & 3.5 MANHOLE	PLANTA DE DEMOLICIÓN ELÉCTRICA DE POZO VP-3 Y 3.5
ELECTRIC: ELECTRIC SITE / ELECTRICIDAD: EXTERIOR DE ELECTRICIDAD			
E-401	14142826	ELECTRIC VAULT PLAN BPH-VP-1	PLANTA DE ARQUETA ELÉCTRICA BPH-VP-1
E-404	14142827	ELECTRIC VAULT BPH-VP-3 & 3.5	ARQUETA ELÉCTRICA BPH-VP-3 Y 3.5
E-501	14142828	ELECTRIC DETAILS	DETALLES DE ELECTRICIDAD
E-601	14142829	P3-1-VP1 CONTROL PANEL 1 LINE DIAGRAM	DIAGRAMA UNIFILAR DEL PANEL 1 DE CONTROL P3-1-VP1
E-607	14142830	BPH-VP-3 & 3.5 1-LINE DIAGRAM	DIAGRAMA UNIFILAR DEL PANEL DE CONTROL RÍO SALADO ESTE
ELECTRIC: CATHODIC PROTECTION / ELECTRICIDAD: PROTECCIÓN CATÓDICA			
EC101	14142831	CATHODIC PROTECTION KEY PLAN	PLANTA DE REFERENCIA DE PROTECCIÓN CATÓDICA
EC106	14142832	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC107	14142833	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC108	14142834	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC109	14142835	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC110	14142836	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC111	14142837	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC112	14142838	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA

EC113	14142839	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC114	14142840	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC115	14142841	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC116	14142842	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC117	14142843	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC118	14142844	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC119	14142845	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC120	14142846	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC501	14142847	CATHODIC PROTECTION DETAILS	DETALLES DE PROTECCIÓN CATÓDICA
ELECTRIC GROUNDING/			
EG101	14142848	ELECTRICAL GROUNDING SITE PLAN AND KEY MAP	PLANTA DE REFERENCIA DE EXTERIORES GENERAL
EG401	14142849	ELECTRICAL ENLARGED GROUNDING PLANS	PLANTAS ELÉCTRICAS AMPLIADAS DE PUESTA A TIERRA
EG501	14142850	ELECTRICAL GROUNDING DETAILS	DETALLES
REFERENCE SHEETS/ PLANOS DE REFERENCIA			
SURVEY / TOPOGRAFÍA			
V-001	N/A	NOTES, LEGEND AND SYMBOLS	NOTAS, LEYENDAS Y SIMBOLOS
V-002	N/A	BENCHMARKS	BASES DE REPLANTEO

V-003	N/A	BENCHMARKS	BASES DE REPLANTEO
V-004	N/A	BENCHMARK SCHEDULES	CUADROS DE BASES DE REPLANTEO
V-005	N/A	BENCHMARK SCHEDULES	CUADROS DE BASES DE REPLANTEO
V-006	N/A	BENCHMARK SCHEDULES	CUADROS DE BASES DE REPLANTEO
V-101	N/A	GENERAL SURVEY - EN -ROUTE SITE	PLANTA GENERAL DE TOPOGRAFIA - SITIO EN-ROUTE
V-102	N/A	GENERAL SURVEY - NEW PIPE	PLANTA GENERAL DE TOPOGRAFÍA - NUEVA TUBERÍA
V-401	N/A	ENLARGED SURVEY - AREA I - EN-ROUTE SITE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA I - SITIO EN-ROUTE
V-402	N/A	ENLARGED SURVEY - AREA II - EN-ROUTE SITE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA II - SITIO EN-ROUTE
V-403	N/A	ENLARGED SURVEY - AREA III - EN-ROUTE SITE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA III - SITIO EN-ROUTE
V-404	N/A	ENLARGED SURVEY - AREA IV - EN-ROUTE SITE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA IV - SITIO EN-ROUTE
V-405	N/A	ENLARGED SURVEY - AREA I - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA I - NUEVA TUBERÍA
V-406	N/A	ENLARGED SURVEY - AREA II - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA II - NUEVA TUBERÍA
V-407	N/A	ENLARGED SURVEY - AREA III - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA III - NUEVA TUBERÍA
V-408	N/A	ENLARGED SURVEY - AREA IV - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA IV - NUEVA TUBERÍA

V-409	N/A	ENLARGED SURVEY - AREA V - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA V - NUEVA TUBERÍA
V-410	N/A	ENLARGED SURVEY - AREA VI - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA VI - NUEVA TUBERÍA
V-411	N/A	ENLARGED SURVEY - AREA VII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA VII - NUEVA TUBERÍA
V-412	N/A	ENLARGED SURVEY - AREA VIII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA VIII - NUEVA TUBERÍA
V-413	N/A	ENLARGED SURVEY - AREA IX - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA IX - NUEVA TUBERÍA
V-414	N/A	ENLARGED SURVEY - AREA X - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA X - NUEVA TUBERÍA
V-415	N/A	ENLARGED SURVEY - AREA XI - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XI - NUEVA TUBERÍA
V-416	N/A	ENLARGED SURVEY - AREA XII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XII - NUEVA TUBERÍA
V-417	N/A	ENLARGED SURVEY - AREA XIII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XIII - NUEVA TUBERÍA
V-418	N/A	ENLARGED SURVEY - AREA XIV - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XIV - NUEVA TUBERÍA
V-419	N/A	ENLARGED SURVEY - AREA XV - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XV - NUEVA TUBERÍA
V-420	N/A	ENLARGED SURVEY - AREA XVI - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XVI - NUEVA TUBERÍA
V-421	N/A	ENLARGED SURVEY - AREA XVII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XVII - NUEVA TUBERÍA

V-422	N/A	ENLARGED SURVEY - AREA XVIII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XVIII - NUEVA TUBERÍA
V-423	N/A	ENLARGED SURVEY - AREA XIX - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XIX - NUEVA TUBERÍA
V-424	N/A	ENLARGED SURVEY - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - NUEVA TUBERÍA
V-425	N/A	ENLARGED SURVEY - DETAIL BPH-VP-1.5 MANHOLE	PLANTA AMPLIADA - DETALLE DE ARQUETA BPH-VP-1.5
V-426	N/A	ENLARGED SURVEY - DETAIL BPH-VP-2 MANHOLE	PLANTA AMPLIADA - DETALLE DE ARQUETA BPH-VP-2
V-427	N/A	ENLARGED SURVEY - DETAIL BPH-VP-3 MANHOLE	PLANTA AMPLIADA - DETALLE DE ARQUETA BPH-VP-3
V-428	N/A	ENLARGED SURVEY - DETAIL ESPAGUETI MANHOLE	PLANTA AMPLIADA - DETALLE DE ARQUETA ESPAGUETI
V-429	N/A	ENLARGED SURVEY - PIPING DETAILS SALADO RIVER CROSSING	PLANTA AMPLIADA - DETALLE PASO DE TUBERIAS RIO SALADO
V-430	N/A	ENLARGED SURVEY - AREA II - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA II - NUEVA TUBERÍA
V-431	N/A	ENLARGED SURVEY - ML-1	PLANTA AMPLIADA DE TOPOGRAFÍA - ML-1
V-432	N/A	ENLARGED SURVEY - ML-2	PLANTA AMPLIADA DE TOPOGRAFÍA - ML-2
V-433	N/A	ENLARGED SURVEY - ML-3	PLANTA AMPLIADA DE TOPOGRAFÍA - ML-3
V-434	N/A	ENLARGED SURVEY - ML-4	PLANTA AMPLIADA DE TOPOGRAFÍA - ML-4

GEOTECHNICAL / GEOTECNIA			
B-001	N/A	GENERAL NOTES	NOTAS GENERALES
BJ101	N/A	BORINGS LOCATION	SITUACIÓN DE SONDEOS
BJ102	N/A	BORINGS LOCATION - KEY PLAN - NEW PIPE	SITUACIÓN DE SONDEOS - PLANTA DE REFERENCIA - NUEVA TUBERÍA
BJ103	N/A	BORINGS LOCATION - AREA I - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA I - NUEVA TUBERÍA
BJ104	N/A	BORINGS LOCATION - AREA II - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA II - NUEVA TUBERÍA
BJ105	N/A	BORINGS LOCATION - AREA III - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA III - NUEVA TUBERÍA
BJ106	N/A	BORINGS LOCATION - AREA IV - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA IV - NUEVA TUBERÍA
BJ107	N/A	BORINGS LOCATION - AREA V - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA V - NUEVA TUBERÍA
BJ108	N/A	BORINGS LOCATION - AREA VI - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA VI - NUEVA TUBERÍA
BJ301	N/A	PROFILES	PERFILES
BJ302	N/A	PROFILES	PERFILES
BJ601	N/A	BORINGS LOGOS - EN ROUTE SITE	REGISTROS DE SONDEOS - SITIO EN-ROUTE
BJ602	N/A	BORINGS LOGOS - EN ROUTE SITE	REGISTROS DE SONDEOS - SITIO EN-ROUTE

BJ603	N/A	BORINGS LOGOS - EN ROUTE SITE	REGISTROS DE SONDEOS - SITIO EN-ROUTE
BJ604	N/A	BORINGS LOGOS - EN ROUTE SITE	REGISTROS DE SONDEOS - SITIO EN-ROUTE
BJ605	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ606	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ607	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ608	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ609	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ610	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ611	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ612	N/A	SCHEDULES	CUADROS
BJ613	N/A	SCHEDULES	CUADROS
BJ614	N/A	SCHEDULES	CUADROS
BJ615	N/A	SCHEDULES	CUADROS
ELECTRICAL / ELECTRICIDAD			

P8	N/A	POWER CABLE FLOW DIAGRAMS - PIER 3, METER RUN AND SHORE BLOCKING VALVES	DIAGRAMAS DE FLUJO DEL CABLE DE FUERZA - MUELLE 3, CONTADOR Y VALVULAS DE BLOQUEO
E3	N/A	ELECTRICAL INSTALLATION DETAILS - 480V PANEL SCHEDULE - NEW PANELS	DETALLES DE LA INSTALACIÓN ELÉCTRICA - CUADRO DEL PANEL DE 480V-NUEVOS PANELES
PD24	N/A	PIPING AND INSTRUMENTATION DIAGRAMS - PIER 3 VALVE PIT 1	DIAGRAMAS DE TUBERÍA INSTRUMENTACIÓN - FOSA DE LA VÁLVULA 1 MUELLE 3
T21	N/A	TRENCHING AND CONDUIT ROUTING - VALVE PIT P3-1 AND PIER 3 METER RUN AREA	TRAZADO DE ZANJA Y TENDIDO DE CONDUCTOS - FOSA DE LA VÁLVULA P3-1 Y ZONA DEL CONTADOR MUELLE 3
PD31	N/A	PIPING AND INSTRUMENTATION DIAGRAMS - BOOSTER PUMP HOUSE VALVE PIT PIER 3	DIAGRAMAS DE TUBERÍA INSTRUMENTACIÓN - FOSA DE LA VÁLVULA 1 MUELLE 3
P6	N/A	POWER CABLE FLOW DIAGRAMS - RIO SALADO EAST AND WEST VALVE PITS, VP 2 AND VALVE PIT PP	DIAGRAMAS DE FLUJO DEL CABLE DE FUERZA - - FOSAS DE LA VÁLVULA DE RÍO SALADO ESTE Y OESTE, VP2 Y FOSA VÁLVULA PP
E4	N/A	ELECTRICAL INSTALLATION DETAILS - 480V PANEL SCHEDULE - NEW PANELS	DETALLES DE LA INSTALACIÓN ELÉCTRICA - CUADRO DEL PANEL DE 480V - NUEVOS PLANELES
PD40	N/A	PIPING AND INSTRUMENTATION DIAGRAMS - BOSTER PUMP HOUSE RIO SALADO VALVE PIT	DIAGRAMAS DE TUBERÍA INSTRUMENTACIÓN - FOSAS DE VÁLVULA RÍO SALADO DEL CUARTO DE BOMBAS DE REFUERZO

T19	N/A	TRENCHING AND CONDUIT ROUTING - BOOSTER PUMP HOUSE TANKS 203 THRU 210 AND 212	TRAZADO DE ZANJA Y TENDIDO DE CONDUCTOS - TANQUES DEL CUARTO DE BOMBAS DE REFUERZO DE 203 A 210 Y 212
-----	-----	---	--

1.3 SUPPLEMENTARY DRAWINGS

These supplementary drawings may not be a part of the contract but are included with the drawings for information.

-- End of Document --

SECTION TABLE OF CONTENTS

DIVISION 00 - PROCUREMENT AND CONTRACTING REQUIREMENTS

DOCUMENT 00 73 04.00 41

SUPPLEMENTARY CONDITIONS FOR PROJECTS IN MEDITERRANEAN AREA

08/08

PART 1 GENERAL

- 1.1 SUBMITTALS
- 1.2 COOPERATION WITH OTHER CONTRACTORS
- 1.3 REFERENCE TO GENERAL PROVISIONS CLAUSES
- 1.4 PUBLIC RELEASE OF INFORMATION
 - 1.4.1 Prohibition
 - 1.4.2 Subcontracts and Purchase Orders
- 1.5 ENGLISH SPEAKING REPRESENTATIVE
- 1.6 CORRESPONDENCE IN THE ENGLISH LANGUAGE
- 1.7 NORMS AND LAWS
- 1.8 INDUSTRY STANDARDS
- 1.9 PROHIBITED ITEMS
- 1.10 PROPRIETARY NAMES
- 1.11 SUBMITTAL OF PROOF OF QUALIFICATIONS AND EXPERIENCE
- 1.12 ORAL MODIFICATION
- 1.13 NO WAIVER BY GOVERNMENT
- 1.14 PRODUCTS SPECIFIED USING USA STANDARDS
 - 1.14.1 Local Products Specified Using United States of America (USA) Standards
 - 1.14.2 USA Manufactured Products Specified Using USA Standards
 - 1.14.3 Consignment, Marking, and Customs Clearance
 - 1.14.4 Excess
- 1.15 FINAL GOVERNING STANDARDS (FGS)
- 1.16 EXCLUSION OF PRICE REVISION LAWS
- 1.17 OPTIONAL PURCHASE
- 1.18 SECURITY REQUIREMENTS
 - 1.18.1 Contractor Employee Base Access Pass:
 - 1.18.2 Responsibility for Physical Security
 - 1.18.3 Access to Operational Areas
 - 1.18.4 Employment
 - 1.18.5 Vehicle Security Inspections
 - 1.18.6 Vehicle List
 - 1.18.7 Control
 - 1.18.8 Photographs
 - 1.18.9 Compliance

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

DOCUMENT 00 73 04.00 41

SUPPLEMENTARY CONDITIONS FOR PROJECTS IN MEDITERRANEAN AREA
08/08

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contractor personnel identification; G

1.2 COOPERATION WITH OTHER CONTRACTORS

Other contractors may be engaged in similar and/or supporting work requiring close cooperation. The Contractor shall cooperate and schedule his work to avoid conflict with and interruption of the work of others insofar as practicable. In the case of conflicts with other contractors that cannot be resolved satisfactorily, the matter shall be referred to the Contracting Officer for decision, and such decision shall be final, subject to right of appeal in accordance with terms of the contract.

1.3 REFERENCE TO GENERAL PROVISIONS CLAUSES

Wherever a reference to a clause of the General Provisions occurs in the technical sections of the specifications, it shall be taken to mean reference to a clause of the Contract Clauses having the same title as the referenced General Provisions clause.

1.4 PUBLIC RELEASE OF INFORMATION

1.4.1 Prohibition

There shall be no public release of information or photographs concerning any aspect of the materials or services relating to this bid, contract, or purchase order or other documents resulting therefrom without the prior written approval of the Contracting Officer.

1.4.2 Subcontracts and Purchase Orders

The Contractor agrees to insert the substance of above paragraph "Prohibition" in each subcontract and purchase order generated for this contract.

1.5 ENGLISH SPEAKING REPRESENTATIVE

At all times when any performance of the work at the site is being conducted by any employee of the Contractor or his subcontractors, the Contractor shall have a representative present on the site who is capable of explaining the work operations and receiving instructions in the English language. The Contracting Officer shall have the right to

determine without appeal of such decision, whether the proposed representative has sufficient technical and lingual capabilities and the Contractor shall immediately replace any individual not acceptable to the Contracting Officer. The Construction Contractor's qualified Project Manager, on-site Superintendent, Quality Control Manager, and Quality Control Specialist shall be fluent (can read, speak, and write) in both English and Spanish languages.

1.6 CORRESPONDENCE IN THE ENGLISH LANGUAGE

It shall be the responsibility of the Contractor to prepare all proposals, change orders, shop drawings and submittals, quality control reports, computations, and all correspondence pertaining to this contract, in the English language; but the Contractor may, for his own record purposes, prepare them in the local language. All correspondence to the Contracting Officer shall be in the English language. In case of dispute or claim, the English version will govern.

1.7 NORMS AND LAWS

In case of differences between project specifications or the accompanying drawings and referenced norms and laws, norms and laws shall govern. In case of differences between project specifications and the accompanying drawings, the specifications shall govern.

1.8 INDUSTRY STANDARDS

When both U.S. and European, or Spanish, Industry Standards are applicable to this project and the standards are in conflict with each other, the most stringent of the industry shall govern.

1.9 PROHIBITED ITEMS

Use of the following items in this construction project is prohibited:

- a. Use of aluminum for electrical conductors.
- b. Embedding aluminum conduit in concrete.
- c. Use of fluorescent light ballasts and other products containing PCB's.
- d. Use of urea-formaldehyde foam insulation products.
- e. Use of any paint/coatings having a lead content of over 0.06 percent by weight of non-volatile content. The use of ozone depleting chemicals is prohibited. The use of zinc-chromate is prohibited.
- f. The use of materials containing asbestos is prohibited.

1.10 PROPRIETARY NAMES

Manufacturer's proprietary names indicated for colors, textures and patterns of materials are for the purpose of color, texture and pattern selection only. Other manufacturers materials are acceptable and provided they closely approximate colors, textures and patterns indicated and provided they conform to all other requirements subject to Contracting Officer's approval.

1.11 SUBMITTAL OF PROOF OF QUALIFICATIONS AND EXPERIENCE

Where qualifications or experience requirements are set forth in the specifications with respect to equipment and equipment installers, written proof of such qualifications or experience must be provided within thirty (30) calendar days after Contract award, and before placing any order for such equipment or before dispatching equipment installers to the project site.

1.12 ORAL MODIFICATION

No oral statement of any person other than the Contracting Officer as provided in the clause in this Contract entitled FAR 52.243-4 "Changes" (AUG 1987), shall in any manner or degree modify or otherwise affect the terms of this Contract.

1.13 NO WAIVER BY GOVERNMENT

The failure of the Government, in any one or more instances, to insist upon the strict performance of any of the terms of this Contract or to exercise any option herein conferred shall not be construed as a waiver or relinquishment to any extent of the right to assert or rely upon such terms or option on any future occasion.

1.14 PRODUCTS SPECIFIED USING USA STANDARDS

1.14.1 Local Products Specified Using United States of America (USA) Standards

There are locally available products specified in this contract that meet USA standards. "Locally available products that meet USA standards" are defined as:

- a. "Products that meet the referenced USA standards, and are manufactured and distributed in-country, or manufactured and distributed in the European Union, and are readily available locally from in-country sources or from the European Union sources"; or
- b. "Products manufactured in the USA and distributed in-country or in the European Union or in the USA, and are readily available locally from in-country sources or from the European Union sources, or from USA sources."

1.14.2 USA Manufactured Products Specified Using USA Standards

There are products to be provided in this contract that are not readily available locally and must be manufactured in the United States of America (USA) and in compliance with USA standards. Products manufactured in another country are considered USA manufactured, if the manufacturers or suppliers are under license by USA manufacturers and the products meet the same applicable specifications and standards.

1.14.3 Consignment, Marking, and Customs Clearance

In order for the Contractor to be exempt from paying duty on material or equipment to be imported for incorporation into the contract work, all shipments shall be consigned to the Contracting Officer and marked for the contract by contract number. Additionally, the Contractor shall supply to the Contracting Officer with certified copies of the confirmed

material/equipment orders and invoices in order that the duty free customs clearance may be accomplished. Failure on the part of the Contractor to comply with the requirements herein will not be grounds for non-enforcement of the provisions of the Clause, "Taxes" of the Contract Clauses.

1.14.4 Excess

The Contractor shall furnish to the Contracting Officer, just prior to completion of this contract, a consolidated inventory of all excess supplies, materials, and equipment imported duty free for use under this contract. The Contract shall either pay required duties on the excesses or re-export the excesses.

1.15 FINAL GOVERNING STANDARDS (FGS)

Final Governing Standards (FGS) for environmental protection at DOD installations and facilities in Spain went into effect June 1994. These standards were developed by comparing and adopting the more protective requirements of the Overseas Environmental Baseline Guidance Document (OEBGD), European Union and national, regional, and local environmental laws and regulations. The FGS are applicable to work under this contract. The FGS (both English and local language versions) may be reviewed at the NAVFAC Rota Office during normal business hours.

1.16 EXCLUSION OF PRICE REVISION LAWS

The Contractor and the United States Government, as the sole parties to this contract, do specifically and irrevocably agree that this contract is a FIRM FIXED-PRICE CONTRACT, and do expressly waive and exclude the application of any and all local Laws, Decrees, Orders, and Regulations, which might mandate, authorize, or permit an increase in the contract price, revision or any other mechanism as a result of increased labor costs, equipment costs, material costs, supplies, subcontractor costs, other overhead costs, direct or indirect, or otherwise.

The risks and consequences of this exclusion of the application of Laws as aforementioned are understood by the parties, and have been taken into consideration in arriving at the contract price set forth herein.

It is the parties' intention that the work contracted for, including any subsequent modifications to this contract, shall be performed on a firm fixed-price basis without price revision of any kind.

Any dispute in this respect arising between the parties will be resolved by strict interpretation of the provision of the contract itself.

1.17 OPTIONAL PURCHASE

All materials and equipment not specifically required to be purchased in the United States of America shall be purchased in Spain and may not be purchased elsewhere without specific written approval of the Contracting Officer. If purchase of material and equipment is not feasible in Spain or it is specified that material and equipment be purchased in the United States, the Clause, "STATUS OF CONTRACTOR AS DIRECT CONTRACTOR" of this Section, will apply.

1.18 SECURITY REQUIREMENTS

1.18.1 Contractor Employee Base Access Pass:

The Contractor shall be responsible for furnishing each employee, and for requiring that each employee engaged in the work display, such identification as may be approved and directed by the Contracting Officer. All prescribed identification shall immediately be delivered to the Contracting Officer for cancellation upon release of any employee.

Contractor personnel identification, including names and photographs, D.N.I. (Documento Nacional de Identidad) numbers, vehicles identification and license plate numbers, of all personnel to be employed in the performance of the work at the site, shall be submitted to the Contracting Officer at least seven (7) days prior to commencement of the work.

Contractor personnel shall be restricted from entering operational buildings or areas without the specific authorization of the Contracting Officer.

1.18.2 Responsibility for Physical Security

The Contractor shall be responsible for physical security of all construction materials, supplies, and equipment of every description (including property which may be Government furnished or owned) provided and/or utilized in the execution of this contract.

1.18.3 Access to Operational Areas

Contractor personnel are expressly prohibited and shall be restricted from entering operational buildings or areas without the specific authorization of the Contracting Officer. To perform work, where "security escorts" are required for access to a facility, the Contractor shall coordinate with the Contracting Officer for access to such facilities.

1.18.4 Employment

Should the continued employment of any person in connection with this contract, or any subcontract hereunder, be deemed by the Contracting Officer to be prejudicial to the interests of the Government, that person shall immediately be removed from the work. In this connection the Contractor agrees that:

a. Cases which may involve disciplinary action against such persons, or the necessity of reassignment or termination of their services, shall be investigated, processed, reported and disposed of, as directed by the Contracting Officer.

b. Employment contracts of all persons employed in connection with this contract, or any subcontract thereunder shall include clauses containing the substance of this clause.

1.18.5 Vehicle Security Inspections

All Contractor and subcontractor vehicles entering the Station including, but not limited to, service vehicles, material delivery vehicles, equipment delivery vehicles, and various other support vehicles, shall first undergo a security inspection.

1.18.6 Vehicle List

Submit an original list of vehicles to be utilized at the work site with the following information for each vehicle:

- a. Make.
- b. Year.
- c. Model.
- d. License number and insurance data's.
- e. Registered owner.

1.18.7 Control

Maintain strict accountability over passes. Immediately report to the source of issue, passes missing or lost and circumstances. If the Contractor has another active contract and one commencing immediately, employee's names may be transferred from one contract to the other. Final payment will not be effected until employees are transferred to another contract or the records are cleared. Furnish a signed letter, countersigned by the source of issue, stating that passes have been turned in.

1.18.8 Photographs

Photographs are prohibited in and around the building unless a written request containing specific justification and details is approved by the security officer via the Contracting Officer. Four days' advance notice is required. Photographs, when approved, shall be taken in the presence of an activity escort.

1.18.9 Compliance

Compliance with the foregoing provisions of this clause by subcontractors shall be the responsibility of the Contractor.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Document --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 11 00

SUMMARY OF WORK

08/15, CHG 2: 08/21

PART 1 GENERAL

- 1.1 SUBMITTALS
- 1.2 WORK COVERED BY CONTRACT DOCUMENTS
 - 1.2.1 Project Description
 - 1.2.2 Location
- 1.3 CONTRACT DRAWINGS
- 1.4 LOCATION OF UNDERGROUND UTILITIES
 - 1.4.1 Notification Prior to Excavation
- 1.5 SALVAGE MATERIAL AND EQUIPMENT

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 11 00

SUMMARY OF WORK
08/15, CHG 2: 08/21

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals
Salvage Plan; G

1.2 WORK COVERED BY CONTRACT DOCUMENTS

1.2.1 Project Description

This work replaces existing aged fuel piping from bulk storage (ML-2) to Pier #3 with a new 20-inch transfer pipeline (approx. pipeline length 13,800 LF). The piers are currently limited in flow and/or concurrent delivery due to existing main line characteristics. The pipeline will be routed underground parallel or nearby to the existing transfer pipelines. This pipeline shall replace the existing 24 inch fuel oil line in kind and shall connect to existing fuel lines in ML-1 and ML-2 where existing 24 inch fuel oil line connects currently. The existing metering assembly near Pier 3 need not be replaced. The new pipe shall connect to the marine loading arms at Pier 3 where the 24 inch line connects currently. The new line shall replace all of the risers from the existing 24 inch fuel oil line on Pier 3 and shall connect to auxiliary and stripping pipelines near building 612 where the existing 24 inch line connects currently. The project will include impressed current cathodic protection to the pipeline and will require the appropriate isolation valves, equipment, coating systems for aboveground and underground piping, and accessories similar to what has been provided for the existing transfer pipelines. The project also provides high point vent and low point drain connection, thermal relief, and all other appurtenances required by UFC 3-460-01. Valves in the 20 inch line shall be full bore plug valves (dual block and bleed). Valves connecting the new pipeline to laterals in ML1 and ML-2 shall match the diameter of the lateral pipeline.

1.2.2 Location

The work is located at the NAVSTA Rota, Spain, approximately as indicated. The exact location will be shown by the Contracting Officer.

1.3 CONTRACT DRAWINGS

The following drawings accompany this specification and are a part thereof.

SHEET NUMBER / NÚMERO DE PLANO	NAVFAC NO.	SHEET TITLE	TÍTULO DE PLANO
GENERAL / GENERAL			
G-001	14142702	COVER SHEET	PORTADA
G-002	14142703	INDEX - SHEET 1	ÍNDICE - PLANO 1
G-003	14142704	INDEX - SHEET 2	ÍNDICE - PLANO 2
G-004	14142705	GENERAL ABBREVIATION SHEET	PLANO DE ABREVIATURAS GENERALES
CIVIL: CIVIL-GENERAL / URBANIZACIÓN: URBANIZACIÓN - GENERAL			
C-001	14142706	CIVIL LEGEND - SHEET 1	LEYENDA DE URBANIZACIÓN - PLANO 1
C-002	14142707	CIVIL LEGEND - SHEET 2	LEYENDA DE URBANIZACIÓN - PLANO 2
CIVIL: CIVIL DEMO / URBANIZACIÓN: URBANIZACIÓN - DEMOLICIONES			
CD101	14142708	SITE DEMOLITION KEY PLAN	PLANTA DE REFERENCIA DE DEMOLICIÓN DE EXTERIORES
CD106	14142709	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD107	14142710	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD108	14142711	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD109	14142712	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD110	14142713	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN

CD111	14142714	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD112	14142715	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD113	14142716	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD114	14142717	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD115	14142718	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD116	14142719	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD117	14142720	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD118	14142721	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD119	14142722	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CD120	14142723	DEMOLITION PLAN	PLANTA DE DEMOLICIÓN
CIVIL: CIVIL-SITE / URBANIZACIÓN: URBANIZACIÓN - EXTERIORES			
C-101	14142724	OVERALL SITE KEY PLAN	PLANTA DE REFERENCIA DE EXTERIORES GENERAL
C-106	14142725	SITE PLAN	PLANTA DE EXTERIORES
C-107	14142726	SITE PLAN	PLANTA DE EXTERIORES
C-108	14142727	SITE PLAN	PLANTA DE EXTERIORES
C-109	14142728	SITE PLAN	PLANTA DE EXTERIORES
C-110	14142729	SITE PLAN	PLANTA DE EXTERIORES
C-111	14142730	SITE PLAN	PLANTA DE EXTERIORES

C-112	14142731	SITE PLAN	PLANTA DE EXTERIORES
C-113	14142732	SITE PLAN	PLANTA DE EXTERIORES
C-114	14142733	SITE PLAN	PLANTA DE EXTERIORES
C-115	14142734	SITE PLAN	PLANTA DE EXTERIORES
C-116	14142735	SITE PLAN	PLANTA DE EXTERIORES
C-117	14142736	SITE PLAN	PLANTA DE EXTERIORES
C-118	14142737	SITE PLAN	PLANTA DE EXTERIORES
C-119	14142738	SITE PLAN	PLANTA DE EXTERIORES
C-120	14142739	SITE PLAN	PLANTA DE EXTERIORES
C-401	14142740	ENLARGED SITE PLAN - SHEET 1	PLANTA AMPLIADA DE EXTERIORES - PLANO 1
C-402	14142741	ENLARGED SITE PLAN - SHEET 2	PLANTA AMPLIADA DE EXTERIORES - PLANO 2
C-403	14142742	ENLARGED SITE PLAN - SHEET 3	PLANTA AMPLIADA DE EXTERIORES - PLANO 3
C-404	14142743	ENLARGED SITE PLAN - SHEET 4	PLANTA AMPLIADA DE EXTERIORES - PLANO 4
C-501	14142744	CIVIL DETAILS	DETALLES DE URBANIZACIÓN
C-502	14142745	CIVIL DETAILS	DETALLES DE URBANIZACIÓN
C-503	14142746	CIVIL DETAILS	DETALLES DE URBANIZACIÓN
C-504	14142747	DETAILS - FENCING	DETALLES - VALLA

C-505	14142748	DETAILS - FENCING	DETALLES - VALLA
CIVIL: CIVIL GRADING / URBANIZACIÓN: NIVELACIÓN DE URBANIZACIÓN			
CG401	14142749	ENLARGED GRADING PLAN - SHEET 1	PLANTA AMPLIADA DE NIVELACIÓN - PLANO 1
CG402	14142750	ENLARGED GRADING PLAN - SHEET 2	PLANTA AMPLIADA DE NIVELACIÓN - PLANO 2
STRUCTURAL / ESTRUCTURA			
S-021	14142751	STRUCTURAL NOTES - SHEET 1	NOTAS DE ESTRUCTURA - PLANO 1
S-022	14142752	STRUCTURAL NOTES - SHEET 2	NOTAS DE ESTRUCTURA - PLANO 2
S-521	14142753	TYPICAL REINFORCING DETAILS	DETALLES TÍPICOS DE ARMADO
S-522	14142754	TYPICAL FOUNDATION DETAILS	DETALLES TÍPICOS DE CIMENTACIÓN
S-523	14142755	MISCELLANEOUS DETAILS	DETALLES VARIOS
S-524	14142756	MISCELLANEOUS DETAILS	DETALLES VARIOS
S-530	14142757	TYPICAL PIPE SUPPORT DETAILS	DETALLES TÍPICOS DE SOPORTES DE TUBERÍAS
MECHANICAL: MECHANICAL-GENERAL / MECÁNICA: MECÁNICA - GENERAL			
M-001	14142758	MECHANICAL LEGEND	LEYENDA DE MECÁNICA
MECHANICAL: MECHANICAL - DEMOLITION / MECÁNICA: MECÁNICA - DEMOLICIONES			
MD101	14142759	MECHANICAL DEMOLITION KEY PLAN	PLANTA DE REFERENCIA DE DEMOLICIÓN

MD102	14142760	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD103	14142761	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD104	14142762	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD105	14142763	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD106	14142764	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD107	14142765	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD108	14142766	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD109	14142767	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD110	14142768	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD111	14142769	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD119	14142770	MECHANICAL DEMOLITION PLAN	PLANTA DE DEMOLICIONES DE MECÁNICA
MD120	14142771	MECHANICAL DEMOLITION PLAN ML-1 TO ML-2	PLANTA DE DEMOLICIONES DE MECÁNICA DE ML1 A ML-2
MD401	14142772	DEMOLITION PLAN VP-1 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO VP-1

MD402	14142773	DEMOLITION PLAN VP-1.5 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO VP-1.5
MD403	14142774	DEMOLITION PLAN BPH-VP-1 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO BPH-VP-1
MD404	14142775	DEMOLITION PLAN VP-3 & 3.5 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO VP-3 Y 3.5
MD410	14142776	DEMOLITION PLAN ML-1 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO ML-1
MD411	14142777	DEMOLITION PLAN ML-2 MANHOLE	PLANTA DE DEMOLICIÓN DE POZO ML-2
MECHANICAL: MECHANICAL - SITE / MECÁNICA: MECÁNICA - EXTERIORES			
M-101	14142778	OVERALL MECHANICAL SITE & KEY PLAN	PLANTA GENERAL EXTERIOR Y DE REFERENCIA
M-102	14142779	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-103	14142780	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-104	14142781	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-105	14142782	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-106	14142783	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-107	14142784	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-108	14142785	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-109	14142786	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-110	14142787	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA

M-111	14142788	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-112	14142789	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-113	14142790	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-114	14142791	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-115	14142792	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-116	14142793	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-117	14142794	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-118	14142795	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-119	14142796	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-120	14142797	MECHANICAL SITE PLAN	PLANTA EXTERIOR DE MECÁNICA
M-401	14142798	ENLARGED SITE PLAN VP-1 MANHOLE	PLANTA EXTERIOR AMPLIADA DEL POZO VP-1
M-402	14142799	ENLARGED SITE PLAN BPH-VP-1.5 MANHOLE	PLANTA EXTERIOR AMPLIADA DEL POZO BPH-VP-1.5
M-403	14142800	ENLARGED SITE PLAN - BPH-VP-1 MANHOLE	PLANTA EXTERIOR AMPLIADA DEL POZO BPH-VP-1
M-404	14142801	ENLARGED SITE PLAN BPH-VP- 3 & 3.5 MANHOLE	PLANTA EXTERIOR AMPLIADA DEL POZO BPH-VP-3 Y 3.5
M-410	14142802	ENLARGED SITE PLAN - ML1	PLANTA EXTERIOR AMPLIADA - ML1
M-411	14142803	ENLARGED SITE PLAN - ML2	PLANTA EXTERIOR AMPLIADA - ML2
M-501	14142804	MECHANICAL DETAILS - SHEET 1	DETALLES DE MECÁNICA - PLANO 1

M-502	14142805	MECHANICAL DETAILS - SHEET 2	DETALLES DE MECÁNICA - PLANO 2
M-503	14142806	MECHANICAL FUELING DETAILS	DETALLES DE COMBUSTIBLE MECÁNICO
M-504	14142807	MECHANICAL FUELING DETAILS	DETALLES DE COMBUSTIBLE MECÁNICO
M-505	14142808	TYPICAL PIPING PLAN	PLANTA TÍPICA DE TUBERÍAS
M-506	14142809	MECHANICAL SECTION	SECCIÓN MECÁNICA
M-601	14142810	ALIGNMENT & PIPE SCHEDULES - SHEET 1	CUADROS DE ALINEACIONES Y TUBERÍAS - PLANO 1
M-602	14142811	ALIGNMENT & PIPE SCHEDULES - SHEET 2	CUADROS DE ALINEACIONES Y TUBERÍAS - PLANO 2
M-611	14142812	PROCESS FLOW DIAGRAM - SHEET 1	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 1
M-612	14142813	PROCESS FLOW DIAGRAM - SHEET 2	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 2
M-613	14142814	PROCESS FLOW DIAGRAM - SHEET 3	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 3
M-614	14142815	PROCESS FLOW DIAGRAM - SHEET 4	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 4
M-615	14142816	PROCESS FLOW DIAGRAM - SHEET 5	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 5
M-616	14142817	PROCESS FLOW DIAGRAM - SHEET 6	DIAGRAMA DE FLUJO DEL PROCESO - PLANO 6
ELECTRIC: ELECTRIC GENERAL / ELECTRICIDAD: ELECTRICIDAD GENERAL			
E-001	14142818	ELECTRICAL SYMBOLS & LEGEND SHEET	PLANO DE SÍMBOLOS ELÉCTRICOS Y LEYENDA

E-002	14142819	ELECTRICAL SYMBOLS & LEGEND SHEET	PLANO DE SÍMBOLOS ELÉCTRICOS Y LEYENDA
E-003	14142820	ELECTRICAL SYMBOLS & LEGEND SHEET	PLANO DE SÍMBOLOS ELÉCTRICOS Y LEYENDA
E-004	14142821	ELECTRICAL SYMBOLS & LEGEND SHEET	PLANO DE SÍMBOLOS ELÉCTRICOS Y LEYENDA
E-005	14142822	GENERAL NOTES SHEET	PLANO DE NOTAS GENERALES
E-006	14142823	GENERAL NOTES SHEET	PLANO DE NOTAS GENERALES
ELECTRIC: ELECTRIC DEMOLITION / ELECTRICIDAD: DEMOLICIONES ELÉCTRICAS			
ED401	14142824	ELECTRIC DEMOLITION PLAN - VP-1 MANHOLE	PLANTA DE DEMOLICIÓN ELÉCTRICA DE POZO VP-1
ED404	14142825	ELECTRIC DEMOLITION PLAN - VP-3 & 3.5 MANHOLE	PLANTA DE DEMOLICIÓN ELÉCTRICA DE POZO VP-3 Y 3.5
ELECTRIC: ELECTRIC SITE / ELECTRICIDAD: EXTERIOR DE ELECTRICIDAD			
E-401	14142826	ELECTRIC VAULT PLAN BPH-VP-1	PLANTA DE ARQUETA ELÉCTRICA BPH-VP-1
E-404	14142827	ELECTRIC VAULT BPH-VP-3 & 3.5	ARQUETA ELÉCTRICA BPH-VP-3 Y 3.5
E-501	14142828	ELECTRIC DETAILS	DETALLES DE ELECTRICIDAD
E-601	14142829	P3-1-VP1 CONTROL PANEL 1 LINE DIAGRAM	DIAGRAMA UNIFILAR DEL PANEL 1 DE CONTROL P3-1-VP1
E-607	14142830	BPH-VP-3 & 3.5 1-LINE DIAGRAM	DIAGRAMA UNIFILAR DEL PANEL DE CONTROL RÍO SALADO ESTE
ELECTRIC: CATHODIC PROTECTION / ELECTRICIDAD: PROTECCIÓN CATÓDICA			

EC101	14142831	CATHODIC PROTECTION KEY PLAN	PLANTA DE REFERENCIA DE PROTECCIÓN CATÓDICA
EC106	14142832	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC107	14142833	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC108	14142834	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC109	14142835	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC110	14142836	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC111	14142837	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC112	14142838	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC113	14142839	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC114	14142840	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC115	14142841	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC116	14142842	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC117	14142843	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC118	14142844	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC119	14142845	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC120	14142846	CATHODIC PROTECTION PLAN	PLANTA DE PROTECCIÓN CATÓDICA
EC501	14142847	CATHODIC PROTECTION DETAILS	DETALLES DE PROTECCIÓN CATÓDICA
ELECTRIC GROUNDING/			

EG101	14142848	ELECTRICAL GROUNDING SITE PLAN AND KEY MAP	PLANTA DE REFERENCIA DE EXTERIORES GENERAL
EG401	14142849	ELECTRICAL ENLARGED GROUNDING PLANS	PLANTAS ELÉCTRICAS AMPLIADAS DE PUESTA A TIERRA
EG501	14142850	ELECTRICAL GROUNDING DETAILS	DETALLES
REFERENCE SHEETS/ PLANOS DE REFERENCIA			
SURVEY / TOPOGRAFÍA			
V-001	N/A	NOTES, LEGEND AND SYMBOLS	NOTAS, LEYENDAS Y SIMBOLOS
V-002	N/A	BENCHMARKS	BASES DE REPLANTEO
V-003	N/A	BENCHMARKS	BASES DE REPLANTEO
V-004	N/A	BENCHMARK SCHEDULES	CUADROS DE BASES DE REPLANTEO
V-005	N/A	BENCHMARK SCHEDULES	CUADROS DE BASES DE REPLANTEO
V-006	N/A	BENCHMARK SCHEDULES	CUADROS DE BASES DE REPLANTEO
V-101	N/A	GENERAL SURVEY - EN -ROUTE SITE	PLANTA GENERAL DE TOPOGRAFIA - SITIO EN-ROUTE
V-102	N/A	GENERAL SURVEY - NEW PIPE	PLANTA GENERAL DE TOPOGRAFÍA - NUEVA TUBERÍA
V-401	N/A	ENLARGED SURVEY - AREA I - EN-ROUTE SITE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA I - SITIO EN-ROUTE
V-402	N/A	ENLARGED SURVEY - AREA II - EN-ROUTE SITE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA II - SITIO EN-ROUTE
V-403	N/A	ENLARGED SURVEY - AREA III - EN-ROUTE SITE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA III - SITIO EN-ROUTE

V-404	N/A	ENLARGED SURVEY - AREA IV - EN-ROUTE SITE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA IV - SITIO EN-ROUTE
V-405	N/A	ENLARGED SURVEY - AREA I - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA I - NUEVA TUBERÍA
V-406	N/A	ENLARGED SURVEY - AREA II - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA II - NUEVA TUBERÍA
V-407	N/A	ENLARGED SURVEY - AREA III - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA III - NUEVA TUBERÍA
V-408	N/A	ENLARGED SURVEY - AREA IV - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA IV - NUEVA TUBERÍA
V-409	N/A	ENLARGED SURVEY - AREA V - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA V - NUEVA TUBERÍA
V-410	N/A	ENLARGED SURVEY - AREA VI - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA VI - NUEVA TUBERÍA
V-411	N/A	ENLARGED SURVEY - AREA VII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA VII - NUEVA TUBERÍA
V-412	N/A	ENLARGED SURVEY - AREA VIII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA VIII - NUEVA TUBERÍA
V-413	N/A	ENLARGED SURVEY - AREA IX - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA IX - NUEVA TUBERÍA
V-414	N/A	ENLARGED SURVEY - AREA X - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA X - NUEVA TUBERÍA
V-415	N/A	ENLARGED SURVEY - AREA XI - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XI - NUEVA TUBERÍA
V-416	N/A	ENLARGED SURVEY - AREA XII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XII - NUEVA TUBERÍA

V-417	N/A	ENLARGED SURVEY - AREA XIII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XIII - NUEVA TUBERÍA
V-418	N/A	ENLARGED SURVEY - AREA XIV - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XIV - NUEVA TUBERÍA
V-419	N/A	ENLARGED SURVEY - AREA XV - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XV - NUEVA TUBERÍA
V-420	N/A	ENLARGED SURVEY - AREA XVI - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XVI - NUEVA TUBERÍA
V-421	N/A	ENLARGED SURVEY - AREA XVII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XVII - NUEVA TUBERÍA
V-422	N/A	ENLARGED SURVEY - AREA XVIII - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XVIII - NUEVA TUBERÍA
V-423	N/A	ENLARGED SURVEY - AREA XIX - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA XIX - NUEVA TUBERÍA
V-424	N/A	ENLARGED SURVEY - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - NUEVA TUBERÍA
V-425	N/A	ENLARGED SURVEY - DETAIL BPH-VP-1.5 MANHOLE	PLANTA AMPLIADA - DETALLE DE ARQUETA BPH-VP-1.5
V-426	N/A	ENLARGED SURVEY - DETAIL BPH-VP-2 MANHOLE	PLANTA AMPLIADA - DETALLE DE ARQUETA BPH-VP-2
V-427	N/A	ENLARGED SURVEY - DETAIL BPH-VP-3 MANHOLE	PLANTA AMPLIADA - DETALLE DE ARQUETA BPH-VP-3
V-428	N/A	ENLARGED SURVEY - DETAIL ESPAGUETI MANHOLE	PLANTA AMPLIADA - DETALLE DE ARQUETA ESPAGUETI
V-429	N/A	ENLARGED SURVEY - PIPING DETAILS SALADO RIVER CROSSING	PLANTA AMPLIADA - DETALLE PASO DE TUBERIAS RIO SALADO

V-430	N/A	ENLARGED SURVEY - AREA II - NEW PIPE	PLANTA AMPLIADA DE TOPOGRAFÍA - AREA II - NUEVA TUBERÍA
V-431	N/A	ENLARGED SURVEY - ML-1	PLANTA AMPLIADA DE TOPOGRAFÍA - ML-1
V-432	N/A	ENLARGED SURVEY - ML-2	PLANTA AMPLIADA DE TOPOGRAFÍA - ML-2
V-433	N/A	ENLARGED SURVEY - ML-3	PLANTA AMPLIADA DE TOPOGRAFÍA - ML-3
V-434	N/A	ENLARGED SURVEY - ML-4	PLANTA AMPLIADA DE TOPOGRAFÍA - ML-4
GEOTECHNICAL / GEOTECNIA			
B-001	N/A	GENERAL NOTES	NOTAS GENERALES
BJ101	N/A	BORINGS LOCATION	SITUACIÓN DE SONDEOS
BJ102	N/A	BORINGS LOCATION - KEY PLAN - NEW PIPE	SITUACIÓN DE SONDEOS - PLANTA DE REFERENCIA - NUEVA TUBERÍA
BJ103	N/A	BORINGS LOCATION - AREA I - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA I -NUEVA TUBERÍA
BJ104	N/A	BORINGS LOCATION - AREA II - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA II - NUEVA TUBERÍA
BJ105	N/A	BORINGS LOCATION - AREA III - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA III - NUEVA TUBERÍA
BJ106	N/A	BORINGS LOCATION - AREA IV - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA IV - NUEVA TUBERÍA
BJ107	N/A	BORINGS LOCATION - AREA V - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA V - NUEVA TUBERÍA

BJ108	N/A	BORINGS LOCATION - AREA VI - NEW PIPE	SITUACIÓN DE SONDEOS - ÁREA VI - NUEVA TUBERÍA
BJ301	N/A	PROFILES	PERFILES
BJ302	N/A	PROFILES	PERFILES
BJ601	N/A	BORINGS LOGOS - EN ROUTE SITE	REGISTROS DE SONDEOS - SITIO EN-ROUTE
BJ602	N/A	BORINGS LOGOS - EN ROUTE SITE	REGISTROS DE SONDEOS - SITIO EN-ROUTE
BJ603	N/A	BORINGS LOGOS - EN ROUTE SITE	REGISTROS DE SONDEOS - SITIO EN-ROUTE
BJ604	N/A	BORINGS LOGOS - EN ROUTE SITE	REGISTROS DE SONDEOS - SITIO EN-ROUTE
BJ605	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ606	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ607	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ608	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ609	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ610	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA
BJ611	N/A	BORING LOGOS - NEW PIPE SITE	REGISTROS DE SONDEOS - SITIO NUEVA TUBERÍA

BJ612	N/A	SCHEDULES	CUADROS
BJ613	N/A	SCHEDULES	CUADROS
BJ614	N/A	SCHEDULES	CUADROS
BJ615	N/A	SCHEDULES	CUADROS
ELECTRICAL / ELECTRICIDAD			
P8	N/A	POWER CABLE FLOW DIAGRAMS - PIER 3, METER RUN AND SHORE BLOCKING VALVES	DIAGRAMAS DE FLUJO DEL CABLE DE FUERZA - MUELLE 3, CONTADOR Y VALVULAS DE BLOQUEO
E3	N/A	ELECTRICAL INSTALLATION DETAILS - 480V PANEL SCHEDULE - NEW PANELS	DETALLES DE LA INSTALACIÓN ELÉCTRICA - CUADRO DEL PANEL DE 480V-NUEVOS PANELES
PD24	N/A	PIPING AND INSTRUMENTATION DIAGRAMS - PIER 3 VALVE PIT 1	DIAGRAMAS DE TUBERÍA INSTRUMENTACIÓN - FOSA DE LA VÁLVULA 1 MUELLE 3
T21	N/A	TRENCHING AND CONDUIT ROUTING - VALVE PIT P3-1 AND PIER 3 METER RUN AREA	TRAZADO DE ZANJA Y TENDIDO DE CONDUCTOS - FOSA DE LA VÁLVULA P3-1 Y ZONA DEL CONTADOR MUELLE 3
PD31	N/A	PIPING AND INSTRUMENTATION DIAGRAMS - BOOSTER PUMP HOUSE VALVE PIT PIER 3	DIAGRAMAS DE TUBERÍA INSTRUMENTACIÓN - FOSA DE LA VÁLVULA 1 MUELLE 3
P6	N/A	POWER CABLE FLOW DIAGRAMS - RIO SALADO EAST AND WEST VALVE PITS, VP 2 AND VALVE PIT PP	DIAGRAMAS DE FLUJO DEL CABLE DE FUERZA - - FOSAS DE LA VÁLVULA DE RÍO SALADO ESTE Y OESTE, VP2 Y FOSA VÁLVULA PP
E4	N/A	ELECTRICAL INSTALLATION DETAILS - 480V PANEL SCHEDULE - NEW PANELS	DETALLES DE LA INSTALACIÓN ELÉCTRICA - CUADRO DEL PANEL DE 480V - NUEVOS PLANELES

PD40	N/A	PIPING AND INSTRUMENTATION DIAGRAMS - BOSTER PUMP HOUSE RIO SALADO VALVE PIT	DIAGRAMAS DE TUBERÍA INSTRUMENTACIÓN - FOSAS DE VÁLVULA RÍO SALADO DEL CUARTO DE BAMBAS DE REFUERZO
T19	N/A	TRENCHING AND CONDUIT ROUTING - BOOSTER PUMP HOUSE TANKS 203 THRU 210 AND 212	TRAZADO DE ZANJA Y TENDIDO DE CONDUCTOS - TANQUES DEL CUARTO DE BOMBAS DE REFUERZO DE 203 A 210 Y 212

Five sets of full size contract drawings, maps, and specifications will be furnished to the Contractor without charge. Reference publications will not be furnished. Immediately check furnished drawings and notify the Government of any discrepancies.

1.4 LOCATION OF UNDERGROUND UTILITIES

Obtain digging permits prior to start of excavation, and comply with Installation requirements for locating and marking underground utilities. Contact local utility locating service a minimum of 48 hours prior to excavating, to mark utilities, and within sufficient time required if work occurs on a Monday or after a Holiday. Verify existing utility locations indicated on contract drawings, within area of work.

Identify and mark all other utilities not managed and located by the local utility companies. Scan the construction site with Ground Penetrating Radar (GPR), electromagnetic, or sonic equipment, and mark the surface of the ground or paved surface where existing underground utilities or utilities encased in pier structures are discovered. Verify the elevations of existing piping, utilities, and any type of underground or

encased obstruction not indicated, or specified to be removed, that is indicated or discovered during scanning, in locations to be traversed by piping, ducts, and other work to be conducted or installed. Verify elevations before installing new work closer than nearest manhole or other structure at which an adjustment in grade can be made.

1.4.1 Notification Prior to Excavation

Notify the Contracting Officer at least 48 hours prior to starting excavation work.

1.5 SALVAGE MATERIAL AND EQUIPMENT

Items designated by the Contracting Officer to be salvaged remain the property of the Government. Segregate, itemize, deliver and off-load the salvaged property at the Government designated storage area located within 5.0 kilometers of the construction site.

Provide a salvage plan, listing material and equipment to be salvaged, and their storage location. Maintain property control records for material or equipment designated as salvage. Provide a system for property control in the salvage plan. Store and protect salvaged materials and equipment until disposition by the Contracting Officer.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 14 00

WORK RESTRICTIONS

11/11, CHG 14: 02/22

PART 1 GENERAL

- 1.1 SUBMITTALS
- 1.2 SPECIAL SCHEDULING REQUIREMENTS
- 1.3 CONTRACTOR ACCESS AND USE OF PREMISES
 - 1.3.1 Activity Regulations
 - 1.3.1.1 Subcontractors and Personnel Contacts
 - 1.3.1.2 Additional Personnel Requirements
 - 1.3.1.3 Employee List
 - 1.3.1.4 Personnel Entry Approval
 - 1.3.1.5 No Smoking Policy
 - 1.3.2 Working Hours
 - 1.3.3 Work Outside Regular Hours
 - 1.3.4 Exclusionary Period
 - 1.3.5 Occupied and Existing Building(s)
 - 1.3.6 Utility Cutovers and Interruptions
- 1.4 NAVFAC EURAFCENT REQUIREMENTS AND RESTRICTIONS
 - 1.4.1 Building Entrance Restrictions
 - 1.4.2 Public Release of Information
 - 1.4.3 Responsibility for Physical Security
 - 1.4.4 Employment
 - 1.4.5 Compliance
 - 1.4.6 Cultural Resources
 - 1.4.7 Unexploded Ordnance
 - 1.4.7.1 Ordnance Removal Requirements
 - 1.4.7.2 Perimeter Fence Restrictions

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 14 00

WORK RESTRICTIONS
11/11, CHG 14: 02/22

PART 1 GENERAL

1.1 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

List of Contact Personnel

SD-07 Certificates

Proof of IMD Ordnance Work Authorization; G

1.2 SPECIAL SCHEDULING REQUIREMENTS

- a. Permission to interrupt any Activity roads, railroads, or utility service must be requested in writing a minimum of 15 calendar days prior to the desired date of interruption.
- b. The work under this contract requires special attention to the scheduling and conduct of the work in connection with existing operations. Identify on the construction schedule each factor which constitutes a potential interruption to operations.

The following conditions apply:

- (1) Maintain existing sub-grade piping and surface valves in working condition at all times.

1.3 CONTRACTOR ACCESS AND USE OF PREMISES

1.3.1 Activity Regulations

1.3.1.1 Subcontractors and Personnel Contacts

Provide a list of contact personnel of the Contractor and subcontractors including addresses and telephone numbers for use in the event of an emergency. As changes occur and additional information becomes available, correct and change the information contained in previous lists.

1.3.1.2 Additional Personnel Requirements

1.3.1.3 Employee List

The Contractor must provide to the Contracting officer, in writing, the names of two designated representatives authorized to request personnel and vehicle passes for employees and subcontractor's employees prior to commencement of work under this contract. The Contractor must adhere to

the requirements of "Important Clarifications - Contractors - How to Gain Access," dated 31 October 1995, in obtaining access to the Naval Air Station complex for the life of the contract. A copy of these requirements will be provided at the preconstruction meeting.

1.3.1.4 Personnel Entry Approval

Failure to obtain entry approval will not affect the contract price or time of completion.

1.3.1.5 No Smoking Policy

Smoking is prohibited within and outside of all buildings on installation, except in designated smoking areas. This applies to existing buildings, buildings under construction and buildings under renovation. Discarding tobacco materials other than into designated tobacco receptacles is considered littering and is subject to fines. The Contracting Officer will identify designated smoking areas.

1.3.2 Working Hours

Regular working hours will consist of an 9 hour period established by the Contracting Officer, excluding Government holidays.

1.3.3 Work Outside Regular Hours

Work outside regular working hours requires Contracting Officer approval. Make application 15 calendar days prior to such work to allow arrangements to be made by the Government for inspecting the work in progress, giving the specific dates, hours, location, type of work to be performed, contract number and project title. Based on the justification provided, the Contracting Officer may approve work outside regular hours. During periods of darkness, the different parts of the work must be lighted in a manner approved by the Contracting Officer. Make utility cutovers after normal working hours or on Saturdays, Sundays, and Government holidays unless directed otherwise.

1.3.4 Exclusionary Period

No work may be performed during the period 8:00 p.m. to 5:00 a.m., inclusive, without prior written approval of the Contracting Officer. This period has not been considered in computing the time allowed for the performance of this contract.

1.3.5 Occupied and Existing Building(s)

The Contractor shall be working around existing buildings which are occupied. Do not enter the building(s) without prior approval of the Contracting Officer.

The existing buildings and their contents must be kept secure at all times. Provide temporary closures as required to maintain security as directed by the Contracting Officer.

1.3.6 Utility Cutovers and Interruptions

- a. Make utility cutovers and interruptions after normal working hours or on Saturdays, Sundays, and Government holidays. Conform to procedures required in paragraph WORK OUTSIDE REGULAR HOURS.

- b. Ensure that new utility lines are complete, except for the connection, before interrupting existing service.
- c. Interruption to water, sanitary sewer, storm sewer, telephone service, electric service, air conditioning, heating, fire alarm, compressed air, are considered utility cutovers pursuant to the paragraph WORK OUTSIDE REGULAR HOURS. Interruptions are to be coordinated with the Contracting Officer.

1.4 NAVFAC EURAFCENT REQUIREMENTS AND RESTRICTIONS

1.4.1 Building Entrance Restrictions

Contractor personnel are restricted from entering operational buildings or areas without the specific authorization of the Contracting Officer. To perform work, where "security escorts" are required for access to a facility, coordinate with the Contracting Officer for access to such facilities.

1.4.2 Public Release of Information

There must be no public release of information or photographs concerning any aspect of the materials or services relating to this contract or other documents resulting therefrom without the prior written approval of the Contracting Officer.

The Contractor agrees to insert the substance of above paragraph in each subcontract and purchase order generated for this contract.

1.4.3 Responsibility for Physical Security

The Contractor is responsible for physical security of construction materials, supplies, and equipment of every description (including property which may be Government furnished or owned) provided or utilized in the execution of this contract.

1.4.4 Employment

Should the continued employment of any person in connection with this contract, or any subcontract thereunder, be deemed by the Contracting Officer to be prejudicial to the interests of the Government, that person must immediately be removed from the work. In this connection the Contractor agrees that:

- a. Cases which may involve disciplinary action against such persons, or the necessity of reassignment or termination of their services, shall be investigated, processed, reported and disposed of, as directed by the Contracting Officer.
- b. Employment contracts of all persons employed in connection with this contract, or any subcontract thereunder shall include clauses containing the substance of this clause.

1.4.5 Compliance

Compliance with the foregoing provisions of this clause by subcontractors is the responsibility of the Contractor.

1.4.6 Cultural Resources

Cultural resources such as archaeological remains or villa remains may be uncovered during construction excavation operations on and off the Installation. If during excavations, suspected cultural remains are found, excavation operations must immediately cease and Contractor must notify the Contracting Officer who will contact the Installation Environmental office.

1.4.7 Unexploded Ordnance

The Contractor is advised that unexploded ordnance may exist in areas where excavation work is required. During the entire life of this Contract, perform both visual and electronic inspection of the work areas to prevent danger to personnel and loss of equipment from inadvertently exploding old ordnance.

During excavation work, use caution when unknown objects are encountered and thoroughly identify the object prior to removal. Upon discovery of suspected unexploded ordnance in the course of contract work, immediately clear the area and contact the Contracting Officer. Resultant exploratory and removal work may be subject to negotiations for a contract change in accordance with the Contract Clauses.

1.4.7.1 Ordnance Removal Requirements

Upon issuance of contract change and notice to proceed by the Contracting Officer, provide exploratory excavation and removal of suspected unexploded ordnance. This work must be conducted only by a firm authorized by the Italian Ministry of Defense (IMD) to perform such work, and approved by the Contracting Officer. Submit proof of IMD ordnance work authorization to the Contracting Officer for review and approval prior to conducting ordnance work.

Have the approved ordnance removal firm committed to an on-call basis in the event that suspected unexploded ordnance is encountered during contract work.

1.4.7.2 Perimeter Fence Restrictions

A 9.2 meter (minimum) clear zone must be maintained inside the base perimeter fence. No equipment, material, vehicles, etc may be within the clear zone.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 20 00

PRICE AND PAYMENT PROCEDURES

11/20, CHG 2: 08/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 SCHEDULE OF PRICES
 - 1.3.1 Data Required
 - 1.3.2 Payment Schedule Instructions
- 1.4 CONTRACT MODIFICATIONS
- 1.5 CONTRACTOR'S INVOICE AND CONTRACT PERFORMANCE STATEMENT
 - 1.5.1 Content of Invoice
 - 1.5.2 Submission of Invoices
 - 1.5.3 Final Invoice
- 1.6 PAYMENTS TO THE CONTRACTOR
 - 1.6.1 Obligation of Government Payments
 - 1.6.2 Payment for Onsite and Offsite Materials
- 1.7 UNIT PRICE PAYMENT ITEMS
 - 1.7.1 Excavation
 - 1.7.1.1 Payment
 - 1.7.1.2 Measurement
 - 1.7.1.3 Unit of Measure
 - 1.7.2 Deformed Steel Bars for Concrete Reinforcement
 - 1.7.2.1 Payment
 - 1.7.2.2 Measurement
 - 1.7.2.3 Unit of Measure
- 1.8 CONTAMINATED SOIL REMOVAL
 - 1.8.1 Unit of Measure

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 20 00

PRICE AND PAYMENT PROCEDURES
11/20, CHG 2: 08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A615/A615M	(2020) Standard Specification for Deformed and Plain Carbon-Steel Bars for Concrete Reinforcement
-----------------	---

U.S. ARMY CORPS OF ENGINEERS (USACE)

EP 1110-1-8	(2021) Construction Equipment Ownership and Operating Expense Schedule
-------------	--

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Schedule of Prices; G

1.3 SCHEDULE OF PRICES

1.3.1 Data Required

Within 15 calendar days of Contract Award, prepare and deliver to the Contracting Officer a Schedule of Prices as directed by the Contracting Officer. Schedule of Prices must have cost summarized and totals provided for each construction category. Provide a detailed breakdown of the Contract price, giving quantities for each of the various kinds of work, unit prices and extended prices. Contractor overhead and profit including salaries for field office personnel, if applicable, must be proportionately spread over all pay items and not included as individual pay items.

1.3.2 Payment Schedule Instructions

Payments will not be made until the Schedule of Prices has been submitted to and accepted by the Contracting Officer.

a. Primary Facilities Cost Breakdown:

Defined as work on the primary facilities out to the 1.5 m line. Work

out to the 1.5 m line includes construction encompassed within a theoretical line 1.5 m from the face of exterior walls and includes attendant construction, such as pad mounted HVAC cooling equipment, cooling towers, and transformers placed beyond the 1.5 m line.

b. Supporting Facilities Cost Breakdown:

Defined as site work, including incidental work, outside the 1.5 m line.

1.4 CONTRACT MODIFICATIONS

In conjunction with the Contract Clause DFARS 252.236-7000 Modification Proposals-Price Breakdown, and where actual ownership and operating costs of construction equipment cannot be determined from Contractor accounting records, base equipment use rates upon the applicable provisions of the EP 1110-1-8.

1.5 CONTRACTOR'S INVOICE AND CONTRACT PERFORMANCE STATEMENT

1.5.1 Content of Invoice

Requests for payment will be processed in accordance with the Contract Clause FAR 52.232-27 Prompt Payment for Construction Contracts and FAR 52.232-5 Payments Under Fixed-Price Construction Contracts. Invoices not completed in accordance with contract requirements will be returned to the Contractor for correction of the deficiencies. The requests for payment shall include the documents listed below.

- a. The Contractor's invoice, on NAVFAC Form 7300/30 furnished by the Government, showing in summary form, the basis for arriving at the amount of the invoice. Form 7300/30 must include certification by Quality Control (QC) Manager as required by the Contract.
- b. Contractor's Monthly Estimate for Voucher and Contractors Certification (NAVFAC Form 4330) with Subcontractor and supplier payment certification. Other documents, including but not limited to, that need to be received prior to processing payment include the following submittals as required. These items are still required monthly even when a pay voucher is not submitted.
- c. Monthly Work-hour report.
- d. Updated Construction Progress Schedule and tabular reports required by the contract.
- e. Contractor Safety Self Evaluation Checklist.
- f. Updated submittal register.
- g. Solid Waste Disposal Report.
- h. Certified payrolls.
- i. Updated testing logs.
- j. Other supporting documents as requested.

1.5.2 Submission of Invoices

If DFARS Clause 252.232-7006 Wide Area WorkFlow Payment Instructions is included in the Contract, provide the documents listed in above paragraph CONTENT OF INVOICE in their entirety as attachments in Wide Area Work Flow (WAWF) for each invoice submitted. The maximum size of each WAWF attachment is two megabytes, but there are no limits on the number of attachments. If a document cannot be attached in WAWF due to system or size restriction, provide it as instructed by the Contracting Officer.

Monthly invoices and supporting forms for work performed through the anniversary award date of the Contract must be submitted to the Contracting Officer within 5 calendar days of the date of invoice. For example, if Contract award date is the 7th of the month, the date of each monthly invoice must be the 7th and the invoice must be submitted by the 12th of the month.

1.5.3 Final Invoice

- a. A final invoice must be accompanied by the certification required by DFARS 252.247.7023 Transportation of Supplies by Sea, and the Contractor's Final Release. If the Contractor is incorporated, the Final Release must contain the corporate seal. An officer of the corporation must sign and the corporate secretary must certify the Final Release.
- b. For final invoices being submitted via WAWF, the original Contractor's Final Release Form and required certification of Transportation of Supplies by Sea must be provided directly to the respective Contracting Officer prior to submission of the final invoice. Once receipt of the original Final Release Form and required certification of Transportation of Supplies by Sea has been confirmed by the Contracting Officer, the Contractor must then submit final invoice and attach a copy of the Final Release Form and required certification of Transportation of Supplies by Sea in WAWF.
- c. Final invoices not accompanied by the Contractor's Final Release and required certification of Transportation of Supplies by Sea will be considered incomplete and will be returned to the Contractor.

1.6 PAYMENTS TO THE CONTRACTOR

Payments will be made on submission of itemized requests by the Contractor which comply with the requirements of this section, and will be subject to reduction for overpayments or increase for underpayments made on previous payments to the Contractor.

1.6.1 Obligation of Government Payments

The obligation of the Government to make payments required under the provisions of this Contract will, at the discretion of the Contracting Officer, be subject to reductions and suspensions permitted under the FAR and agency regulations including the following in accordance with FAR FAR 32.103 Progress Payments Under Construction Contracts:

- a. Reasonable deductions due to defects in material or workmanship;
- b. Claims which the Government may have against the Contractor under or in connection with this Contract;

- c. Unless otherwise adjusted, repayment to the Government upon demand for overpayments made to the Contractor; and
- d. Failure to maintain accurate "as-built" or record drawings in accordance with FAR 52.236.21.

1.6.2 Payment for Onsite and Offsite Materials

Progress payments may be made to the Contractor for materials delivered on the site, for materials stored off construction sites, or materials that are in transit to the construction sites under the following conditions:

- a. FAR 52.232-5(b) Payments Under Fixed Price Construction Contracts.
- b. Materials delivered on the site but not installed, including completed preparatory work, and off-site materials to be considered for progress payment must be major high cost, long lead, special order, or specialty items, not susceptible to deterioration or physical damage in storage or in transit to the construction site. Examples of materials acceptable for payment consideration include, but are not limited to, structural steel, non-magnetic steel, non-magnetic aggregate, equipment, machinery, large pipe and fittings, precast/prestressed concrete products, plastic lumber (e.g., fender piles/curbs), and high-voltage electrical cable. Materials not acceptable for payment include consumable materials such as nails, fasteners, conduits, gypsum board, glass, insulation, and wall coverings.
- c. Materials to be considered for progress payment prior to installation must be specifically and separately identified in the Contractor's estimates of work submitted for the Contracting Officer's approval in accordance with Schedule of Prices requirement of this Contract. Requests for progress payment consideration for such items must be supported by documents establishing their value and that the title requirements of the clause at FAR 52.232-5 Payments Under Fixed-Price Construction Contracts have been met.
- d. Materials are adequately insured and protected from theft and exposure.
- e. Provide a written consent from the surety company with each payment request for offsite materials.
- f. Materials to be considered for progress payments prior to installation must be stored either in Hawaii, Guam, Puerto Rico, or the Continental United States. Other locations are subject to written approval by the Contracting Officer.
- g. Materials in transit to the job site or storage site are not acceptable for payment.

1.7 UNIT PRICE PAYMENT ITEMS

Payment items for the work of this Contract on which the Contract unit price payments will be made are listed in the BIDDING PROPOSAL SCHEDULE and described below. The unit price and payment made for each item listed constitutes full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports,

and for performing all work required for each of the unit price items.

1.7.1 Excavation

1.7.1.1 Payment

Payment will be made for costs associated with excavation for the channel, which includes performing required excavation and other operations incidental thereto, Contractor-furnished disposal area(s) and disposition of excess excavated material and unsuitable and frozen materials.

1.7.1.2 Measurement

The total quantity of excavated material for which payment will be made will be the theoretical quantity between the ground surface as determined by a survey and the grade and slope of the theoretical cross sections indicated. No allowance will be made for overdepth excavation or for the removal of any material outside the required slope lines unless authorized.

1.7.1.3 Unit of Measure

Unit of measure: cubic meter.

1.7.2 Deformed Steel Bars for Concrete Reinforcement

1.7.2.1 Payment

Payment will be made for costs associated with furnishing, transporting, delivering, and placing deformed steel bars for concrete reinforcement, which includes steel in laps as indicated or as required. No payment will be made for the additional steel in laps which are authorized for the convenience of the Contractor. No separate payment will be made for accessories; include payment in the Contract unit price for the items of work to which the accessories are incidental.

1.7.2.2 Measurement

The measured lengths of deformed steel bars for concrete reinforcement will be converted to weights for the size of bars listed by the use of the nominal weights per lineal meter specified in ASTM A615/A615M.

1.7.2.3 Unit of Measure

Unit of measure: per kilogram in place.

1.8 CONTAMINATED SOIL REMOVAL

Payment items for the work of this Contract on which the Contract unit price payments will be made are listed in the BIDDING and described below. The unit price and payment made for each item listed must constitute full compensation for furnishing all plant, labor, materials, and equipment, and performing any associated Contractor quality control, environmental protection, meeting safety requirements, tests and reports, and for performing all work required for each of the unit price items.

1.8.1 Unit of Measure

Unit of measure: cubic meter.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 30 00

ADMINISTRATIVE REQUIREMENTS

11/20, CHG 2: 05/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 VIEW LOCATION MAP
- 1.4 PROGRESS AND COMPLETION PICTURES
- 1.5 MINIMUM INSURANCE REQUIREMENTS
- 1.6 SUPERVISION
 - 1.6.1 Superintendent Qualifications
 - 1.6.2 Minimum Communication Requirements
 - 1.6.3 Duties
 - 1.6.4 Non-Compliance Actions
- 1.7 PRECONSTRUCTION MEETING CONFERENCE
 - 1.7.1 Attendees
- 1.8 FACILITY TURNOVER PLANNING MEETINGS (Red Zone Meetings)
 - 1.8.1 Red Zone Checklist
 - 1.8.2 Meetings
- 1.9 EXPORT LICENSES FOR OVERSEAS PROJECTS
- 1.10 WAIVER FOR WORKER'S COMPENSATION
- 1.11 PARTNERING
 - 1.11.1 Facilitated (Formal) Partnering
- 1.12 MOBILIZATION

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 30 00

ADMINISTRATIVE REQUIREMENTS
11/20, CHG 2: 05/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety and Health Requirements
Manual

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

15 CFR 772 Definition of Terms

15 CFR 773 Special Licensing Procedures

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

View Location Map

Progress and Completion Pictures

1.3 VIEW LOCATION MAP

Submit, prior to or with the first digital photograph submittals, a sketch or drawing indicating the required photographic locations. Update as required if the locations are moved.

1.4 PROGRESS AND COMPLETION PICTURES

Photographically document site conditions prior to start of construction operations. Provide monthly, and within one month of the completion of work, digital photographs, 1600x1200x24 bit true color minimum resolution in JPEG file format showing the sequence and progress of work. Take a minimum of 20 digital photographs each week throughout the entire project from a minimum of ten different viewpoints selected by the Contractor unless otherwise directed by the Contracting Officer. Submit with the monthly invoice two sets of digital photographs, each set on a separate compact disc (CD) or data versatile disc (DVD), cumulative of all photos to date. Indicate photographs demonstrating environmental procedures. Provide photographs for each month in a separate monthly directory and name each file to indicate its location on the view location sketch. Also

provide the view location sketch on the CD or DVD as a digital file. Include a date designator in file names. Photographs provided are for unrestricted use by the Government.

1.5 MINIMUM INSURANCE REQUIREMENTS

Provide the minimum insurance coverage required by FAR 28.307-2 Liability, during the entire period of performance under this contract. Provide other insurance coverage as required by Spanish law.

1.6 SUPERVISION

1.6.1 Superintendent Qualifications

Provide project superintendent with a minimum of 10 years experience in construction with at least 5 of those years as a superintendent on projects similar in size and complexity. The individual must be familiar with the requirements of EM 385-1-1 and have experience in the areas of hazard identification and safety compliance. The individual must be capable of interpreting a critical path schedule and construction drawings. The qualification requirements for the alternate superintendent are the same as for the project superintendent. The Contracting Officer may request proof of the superintendent's qualifications at any point in the project if the performance of the superintendent is in question.

For projects where the superintendent is permitted to also serve as the Quality Control (QC) Manager as established in Section 01 45 00.00 10 01 45 00.00 20 01 45 00.00 40 QUALITY CONTROL, the superintendent must have qualifications in accordance with that section.

1.6.2 Minimum Communication Requirements

Have at least one qualified superintendent, or competent alternate, capable of reading, writing, and conversing fluently in the English language, on the job-site at all times during the performance of Contract work. In addition, if a Quality Control (QC) representative is required on the Contract, then that individual must also have fluent English communication skills.

1.6.3 Duties

The project superintendent is primarily responsible for managing subcontractors and coordinating day-to-day production and schedule adherence on the project. The superintendent is required to attend partnering meetings, and quality control meetings. The superintendent or qualified alternative must be on-site at all times during the performance of this contract until the work is completed and accepted.

1.6.4 Non-Compliance Actions

The Project Superintendent is subject to removal by the Contracting Officer for non-compliance with requirements specified in the contract and for failure to manage the project to ensure timely completion. Furthermore, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders is acceptable as the subject of claim for extension of time for excess costs or damages by the Contractor.

1.7 PRECONSTRUCTION MEETING CONFERENCE

Immediately after award, prior to commencing any work at the site, coordinate with the Contracting Officer a time and place to meet for the Preconstruction Meeting Conference. The meeting conference must take place within 14 calendar days after award of the contract, but prior to commencement of any work at the site. The purpose of this meeting conference is to discuss and develop a mutual understanding of the administrative requirements of the Contract including but not limited to: daily reporting, invoicing, value engineering, safety, base-access, outage requests, hot work permits, schedule requirements, quality control, schedule of prices, shop drawings, submittals, cybersecurity, prosecution of the work, government acceptance, final inspections and contract close-out. Contractor must present and discuss their basic approach to scheduling the construction work and any required phasing.

1.7.1 Attendees

Contractor attendees must include the Project Manager, Superintendent, Site Safety and Health Officer (SSHO), Quality Control Manager and major subcontractors.

1.8 FACILITY TURNOVER PLANNING MEETINGS (Red Zone Meetings)

Meet with the Government to identify strategies to ensure the project is carried to expeditious closure and turnover to the Client. Start planning the turnover process at the Pre-Construction Conference meeting with a discussion of the Red Zone process and convene at regularly scheduled NRZ Meetings beginning at approximately 75 percent of project completion. Include the following in the facility Turnover effort:

1.8.1 Red Zone Checklist

- a. Contracting Officer's Technical Representative (COTR) will provide the Contractor a copy of the Red Zone Checklist template.
- b. Prior to 75 percent completion, modify the Red Zone Checklist template by adding or deleting critical activities applicable to the project and assign planned completion dates for each activity. Submit the modified Red Zone Checklist to the Contracting Officer. The Contracting Officer may request additional activities be added to the Red Zone Checklist at any time as necessary.

1.8.2 Meetings

- a. Conduct regular Red Zone Meetings beginning at approximately 75 percent project completion, or three to six months prior to Beneficial Occupancy Date (BOD), whichever comes first.
- b. The Contracting Officer will establish the frequency of the meetings, which is expected to increase as the project completion draws nearer. At the beginning, Red Zone meetings may be every two weeks then increase to weekly towards the final month of the project.
- c. Using the Red Zone Checklist as a Plan of Action and Milestones (POAM) and basis for discussion, review upcoming critical activities and strategies to ensure work is completed on time.
- d. During the Red Zone Meetings discuss with the COTR any upcoming

activities that require Government involvement.

- e. Maintain the Red Zone Checklist by documenting the actual completion dates as work is completed and update the Red Zone Checklist with revised planned completion dates as necessary to match progress. Distribute copies of the current Red Zone Checklist to attendees at each Red Zone Meeting.

1.9 EXPORT LICENSES FOR OVERSEAS PROJECTS

Obtain individual export licenses and project export licenses required by the Department of Commerce regulations (15 CFR 772 and 15 CFR 773) so that no delays are experienced in shipping from the United States of America to a foreign country. For additional information, the Contractor may contact one of the U.S. and Foreign Commercial Service District Offices of the Department of Commerce which are located in almost every State.

1.10 WAIVER FOR WORKER'S COMPENSATION

In addition to FAR 52.228-4 Workers' Compensation and War-Hazard Insurance Overseas, the Secretary of Labor has granted a waiver. The waiver does not apply to employees who are hired in the United States, or who are residents, or citizens of the United States.

1.11 PARTNERING

Contractor shall host the partnering session within 45 calendar days of contract award. To most effectively accomplish this Contract, the Contractor and Government must form a cohesive partnership with the common goal of drawing on the strength of each organization in an effort to achieve a successful project without safety mishaps, conforming to the Contract, within budget and on schedule. The partnering team must consist of personnel from both the Government and Contractor including project level and corporate level leadership positions. Key Personnel from the supported command, end user, NAVFAC, PWD, Contractor, key subcontractors and the Designer of Record are required to participate in the Partnering process.

1.11.1 Facilitated (Formal) Partnering

- a. Within 35 calendar days after award and prior to the start of work, host a Formal Partnering session with key personnel from the project team including both Contractor and Government personnel. All costs associated with the Partnering session including the third-party independent Facilitator Consultant, meeting room and other incidental items are the responsibility of the Contractor.
- b. Before the Facilitated (Formal) Partnering session, coordinate with the Facilitator all requirements for incidental items (such as audio-visual equipment, easels, flipchart paper, colored markers, note pads, pens/pencils, colored flash cards) and have these items available at the Partnering session. Provide copies of any documents required for distribution to all attendees. Participants will bear their own costs for meals, lodging and transportation associated with Partnering.
- c. The Initial Partnering Session must be a duration of one day and be held at a location off base as agreed to by the Contracting Officer. Partnering session may take place concurrently with the

Pre-Construction Post-Award Kickoff Meeting.

- d. Facilitator must be experienced in conducting corporate Partnering sessions and must be a third-party independent facilitating consultant - not an employee of the Contractor. The Facilitator is responsible for leading all aspects of the Partnering session necessary to achieve the Partnering goal.
- e. An outcome of the Partnering session must be an escalation matrix agreed upon by both the Government and Contractor, which identifies key Government and Contractor decision makers by name and anticipated decision durations.
- f. Host follow-on Partnering Sessions at three- to six-month intervals or more frequently if needed and lasting generally a half day or less. Attendees need only be those required to resolve current issues. The same Facilitator used in the Initial Partnering session must lead the follow-on sessions unless an alternative is permitted by the Contractor Officer. All costs associated with follow-on Partnering sessions are the responsibility of the Contractor.

1.12 MOBILIZATION

Contractor shall mobilize to the jobsite within 45 calendar days after contract award. Mobilize is defined as having equipment AND having a physical presence of at least one person from the contractor's team on the jobsite.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 32 17.00 20

COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS)

05/18, CHG 3: 08/20

PART 1 GENERAL

- 1.1 DEFINITIONS
- 1.2 SCHEDULE REQUIREMENTS PRIOR TO THE START OF WORK
 - 1.2.1 Preliminary Scheduling Meeting
 - 1.2.2 Project Baseline Schedule
 - 1.2.3 Project Baseline Schedule
 - 1.2.3.1 Baseline NAS
 - 1.2.3.2 Post-Award Kickoff (PAK) Meeting and Baseline NAS
 - 1.2.3.3 Construction Baseline
- 1.3 THREE-WEEK LOOK AHEAD SCHEDULE
 - 1.3.1 Weekly CQC Coordination and Production Meeting
 - 1.3.2 Look Ahead Schedule Requirements
- 1.4 MONTHLY NETWORK ANALYSIS
 - 1.4.1 Monthly Network Analysis Updates
 - 1.4.2 As-Built Schedule
- 1.5 CORRESPONDENCE AND TEST REPORTS
- 1.6 ADDITIONAL SCHEDULING REQUIREMENTS
- 1.7 SUBMITTALS
- 1.8 SOFTWARE
- 1.9 DESIGNATED PROJECT SCHEDULER
 - 1.9.1 Qualifications
 - 1.9.2 Duties
- 1.10 NETWORK SYSTEM FORMAT
 - 1.10.1 Schedule Activity Properties and Level of Detail
 - 1.10.1.1 Activity Identification and Organization
 - 1.10.1.2 Activity Logic
 - 1.10.1.3 Longest Path Activity Baseline Limitation
 - 1.10.1.4 Assigned Calendars
 - 1.10.1.5 Activity Categories
 - 1.10.1.5.1 Design Activities
 - 1.10.1.5.2 Pre-construction Activities
 - 1.10.1.5.3 Procurement Activities
 - 1.10.1.5.4 Government Activities
 - 1.10.1.5.5 Construction Quality Management (CQM) Activities
 - 1.10.1.5.6 Construction Activities
 - 1.10.1.5.7 Turnover and Closeout Activities
 - 1.10.1.5.8 Commissioning Activities
 - 1.10.1.6 Contract Milestones and Constraints
 - 1.10.1.6.1 Project Start Date Milestones
 - 1.10.1.6.1.1 Post-Award Kickoff (PAK) meeting Milestone
 - 1.10.1.6.2 Pre-Construction Meeting Milestone
 - 1.10.1.6.3 Preconstruction Submittals Finish Milestone
 - 1.10.1.6.4 Contractor Mobilization Finish Milestone
 - 1.10.1.6.5 NAVFAC Red Zone - Facility Turnover Planning Meeting Milestones

- 1.10.1.6.6 Substantial Completion Milestone
- 1.10.1.6.7 DD-1354 Finish Milestone
- 1.10.1.6.8 Projected Completion Milestone
- 1.10.1.6.9 Contract Completion Date (CCD) Milestone
- 1.10.1.6.10 Additional Milestones
- 1.10.1.7 Work Breakdown Structure & Activity Code
 - 1.10.1.7.1 Work Breakdown Structure (WBS)
 - 1.10.1.7.2 Responsibility Code
 - 1.10.1.7.3 Activity Category Code
 - 1.10.1.7.4 Construction Specification Institute (CSI) Masterformat Code
 - 1.10.1.7.5 Drawing Code
- 1.10.1.8 Adverse Weather Lost Work Days
- 1.10.1.9 Anticipated Restricted Delays
- 1.10.1.10 Cost Loading
 - 1.10.1.10.1 Cost Loading Activities
 - 1.10.1.10.2 Partial Payment
- 1.10.2 Schedule Software Settings and Restrictions
- 1.10.3 Required Tabular Reports and Native P6 XER Files
- 1.11 CONTRACT MODIFICATION
 - 1.11.1 Time Impact Analysis (TIA)
- 1.12 PROJECT FLOAT

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 32 17.00 20

COST-LOADED NETWORK ANALYSIS SCHEDULES (NAS)
05/18, CHG 3: 08/20

PART 1 GENERAL

1.1 DEFINITIONS

The cost-loaded Network Analysis Schedule (NAS) is a tool to manage the project, both for Contractor and Government activities. The NAS is also used to report progress, evaluate time extensions, and provide the basis for progress payments.

For consistency, when scheduling software terminology is used in this section, the terms in Primavera's scheduling programs are used.

1.2 SCHEDULE REQUIREMENTS PRIOR TO THE START OF WORK

1.2.1 Preliminary Scheduling Meeting

Before preparation of the Project Baseline Schedule, and prior to the start of work, meet with the Contracting Officer to discuss the proposed schedule and the requirements of this section. Propose projected data dates for monthly update schedules for the project and incorporate each monthly update submittal into submittal register. Discuss required forms, terminology, and submittal requirements of this section and other requirements related to schedule management for this contract.

1.2.2 Project Baseline Schedule

Submit the Baseline NAS within 30 calendar days after contract award. Data date must be set to contract award date and no progress statused for any activity. Only bonds may be paid prior to acceptance of the Baseline NAS. The acceptance of a Baseline NAS is a condition precedent to:

- a. The Contractor starting demolition work or construction stage(s) of the contract.
- b. Processing Contractor's invoices(s) for any items other than bonds.
- c. Review of any schedule updates.

Submittal of the Baseline NAS is the Contractor's certification that the submitted schedule meets the requirements of the Contract Documents and represents the Contractor's plan on how the work will be accomplished. Provide all items listed in paragraph REQUIRED TABULAR REPORTS AND NATIVE P6 XER FILES with baseline NAS submittal.

1.2.3 Project Baseline Schedule

1.2.3.1 Baseline NAS

Submit the Baseline NAS at the Post-Award Kickoff (PAK) Meeting or within 30 calendar days after contract award whichever occurs first. The Baseline schedule must include detailed design activities and a general approach to construction, including summary activities for required

phasing and definable areas. Summary Construction activities must not exceed duration of 60 calendar days, unless approved otherwise by Contracting Officer. Data date must be set to contract award date and no progress statused for any activity.

Only bonds may be paid prior to acceptance of the Baseline NAS. The acceptance of a Baseline NAS is a condition precedent to:

- a. The Contractor submitting the first design submittal.
- b. Processing Contractor's invoices(s) other than that for the bonds.
- c. Review of any schedule updates.

1.2.3.2 Post-Award Kickoff (PAK) Meeting and Baseline NAS

Present the Draft Baseline NAS at the PAK Meeting. Be prepared to discuss the schedule logic emphasizing how the schedule satisfies the design package requirements and incorporates the required government review periods for each design submittal.

1.2.3.3 Construction Baseline

Submit the Construction Baseline NAS prior to the pre-final design submittal. The pre-final design submittal will not be reviewed until a Construction Baseline NAS is submitted.

The acceptance of the Construction Baseline NAS is a condition precedent to:

- a. The contractor starting demolition work or construction stage(s) of the contract.
- b. Processing Contractor's invoices for demolition or construction activities.
- c. Review of any construction phase schedule updates.

Submittal of the Construction Baseline NAS must be the Contractor's certification that the submitted schedule meets the requirements of the Contract Documents, and represents the Contractor's plan on how the work will be accomplished. Provide all items listed in paragraph REQUIRED TABULAR REPORTS AND NATIVE P6 XER FILES with baseline NAS submittal.

1.3 THREE-WEEK LOOK AHEAD SCHEDULE

1.3.1 Weekly CQC Coordination and Production Meeting

Deliver electronic file of 3-Week Look Ahead Schedule to the Contracting Office at least 24 hours prior to the weekly scheduled CQC Coordination and Production Meeting. Contractor is required to provide all attendees at the CQC Coordination and Production Meeting with a hard copy of the 3-Week Look Ahead Schedule.

1.3.2 Look Ahead Schedule Requirements

Prepare and issue a 3-Week Look Ahead schedule to provide a more detailed day-to-day plan of upcoming work identified on the Project Network Analysis Schedule. Requirements include:

- a. For each Look Ahead schedule activity, identify parent NAS activity number(s). The parent NAS activity is the activity in the NAS that would incorporate the Look Ahead schedule activity requirement and or scope of work.
- b. Update schedule each week to show the planned work for the current and following two-week period. Also include previous week, as-built work, showing actual start and finish dates.
- c. Include upcoming outages, closures, preparatory meetings, and initial meetings, testing and inspections.
- d. Clearly identify longest path activities on the Three-Week Look Ahead Schedule. Include a key or legend that distinguishes longest path activities. Include all Longest Path activity NAS start/finish dates exceeded and/or occurring during this period.
- e. The detail work plans are to be bar chart type schedules, derived from but maintained separately from the Project NAS on an electronic spreadsheet program and printed on 299 by 432 mm sheets as directed by the Contracting Officer.
- f. Activities must not exceed 5 working days in duration and have sufficient level of detail to assign crews, tools and equipment required to complete the work.

1.4 MONTHLY NETWORK ANALYSIS

Submittal of Monthly NAS is the Contractor's certification that the submitted schedule meets the requirements of the Contract Documents and represents the Contractor's plan on how the work will be accomplished. Provide all items listed in paragraph REQUIRED TABULAR REPORTS AND NATIVE P6 XER FILES with the monthly NAS submittal.

1.4.1 Monthly Network Analysis Updates

- a. Regardless of whether an invoice is being submitted monthly, an updated schedule must be submitted monthly to the Government. The Monthly NAS update must be submitted within 10 calendar days of the data date.
- b. Provide all items listed in paragraph REQUIRED TABULAR REPORTS AND NATIVE P6 XER FILES, with each monthly NAS update submittal.
- c. Meet with Government representative(s) at monthly intervals to review and agree on the information presented in the updated project schedule. The submission of an accepted, updated schedule to the Government is a condition precedent to the processing of the Contractor's invoice.
- d. Activity progress must incorporate as-built events as they occurred and correspond to records including but not limited to submittals and daily production and quality control reports. Software Settings: Handle schedule calculations and Out-of-Sequence progress (if applicable) through Retained Logic, not Progress Override. Show all activity durations and float values in days. Show activity progress using Remaining Duration. Set default activity type to "Task Dependent".

- e. Update schedule must reflect current Contract Completion Date and contract value in accordance with all conformed contract modifications issued prior to data date of NAS update.

1.4.2 As-Built Schedule

As a condition precedent to the release of retention and making final payment, submit an "As-Built Schedule," as the last schedule update showing all activities at 100 percent completion. This schedule must reflect the exact manner in which the project was actually constructed.

1.5 CORRESPONDENCE AND TEST REPORTS

Reference Schedule activity IDs that are being addressed in each correspondence (e.g., letters, Requests for Information (RFIs), e-mails, meeting minute items, Production and QC Daily Reports, material delivery tickets, photographs) and test report (e.g., concrete, soil compaction, weld, pressure).

1.6 ADDITIONAL SCHEDULING REQUIREMENTS

Other specification sections may include additional scheduling requirements, including systems to be inspected, tested and commissioned, and submittal procedures. Those schedule requirements must be incorporated into the NAS schedule.

1.7 SUBMITTALS

Government approval/acceptance is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Baseline NAS; G

Construction Baseline NAS; G

Designated Project Scheduler; G

SD-07 Certificates

Three-Week Look Ahead Schedule; G

Monthly Network Analysis Updates; G

SD-11 Closeout Submittals

As-Built Schedule; G

1.8 SOFTWARE

Prepare and maintain project schedules using Primavera P6 software in a version compatible with Government's current version. Importing data into P6 using data conversion techniques or third party software is cause for rejection of the submitted schedule. Schedules with Performing Organizational Breakdown Structure (POBS) data is cause for rejection.

1.9 DESIGNATED PROJECT SCHEDULER

Within 30 calendar days of contract award, submit to the Contracting Officer for approval an individual who will serve as the Designated Project Scheduler. Include a copy of the candidate's resume with qualifications. The Contracting Officer may remove the Designated Project Scheduler, and require replacement, if the scheduler does not effectively fulfill their duties in accordance with the contract requirements. Payment request will not be processed without an approved Designated Project Scheduler.

1.9.1 Qualifications

The Designated Project Scheduler must have prepared and maintained at least three previous construction schedules, of similar size and complexity to this contract, using Primavera P6.

1.9.2 Duties

Duties of the Designated Project Scheduler:

- a. Prepare Baseline NAS.
- b. Prepare monthly schedule updates.
- c. Prepare tabular reports.
- d. Prepare Time Impact Analysis (TIA) as necessary.
- e. Provide certification that NAS and TIA submittals conform to the contract requirements.
- f. Participate with the Prime Contractor and Government Representative in a monthly teleconference call, and scheduled with sufficient time to support the Monthly Network Analysis Updates process, to discuss project status, schedule updates, critical activities, potential delays, and contract modifications impacting the schedule. Have a computer with P6 software available during the meeting.

1.10 NETWORK SYSTEM FORMAT

Prepare the schedule in accordance with the following Primavera P6 settings and parameters. Deviation from these settings and parameters, without prior consent of the Contracting Officer, is cause for rejection of schedule submission.

1.10.1 Schedule Activity Properties and Level of Detail

1.10.1.1 Activity Identification and Organization

- a. Identify design and construction activities planned for the project and other activities that could impact project completion if delayed in the NAS.
- b. Each activity must have a unique name.
- c. Identify administrative type activity/milestones, including all pre-construction submittal and permit requirements prior to demolition

or construction stage.

- d. Include times for procurement, Contractor quality control and construction, acceptance testing and training in the schedule.
- e. Include the Government approval time required for the submittals that require Government Approval prior to construction, as indicated in Section 01 33 00 SUBMITTAL PROCEDURES Section 01 33 00.05 20 CONSTRUCTION SUBMITTAL PROCEDURES.
- f. Create separate activities for each Phase, Area, Floor Level and Location the activity is occurring.
- g. Do not use construction category activity to represent non-work type reference (e.g. Serial Letter, Request for Information) in NAS. Place Non-work reference within the P6 activity details notebook.

Activity categories included in the schedule are specified below.

1.10.1.2 Activity Logic

- a. With the exception of the Contract Award and Contract Completion Date (CCD) milestone activities, activity must not be open-ended; each activity must have at least one predecessor and at least one successor.
- b. Activities must not have open start or open finish (dangling) logic.
- c. Do not use lead or lag logic without Contracting Officer prior approval.
- d. Minimize redundant logic ties.
- e. Once an activity exists on the schedule it must not be deleted or renamed to change the scope of the activity and must not be removed from the schedule logic without approval from the Contracting Officer.
 - (1) While an activity cannot be deleted, where said activity is no longer applicable to the schedule, but must remain within the logic stream for historical record, change the activity original and remaining duration to zero and clearly label "(NO LONGER REQUIRED)" after the activity name. Actual finish date for activity that falls behind the data date. Redistribute accordingly any remaining budget associated with that activity, to other remaining appropriate activity.
 - (2) Document any such change in the activities' "Notebook," including a date and explanation for the change.
 - (3) The ID number for a "NO LONGER REQUIRED" activity must not be re-used for another activity.

1.10.1.3 Longest Path Activity Baseline Limitation

For P6 settings, critical activities are defined as being on the Longest Path. Longest Path (Critical) Activities must not make up more than 30 percent of all activity within the Construction Baseline Schedule.

1.10.1.4 Assigned Calendars

All NAS activity must be assigned calendars that reflect required and anticipated non-work days.

1.10.1.5 Activity Categories

1.10.1.5.1 Design Activities

Design activities must include design decision points and design submittal packages, including critical path submittals for Fast Tracked Phases. Review times for design development packages must be included in the schedule. Refer to Section 01 33 10.05 20 DESIGN SUBMITTAL PROCEDURES, for specific requirements.

1.10.1.5.2 Pre-construction Activities

Examples of pre-construction activities include, but are not limited to, bond approval, permits, pre-construction submittals and approvals. Include pre-construction activities that are required to be completed prior to the Contractor starting the demolition or construction stage of work.

1.10.1.5.3 Procurement Activities

Examples of procurement activities include, but are not limited to: Material/equipment submittal preparation, submittal and approval of material/equipment; material/equipment fabrication and delivery, and material/equipment on-site. As a minimum, separate procurement activities must be provided for critical items, long lead items, items requiring Government approval and material/equipment procurement for which payment will be requested in advance of installation. Show each delivery with relationship tie to the Construction Activity specifically for the delivery.

1.10.1.5.4 Government Activities

Government and other agency activities that could impact progress must be clearly identified. Government activities include, but are not limited to; Government approved submittal reviews, Government conducted inspections/tests, environmental permit approvals by State regulators, utility outages, Construction Start, and delivery of Government Furnished Material/Equipment.

1.10.1.5.5 Construction Quality Management (CQM) Activities

The Preparatory and Initial Phase meetings for each Definable Feature of Work identified in the Contractor's Quality Control Plan must be included in the Three-Week Look Ahead Schedule. Preparatory and Initial phase meetings are not required in the NAS, but can be represented by a start milestone linked to successor parent Construction Activity. The Follow-up Phase must be represented by the Construction Activities themselves in the NAS.

1.10.1.5.6 Construction Activities

On-site construction activities must not have a duration in excess of 20 working days. Contractor activities must be driven by calendars that reflect Saturdays, Sundays and all Federal Holidays as non-work days,

unless otherwise defined in this contract. Federal Holidays are as defined in 5 USC 6103.

1.10.1.5.7 Turnover and Closeout Activities

Include activities or milestones for items on the NAVFAC Red Zone Checklist/POAM that are applicable to this project. As a minimum, include required Contractor testing, required Government acceptance inspections on equipment, Pre-Final Inspection, Punch List Completion, Final Inspection and Acceptance. Add an unconstrained start milestone for the initial NAVFAC Red Zone - Facility Turnover Planning Meeting at approximately 75 percent construction contract completion or six months prior to Contract Completion Date (CCD), whichever is sooner.

1.10.1.5.8 Commissioning Activities

Include in the baseline schedule activities and milestones associated with Commissioning.

- a. Identify the general area or location(s) of systems for Commissioning Inspection and Testing
- b. Incorporate into the baseline schedule time periods for Government submittal review

1.10.1.6 Contract Milestones and Constraints

1.10.1.6.1 Project Start Date Milestones

Include as the first activity on the schedule a start milestone titled, "Contract Award", which must have a Mandatory Start constraint equal to the Contract Award Date.

1.10.1.6.1.1 Post-Award Kickoff (PAK) meeting Milestone

Include an unconstrained finish milestone on the schedule titled, "Post-Award Kickoff Meeting". The Post Award Kickoff Meeting may be a single day, or it may range over several days. The intent is to cover all PAK topics, including Partnering and Concept Design Workshop (if required) in one continuous session.

1.10.1.6.2 Pre-Construction Meeting Milestone

Include an unconstrained finish milestone on the schedule titled, "Pre-Construction Meeting". The Pre-Construction meeting may be a single day, or it may range over several days. The intent to cover all the Pre-Con topics, including Partnering and DD1354.

1.10.1.6.3 Preconstruction Submittals Finish Milestone

Include an unconstrained finish milestone on the schedule titled, "Preconstruction Submittals". This milestone is complete when all required preconstruction submittals have been reviewed and approved by the Government.

1.10.1.6.4 Contractor Mobilization Finish Milestone

Include an unconstrained finish milestone on the schedule titled, "Contractor Mobilization".

1.10.1.6.5 NAVFAC Red Zone - Facility Turnover Planning Meeting Milestones

See paragraph TURNOVER AND CLOSEOUT ACTIVITIES above.

1.10.1.6.6 Substantial Completion Milestone

Include an unconstrained finish milestone on the schedule titled "Substantial Completion." Substantial Completion is defined as the point in time the Government would consider the project ready for beneficial occupancy wherein by mutual agreement of the Government and Contractor, Government use of the facility is allowed while construction access continues in order to complete remaining items (e.g. punch list and other close out submittals). Include a separate Substantial Completion Milestone for each phase if the contract requires construction to be completed in phases.

1.10.1.6.7 DD-1354 Finish Milestone

Add unconstrained finish milestone, titled "DD-1354" and scheduled 30 calendar days prior to Substantial Completion, whenever a Form DD-1354 is required in accordance with Section 01 78 00 CLOSEOUT SUBMITTALS.

1.10.1.6.8 Projected Completion Milestone

Include an unconstrained finish milestone on the schedule titled "Projected Completion." Projected Completion is defined as the point in time all contract requirements are complete and verified by the Government with a successful Final Inspection in accordance with Section 01 45 00.00 20 QUALITY CONTROL. This milestone must have the Contract Completion Date (CCD) milestone as its only successor.

1.10.1.6.9 Contract Completion Date (CCD) Milestone

Last schedule entry must be an unconstrained finish milestone titled "Contract Completion (CCD: DD-MM-YY)." DD-MM-YYYY is the current contract completion date at data date, day-month-year corresponding to P6 Must Finish By Date. NAS milestone updates of Project Completion finish date for longest path must reflect calculated float as positive or negative based on CCD. Calculation of schedule updates must be such that if the finish of the "Projected Completion" milestone falls after the contract completion date, then negative float is calculated on the longest path. If the finish of the "Projected Completion" milestone falls before the contract completion date, the float calculation must reflect positive float on the longest path.

1.10.1.6.10 Additional Milestones

Provide up to 5 additional milestones as required by Contracting Officer.

1.10.1.7 Work Breakdown Structure & Activity Code

At a minimum, establish a Work Breakdown Structure (WBS) and provide activity codes identified as follows:

1.10.1.7.1 Work Breakdown Structure (WBS)

Group all activities and milestones within appropriate WBS categories including, at a minimum, the following:

- a. Project Milestones:
 - (1) Management Milestones
 - (2) Project Administrative Meetings
 - (3) Permits
- b. Pre-Construction Phase:
 - (1) Submittals and Reviews
 - (2) Procurement
 - (3) Mobilization
- c. Construction Phase: Create multiple sub-sections in accordance with project specific categories of work including in WBS descending order as follows:
 - (1) General Area
 - (a) Type of Work Item
 - 1. Location
- d. Project Closeout: Include activity items such as, but not limited to, Punchlist, Demobilization, O&M, As-built Drawings, Training, and As-built NAS.
- e. Modifications: Create sub-category of Conformed and Non-Conformed under Modification WBS. Create multiple sub-sections as the project progresses identified by issue and Fragnet placed in Conformed for modifications issued prior data date, or Non-Conformed for issues not modified to contract prior data date.
- f. Removed Activity: Activity is "removed" by remaining within logic sequence, eliminating duration and adding "(NO LONGER REQUIRED)" after Activity Name in Activity Table.
- g. Commissioning & Testing:
 - (1) Specific area/locations of commissioning
 - (2) Final Testing
 - (3) Training

1.10.1.7.2 Responsibility Code

All activities in the project schedule must be identified with the resource for completing the task. Activities must not belong to more than one responsible party.

1.10.1.7.3 Activity Category Code

Provide user defined "CAT" codes for Project Level activity codes. Use the following codes:

- a. Assign "PROC" value to Procurement type activity
- b. Assign "PRE-CON" value to Pre-construction activity
- c. Assign "CONS" value to Construction type activity
- d. Assign "TEST" value to dedicated testing type activities
- e. Assign "CX" value to dedicated Commissioning type activities
- f. Assign "CLOS" value to dedicated Close Out type activity
- g. Assign "OTHR" to other activity not otherwise designated
- h. Assign "DSGN" value to Design type activity

1.10.1.7.4 Construction Specification Institute (CSI) Masterformat Code

Provide up to an additional five activity codes as required by the Contracting Officer.

1.10.1.7.5 Drawing Code

Identify all activities in the project schedule with its respective Drawing Code. The Drawing Code is the Sheet Number on the primary project drawing which indicates work to be performed. If an activity does not have an applicable Drawing Code (e.g. Mobilize), the code must be "0000".

1.10.1.8 Adverse Weather Lost Work Days

Use the National Oceanic and Atmospheric Administration's (NOAA) Summary of Monthly Normals report to obtain the historical average number of days each month with precipitation, using a nominal 30-year, greater than 2.5 mm precipitation amount parameter, as indicated on the Station Report for the NOAA location closest to the project site as the basis for establishing a "Weather Calendar" showing the number of anticipated non-workdays for each month due to adverse weather, in addition to Saturdays, Sundays and all Federal Holidays as non-work days.

Use the following schedule of anticipated monthly non-work days due to adverse weather as the basis for establishing a "Weather Calendar" showing the number of anticipated non-workdays for each month due to adverse weather, in addition to Saturdays, Sundays and all Federal Holidays as non-work days.

MONTHLY ANTICIPATED ADVERSE WEATHER LOST WORK DAY											
JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEP	OCT	NOV	DEC
2	2	2	2	2	2	2	2	2	2	2	2

Assign the Weather Calendar to any activity that could be impacted by adverse weather. The Contracting Officer will issue a modification in accordance with the contract clauses, giving the Contractor a time only extension for the difference of days between the anticipated and actual adverse weather delay if the number of actual adverse weather delay days exceeds the number of days anticipated for the month in which the delay

occurs and the adverse weather delayed activities are on the longest path to contract completion in the period when delay occurred. A lost workday due to weather conditions is defined as a day in which the Contractor cannot work at least 50 percent of the day on the impacted activity. Impacts resulting from adverse weather must be documented in Narrative Report for the month that it occurred.

Make changes to P6 project calendars to reflect as-built conditions where work occurred where originally anticipated as non-work days, and where work did not occur (lost work day).

1.10.1.9 Anticipated Restricted Delays

Unless otherwise noted or defined in Section 01 14 00 WORK RESTRICTIONS, allow in the schedule one lost workday for every two months of project duration for instances where base access is not permitted or where work areas are temporarily not accessible for security reasons which causes a delay in the work. Use Anticipated Restricted Delays as basis for establishing a "Security Calendar" showing the number of anticipated non-workdays for each month due to anticipated restrictions, in addition to anticipated adverse weather, Saturdays, Sundays and all Federal Holidays as non-work days. Assign the Security Calendar to any activity that could be impacted by restriction delays. The Contracting Officer will issue a modification in accordance with the contract clauses, giving the Contractor a time extension for the difference of days between the anticipated and actual lost work days if the number of actual restriction delay days exceeds the number of anticipated for the month in which the delay occurs and the restriction delayed activities are critical to contract completion. A lost workday due to restriction delay is defined as a day in which the Contractor cannot work at least 50 percent of the day on the impacted activity.

Impacts resulting from restriction delays must be documented in Narrative Report for the month that it occurred.

Make changes to P6 project calendars to reflect as-built conditions where work occurred where originally anticipated as non-work days, and where work did not occur (lost work day).

1.10.1.10 Cost Loading

The Project Network Analysis Schedule (NAS) must be cost-loaded and will provide the basis for progress payments. Earned Value Reports must be derived from and correspond to cost loaded NAS. Use the Critical Path Method (CPM) and the Precedence Diagram Method (PDM) to satisfy time and cost applications.

1.10.1.10.1 Cost Loading Activities

Assign material and equipment costs, including their quantities, for which payment will be requested in advance of installation, to their respective procurement activity. Assign labor costs, including their quantities, for material and equipment paid for after installation to their respective construction activities. Include all typical mobilization costs dispersed over early construction activities. Costs for mobilization will not be paid as individual pay items with the exception of batch plant set-up, mobilization of dredging equipment or other similar labor-intensive situations. The value of commissioning, testing and closeout WBS section may not be less than 10 percent of the total costs for procurement and

construction activities. ALL activities assigned Government responsibility will have Zero Cost. No contractor cost should be assigned to an activity designated as a Government responsibility. Do not include field overhead positions as individual pay items. Evenly disperse overhead costs and profit to each activity over the duration of the project.

1.10.1.10.2 Partial Payment

Breakdown unit of measure and cost must be defined within P6 Activity Detail Expenses for partial payment of any cost loaded activity. Lump sum cost loaded activity will not be partially paid.

1.10.2 Schedule Software Settings and Restrictions

- a. Activity Constraints: Date/time constraint(s), other than those required by the contract, are not allowed unless accepted by the Contracting Officer. Identify any constraints proposed and provide an explanation for the purpose of the constraint in the Narrative Report as described in paragraph REQUIRED TABULAR REPORTS.
- b. Default Progress Data Disallowed: Actual Start is date work begins on activity with intent to pursue work to substantial completion. Actual Finish is date work is substantially complete to point where successor activity can begin. Actual dates on the CPM schedule must correspond with activity dates reported on the Contractor Quality Control and Production Reports.
- c. At a minimum, include the following settings and parameters in P6 Schedule preparation:
 - (1) General: Define or establish Calendars and Activity Codes at the "Project" level, not the "Global" level.
 - (2) Admin Drop-Down Menu, Admin Preferences, Time Periods Tab:
 - (a) Set time periods for P6 to 8.0 Hours/Day, 40.0 Hours/Week, 172.0 Hours/Month and 2000.0 Hours/Year.
 - (b) Use assigned calendar to specify the number of work hours for each time period: Must be checked.
 - (3) Admin Drop-Down Menu, Admin Preferences, Earned Value Tab:
 - (a) Earned Value Calculation: Use "Budgeted values with current dates".
 - (4) Project Level, Dates Tab:
 - (a) Set "Must Finish By" date to "Contract Completion Date", and set "Must Finish By" time to 05:00pm.
 - (5) Project Level, Defaults Tab:
 - (a) Duration Type: Set to "Fixed Duration & Units".
 - (b) Percent Complete Type: Set to "Physical".
 - (c) Activity Type: Set to "Task Dependent".

(d) Calendar: Set to "Standard 5 Day Workweek". Calendar must reflect Saturday, Sunday and all Federal holidays as non-work days. Alternative calendars may be used with Contracting Officer approval.

(6) Project Level, Calculations Tab:

(a) Default Price/Unit for activities without resource or role
Price/Units: Set to "\$1/h".

(b) Activity percent complete based on activity steps: Must be Checked.

(c) Link Budget and At Completion for not started activities:
Must be Checked.

(d) Reset Remaining Duration and Units to Original: Must be Selected.

(e) Subtract Actual from At Completion: Must be Selected.

(f) Recalculate Actual units and Cost when duration percent complete changes: Must be Checked.

(g) Update units when costs change on resource assignments: Must be Unchecked.

(h) Link Actual to Date and Actual This Period Units and Cost:
Must be Checked.

(7) Project Level, Settings Tab:

(a) Define Critical Activities: Check "Longest Path".

(8) Work Breakdown Structure Level, Earned Value Tab:

(a) Technique for Computing Performance Percent Complete:
"Activity percent complete" is selected.

(b) Technique for Computing Estimate to Complete (ETC): "PF = 1" is selected.

1.10.3 Required Tabular Reports and Native P6 XER Files

Include the following reports with the Baseline, Monthly Update and any other required schedule submittals:

a. Time Scaled Logic Schedule

Provide formatted 11 by 17-inch Time-scaled Logic Schedule in color and landscape-oriented with each schedule submittal. Clearly show activities on the longest path setting Gantt chart longest path activity bars to red. Group activities by WBS and sort by finish date in ascending order. Include the following information in column form for each activity and include accompanying Gantt chart:

(1) Activity ID

- (2) Activity Name
- (3) Original Duration
- (4) Remaining duration
- (5) Physical Percent Complete
- (6) Start Date
- (7) Finish Date
- (8) Total Float

- b. Previous Monthly Update Comparison Time Scaled Logic Schedule (Submit with all Monthly Update Schedule Submittals.)

Provide formatted 11 by 17-inch Time-scaled Logic Schedule in color and landscape-oriented with each monthly update schedule submittal. Clearly show activities on the current month longest path setting Gantt chart longest path activities bars to red. Show previous month activities as yellow bars and previous month milestones in yellow within Gantt chart. Sort by finish date in ascending order. Filter activities for longest path. Maintain and assign the accepted previous month update or the accepted baseline schedule for the first update submittal as the baseline and primary baseline in P6 before printing the schedule. Include the following information in column form for each activity and include accompanying Gantt chart:

- (1) Activity ID
- (2) Activity Name
- (3) Original Duration
- (4) Current Month Remaining Duration
- (5) Current Month Start Date
- (6) Previous Month Update Start Date (BL Project Start)
- (7) Start Date Delta between Current Month and Previous Month
(Variance - BL Project Start Date)
- (8) Current Month Finish Date
- (9) Previous Month Finish Date (BL Project Finish)
- (10) Finish Date Delta between Current Month and Previous Month
(Variance - BL Project Start Date)
- (11) Current Month Total Float

- c. P6 native XER file: Include the back-up native .xer program file compatible with the Government version of P6. Each native schedule file must have a unique file name to include project name and data date using (yyyy-mm-dd) convention. Each native schedule must have a unique Project ID and Project Name.

d. Log Report: P6 Scheduling/Leveling Report.

e. Narrative Report: Identify and justify:

(1) Provide Project Summary Data in format below:

(a) Data Date _____

(b) Award Date: _____

(c) Original Project Duration: _____ days post Award Date

(d) Current Project Duration: _____ days post Award Date

(e) Time percent elapsed: _____ percent at data date

(f) Original CCD: _____

(g) Current CCD: _____ (thru MOD _____)

(h) Anticipated CCD: _____ (____ calendar days early/late)

(i) Original Contract Value: \$_____

(j) Current Contract Value: \$_____

(k) Invoiced Amount: \$_____ (____ percent)

(l) Cost Growth: _____ percent

(m) Schedule Growth: _____ percent

(n) There are a total of _____ activities, _____ activities complete (____ percent), _____ activities in progress (____ percent), _____ activities not started (____ percent). Of the in progress and not started activities; _____ (____ percent) are on the longest path. The longest path has duration of _____ calendar days from data-date to anticipated project completion.

(2) Progress made in each area of the project;

(3) Longest Path;

(4) Date/time constraint(s), other than those required by the contract

(5) Listing of all changes made between the previous schedule and current updated schedule include: added or deleted activities, original and remaining durations for activities that have not started, logic (sequence constraint lag/lead), milestones, planned sequence of operations, longest path, calendars or calendar assignments, and cost loading;

(6) Any decrease in previously reported activity Earned Amount;

(7) Pending items and status thereof, including permits, changes orders, and time extensions;

(8) Status of Contract Completion Date and interim milestones;

- (9) Status of Projected Completion Milestone and account of difference in calendar days between previous update Projected Completion Milestone
- (10) Current and anticipated delays listing Activity Names and IDs for impacted activities(describe cause of delay and corrective actions(s) and mitigation measures to minimize);
- (11) Description of current and potential future schedule problem areas.
- (12) Identification of any weather and restricted lost time as compared to anticipated weather for the month and anticipated restricted days for which the update is submitted. Impacts resulting from adverse weather must be documented in tabular form showing the calendar month (or billing period) with the days on which construction activity incurred Lost Work Days due to adverse weather. In narrative form, describe the adverse weather cause such as precipitation measurement, temperature, wind or other influencing factors, and why work was impacted. Describe the construction activity(s) that was (were) scheduled, impacted.

Each entry in the narrative report must cite the respective Activity ID and Activity Name, the date and reason for the change, and description of the change.

- f. Earned Value Report: Derive from and correspond to P6 cost loaded schedule. List all activities having a budget amount cost loaded. Compile total earnings on the project from notice to proceed to current progress payment request. Show current budget, previous physical percent complete, to-date physical percent complete, previous earned value, to-date earned value, cost this period and cost to complete on the report for each activity.
- g. Schedule Variance Control (SVC) Diagram: With each schedule submission, provide a SVC diagram showing 1) A Cash Flow Curve indicating planned project cost based on each of projected early and projected late activity finish dates and 2) one curve for Earned Value to-date. Revise Cash Flow Curves when the contract is modified, or as directed by the Contracting Officer Include a legend on report clearly indication 3 curves: early finish, late finish, and earned-value to date.

Use the following settings in Activity Usage Profile Options:

- (1) In the Data section, under Display, the radio box for Cost must be selected.
- (2) In the Data section, under Filter for Bars/Graphs, the checkbox for Total must be checked.
- (3) In the Show Bars/Curves section:
 - (a) Under the By Date column, the checkboxes for Baseline, Actual and Remaining Late must be checked. The checkboxes for Budgeted and Remaining Early must be unchecked.
 - (b) Under the Cumulative column, the checkboxes for Baseline, Actual and Remaining Late must be checked. The checkboxes for

Budgeted and Remaining Early must be unchecked.

(c) Set the color for Baseline to green.

(d) Set the color for Actual to blue.

(e) Set the color for Remaining Late to red.

(4) In the Show Earned Value Curves section, the checkboxes for Planned Value Cost, Earned Value Cost and Estimate at Completion must be unchecked.

h. Logic Diagram showing timescale from data date to 60 days after data date with filter for longest path. Leave Group By selection blank and sort by finish date in ascending order.

i. Baseline or Monthly Update Checklist as applicable completed and certified by Qualified Scheduler. Baseline Project Schedule and Monthly Update Schedule Checklists can be found on the Whole Building Design Guide website at <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-32-17-00-20>

j. Screen shot PDF of P6 Time Periods Settings referenced in paragraph SCHEDULE SOFTWARE SETTINGS AND RESTRICTIONS, list item d.(2): ADMIN DROP-DOWN MENU, ADMIN PREFERENCES, TIME PERIODS TAB

k. Daily Reported Production Activity: Submit on a monthly basis, in electronic spreadsheet (format provided by the Government), summary of daily reported production activity for the reporting month in the update schedule. Use the following columns for reporting:

(1) Date

(2) Activity ID

(3) Work Description

(4) Contractor

(5) Billable Hours

1.11 CONTRACT MODIFICATION

1.11.1 Time Impact Analysis (TIA)

Submit a Time Impact Analysis with each cost and time proposal for a proposed change. TIA must illustrate the influence of each change or delay on the Contract Completion Date or milestones. No time extensions will be granted nor delay damages paid unless a delay occurs which consumes all available Project Float, impacts the longest path, and extends the Projected Completion beyond the Contract Completion Date.

a. Each TIA must be in both narrative and schedule form. The narrative must define the scope and conditions of the change; provide start and finish dates of impact, successor and predecessor activity to impact period, responsible party; describe how it originated, and how it impacts the schedule's longest path. The schedule submission must consist of three native XER files:

- (1) Fragnet used to define the scope of the changed condition
 - (2) Most recent accepted schedule update as of the time of the impact start date. Update this schedule to show all activity progress as of the time of the impact start date. The impact start date is identified as the time when an existing activity is impeded for either starting or finishing.
 - (3) The impacted schedule that has the fragnet inserted in the updated schedule and the schedule "run" so that the new completion date is determined.
- b. For claimed as-built project delay, the inserted fragnet TIA method must be modified to account for as-built events known to occur after the data date of schedule update used. Updated schedules for periods following the impact start date will be used to evaluate how the project progressed (as-built) through the finish of impact. Impact to longest path must be determined for each following update period.
- c. All TIAs must include any mitigation, and must determine the apportionment of the overall delay assignable to each individual delay. Apportionment must provide identification of delay type and classification of delay by compensable and non-compensable events. The associated narrative must clearly describe analysis methodology used, and the findings in a chronological listing beginning with the earliest delay event.
- (1) Identify and classify types of delay defined as follows:
 - (a) Force majeure delay (e.g. weather delay): Any delay event caused by something or someone other than the Government or the Contractor, or the risk of which has not been assigned solely to the Government or the Contractor. If the force majeure delay is on the longest path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, classified as a non-compensable event.
 - (b) A Contractor-delay: Any delay event caused by the Contractor, or the risk of which has been assigned solely to the Contractor. If the contractor-delay is on the longest path, in absence of other types of concurrent delays, Contractor is not granted extension of contract time, and classified as a non-compensable event. Where absent other types of delays, and having impact to project completion, Contractor must provide to Contracting Officer a Corrective Action Plan identifying plan to mitigate delay.
 - (c) A Government-delay: Any delay event caused by the Government, or the risk of which has been assigned solely to the Government. If the Government-delay is on the longest path, in absence of other types of concurrent delays, the Contractor is granted an extension of contract time, and classified as a compensable event.
 - (2) Functional concurrency must be used to analyze concurrent delays, where: separate delay issues delay project completion, do not necessarily occur at same time, rather occur within same monthly schedule update period at minimum, or within same as-built period under review. If a combination of functionally concurrent delay types occurs, it is considered Concurrent Delay, which is defined

in the following combinations:

(a) Government-delay concurrent with contractor-delay: excusable time extension, classified non-compensable event.

(b) Government-delay concurrent with force majeure delay: excusable time extension, classified non-compensable event.

(c) Contractor-delay concurrent with force majeure delay: excusable time extension, classified non-compensable event.

(3) Pacing delay reacting to another delay (parent delay) equally or more critical than paced activity must be identified prior to pacing. Contracting Officer will notify Contractor prior to pacing. Contractor must notify Contracting Officer prior to pacing. Notification must include identification of parent delay issue, estimated parent delay time period, paced activity(s) identity, and pacing reason(s). Pacing Concurrency is defined as follows:

(a) Government-delay concurrent with contractor-pacing: excusable time extension, classified compensable event.

(b) Contractor-delay concurrent with Government-pacing: inexcusable time extension, classified non-compensable event

d. Submit electronic file containing the narrative and the source schedule files used in the time impact analysis.

1.12 PROJECT FLOAT

Project Float is the length of time between the Contractor's Projected Completion Milestone and the Contract Completion Date. Project Float available in the schedule will not be for the exclusive use of either the Government or the Contractor.

The use of Resource Leveling or other techniques used for the purpose of artificially adjusting activity durations to consume float and influence longest path is prohibited.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 33 00

SUBMITTAL PROCEDURES

08/18, CHG 4: 02/21

PART 1 GENERAL

- 1.1 DEFINITIONS
 - 1.1.1 DEFINITIONS
 - 1.1.2 DEFINITIONS
 - 1.1.3 DEFINITIONS
- 1.2 DEFINITIONS
 - 1.2.1 Submittal Descriptions (SD)
 - 1.2.2 Approving Authority
 - 1.2.3 Work
- 1.3 SUBMITTALS
- 1.4 SUBMITTAL CLASSIFICATION
 - 1.4.1 Government Approved (G)
 - 1.4.2 For Information Only
 - 1.4.2.1 For Information Only
 - 1.4.2.2 For Information Only
 - 1.4.2.3 For Information Only
 - 1.4.2.3.1 For Information Only
 - 1.4.2.3.2 For Information Only
 - 1.4.2.4 For Information Only
 - 1.4.3 For Information Only
 - 1.4.4 FORWARDING SUBMITTALS REQUIRING GOVERNMENT APPROVAL
- 1.5 FORWARDING SUBMITTALS REQUIRING GOVERNMENT APPROVAL
 - 1.5.1 O&M Data
 - 1.5.2 Submittals Reserved for NAVFAC Approval
 - 1.5.3 Overseas Shop Drawing Submittals
- 1.6 PREPARATION
 - 1.6.1 Transmittal Form
 - 1.6.2 Identifying Submittals
 - 1.6.3 Submittal Format
 - 1.6.3.1 Format of SD-01 Preconstruction Submittals
 - 1.6.3.2 Format for SD-02 Shop Drawings
 - 1.6.3.2.1 Drawing Identification
 - 1.6.3.3 Format of SD-03 Product Data
 - 1.6.3.3.1 Product Information
 - 1.6.3.3.2 Standards
 - 1.6.3.3.3 Data Submission
 - 1.6.3.4 Format of SD-04 Samples
 - 1.6.3.4.1 Sample Characteristics
 - 1.6.3.4.2 Sample Incorporation
 - 1.6.3.4.3 Comparison Sample
 - 1.6.3.5 Format of SD-05 Design Data
 - 1.6.3.6 Format of SD-06 Test Reports
 - 1.6.3.7 Format of SD-07 Certificates
 - 1.6.3.8 Format of SD-08 Manufacturer's Instructions
 - 1.6.3.8.1 Standards

- 1.6.3.9 Format of SD-09 Manufacturer's Field Reports
- 1.6.3.10 Format of SD-10 Operation and Maintenance Data (O&M)
- 1.6.3.11 Format of SD-11 Closeout Submittals
- 1.6.4 Source Drawings for Shop Drawings
 - 1.6.4.1 Source Drawings
 - 1.6.4.2 Terms and Conditions
- 1.6.5 Electronic File Format
- 1.7 QUANTITY OF SUBMITTALS
 - 1.7.1 Number of SD-01 Preconstruction Submittal Copies
 - 1.7.2 Number of SD-02 Shop Drawing Copies
 - 1.7.3 Number of SD-03 Product Data Copies
 - 1.7.4 Number of SD-04 Samples
 - 1.7.5 Number of SD-05 Design Data Copies
 - 1.7.6 Number of SD-06 Test Report Copies
 - 1.7.7 Number of SD-07 Certificate Copies
 - 1.7.8 Number of SD-08 Manufacturer's Instructions Copies
 - 1.7.9 Number of SD-09 Manufacturer's Field Report Copies
 - 1.7.10 Number of SD-10 Operation and Maintenance Data Copies
 - 1.7.11 Number of SD-11 Closeout Submittals Copies
- 1.8 INFORMATION ONLY SUBMITTALS
- 1.9 PROJECT SUBMITTAL REGISTER AND DATABASE
 - 1.9.1 Submittal Management
 - 1.9.2 Preconstruction Use of Submittal Register
 - 1.9.3 Preconstruction Use of Submittal Register
 - 1.9.4 Contractor Use of Submittal Register
 - 1.9.5 Approving Authority Use of Submittal Register
 - 1.9.6 Action Codes
 - 1.9.6.1 Government Review Action Codes
 - 1.9.6.2 Government Review Action Codes
 - 1.9.6.3 VARIATIONS
 - 1.9.6.4 VARIATIONS
 - 1.9.7 VARIATIONS
- 1.10 VARIATIONS
 - 1.10.1 Warranting that Variations are Compatible
 - 1.10.2 Warranting that Variations are Compatible
 - 1.10.3 Warranting that Variations are Compatible
 - 1.10.4 Review Schedule Extension
- 1.11 SCHEDULING
 - 1.11.1 Reviewing, Certifying, and Approving Authority
 - 1.11.2 Constraints
 - 1.11.3 QC Organization Responsibilities
 - 1.11.4 GOVERNMENT APPROVING AUTHORITY
- 1.12 GOVERNMENT APPROVING AUTHORITY
 - 1.12.1 Review Notations
- 1.13 DISAPPROVED SUBMITTALS
- 1.14 APPROVED SUBMITTALS
- 1.15 APPROVED SAMPLES
- 1.16 WITHHOLDING OF PAYMENT
- 1.17 CERTIFICATION OF SUBMITTAL DATA

PART 2 PRODUCTS

PART 3 EXECUTION

ATTACHMENTS:

ENG Form 4025-R

Appendix A - Submittal Register

-- End of Section Table of Contents --

SECTION 01 33 00

SUBMITTAL PROCEDURES
08/18, CHG 4: 02/21

PART 1 GENERAL

1.1 DEFINITIONS

1.1.1 Submittal Descriptions (SD)

Submittal requirements are specified in the technical sections. Examples and descriptions of submittals identified by the Submittal Description (SD) numbers and titles follow:

SD-01 Preconstruction Submittals

Submittals that are required prior to or commencing with the start of work on site. Submittals that are required prior to or at the start of construction (work) or the next major phase of the construction on a multiphase contract.

Preconstruction Submittals include schedules and a tabular list of locations, features, and other pertinent information regarding products, materials, equipment, or components to be used in the work.

Certificates Of Insurance

Surety Bonds

List Of Proposed Subcontractors

List Of Proposed Products

Baseline Network Analysis Schedule (NAS)

Submittal Register

Schedule Of Prices Or Earned Value Report

Accident Prevention Plan Health And Safety Plan

Work Plan

Quality Control (QC) plan

Environmental Protection Plan

SD-02 Shop Drawings

Drawings, diagrams and schedules specifically prepared to illustrate some portion of the work.

Diagrams and instructions from a manufacturer or fabricator for use in producing the product and as aids to the Contractor for integrating the product or system into the project.

Drawings prepared by or for the Contractor to show how multiple systems and interdisciplinary work will be coordinated.

SD-03 Product Data

Catalog cuts, illustrations, schedules, diagrams, performance charts, instructions and brochures illustrating size, physical appearance and other characteristics of materials, systems or equipment for some portion of the work.

Samples of warranty language when the contract requires extended product warranties.

SD-04 Samples

Fabricated or unfabricated physical examples of materials, equipment or workmanship that illustrate functional and aesthetic characteristics of a material or product and establish standards by which the work can be judged.

Color samples from the manufacturer's standard line (or custom color samples if specified) to be used in selecting or approving colors for the project.

Field samples and mock-ups constructed on the project site establish standards ensuring work can be judged. Includes assemblies or portions of assemblies that are to be incorporated into the project and those that will be removed at conclusion of the work.

SD-05 Design Data

Design calculations, mix designs, analyses or other data pertaining to a part of work.

Design submittals, design substantiation submittals and extensions of design submittals.

SD-06 Test Reports

Report signed by authorized official of testing laboratory that a material, product or system identical to the material, product or system to be provided has been tested in accord with specified requirements. Unless specified in another section, testing must have been within three years of date of contract award for the project.

Report that includes findings of a test required to be performed on an actual portion of the work or prototype prepared for the project before shipment to job site.

Report that includes finding of a test made at the job site or on sample taken from the job site, on portion of work during or after installation.

Investigation reports

Daily logs and checklists

Final acceptance test and operational test procedure

SD-07 Certificates

Statements printed on the manufacturer's letterhead and signed by responsible officials of manufacturer of product, system or material attesting that the product, system, or material meets specification requirements. Must be dated after award of project contract and clearly name the project.

Document required of Contractor, or of a manufacturer, supplier, installer or Subcontractor through Contractor. The document purpose is to further promote the orderly progression of a portion of the work by documenting procedures, acceptability of methods, or personnel qualifications.

Confined space entry permits

Text of posted operating instructions

SD-08 Manufacturer's Instructions

Preprinted material describing installation of a product, system or material, including special notices and (SDS)concerning impedances, hazards and safety precautions.

SD-09 Manufacturer's Field Reports

Documentation of the testing and verification actions taken by manufacturer's representative at the job site, in the vicinity of the job site, or on a sample taken from the job site, on a portion of the work, during or after installation, to confirm compliance with manufacturer's standards or instructions. The documentation must be signed by an authorized official of a testing laboratory or agency and state the test results; and indicate whether the material, product, or system has passed or failed the test.

Factory test reports.

SD-10 Operation and Maintenance Data

Data provided by the manufacturer, or the system provider, including manufacturer's help and product line documentation, necessary to maintain and install equipment, for operating and maintenance use by facility personnel.

Data required by operating and maintenance personnel for the safe and efficient operation, maintenance and repair of the item.

Data incorporated in an operations and maintenance manual or control system.

SD-11 Closeout Submittals

Documentation to record compliance with technical or administrative requirements or to establish an administrative mechanism.

Submittals required for Guiding Principle Validation (GPV) or Third Party Certification (TPC).

Special requirements necessary to properly close out a construction

contract. For example, Record Drawings and as-built drawings. Also, submittal requirements necessary to properly close out a major phase of construction on a multi-phase contract.

1.1.2 Approving Authority

Office or designated person authorized to approve the submittal.

1.1.3 Work

As used in this section, on-site and off-site construction required by contract documents, including labor necessary to produce submittals, construction, materials, products, equipment, and systems incorporated or to be incorporated in such construction. In exception, excludes work to produce SD-01 submittals.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Submittal Register; G

1.3 SUBMITTAL CLASSIFICATION

1.3.1 Government Approved (G)

Government approval is required for extensions of design, critical materials, variations, equipment whose compatibility with the entire system must be checked, and other items as designated by the Government.

Government approval is required for any variations from the Solicitation or the Accepted Proposal and for other items as designated by the Government.

Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, submittals are considered to be "shop drawings."

1.3.2 For Information Only

Submittals not requiring Government approval will be for information only. For Design-build construction all submittals not requiring DOR or Government approval will be for information only. Within the terms of the Contract Clause SPECIFICATIONS AND DRAWINGS FOR CONSTRUCTION, they are not considered to be "shop drawings."

1.4 FORWARDING SUBMITTALS REQUIRING GOVERNMENT APPROVAL

As soon as practicable after award of contract, and before procurement or fabrication, forward to the Architect-Engineer: Stanley Consultants, submittals required in the technical sections of this specification, including shop drawings, product data and samples. In addition, forward a copy of the submittals to the Contracting Officer.

1.4.1 O&M Data

Submit data specified for a given item within 30 calendar days after the item is delivered to the contract site.

In the event the Contractor fails to deliver O&M data within the time limits specified, the Contracting Officer may withhold from progress payments 50 percent of the price of the items to which such O&M data apply.

1.4.2 Submittals Reserved for NAVFAC Approval

As an exception to the standard submittal procedure for Government Approval, submit the following to the Commander, NAVFAC:

1.4.3 Overseas Shop Drawing Submittals

Send submittals via overnight express mail service. All costs associated with the overnight express mail service are borne by the Contractor. Costs associated with the overnight express mail of submittals related to proposed submittal variances of resubmittals necessary as a result of noncompliant or incomplete Contractor submittals are the responsibility of the Contractor.

1.5 PREPARATION

1.5.1 Transmittal Form

Transmit each submittal, except sample installations and sample panels to the office of the approving authority using the transmittal form prescribed by the Contracting Officer. Include all information prescribed by the transmittal form and required in paragraph IDENTIFYING SUBMITTALS. Use the submittal transmittal forms to record actions regarding samples.

Use the ENG Form 4025-R transmittal form for submitting both Government-approved and information-only submittals. Submit in accordance with the instructions on the reverse side of the form. These forms or similar forms will be furnished to the Contractor. Properly complete this form by filling out all the heading blank spaces and identifying each item submitted. Exercise special care to ensure proper listing of the specification paragraph and sheet number of the contract drawings pertinent to the data submitted for each item.

1.5.2 Identifying Submittals

The Contractor's approving authority must prepare, review and stamp submittals, including those provided by a subcontractor, before submittal to the Government.

Identify submittals, except sample installations and sample panels, with the following information permanently adhered to or noted on each separate component of each submittal and noted on transmittal form. Mark each copy of each submittal identically, with the following:

- a. Project title and location
- b. Construction contract number
- c. Dates of the drawings and revisions

- d. Name, address, and telephone number of Subcontractor, supplier, manufacturer, and any other Subcontractor associated with the submittal.
- e. Section number of the specification by which submittal is required
- f. Submittal description (SD) number of each component of submittal
- g. For a resubmission, add alphabetic suffix on submittal description, for example, submittal 18 would become 18A, to indicate resubmission
- h. Product identification and location in project.

1.5.3 Submittal Format

1.5.3.1 Format of SD-01 Preconstruction Submittals

When the submittal includes a document that is to be used in the project, or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

1.5.3.2 Format for SD-02 Shop Drawings

Provide shop drawings not less than 210 by 297 mm nor more than 1189 by 841 mm, except for full-size patterns or templates. Prepare drawings to accurate size, with scale indicated, unless another form is required. Ensure drawings are suitable for reproduction and of a quality to produce clear, distinct lines and letters, with dark lines on a white background.

- a. Include the nameplate data, size, and capacity on drawings. Also include applicable federal, military, industry, and technical society publication references.
- b. Dimension drawings, except diagrams and schematic drawings. Prepare drawings demonstrating interface with other trades to scale. Use the same unit of measure for shop drawings as indicated on the contract drawings. Identify materials and products for work shown.

Submit an electronic copy of drawings in PDF format.

1.5.3.2.1 Drawing Identification

Include on each drawing the drawing title, number, date, and revision numbers and dates, in addition to information required in paragraph IDENTIFYING SUBMITTALS.

Number drawings in a logical sequence. Each drawing is to bear the number of the submittal in a uniform location next to the title block. Place the Government contract number in the margin, immediately below the title block, for each drawing.

Reserve a blank space, no smaller than 60 millimeter on the right-hand side of each sheet for the Government disposition stamp.

1.5.3.3 Format of SD-03 Product Data

Present product data submittals for each section as a complete, bound volume. Include a table of contents, listing the page and catalog item numbers for product data.

Indicate, by prominent notation, each product that is being submitted; indicate the specification section number and paragraph number to which it pertains.

1.5.3.3.1 Product Information

Supplement product data with material prepared for the project to satisfy the submittal requirements where product data does not exist. Identify this material as developed specifically for the project, with information and format as required for submission of SD-07 Certificates.

Provide product data in metric dimensions. Where product data are included in preprinted catalogs with English units only, submit metric dimensions on separate sheet.

Provide product data in units used in the Contract documents. Where product data are included in preprinted catalogs with another unit, submit the dimensions in contract document units, on a separate sheet.

1.5.3.3.2 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.3.3.3 Data Submission

Collect required data submittals for each specific material, product, unit of work, or system into a single submittal that is marked for choices, options, and portions applicable to the submittal. Mark each copy of the product data identically. Partial submittals will not be accepted for expedition of the construction effort.

Submit the manufacturer's instructions before installation.

1.5.3.4 Format of SD-04 Samples

1.5.3.4.1 Sample Characteristics

Furnish samples in the following sizes, unless otherwise specified or unless the manufacturer has prepackaged samples of approximately the same size as specified:

- a. Sample of Equipment or Device: Full size.
- b. Sample of Materials Less Than 50 by 75 mm: Built up to 210 by 297 mm.
- c. Sample of Materials Exceeding 210 by 297 mm: Cut down to 210 by 297 mm and adequate to indicate color, texture, and material variations.
- d. Sample of Linear Devices or Materials: 250 mm length or length to be supplied, if less than 250 mm. Examples of linear devices or materials are conduit and handrails.
- e. Sample Volume of Nonsolid Materials: 750 ml. Examples of nonsolid materials are sand and paint.
- f. Color Selection Samples: 50 by 100 mm. Where samples are specified for selection of color, finish, pattern, or texture, submit the full set of available choices for the material or product specified. Sizes and quantities of samples are to represent their respective standard unit.
- g. Sample Panel: 1200 by 1200 mm.
- h. Sample Installation: 10 square meters.

1.5.3.4.2 Sample Incorporation

Reusable Samples: Incorporate returned samples into work only if so specified or indicated. Incorporated samples are to be in undamaged condition at the time of use.

Recording of Sample Installation: Note and preserve the notation of any area constituting a sample installation, but remove the notation at the final clean-up of the project.

1.5.3.4.3 Comparison Sample

Samples Showing Range of Variation: Where variations in color, finish, pattern, or texture are unavoidable due to nature of the materials, submit sets of samples of not less than three units showing extremes and middle of range. Mark each unit to describe its relation to the range of the variation.

When color, texture, or pattern is specified by naming a particular manufacturer and style, include one sample of that manufacturer and style, for comparison.

1.5.3.5 Format of SD-05 Design Data

Provide design data and certificates on 210 by 297 mm paper. Provide a bound volume for submittals containing numerous pages.

1.5.3.6 Format of SD-06 Test Reports

Provide reports on 210 by 297 mm paper in a complete bound volume.

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.3.7 Format of SD-07 Certificates

Provide design data and certificates on 210 by 297 mm paper. Provide a bound volume for submittals containing numerous pages.

1.5.3.8 Format of SD-08 Manufacturer's Instructions

Present manufacturer's instructions submittals for each section as a complete, bound volume. Include the manufacturer's name, trade name, place of manufacture, and catalog model or number on product data. Also include applicable federal, military, industry, and technical-society publication references. If supplemental information is needed to clarify the manufacturer's data, submit it as specified for SD-07 Certificates.

Submit the manufacturer's instructions before installation.

1.5.3.8.1 Standards

Where equipment or materials are specified to conform to industry or technical-society reference standards of such organizations as the American National Standards Institute (ANSI), ASTM International (ASTM), National Electrical Manufacturer's Association (NEMA), Underwriters Laboratories (UL), or Association of Edison Illuminating Companies (AEIC), submit proof of such compliance. The label or listing by the specified organization will be acceptable evidence of compliance. In lieu of the label or listing, submit a certificate from an independent testing organization, competent to perform testing, and approved by the Contracting Officer. State on the certificate that the item has been tested in accordance with the specified organization's test methods and that the item complies with the specified organization's reference standard.

1.5.3.9 Format of SD-09 Manufacturer's Field Reports

Provide reports on 210 by 297 mm paper in a complete bound volume.

By prominent notation, indicate each report in the submittal. Indicate the specification number and paragraph number to which each report pertains.

1.5.3.10 Format of SD-10 Operation and Maintenance Data (O&M)

Comply with the requirements specified in Section 01 78 23 OPERATION AND MAINTENANCE DATA for O&M Data format.

1.5.3.11 Format of SD-11 Closeout Submittals

When the submittal includes a document that is to be used in the project or is to become part of the project record, other than as a submittal, do not apply the Contractor's approval stamp to the document itself, but to a separate sheet accompanying the document.

Provide data in the unit of measure used in the contract documents.

Provide all dimensions in administrative submittals in metric. Where data are included in preprinted material with English units only, submit metric dimensions on separate sheet.

1.5.4 Source Drawings for Shop Drawings

1.5.4.1 Source Drawings

The entire set of source drawing files (DWG) will not be provided to the Contractor. Request the specific Drawing Number for the preparation of shop drawings. Only those drawings requested to prepare shop drawings will be provided. These drawings are provided only after award.

1.5.4.2 Terms and Conditions

Data contained on these electronic files must not be used for any purpose other than as a convenience in the preparation of construction data for the referenced project. Any other use or reuse is at the sole risk of the Contractor and without liability or legal exposure to the Government. The Contractor must make no claim, and waives to the fullest extent permitted by law any claim or cause of action of any nature against the Government, its agents, or its subconsultants that may arise out of or in connection with the use of these electronic files. The Contractor must, to the fullest extent permitted by law, indemnify and hold the Government harmless against all damages, liabilities, or costs, including reasonable attorney's fees and defense costs, arising out of or resulting from the use of these electronic files.

These electronic source drawing files are not construction documents. Differences may exist between the source drawing files and the corresponding construction documents. The Government makes no representation regarding the accuracy or completeness of the electronic source drawing files, nor does it make representation to the compatibility of these files with the Contractor hardware or software. The Contractor is responsible for determining if any conflict exists. In the event that a conflict arises between the signed and sealed construction documents prepared by the Government and the furnished source drawing files, the signed and sealed construction documents govern. Use of these source drawing files does not relieve the Contractor of the duty to fully comply with the contract documents, including and without limitation the need to check, confirm and coordinate the work of all contractors for the project. If the Contractor uses, duplicates or modifies these electronic source drawing files for use in producing construction data related to this contract, remove all previous indication of ownership (seals, logos, signatures, initials and dates).

1.5.5 Electronic File Format

Provide submittals in electronic format, with the exception of material samples required for SD-04 Samples items. Compile the submittal file as a single, complete document, to include the Transmittal Form described within. Name the electronic submittal file specifically according to its contents, and coordinate the file naming convention with the Contracting Officer. Electronic files must be of sufficient quality that all information is legible. Use PDF as the electronic format, unless otherwise specified or directed by the Contracting Officer. Generate PDF files from original documents with bookmarks so that the text included in the PDF file is searchable and can be copied. If documents are scanned, optical character resolution (OCR) routines are required. Index and bookmark files exceeding 30 pages to allow efficient navigation of the file. When required, the electronic file must include a valid electronic signature or a scan of a signature.

E-mail electronic submittal documents smaller than 10MB to an e-mail address as directed by the Contracting Officer. Provide electronic documents over 10 MB on an optical disc or through an electronic file sharing system such as the AMRDEC SAFE Web Application located at the following website: <https://safe.amrdec.army.mil/safe/>.

1.6 QUANTITY OF SUBMITTALS

1.6.1 Number of SD-01 Preconstruction Submittal Copies

Unless otherwise specified, submit three sets of administrative submittals.

1.6.2 Number of SD-02 Shop Drawing Copies

Submit six copies of submittals of shop drawings requiring review and approval by a QC organization. Submit seven copies of shop drawings requiring review and approval by the Contracting Officer.

1.6.3 Number of SD-03 Product Data Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.4 Number of SD-04 Samples

- a. Submit one (1) sample, or one (1) set of samples showing the range of variation, of each required item.
- b. Submit one sample panel or provide one sample installation where directed. Include components listed in the technical section or as directed.
- c. Submit one sample installation, where directed.

1.6.5 Number of SD-05 Design Data Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.6 Number of SD-06 Test Report Copies

Submit in compliance with quantity and quality requirements specified for shop drawings, other than field test results that will be submitted with QC reports.

1.6.7 Number of SD-07 Certificate Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.8 Number of SD-08 Manufacturer's Instructions Copies

Submit in compliance with quantity requirements specified for shop drawings.

1.6.9 Number of SD-09 Manufacturer's Field Report Copies

Submit in compliance with quantity and quality requirements specified for shop drawings other than field test results that will be submitted with QC

reports.

1.6.10 Number of SD-10 Operation and Maintenance Data Copies

Submit three copies of O&M data to the Contracting Officer for review and approval.

1.6.11 Number of SD-11 Closeout Submittals Copies

Unless otherwise specified, submit three sets of administrative submittals.

1.7 INFORMATION ONLY SUBMITTALS

Submittals without a "G" designation must be certified by the QC manager and submitted to the Contracting Officer for information-only. Provide information-only submittals to the Contracting Officer a minimum of 14 calendar days prior to the Preparatory Meeting for the associated Definable Feature of Work (DFOW). Approval of the Contracting Officer is not required on information only submittals. The Contracting Officer will mark "receipt acknowledged" on submittals for information and will return only the transmittal cover sheet to the Contractor. Normally, submittals for information only will not be returned. However, the Government reserves the right to return unsatisfactory submittals and require the Contractor to resubmit any item found not to comply with the contract. This does not relieve the Contractor from the obligation to furnish material conforming to the plans and specifications; will not prevent the Contracting Officer from requiring removal and replacement of nonconforming material incorporated in the work; and does not relieve the Contractor of the requirement to furnish samples for testing by the Government laboratory or for check testing by the Government in those instances where the technical specifications so prescribe.

1.8 PROJECT SUBMITTAL REGISTER AND DATABASE

A sample Project Submittal Register showing items of equipment and materials for when submittals are required by the specifications is provided as "Appendix A - Submittal Register."

1.8.1 Submittal Management

Prepare and maintain a submittal register, as the work progresses. Use an electronic submittal register program furnished by the Government. Do not change data that is output in columns (c), (d), (e), and (f) as delivered by Government; retain data that is output in columns (a), (g), (h), and (i) as approved. As an attachment, provide a submittal register showing items of equipment and materials for which submittals are required by the specifications. This list may not be all-inclusive and additional submittals may be required. Maintain a submittal register for the project in accordance with contract requirements. The Government will provide the initial submittal register in electronic format.

Column (c): Lists specification section in which submittal is required.

Column (d): Lists each submittal description (SD Number. and type, e.g., SD-02 Shop Drawings) required in each specification section.

Column (e): Lists one principal paragraph in each specification

section where a material or product is specified. This listing is only to facilitate locating submitted requirements. Do not consider entries in column (e) as limiting the project requirements.

Column (f): Lists the approving authority for each submittal.

1.8.2 Preconstruction Use of Submittal Register

Submit the submittal register as an electronic database, using the submittal management program furnished to Contractor. Include the QC plan and the project schedule. Verify that all submittals required for the project are listed and add missing submittals. Coordinate and complete the following fields on the register database submitted with the QC plan and the project schedule:

Column (a) Activity Number: Activity number from the project schedule.

Column (g) Contractor Submit Date: Scheduled date for the approving authority to receive submittals.

Column (h) Contractor Approval Date: Date that Contractor needs approval of submittal.

Column (i) Contractor Material: Date that Contractor needs material delivered to Contractor control.

1.8.3 Contractor Use of Submittal Register

Update the following fields in the Government-furnished submittal register program or equivalent fields in the program used by the Contractor with each submittal throughout the contract.

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (j) Action Code (k): Date of action used to record Contractor's review when forwarding submittals to QC.

Column (l) Date submittal transmitted.

Column (q) Date approval was received.

1.8.4 Approving Authority Use of Submittal Register

Update the following fields:

Column (b) Transmittal Number: List of consecutive, Contractor-assigned numbers.

Column (l) Date submittal was received.

Column (m) through (p) Dates of review actions.

Column (q) Date of return to Contractor.

1.8.5 Action Codes

Entries for columns (j) and (o) are to be used as follows (others may be prescribed by the Transmittal Form):

1.8.5.1 Government Review Action Codes

"A" - "Approved as submitted"; "Completed"

"B" - "Approved, except as noted on drawings"; "Completed"

"C" - "Approved, except as noted on drawings; resubmission required"; "Resubmit"

"D" - "Returned by separate correspondence"; "Completed"

"E" - "Disapproved (See attached)"; "Resubmit"

"F" - "Receipt acknowledged"; "Completed"

"G" - "Other (Specify)"; "Resubmit"

"X" - "Receipt acknowledged, does not comply with contract requirements"; "Resubmit"

1.8.5.2 Government Review Action Codes

"A" - "Approved as submitted"

"AN" - "Approved as noted"

"RR" - "Disapproved as submitted"; "Completed"

"NR" - "Not Reviewed"

"RA" - "Receipt Acknowledged"

1.9 VARIATIONS

Variations from contract requirements require Contracting Officer approval pursuant to contract Clause FAR 52.236-21 Specifications and Drawings for Construction, and will be considered where advantageous to the Government.

1.9.1 Warranting that Variations are Compatible

When delivering a variation for approval, the Contractor warrants that this contract has been reviewed to establish that the variation, if incorporated, will be compatible with other elements of work.

1.9.2 Review Schedule Extension

In addition to the normal submittal review period, a period of 14 calendar working days will be allowed for the Government to consider submittals with variations.

1.10 SCHEDULING

Schedule and submit concurrently product data and shop drawings covering component items forming a system or items that are interrelated. Submit

pertinent certifications at the same time. No delay damages or time extensions will be allowed for time lost in late submittals.

- a. Coordinate scheduling, sequencing, preparing, and processing of submittals with performance of work so that work will not be delayed by submittal processing. The Contractor is responsible for additional time required for Government reviews resulting from required resubmittals. The review period for each resubmittal is the same as for the initial submittal.
- b. Submittals required by the contract documents are listed on the submittal register. If a submittal is listed in the submittal register but does not pertain to the contract work, the Contractor is to include the submittal in the register and annotate it "N/A" with a brief explanation. Approval by the Contracting Officer does not relieve the Contractor of supplying submittals required by the contract documents but that have been omitted from the register or marked "N/A."
- c. Resubmit the submittal register and annotate it monthly with actual submission and approval dates. When all items on the register have been fully approved, no further resubmittal is required.

Contracting Officer review will be completed within 14 calendar working days after the date of submission.

- d. Except as specified otherwise, allow a review period, beginning with receipt by the approving authority, that includes at least 15 working days for submittals for QC manager approval and 20 working days for submittals where the Contracting Officer is the approving authority. The period of review for submittals with Contracting Officer approval begins when the Government receives the submittal from the QC organization.

1.10.1 Reviewing, Certifying, and Approving Authority

The QC Manager is responsible for reviewing all submittals and certifying that they are in compliance with contract requirements. The approving authority on submittals is the QC Manager unless otherwise specified. At each "Submittal" paragraph in individual specification sections, a notation "G" following a submittal item indicates that the Contracting Officer is the approving authority for that submittal item. Provide an additional copy of the submittal to the Government Approving authority

1.10.2 Constraints

Conform to provisions of this section, unless explicitly stated otherwise for submittals listed or specified in this contract.

Submit complete submittals for each definable feature of the work. At the same time, submit components of definable features that are interrelated as a system.

When acceptability of a submittal is dependent on conditions, items, or materials included in separate subsequent submittals, the submittal will be returned without review.

Approval of a separate material, product, or component does not imply

approval of the assembly in which the item functions.

1.10.3 QC Organization Responsibilities

- a. Review submittals for conformance with project design concepts and compliance with contract documents.
- b. Process submittals based on the approving authority indicated in the submittal register.
 - (1) When the QC manager is the approving authority, take appropriate action on the submittal from the possible actions defined in paragraph APPROVED SUBMITTALS.
 - (2) When the Contracting Officer is the approving authority or when variation has been proposed, forward the submittal to the Government, along with a certifying statement, or return the submittal marked "not reviewed" or "revise and resubmit" as appropriate. The QC organization's review of the submittal determines the appropriate action.
- c. Ensure that material is clearly legible.
- d. Stamp each sheet of each submittal with a QC certifying statement or an approving statement, except that data submitted in a bound volume or on one sheet printed on two sides may be stamped on the front of the first sheet only.
 - (1) When the approving authority is the Contracting Officer, the QC organization will certify submittals forwarded to the Contracting Officer with the following certifying statement:

"I hereby certify that the (equipment) (material) (article) shown and marked in this submittal is that proposed to be incorporated with Contract Number (contract number to be determined) is in compliance with the contract drawings and specification, can be installed in the allocated spaces, and is submitted for Government approval.

Certified by Submittal Reviewer _____, Date _____
(Signature when applicable)

Certified by QC Manager _____, Date _____"
(Signature)

- (2) When approving authority is the QC manager, the QC manager will use the following approval statement when returning submittals to the Contractor as "Approved" or "Approved as Noted."

"I hereby certify that the (material) (equipment) (article) shown and marked in this submittal and proposed to be incorporated with Contract Number (contract number to be determined) is in compliance with the contract drawings and specification, can be installed in the allocated spaces, and is approved for use.

Certified by Submittal Reviewer _____, Date _____
(Signature when applicable)

Approved by QC Manager _____, Date _____"
(Signature)

- e. Sign the certifying statement or approval statement. The QC organization member designated in the approved QC plan is the person signing certifying statements. The use of original ink for signatures is required. Stamped signatures are not acceptable.
- f. Update the submittal register as submittal actions occur, and maintain the submittal register at the project site until final acceptance of all work by the Contracting Officer.
- g. Retain a copy of approved submittals and approved samples at the project site.
- h. For "S" submittals, provide a copy of the approved submittal to the Government Approving authority.

1.11 GOVERNMENT APPROVING AUTHORITY

When the approving authority is the Contracting Officer, the Government will:

- a. Note the date on which the submittal was received from the QC manager.
- b. Review submittals for approval within the scheduling period specified and only for conformance with project design concepts and compliance with contract documents.
- c. Identify returned submittals with one of the actions defined in paragraph REVIEW NOTATIONS and with comments and markings appropriate for the action indicated.

Upon completion of review of submittals requiring Government approval, stamp and date submittals. Three copies of the submittal will be retained by the Contracting Officer and two copies of the submittal will be returned to the Contractor. If the Government performs a conformance review of other Designer of Record approved submittals, the submittals will be identified and returned, as described above.

1.11.1 Review Notations

Submittals will be returned to the Contractor with the following notations:

- a. Submittals marked "approved" or "accepted" authorize proceeding with the work covered.
- b. Submittals marked "approved as noted" or "approved, except as noted, resubmittal not required," authorize proceeding with the work covered provided that the Contractor takes no exception to the corrections.
- c. Submittals marked "not approved," "disapproved," or "revise and resubmit" indicate incomplete submittal or noncompliance with the contract requirements or design concept. Resubmit with appropriate changes. Do not proceed with work for this item until the resubmittal is approved.
- d. Submittals marked "not reviewed" indicate that the submittal has been previously reviewed and approved, is not required, does not have evidence of being reviewed and approved by Contractor, or is not complete. A submittal marked "not reviewed" will be returned with an

explanation of the reason it is not reviewed. Resubmit submittals returned for lack of review by Contractor or for being incomplete, with appropriate action, coordination, or change.

- e. Submittals marked "receipt acknowledged" indicate that submittals have been received by the Government. This applies only to "information-only submittals" as previously defined.

1.12 DISAPPROVED SUBMITTALS

Make corrections required by the Contracting Officer. If the Contractor considers any correction or notation on the returned submittals to constitute a change to the contract drawings or specifications, give notice to the Contracting Officer as required under the FAR clause titled CHANGES. The Contractor is responsible for the dimensions and design of connection details and the construction of work. Failure to point out variations may cause the Government to require rejection and removal of such work at the Contractor's expense.

If changes are necessary to submittals, make such revisions and resubmit in accordance with the procedures above. No item of work requiring a submittal change is to be accomplished until the changed submittals are approved.

1.13 APPROVED SUBMITTALS

The Contracting Officer's approval of submittals is not to be construed as a complete check, and indicates only that the general method of construction, materials, detailing, and other information are satisfactory. The design, general method of construction, materials, detailing, and other information appear to meet the Solicitation and Accepted Proposal.

Approval or acceptance by the Government for a submittal does not relieve the Contractor of the responsibility for meeting the contract requirements or for any error that may exist, because under the Quality Control (QC) requirements of this contract, the Contractor is responsible for ensuring information contained within each submittal accurately conforms with the requirements of the contract documents.

After submittals have been approved or accepted by the Contracting Officer, no resubmittal for the purpose of substituting materials or equipment will be considered unless accompanied by an explanation of why a substitution is necessary.

1.14 APPROVED SAMPLES

Approval of a sample is only for the characteristics or use named in such approval and is not to be construed to change or modify any contract requirements. Before submitting samples, provide assurance that the materials or equipment will be available in quantities required in the project. No change or substitution will be permitted after a sample has been approved.

Match the approved samples for materials and equipment incorporated in the work. If requested, approved samples, including those that may be damaged in testing, will be returned to the Contractor, at its expense, upon completion of the contract. Unapproved samples will also be returned to the Contractor at its expense, if so requested.

Failure of any materials to pass the specified tests will be sufficient cause for refusal to consider, under this contract, any further samples of the same brand or make as that material. The Government reserves the right to disapprove any material or equipment that has previously proved unsatisfactory in service.

Samples of various materials or equipment delivered on the site or in place may be taken by the Contracting Officer for testing. Samples failing to meet contract requirements will automatically void previous approvals. Replace such materials or equipment to meet contract requirements.

1.15 WITHHOLDING OF PAYMENT

Payment for materials incorporated in the work will not be made if required approvals have not been obtained. No payment for materials incorporated in the work will be made unless all required DOR approvals or required Government approvals have been obtained. No payment will be made for any materials incorporated into the work for any conformance review submittals or information-only submittals found to contain errors or deviations from the Solicitation or Accepted Proposal.

1.16 CERTIFICATION OF SUBMITTAL DATA

Certify the submittal data as follows on Form ENG 4025: "I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the contract drawings and specifications except as otherwise stated.

____NAME OF CONTRACTOR _____ SIGNATURE OF CONTRACTOR

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

Not Used

-- End of Section --

TRANSMITTAL OF SHOP DRAWINGS, EQUIPMENT DATA, MATERIAL SAMPLES, OR MANUFACTURER'S CERTIFICATES OF COMPLIANCE For use of this form, see ER 415-1-10; the proponent agency is CECW-CE.					DATE		TRANSMITTAL NO.	
SECTION I - REQUEST FOR APPROVAL OF THE FOLLOWING ITEMS <i>(This section will be initiated by the contractor)</i>								
TO:		FROM:		CONTRACT NO.			CHECK ONE: ☐ THIS IS A NEW TRANSMITTAL ☐ THIS IS A RESUBMITTAL OF TRANSMITTAL _____	
SPECIFICATION SEC. NO. <i>(Cover only one section with each transmittal)</i>		PROJECT TITLE AND LOCATION			THIS TRANSMITTAL IS FOR: <i>(Check one)</i> ☐ FIO ☐ GA ☐ DA ☐ CR ☐ DA/CR ☐ DA/GA			
ITEM NO. <i>(See Note 3)</i> a.	DESCRIPTION OF SUBMITTAL ITEM <i>(Type size, model number/etc.)</i> b.	SUBMITTAL TYPE CODE <i>(See Note 8)</i> c.	NO. OF COPIES d.	CONTRACT DOCUMENT REFERENCE		CONTRACTOR REVIEW CODE g.	VARIATION Enter "Y" if requesting a variation <i>(See Note 6)</i> h.	USACE ACTION CODE <i>(Note 9)</i> i.
e.	f.							
REMARKS				I certify that the above submitted items had been reviewed in detail and are correct and in strict conformance with the contract drawings and specifications except as otherwise stated.				
				NAME OF CONTRACTOR			SIGNATURE OF CONTRACTOR	
SECTION II - APPROVAL ACTION								
ENCLOSURES RETURNED <i>(List by item No.)</i>		NAME AND TITLE OF APPROVING AUTHORITY			SIGNATURE OF APPROVING AUTHORITY		DATE	

INSTRUCTIONS

1. Section I will be initiated by the Contractor in the required number of copies.
2. Each Transmittal shall be numbered consecutively. The Transmittal Number typically includes two parts separated by a dash (-). The first part is the specification section number. The second part is a sequential number for the submittals under that spec section. If the Transmittal is a resubmittal, then add a decimal point to the end of the original Transmittal Number and begin numbering the resubmittal packages sequentially after the decimal.
3. The "Item No." for each entry on this form will be the same "Item No." as indicated on ENG FORM 4288-R.
4. Submittals requiring expeditious handling will be submitted on a separate ENG Form 4025-R.
5. Items transmitted on each transmittal form will be from the same specification section. Do not combine submittal information from different specification sections in a single transmittal.
6. If the data submitted are intentionally in variance with the contract requirements, indicate a variation in column h, and enter a statement in the Remarks block describing the detailed reason for the variation.
7. ENG Form 4025-R is self-transmitting - a letter of transmittal is not required.
8. When submittal items are transmitted, indicate the "Submittal Type" (*SD-01 through SD-11*) in column c of Section I.
Submittal types are the following:

SD-01 - Preconstruction	SD-02 - Shop Drawings	SD-03 - Product Data	SD-04 - Samples	SD-05 - Design Data	SD-06 - Test Reports
SD-07 - Certificates	SD-08 - Manufacturer's Instructions	SD-09 - Manufacturer's Field Reports	SD-10 - O&M Data	SD-11 - Closeout	
9. For each submittal item, the Contractor will assign Submittal Action Codes in column g of Section I. The U.S. Army Corps of Engineers approving authority will assign Submittal Action Codes in column i of Section I. The Submittal Action Codes are:

A -- Approved as submitted.	F -- Receipt acknowledged.
B -- Approved, except as noted on drawings. Resubmission not required.	X -- Receipt acknowledged, does not comply with contract requirements, as noted.
C -- Approved, except as noted on drawings. Refer to attached comments. Resubmission required.	G -- Other action required (<i>Specify</i>)
D -- Will be returned by separate correspondence.	K -- Government concurs with intermediate design. (<i>For D-B contracts</i>)
E -- Disapproved. Refer to attached comments.	R -- Design submittal is acceptable for release for construction. (<i>For D-B contracts</i>)
10. Approval of items does not relieve the contractor from complying with all the requirements of the contract.

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		00 73 04.00 41	SD-01 Preconstruction Submittals														
			Contractor personnel	1.18.1	G												
			identification														
		01 11 00	SD-01 Preconstruction Submittals														
			Salvage Plan	1.5	G												
		01 14 00	SD-01 Preconstruction Submittals														
			List of Contact Personnel	1.3.1.1													
			SD-07 Certificates														
			Proof of IMD Ordnance Work	1.4.7.1													
			Authorization														
		01 20 00	SD-01 Preconstruction Submittals														
			Schedule of Prices	1.3	G												
		01 30 00	SD-01 Preconstruction Submittals														
			View Location Map	1.3													
			Progress and Completion	1.4													
			Pictures														
		01 32 17.00 20	SD-01 Preconstruction Submittals														
			Baseline NAS	1.2.2	G												
			Baseline NAS	1.2.3.1	G												
			Construction Baseline NAS	1.2.3.3	G												
			Designated Project Scheduler	1.9	G												
			SD-07 Certificates														
			Three-Week Look Ahead	1.3	G												
			Schedule														
			Monthly Network Analysis	1.4.1	G												
			Updates														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 32 17.00 20	SD-11 Closeout Submittals														
			As-Built Schedule	1.4.2	G												
		01 33 00	SD-01 Preconstruction Submittals														
			Submittal Register	1.8	G												
		01 35 26	SD-01 Preconstruction Submittals														
			Accident Prevention Plan (APP)	1.8	G												
			Accident Prevention Plan (APP)	1.8	G												
			APP - Construction	1.8.1	G												
			SD-06 Test Reports														
			Monthly Exposure Reports	1.4													
			Notifications and Reports	1.13													
			Accident Reports	1.13.2	G												
			LHE Inspection Reports	1.13.3													
			Gas Protection	1.17													
			SD-07 Certificates														
			Contractor Safety Self-Evaluation	1.5													
			Checklist														
			Crane Operators/Riggers	1.7.1.6													
			Standard Lift Plan	1.8.3.2	G												
			Critical Lift Plan	1.8.3.3	G												
			Activity Hazard Analysis (AHA)	1.9													
			Confined Space Entry Permit	1.10.1													
			Hot Work Permit	1.10.1													
			Certificate of Compliance	1.13.4													
			License Certificates	1.15													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS S I F I C A T I O N A / E R E V I O W N R	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 35 26	Radiography Operation Planning Work Sheet	1.15.1	G												
			Portable Gauge Operations Planning Worksheet	1.15.1	G												
		01 45 00.00 20	SD-01 Preconstruction Submittals														
			Construction Quality Control (QC) Plan	1.6.1	G												
			Indoor Air Quality (IAQ) Management Plan	1.16	G												
		01 50 00	SD-01 Preconstruction Submittals														
			Construction Site Plan	1.3	G												
			Traffic Control Plan	3.4.1	G												
			Haul Road Plan	2.2.1	G												
			Contractor Computer Cybersecurity Compliance Statements	1.5.1.4	G												
			Contractor Temporary Network Cybersecurity Compliance Statements	1.5.6	G												
		01 57 19.01 21	SD-01 Preconstruction Submittals														
			Preconstruction Environmental, Including Common Chameleon and Vegetation, Survey		G												
			Preconstruction Archaeological Survey		G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH #	GOVT CLASS SIF CATT ION R	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 19.01 21	Environmental Protection Plan (EPP)		G												
			Spill Prevention, Control and Response Plan (for projects at or near the Bay of Cadiz)		G												
			Estimate of Solid Waste and Recyclables weights and disposal/recycle locations.		G												
			List of Employee Environmental Protection Trainings		G												
			Environmental Manager Qualifications		G												
			Safety Data Sheets (SDS) for Hazardous Materials														
			SD-07 Certificates														
			List of Employee Environmental Training Records		G												
			List of ECATTS Environmental on-line course Certificates of Completion		G												
			SD-11 Closeout Submittals														
			Stormwater Pollution Prevention Plan Compliance Notebook		G												
			Solid Waste and Recycling Report (by weight)		G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 57 19.01 21	Disposal Documentation for Hazardous and Regulated Waste		G												
			Contractor Hazardous Material Inventory Log		G												
			Regulatory Notifications		G												
			Sales Documentation		G												
			Construction & Demolition (C & D) Debris data and management cost														
		01 74 19	SD-01 Preconstruction Submittals														
			Construction Waste Management Plan	1.7	G												
			SD-11 Closeout Submittals														
			Final Construction Waste Diversion Report	1.10	S												
		01 78 00	SD-03 Product Data														
			Warranty Management Plan	1.5.1													
			Warranty Tags	1.5.4													
			Final Cleaning	3.2													
			Spare Parts Data	1.4													
			SD-08 Manufacturer's Instructions														
			Instructions	1.5.1													
			SD-11 Closeout Submittals														
			As-Built Drawings	3.1	G												
			As-Built Record of Equipment and Materials	1.5.1													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 78 00	Certification of EPA Designated Items	2.1	G												
			Interim DD FORM 1354	3.3.1	G												
			Checklist for DD FORM 1354	3.3.2	G												
		01 78 23	SD-10 Operation and Maintenance Data														
			O&M Database	1.4	G												
			Training Plan	3.1.1	G												
			Training Outline	3.1.3	G												
			Training Content	3.1.2	G												
			SD-11 Closeout Submittals														
			Training Video Recording	3.1.4	G												
			Validation of Training Completion	3.1.6	G												
		01 78 24.00 10	SD-01 Preconstruction Submittals														
			Facility Data Project Execution Plan	1.5.1													
			SD-10 Operation and Maintenance Data														
			Facility Data Workbook, Construction Progress	3.1	G												
			Facility Document Set, Construction Progress	3.1	G												
			SD-11 Closeout Submittals														
			Facility Data Workbook, Construction Final	3.2	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		01 78 24.00 10	Facility Document Set, Construction Final	1.5.3	G												
			Facility Document Set, Construction Final	3.2	G												
		02 41 00	SD-01 Preconstruction Submittals														
			Demolition Plan	1.2.2	G												
			Existing Conditions	1.10													
		02 61 13	SD-01 Preconstruction Submittals														
			Work Plan	1.3	G												
			SD-02 Shop Drawings														
			Surveys	3.1													
			SD-06 Test Reports														
			Closure Report	3.10	G												
		03 30 00	SD-01 Preconstruction Submittals														
			Concrete Curing Plan	1.6.3.1													
			Quality Control Plan	1.6.5	G												
			Quality Control Personnel	1.6.6	G												
			Certifications														
			Quality Control Organizational	1.6.6													
			Chart														
			Laboratory Accreditation	1.6.8	G												
			Form Removal Schedule	1.6.2.1	G												
			SD-02 Shop Drawings														
			Formwork	1.6.2.1													
			Reinforcing Steel	1.6.2.2	G												
			SD-03 Product Data														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		03 30 00	Joint Sealants	2.4.4													
			Joint Filler	2.4.3													
			Formwork Materials	2.1													
			Recycled Aggregate Materials	2.3.3.2													
			Cementitious Materials	2.3.1													
			Concrete Curing Materials	2.4.1													
			Reinforcement	2.6													
			Admixtures	2.3.4													
			Waterstops	2.2.2													
			Local/Regional Materials	1.8.1													
			Biodegradable Form Release Agent	2.2.3													
			Pumping Concrete	1.6.3.2													
			Nonshrink Grout	2.4.2													
			SD-05 Design Data														
			Concrete Mix Design	1.6.1.2	G												
			Formwork Calculations	1.6.1.1													
			SD-06 Test Reports														
			Concrete Mix Design	1.6.1.2	G												
			Fly Ash	1.6.4.1													
			Pozzolan	1.6.4.1													
			Slag Cement	1.6.4.2													
			Aggregates	1.6.4.3													
			Compressive Strength Tests	3.13.3.3	G												
			Slump Tests	3.13.3.1													
			Water	2.3.2													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		03 30 00	SD-07 Certificates														
			Reinforcing Bars	2.6.1													
			VOC Content for Form Release	1.6.3.3													
			Agents, Curing Compounds, and														
			Concrete Penetrating Sealers														
			Safety Data Sheets	1.6.3.4													
			Field Testing Technician and	1.6.6.2													
			Testing Agency														
			SD-08 Manufacturer's Instructions														
			Curing Compound	2.4.1													
		05 50 13	SD-02 Shop Drawings														
			Floor Gratings	2.3	G												
			Bollards	2.4	G												
			Angles and Plates	2.5	G												
			SD-03 Product Data														
			Floor Gratings	2.3	G												
			SD-07 Certificates														
			Certified Mill	2.1	G												
		09 97 13.27	SD-05 Design Data														
			Containment System	1.4.4.1													
			SD-06 Test Reports														
			Joint Sealant Qualification Test	1.4.5.1													
			Reports														
			Coatings Qualification Test	1.4.5.2													
			Reports														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		09 97 13.27	Metallic Abrasive Qualification Test Reports	1.4.5.3													
			Coating Sample Test Reports	3.1.3													
			Abrasive Sample Test Reports	3.1.4													
			Inspection Report Forms	3.7.2.2													
			Daily Inspection Reports	3.7.2.3													
			Recycled Metallic Abrasive Field Test Reports (Daily and Weekly)	1.4.5.4													
			SD-07 Certificates														
			Contract Errors, Omissions, and Other Discrepancies	1.4.1													
			Corrective Action Procedures	1.4.2.1													
			Coating Work Plan	1.4.3													
			Qualifications of Certified Industrial Hygienist (CIH)	1.4.6.1													
			Qualifications Of Individuals Performing Abrasive Blasting	1.4.6.5													
			Qualifications of Certified Protective Coatings Specialist (PCS)	1.4.6.2													
			Qualifications of Coating Inspection Company	1.4.6.3													
			Qualifications of QC Specialist Coating Inspector	1.4.6.4													
			Qualifications of Testing Laboratory for Coatings	1.4.6.6													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		09 97 13.27	Qualifications of Testing Laboratory for Abrasive	1.4.6.7													
			Qualifications of Coating Contractors	1.4.6.8													
			Joint Sealant Materials	1.4.6.9													
			Coating Materials	1.4.6.10													
			Coating System Component Compatibility	1.4.6.11													
			Non-metallic Abrasive	1.4.6.12													
			Metallic Abrasive	1.4.6.13													
			SD-08 Manufacturer's Instructions														
			Coating System Instructions	1.5.1													
			SD-11 Closeout Submittals														
			Disposal of Used Abrasive	3.5.6													
			Inspection Logbook	3.7.2.4	G												
		25 05 11	SD-01 Preconstruction Submittals														
			Contractor Computer Cybersecurity Compliance Statements	1.7.1.6	G												
			Contractor Temporary Network Cybersecurity Compliance Statements	1.7.6	G												
			SD-02 Shop Drawings														
			Cybersecurity Riser Diagram	1.6.2	G												
			SD-03 Product Data														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION						CONTRACTOR											
Replace Fuel Pipeline from Pier 3 to Bulk Storage																	
ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		25 05 11	Control System Cybersecurity Documentation	1.6.3	G												
			SD-06 Test Reports														
			Control System Cybersecurity Testing Procedures	3.6.1	G												
			Control System Cybersecurity Testing Report	3.6.3	G												
			SD-11 Closeout Submittals														
			Password Change Summary Report	3.4.2.2	G												
			Enclosure Keys	3.3.2	G												
			Control System Inventory Report	1.6.1	G												
		26 08 00	SD-06 Test Reports														
			Acceptance Tests and Inspections	3.1	G												
			SD-07 Certificates														
			Qualifications	1.4.1	G												
			Acceptance Test and Inspections Procedure	1.4.3	G												
		26 20 00	SD-02 Shop Drawings														
			Panelboards	2.7	G												
			Marking Strips	3.1.7.1	G												
			SD-03 Product Data														
			Circuit Breakers	2.7.1	G												
			Motor Controllers	2.10	G												
			Primary Bonding Busbar	2.12.3	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		26 20 00	SD-06 Test Reports														
			600-volt Wiring Test	3.5.2	G												
			Grounding System Test	3.5.4	G												
			Ground-fault Receptacle Test	3.5.3	G												
			SD-09 Manufacturer's Field Reports														
			Transformer Factory Tests	2.18.1													
			SD-10 Operation and Maintenance Data														
			Electrical Systems	1.5.1	G												
		26 28 01.00 10	SD-03 Product Data														
			Fault Current Analysis	2.1													
			Protective Device Coordination Study	2.1													
			System Coordinator	1.4.1													
		26 42 17	SD-01 Preconstruction Submittals														
			Preconstruction Survey	1.8.2													
			SD-02 Shop Drawings														
			Drawings	2.1.2.6	G AE												
			Test stations	2.3.4													
			Joint bonds	3.5.2													
			Contractor's Modifications	2.1.2.8	G AE												
			SD-03 Product Data														
			Qualifications	1.6.2													
			Equipment	1.5.1	G AE												
			Test stations	2.3.4													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		26 42 17	Isolation flange kits	2.5.9													
			Wires	2.1.2.2													
			Cable and wire	2.4.2													
			Shunts	2.3.2													
			Permanent reference electrodes	2.4.3	G AE												
			Spare Parts	1.5.1													
			SD-06 Test Reports														
			Tests and Measurements	3.6.1	G AE												
			Contractor's Modifications	2.1.2.8	G AE												
			SD-10 Operation and Maintenance														
			Data														
			Cathodic Protection System	3.6	G AE												
			Cathodic Protection System	3.6	G AE												
			Training Course	1.5.1	G AE												
			SD-11 Closeout Submittals														
			Initial Cathodic Protection System	3.6.1.13	G AE												
			Field Testing														
			One Year Warranty Period	3.6.1.15	G AE												
			Cathodic Protection System Field														
			Test Report														
			Final Cathodic Protection System	3.6.1.16	G AE												
			Field Test Report														
		31 23 00.00 20	SD-01 Preconstruction Submittals														
			Shoring and Sheeting Plan	1.7.1													
			Dewatering work plan	1.7.2													
			SD-06 Test Reports														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		31 23 00.00 20	Borrow Site Testing	1.6	G												
			Fill and backfill	3.12.2.1													
			Select material	3.12.2.2													
			Density tests	3.12.2.3													
		32 11 23	SD-03 Product Data														
			Plant, Equipment, and Tools	2.4	G												
			SD-06 Test Reports														
			Initial Tests	2.3.1	G												
			In-Place Tests	3.12.1	G												
		32 12 13	SD-03 Product Data														
			Prime Coat	2.1	G												
			Tack Coat	2.2	G												
			SD-06 Test Reports														
			Sampling and Testing	3.7													
		32 12 16.16	SD-02 Shop Drawings														
			Placement Plan	2.1	G												
			SD-03 Product Data														
			Diamond Grinding Plan	2.1.5	G												
			Mix Design	2.4	G												
			Contractor Quality Control	3.1	G												
			SD-04 Samples														
			Aggregates	2.2													
			Asphalt Cement Binder	2.3													
			SD-06 Test Reports														
			Aggregates	2.2	G												
			QC Monitoring	3.1.3.9													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION REVIEW	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		32 12 16.16	SD-07 Certificates														
			Asphalt Cement Binder	2.3	G												
			Laboratory Accreditation and Validation	1.5.11													
		32 15 00	SD-03 Product Data														
			Plant, Equipment, and Tools	1.4	G												
			SD-06 Test Reports														
			Initial Tests	2.3.1	G												
			In-Place Tests	3.9.1	G												
		32 16 19	SD-03 Product Data														
			Concrete	2.1													
			SD-04 Samples														
			Concrete Curbs	3.4.4													
			SD-06 Test Reports														
			Field Quality Control	3.7													
		32 17 23	SD-03 Product Data														
			Surface Preparation Equipment List	2.1.1.1	G												
			Application Equipment List	2.1.2	G												
			Exterior Surface Preparation	3.2													
			Safety Data Sheets	1.3.1	G												
			Reflective media for roads	2.2.2.1	G												
			Waterborne Paint	2.2.1	G												
			SD-06 Test Reports														
			Reflective Media for Roads	2.2.2.1	G												
			Waterborne Paint	2.2.1	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS S I F I C A T I O N A / E R E V W R	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		32 17 23	SD-07 Certificates														
			Qualifications	1.3.2	G												
			Reflective Media for Roads	2.2.2.1													
			Waterborne Paint	2.2.1													
			Volatile Organic Compound	1.3.1	G												
			SD-08 Manufacturer's Instructions														
			Waterborne Paint	2.2.1	G												
		32 31 13.53	SD-02 Shop Drawings														
			Fence Installation Drawings	3.2.1	G												
			SD-03 Product Data														
			Fabric	2.1.1.1													
			Posts	2.1.2													
			Post Caps	2.1.2.2													
			Chain Link Braces	2.1.3													
			Rails	2.1.3													
			Tension Wire	2.1.2.2													
			Barbed Wire	2.2.1.2													
			Barbed Wire Supporting Arms	2.1.2.2													
			Latches	2.1.4.3													
			Hinges	2.1.4.3													
			Stops	2.1.4.3													
			Keepers	2.1.4.3													
			Padlocks	2.1.5													
			Wire Ties	2.2.1.1													
			SD-07 Certificates														
			Chain Link Fence	2.1.2.1													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		32 31 13.53	Fabric	2.1.1.1													
			Barbed Wire	2.2.1.2													
			Gate Hardware and Accessories	3.2.1													
			Concrete	2.2.2													
			SD-10 Operation and Maintenance Data														
			Operating and maintenance instructions	3.2.7													
		32 92 19	SD-03 Product Data														
			Wood Cellulose Fiber Mulch	2.5.1													
			Fertilizer	2.4													
			SD-06 Test Reports														
			Topsoil Composition Tests	2.2.3													
			SD-07 Certificates														
			Seed	2.1													
			SD-08 Manufacturer's Instructions														
			Erosion Control Materials	2.7													
		33 08 55	SD-01 Preconstruction Submittals														
			Copies of API RP 1595	1.3.2	G												
			Existing System Cleanliness	1.3.3	G												
			Confirmation Plan														
			System Start-Up Plan	1.3.5	G												
			Performance Testing Plan	1.3.1	G												
			Fuel Provisioning Plan	1.3.1	G												
			SD-06 Test Reports														
			Final Reports	1.5.1	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		33 08 55	Piping Flushing Checklist	3.5.3	G												
			SD-07 Certificates														
			Fuel Testing Laboratory	1.3.4	G												
			Qualifications														
			Certification (Ready for Start-Up and Performance Testing)	1.3.8	G												
			Contractor Start-Up Qualifications	1.6.1	G												
			Certificate of Calibration	2.2.1	G												
			Disposal Of Waste Materials	1.7													
			SD-11 Closeout Submittals														
			Certification of Entire System	1.6.2													
		33 52 23.15	SD-01 Preconstruction Submittals														
			Welding Procedure Specifications (WPS)	1.6	G												
			Welder Performance Qualification (WPQ)	1.6	G												
			Procedure Qualification Record (PQR)	1.6	G												
			Welding Operations	3.1	G												
			SD-02 Shop Drawings														
			POL Service Piping	1.6.2	G												
			SD-06 Test Reports														
			Examinations, Inspections and Tests	3.2													
			SD-07 Certificates														
			Qualifications	1.6	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS SIF CATION OR A/E REVIEW WR	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		33 52 40	SD-01 Preconstruction Submittals														
			Work Plan	1.5.2	G												
			Pigging Plan	1.5.3	G												
			Hydrotesting Plan	1.5.4	G												
			SD-02 Shop Drawings														
			Grounding and Bonding	2.1.2													
			Pipe Supports	2.5.5	G												
			Pigging System Components	2.6	G												
			SD-03 Product Data														
			Carbon Steel Piping	2.2.2	G												
			Stainless Steel Piping	2.2.3	G												
			External Protective Coatings For Aboveground Piping	2.2.4	G												
			External Protective Coatings For Buried Steel Piping	2.2.5	G												
			Fittings	2.2.6	G												
			Carbon Steel Fittings	2.2.6.2	G												
			Insulating Flange Kits	2.2.7	G												
			Lightning Surge Arrester	2.2.7	G												
			Bolts, Nuts And Washers	2.2.8	G												
			Ball Valves	2.3.1	G												
			Plug (Double Block and Bleed) Valves	2.3.2	G												
			Electric Valve Actuator	2.3.2.5													
			Thermal Relief Valve	2.4	G												
			Thermal Relief Valve	2.4.2	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION						CONTRACTOR											
Replace Fuel Pipeline from Pier 3 to Bulk Storage						CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION			APPROVING AUTHORITY					
ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION OR REVIEW	SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		33 52 40	Thermal Relief Valve	2.4.3	G												
			Mechanically Adjustable	2.5.1	G												
			Segmented Elastomeric Seal														
			Pipe Sleeves	2.5.2	G												
			Sample Connections	2.5.6	G												
			Buried Utility Tape	2.7.5	G												
			Pipeline Markers	2.7.6	G												
			Flange Gaskets, Non-Metallic,	2.2.9	G												
			Non-Electrically Isolating														
			SD-06 Test Reports														
			Exterior Coating Holiday Test	3.12.1.1													
			Pneumatic Test	3.12.2													
			Hydrostatic Test	3.12.3													
			Performance Testing	3.12.5													
			Pipe Pigging Verification	3.14													
			SD-07 Certificates														
			Contractor Qualifications	1.5.1	G												
			Pipeline Inventory	1.5.5.1	G												
			Survey Final Elevations	3.5.7	G												
			SD-10 Operation and Maintenance														
			Data														
			Insulating Flange Kits	2.2.7	G												
			Lightning Surge Arrester	2.2.7	G												
			Ball Valves	2.3.1	G												
			Plug (Double Block and Bleed)	2.3.2	G												
			Valves														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION						CONTRACTOR											
Replace Fuel Pipeline from Pier 3 to Bulk Storage																	
ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION OR REVIEW	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		33 52 40	Electric Valve Actuator	2.3.2.5	G												
			Thermal Relief Valve	2.4	G												
			Thermal Relief Valve	2.4.2	G												
			Thermal Relief Valve	2.4.3	G												
			Mechanically Adjustable	2.5.1	G												
			Segmented Elastomeric Seal														
			Sample Connections	2.5.6	G												
		33 52 80	SD-01 Preconstruction Submittals														
			Contract Errors, Omissions, and	1.5.1													
			Other Discrepancies														
			Corrective Action Procedures	1.5.2.1													
			Corrective Action Request (CAR)	1.5.2.2													
			Form														
			Coatings Work Plan	1.5.3													
			Inspection Report Forms	3.7.1.2													
			SD-05 Design Data														
			Containment System	1.5.4.1													
			SD-06 Test Reports														
			Daily Inspection Reports	3.7.1.3													
			Ferrous Metallic Abrasive	1.5.5.1													
			Qualification Test Reports														
			Non-Metallic Abrasive	1.5.5.2													
			Qualification Test Reports														
			Abrasive Field Test Reports	3.1.3													

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASS SIF CATION OR A/E REVIEW WR	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/ DATE RCD FROM CONTR	APPROVING AUTHORITY				MAILED TO CONTR/ DATE RCD FRM APPR AUTH	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION		DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		33 52 80	Recycled Ferrous Metallic	1.5.5.3													
			Abrasive Field Test Reports (Daily														
			and Weekly)														
			SD-07 Certificates														
			Qualifications of Certified	1.5.6.1													
			Industrial Hygienist (CIH)														
			Qualifications of Pipe Coating	1.5.6.2													
			Shop														
			Qualifications of Certified	1.5.6.3													
			Protective Coatings Specialist														
			(PCS)														
			Qualifications of Coatings	1.5.6.4													
			Inspection Company														
			Qualifications of Quality	1.5.6.5													
			Assurance Coatings Inspector														
			Qualifications of Coatings	1.5.6.6													
			Contractors for Field Coating														
			Qualifications of Individuals	1.5.6.7													
			Performing Abrasive Blasting for														
			Field Coating														
			Qualifications of Individuals	1.5.6.8													
			Performing Coating Application for														
			Field Coating														
			Qualifications of Testing	1.5.6.9													
			Laboratory for Abrasive														

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

ACTIVITY NO	TRANSMITTAL NO	SPEC SECT	DESCRIPTION ITEM SUBMITTED	PARAGRAPH	GOVT CLASSIFICATION	CONTRACTOR: SCHEDULE DATES			CONTRACTOR ACTION		DATE FWD TO APPR AUTH/	APPROVING AUTHORITY				MAILED TO CONTR/	REMARKS
						SUBMIT	APPROVAL NEEDED BY	MATERIAL NEEDED BY	ACTION CODE	DATE OF ACTION	DATE RCD FROM CONTR	DATE FWD TO OTHER REVIEWER	DATE RCD FROM OTH REVIEWER	ACTION CODE	DATE OF ACTION	DATE RCD FRM APPR AUTH	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)	(r)
		33 52 80	Ferrous Metallic Abrasive	1.5.6.10													
			Certificate of Conformance														
			Non-Metallic Abrasive Certificate	1.5.6.11													
			of Conformance														
			SD-08 Manufacturer's Instructions														
			Coating System Instructions	1.6.1													
			SD-11 Closeout Submittals														
			Disposal of Used Abrasive Shop	3.4.1.5													
			Applied Coating														
			Disposal of Used Abrasive Field	3.6.1.6													
			Applied Coating														
			Inspection Logbook	3.7.1.4													
			Corrective Action Log	1.5.2.3	G												
		33 71 02	SD-02 Shop Drawings														
			Precast Underground Structures	1.6.1	G												
			SD-03 Product Data														
			Precast Concrete Structures	2.9.2.1	G												
			Sealing Material	2.9.2.4													
			Pulling-In Irons	3.5.3													
			Cable Supports	2.10	G												
			Protective Devices and	2.12	G												
			Coordination														
			SD-06 Test Reports														
			Field Acceptance Checks and	3.16.1	G												
			Tests														
			Arc-proofing Test	2.13.1	G												

SUBMITTAL REGISTER

CONTRACT NO.

TITLE AND LOCATION

Replace Fuel Pipeline from Pier 3 to Bulk Storage

CONTRACTOR

[illegible]

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS

11/20, CHG 3: 02/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Competent Person (CP)
 - 1.2.2 Competent Person, Confined Space
 - 1.2.3 Competent Person, Cranes and Rigging
 - 1.2.4 Competent Person, Excavation/Trenching
 - 1.2.5 Competent Person, Fall Protection
 - 1.2.6 Competent Person, Scaffolding
 - 1.2.7 Competent Person (CP) Trainer
 - 1.2.8 High Risk Activities
 - 1.2.9 High Visibility Accident
 - 1.2.10 Load Handling Equipment (LHE)
 - 1.2.11 Medical Treatment
 - 1.2.12 Near Miss
 - 1.2.13 Operating Envelope
 - 1.2.14 Qualified Person (QP)
 - 1.2.15 Qualified Person, Fall Protection (QP for FP)
 - 1.2.16 Recordable Injuries or Illnesses
 - 1.2.17 Government Property and Equipment
 - 1.2.18 Load Handling Equipment (LHE) Accident or Load Handling Equipment Mishap
- 1.3 SUBMITTALS
- 1.4 MONTHLY EXPOSURE REPORTS
- 1.5 CONTRACTOR SAFETY SELF-EVALUATION CHECKLIST
- 1.6 REGULATORY REQUIREMENTS
 - 1.6.1 Subcontractor Safety Requirements
 - 1.6.1.1 Experience Modification Rate (EMR)
 - 1.6.1.2 OSHA Days Away from Work, Restricted Duty, or Job Transfer (DART) Rate
- 1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS
 - 1.7.1 Personnel Qualifications
 - 1.7.1.1 Site Safety and Health Officer (SSHO)
 - 1.7.1.1.1 Additional Site Safety and Health Officer (SSHO) Requirements and Duties
 - 1.7.1.2 Competent Person Qualifications
 - 1.7.1.2.1 Competent Person for Confined Space Entry
 - 1.7.1.2.2 Competent Person for Scaffolding
 - 1.7.1.2.3 Competent Person for Fall Protection
 - 1.7.1.3 Qualified Trainer Requirements
 - 1.7.1.4 Requirements for all Contractor Jobsite Personnel Holding H-1B or H-2B Visas:
 - 1.7.1.5 Dredging Contract Requirements
 - 1.7.1.6 Crane Operators/Riggers
 - 1.7.2 Personnel Duties

- 1.7.2.1 Duties of the Site Safety and Health Officer (SSHO)
- 1.7.3 Meetings
 - 1.7.3.1 Preconstruction Meeting
 - 1.7.3.2 Safety Meetings
- 1.8 ACCIDENT PREVENTION PLAN (APP)
 - 1.8.1 APP - Construction
 - 1.8.2 Names and Qualifications
 - 1.8.3 Plans
 - 1.8.3.1 Confined Space Entry Plan
 - 1.8.3.2 Standard Lift Plan (SLP)
 - 1.8.3.3 Critical Lift Plan - Crane or Load Handling Equipment
 - 1.8.3.3.1 Critical Lift Plan Planning and Schedule
 - 1.8.3.3.2 Lifts of Personnel
 - 1.8.3.4 Multi-Purpose Machines, Material Handling Equipment, and Construction Equipment Lift Plan
 - 1.8.3.5 Fall Protection and Prevention (FP&P) Plan
 - 1.8.3.6 Rescue and Evacuation Plan
 - 1.8.3.7 Hazardous Energy Control Program (HECP)
 - 1.8.3.8 Excavation Plan
 - 1.8.3.9 Site Demolition Plan
- 1.9 ACTIVITY HAZARD ANALYSIS (AHA)
 - 1.9.1 AHA Management
 - 1.9.2 AHA Signature Log
- 1.10 DISPLAY OF SAFETY INFORMATION
 - 1.10.1 Safety Bulletin Board
 - 1.10.2 Safety and Occupational Health (SOH) Deficiency Tracking System
- 1.11 SITE SAFETY REFERENCE MATERIALS
- 1.12 EMERGENCY MEDICAL TREATMENT
- 1.13 NOTIFICATIONS and REPORTS
 - 1.13.1 Mishap Notification
 - 1.13.2 Accident Reports
 - 1.13.3 LHE Inspection Reports
 - 1.13.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging
- 1.14 HOT WORK
 - 1.14.1 Permit and Personnel Requirements
 - 1.14.2 Work Around Flammable Materials
- 1.15 RADIATION SAFETY REQUIREMENTS
 - 1.15.1 Radiography Operation Planning Work Sheet
 - 1.15.2 Site Access and Security
 - 1.15.3 Loss or Release and Unplanned Personnel Exposure
 - 1.15.4 Site Demarcation and Barricade
 - 1.15.5 Security of Material and Equipment
 - 1.15.6 Transportation of Material
 - 1.15.7 Schedule for Exposure or Unshielding
 - 1.15.8 Transmitter Requirements
- 1.16 CONFINED SPACE ENTRY REQUIREMENTS
 - 1.16.1 Entry Procedures
 - 1.16.2 Forced Air Ventilation
 - 1.16.3 Sewer Wet Wells
 - 1.16.4 Rescue Procedures and Coordination with Local Emergency Responders
- 1.17 GAS PROTECTION
 - 1.17.1 Special Requirements
- 1.18 SEVERE STORM PLAN

PART 2 PRODUCTS

2.1 CONFINED SPACE SIGNAGE

PART 3 EXECUTION

- 3.1 CONSTRUCTION AND OTHER WORK
 - 3.1.1 Worksite Communication
 - 3.1.2 Hazardous Material Use
 - 3.1.3 Hazardous Material Exclusions
 - 3.1.4 Unforeseen Hazardous Material
- 3.2 UTILITY OUTAGE REQUIREMENTS
- 3.3 OUTAGE COORDINATION MEETING
- 3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)
 - 3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility
 - 3.4.2 Lockout/Tagout Isolation
 - 3.4.3 Lockout/Tagout Removal
- 3.5 FALL PROTECTION PROGRAM
 - 3.5.1 Training
 - 3.5.2 Fall Protection Equipment and Systems
 - 3.5.2.1 Personal Fall Protection Equipment
 - 3.5.3 Fall Protection for Roofing Work
 - 3.5.4 Horizontal Lifelines (HLL)
 - 3.5.5 Guardrails and Safety Nets
 - 3.5.6 Rescue and Evacuation Plan and Procedures
- 3.6 WORK PLATFORMS
 - 3.6.1 Scaffolding
 - 3.6.2 Elevated Aerial Work Platforms (AWPs)
- 3.7 EQUIPMENT
 - 3.7.1 Material Handling Equipment (MHE)
 - 3.7.2 Load Handling Equipment (LHE)
 - 3.7.3 Machinery and Mechanized Equipment
 - 3.7.4 Base Mounted Drum Hoists
- 3.8 EXCAVATIONS
 - 3.8.1 Utility Locations
 - 3.8.2 Utility Location Verification
 - 3.8.3 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces
- 3.9 ELECTRICAL
 - 3.9.1 Conduct of Electrical Work
 - 3.9.2 Qualifications
 - 3.9.3 Arc Flash
 - 3.9.4 Grounding
 - 3.9.5 Testing

-- End of Section Table of Contents --

SECTION 01 35 26

GOVERNMENTAL SAFETY REQUIREMENTS
11/20, CHG 3: 02/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B30.3	(2020) Tower Cranes
ASME B30.5	(2021) Mobile and Locomotive Cranes
ASME B30.7	(2021) Winches
ASME B30.8	(2020) Floating Cranes and Floating Derricks
ASME B30.9	(2018) Slings
ASME B30.20	(2018) Below-the-Hook Lifting Devices
ASME B30.22	(2016) Articulating Boom Cranes
ASME B30.23	(2016) Personnel Lifting Systems Safety Standard for Cableways, Cranes, Derricks, Hoists, Hooks, Jacks, and Slings
ASME B30.26	(2015; R 2020) Rigging Hardware

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.22	(2007; R 2017) Safety Requirements for Rope-Guided and Non-Guided Workers' Hoists
ASSP A10.34	(2021) Protection of the Public on or Adjacent to Construction Sites
ASSP A10.44	(2020) Control of Energy Sources (Lockout/Tagout) for Construction and Demolition Operations
ASSP Z244.1	(2016) The Control of Hazardous Energy Lockout, Tagout and Alternative Methods
ASSP Z359.0	(2018) Definitions and Nomenclature Used for Fall Protection and Fall Arrest
ASSP Z359.1	(2020) The Fall Protection Code
ASSP Z359.2	(2017) Minimum Requirements for a

Comprehensive Managed Fall Protection
Program

ASSP Z359.3	(2019) Safety Requirements for Lanyards and Positioning Lanyards
ASSP Z359.4	(2013) Safety Requirements for Assisted-Rescue and Self-Rescue Systems, Subsystems and Components
ASSP Z359.6	(2016) Specifications and Design Requirements for Active Fall Protection Systems
ASSP Z359.7	(2019) Qualification and Verification Testing of Fall Protection Products
ASSP Z359.11	(2014) Safety Requirements for Full Body Harnesses
ASSP Z359.12	(2019) Connecting Components for Personal Fall Arrest Systems
ASSP Z359.13	(2013) Personal Energy Absorbers and Energy Absorbing Lanyards
ASSP Z359.14	(2014) Safety Requirements for Self-Retracting Devices for Personal Fall Arrest and Rescue Systems
ASSP Z359.15	(2014) Safety Requirements for Single Anchor Lifelines and Fall Arresters for Personal Fall Arrest Systems
ASSP Z359.16	(2016) Safety Requirements for Climbing Ladder Fall Arrest Systems
ASSP Z359.18	(2017) Safety Requirements for Anchorage Connectors for Active Fall Protection Systems

ASTM INTERNATIONAL (ASTM)

ASTM F855	(2019) Standard Specifications for Temporary Protective Grounds to Be Used on De-energized Electric Power Lines and Equipment
-----------	---

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 1048	(2016) Guide for Protective Grounding of Power Lines
IEEE C2	(2017; Errata 1-2 2017; INT 1 2017) National Electrical Safety Code

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA)

NASA NPR 8715.3	(2017d; Change 1) NASA General Safety
-----------------	---------------------------------------

Program Requirements

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

NEMA Z535.2 (2011; R 2017) Environmental and Facility
Safety Signs

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 10 (2022; ERTA 1 2021) Standard for Portable
Fire Extinguishers

NFPA 51B (2019; TIA 20-1) Standard for Fire
Prevention During Welding, Cutting, and
Other Hot Work

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020;
ERTA 20-3 2020; TIA 20-1; TIA 20-2; TIA
20-3; TIA 20-4; TIA 20-5; TIA 20-6; TIA
20-7; TIA 20-8; TIA 20-9; TIA 20-10; TIA
20-11; TIA 20-12; TIA 20-13; TIA 20-14;
TIA 20-15; TIA 20-16; ERTA 20-4 2022)
National Electrical Code

NFPA 70E (2021) Standard for Electrical Safety in
the Workplace

NFPA 241 (2022) Standard for Safeguarding
Construction, Alteration, and Demolition
Operations

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-222 (2018H; Add 1 2019) Structural Standard
for Antenna Supporting Structures and
Antennas and Small Wind Turbine Support
Structures

TIA-1019 (2012; R 2016) Standard for Installation,
Alteration and Maintenance of Antenna
Supporting Structures and Antennas

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety and Health Requirements
Manual

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

10 CFR 20 Standards for Protection Against Radiation

29 CFR 1910 Occupational Safety and Health Standards

29 CFR 1910.146 Permit-required Confined Spaces

29 CFR 1910.147 The Control of Hazardous Energy (Lock
Out/Tag Out)

29 CFR 1910.333 Selection and Use of Work Practices

29 CFR 1915	Confined and Enclosed Spaces and Other Dangerous Atmospheres in Shipyard Employment
29 CFR 1915.89	Control of Hazardous Energy (Lockout/Tags-Plus)
29 CFR 1926	Safety and Health Regulations for Construction
29 CFR 1926.16	Rules of Construction
29 CFR 1926.450	Scaffolds
29 CFR 1926.500	Fall Protection
29 CFR 1926.552	Material Hoists, Personal Hoists, and Elevators
29 CFR 1926.553	Base-Mounted Drum Hoists
29 CFR 1926.1400	Cranes and Derricks in Construction
49 CFR 173	Shippers - General Requirements for Shipments and Packagings
CPL 02-01-056	(2014) Inspection Procedures for Accessing Communication Towers by Hoist
CPL 2.100	(1995) Application of the Permit-Required Confined Spaces (PRCS) Standards, 29 CFR 1910.146

1.2 DEFINITIONS

1.2.1 Competent Person (CP)

The CP is a person designated in writing, who, through training, knowledge and experience, is capable of identifying, evaluating, and addressing existing and predictable hazards in the working environment or working conditions that are dangerous to personnel, and who has authorization to take prompt corrective measures with regards to such hazards.

1.2.2 Competent Person, Confined Space

The CP, Confined Space, is a person meeting the competent person requirements as defined EM 385-1-1 Appendix Q, with thorough knowledge of OSHA's Confined Space Standard, 29 CFR 1910.146, and designated in writing to be responsible for the immediate supervision, implementation and monitoring of the confined space program, who through training, knowledge and experience in confined space entry is capable of identifying, evaluating and addressing existing and potential confined space hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.3 Competent Person, Cranes and Rigging

The CP, Cranes and Rigging, as defined in EM 385-1-1 Appendix Q, is a

person meeting the competent person requirements, who has been designated in writing to be responsible for the immediate supervision, implementation and monitoring of the Crane and Rigging Program, who through training, knowledge and experience in crane and rigging is capable of identifying, evaluating and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.4 Competent Person, Excavation/Trenching

A CP, Excavation/Trenching, is a person meeting the competent person requirements as defined in EM 385-1-1 Appendix Q and 29 CFR 1926, who has been designated in writing to be responsible for the immediate supervision, implementation and monitoring of the excavation/trenching program, who through training, knowledge and experience in excavation/trenching is capable of identifying, evaluating and addressing existing and potential hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.5 Competent Person, Fall Protection

The CP, Fall Protection, is a person meeting the competent person requirements as defined in EM 385-1-1 Appendix Q and in accordance with ASSP Z359.0, who has been designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the fall protection program, who through training, knowledge and experience in fall protection and rescue systems and equipment, is capable of identifying, evaluating and addressing existing and potential fall hazards and, who has the authority to take prompt corrective measures with regard to such hazards.

1.2.6 Competent Person, Scaffolding

The CP, Scaffolding is a person meeting the competent person requirements in EM 385-1-1 Appendix Q, and designated in writing by the employer to be responsible for immediate supervising, implementing and monitoring of the scaffolding program. The CP for Scaffolding has enough training, knowledge and experience in scaffolding to correctly identify, evaluate and address existing and potential hazards and also has the authority to take prompt corrective measures with regard to these hazards. CP qualifications must be documented including experience on the specific scaffolding systems/types being used, assessment of the base material that the scaffold will be erected upon, load calculations for materials and personnel, and erection and dismantling. The CP for scaffolding must have a documented minimum of 8-hours of scaffold training to include training on the specific type of scaffold being used (e.g. mast-climbing, adjustable, tubular frame), in accordance with EM 385-1-1 Section 22.B.02.

1.2.7 Competent Person (CP) Trainer

A competent person trainer as defined in EM 385-1-1 Appendix Q, who is qualified in the training material presented, and who possesses a working knowledge of applicable technical regulations, standards, equipment and systems related to the subject matter on which they are training Competent Persons. A competent person trainer must be familiar with the typical hazards and the equipment used in the industry they are instructing. The training provided by the competent person trainer must be appropriate to that specific industry. The competent person trainer must evaluate the knowledge and skills of the competent persons as part of the training process.

1.2.8 High Risk Activities

High Risk Activities are activities that involve work at heights, crane and rigging, excavations and trenching, scaffolding, electrical work, and confined space entry.

1.2.9 High Visibility Accident

A High Visibility Accident is any mishap which may generate publicity or high visibility.

1.2.10 Load Handling Equipment (LHE)

LHE is a term used to describe cranes, hoists and all other hoisting equipment (hoisting equipment means equipment, including crane, derricks, hoists and power operated equipment used with rigging to raise, lower or horizontally move a load).

1.2.11 Medical Treatment

Medical Treatment is treatment administered by a physician or by registered professional personnel under the standing orders of a physician. Medical treatment does not include first aid treatment even when provided by a physician or registered personnel.

1.2.12 Near Miss

A Near Miss is a mishap resulting in no personal injury and zero property damage, but given a shift in time or position, damage or injury may have occurred (e.g., a worker falls off a scaffold and is not injured; a crane swings around to move the load and narrowly misses a parked vehicle).

1.2.13 Operating Envelope

The Operating Envelope is the area surrounding any crane or load handling equipment. Inside this "envelope" is the crane, the operator, riggers and crane walkers, other personnel involved in the operation, rigging gear between the hook, the load, the crane's supporting structure (i.e. ground or rail), the load's rigging path, the lift and rigging procedure.

1.2.14 Qualified Person (QP)

The QP is a person designated in writing, who, by possession of a recognized degree, certificate, or professional standing, or extensive knowledge, training, and experience, has successfully demonstrated their ability to solve or resolve problems related to the subject matter, the work, or the project.

1.2.15 Qualified Person, Fall Protection (QP for FP)

A QP for FP is a person meeting the definition requirements of EM 385-1-1 Appendix Q, and ASSP Z359.2 standard, having a recognized degree or professional certificate and with extensive knowledge, training and experience in the fall protection and rescue field who is capable of designing, analyzing, and evaluating and specifying fall protection and rescue systems.

1.2.16 Recordable Injuries or Illnesses

Recordable Injuries or Illnesses are any work-related injury or illness that results in:

- a. Death, regardless of the time between the injury and death, or the length of the illness;
- b. Days away from work (any time lost after day of injury/illness onset);
- c. Restricted work;
- d. Transfer to another job;
- e. Medical treatment beyond first aid;
- f. Loss of consciousness; or
- g. A significant injury or illness diagnosed by a physician or other licensed health care professional, even if it did not result in (a) through (f) above

1.2.17 Government Property and Equipment

Interpret "USACE" property and equipment specified in USACE EM 385-1-1 as Government property and equipment.

1.2.18 Load Handling Equipment (LHE) Accident or Load Handling Equipment Mishap

A LHE accident occurs when any one or more of the eight elements in the operating envelope fails to perform correctly during operation, including operation during maintenance or testing resulting in personnel injury or death; material or equipment damage; dropped load; derailment; two-blocking; overload; or collision, including unplanned contact between the load, crane, or other objects. A dropped load, derailment, two-blocking, overload and collision are considered accidents, even though no material damage or injury occurs. A component failure (e.g., motor burnout, gear tooth failure, bearing failure) is not considered an accident solely due to material or equipment damage unless the component failure results in damage to other components (e.g., dropped boom, dropped load, or roll over). Document an LHE mishap or accident using the NAVFAC prescribed Navy Crane Center (NCC) accident form.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Accident Prevention Plan (APP); G

APP - Construction; G

Accident Prevention Plan (APP); G

SD-06 Test Reports

- Monthly Exposure Reports
- Notifications and Reports
- Accident Reports; G
- LHE Inspection Reports
- Gas Protection for NASA projects

SD-07 Certificates

- Contractor Safety Self-Evaluation Checklist
- Crane Operators/Riggers
- Standard Lift Plan; G
- Critical Lift Plan ; G
- Activity Hazard Analysis (AHA)
- Confined Space Entry Permit
- Hot Work Permit
- Certificate of Compliance
- License Certificates
- Radiography Operation Planning Work Sheet; G
- Portable Gauge Operations Planning Worksheet; G

1.4 MONTHLY EXPOSURE REPORTS

Provide a Monthly Exposure Report and attach to the monthly billing request. This report is a compilation of employee-hours worked each month for all site workers, both Prime and subcontractor. Failure to submit the report may result in retention of up to 10 percent of the voucher.

1.5 CONTRACTOR SAFETY SELF-EVALUATION CHECKLIST

Contracting Officer will provide a "Contractor Safety Self-Evaluation checklist" to the Contractor at the pre-construction meeting. Complete the checklist monthly and submit with each request for payment voucher. An acceptable score of 90 or greater is required. Failure to submit the completed safety self-evaluation checklist or achieve a score of at least 90 may result in retention of up to 10 percent of the voucher. The Contractor Safety Self-Evaluation checklist can be found on the Whole Building Design Guide website at www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-01-35-26

1.6 REGULATORY REQUIREMENTS

In addition to the detailed requirements included in the provisions of this Contract, comply with the most recent edition of USACE EM 385-1-1,

and the following host nation laws, ordinances, criteria, rules and regulations. Submit matters of interpretation of standards to the appropriate administrative agency for resolution before starting work. Where the requirements of this specification, applicable laws, criteria, ordinances, regulations, and referenced documents vary, the most stringent requirements govern.

1.6.1 Subcontractor Safety Requirements

1.6.1.1 Experience Modification Rate (EMR)

Subcontractors on this Contract must have an effective EMR less than or equal to 1.10, as computed by the National Council on Compensation Insurance (NCCI) or if not available, as computed by the state agency's rating bureau in the state where the subcontractor is registered, when entering into a subcontract agreement with the Prime Contractor or a subcontractor at any tier. The Prime Contractor may submit a written request for additional consideration to the Contracting Officer where the specified acceptable EMR range cannot be achieved. Relaxation of the EMR range will only be considered for approval on a case-by-case basis for special conditions and must not be anticipated as tacit approval. Contractor's Site Safety and Health Officer (SSHO) must collect and maintain the certified EMR ratings for all subcontractors on the project and make them available to the Government at the Government's request.

1.6.1.2 OSHA Days Away from Work, Restricted Duty, or Job Transfer (DART) Rate

Subcontractors on this Contract must have a DART rate, calculated from the most recent, complete calendar year, less than or equal to 3.4 when entering into a subcontract agreement with the Prime Contractor or a subcontractor at any tier. The OSHA Dart Rate is calculated using the following formula:

$$(N/EH) \times 200,000$$

where:

N = number of injuries and illnesses with days away, restricted work, or job transfer

EH = total hours worked by all employees during most recent, complete calendar year

200,000 = base for 100 full-time equivalent workers (working 40 hours per week, 50 weeks per year)

The Prime Contractor may submit a written request for additional consideration to the Contracting Officer where the specified acceptable OSHA Dart rate range cannot be achieved for a particular subcontractor. Relaxation of the OSHA DART rate range will only be considered for approval on a case-by-case basis for special conditions and must not be anticipated as tacit approval. Contractor's Site Safety and Health Officer (SSHO) must collect and maintain self-certified OSHA DART rates for all subcontractors on the project and make them available to the Government at the Government's request.

1.7 SITE QUALIFICATIONS, DUTIES, AND MEETINGS

1.7.1 Personnel Qualifications

1.7.1.1 Site Safety and Health Officer (SSHO)

Provide an SSHO that meets the requirements of EM 385-1-1 Section 1. The SSHO must ensure that the requirements of 29 CFR 1926.16 are met for the project. Provide a Safety oversight team that includes a minimum of one person at each project site to function as the Site Safety and Health Officer (SSHO). The SSHO or an equally-qualified Alternate SSHO must be at the work site at all times to implement and administer the Contractor's safety program and Government-accepted Accident Prevention Plan. The SSHO and Alternate SSHO must have the required training, experience, and qualifications in accordance with EM 385-1-1 Section 01.A.17, and all associated sub-paragraphs.

If the SSHO is off-site for a period longer than 24 hours, an equally-qualified alternate SSHO must be provided and must fulfill the same roles and responsibilities as the primary SSHO. When the SSHO is temporarily (up to 24 hours) off-site, a Designated Representative (DR), as identified in the AHA may be used in lieu of an Alternate SSHO, and must be on the project site at all times when work is being performed. Note that the DR is a collateral duty safety position, with safety duties in addition to their full time occupation.

1.7.1.1.1 Additional Site Safety and Health Officer (SSHO) Requirements and Duties

The SSHO may also serve as the Quality Control Manager. The SSHO may also serve as the Superintendent.

1.7.1.2 Competent Person Qualifications

Provide Competent Persons in accordance with EM 385-1-1, Appendix Q and herein. Competent Persons for high risk activities include confined space, cranes and rigging, excavation/trenching, fall protection, and electrical work. The CP for these activities must be designated in writing, and meet the requirements for the specific activity (i.e. competent person, fall protection).

The Competent Person identified in the Contractor's Safety and Health Program and accepted Accident Prevention Plan, must be on-site at all times when the work that presents the hazards associated with their professional expertise is being performed. Provide the credentials of the Competent Persons(s) to the Contracting Officer for information in consultation with the Safety Office.

1.7.1.2.1 Competent Person for Confined Space Entry

Provide a Confined Space (CP) Competent Person who meets the requirements of EM 385-1-1, Appendix Q, and herein. The CP for Confined Space Entry must supervise the entry into each confined space in accordance with EM 385-1-1, Section 34.

1.7.1.2.2 Competent Person for Scaffolding

Provide a Competent Person for Scaffolding who meets the requirements of EM 385-1-1, Section 22.B.02 and herein.

1.7.1.2.3 Competent Person for Fall Protection

Provide a Competent Person for Fall Protection who meets the requirements of EM 385-1-1, Section 21.C.04, 21.B.03, and herein.

1.7.1.3 Qualified Trainer Requirements

Individuals qualified to instruct the 40 hour contract safety awareness course, or portions thereof, must meet the definition of a Competent Person Trainer, and, at a minimum, possess a working knowledge of the following subject areas: EM 385-1-1, Electrical Standards, Lockout/Tagout, Fall Protection, Confined Space Entry for Construction; Excavation, Trenching and Soil Mechanics, and Scaffolds in accordance with 29 CFR 1926.450, Subpart L.

Instructors are required to:

- a. Prepare class presentations that cover construction-related safety requirements and includes topics covered in the NAVFAC Construction Safety Hazard Awareness Course for Contractors.
- b. Ensure that all attendees attend all sessions by using a class roster signed daily by each attendee. Maintain copies of the roster for at least five years. This is a certification class and must be attended 100 percent. In cases of emergency where an attendee cannot make it to a session, the attendee can make it up in another class session for the same subject.
- c. Update training course materials whenever an update of the EM 385-1-1 becomes available.
- d. Provide a written exam of at least 50 questions. Students are required to answer 80 percent correctly to pass.
- e. Request, review and incorporate student feedback into a continuous course improvement program.

1.7.1.4 Requirements for all Contractor Jobsite Personnel Holding H-1B or H-2B Visas:

Submit a list of workers who have completed the training to the Contracting Officer prior to them reporting to the jobsite. Update the list as additional workers are added. Maintain the updated list at the jobsite for review by the Government's designated authority. Include the name and qualifications of qualified trainer(s) that provided the training. Personnel who have taken the 40 Hour Construction Safety Hazard Awareness Training Course for Contractors or similar course that includes emphasis on EM 385-1-1 compliance, are not required to take the 16 hours of classroom training on the requirements of the latest version of the EM 385-1-1. The 16 hours classroom training may be provided by the Guam Contractors Association (GCA), Trades Academy, or other qualified trainers as outlined in paragraph QUALIFIED TRAINER REQUIREMENTS.

1.7.1.5 Dredging Contract Requirements

1.7.1.6 Crane Operators/Riggers

Provide Operators, Signal Persons, and Riggers meeting the requirements in

EM 385-1-1, Section 15.B for Riggers and Section 16.B for Crane Operators and Signal Persons. In addition, for mobile cranes with Original Equipment Manufacturer (OEM) rated capacities of 22,680 kg or greater, designate crane operators qualified by a source that qualifies crane operators (i.e., union, a Government agency, or an organization that tests and qualifies crane operators). Provide proof of current qualification.

1.7.2 Personnel Duties

1.7.2.1 Duties of the Site Safety and Health Officer (SSHO)

The SSHO must:

- a. Conduct daily safety and health inspections and maintain a written log which includes area/operation inspected, date of inspection, identified hazards, recommended corrective actions, estimated and actual dates of corrections. Attach safety inspection logs to the Contractors' daily production report.
- b. Conduct mishap investigations and complete required accident reports. Report mishaps and near misses.
- c. Use and maintain OSHA's Form 300 to log work-related injuries and illnesses occurring on the project site for Prime Contractors and subcontractors, and make available to the Contracting Officer upon request. Post and maintain the Form 300A on the site Safety Bulletin Board.
- d. Maintain applicable safety reference material on the job site.
- e. Attend the pre-construction meeting, pre-work meetings including preparatory meetings, and periodic in-progress meetings.
- f. Review the APP and AHAs for compliance with EM 385-1-1, and approve, sign, implement and enforce them.
- g. Establish a Safety and Occupational Health (SOH) Deficiency Tracking System that lists and monitors outstanding deficiencies until resolution.
- h. Ensure subcontractor compliance with safety and health requirements.
- i. Maintain a list of hazardous chemicals on site and their material Safety Data Sheets (SDS).
- j. Maintain a weekly list of high hazard activities involving energy, equipment, excavation, entry into confined space, and elevation, and be prepared to discuss details during QC Meetings.
- k. Provide and keep a record of site safety orientation and indoctrination for Contractor employees, subcontractor employees, and site visitors.

Superintendent, QC Manager, and SSHO are subject to dismissal if the above or any other required duties are not being effectively carried out. If either the Superintendent, QC Manager, or SSHO are dismissed, project work will be stopped and will not be allowed to resume until a suitable replacement is approved and the above duties are again being effectively carried out.

1.7.3 Meetings

1.7.3.1 Preconstruction Meeting

- a. Contractor representatives who have a responsibility or significant role in accident prevention on the project must attend the preconstruction meeting. This includes the project superintendent, Site Safety and Occupational Health Officer, quality control manager, or any other assigned safety and health professionals who participated in the development of the APP (including the Activity Hazard Analyses (AHAs) and special plans, program and procedures associated with it).
- b. Discuss the details of the submitted APP to include incorporated plans, programs, procedures and a listing of anticipated AHAs that will be developed and implemented during the performance of the Contract. This list of proposed AHAs will be reviewed and an agreement will be reached between the Contractor and the Contracting Officer as to which phases will require an analysis. In addition, establish a schedule for the preparation, submittal, and Government review of AHAs to preclude project delays.
- c. Deficiencies in the submitted APP, identified during the Contracting Officer's review, must be corrected, and the APP re-submitted for review prior to the start of construction. Work is not permitted to begin until an APP is established that is acceptable to the Contracting Officer.

1.7.3.2 Safety Meetings

Conduct safety meetings to review past activities, plan for new or changed operations, review pertinent aspects of appropriate AHA (by trade), establish safe working procedures for anticipated hazards, and provide pertinent Safety and Occupational Health (SOH) training and motivation. Conduct meetings at least once a month for all supervisors at the project location. The SSHO, supervisors, foremen, or CDSOs must conduct meetings at least once a week for the trade workers. Document meeting minutes to include the date, persons in attendance, subjects discussed, and names of individual(s) who conducted the meeting. Maintain documentation on-site and furnish copies to the Contracting Officer on request. Notify the Contracting Officer of all scheduled meetings 7 calendar days in advance.

1.8 ACCIDENT PREVENTION PLAN (APP)

1.8.1 APP - Construction

A qualified person must prepare the written site-specific APP. Prepare the APP in accordance with the format and requirements of EM 385-1-1, Appendix A, and as supplemented herein. Cover all paragraph and subparagraph elements in EM 385-1-1, Appendix A and show compliance with NASA NPR 8715.3. The APP must be job-specific and address any unusual or unique aspects of the project or activity for which it is written. The APP must interface with the Contractor's overall safety and health program referenced in the APP in the applicable APP element, and made site-specific. Describe the methods to evaluate past safety performance of potential subcontractors in the selection process. Also, describe innovative methods used to ensure and monitor safe work practices of subcontractors. The Government considers the Prime Contractor to be the "controlling authority" for all work site safety and health of the

subcontractors. Contractors are responsible for informing their subcontractors of the safety provisions under the terms of the Contract and the penalties for noncompliance, coordinating the work to prevent one craft from interfering with or creating hazardous working conditions for other crafts, and inspecting subcontractor operations to ensure that accident prevention responsibilities are being carried out. The APP must be signed by an officer of the firm (Prime Contractor senior person), the individual preparing the APP, the on-site superintendent, the designated SSHO, the Contractor Quality Control Manager, and any designated Certified Safety Professional (CSP) or Certified Health Physicist (CIH). The SSHO must provide and maintain the APP and a log of signatures by each subcontractor foreman, attesting that they have read and understand the APP, and make the APP and log available on-site to the Contracting Officer. If English is not the foreman's primary language, the Prime Contractor must provide an interpreter.

Submit the APP to the Contracting Officer 15 calendar days prior to the date of the preconstruction meeting for acceptance. Work cannot proceed without an accepted APP. Once reviewed and accepted by the Contracting Officer, the APP and attachments will be enforced as part of the Contract. Disregarding the provisions of this Contract or the accepted APP is cause for stopping of work, at the discretion of the Contracting Officer, until the matter has been rectified. Continuously review and amend the APP, as necessary, throughout the life of the Contract. Changes to the accepted APP must be made with the knowledge and concurrence of the Contracting Officer, project superintendent, SSHO and Quality Control Manager. Incorporate unusual or high-hazard activities not identified in the original APP as they are discovered. Should any severe hazard exposure (i.e. imminent danger) become evident, stop work in the area, secure the area, and develop a plan to remove the exposure and control the hazard. Notify the Contracting Officer within 24 hours of discovery. Eliminate and remove the hazard. In the interim, take all necessary action to restore and maintain safe working conditions in order to safeguard onsite personnel, visitors, the public (as defined by ASSP A10.34), and the environment.

1.8.2 Names and Qualifications

Provide plans in accordance with the requirements outlined in Appendix A of EM 385-1-1, including the following:

- a. Names and qualifications (resumes including education, training, experience and certifications) of site safety and health personnel designated to perform work on this project to include the designated Site Safety and Health Officer and other competent and qualified personnel to be used. Specify the duties of each position.
- b. Qualifications of competent and of qualified persons. As a minimum, designate and submit qualifications of competent persons for each of the following major areas: excavation; scaffolding; fall protection; hazardous energy; confined space; health hazard recognition, evaluation and control of chemical, physical and biological agents; and personal protective equipment and clothing to include selection, use and maintenance.

1.8.3 Plans

Provide plans in the APP in accordance with the requirements outlined in Appendix A of EM 385-1-1, including the following:

1.8.3.1 Confined Space Entry Plan

Develop a confined or enclosed space entry plan in accordance with EM 385-1-1, applicable OSHA standards 29 CFR 1910, 29 CFR 1915, and 29 CFR 1926, OSHA Directive CPL 2.100, and any other federal, state and local regulatory requirements identified in this Contract. Identify the qualified person's name and qualifications, training, and experience. Delineate the qualified person's authority to direct work stoppage in the event of hazardous conditions. Include procedure for rescue by Contractor personnel and the coordination with emergency responders. (If there is no confined space work, include a statement that no confined space work exists and none will be created.)

1.8.3.2 Standard Lift Plan (SLP)

Plan lifts to avoid situations where the operator cannot maintain safe control of the lift. Prepare a written SLP in accordance with EM 385-1-1, Section 16.A.03, using Form 16-2 for every lift or series of lifts (if duty cycle or routine lifts are being performed). The SLP must be developed, reviewed and accepted by all personnel involved in the lift in conjunction with the associated AHA. Signature on the AHA constitutes acceptance of the plan. Maintain the SLP on the LHE for the current lift(s) being made. Maintain historical SLPs for a minimum of three months.

1.8.3.3 Critical Lift Plan - Crane or Load Handling Equipment

Provide a Critical Lift Plan as required by EM 385-1-1, Section 16.H.01, using Form 16-3. In addition, Critical Lift Plans are required for the following:

- a. Lifts over 50 percent of the capacity of barge mounted mobile crane's hoist.
- b. When working around energized power lines where the work will get closer than the minimum clearance distance in EM 385-1-1 Table 16-1.
- c. For lifts with anticipated binding conditions.
- d. When erecting cranes.

1.8.3.3.1 Critical Lift Plan Planning and Schedule

Critical lifts require detailed planning and additional or unusual safety precautions. Develop and submit a critical lift plan to the Contracting Officer 30 calendar days prior to critical lift. Comply with load testing requirements in accordance with EM 385-1-1, Section 16.F.03.

1.8.3.3.2 Lifts of Personnel

In addition to the requirements of EM 385-1-1, Section 16.H.02, for lifts of personnel, demonstrate compliance with the requirements of 29 CFR 1926.1400 and EM 385-1-1, Section 16.T.

1.8.3.4 Multi-Purpose Machines, Material Handling Equipment, and Construction Equipment Lift Plan

Multi-purpose machines, material handling equipment, and construction

equipment used to lift loads that are suspended by rigging gear, require proof of authorization from the machine OEM that the machine is capable of making lifts of loads suspended by rigging equipment. Written approval from a qualified registered professional engineer, after a safety analysis is performed, is allowed in lieu of the OEM's approval. Demonstrate that the operator is properly trained and that the equipment is properly configured to make such lifts and is equipped with a load chart.

1.8.3.5 Fall Protection and Prevention (FP&P) Plan

The plan must be in accordance with the requirements of EM 385-1-1, Section 21.D and ASSP Z359.2, be site specific, and address all fall hazards in the work place and during different phases of construction. Address how to protect and prevent workers from falling to lower levels when they are exposed to fall hazards above 1.8 m. A competent person or qualified person for fall protection must prepare and sign the plan documentation. Include fall protection and prevention systems, equipment and methods employed for every phase of work, roles and responsibilities, assisted rescue, self-rescue and evacuation procedures, training requirements, and monitoring methods. Review and revise, as necessary, the Fall Protection and Prevention Plan documentation as conditions change, but at a minimum every six months, for lengthy projects, reflecting any changes during the course of construction due to changes in personnel, equipment, systems or work habits. Keep and maintain the accepted Fall Protection and Prevention Plan documentation at the job site for the duration of the project. Include the Fall Protection and Prevention Plan documentation in the Accident Prevention Plan (APP).

1.8.3.6 Rescue and Evacuation Plan

Provide a Rescue and Evacuation Plan in accordance with EM 385-1-1 Section 21.N and ASSP Z359.2, and include in the FP&P Plan and as part of the APP. Include a detailed discussion of the following: methods of rescue; methods of self-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility.

1.8.3.7 Hazardous Energy Control Program (HECP)

Develop a HECP in accordance with EM 385-1-1 Section 12, 29 CFR 1910.147, 29 CFR 1910.333, 29 CFR 1915.89, ASSP Z244.1, and ASSP A10.44. Submit this HECP as part of the Accident Prevention Plan (APP). Conduct a preparatory meeting and inspection with all effected personnel to coordinate all HECP activities. Document this meeting and inspection in accordance with EM 385-1-1, Section 12.A.02. Ensure that each employee is familiar with and complies with these procedures.

1.8.3.8 Excavation Plan

Identify the safety and health aspects of excavation, and provide and prepare the plan in accordance with EM 385-1-1.

1.8.3.9 Site Demolition Plan

Identify the safety and health aspects, and prepare in accordance with Section 02 41 00 DEMOLITION and referenced sources. Include engineering survey as applicable.

1.9 ACTIVITY HAZARD ANALYSIS (AHA)

Before beginning each activity, task or Definable Feature of Work (DFOW) involving a type of work presenting hazards not experienced in previous project operations, or where a new work crew or subcontractor is to perform the work, the Contractor(s) performing that work activity must prepare an AHA. AHAs must be developed by the Prime Contractor, subcontractor, or supplier performing the work, and provided for Prime Contractor review and approval before submitting to the Contracting Officer. AHAs must be signed by the SSHO, Superintendent, QC Manager and the subcontractor Foreman performing the work. Format the AHA in accordance with EM 385-1-1, Section 1 or as directed by the Contracting Officer. Submit the AHA for review at least 15 working days prior to the start of each activity task, or DFOW. The Government reserves the right to require the Contractor to revise and resubmit the AHA if it fails to effectively identify the work sequences, specific anticipated hazards, site conditions, equipment, materials, personnel and the control measures to be implemented.

AHAs must identify competent persons required for phases involving high risk activities, including confined entry, crane and rigging, excavations, trenching, electrical work, fall protection, and scaffolding.

1.9.1 AHA Management

Review the AHA list periodically (at least monthly) at the Contractor supervisory safety meeting, and update as necessary when procedures, scheduling, or hazards change. Use the AHA during daily inspections by the SSHO to ensure the implementation and effectiveness of the required safety and health controls for that work activity.

1.9.2 AHA Signature Log

Each employee performing work as part of an activity, task or DFOW must review the AHA for that work and sign a signature log specifically maintained for that AHA prior to starting work on that activity. The SSHO must maintain a signature log on site for every AHA. Provide employees whose primary language is other than English, with an interpreter to ensure a clear understanding of the AHA and its contents.

1.10 DISPLAY OF SAFETY INFORMATION

1.10.1 Safety Bulletin Board

Prior to commencement of work, erect a safety bulletin board at the job site. Where size, duration, or logistics of project do not facilitate a bulletin board, an alternative method, acceptable to the Contracting Officer, that is accessible and includes all mandatory information for employee and visitor review, may be deemed as meeting the requirement for a bulletin board. Include and maintain information on safety bulletin board as required by EM 385-1-1, Section 01.A.07. Additional items required to be posted include:

- a. Confined space entry permit.
- b. Hot work permit.

1.10.2 Safety and Occupational Health (SOH) Deficiency Tracking System

Establish a SOH deficiency tracking system that lists and monitors the status of SOH deficiencies in chronological order. Use the tracking system to evaluate the effectiveness of the APP. A monthly evaluation of the data must be discussed in the QC or SOH meeting with everyone on the project. The list must be posted on the project bulletin board and updated daily, and provide the following information:

- a. Date deficiency identified;
- b. Description of deficiency;
- c. Name of person responsible for correcting deficiency;
- d. Projected resolution date;
- e. Date actually resolved.

1.11 SITE SAFETY REFERENCE MATERIALS

Maintain safety-related references applicable to the project, including those listed in paragraph REFERENCES. Maintain applicable equipment manufacturer's manuals.

1.12 EMERGENCY MEDICAL TREATMENT

Contractors must arrange for their own emergency medical treatment in accordance with EM 385-1-1. Government has no responsibility to provide emergency medical treatment.

1.13 NOTIFICATIONS and REPORTS

1.13.1 Mishap Notification

Notify the Contracting Officer as soon as practical, but no more than twenty-four hours, after any mishaps, including recordable accidents, incidents, and near misses, as defined in EM 385-1-1 Appendix Q, any report of injury, illness, or any property damage. For LHE or rigging mishaps, notify the Contracting Officer as soon as practical but not more than four hours after mishap. The Contractor is responsible for obtaining appropriate medical and emergency assistance and for notifying fire, law enforcement, and regulatory agencies. Immediate reporting is required for electrical mishaps, to include Arc Flash; shock; uncontrolled release of hazardous energy includes electrical and non-electrical; load handling equipment or rigging; fall from height (any level other than same surface); and underwater diving. These mishaps must be investigated in depth to identify all causes and to recommend hazard control measures.

Within notification include Contractor name; Contract title; type of Contract; name of activity, installation or location where accident occurred; date and time of accident; names of personnel injured; extent of property damage, if any; extent of injury, if known, and brief description of accident (for example, type of construction equipment used and PPE used). Preserve the conditions and evidence on the accident site until the Government investigation team arrives on-site and Government investigation is conducted. Assist and cooperate fully with the Government's investigation(s) of any mishap.

1.13.2 Accident Reports

- a. Conduct an accident investigation for recordable injuries and illnesses, property damage, and near misses as defined in EM 385-1-1, to establish the root cause(s) of the accident. Complete the applicable NAVFAC Contractor Incident Reporting System (CIRS), and electronically submit via the NAVFAC Enterprise Safety Applications Management System (ESAMS). Complete and submit an accident investigation report in ESAMS within 5 days for mishaps defined in EM 385-1-1 01.D.03 and 10 days for accidents defined by EM 385-1-1 01.D.05. Complete an investigation report within 30 days for those mishaps defined by EM 385-1-1 01.D.04. Mishaps defined by EM 385-1-1 01.D.04 and 01.D.05 must include a written report submitted as an attachment in ESAMS using the following outline: (1) Mishap summary description to include process, findings and outcomes; (2) Root Cause; (3) Direct Factors; (4) Indirect and Contributing Factors; (5) Corrective Actions; and (6) Recommendations. The Contracting Officer will provide copies of any required or special forms.
- b. Near Misses: For Navy Projects, complete the applicable documentation in NAVFAC Contractor Incident Reporting System (CIRS), and electronically submit via the NAVFAC Enterprise Safety Applications Management System (ESAMS). Near miss reports are considered positive and proactive Contractor safety management actions.
- c. Conduct an accident investigation for any load handling equipment accident (including rigging accidents) to establish the root cause(s) of the accident. Complete the LHE Accident Report (Crane and Rigging Accident Report) form and provide the report to the Contracting Officer within 30 calendar days of the accident. Do not proceed with crane operations until cause is determined and corrective actions have been implemented to the satisfaction of the Contracting Officer. The Contracting Officer will provide a blank copy of the accident report form.

1.13.3 LHE Inspection Reports

Submit LHE inspection reports required in accordance with EM 385-1-1 and as specified herein with Daily Reports of Inspections.

1.13.4 Certificate of Compliance and Pre-lift Plan/Checklist for LHE and Rigging

Provide a FORM 16-1 Certificate of Compliance for LHE entering an activity under this Contract and in accordance with EM 385-1-1. Post certifications on the crane.

Develop a Standard Lift Plan (SLP) in accordance with EM 385-1-1, Section 16.H.03 using Form 16-2 Standard Pre-Lift Crane Plan/Checklist for each lift planned. Submit SLP to the Contracting Officer for approval within 15 calendar days in advance of planned lift.

1.14 HOT WORK

1.14.1 Permit and Personnel Requirements

Submit and obtain a written permit prior to performing "Hot Work" (i.e. welding or cutting) or operating other flame-producing/spark producing devices, from the Base Fire Department. A permit is required from the

Explosives Safety Office for work in and around where explosives are processed, stored, or handled. CONTRACTORS ARE REQUIRED TO MEET ALL CRITERIA BEFORE A PERMIT IS ISSUED. Provide at least two 9 kg 4A:20 BC rated extinguishers for normal "Hot Work". The extinguishers must be current inspection tagged, and contain an approved safety pin and tamper resistant seal. It is also mandatory to have a designated FIRE WATCH for any "Hot Work" done at this activity. The Fire Watch must be trained in accordance with NFPA 51B and remain on-site for a minimum of one hour after completion of the task or as specified on the hot work permit.

When starting work in the facility, require personnel to familiarize themselves with the location of the nearest fire alarm boxes and place in memory the emergency Base Fire Department phone number. REPORT ANY FIRE, NO MATTER HOW SMALL, TO THE RESPONSIBLE BASE FIRE DEPARTMENT IMMEDIATELY.

1.14.2 Work Around Flammable Materials

Obtain permit approval from a NFPA Certified Marine Chemist, or Certified Industrial Hygienist for "HOT WORK" within or around flammable materials (such as fuel systems or welding/cutting on fuel pipes) or confined spaces (such as sewer wet wells, manholes, or vaults) that have the potential for flammable or explosive atmospheres.

Whenever these materials, except beryllium and chromium (VI), are encountered in indoor operations, local mechanical exhaust ventilation systems that are sufficient to reduce and maintain personal exposures to within acceptable limits must be used and maintained in accordance with manufacturer's instruction and supplemented by exceptions noted in EM 385-1-1, Section 06.H

1.15 RADIATION SAFETY REQUIREMENTS

Submit License Certificates, employee training records, and Leak Test Reports for radiation materials and equipment to the Contracting Officer and Radiation Safety Office (RSO), and Contracting Oversight Technician (COT) for all specialized and licensed material and equipment proposed for use on the construction project (excludes portable machine sources of ionizing radiation including moisture density and X-Ray Fluorescence (XRF)). Maintain on-site records whenever licensed radiological materials or ionizing equipment are on Government property.

Protect workers from radiation exposure in accordance with 10 CFR 20, ensuring any personnel exposures are maintained As Low As Reasonably Achievable.

1.15.1 Radiography Operation Planning Work Sheet

Submit a Gamma and X-Ray Radiography Operation Planning Work Sheet to Contracting Officer 14 days prior to commencement of operations involving radioactive materials or radiation generating devices. For portable machine sources of ionizing radiation, including moisture density and XRF, use and submit the Portable Gauge Operations Planning Worksheet instead. The Contracting Officer and COT will review the submitted worksheet and provide questions and comments.

Contractors must use primary dosimeters process by a National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratory.

1.15.2 Site Access and Security

Coordinate site access and security requirements with the Contracting Officer and COT for all radiological materials and equipment containing ionizing radiation that are proposed for use on a government facility. For gamma radiography materials and equipment, a Government escort is required for any travels on the Installation. The Navy COT or Government authorized representative will meet the Contractor at a designated location outside the Installation, ensure safety of the materials being transported, and will escort the Contractor for gamma sources onto the Installation, to the job site, and off the Installation. For portable machine sources of ionizing radiation, including moisture density and XRF, the Navy COT or Government authorized representative will meet the Contractor at the job site.

Provide a copy of all calibration records, and utilization records to the COT for radiological operations performed on the site.

1.15.3 Loss or Release and Unplanned Personnel Exposure

Loss or release of radioactive materials, and unplanned personnel exposures must be reported immediately to the Contracting Officer, RSO, and Base Security Department Emergency Number.

1.15.4 Site Demarcation and Barricade

Properly demark and barricade an area surrounding radiological operations to preclude personnel entrance, in accordance with EM 385-1-1, Nuclear Regulatory Commission, and Applicable State regulations and license requirements, and in accordance with requirements established in the accepted Radiography Operation Planning Work Sheet.

Do not close or obstruct streets, walks, and other facilities occupied and used by the Government without written permission from the Contracting Officer.

1.15.5 Security of Material and Equipment

Properly secure the radiological material and ionizing radiation equipment at all times, including keeping the devices in a properly marked and locked container, and secondarily locking the container to a secure point in the Contractor's vehicle or other approved storage location during transportation and while not in use. While in use, maintain a continuous visual observation on the radiological material and ionizing radiation equipment. In instances where radiography is scheduled near or adjacent to buildings or areas having limited access or one-way doors, make no assumptions as to building occupancy. Where necessary, the Contracting Officer will direct the Contractor to conduct an actual building entry, search, and alert. Where removal of personnel from such a building cannot be accomplished and it is otherwise safe to proceed with the radiography, position a fully instructed employee inside the building or area to prevent exiting while external radiographic operations are in process.

1.15.6 Transportation of Material

Comply with 49 CFR 173 for Transportation of Regulated Amounts of Radioactive Material. Notify Local Fire authorities and the site Radiation Safety Officer (RSO) of any Radioactive Material use.

1.15.7 Schedule for Exposure or Unshielding

Actual exposure of the radiographic film or unshielding the source must not be initiated until after 5 p.m. on weekdays.

1.15.8 Transmitter Requirements

Adhere to the base policy concerning the use of transmitters, such as radios and cell phones. Obey Emissions control (EMCON) restrictions.

1.16 CONFINED SPACE ENTRY REQUIREMENTS

Confined space entry must comply with Section 34 of EM 385-1-1, OSHA 29 CFR 1926, OSHA 29 CFR 1910, OSHA 29 CFR 1910.146, and OSHA Directive CPL 2.100. Any potential for a hazard in the confined space requires a permit system to be used.

1.16.1 Entry Procedures

Prohibit entry into a confined space by personnel for any purpose, including hot work, until the qualified person has conducted appropriate tests to ensure the confined or enclosed space is safe for the work intended and that all potential hazards are controlled or eliminated and documented. Comply with EM 385-1-1, Section 34 for entry procedures. Hazards pertaining to the space must be reviewed with each employee during review of the AHA.

1.16.2 Forced Air Ventilation

Forced air ventilation is required for all confined space entry operations and the minimum air exchange requirements must be maintained to ensure exposure to any hazardous atmosphere is kept below its action level.

1.16.3 Sewer Wet Wells

Sewer wet wells require continuous atmosphere monitoring with audible alarm for toxic gas detection.

1.16.4 Rescue Procedures and Coordination with Local Emergency Responders

Develop and implement an on-site rescue and recovery plan and procedures. The rescue plan must not rely on local emergency responders for rescue from a confined space.

1.17 GAS PROTECTION

1.17.1 Special Requirements

Special requirements, coordination, and precautions will apply to areas that contain a hazardous atmosphere or, by virtue of their use or physical character, may be oxygen deficient. A check by Government is required prior to entering confined space. Surveillance and monitoring are required in these types of work spaces by both Contractor and Government personnel.

1.18 SEVERE STORM PLAN

In the event of a severe storm warning, the Contractor must comply with the applicable Storm Plan and:

- a. Secure outside equipment and materials and place materials that could be damaged in protected areas.
- b. Check surrounding area, including roof, for loose material, equipment, debris, and other objects that could be blown away or against existing facilities.
- c. Ensure that temporary erosion controls are adequate.

PART 2 PRODUCTS

2.1 CONFINED SPACE SIGNAGE

Provide permanent signs integral to or securely attached to access covers for new permit-required confined spaces. Signs for confined spaces must comply with NEMA Z535.2. Provide signs with wording: "DANGER--PERMIT-REQUIRED CONFINED SPACE, DO NOT ENTER" in bold letters a minimum of 25 mm in height and constructed to be clearly legible with all paint removed. The signal word "DANGER" must be red and readable from 1520 mm.

PART 3 EXECUTION

3.1 CONSTRUCTION AND OTHER WORK

Comply with EM 385-1-1, NFPA 70, NFPA 70E, NFPA 241, the APP, the AHA, Federal and State OSHA regulations, and other related submittals and activity fire and safety regulations. The most stringent standard prevails.

PPE is governed in all areas by the nature of the work the employee is performing. Use personal hearing protection at all times in designated noise hazardous areas or when performing noise hazardous tasks. Safety glasses must be worn or carried/available on each person. Mandatory PPE includes:

- a. Hard Hat
- b. Long Pants
- c. Appropriate Safety Shoes
- d. Appropriate Class Reflective Vests

3.1.1 Worksite Communication

Employees working alone in a remote location or away from other workers must be provided an effective means of emergency communications (i.e., cellular phone, two-way radios, land-line telephones or other acceptable means). The selected communication must be readily available (easily within the immediate reach) of the employee and must be tested prior to the start of work to verify that it effectively operates in the area/environment. Develop an employee check-in/check-out communication procedure to ensure employee safety.

3.1.2 Hazardous Material Use

Each hazardous material must receive approval from the Contracting Office

or their designated representative prior to being brought onto the job site or prior to any other use in connection with this Contract. Allow a minimum of 10 working days for processing of the request for use of a hazardous material.

3.1.3 Hazardous Material Exclusions

Notwithstanding any other hazardous material used in this Contract, radioactive materials or instruments capable of producing ionizing/non-ionizing radiation (with the exception of radioactive material and devices used in accordance with EM 385-1-1 such as nuclear density meters for compaction testing and laboratory equipment with radioactive sources) as well as materials which contain asbestos, mercury or polychlorinated biphenyls, di-isocyanates, lead-based paint, and hexavalent chromium, are prohibited. The Contracting Officer, upon written request by the Contractor, may consider exceptions to the use of any of the above excluded materials. Low mercury lamps used within fluorescent lighting fixtures are allowed as an exception without further Contracting Officer approval. Notify the Radiation Safety Officer (RSO) prior to excepted items of radioactive material and devices being brought on base.

3.1.4 Unforeseen Hazardous Material

Contract documents identify materials such as PCB, lead paint, and friable and non-friable asbestos and other OSHA regulated chemicals (i.e. 29 CFR Part 1910.1000). If material(s) that may be hazardous to human health upon disturbance are encountered during construction operations, stop that portion of work and notify the Contracting Officer immediately. Within 14 calendar days the Government will determine if the material is hazardous. If material is not hazardous or poses no danger, the Government will direct the Contractor to proceed without change. If material is hazardous and handling of the material is necessary to accomplish the work, the Government will issue a modification pursuant to FAR 52.243-4 Changes and FAR 52.236-2 Differing Site Conditions.

3.2 UTILITY OUTAGE REQUIREMENTS

Apply for utility outages at least 10 calendar days in advance. At a minimum, the written request must include the location of the outage, utilities being affected, duration of outage, any necessary sketches, and a description of the means to fulfill energy isolation requirements in accordance with EM 385-1-1, Section 11.A.02 (Isolation). Some examples of energy isolation devices and procedures are highlighted in EM 385-1-1, Section 12.D. In accordance with EM 385-1-1, Section 12.A.01, where outages involve Government or Utility personnel, coordinate with the Government on all activities involving the control of hazardous energy.

These activities include, but are not limited to, a review of HECF and HEC procedures, as well as applicable Activity Hazard Analyses (AHAs). In accordance with EM 385-1-1, Section 11.A.02 and NFPA 70E, work on energized electrical circuits must not be performed without prior Government authorization. Government permission is considered through the permit process and submission of a detailed AHA. Energized work permits are considered only when de-energizing introduces additional or increased hazard or when de-energizing is infeasible.

3.3 OUTAGE COORDINATION MEETING

After the utility outage request is approved and prior to beginning work on the utility system requiring shut-down, conduct a pre-outage coordination meeting in accordance with EM 385-1-1, Section 12.A. This meeting must include the Prime Contractor, the Prime and subcontractors performing the work, the Contracting Officer, and the Installation representative. All parties must fully coordinate HEC activities with one another. During the coordination meeting, all parties must discuss and coordinate on the scope of work, HEC procedures (specifically, the lock-out/tag-out procedures for worker and utility protection), the AHA, assurance of trade personnel qualifications, identification of competent persons, and compliance with HEC training in accordance with EM 385-1-1, Section 12.C. Clarify when personal protective equipment is required during switching operations, inspection, and verification.

3.4 CONTROL OF HAZARDOUS ENERGY (LOCKOUT/TAGOUT)

Provide and operate a Hazardous Energy Control Program (HECP) in accordance with EM 385-1-1 Section 12, 29 CFR 1910.333, 29 CFR 1915.89, ASSP A10.44, NFPA 70E, and paragraph HAZARDOUS ENERGY CONTROL PROGRAM (HECP).

3.4.1 Safety Preparatory Inspection Coordination Meeting with the Government or Utility

For electrical distribution equipment that is to be operated by Government or Utility personnel, the Prime Contractor and the subcontractor performing the work must attend the safety preparatory inspection coordination meeting, which will also be attended by the Contracting Officer's Representative, and required by EM 385-1-1, Section 12.A.02. The meeting will occur immediately preceding the start of work and following the completion of the outage coordination meeting. Both the safety preparatory inspection coordination meeting and the outage coordination meeting must occur prior to conducting the outage and commencing with lockout/tagout procedures.

3.4.2 Lockout/Tagout Isolation

Where the Government or Utility performs equipment isolation and lockout/tagout, the Contractor must place their own locks and tags on each energy-isolating device and proceed in accordance with the HECP. Before any work begins, both the Contractor and the Government or Utility must perform energy isolation verification testing while wearing required PPE detailed in the Contractor's AHA and required by EM 385-1-1, Sections 05.I and 11.B. Install personal protective grounds, with tags, to eliminate the potential for induced voltage in accordance with EM 385-1-1, Section 12.E.06.

3.4.3 Lockout/Tagout Removal

Upon completion of work, conduct lockout/tagout removal procedure in accordance with the HECP. In accordance with EM 385-1-1, Section 12.E.08, each lock and tag must be removed from each energy isolating device by the authorized individual or systems operator who applied the device. Provide formal notification to the Government (by completing the Government form if provided by Contracting Officer's Representative), confirming that steps of de-energization and lockout/tagout removal procedure have been conducted and certified through inspection and verification. Government

or Utility locks and tags used to support the Contractor's work will not be removed until the authorized Government employee receives the formal notification.

3.5 FALL PROTECTION PROGRAM

Establish a fall protection program, for the protection of all employees exposed to fall hazards. Within the program include company policy, identify roles and responsibilities, education and training requirements, fall hazard identification, prevention and control measures, inspection, storage, care and maintenance of fall protection equipment and rescue and evacuation procedures in accordance with ASSP Z359.2 and EM 385-1-1, Sections 21.A and 21.D.

3.5.1 Training

Institute a fall protection training program. As part of the Fall Protection Program, provide training for each employee who might be exposed to fall hazards and using personal fall protection equipment. Provide training by a competent person for fall protection in accordance with EM 385-1-1, Section 21.C. Document training and practical application of the competent person in accordance with EM 385-1-1, Section 21.C.04 and ASSP Z359.2 in the AHA.

3.5.2 Fall Protection Equipment and Systems

Enforce use of personal fall protection equipment and systems designated (to include fall arrest, restraint, and positioning) for each specific work activity in the Site Specific Fall Protection and Prevention Plan and AHA at all times when an employee is exposed to a fall hazard. Protect employees from fall hazards as specified in EM 385-1-1, Section 21.

Provide personal fall protection equipment, systems, subsystems, and components that comply with EM 385-1-1 Section 21.I, 29 CFR 1926.500 Subpart M, ASSP Z359.0, ASSP Z359.1, ASSP Z359.2, ASSP Z359.3, ASSP Z359.4, ASSP Z359.6, ASSP Z359.7, ASSP Z359.11, ASSP Z359.12, ASSP Z359.13, ASSP Z359.14, ASSP Z359.15, ASSP Z359.16 and ASSP Z359.18.

3.5.2.1 Personal Fall Protection Equipment

Only a full-body harness with a shock-absorbing lanyard or self-retracting lanyard is an acceptable personal fall arrest body support device. The use of body belts is not acceptable. Harnesses must have a fall arrest attachment affixed to the body support (usually a Dorsal D-ring) and specifically designated for attachment to the rest of the system. Snap hooks and carabineers must be self-closing and self-locking, capable of being opened only by at least two consecutive deliberate actions and have a minimum gate strength of 1633 kg in all directions. Use webbing, straps, and ropes made of synthetic fiber. The maximum free fall distance when using fall arrest equipment must not exceed 1.8 m, unless the proper energy absorbing lanyard is used. Always take into consideration the total fall distance and any swinging of the worker (pendulum-like motion), that can occur during a fall, when attaching a person to a fall arrest system. Equip all full body harnesses with Suspension Trauma Preventers such as stirrups, relief steps, or similar in order to provide short-term relief from the effects of orthostatic intolerance in accordance with EM 385-1-1, Section 21.I.06.

3.5.3 Fall Protection for Roofing Work

Implement fall protection controls based on the type of roof being constructed and work being performed. Evaluate the roof area to be accessed for its structural integrity including weight-bearing capabilities for the projected loading.

a. Low Sloped Roofs:

- (1) For work within 1.8 m from unprotected edge of a roof having a slope less than or equal to 4:12 (vertical to horizontal), protect personnel from falling by the use of conventional fall protection systems (personal fall arrest/restraint systems, guardrails, or safety nets) in accordance with EM 385-1-1, Section 21 and 29 CFR 1926.500. A safety monitoring system is not adequate fall protection and is not authorized.
- (2) For work greater than 1.8 m from the unprotected roof edge, addition to the use of conventional fall protection systems the use of a warning line system is also permitted, in accordance with 29 CFR 1926.500 and EM 385-1-1, Section 21.L.

b. Steep-Sloped Roofs: Work on a roof having a slope greater than 4:12 (vertical to horizontal) requires a personal fall arrest system, guardrails with toe-boards, or safety nets. This requirement also applies to residential or housing type construction.

3.5.4 Horizontal Lifelines (HLL)

Provide HLL in accordance with EM 385-1-1, Section 21.I.08.d.2. Commercially manufactured horizontal lifelines (HLL) must be designed, installed, certified and used, under the supervision of a qualified person, for fall protection as part of a complete fall arrest system which maintains a safety factor of 2 (29 CFR 1926.500). The competent person for fall protection may (if deemed appropriate by the qualified person) supervise the assembly, disassembly, use and inspection of the HLL system under the direction of the qualified person. Locally manufactured HLLs are not acceptable unless they are custom designed for limited or site specific applications by a Registered Professional Engineer who is qualified in designing HLL systems.

3.5.5 Guardrails and Safety Nets

Design, install and use guardrails and safety nets in accordance with EM 385-1-1, Section 21.F.01 and 29 CFR 1926 Subpart M.

3.5.6 Rescue and Evacuation Plan and Procedures

When personal fall arrest systems are used, ensure that the mishap victim can self-rescue or can be rescued promptly should a fall occur. Prepare a Rescue and Evacuation Plan and include a detailed discussion of the following: methods of rescue; methods of self-rescue or assisted-rescue; equipment used; training requirement; specialized training for the rescuers; procedures for requesting rescue and medical assistance; and transportation routes to a medical facility. Include the Rescue and Evacuation Plan within the Activity Hazard Analysis (AHA) for the phase of work, in the Fall Protection and Prevention (FP&P) Plan, and the Accident Prevention Plan (APP). The plan must be in accordance with the requirements of EM 385-1-1, ASSP Z359.2, and ASSP Z359.4.

3.6 WORK PLATFORMS

3.6.1 Scaffolding

Provide employees with a safe means of access to the work area on the scaffold. Climbing of any scaffold braces or supports not specifically designed for access is prohibited. Comply with the following requirements:

- a. Scaffold platforms greater than 6 m in height must be accessed by use of a scaffold stair system.
- b. Ladders commonly provided by scaffold system manufacturers are prohibited for accessing scaffold platforms greater than 6 m maximum in height.
- c. An adequate gate is required.
- d. Employees performing scaffold erection and dismantling must be qualified.
- e. Scaffold must be capable of supporting at least four times the maximum intended load, and provide appropriate fall protection as delineated in the accepted fall protection and prevention plan.
- f. Stationary scaffolds must be attached to structural building components to safeguard against tipping forward or backward.
- g. Special care must be given to ensure scaffold systems are not overloaded.
- h. Side brackets used to extend scaffold platforms on self-supported scaffold systems for the storage of material are prohibited. The first tie-in must be at the height equal to 4 times the width of the smallest dimension of the scaffold base.
- i. Scaffolding other than suspended types must bear on base plates upon wood mudsills (51 mm x 254 mm x 203 mm minimum) or other adequate firm foundation.
- j. Scaffold or work platform erectors must have fall protection during the erection and dismantling of scaffolding or work platforms that are more than 1.83 meters.
- k. Delineate fall protection requirements when working above 1.83 meters or above dangerous operations in the Fall Protection and Prevention (FP&P) Plan and Activity Hazard Analysis (AHA) for the phase of work.

3.6.2 Elevated Aerial Work Platforms (AWPs)

Workers must be anchored to the basket or bucket in accordance with manufacturer's specifications and instructions (anchoring to the boom may only be used when allowed by the manufacturer and permitted by the CP). Lanyards used must be sufficiently short to prohibit worker from climbing out of basket. The climbing of rails is prohibited. Lanyards with built-in shock absorbers are acceptable. Self-retracting devices are not acceptable. Tying off to an adjacent pole or structure is not permitted unless a safe device for 100 percent tie-off is used for the transfer.

Use of AWP's must be operated, inspected, and maintained as specified in the operating manual for the equipment and delineated in the AHA. Operators of AWP's must be designated as qualified operators by the Prime Contractor. Maintain proof of qualifications on site for review and include in the AHA.

3.7 EQUIPMENT

3.7.1 Material Handling Equipment (MHE)

- a. Material handling equipment such as forklifts must not be modified with work platform attachments for supporting employees unless specifically delineated in the manufacturer's printed operating instructions. Material handling equipment fitted with personnel work platform attachments are prohibited from traveling or positioning while personnel are working on the platform.
- b. The use of hooks on equipment for lifting of material must be in accordance with manufacturer's printed instructions. Material Handling Equipment Operators must be trained in accordance with OSHA 29 CFR 1910, Subpart N.
- c. Operators of forklifts or power industrial trucks must be licensed in accordance with OSHA.

3.7.2 Load Handling Equipment (LHE)

The following requirements apply. In exception, these requirements do not apply to commercial truck mounted and articulating boom cranes used solely to deliver material and supplies (not prefabricated components, structural steel, or components of a systems-engineered metal building) where the lift consists of moving materials and supplies from a truck or trailer to the ground; to cranes installed on mechanics trucks that are used solely in the repair of shore-based equipment; to crane that enter the activity but are not used for lifting; nor to other machines not used to lift loads suspended by rigging equipment. However, LHE accidents occurring during such operations must be reported.

- a. Equip cranes and derricks as specified in EM 385-1-1, Section 16.
- b. Notify the Contracting Officer 15 working days in advance of any LHE entering the activity, in accordance with EM 385-1-1, Section 16.A.02, so that necessary quality assurance spot checks can be coordinated. Prior to cranes entering federal activities, a Crane Access Permit must be obtained from the Contracting Officer. A copy of the permitting process will be provided at the Preconstruction Meeting. Contractor's operator must remain with the crane during the spot check. Rigging gear must be in accordance with OSHA, ASME B30.9 Standards and host country safety standards.
- c. Comply with the LHE manufacturer's specifications and limitations for erection and operation of cranes and hoists used in support of the work. Perform erection under the supervision of a designated person (as defined in ASME B30.5). Perform all testing in accordance with the manufacturer's recommended procedures.
- d. Comply with ASME B30.5 for mobile and locomotive cranes, ASME B30.22 for articulating boom cranes, ASME B30.3 for construction tower cranes, ASME B30.8 for floating cranes and floating derricks,

ASME B30.9 for slings, ASME B30.20 for below the hook lifting devices and ASME B30.26 for rigging hardware.

- e. When operating in the vicinity of overhead transmission lines, operators and riggers must be alert to this special hazard and follow the requirements of EM 385-1-1 Section 11, and ASME B30.5 or ASME B30.22 as applicable.
- f. Do not use crane suspended personnel work platforms (baskets) unless the Contractor proves that using any other access to the work location would provide a greater hazard to the workers or is impossible. Do not lift personnel with a line hoist or friction crane. Additionally, submit a specific AHA for this work to the Contracting Officer. Ensure the activity and AHA are thoroughly reviewed by all involved personnel.
- g. Inspect, maintain, and recharge portable fire extinguishers as specified in NFPA 10, Standard for Portable Fire Extinguishers.
- h. All employees must keep clear of loads about to be lifted and of suspended loads, except for employees required to handle the load.
- i. Use cribbing when performing lifts on outriggers.
- j. The crane hook/block must be positioned directly over the load. Side loading of the crane is prohibited.
- k. A physical barricade must be positioned to prevent personnel access where accessible areas of the LHE's rotating superstructure poses a risk of striking, pinching or crushing personnel.
- l. Maintain inspection records in accordance by EM 385-1-1, Section 16.D, including shift, monthly, and annual inspections, the signature of the person performing the inspection, and the serial number or other identifier of the LHE that was inspected. Records must be available for review by the Contracting Officer.
- m. Maintain written reports of operational and load testing in accordance with EM 385-1-1, Section 16.F, listing the load test procedures used along with any repairs or alterations performed on the LHE. Reports must be available for review by the Contracting Officer.
- n. Certify that all LHE operators have been trained in proper use of all safety devices (e.g. anti-two block devices).
- o. Take steps to ensure that wind speed does not contribute to loss of control of the load during lifting operations. At wind speeds greater than 9 m/s, the operator, rigger and lift supervisor must cease all crane operations, evaluate conditions and determine if the lift may proceed. Base the determination to proceed or not on wind calculations per the manufacturer and a reduction in LHE rated capacity if applicable. Include this maximum wind speed determination as part of the activity hazard analysis plan for that operation.
- p. On mobile cranes, lifts where the load weight is greater than 90 percent of the equipment's capacity are prohibited.
- q. Follow FAA guidelines when required based on project location.

3.7.3 Machinery and Mechanized Equipment

- a. Proof of qualifications for operator must be kept on the project site for review.
- b. Manufacture specifications or owner's manual for the equipment must be on-site and reviewed for additional safety precautions or requirements that are sometimes not identified by OSHA or USACE EM 385-1-1. Incorporate such additional safety precautions or requirements into the AHAs.

3.7.4 Base Mounted Drum Hoists

- a. Operation of base mounted drum hoists must be in accordance with EM 385-1-1 and ASSP A10.22.
- b. Rigging gear must be in accordance with applicable ASME/OSHA standards.
- c. When used on telecommunication towers, base mounted drum hoists must be in accordance with TIA-1019, TIA-222, ASME B30.7, 29 CFR 1926.552, and 29 CFR 1926.553.
- d. When used to hoist personnel, the AHA must include a written standard operating procedure. Operators must have a physical examination in accordance with EM 385-1-1 Section 16.B.05 and trained, at a minimum, in accordance with EM 385-1-1 Section 16.U and 16.T. The base mounted drum hoist must also comply with OSHA Instruction CPL 02-01-056 and ASME B30.23.
- e. Material and personnel must not be hoisted simultaneously.
- f. Personnel cage must be marked with the capacity (in number of persons) and load limit in kg.
- g. Construction equipment must not be used for hoisting material or personnel or with trolley/tag lines. Construction equipment may be used for towing and assisting with anchoring guy lines.

3.8 EXCAVATIONS

Soil classification must be performed by a competent person in accordance with 29 CFR 1926 and EM 385-1-1.

3.8.1 Utility Locations

Provide a third party, independent, private utility locating company to positively identify underground utilities in the work area in addition to any station locating service and coordinated with the station utility department.

3.8.2 Utility Location Verification

Physically verify underground utility locations, including utility depth, by hand digging using wood or fiberglass handled tools when any adjacent construction work is expected to come within one meter of the underground system.

3.8.3 Utilities Within and Under Concrete, Bituminous Asphalt, and Other Impervious Surfaces

Utilities located within and under concrete slabs or pier structures, bridges, parking areas, and the like, are extremely difficult to identify. Whenever Contract work involves chipping, saw cutting, or core drilling through concrete, bituminous asphalt or other impervious surfaces, the existing utility location must be coordinated with station utility departments in addition to location and depth verification by a third party, independent, private locating company. The third party, independent, private locating company must locate utility depth by use of Ground Penetrating Radar (GPR), X-ray, bore scope, or ultrasound prior to the start of demolition and construction. Outages to isolate utility systems must be used in circumstances where utilities are unable to be positively identified. The use of historical drawings does not alleviate the Contractor from meeting this requirement.

3.9 ELECTRICAL

Perform electrical work in accordance with EM 385-1-1, Sections 11 and 12.

3.9.1 Conduct of Electrical Work

As delineated in EM 385-1-1, electrical work is to be conducted in a de-energized state unless there is no alternative method for accomplishing the work. In those cases obtain an energized work permit from the Contracting Officer. The energized work permit application must be accompanied by the AHA and a summary of why the equipment/circuit needs to be worked energized. Underground electrical spaces must be certified safe for entry before entering to conduct work. Cables that will be cut must be positively identified and de-energized prior to performing each cut. Attach temporary grounds in accordance with ASTM F855 and IEEE 1048. Perform all high voltage cable cutting remotely using hydraulic cutting tool. When racking in or live switching of circuit breakers, no additional person other than the switch operator is allowed in the space during the actual operation. Plan so that work near energized parts is minimized to the fullest extent possible. Use of electrical outages clear of any energized electrical sources is the preferred method.

When working in energized substations, only qualified electrical workers are permitted to enter. When work requires work near energized circuits as defined by NFPA 70, high voltage personnel must use personal protective equipment that includes, as a minimum, electrical hard hat, safety shoes, insulating gloves and electrical arc flash protection for personnel as required by NFPA 70E. Insulating blankets, hearing protection, and switching suits may also be required, depending on the specific job and as delineated in the Contractor's AHA. Ensure that each employee is familiar with and complies with these procedures and 29 CFR 1910.147.

3.9.2 Qualifications

Electrical work must be performed by QP with verifiable credentials who are familiar with applicable code requirements. Verifiable credentials consist of State, National and Local Certifications or Licenses that a Master or Journeyman Electrician may hold, depending on work being performed, and must be identified in the appropriate AHA. Journeyman/Apprentice ratio must be in accordance with State, Local and Host Nation requirements applicable to where work is being performed.

3.9.3 Arc Flash

Conduct a hazard analysis/arc flash hazard analysis whenever work on or near energized parts greater than 50 volts is necessary, in accordance with NFPA 70E.

All personnel entering the identified arc flash protection boundary must be QPs and properly trained in NFPA 70E requirements and procedures. Unless permitted by NFPA 70E, no Unqualified Person is permitted to approach nearer than the Limited Approach Boundary of energized conductors and circuit parts. Training must be administered by an electrically qualified source and documented.

3.9.4 Grounding

Ground electrical circuits, equipment and enclosures in accordance with NFPA 70 and IEEE C2 to provide a permanent, continuous and effective path to ground unless otherwise noted by EM 385-1-1.

Check grounding circuits to ensure that the circuit between the ground and a grounded power conductor has a resistance low enough to permit sufficient current flow to allow the fuse or circuit breaker to interrupt the current.

3.9.5 Testing

Temporary electrical distribution systems and devices must be inspected, tested and found acceptable for Ground-Fault Circuit Interrupter (GFCI) protection, polarity, ground continuity, and ground resistance before initial use, before use after modification and at least monthly. Monthly inspections and tests must be maintained for each temporary electrical distribution system, and signed by the electrical CP or QP.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS

02/19

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 ORDERING INFORMATION

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 42 00

SOURCES FOR REFERENCE PUBLICATIONS

02/19

PART 1 GENERAL

1.1 REFERENCES

Various publications are referenced in other sections of the specifications to establish requirements for the work. These references are identified in each section by document number, date and title. The document number used in the citation is the number assigned by the standards producing organization (e.g., ASTM B564 Standard Specification for Nickel Alloy Forgings). However, when the standards producing organization has not assigned a number to a document, an identifying number has been assigned for reference purposes.

1.2 ORDERING INFORMATION

The addresses of the standards publishing organizations whose documents are referenced in other sections of these specifications are listed below, and if the source of the publications is different from the address of the sponsoring organization, that information is also provided.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)
444 North Capital Street, NW, Suite 249
Washington, DC 20001
Ph: 202-624-5800
Fax: 202-624-5806
E-Mail: info@aaashto.org
Internet: <https://www.transportation.org/>

AMERICAN CONCRETE INSTITUTE (ACI)
38800 Country Club Drive
Farmington Hills, MI 48331-3439
Ph: 248-848-3700
Fax: 248-848-3701
Internet: <https://www.concrete.org/>

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)
130 East Randolph, Suite 2000
Chicago, IL 60601
Ph: 312-670-5444
Fax: 312-670-5403
Steel Solutions Center: 866-275-2472
E-mail: solutions@aisc.org
Internet: <https://www.aisc.org/>

AMERICAN PETROLEUM INSTITUTE (API)
1220 L Street, NW
Washington, DC 20005-4070
Ph: 202-682-8000
Internet: <https://www.api.org/>

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)
P.O. Box 28518
1711 Arlingate Lane
Columbus, OH 43228-0518
Ph: 800-222-2768 or 614-274-6003
Fax: 614-274-6899
E-mail: tjones@asnt.org
Internet: <https://www.asnt.org/>

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)
1791 Tullie Circle, NE
Atlanta, GA 30329
Ph: 404-636-8400 or 800-527-4723
Fax: 404-321-5478
E-mail: ashrae@ashrae.org
Internet: <https://www.ashrae.org/>

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)
Two Park Avenue
New York, NY 10016-5990
Ph: 800-843-2763
Fax: 973-882-1717
E-mail: customercare@asme.org
Internet: <https://www.asme.org/>

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)
520 N. Northwest Highway
Park Ridge, IL 60068
Ph: 847-699-2929
E-mail: customerservice@assp.org
Internet: <https://www.assp.org/>

AMERICAN WATER WORKS ASSOCIATION (AWWA)
6666 W. Quincy Avenue
Denver, CO 80235 USA
Ph: 303-794-7711 or 800-926-7337
Fax: 303-347-0804
Internet: <https://www.awwa.org/>

AMERICAN WELDING SOCIETY (AWS)
8669 NW 36 Street, #130
Miami, FL 33166-6672
Ph: 800-443-9353
Internet: <https://www.aws.org/>

ASPHALT INSTITUTE (AI)
2696 Research Park Drive
Lexington, KY 40511-8480
Ph: 859-288-4960
Fax: 859-288-4999
E-mail: info@asphaltinstitute.org
Internet: <http://www.asphaltinstitute.org>

ASTM INTERNATIONAL (ASTM)
100 Barr Harbor Drive, P.O. Box C700
West Conshohocken, PA 19428-2959
Ph: 610-832-9500
Fax: 610-832-9555

E-mail: service@astm.org
Internet: <https://www.astm.org/>

GREEN SEAL (GS)
1001 Connecticut Avenue, NW
Suite 827
Washington, DC 20036-5525
Ph: 202-872-6400
Fax: 202-872-4324
E-mail: green seal@green seal.org
Internet: <https://www.green seal.org/>

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)
445 and 501 Hoes Lane
Piscataway, NJ 08854-4141
Ph: 732-981-0060 or 800-701-4333
Fax: 732-981-9667
E-mail: onlinesupport@ieee.org
Internet: <https://www.ieee.org/>

INTERNATIONAL CODE COUNCIL (ICC)
500 New Jersey Avenue, NW
6th Floor, Washington, DC 20001
Ph: 800-786-4452 or 888-422-7233
Fax: 202-783-2348
E-mail: order@iccsafe.org
Internet: <https://www.iccsafe.org/>

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)
3050 Old Centre Ave. Suite 101
Portage, MI 49024
Ph: 269-488-6382
Fax: 269-488-6383
Internet: <https://www.netaworld.org/>

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)
ISO Central Secretariat
BIBC II
Chemin de Blandonnet 8
CP 401 - 1214 Vernier, Geneva
Switzerland
Ph: 41-22-749-01-11
E-mail: central@iso.ch
Internet: <https://www.iso.org>

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS
INDUSTRY (MSS)
127 Park Street, NE
Vienna, VA 22180-4602
Ph: 703-281-6613
E-mail: info@msshq.org
Internet: <http://msshq.org>

MASTER PAINTERS INSTITUTE (MPI)
2800 Ingleton Avenue
Burnaby, BC CANADA V5C 6G7
Ph: 1-888-674-8937
Fax: 1-888-211-8708

E-mail: info@paintinfo.com or techservices@mpi.net
Internet: <http://www.mpi.net/>

NACE INTERNATIONAL (NACE)
15835 Park Ten Place
Houston, TX 77084
Ph: 281-228-6200
Fax: 281-228-6300
E-mail: firstservice@nace.org
Internet: <https://www.nace.org>

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA)
NASA Headquarters
300 E. Street SW, Suite 5R30
Washington, DC 20546
Ph: 202-358-0001
Fax: 202-358-4338
Internet: <https://www.nasa.gov/>

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)
800 Roosevelt Road, Bldg C, Suite 312
Glen Ellyn, IL 60137
Ph: 630-942-6591
Fax: 630-790-3095
E-mail: info@naamm.org
Internet: <http://www.naamm.org>

NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)
3 Bethesda Metro Center, Suite 1100
Bethesda, MD 20814
Ph: 301-657-3110
Fax: 301-215-4500
Internet: <https://www.necanet.org/>

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)
1300 North 17th Street, Suite 900
Arlington, VA 22209
Ph: 703-841-3200
Internet: <https://www.nema.org>

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)
1 Batterymarch Park
Quincy, MA 02169-7471
Ph: 800-344-3555
Fax: 800-593-6372
Internet: <https://www.nfpa.org>

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION
(SMACNA)
4201 Lafayette Center Drive
Chantilly, VA 20151-1219
Ph: 703-803-2980
Fax: 703-803-3732
Internet: <https://www.smacna.org/>

SOCIETY FOR PROTECTIVE COATINGS (SSPC)
800 Trumbull Drive
Pittsburgh, PA 15205

Ph: 877-281-7772 or 412-281-2331
Fax: 412-444-3591
E-mail: customerservice@sspc.org
Internet: <http://www.sspc.org>

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)
400 Commonwealth Drive
Warrendale, PA 15096
Ph: 877-606-7323 or 724-776-4841
Fax: 724-776-0790
E-mail: customerservice@sae.org
Internet: <https://www.sae.org/>

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)
1320 North Courthouse Road, Suite 200
Arlington, VA 22201
Ph: 703-907-7700
Fax: 703-907-7727
E-mail: marketing@tiaonline.org
Internet: <https://www.tiaonline.org/>

U.S. ARMY CORPS OF ENGINEERS (USACE)
CRD-C DOCUMENTS available on Internet:
<http://www.wbdg.org/ffc/army-coe/standards>
Order Other Documents from:
Official Publications of the Headquarters, USACE
E-mail: hqpublications@usace.army.mil
Internet: <http://www.publications.usace.army.mil/>
or
<https://www.hnc.usace.army.mil/Missions/Engineering-Directorate/TECHINFO/>

U.S. DEPARTMENT OF AGRICULTURE (USDA)
Order AMS Publications from:
AGRICULTURAL MARKETING SERVICE (AMS)
Seed Regulatory and Testing Branch
801 Summit Crossing Place, Suite C
Gastonia, NC 28054-2193
Ph: 704-810-8884
E-mail: PA@ams.usda.gov
Internet: <https://www.ams.usda.gov/>
Order Other Publications from:
USDA Rural Development
Rural Utilities Service
STOP 1510, Rm 5135
1400 Independence Avenue SW
Washington, DC 20250-1510
Phone: (202) 720-9540
Internet:
<https://www.rd.usda.gov/about-rd/agencies/rural-utilities-service>

U.S. DEPARTMENT OF DEFENSE (DOD)
Order DOD Documents from:
Room 3A750-The Pentagon
1400 Defense Pentagon
Washington, DC 20301-1400
Ph: 703-571-3343
Fax: 215-697-1462
E-mail: customerservice@ntis.gov
Internet: <https://www.ntis.gov/>

Obtain Military Specifications, Standards and Related Publications
from:

Acquisition Streamlining and Standardization Information System
(ASSIST)

Department of Defense Single Stock Point (DODSSP)

Document Automation and Production Service (DAPS)

Building 4/D

700 Robbins Avenue

Philadelphia, PA 19111-5094

Ph: 215-697-6396 - for account/password issues

Internet: <https://assist.dla.mil/online/start/>; account
registration required

Obtain Unified Facilities Criteria (UFC) from:

Whole Building Design Guide (WBDG)

National Institute of Building Sciences (NIBS)

1090 Vermont Avenue NW, Suite 700

Washington, DC 20005

Ph: 202-289-7800

Fax: 202-289-1092

Internet:

<https://www.wbdg.org/ffc/dod/unified-facilities-criteria-ufc>

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)

1200 New Jersey Ave., SE

Washington, DC 20590

Ph: 202-366-4000

E-mail: ExecSecretariat.FHWA@dot.gov

Internet: <https://www.fhwa.dot.gov/>

Order from:

Superintendent of Documents

U.S. Government Publishing Office (GPO)

732 N. Capitol Street, NW

Washington, DC 20401

Ph: 202-512-1800 or 866-512-1800

Bookstore: 202-512-0132

Internet: <https://www.gpo.gov/>

U. S. GREEN BUILDING COUNCIL (USGBC)

2101 L St NW, Suite 500

Washington, DC 20037

Ph: 202-828-7422

Internet: <https://new.usgbc.org/>

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

8601 Adelphi Road

College Park, MD 20740-6001

Ph: 866-272-6272

Internet: <https://www.archives.gov/>

Order documents from:

Superintendent of Documents

U.S. Government Publishing Office (GPO)

732 N. Capitol Street, NW

Washington, DC 20401

Ph: 202-512-1800 or 866-512-1800

Bookstore: 202-512-0132

Internet: <https://www.gpo.gov/>

UNDERWRITERS LABORATORIES (UL)

2600 N.W. Lake Road

Camas, WA 98607-8542
Ph: 877-854-3577 or 360-817-5500
E-mail: CustomerExperienceCenter@ul.com
Internet: <https://www.ul.com/>
UL Directories available through IHS at <https://ihsmarkit.com/>

JUNTA DE ANDALUCIA (JA)
Servicios Centrales
Archivo Central de la Consejería de Fomento, Infraestructuras y
Ordenación del Territorio
Calle Pablo Picasso, nº 6
41018 Sevilla
Ph: 955 92 67 37 / 955 92 66 73
E-mail: archivocentral.sscs.cfiot@juntadeandalucia.es
internet: <https://www.juntadeandalucia.es/organismos.html>

MINISTERIO DE MEDIO AMBIENTE (MA)
Secretaría de Estado de Medio Ambiente
Plaza San Juan de la Cruz, s/n
Madrid 28071
Ph: 91 597.6060/5669
E-mail: sema@mapama.es
internet: <https://www.miteco.gob.es/es/>

AENOR - UNA NORMA ESPAÑOLA (UNE)
Calle Génova, nº 6
28004 Madrid
Ph: 91 4326 000
E-mail: info@aenor.com
internet: <https://www.aenor.com/>

MINISTERIO DE FOMENTO (MF)
Paseo de la Castellana, 67
28046 Madrid
Ph: 915 97 70 00
internet: <https://www.mitma.gob.es/>

MINISTERIO DE VIVIENDA (MV)
Paseo de la Castellana, 67
28046 Madrid
Ph: 915 97 70 00
internet: <https://www.mitma.gob.es/vivienda>

CENTRO DE ESTUDIOS Y EXPERIMENTACION DE OBRAS PUBLICAS (CEDEX)
Venta de Publicaciones
Alfonso XII, 3
28014 MADRID
Ph: +(34) 91 335 72 95
E-mail: publicaciones@cedex.es
internet: http://www.cedex.es/CEDEX/lang_castellano/

MINISTERIO DE LA PRESIDENCIA (MP)
Complejo de la Moncloa, Av Puerta de Hierro, S/N
28071 Madrid
Ph: 91 335 35 35
internet: <https://www.mpr.gob.es/Paginas/index.aspx>

NAVSTA ROTA ENVIRONMENTAL REFERENCE DOCUMENTS

PSC 819

Box 1

Rota, Spain 09645

internet:

https://www.cnmc.navy.mil/regions/cnreurfcent/installations/ns_rota.html

JEFATURA DEL ESTADO (JE)

c/ Manuel Cortina, 2. 28071 Madrid

Ph: 060

internet:

https://administracion.gob.es/pag_Home/espanaAdmon/comoSeOrganizaEstado/Instit

PART 2 PRODUCTS

Not used

PART 3 EXECUTION

Not used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 42 15

METRIC MEASUREMENTS

02/14

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 GENERAL
- 1.3 USE OF MEASUREMENTS IN SPECIFICATIONS
 - 1.3.1 Hard Metric
 - 1.3.2 Soft Metric
 - 1.3.3 Neutral
- 1.4 COORDINATION
- 1.5 RELATIONSHIP TO SUBMITTALS

-- End of Section Table of Contents --

SECTION 01 42 15

METRIC MEASUREMENTS 02/14

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM SI10	(2016) American National Standard for Use of the International System of Units (SI): The Modern Metric System
-----------	---

1.2 GENERAL

This project includes metric units of measurements. The metric units used are the International System of Units (SI) developed and maintained by the General Conference on Weights and Measures (CGPM); the name International System of Units and the international abbreviation SI were adopted by the 11th CGPM in 1960. When both metric and English inch-pound (I-P) measurements are included the specification may contain measurements for products that are manufactured to an industry recognized rounded metric (hard metric) dimensions but are allowed to be substituted by I-P products, to indicate industry and/or Government standards, test values or other controlling factors, such as the code requirements where I-P values are needed for clarity, or to trace back to the referenced standards, test values or codes.

1.3 USE OF MEASUREMENTS IN SPECIFICATIONS

Measurements in specifications are either in SI or I-P units as indicated, except as otherwise authorized. When only SI or I-P measurements are specified for a product, procure the product in the specified units (SI or I-P) unless otherwise authorized by the Contracting Officer. The Contractor is responsible for all associated labor and materials when authorized to substitute one system of units for another and for the final assembly and performance of the specified work and/or products.

1.3.1 Hard Metric

Hard metric measurements are often used for field data such as distance from one point to another or distance above the floor. Products are considered to be hard metric when they are manufactured to metric dimensions or have an industry recognized metric designation.

1.3.2 Soft Metric

- a. A soft metric measurement is a non-mathematical, industry related conversion. Soft metric measurements are used for measurements pertaining to products, test values, and other situations where the I-P units are the standard for manufacture, verification, or other

controlling factor.

- b. A soft metric measurement is also indicated for products that are manufactured in industry designated metric dimensions but are required by law to allow substitute I-P products.

1.3.3 Neutral

A neutral measurement is indicated by an identifier which has no expressed relation to either an SI or an I-P value (e.g., American Wire Gage (AWG) which indicates thickness but in itself is neither SI nor I-P).

1.4 COORDINATION

Bring discrepancies, such as mismatches or product unavailability, arising from use of both metric and non-metric measurements and discrepancies between the measurements in the specifications and the measurements in the drawings to the attention of the Contracting Officer for resolution.

1.5 RELATIONSHIP TO SUBMITTALS

Submittals for Government approval or for information only covers the SI or I-P products actually being furnished for the project. Submit the required drawings and calculations in the same units used in the contract documents describing the product or requirement unless otherwise instructed or approved. Use ASTM SI10 as the basis for establishing metric measurements required to be used in submittals.

-- End of Document --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 45 00.00 20

QUALITY CONTROL

11/11, CHG 8: 02/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 INFORMATION FOR THE CONTRACTING OFFICER
- 1.4 QC PROGRAM REQUIREMENTS
 - 1.4.1 Acceptance of the Construction Quality Control (QC) Plan
 - 1.4.2 Preliminary Construction Work Authorized Prior to Acceptance
 - 1.4.3 Notification of Changes
- 1.5 QC ORGANIZATION
 - 1.5.1 QC Manager
 - 1.5.1.1 Duties
 - 1.5.1.2 Qualifications
 - 1.5.2 Construction Quality Management Training
 - 1.5.3 Alternate QC Manager Duties and Qualifications
- 1.6 QUALITY CONTROL (QC) PLAN
 - 1.6.1 Construction Quality Control (QC) Plan
 - 1.6.1.1 Requirements
- 1.7 COORDINATION AND MUTUAL UNDERSTANDING MEETING
 - 1.7.1 Purpose
 - 1.7.2 Coordination of Activities
 - 1.7.3 Attendees
- 1.8 QC MEETINGS
- 1.9 THREE PHASES OF CONTROL
 - 1.9.1 Preparatory Phase
 - 1.9.2 Initial Phase
 - 1.9.3 Follow-Up Phase
 - 1.9.4 Additional Preparatory and Initial Phases
 - 1.9.5 Notification of Three Phases of Control for Off-Site Work
- 1.10 SUBMITTAL REVIEW AND APPROVAL
- 1.11 TESTING
 - 1.11.1 Accreditation Requirements
 - 1.11.2 Laboratory Accreditation Authorities
 - 1.11.3 Capability Check
 - 1.11.4 Test Results
 - 1.11.5 Test Reports and Monthly Summary Report of Tests
- 1.12 QC CERTIFICATIONS
 - 1.12.1 CQC Report Certification
 - 1.12.2 Invoice Certification
 - 1.12.3 Completion Certification
- 1.13 COMPLETION INSPECTIONS
 - 1.13.1 Punch-Out Inspection
 - 1.13.2 Pre-Final Inspection
 - 1.13.3 Final Acceptance Inspection
- 1.14 DOCUMENTATION
 - 1.14.1 Construction Documentation

- 1.14.2 Quality Control Validation
- 1.14.3 Testing Plan and Log
- 1.14.4 Rework Items List
- 1.14.5 As-Built Drawings
- 1.15 NOTIFICATION ON NON-COMPLIANCE
- 1.16 CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN
 - 1.16.1 Requirements During Construction
 - 1.16.1.1 Control Measures
 - 1.16.1.2 Moisture Contamination
 - 1.16.2 Requirements after Construction

PART 2 PRODUCTS

PART 3 EXECUTION

3.1 PREPARATION

-- End of Section Table of Contents --

SECTION 01 45 00.00 20

QUALITY CONTROL
11/11, CHG 8: 02/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

AMERICAN SOCIETY OF HEATING, REFRIGERATING AND AIR-CONDITIONING
ENGINEERS (ASHRAE)

ASHRAE 52.2 (2012) Method of Testing General
Ventilation Air-Cleaning Devices for
Removal Efficiency by Particle Size

SHEET METAL AND AIR CONDITIONING CONTRACTORS' NATIONAL ASSOCIATION
(SMACNA)

ANSI/SMACNA 008 (2007) IAQ Guidelines for Occupied
Buildings Under Construction, 2nd Edition

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety and Health Requirements
Manual

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES.

SD-01 Preconstruction Submittals

Construction Quality Control (QC) Plan; G

Indoor Air Quality (IAQ) Management Plan; G

1.3 INFORMATION FOR THE CONTRACTING OFFICER

Prior to commencing work on construction, the Contractor can obtain a single copy set of the current report forms from the Contracting Officer. The report forms will consist of the Contractor Production Report, Contractor Production Report (Continuation Sheet), Contractor Quality Control (CQC) Report, CQC Report (Continuation Sheet), Preparatory Phase Checklist, Initial Phase Checklist, Rework Items List, and Testing Plan and Log.

Deliver the following to the Contracting Officer during Construction:

- a. CQC Report: Submit the report electronically or hand-carry the

original (wet signatures) and one copy by 10:00 AM the next working day after each day that work is performed and for every seven consecutive calendar days of no-work.

- b. Contractor Production Report: Submit the report electronically by 10:00 AM the next working day after each day that work is performed and for every seven consecutive calendar days of no-work. Hand-carry the original (wet signatures) and one copy by 10:00 AM the next working day after each day that work is performed and for every seven consecutive calendar days of no-work, attached to the CQC Report.
- c. Preparatory Phase Checklist: Submit the report electronically in the same manner as the CQC Report for each Preparatory Phase held.
- d. Initial Phase Checklist: Submit the report electronically in the same manner as the CQC Report for each Initial Phase held.
- e. Field Test Reports: Within two working days after the test is performed, submit the report as an electronic attachment to the CQC Report.
- f. Monthly Summary Report of Tests: Submit the report as an electronic attachment to the CQC Report at the end of each month.
- g. Testing Plan and Log: Submit the report as an electronic attachment to the CQC Report, at the end of each month. Provide a copy of the final Testing Plan and Log to the preparer of the Operation & Maintenance (O&M) documentation.
- h. Rework Items List: Submit lists containing new entries daily, in the same manner as the CQC Report.
- i. CQC Meeting Minutes: Within two working days after the meeting is held, submit the report as an electronic attachment to the CQC Report.
- j. QC Certifications: As required by the paragraph QC CERTIFICATIONS.
- k. Special Inspection Report: Submit the Special Inspection reports, in the same manner as the CQC Report.

1.4 QC PROGRAM REQUIREMENTS

Establish and maintain a QC program as described in this section. This QC program is a key element in meeting the objectives of NAVFAC Commissioning. The QC program consists of a QC Organization, QC Plan, QC Plan Meeting(s), a Coordination and Mutual Understanding Meeting, QC meetings, three phases of control, submittal review and approval, testing, completion inspections, QC certifications, and documentation necessary to provide materials, equipment, workmanship, fabrication, construction and operations which comply with the requirements of this Contract. The QC program must cover on-site and off-site work and be keyed to the work sequence. No construction work or testing may be performed unless the QC Manager is on the work site. The QC Manager must report to an officer of the firm and not be subordinate to the Project Superintendent or the Project Manager. The QC Manager, Project Superintendent and Project Manager must work together effectively. Although the QC Manager is the primary individual responsible for quality control, all individuals will be held responsible for the quality of work on the job.

1.4.1 Acceptance of the Construction Quality Control (QC) Plan

Acceptance of the QC Plan is required prior to the start of construction. The Contracting Officer reserves the right to require changes in the QC Plan and operations as necessary, including removal of personnel, to ensure the specified quality of work. The Contracting Officer reserves the right to interview any member of the QC organization at any time in order to verify the submitted qualifications. All QC organization personnel are subject to acceptance by the Contracting Officer. The Contracting Officer may require the removal of any individual for non-compliance with quality requirements specified in the Contract.

1.4.2 Preliminary Construction Work Authorized Prior to Acceptance

The only construction work that is authorized to proceed prior to the acceptance of the QC Plan is mobilization of storage and office trailers, temporary utilities, and surveying.

1.4.3 Notification of Changes

Notify the Contracting Officer, in writing, of any proposed changes in the QC Plan or changes to the QC organization personnel, a minimum of 10 work days prior to a proposed change. Proposed changes are subject to acceptance by the Contracting Officer.

1.5 QC ORGANIZATION

1.5.1 QC Manager

1.5.1.1 Duties

In addition to implementing and managing the QC program, the QC Manager may perform the duties of Project Superintendent. The QC Manager is required to attend the partnering meetings, QC Plan Meetings, Coordination and Mutual Understanding Meeting, conduct the QC meetings, perform the three phases of control, perform submittal review and approval, ensure testing is performed and provide QC certifications and documentation required in this Contract. The QC Manager is responsible for managing and coordinating the three phases of control and documentation performed by testing laboratory personnel and any other inspection and testing personnel required by this Contract. The QC Manager is the manager of all QC activities. The QC manager is responsible for notifying the Special Inspector of activities which require their review.

1.5.1.2 Qualifications

An individual with a minimum of 10 years combined experience in the following positions: Project Superintendent, QC Manager, Project Manager, Project Engineer or Construction Manager on similar size and type construction contracts which included the major trades that are part of this Contract. The individual must have at least two years experience as a QC Manager. The individual must be familiar with the requirements of EM 385-1-1, and have experience in the areas of hazard identification, safety compliance.

1.5.2 Construction Quality Management Training

In addition to the above experience and education requirements, the QC Manager must have completed the course entitled "Construction Quality

Management (CQM) for Contractors." If the QC Manager does not have a current certification, they must obtain the CQM for Contractors course certification within 90 days of award. This course is periodically offered by the Naval Facilities Engineering Command and the Army Corps of Engineers. Contact the Contracting Officer for information on the next scheduled class.

1.5.3 Alternate QC Manager Duties and Qualifications

Designate an alternate for the QC Manager at the work site to serve in the event of the designated QC Manager's absence. The period of absence may not exceed two weeks at one time, and not more than 30 workdays during a calendar year. The qualification requirements for the Alternate QC Manager must be the same as for the QC Manager.

1.6 QUALITY CONTROL (QC) PLAN

1.6.1 Construction Quality Control (QC) Plan

Submit a Construction QC Plan prior to start of construction. The Accepted QC plan is required prior to start of construction.

1.6.1.1 Requirements

Provide a Construction QC Plan, prior to start of construction, that includes a table of contents, with major sections identified, with pages numbered sequentially, and that documents the proposed methods and responsibilities for accomplishing quality control during the construction of the project:

- a. QC ORGANIZATION: A chart showing the QC organizational structure.
- b. NAMES AND QUALIFICATIONS: Names and qualifications, in resume format, for each person in the QC organization. Include the CQM for Contractors course certifications for the QC Manager and Alternate QC Manager as required by the paragraphs CONSTRUCTION QUALITY MANAGEMENT TRAINING and ALTERNATE QC MANAGER DUTIES AND QUALIFICATIONS.
- c. DUTIES, RESPONSIBILITY AND AUTHORITY OF QC PERSONNEL: Duties, responsibilities, and authorities of each person in the QC organization.
- d. OUTSIDE ORGANIZATIONS: A listing of outside organizations, such as architectural and consulting engineering firms, that will be employed by the Contractor and a description of the services these firms will provide.
- e. APPOINTMENT LETTERS: Letters signed by an officer of the firm appointing the QC Manager and Alternate QC Manager and stating that they are responsible for implementing and managing the QC program as described in this Contract. Include in this letter the responsibility of the QC Manager and Alternate QC Manager to implement and manage the three phases of control, and their authority to stop work which is not in compliance with the Contract. Letters of direction are to be issued by the QC Manager to the Assistant QC Manager and all other QC Specialists outlining their duties, authorities, and responsibilities. Include copies of the letters in the QC Plan.
- f. SUBMITTAL PROCEDURES AND INITIAL SUBMITTAL REGISTER: Procedures for

reviewing, approving, and managing submittals. Provide the name(s) of the person(s) in the QC organization authorized to review and certify submittals prior to approval. Provide the initial submittal of the Submittal Register as specified in Section 01 33 00 SUBMITTAL PROCEDURES.

- g. TESTING LABORATORY INFORMATION: Testing laboratory information required by the paragraphs ACCREDITATION REQUIREMENTS, as applicable.
- h. TESTING PLAN AND LOG: A Testing Plan and Log that includes the tests required, referenced by the specification paragraph number requiring the test, the frequency, and the person responsible for each test.
- i. PROCEDURES TO COMPLETE REWORK ITEMS: Procedures to identify, record, track, and complete rework items.
- j. LIST OF DEFINABLE FEATURES: A Definable Feature of Work (DFOW) is a task that is separate and distinct from other tasks and has control requirements and work crews unique to that task. A DFOW is identified by different trades or disciplines and is an item or activity on the construction schedule. Include in the list of DFOWs, but not be limited to, all critical path activities on the NAS. Include all activities for which this specification requires QC Specialists or specialty inspection personnel. Provide separate DFOWs in the Network Analysis Schedule for each submittal package.
- k. PROCEDURES FOR PERFORMING THE THREE PHASES OF CONTROL: Identify procedures used to ensure the three phases of control to manage the quality on this project. For each DFOW, a Preparatory and Initial phase checklist will be filled out during the Preparatory and Initial phase meetings. Conduct the Preparatory and Initial Phases and meetings with a view towards obtaining quality construction by planning ahead and identifying potential problems for each DFOW.
- l. PROCEDURES FOR COMPLETION INSPECTION: Procedures for identifying and documenting the completion inspection process. Include in these procedures the responsible party for punch out inspection, pre-final inspection, and final acceptance inspection.
- m. TRAINING PROCEDURES AND TRAINING LOG: Procedures for coordinating and documenting the training of personnel required by the Contract.
- n. ORGANIZATION AND PERSONNEL CERTIFICATIONS LOG: Procedures for coordinating, tracking and documenting all certifications on subcontractors, testing laboratories, suppliers, personnel, etc. QC Manager will ensure that certifications are current, appropriate for the work being performed, and will not lapse during any period of the contract that the work is being performed.

1.7 COORDINATION AND MUTUAL UNDERSTANDING MEETING

After submission of the QC Plan, and prior to Government approval and the start of construction, the QC Manager will meet with the Contracting Officer to present the QC program required by this Contract. When a new QC Manager is appointed, the coordination and mutual understanding meeting must be repeated.

1.7.1 Purpose

The purpose of this meeting is to develop a mutual understanding of the QC details, including documentation, administration for on-site and off-site work, design intent, environmental requirements and procedures, coordination of activities to be performed, and the coordination of the Contractor's management, production, and QC personnel. At the meeting, the Contractor will be required to explain in detail how three phases of control will be implemented for each DFOW, as well as how each DFOW will be affected by each management plan or requirement as listed below:

- a. Waste Management Plan.
- b. Environmental Protection Plan.
- c. Environmental regulatory requirements.

1.7.2 Coordination of Activities

Coordinate activities included in various sections to assure efficient and orderly installation of each component. Coordinate operations included under different sections that are dependent on each other for proper installation and operation. Schedule construction operations with consideration for coordinating any special inspections.

1.7.3 Attendees

As a minimum, the Contractor's personnel required to attend include an officer of the firm, the Project Manager, Project Superintendent, QC Manager, Alternate QC Manager, Environmental Manager, and subcontractor representatives. Each subcontractor who will be assigned QC responsibilities must have a principal of the firm at the meeting. Minutes of the meeting will be prepared by the QC Manager and signed by the Contractor, the A/E and the Contracting Officer. Provide a copy of the signed minutes to all attendees and include in the QC Plan.

1.8 QC MEETINGS

After the start of construction, conduct QC meetings once every two weeks by the QC Manager at the work site with the Project Superintendent, the QC Specialists, and the foremen who are performing the work of the DFOWs. The QC Manager is to prepare the minutes of the meeting and provide a copy to the Contracting Officer within two working days after the meeting. The Contracting Officer may attend these meetings. As a minimum, accomplish the following at each meeting:

- a. Review the minutes of the previous meeting.
- b. Review the schedule and the status of work and rework.
- c. Review the status of submittals.
- d. Review the work to be accomplished in the next two weeks and documentation required.
- e. Resolve QC and production problems (RFI, etc.).
- f. Address items that may require revising the QC Plan.

- g. Review Accident Prevention Plan (APP).
- h. Review environmental requirements and procedures.
- i. Review Waste Management Plan.
- j. Review IAQ Management Plan.
- k. Review Environmental Management Plan.
- l. Review the status of training completion.

1.9 THREE PHASES OF CONTROL

Adequately cover both on-site and off-site work with the Three Phases of Control and include the following for each DFOV.

1.9.1 Preparatory Phase

Notify the Contracting Officer at least two work days in advance of each preparatory phase meeting. The meeting will be conducted by the QC Manager and attended by the Project Superintendent, and the foreman responsible for the DFOV. When the DFOV will be accomplished by a subcontractor, that subcontractor's foreman must attend the preparatory phase meeting. Document the results of the preparatory phase actions in the daily Contractor Quality Control Report and in the Preparatory Phase Checklist. Perform the following prior to beginning work on each DFOV:

- a. Review each paragraph of the applicable specification sections.
- b. Review the Contract drawings.
- c. Verify that field measurements are as indicated on construction and/or shop drawings before confirming product orders, in order to minimize waste due to excessive materials.
- d. Verify that appropriate shop drawings and submittals for materials and equipment have been submitted and approved. Verify receipt of approved factory test results, when required.
- e. Review the testing plan and ensure that provisions have been made to provide the required QC testing.
- f. Examine the work area to ensure that the required preliminary work has been completed.
- hg Coordinate the schedule of product delivery to designated prepared areas in order to minimize site storage time and potential damage to stored materials.
- h. Arrange for the return of shipping/packaging materials, such as wood pallets, where economically feasible.
- i. Examine the required materials, equipment and sample work to ensure that they are on hand and conform to the approved shop drawings and submitted data and are properly stored.
- j. Discuss specific controls used and construction methods, construction tolerances, workmanship standards, and the approach that will be used

to provide quality construction by planning ahead and identifying potential problems for each DFOW.

- k. Review the APP and appropriate Activity Hazard Analysis (AHA) to ensure that applicable safety requirements are met, and that required Safety Data Sheets (SDS) are submitted.

1.9.2 Initial Phase

Notify the Contracting Officer at least two work days in advance of each initial phase. When construction crews are ready to start work on a DFOW, conduct the initial phase with the Project Superintendent, and the foreman responsible for that DFOW. Observe the initial segment of the DFOW to ensure that the work complies with Contract requirements. Document the results of the initial phase in the daily CQC Report and in the Initial Phase Checklist. Repeat the initial phase for each new crew to work on-site, or when acceptable levels of specified quality are not being met. Perform the following for each DFOW:

- a. Establish level of workmanship and verify that it meets the minimum acceptable workmanship standards. Compare with required sample panels as appropriate.
- b. Resolve any workmanship issues.
- c. Ensure that testing is performed by the approved laboratory.
- d. Check work procedures for compliance with the APP and the appropriate AHA to ensure that applicable safety requirements are met.
- e. Review project specific work plans (i.e. Cx, HAZMAT Abatement, Stormwater Management) to ensure all preparatory work items have been completed and documented.

1.9.3 Follow-Up Phase

Perform the following for on-going work daily, or more frequently as necessary, until the completion of each DFOW and document in the daily CQC Report:

- a. Ensure the work is in compliance with Contract requirements.
- b. Maintain the quality of workmanship required.
- c. Ensure that testing is performed by the approved laboratory.
- d. Ensure that rework items are being corrected.
- e. Assure manufacturers representatives have performed necessary inspections if required and perform safety inspections.

1.9.4 Additional Preparatory and Initial Phases

Conduct additional preparatory and initial phases on the same DFOW if the quality of on-going work is unacceptable, if there are changes in the applicable QC organization, if there are changes in the on-site production supervision or work crew, if work on a DFOW is resumed after substantial period of inactivity, or if other problems develop.

1.9.5 Notification of Three Phases of Control for Off-Site Work

Notify the Contracting Officer at least two weeks prior to the start of the preparatory and initial phases.

1.10 SUBMITTAL REVIEW AND APPROVAL

Procedures for submission, review and approval of submittals are described in Section 01 33 00 SUBMITTAL PROCEDURES.

1.11 TESTING

Except as stated otherwise in the specification sections, perform sampling and testing required under this Contract.

1.11.1 Accreditation Requirements

Construction materials testing laboratories must be accredited by a laboratory accreditation authority and will be required to submit a copy of the Certificate of Accreditation and Scope of Accreditation. The laboratory's scope of accreditation must include the appropriate ASTM standards (E 329, C 1077, D 3666, D 3740, E 543) listed in the technical sections of the specifications. Laboratories engaged in Hazardous Materials Testing must meet the requirements of OSHA and EPA. The policy applies to the specific laboratory performing the actual testing, not just the Corporate Office.

1.11.2 Laboratory Accreditation Authorities

Laboratory Accreditation Authorities include the National Voluntary Laboratory Accreditation Program (NVLAP) administered by the National Institute of Standards and Technology at <https://www.nist.gov/nvlap>, the American Association of State Highway and Transportation Officials (AASHTO) Accreditation Program at <http://www.aashtoresource.org/aap/overview>, International Accreditation Services, Inc. (IAS) at <http://www.iasonline.org>, U.S. Army Corps of Engineers Materials Testing Center (MTC) at <http://www.erdc.usace.army.mil/Media/FactSheets/FactSheetArticleView/tabid/9254/Article/476661/materials-testing-center.aspx>, the American Association for Laboratory Accreditation (A2LA) program at <http://www.a2la.org/>, the Washington Association of Building Officials (WABO) at <http://www.wabo.org/> (Approval authority for WABO is limited to projects within Washington State), and the Washington Area Council of Engineering Laboratories (WACEL) at <https://www.wacel.org/lab-accreditation-and-inspection-agency-audit-programs/laboratory-accreditation-program/> (Approval authority by WACEL is limited to projects within Facilities Engineering Command (FEC) Washington geographical area).

1.11.3 Capability Check

The Contracting Officer retains the right to check laboratory equipment in the proposed laboratory and the laboratory technician's testing procedures, techniques, and other items pertinent to testing, for compliance with the standards set forth in this Contract.

1.11.4 Test Results

Cite applicable Contract requirements, tests or analytical procedures

used. Provide actual results and include a statement that the item tested or analyzed conforms or fails to conform to specified requirements. If the item fails to conform, notify the Contracting Officer immediately. Conspicuously stamp the cover sheet for each report in large red letters "CONFORMS" or "DOES NOT CONFORM" to the specification requirements, whichever is applicable. Test results must be signed by a testing laboratory representative authorized to sign certified test reports. Furnish the signed reports, certifications, and other documentation to the Contracting Officer via the QC Manager. Furnish a summary report of field tests at the end of each month, in accordance with paragraph INFORMATION FOR THE CONTRACTING OFFICER.

1.11.5 Test Reports and Monthly Summary Report of Tests

Furnish the signed reports, certifications, and a summary report of field tests at the end of each month to the Contracting Officer. Attach a copy of the summary report to the last daily Contractor Quality Control Report of each month.

1.12 QC CERTIFICATIONS

1.12.1 CQC Report Certification

Contain the following statement within the CQC Report: "On behalf of the Contractor, I certify that this report is complete and correct and equipment and material used and work performed during this reporting period is in compliance with the contract drawings and specifications to the best of my knowledge, except as noted in this report."

1.12.2 Invoice Certification

Furnish a certificate to the Contracting Officer with each payment request, signed by the QC Manager, attesting that as-built drawings are current, coordinated and attesting that the work for which payment is requested, including stored material, is in compliance with Contract requirements.

1.12.3 Completion Certification

Upon completion of work under this Contract, the QC Manager must furnish a certificate to the Contracting Officer attesting that "the work has been completed, inspected, tested and is in compliance with the Contract." Provide a copy of this final QC Certification for completion to the preparer of the Operation & Maintenance (O&M) documentation.

1.13 COMPLETION INSPECTIONS

1.13.1 Punch-Out Inspection

Near the completion of all work or any increment thereof, established by a completion time stated in the Contract Clause entitled "Commencement, Prosecution, and Completion of Work," or stated elsewhere in the specifications, the QC Manager must conduct an inspection of the work and develop a "punch list" of items which do not conform to the approved drawings, specifications and Contract. Include in the punch list any remaining items on the "Rework Items List", which were not corrected prior to the Punch-Out Inspection. Include within the punch list the estimated date by which the deficiencies will be corrected. Provide a copy of the punch list to the Contracting Officer. The QC Manager, or staff, must

make follow-on inspections to ascertain that all deficiencies have been corrected. Once this is accomplished, notify the Government that the facility is ready for the Government "Pre-Final Inspection".

1.13.2 Pre-Final Inspection

The Government and QC Manager will perform this inspection to verify that the facility is complete and ready to be occupied. A Government "Pre-Final Punch List" will be documented by the QC Manager as a result of this inspection. The QC Manager will ensure that all items on this list are corrected prior to notifying the Government that a "Final" inspection with the Client can be scheduled. Any items noted on the "Pre-Final" inspection must be corrected in a timely manner and be accomplished before the contract completion date for the work, or any particular increment thereof, if the project is divided into increments by separate completion dates.

1.13.3 Final Acceptance Inspection

Notify the Contracting Officer at least 14 calendar days prior to the date a final acceptance inspection can be held. State within the notice that all items previously identified on the pre-final punch list will be corrected and acceptable, along with any other unfinished Contract work, by the date of the final acceptance inspection. The Contractor must be represented by the QC Manager, the Project Superintendent and others deemed necessary. Attendees for the Government will include the Contracting Officer, other FEAD/ROICC personnel, and personnel representing the Client. Failure of the Contractor to have all contract work acceptably complete for this inspection will be cause for the Contracting Officer to bill the Contractor for the Government's additional inspection cost in accordance with the Contract Clause entitled "Inspection of Construction."

1.14 DOCUMENTATION

Maintain current and complete records of on-site and off-site QC program operations and activities.

1.14.1 Construction Documentation

Reports are required for each day that work is performed and must be attached to the Contractor Quality Control Report prepared for the same day. Maintain current and complete records of on-site and off-site QC program operations and activities. The forms identified under the paragraph "INFORMATION FOR THE CONTRACTING OFFICER" will be used. Reports are required for each day work is performed. Account for each calendar day throughout the life of the Contract. Every space on the forms must be filled in. Use N/A if nothing can be reported in one of the spaces. The Project Superintendent and the QC Manager must prepare and sign the Contractor Production and CQC Reports, respectively. The reporting of work must be identified by terminology consistent with the construction schedule. In the "remarks" sections of the reports, enter pertinent information including directions received, problems encountered during construction, work progress and delays, conflicts or errors in the drawings or specifications, field changes, safety hazards encountered, instructions given and corrective actions taken, delays encountered and a record of visitors to the work site, quality control problem areas, deviations from the QC Plan, construction deficiencies encountered, meetings held. For each entry in the report(s), identify the Schedule

Activity No. that is associated with the entered remark.

1.14.2 Quality Control Validation

Establish and maintain the following in an electronic folder. Divide folder into a series of tabbed sections as shown below. Ensure folder is updated at each required progress meeting.

- a. All completed Preparatory and Initial Phase Checklists, arranged by specification section.
- b. All milestone inspections, arranged by Activity Number.
- c. An up-to-date copy of the Testing Plan and Log with supporting field test reports, arranged by specification section.
- d. Copies of all contract modifications, arranged in numerical order. Also include documentation that modified work was accomplished.
- e. An up-to-date copy of the Rework Items List.
- f. Maintain up-to-date copies of all punch lists issued by the QC staff to the Contractor and Sub-Contractors and all punch lists issued by the Government.

1.14.3 Testing Plan and Log

As tests are performed, the QC Manager will record on the "Testing Plan and Log" the date the test was performed and the date the test results were forwarded to the Contracting Officer. Attach a copy of the updated "Testing Plan and Log" to the last daily CQC Report of each month, per the paragraph "INFORMATION FOR THE CONTRACTING OFFICER". Provide a copy of the final "Testing Plan and Log" to the preparer of the Operation & Maintenance (O&M) documentation.

1.14.4 Rework Items List

The QC Manager must maintain a list of work that does not comply with the Contract, identifying what items need to be reworked, the date the item was originally discovered, the date the item will be corrected by, and the date the item was corrected. There is no requirement to report a rework item that is corrected the same day it is discovered. Attach a copy of the "Rework Items List" to the last daily CQC Report of each month. The Contractor is responsible for including those items identified by the Contracting Officer.

1.14.5 As-Built Drawings

The QC Manager is required to ensure the as-built drawings, required by Section 01 78 00 CLOSEOUT SUBMITTALS are kept current on a daily basis and marked to show deviations which have been made from the Contract drawings. Ensure each deviation has been identified with the appropriate modifying documentation (e.g. PC No., Modification No., Request for Information No., etc.). The QC Manager or QC Specialist assigned to an area of responsibility must initial each revision. Upon completion of work, the QC Manager will furnish a certificate attesting to the accuracy of the as-built drawings prior to submission to the Contracting Officer.

1.15 NOTIFICATION ON NON-COMPLIANCE

The Contracting Officer will notify the Contractor of any detected non-compliance with the Contract. Take immediate corrective action after receipt of such notice. Such notice, when delivered to the Contractor at the work site, is deemed sufficient for the purpose of notification. If the Contractor fails or refuses to comply promptly, the Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No part of the time lost due to such stop orders will be made the subject of claim for extension of time for excess costs or damages by the Contractor.

1.16 CONSTRUCTION INDOOR AIR QUALITY (IAQ) MANAGEMENT PLAN

1.16.1 Requirements During Construction

1.16.1.1 Control Measures

Meet or exceed the requirements of ANSI/SMACNA 008, Chapter 3, to help minimize contamination of the building from construction activities. The five requirements of this manual which must be adhered to are described below:

- a. HVAC protection: Isolate return side of HVAC system from surrounding environment to prevent construction dust and debris from entering the duct work and spaces.
- b. Source control: Use low emitting paints and other finishes, sealants, adhesives, and other materials as specified. When available, cleaning products must have a low VOC content and be non-toxic to minimize building contamination. Utilize cleaning techniques that minimize dust generation. Cycle equipment off when not needed. Prohibit idling motor vehicles where emissions could be drawn into building. Designate receiving/storage areas for incoming material that minimize IAQ impacts.
- c. Pathway interruption: When pollutants are generated use strategies such as 100 percent outside air ventilation or erection of physical barriers between work and non-work areas to prevent contamination.
- d. Housekeeping: Clean frequently to remove construction dust and debris. Promptly clean up spills. Remove accumulated water and keep work areas dry to discourage the growth of mold and bacteria. Take extra measures when hazardous materials are involved.
- e. Scheduling: Control the sequence of construction to minimize the absorption of VOCs by other building materials.

1.16.1.2 Moisture Contamination

- a. Remove accumulated water and keep work dry.
- b. Use dehumidification to remove moist, humid air from a work area.
- c. Do not use combustion heaters or generators inside the building.
- d. Protect porous materials from exposure to moisture.
- e. Remove and replace items which remain damp for more than a few hours.

1.16.2 Requirements after Construction

After construction ends and prior to occupancy, conduct a building flush-out or test the indoor air contaminant levels. Flush-out must be a minimum two-weeks with MERV-13 filtration media as determined by ASHRAE 52.2 at 100 percent outside air. Air contamination testing must be consistent with EPA's current Compendium of Methods for the Determination of Air Pollutants in Indoor Air. After building flush-out or testing and prior to occupancy, replace filtration media. Filtration media must have a MERV of 13 as determined by ASHRAE 52.2.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 PREPARATION

Designate receiving/storage areas for incoming material to be delivered according to installation schedule and to be placed convenient to work area in order to minimize waste due to excessive materials handling and misapplication. Store and handle materials in a manner as to prevent loss from weather and other damage. Keep materials, products, and accessories covered and off the ground, and store in a dry, secure area. Prevent contact with material that may cause corrosion, discoloration, or staining. Protect all materials and installations from damage by the activities of other trades.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 50 00

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS

11/20, CHG 2: 08/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 CONSTRUCTION SITE PLAN
- 1.4 DOD CONDITION OF READINESS (COR)
- 1.5 CYBERSECURITY DURING CONSTRUCTION
 - 1.5.1 Contractor Computer Equipment
 - 1.5.1.1 Operating System
 - 1.5.1.2 Anti-Malware Software
 - 1.5.1.3 Passwords and Passphrases
 - 1.5.1.4 Contractor Computer Cybersecurity Compliance Statements
 - 1.5.2 Temporary IP Networks
 - 1.5.2.1 Network Boundaries and Connections
 - 1.5.3 Government Access to Network
 - 1.5.4 Temporary Wireless IP Networks
 - 1.5.5 Passwords and Passphrases
 - 1.5.6 Contractor Temporary Network Cybersecurity Compliance Statements

PART 2 PRODUCTS

- 2.1 TEMPORARY SIGNAGE
 - 2.1.1 Bulletin Board
 - 2.1.2 Project Identification Signs
 - 2.1.3 Warning Signs
- 2.2 TEMPORARY TRAFFIC CONTROL
 - 2.2.1 Haul Roads
 - 2.2.2 Barricades
- 2.3 FENCING
 - 2.3.1 Polyethylene Mesh Safety Fencing
 - 2.3.2 Chain Link Panel Fencing
 - 2.3.3 Post-Driven Chain Link Fencing
- 2.4 TEMPORARY WIRING

PART 3 EXECUTION

- 3.1 EMPLOYEE PARKING
- 3.2 AVAILABILITY AND USE OF UTILITY SERVICES
 - 3.2.1 Temporary Utilities
 - 3.2.2 Payment for Utility Services
 - 3.2.3 Meters and Temporary Connections
 - 3.2.4 Advance Deposit
 - 3.2.5 Final Meter Reading
 - 3.2.6 Utilities at Special Locations
 - 3.2.7 Electricity

- 3.2.8 Water
- 3.2.9 Sanitation
- 3.2.10 Telephone
- 3.2.11 Fire Protection
- 3.3 STATION OPERATION AFFECT ON CONTRACTOR OPERATIONS
- 3.3.1 Restricted Access Areas
- 3.4 TRAFFIC PROVISIONS
- 3.4.1 Maintenance of Traffic
- 3.4.2 Protection of Traffic
- 3.4.3 Rush Hour Restrictions
- 3.4.4 Dust Control
- 3.5 CONTRACTOR'S TEMPORARY FACILITIES
- 3.5.1 Administrative Field Offices
- 3.5.2 Quality Control Manager Records and Field Office
- 3.5.3 Storage Area
- 3.5.4 Supplemental Storage Area
- 3.5.5 Appearance of Trailers
- 3.5.6 Trailers or Storage Buildings
- 3.5.7 Safety Systems
- 3.5.8 Special Storage Requirements
- 3.5.8.1 Storage Size and Location
- 3.5.9 Weather Protection of Temporary Facilities and Stored Materials
- 3.5.9.1 Building and Site Storm Protection
- 3.6 PLANT COMMUNICATIONS
- 3.7 TEMPORARY PROJECT SAFETY FENCING
- 3.8 DUMPSTERS
- 3.9 CLEANUP
- 3.10 RESTORATION OF STORAGE AREA

-- End of Section Table of Contents --

SECTION 01 50 00

TEMPORARY CONSTRUCTION FACILITIES AND CONTROLS
11/20, CHG 2: 08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; ERTA 20-3 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4; TIA 20-5; TIA 20-6; TIA 20-7; TIA 20-8; TIA 20-9; TIA 20-10; TIA 20-11; TIA 20-12; TIA 20-13; TIA 20-14; TIA 20-15; TIA 20-16; ERTA 20-4 2022)
National Electrical Code

NFPA 241 (2022) Standard for Safeguarding Construction, Alteration, and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety -- Safety and Health Requirements Manual

U.S. FEDERAL HIGHWAY ADMINISTRATION (FHWA)

MUTCD (2009; Rev 2012) Manual on Uniform Traffic Control Devices

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Site Plan; G

Traffic Control Plan; G

Haul Road Plan; G

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

1.3 CONSTRUCTION SITE PLAN

Prior to the start of work, submit for Government approval a site plan showing the locations and dimensions of temporary facilities (including layouts and details, equipment and material storage area (onsite and offsite), and access and haul routes, avenues of ingress/egress to the fenced area and details of the fence installation. Identify any areas which may have to be graveled to prevent the tracking of mud. Indicate if the use of a supplemental or other staging area is desired. Show locations of safety and construction fences, site trailers, construction entrances, trash dumpsters, temporary sanitary facilities, and worker parking areas.

1.4 DOD CONDITION OF READINESS (COR)

DOD will set the Condition of Readiness (COR) based on the weather forecast for sustained winds 50 knots (93 km/hr) or greater. Contact the Contracting Officer for the current COR setting.

Monitor weather conditions a minimum of twice a day and take appropriate actions according to the approved Emergency Plan in the accepted Accident Prevention Plan, EM 385-1-1 Section 01 Emergency Planning and the instructions below.

Unless otherwise directed by the Contracting Officer, comply with:

- a. Condition FOUR (Sustained winds of 93 km/hr or greater expected within 72 hours): Normal daily jobsite cleanup and good housekeeping practices. Collect and store in piles or containers scrap lumber, waste material, and rubbish for removal and disposal at the close of each work day. Maintain the construction site including storage areas, free of accumulation of debris. Stack form lumber in neat piles less than one m high. Remove all debris, trash, or objects that could become missile hazards. Review requirements pertaining to "Condition THREE" and continue action as necessary to attain "Condition FOUR" readiness. Contact Contracting Officer for weather and COR updates and completion of required actions.
- b. Condition THREE (Sustained winds of 93 km/hr or greater expected within 48 hours): Maintain "Condition FOUR" requirements and commence securing operations necessary for "Condition ONE" which cannot be completed within 18 hours. Cease all routine activities which might interfere with securing operations. Commence securing and stow all gear and portable equipment. Make preparations for securing buildings. Reinforce or remove formwork and scaffolding. Secure machinery, tools, equipment, materials, or remove from the jobsite. Expend every effort to clear all missile hazards and loose equipment from general base areas. Contact Contracting Officer for weather and COR updates and completion of required actions. Review requirements pertaining to "Condition TWO" and continue action as necessary to attain "Condition THREE" readiness.
- c. Condition TWO (Sustained winds of 93 km/hr or greater expected within 24 hours): Secure the jobsite, and leave Government premises.
- d. Condition ONE. (Sustained winds of 93 km/hr or greater expected within 12 hours): Contractor access to the jobsite and Government premises is prohibited.

1.5 CYBERSECURITY DURING CONSTRUCTION

{For Reference Only: This subpart (and its subparts) relates to AC-18, SA-3, CCI-00258.} Meet the following requirements throughout the construction process.

1.5.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. When used, contractor computers must meet the following requirements:

1.5.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.5.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. All computers used on this project must be scanned using the installed software at least once per day.

1.5.1.3 Passwords and Passphrases

The passwords and passphrases for all computers must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.5.1.4 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Each Statement must be signed by a cybersecurity representative for the relevant company.

1.5.2 Temporary IP Networks

Temporary contractor-installed IP networks may be used during construction. When used, temporary contractor-installed IP networks must meet the following requirements:

1.5.2.1 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than IP networks provided under this project or Government furnished IP networks provided for this purpose. Any and all network access from outside the project site is prohibited.

1.5.3 Government Access to Network

Government personnel, as defined, prescribed, and identified by the Contracting Officer, must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification. Or if there is a Government agency that's responsible, identify that agency.

1.5.4 Temporary Wireless IP Networks

In addition to the other requirements on temporary IP networks, temporary wireless IP (WiFi) networks must not interfere with existing wireless network and must use WPA2 security. Network names (SSID) for wireless networks must be changed from their default values.

1.5.5 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.5.6 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

PART 2 PRODUCTS

2.1 TEMPORARY SIGNAGE

2.1.1 Bulletin Board

Prior to the commencement of work activities, provide a clear weatherproof covered bulletin board not less than 915 by 1220 mm in size for displaying the Equal Employment Opportunity poster, a copy of the wage decision contained in the Contract, Wage Rate Information poster, Safety and Health Information as required by EM 385-1-1 Section 01 and other information approved by the Contracting Officer. Coordinate requirements herein with 01 35 26 GOVERNMENTAL SAFETY REQUIREMENTS. Locate the bulletin board at the project site in a conspicuous place easily accessible to all employees, and in location as approved by the Contracting Officer.

2.1.2 Project Identification Signs

The requirements for the signs, their content, and location are as specified in Section 01 58 00 PROJECT IDENTIFICATION. Erect signs within 15 days after receipt of the notice to proceed. Correct the data required by the safety sign daily, with light colored metallic or non-metallic numerals.

2.1.3 Warning Signs

Post temporary signs, tags, and labels to give workers and the public adequate warning and caution of construction hazards according to the EM 385-1-1 Section 04. Attach signs to the perimeter fencing every 150 feet warning the public of the presence of construction hazards. Signs must require unauthorized persons to keep out of the construction site. Correct the data required by safety signs daily. Post signs at all points of entry designating the construction site as a hard hat area.

2.2 TEMPORARY TRAFFIC CONTROL

2.2.1 Haul Roads

Construct access and haul roads necessary for proper prosecution of the work under this Contract in accordance with EM 385-1-1 Section 04. Construct with suitable grades and widths; avoid sharp curves, blind corners, and dangerous cross traffic. Submit haul road plan for approval. Provide necessary lighting, signs, barricades, and distinctive markings for the safe movement of traffic. The method of dust control, although optional, must be adequate to ensure safe operation at all times. Location, grade, width, and alignment of construction and haul roads are subject to approval by the Contracting Officer. Lighting must be adequate to assure full and clear visibility for full width of haul road and work areas during any night work operations.

2.2.2 Barricades

Erect and maintain temporary barricades to limit public access to hazardous areas. Barricades are required whenever safe public access to paved areas such as roads, parking areas or sidewalks is prevented by construction activities or as otherwise necessary to ensure the safety of both pedestrian and vehicular traffic. Securely place barricades clearly visible with adequate illumination to provide sufficient visual warning of the hazard during both day and night.

2.3 FENCING

Provide fencing along the construction site and at all open excavations and tunnels to control access by unauthorized personnel. Safety fencing must be highly visible to be seen by pedestrians and vehicular traffic. All fencing must meet the requirements of EM 385-1-1. Remove the fence upon completion and acceptance of the work.

2.3.1 Polyethylene Mesh Safety Fencing

Temporary safety fencing must be a high visibility orange colored, high density polyethylene grid, a minimum of 1.2 m high and maximum mesh size of 50 mm. Fencing must extend from the grade to a minimum of 1.2 m above the grade and be tightly secured to T-posts spaced as necessary to maintain a rigid and taut fence. Fencing must remain rigid and taut with a minimum of 90.7 m of force exerted on it from any direction with less than 100 mm of deflection.

2.3.2 Chain Link Panel Fencing

Temporary panel fencing must be galvanized steel chain link panels 1.8 m high. Multiple fencing panels may be linked together at the bases to form long spans as needed. Each panel base must be weighted down using sand

bags or other suitable materials in order for the fencing to withstand anticipated winds while remaining upright. Fencing must remain rigid and taut with a minimum of 90.7 kg of force exerted on it from any direction with less than 100 mm of deflection.

2.3.3 Post-Driven Chain Link Fencing

Temporary post-driven fencing must be galvanized chain link fencing 1.8 m high supported by an tightly secured to galvanized steel posts driven below grade. Fence posts must be located on minimum 3 meter centers. Posts may be set in various surfaces such as sand, soil, asphalt or concrete as necessary. Chain link fencing must remain rigid and taut with a minimum of 90.7 kg of force exerted on it from any direction with less than 100 mm of deflection. Completely remove fencing and posts at the completion of construction and restore surfaces disturbed or damaged to its original condition. Locate and identify underground utilities prior to setting fence posts. Equip fence with a lockable gate. Gate must remain locked when construction personnel are not present.

2.4 TEMPORARY WIRING

Provide temporary wiring in accordance with EM 385-1-1 Section 11, NFPA 241 and NFPA 70. Include monthly inspection and testing of all equipment and apparatus.

PART 3 EXECUTION

3.1 EMPLOYEE PARKING

Construction Contract employees must park privately owned vehicles in an area designated by the Contracting Officer. Employee parking must not interfere with existing and established parking requirements of the Government installation.

3.2 AVAILABILITY AND USE OF UTILITY SERVICES

3.2.1 Temporary Utilities

Provide temporary utilities required for construction. Materials may be new or used, must be adequate for the required usage, not create unsafe conditions, and not violate applicable codes and standards.

3.2.2 Payment for Utility Services

- a. The Government will make all reasonably required utilities available from existing outlets and supplies, as specified in the Contract. Unless otherwise provided in the Contract, the amount of each utility service consumed will be charged to or paid at prevailing rates charged to the Government or, where the utility is produced by the Government, at reasonable rates determined by the Contracting Officer. Carefully conserve utilities furnished without charge.
- b. Reasonable amounts of the following utilities will be made available without charge.

Utility Services		
	Cost (\$) per	Unit
Electricity		
Potable Water		
Salt Water		
Compressed Air		
Steam		
Natural Gas		
Sanitary Sewer		

- c. The point at which the Government will deliver such utilities or services and the quantity available is must be coordinated with the Contracting Officer. Pay all costs incurred in connecting, converting, and transferring the utilities to the work. Make connections, including providing meters; and providing transformers; and make disconnections. Under no circumstances will taps to base fire hydrants be allowed for obtaining domestic water.

3.2.3 Meters and Temporary Connections

Provide and maintain necessary temporary connections, distribution lines, and meter bases (Government will provide meters) required to measure the amount of each utility used for the purpose of determining charges. Notify the Contracting Officer, in writing, 5 working days before final electrical connection is desired so that a utilities contract can be established. The Government will provide a meter and make the final hot connection after inspection and approval of the Contractor's temporary wiring installation. Do not make the final electrical connection.

3.2.4 Advance Deposit

An advance deposit for utilities consisting of an estimated month's usage or a minimum of \$50.00 will be required. The last monthly bills for the fiscal year will normally be offset by the deposit and adjustments will be billed or returned as appropriate. Services to be rendered for the next fiscal year, beginning 1 October, will require a new deposit. Notification of the due date for this deposit will be mailed prior to the end of the current fiscal year.

3.2.5 Final Meter Reading

Before completion of the work and final acceptance of the work by the Government, notify the Contracting Officer, in writing, 5 working days before termination is desired. The Government will take a final meter reading, disconnect service, and remove the meters. Then remove all the temporary distribution lines, meter bases, and associated appurtenances. Pay all outstanding utility bills before final acceptance of the work by the Government.

3.2.6 Utilities at Special Locations

- a. Reasonable amounts of utilities will be made available at the

prevailing Government rates. These rates may be obtained upon application to the Commanding Officer, by way of the Contracting Officer. Make connections, provide transformers and meters, and make disconnections; and provide backflow preventer devices on connections to domestic water lines.

- b. Reasonable amounts of utilities will be made available without charge. Make connections, provide transformers and meters, and make disconnections; and provide backflow preventer devices on connections to domestic water lines. Under no circumstances will taps to base fire hydrants be allowed for obtaining domestic water.

3.2.7 Electricity

Provide connections, sized to provide service required for power and lighting. Locate feeder and branch wiring with area distribution boxes so that power is available throughout the project site by use of power cords. Provide lighting as required for safe and secure operations. Electricity used will be furnished by the Government.

3.2.8 Water

Make connections to existing facilities to provide water for construction purposes. Water used will be furnished by the Government.

3.2.9 Sanitation

Provide and maintain within the construction area minimum field-type sanitary facilities in accordance with EM 385-1-1 Section 02. Locate the facilities behind the construction fence or out of the public view. Clean units and empty wastes at least once a week or more frequently into a municipal, district, or station sanitary sewage system, or remove waste to a commercial facility. Obtain approval from the system owner prior to discharge into a municipal, district, or commercial sanitary sewer system. Penalties or fines associated with improper discharge will be the responsibility of the Contractor. Coordinate with the Contracting Officer and follow station regulations and procedures when discharging into the station sanitary sewer system. Maintain these conveniences at all times. Include provisions for pest control and elimination of odors. Government toilet facilities will not be available to Contractor's personnel.

3.2.10 Telephone

Make arrangements and pay all costs for telephone facilities desired.

3.2.11 Fire Protection

Provide temporary fire protection equipment for the protection of personnel and property during construction. Remove debris and flammable materials daily to minimize potential hazards.

3.3 STATION OPERATION AFFECT ON CONTRACTOR OPERATIONS

3.3.1 Restricted Access Areas

The Government will monitor work in areas listed below. Notify Contracting Officer at least 14 calendar days prior to starting work in these areas.

- a. Pier 3 and related access bridge to Pier 3.

3.4 TRAFFIC PROVISIONS

3.4.1 Maintenance of Traffic

- a. Conduct operations in a manner that will not close a thoroughfare or interfere with traffic on railways or highways except with written permission of the Contracting Officer at least 15 calendar days prior to the proposed modification date, and provide a Traffic Control Plan for Government approval detailing the proposed controls to traffic movement for approval. The plan must be in accordance with State and local regulations and the MUTCD, Part VI. Make all notifications and obtain all permits required for modification to traffic movements outside Station's jurisdiction. Contractor may move oversized and slow-moving vehicles to the worksite provided requirements of the highway authority have been met.
- b. Conduct work so as to minimize obstruction of traffic, and maintain traffic on at least half of the roadway width at all times. Obtain approval from the Contracting Officer prior to starting any activity that will obstruct traffic.
- c. Provide, erect, and maintain, at Contractor's expense, lights, barriers, signals, passageways, detours, and other items, that may be required by the Life Safety Signage, overhead protection authority having jurisdiction.
- d. Provide cones, signs, barricades, lights, or other traffic control devices and personnel required to control traffic. Do not use foil-backed material for temporary pavement marking because of its potential to conduct electricity during accidents involving downed power lines.

3.4.2 Protection of Traffic

Maintain and protect traffic on all affected roads during the construction period except as otherwise specifically directed by the Contracting Officer. Measures for the protection and diversion of traffic, including the provision of watchmen and flagmen, erection of barricades, placing of lights around and in front of equipment the work, and the erection and maintenance of adequate warning, danger, and direction signs, will be as required by the State and local authorities having jurisdiction. Provide self-illuminated (lighted) barricades during hours of darkness. Brightly-colored (orange) vests are required for all personnel working in roadways. Protect the traveling public from damage to person and property. Minimize the interference with public traffic on roads selected for hauling material to and from the site. Investigate the adequacy of existing roads and their allowable load limit. Contractor is responsible for the repair of damage to roads caused by construction operations.

3.4.3 Rush Hour Restrictions

Do not interfere with the peak traffic flows preceding and during normal operations without notification to and approval by the Contracting Officer.

3.4.4 Dust Control

Dust control methods and procedures must be approved by the Contracting

Officer.

3.5 CONTRACTOR'S TEMPORARY FACILITIES

Contractor is responsible for security of their property. Provide adequate outside security lighting at the temporary facilities. Trailers must be anchored to resist high winds and meet applicable state or local standards for anchoring mobile trailers. Coordinate anchoring with EM 385-1-1 Section 04. The Contract Clause entitled "FAR 52.236-10, Operations and Storage Areas" and the following apply:

3.5.1 Administrative Field Offices

Provide and maintain administrative field office facilities within the construction area at the designated site. Government office and warehouse facilities will not be available to the Contractor's personnel.

3.5.2 Quality Control Manager Records and Field Office

Provide on the jobsite an office with approximately 65 square meter of useful floor area for the exclusive use of the QC Manager. Provide a weathertight structure with adequate heating and cooling, toilet facilities, lighting, ventilation, a 1200 by 2400 mm plan table, a standard size office desk and chair, computer station, and working communications facilities. Provide either a 1,500 watt radiant heater and a window-mounted air conditioner rated at 2.6 kW minimum or a window-mounted heat pump of the same minimum heating and cooling ratings. Provide a door with a cylinder lock and windows with locking hardware. Make utility connections. Locate as directed. File quality control records in the office and make available at all times to the Government. After completion of the work, remove the entire structure from the site.

3.5.3 Storage Area

Construct a temporary 1.8 m high chain link fence around trailers and materials. Include plastic strip inserts, colored brown, so that visibility through the fence is obstructed. Fence posts may be driven, in lieu of concrete bases, where soil conditions permit. Do not place or store trailers, materials, or equipment outside the fenced area unless such trailers, materials, or equipment are assigned a separate and distinct storage area by the Contracting Officer away from the vicinity of the construction site but within the installation boundaries. Trailers, equipment, or materials must not be open to public view with the exception of those items which are in support of ongoing work on the current day. Do not stockpile materials outside the fence in preparation for the next day's work. Park mobile equipment, such as tractors, wheeled lifting equipment, cranes, trucks, and like equipment within the fenced area at the end of each work day.

Keep fencing in a state of good repair and proper alignment. If the Contractor elects to traverse grassed or unpaved areas which are not established roadways with construction equipment or other vehicles, cover the grassed or unpaved areas with a layer of gravel as necessary to prevent rutting and the tracking of mud onto paved or established roadways; gravel gradation must be at the Contractor's discretion. Mow and maintain grass located within the boundaries of the construction site for the duration of the project. Grass and vegetation along fences, structures, under trailers, and in areas not accessible to mowers must be

edged or trimmed neatly.

3.5.4 Supplemental Storage Area

Upon request, and pending availability, the Contracting Officer will designate another or supplemental area for the use and storage of trailers, equipment, and materials. This area may not be in close proximity of the construction site but will be within the installation boundaries. Maintain the area in a clean and orderly fashion and secured if needed to protect supplies and equipment. Utilities will not be provided to this area by the Government.

3.5.5 Appearance of Trailers

- a. Trailers must be roadworthy and comply with all appropriate state and local vehicle requirements. Trailers which are rusted, have peeling paint or are otherwise in need of repair will not be allowed on Installation property. Trailers must present a clean and neat exterior appearance and be in a state of good repair.
- b. Maintain the temporary facilities. Failure to do so will be sufficient reason to require their removal at the Contractor's expense.

3.5.6 Trailers or Storage Buildings

- a. Trailers or storage buildings will be permitted, where space is available, subject to the approval of the Contracting Officer.
- b. Mount a sign not smaller than 600 by 600 mm on the trailer or building that shows the company name, business phone number, emergency phone number and conforms to the following requirements and sketch:

Graphic panel	Aluminum, painted blue
Copy	Screen painted or vinyl die-cut, white
Typeface	Univers 65 u/lc
See Sketch No. 01500 (graphic).	

3.5.7 Safety Systems

Protect the integrity of all installed safety systems or personnel safety devices. Obtain prior approval from the Contracting Officer if entrance into systems serving safety devices is required. If it is temporarily necessary to remove or disable personnel safety devices in order to accomplish Contract requirements, provide alternative means of protection prior to removing or disabling any permanently installed safety devices or equipment and obtain approval from the Contracting Officer.

3.5.8 Special Storage Requirements

The following special storage requirements apply:

3.5.8.1 Storage Size and Location

The open site available for storage must be within 300 m of the operations.

3.5.9 Weather Protection of Temporary Facilities and Stored Materials

Take necessary precautions to ensure that roof openings and other critical openings in the building are monitored carefully. Take immediate actions required to seal off such openings when rain or other detrimental weather is imminent, and at the end of each workday. Ensure that the openings are completely sealed off to protect materials and equipment in the building from damage.

3.5.9.1 Building and Site Storm Protection

When a warning of gale force winds is issued, take precautions to minimize danger to persons, and protect the work and nearby Government property. Precautions must include, but are not limited to, closing openings; removing loose materials, tools and equipment from exposed locations; and removing or securing scaffolding and other temporary work. Close openings in the work when storms of lesser intensity pose a threat to the work or any nearby Government property.

3.6 PLANT COMMUNICATIONS

Whenever the individual elements of the plant are located so that operation by normal voice between these elements is not satisfactory, install a satisfactory means of communication, such as telephone or other suitable devices and make available for use by Government personnel.

3.7 TEMPORARY PROJECT SAFETY FENCING

As soon as practicable, but not later than 15 days after the date established for commencement of work, furnish and erect temporary project safety fencing at the work site. Maintain the safety fencing during the life of the Contract and, upon completion and acceptance of the work, remove from the work site.

3.8 DUMPSTERS

Equip dumpsters with a secure cover and paint the standard installation color. Keep dumpster closed, except when being loaded with trash and debris. Empty site dumpsters at least once a week, or as needed to keep the site free of debris and trash. If necessary, provide 200 liter trash containers painted the darker installation color to collect debris in the construction site area. For large demolitions, large dumpsters without lids are acceptable, but must not have debris higher than the sides before emptying.

3.9 CLEANUP

Remove construction debris, waste materials, packaging material and the like from the work site daily. Any dirt or mud which is tracked onto paved or surfaced roadways must be cleaned away. Store all salvageable materials resulting from demolition activities within the fenced area described above or at the supplemental storage area. Neatly stack stored materials not in trailers, whether new or salvaged.

3.10 RESTORATION OF STORAGE AREA

Upon completion of the project remove the bulletin board, signs, barricades, haul roads, and all other temporary products from the site. After removal of trailers, materials, and equipment from within the fenced area, remove the fence. Restore areas used during the performance of the Contract to the original or better condition. Remove gravel used to traverse grassed areas and restore the area to its original condition, including top soil and seeding as necessary.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 57 19.01 21

ENVIRONMENTAL CONTROLS

PART 1 GENERAL

1.1 REFERENCES

1.2 DEFINITIONS

- 1.2.1 Class I and II Ozone Depleting Substance (ODS)
 - 1.2.2 Contractor Generated Hazardous Waste
 - 1.2.3 Cultural Resources
 - 1.2.4 Electronics Waste
 - 1.2.5 Environmental Pollution and Damage
 - 1.2.6 Environmental Protection
 - 1.2.7 Hazardous Debris
 - 1.2.8 Hazardous Materials
 - 1.2.9 Hazardous Waste
 - 1.2.10 Installation Pest Management Coordinator
 - 1.2.11 Land Application
 - 1.2.12 Material not regulated as solid waste
 - 1.2.13 Non-Hazardous Waste
 - 1.2.14 Oily Waste
 - 1.2.15 Pesticide
 - 1.2.16 Pesticide Treatment Plan
 - 1.2.17 Pests
 - 1.2.18 Pest Management Coordinator
 - 1.2.19 Recyclables
 - 1.2.20 Recycling
 - 1.2.21 Regulated Waste
 - 1.2.22 Reportable Release
 - 1.2.23 Scrap Metal
 - 1.2.24 Sediment
 - 1.2.25 Solid Waste
 - 1.2.25.1 Debris
 - 1.2.25.2 Green Waste
 - 1.2.25.3 Bulky Waste
 - 1.2.26 Spill Event
 - 1.2.27 Stormwater
 - 1.2.28 Surface Discharge
 - 1.2.29 Universal Waste
 - 1.2.30 Wastewater
 - 1.2.31 Waters of Spain
 - 1.2.31.1 Continental Waters
 - 1.2.31.2 Coastal Waters
 - 1.2.32 Wetlands
 - 1.2.33 Wood
- 1.3 SUBMITTALS
- 1.4 ENVIRONMENTAL PROTECTION REQUIREMENTS
- 1.4.1 Training in Environmental Compliance Assessment Training and Tracking System (ECATTS)
 - 1.4.1.1 Personnel Requirements
 - 1.4.1.2 Certification

- 1.4.1.3 Refresher Training
- 1.4.2 Conformance with the Environmental Management System
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Preconstruction Survey and Protection of Features
 - 1.5.2 Regulatory Notifications
 - 1.5.3 Environmental Protection Plan (EPP)
 - 1.5.4 Preconstruction Environmental Brief/Discussion
 - 1.5.5 Environmental Coordinator (EC)
 - 1.5.6 Employee Training Records
 - 1.5.6.1 Pest Control Training
 - 1.5.7 Pesticide Delivery, Storage, and Handling
 - 1.5.7.1 Delivery and Storage
 - 1.5.7.2 Handling Requirements
 - 1.5.8 Non-Compliance Notifications
- 1.6 LICENSES AND PERMITS
- 1.7 SOLID WASTE AND RECYCLING MANAGEMENT PLAN
 - 1.7.1 Solid Waste Management Report

PART 2 PRODUCTS

PART 3 EXECUTION

- 3.1 PROTECTION AND MANAGEMENT OF RESOURCES
 - 3.1.1 Natural Resources
 - 3.1.2 Protection of Endangered Species
 - 3.1.3 Vegetation
 - 3.1.4 Surface Waters
 - 3.1.4.1 Flow Ways
 - 3.1.4.2 Rivers and Streams
- 3.2 STORMWATER
 - 3.2.1 Stormwater Pollution Prevention Plan (SWPPP)
 - 3.2.2 Erosion and Sediment Control Measures
- 3.3 SURFACE AND GROUNDWATER
 - 3.3.1 Cofferdams, Diversions, and Dewatering
- 3.4 PROTECTION OF HISTORICAL AND ARCHAEOLOGICAL RESOURCES
- 3.5 WASTE MANAGEMENT AND DISPOSAL AND RECYCLING
 - 3.5.1 Solid Waste
 - 3.5.2 Hazardous Waste
 - 3.5.2.1 Waste Storage/Satellite Accumulation/90 Day Storage Areas
 - 3.5.3 Universal Waste
 - 3.5.4 Electronics End-of-Life Management
- 3.6 PETROLEUM, OIL, LUBRICANT (POL) STORAGE AND FUELING
 - 3.6.1 Fuel Tanks
 - 3.6.2 Used Oil Management
 - 3.6.3 Oil Storage Including Fuel Tanks
- 3.7 RELEASES/SPILLS OF OIL, HAZARDOUS SUBSTANCES AND AQUEOUS FILM FORMING FOAM (AFFF)
- 3.8 WATER AND AIR QUALITY PROTECTION MEASURES
- 3.9 AIR RESOURCES
 - 3.9.1 Oil or Dual-fuel Boilers and Furnaces
 - 3.9.2 Burning
 - 3.9.3 Class I and II ODS Prohibition
 - 3.9.4 Accidental Venting of Refrigerant
 - 3.9.5 Dust Control
 - 3.9.5.1 Particulates
 - 3.9.5.2 Abrasive Blasting
 - 3.9.6 Odors
- 3.10 WASTEWATER

- 3.10.1 Treatment
- 3.10.2 Surface Discharge
- 3.10.3 Land Application
- 3.11 PREVIOUSLY USED EQUIPMENT
- 3.12 CONTROL AND MANAGEMENT OF ASBESTOS-CONTAINING MATERIAL (ACM)
- 3.13 CONTROL AND MANAGEMENT OF LEAD-BASED PAINT (LBP)
- 3.14 CONTROL AND MANAGEMENT OF POLYCHLORINATED BIPHENYLS (PCBS)
- 3.15 CONTROL AND MANAGEMENT OF LIGHTING BALLAST AND LAMPS CONTAINING
- 3.16 MILITARY MUNITIONS
- 3.17 INADVERTENT DISCOVERY OF PETROLEUM-CONTAMINATED SOIL OR HAZARDOUS
WASTES
- 3.18 CHLORDANE
- 3.19 SOUND INTRUSION
- 3.20 POST CONSTRUCTION CLEANUP
- 3.21 LAYDOWN AREA
- 3.22 DRINKING WATER

-- End of Section Table of Contents --

SECTION 01 57 19.01 21

ENVIRONMENTAL CONTROLS
(Naval Station Rota, Spain, March 2021)

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

NAVSTA ROTA ENVIRONMENTAL REFERENCE DOCUMENTS

LG (2016) NAVSTA Rota Landscaping Guidelines

FGS-SPAIN (2014) Final Governing Standards for Spain

1.2 DEFINITIONS

1.2.1 Class I and II Ozone Depleting Substance (ODS)

Class I and II ODS include the chemicals listed in FGS-SPAIN Chapter 2, Table 2.8.

1.2.2 Contractor Generated Hazardous Waste

Hazardous waste as defined in FGS-SPAIN or as defined by applicable Spanish regulations. Any waste which displays hazardous properties and is denoted as an asterisk (*) next to the six digit code in the European List of Wastes.

Excluded from this definition are domestic sewage sludge, and household and medical wastes not possessing the properties in Chapter 6 and Appendix A of FGS-SPAIN. It includes the following:

- a. Discarded materials that contain one or more of the characteristics or properties described in Appendix A of FGS-SPAIN.
- b. All hazardous waste described Chapter 6 of FGS-SPAIN.
- c. Any other form of waste defined as hazardous by the Competent Authority.

These waste streams typically consist of material brought on site by the Contractor to execute work, but are not fully consumed during the course of construction. Examples include, but are not limited to, excess paint thinners (i.e., methyl ethyl ketone, toluene), waste thinners, excess paints, excess solvents, waste solvents, excess pesticides, and contaminated pesticide equipment waste water.

1.2.3 Cultural Resources

Physical remains of any prehistoric or historic district, site, building, structure, or object significant in world, national or local history, architecture, archeology, engineering, or culture. The term includes

artifacts, archeological resources, records, and material remains that are related to such a district, site, building, structure, or object, and also includes natural resources (plants, animals, landscape features, etc.) that may be considered important as a part of a country's traditional culture and history. The term also includes any property listed on the World Heritage List or the Spanish equivalent of the National Register of Historic Places. Spanish lists of properties should be evaluated to determine if they are equivalent with the National Register of Historic Places prior to application.

1.2.4 Electronics Waste

Electronics waste is discarded electronic devices intended for salvage, recycling, or disposal.

1.2.5 Environmental Pollution and Damage

Environmental pollution and damage is the presence of chemical, physical, or biological elements or agents which adversely affect human health or welfare; unfavorably alter ecological balances of importance to human life; affect other species of importance to humankind; or degrade the environment aesthetically, culturally or historically.

1.2.6 Environmental Protection

Environmental protection is the prevention/control of pollution and habitat disruption that may occur to the environment during construction. The control of environmental pollution and damage requires consideration of land, water, and air; biological and cultural resources; and includes management of visual aesthetics; noise; solid, chemical, gaseous, and liquid waste; radiant energy and radioactive material as well as other pollutants.

1.2.7 Hazardous Debris

As defined in paragraph SOLID WASTE, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) in accordance with FGS-SPAIN. Hazardous debris also includes debris that exhibits a characteristic of hazardous waste in accordance with FGS-SPAIN. And as defined in paragraph entitled "Debris" of this section, debris that contains listed hazardous waste (either on the debris surface, or in its interstices, such as pore structure) per FGS-SPAIN and applicable national, regional, and local regulations.

1.2.8 Hazardous Materials

Hazardous materials as defined in FGS-SPAIN or the Country of Spain. Hazardous material is any substance, compound, mixture, or material that is capable of posing an unreasonable risk to health, safety, or the environment if improperly handled, stored, issued, transported, labeled, or disposed because it displays a characteristic listed in Table 5.1 of FGS-SPAIN "Typical Hazardous Materials Characteristics," or the material is listed in Appendix A of FGS-SPAIN, "List of Hazardous Substances and Materials." Or,

- a. Is regulated as a hazardous material or
- b. Requires a Safety Data Sheet (SDS) per EM 385-1-1, or

- c. During end use, treatment, handling, packaging, storage, transportation, or disposal meets or has components that meet or have potential to meet the definition of a hazardous waste.

Designation of a material by this definition, when separately regulated or controlled by other instructions or directives, does not eliminate the need for adherence to that hazard-specific guidance which takes precedence over this instruction for "control" purposes. Such material include ammunition, weapons, explosive actuated devices, propellants, pyrotechnics, chemical and biological warfare materials, medical and pharmaceutical supplies, medical waste and infectious materials, bulk fuels, radioactive materials, and other materials such as asbestos, mercury, and polychlorinated biphenyls (PCBs). Nonetheless, the exposure may occur incident to manufacture, storage, use and demilitarization of these items.

1.2.9 Hazardous Waste

Hazardous waste as defined in FGS-SPAIN or as defined by applicable national, regional and local regulations. Any discarded material, liquid, solid, or gaseous, which meets the definition of hazardous material or is designated hazardous waste by FGS-SPAIN or the Country of Spain. Comply with FGS-SPAIN and the installation Hazardous Waste Management Plan.

1.2.10 Installation Pest Management Coordinator

Installation Pest Management Coordinator (IPMC) is the individual officially designated by the Installation Commander to oversee the Installation Pest Management Program and the Installation Pest Management Plan. All Pesticides to be used at the Installation must be approved by the NAVFAC Atlantic Entomologist/Professional Pest Management Consultant (PPMC).

1.2.11 Land Application

Land Application means spreading or spraying discharge water at a rate that allows the water to percolate into the soil. No sheeting action, soil erosion, discharge into storm sewers, discharge into defined drainage areas, must occur. Comply with FGS-SPAIN and applicable Spanish laws.

1.2.12 Material not regulated as solid waste

Material not regulated as solid waste is nuclear source or byproduct materials regulated under the Federal Atomic Energy Act of 1954 as amended; suspended or dissolved materials in domestic sewage effluent or irrigation return flows, or other regulated point source discharges; regulated air emissions; and fluids or wastes associated with natural gas or crude oil exploration or production.

1.2.13 Non-Hazardous Waste

Non-hazardous waste is waste that is excluded from, or does not meet, hazardous waste criteria in accordance with Chapter 6 of FGS-SPAIN.

1.2.14 Oily Waste

Oily waste are those materials that are, or were, mixed with Petroleum, Oils, and Lubricants (POLs) and have become separated from that POLs. Oily wastes also means materials, including wastewaters, centrifuge

solids, filter residues or sludges, bottom sediments, tank bottoms, and absorbents which have come into contact with and have been contaminated by, POLs and may be appropriately tested and discarded in a manner which is in compliance with Chapter 6 of FGS-SPAIN. A Spanish waste manifest (Hoja de Control y Seguimiento) shall be prepared and approved by authorized firm to properly dispose of hazardous waste in accordance with Spanish Regulations. This definition includes materials such as oily rags, "kitty litter" sorbent clay and organic sorbent material. Centrifuge solids, filter residues or sludge, bottom sediments, tank bottoms, oily rags, kitty litter, sorbents, etc. are to be containerized and turned in for proper disposal to an authorized firm for proper treatment and/or disposal. Chapter 9 of FGS-SPAIN describes the POL related requirements on NAVSTA Rota, Spain.

1.2.15 Pesticide

Pesticide is any substance or mixture of substances intended for preventing, destroying, repelling, or mitigating any pest, or intended for use as a plant regulator, defoliant or desiccant. All Pesticides to be used at the Installation must be approved by the NAVFAC Atlantic Entomologist/Professional Pest Management Consultant (PPMC).

1.2.16 Pesticide Treatment Plan

A plan for the prevention, monitoring, and control to eliminate pest infestation.

1.2.17 Pests

Pests are arthropods, birds, rodents, nematodes, fungi, bacteria, viruses, algae, snails, marine borers, snakes, weeds and other organisms (except for human or animal disease-causing organisms) that adversely affect readiness, military operations, or the well-being of personnel and animals; attack or damage real property, supplies, equipment, or vegetation; or are otherwise undesirable.

1.2.18 Pest Management Coordinator

The Pest Management Coordinator is an individual who works at NAVSTA Rota Public Works Department and who is responsible for implementing the Pest Management Plan.

1.2.19 Recyclables

Recyclables are materials that can be recycled instead of disposed of in landfills, including but not limited to; pipes, doors, windows, door and window frames, plumbing fixtures, glazing and mirrors, wiring, insulated/non-insulated copper wire cable, wire rope, and structural components. It also includes commercial-grade refrigeration equipment with Freon removed, household appliances where the basic material content is metal, clean polyethylene terephthalate bottles, cooking oil, used fuel oil, textiles, high-grade paper products and corrugated cardboard, stackable pallets in good condition, clean crating material, and clean rubber/vehicle tires. Metal meeting the definition of lead contaminated or lead based paint contaminated may not be included as recyclable. Paint cans may not be included as recyclable.

1.2.20 Recycling

Recycling is the process of recovering material from waste and turning it into new products. The original product is destroyed in the process and is used to form new products. Examples are scrap metal, glass, cardboard, paper, plastic and many more items. On NAVSTA Rota all scrap metal, including wiring, remains the property of the U. S. Government and will be brought by the Contractor to the NAVSTA Rota Recycling Center near Tow Way. Other recyclable materials will be taken off-base to a local recycling center.

1.2.21 Regulated Waste

Regulated waste are solid wastes that have specific additional FGS-SPAIN, regional, or local controls for handling, storage, or disposal.

1.2.22 Reportable Release

A reportable release means any spilling, leaking, pumping, pouring, emitting, emptying, discharging, injecting, escaping, leaching, dumping, or disposing into the environment of a known or unknown material or hazardous substance that poses an immediate threat to human health or the environment to the air, soil, or water. In the event of a spill on base call the fire department at 9-1-1.

Report all spills to the NAVSTA Rota Environmental Division.

1.2.23 Scrap Metal

This includes scrap and excess ferrous and non-ferrous metals such as reinforcing steel, structural shapes, pipe, and wire that are recovered or collected and disposed of as scrap. Scrap metal meeting the definition of hazardous material or hazardous waste is not included.

1.2.24 Sediment

Sediment is soil and other debris that has eroded and has been transported by runoff water or wind.

1.2.25 Solid Waste

Solid waste is garbage, refuse, sludge, and other discarded materials, including solid, semi-solid, liquid, and contained gaseous materials resulting from industrial and commercial operations and from community activities. It does not include solids or dissolved material in domestic sewage or other significant pollutants in water resources, such as silt, dissolved or suspended solids in industrial wastewater effluent, dissolved materials in irrigation return flows, or other common water pollutants.

Types of solid waste typically generated at construction sites may include:

1.2.25.1 Debris

Debris is non-hazardous solid material generated during the construction, demolition, or renovation of a structure that exceeds 60 mm (2.5-inch) particle size that is: a manufactured object; plant or animal matter; or natural geologic material (for example, cobbles and boulders), broken or removed concrete, masonry, and rock asphalt paving; ceramics; roofing paper and shingles. Inert materials may not be reinforced with or contain

ferrous wire, rods, accessories and weldments. A mixture of debris and other material such as soil or sludge is also subject to regulation as debris if the mixture is comprised primarily of debris by volume, based on visual inspection.

1.2.25.2 Green Waste

Green waste is the vegetative matter from landscaping, land clearing and grubbing, including, but not limited to, grass, bushes, scrubs, small trees and saplings, tree stumps and plant roots. Marketable trees, grasses and plants that are indicated to remain, be re-located, or be re-used are not included.

1.2.25.3 Bulky Waste

Large items of solid waste such as household appliances, furniture, large auto parts, trees, branches, stumps, and other oversize wastes whose large size precludes or complicates their handling by normal solid waste collection, processing, or disposal methods.

1.2.26 Spill Event

A spill is any release of oil or hazardous substances to the water or ground that is not controlled or permitted. This includes any spilling, leaking, pumping, emitting, discharging, injecting, escaping, leaching, disposing, or dumping of liquid or solid material that is not authorized in writing by the Contracting Officer.

1.2.27 Stormwater

Stormwater is any precipitation in an urban or suburban area that does not evaporate or soak into the ground, but instead collects and flows into storm drains, rivers, and streams.

1.2.28 Surface Discharge

Surface discharge means discharge of water into drainage ditches, storm sewers, creeks or "waters of Spain". Surface discharges are discrete, identifiable sources and require a permit from the governing agency. Comply with FGS-SPAIN and national, regional, and local laws and regulations.

1.2.29 Universal Waste

Universal wastes are certain hazardous wastes in the following categories: batteries, pesticides, mercury-containing equipment (for example, thermostats), and lamps (for example, fluorescent bulbs). These wastes are collected and sent for recycling or proper disposal.

1.2.30 Wastewater

Wastewater is the used water and solids from a community that flow to a treatment plant. Wastewater is also any water which during manufacturing or processing, comes into direct contact with, or results from the production or use of, any raw material, intermediate product, finished product, by-product, or waste product.

1.2.31 Waters of Spain

Waters included in the following categories:

1.2.31.1 Continental Waters

Any water body within the Spanish territory other than coastal waters; this includes groundwater and surface waters (e.g., underground aquifers, rivers, streams, lakes, reservoirs, springs, irrigation channels).

1.2.31.2 Coastal Waters

The zone between the coastline of low waters and a line parallel to the coastline that is approximately 12 miles (20 km) from the shore.

Exclusions to Waters of Spain are domestic or industrial waste treatment systems, including treatment ponds or lagoons designed to meet the requirements of FGS-SPAIN, are not waters of Spain. This exclusion applies only to manmade bodies of water that were neither originally waters of Spain nor resulted from impoundment of waters of Spain.

1.2.32 Wetlands

Wetlands are those areas that are inundated or saturated by surface or groundwater at a frequency and duration sufficient to support, and that under normal circumstances do support, a prevalence of vegetation typically adapted for life in saturated soil conditions.

1.2.33 Wood

Wood is dimension and non-dimension lumber, plywood, chipboard, hardboard. Treated or painted wood that meets the definition of lead contaminated or lead based contaminated paint is not included. Treated wood includes, but is not limited to, lumber, utility poles, crossties, and other wood products with chemical treatment.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. When used, a designation following the "G" designation identifies the office that will review the submittal for the Government.

Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Preconstruction Environmental, Including Common Chameleon and Vegetation, Survey; G (NAVSTA Rota Environmental Division Review).

Preconstruction Archaeological Survey; G (NAVSTA Rota Environmental Division Review).

Environmental Protection Plan (EPP); G (NAVSTA Rota Environmental Division Review). Write the EPP following the template in section 1.5.2.

Spill Prevention, Control and Response Plan (for projects at or

near the Bay of Cadiz); G (NAVSTA Rota Environmental Division Review).

Estimate of Solid Waste and Recyclables weights and disposal/recycle locations.

Stormwater Pollution Prevention Plan; G (NAVSTA Rota Environmental Division Review).

List of Employee Environmental Protection Trainings; G

Environmental Manager Qualifications; G

Safety Data Sheets (SDS) for Hazardous Materials

SD-07 Certificates

List of Employee Environmental Training Records; G

List of ECATTS Environmental on-line course Certificates of Completion; G

SD-11 Closeout Submittals

Stormwater Pollution Prevention Plan Compliance Notebook; G

Solid Waste and Recycling Report (by weight); G (NAVSTA Rota Environmental Division Review).

Disposal Documentation for Hazardous and Regulated Waste; G

Contractor Hazardous Material Inventory Log; G

Regulatory Notifications; G

Sales Documentation; G

Construction & Demolition (C & D) Debris data and management cost

1.4 ENVIRONMENTAL PROTECTION REQUIREMENTS

All work must be performed in accordance with the requirements of FGS-SPAIN. All work must also adhere to the Spanish Navy and the Junta de Andalucia (Regional Spanish Government) natural and cultural resource requirements.

The Contractor must identify, obtain, and comply with all applicable (Spanish and US) environmental requirements, authorizations, and permits. Examples of work elements that typically have environmental requirements, authorizations, and permits are storm water, biological and archaeological resources, hazardous materials, hazardous waste, pesticides, and in-water work requirements. Contractor must coordinate all Spanish Navy and Junta de Andalucia authorization and permitting actions with Installation Environmental Division and NAVFAC Host Nation Liaison.

The Contractor must notify the governing regulatory agency when required by Spanish Law.

Provide and maintain, during the life of the contract, environmental protection as defined and described in the EPP. Plan for and provide

environmental protective measures to control pollution that develops during construction practice. Plan for and provide environmental protective measures required to correct conditions that develop during the construction of permanent or temporary environmental features associated with the project. Protect the environmental resources within the project boundaries and those affected outside the limits of permanent work during the entire duration of this Contract.

Tests and procedures assessing whether construction operations comply with applicable environmental laws may be required. Analytical work must be performed by qualified laboratories; and where required by law, the laboratories must be accredited/certified.

1.4.1 Training in Environmental Compliance Assessment Training and Tracking System (ECATTS)

Coordinate with the NAVSTA Environmental Division to add additional staff that require this on-line training.

1.4.1.1 Personnel Requirements

Appoint in writing to the Environmental Division the Environmental Coordinators (ECs) (Primary and an Alternate) for the project. The ECs and construction workers must complete applicable ECATTS training modules (installation specific or general) prior to starting respective portions of on-site work under this Contract. If personnel changes occur for any of these positions after starting work, replacement personnel must complete applicable ECATTS training within 14 days of assignment to the project.

All personnel assigned to handling Hazardous Waste and Hazardous Materials are properly trained to meet, at a minimum, OSHA standards, 29 CFR 1910.120 Contractor shall assign properly trained Environmental coordinators to ensure proper management of hazardous materials and hazardous waste, and to properly respond to oil and hazardous substance (OHS) spills. Contractor assigned ECs must have at least successfully completed the 24-hr Hazardous Waste Management and Emergency response (HAZWOPER) or its equivalent. Prepare and maintain employee training records throughout the term of the contract.

Refrigerant technicians shall be trained in proper recovery/recycling procedures, leak detection, safety, shipping, and disposal in accordance with recognized industry standards or Spanish equivalent.

1.4.1.2 Certification

Maintain ECATTS certificates of completion for personnel who have completed the required ECATTS training. This training is web-based and can be accessed from any computer with Internet access using the following instructions.

Register for NAVFAC Environmental Compliance Training and Tracking System, by logging on to <https://environmentaltraining.ecatts.com/>. Obtain the password for registration from the Contracting Officer.

1.4.1.3 Refresher Training

This training has been structured to allow Contractor personnel to receive credit under this contract and to carry forward credit to future

contracts. Ensure the Environmental Manager review their training plans for new modules or updated training requirements prior to beginning work.

1.4.2 Conformance with the Environmental Management System

Perform work under this contract consistent with the policy and objectives identified in the installation's Environmental Management System (EMS). Perform work in a manner that conforms to objectives and targets of the environmental programs and operational controls identified by the EMS. Support Government personnel when environmental compliance and EMS audits are conducted by escorting auditors at the Project site, answering questions, and providing proof of records being maintained. Provide monitoring and measurement information as necessary to address environmental performance relative to environmental, energy, and transportation management goals. In the event an EMS nonconformance or environmental noncompliance associated with the contracted services, tasks, or actions occurs, take corrective and preventative actions. In addition, employees must be aware of their roles and responsibilities under the installation EMS and of how these EMS roles and responsibilities affect work performed under the contract.

Coordinate with the installation's EMS manager to identify training needs associated with environmental aspects and the EMS, and arrange training or take other action to meet these needs. Provide training documentation to the Contracting Officer. The Installation Environmental Division will retain associated environmental compliance records. Make EMS Awareness training completion certificates available to Government auditors during EMS audits and include the certificates in the Employee Training Records. See paragraph EMPLOYEE TRAINING RECORDS.

1.5 QUALITY ASSURANCE

1.5.1 Preconstruction Survey and Protection of Features

Prior to start of any onsite construction activities, perform a Preconstruction Survey of the project site with the Contracting Officer, and take photographs showing existing environmental conditions in and adjacent to the site. Submit a report for the record. Include in the report a plan describing the features requiring protection under the provisions of the Contract Clauses, which are not specifically identified on the drawings as environmental features requiring protection along with the condition of trees, shrubs and grassed areas immediately adjacent to the site of work and adjacent to the Contractor's assigned storage area and access route(s), as applicable. The Contractor and the Contracting Officer will sign this survey report upon mutual agreement regarding its accuracy and completeness. Protect those environmental features included in the survey report and any indicated on the drawings, regardless of interference that their preservation may cause to the work under the Contract.

If the project work area include areas of suitable chameleon habitat where indicated on contract drawings, chameleons found during night surveys in these areas must be relocated by a Spanish approved chameleon specialist prior to construction activities. Follow the Protected Species Protocol (PSP) for the common chameleon as applicable.

If the project work area include areas of archaeological concern, the site must be evaluated by an archaeological specialist prior to construction activities. If required by the Junta de Andalucia, an archaeological

monitor may be required to be present during earth moving activities.

If the project work area include areas suitable for bird nests, a bird nest survey will be conducted and if any nests are found to contain eggs or chicks, the nest shall not be disturbed until the chicks have fledged and left the nest.

1.5.2 Regulatory Notifications

Provide regulatory notification requirements in accordance with Spanish regulations. Typically, regulatory notifications must be provided for the following (this listing is not all-inclusive): demolition, renovation, construction, removal or use of an air emissions sources, and remediation of controlled substances (asbestos, hazardous waste, lead paint).

1.5.3 Environmental Protection Plan (EPP)

The purpose of the EPP is to present an overview of known or potential environmental issues that must be considered and addressed during construction or demolition activities.

The EPP (including drawings and figures) addresses environmental categories, as identified below. An EPP must be developed by the Contractor and submitted to the Contracting Officer for review by the NAVSTA Rota Environmental Division. The EPP must be reviewed by the NAVSTA Rota Environmental Division prior to the start of construction. Submit the EPP using the following format:

Environmental Protection Plan (EPP) Template Naval Station Rota Spain

1. Cover Page - Name of project; Navy contract and task order number; name of Navy Base and specific locations; Contractor's name, address, e-mail and phone number. Company's environmental point of contact name and phone number.
2. Table of contents.
3. Project description and general site information. Show work area plan of the proposed activity in each portion of the area and identify the areas of limited use or nonuse, including laydown areas. Include measures for marking the limits of use areas, including methods for protection of features to be preserved within authorized work areas and methods to control runoff and to contain materials on site.
4. Existing environmental conditions of the project site. For example: known archaeological sites; protected species (i.e. common chameleon potential habitat); erosion issues; Andalusian forest trees present such as Stone Pine; nesting birds, and other natural resources present.
5. Conclusions of environmental or archaeological surveys and studies done for the project, if any.
6. Spanish environmental permits needed, if any. Environmental and Historical/Archaeological Resource Permits and Authorizations Required, Including Regulatory Notification. Certification of Environmental and Cultural Permits Obtained by Contractor or HN Installation Commander.

7. Compliance with NAVSTA Rota's Protected Species Protocol for the Common Chameleon (2016), and NAVSTA Rota's Landscaping Guidelines (2016).
8. Measures/Steps to address resources and materials (as are applicable to the project):

Archaeological and historical site protection measures: Include the statement, "If during digging an archaeological artifact is uncovered, work will be stopped and the NAVSTA Rota Environmental Division will be contacted. Work will continue when approved by the Contracting Officer Representative";

Hazardous materials and hazardous waste containment/storage and disposal;

Recycling and Solid waste management/disposal: Include the statement, "Scrap metal and wiring shall remain the property of the U.S. Government and will be taken to the NAVSTA Rota Recycling Center";

Vegetation and wildlife protection and revegetation: including Spanish protected species, Andalusian forest trees, nesting birds, and other sensitive species;

Spill prevention and planned actions in the event of a spill. Notify EV of all spills, regardless of quantity. Outline measures to prevent spills and steps to be taken in the event of a spill.

Storm water management and soil erosion control: storm water pollution prevention plan, if required;

Wastewater management;

Water pollution protection (i.e. ocean/bay, groundwater, streams, rivers, and other water bodies);

Air Quality Protection (e.g., dirt and dust) Submit truck and material haul routes along with a Dirt and Dust Control Plan for controlling dirt, debris, and dust on Installation roadways. At a minimum, identify in the plan the subcontractor and equipment for cleaning along the haul route and measures to reduce dirt, dust, and debris from roadways;

Planned air pollution-generating equipment and processes and management control measures (including, but not limited to, spray painting, abrasive blasting, demolition, material handling, fugitive dust, and fugitive emissions). Log hours of operations and track quantities of materials used;

Identify portable and stationary internal combustion engines that will be supplied, used or serviced. At minimum, include the make, model, serial number, manufacture date, size (engine brake horsepower), and emission certification status of each engine. Maintain applicable records and log hours of operation and fuel use. Logs must include reasons for operation and delineate between emergency and non-emergency operation;

Noise control;

Dirt and Dust Control: include measures to reduce dirt, dust, and debris from roadways, adjacent areas, and building interiors;

Other environmental concerns specific to the project: i.e., in-water dredging and/or pile installments, marine resources, traffic and transportation, etc.;

Hazardous materials and hazardous waste containment/storage and disposal;

Water pollution protection (i.e., groundwater, streams, rivers, lakes and other water bodies);

9. Monitoring and Quality Control Procedures;

10. Employee Environmental Trainings List: Do not submit training certificates, only list the environmental trainings completed by the employees.

1.5.4 Preconstruction Environmental Brief/Discussion

The Construction Manager/Contracting Officer representative, Contractor, and NAVSTA Rota Environmental Division staff must attend an environmental brief/discussion that is to be included in the preconstruction or another meeting before the beginning of construction or demolition activities. Provide the following: types, quantities, use of hazardous materials and copy of Safety Data Sheet (SDS) that will be brought onto the installation for Environmental review and approval; and types and quantities of solid wastes/Hazardous waste/wastewater that may be generated during the Contract. All pesticides must be approved by the NAVFAC Atlantic Entomologist/Professional Pest Management Consultant (PPMC). Discuss the environmental protection plan, environmental protection measures to be taken, and the results of the Preconstruction Survey at this time.

1.5.5 Environmental Coordinator (EC)

The designated Contractor EC will be directly responsible for coordinating Contractor compliance with DoD, Navy and Spanish requirements. This can be a collateral position; however the person in this position must be trained to adequately accomplish the duties and responsibilities. ECs are required to appear for a face-to-face interaction with Environmental Division staff to understand the base environmental policies, regulations and compliance requirements. The ECs are directly responsible for coordinating Contractor compliance with Spanish and NAVSTA Rota requirements. The EC must ensure compliance with Hazardous Waste Program requirements (including hazardous waste handling, storage, manifesting, and disposal); implement the EPP; ensure environmental permits are obtained (as required), maintained, and closed out; ensure compliance with Stormwater Program requirements; ensure compliance with Hazardous Materials (storage, handling, and reporting) requirements; coordinate any remediation of regulated substances (lead, asbestos, PCB transformers); ensure waste segregation and storage compatibility requirements are met; inspect and manage Satellite Accumulation areas; ensure only authorized personnel add wastes to containers; ensure Contractor personnel are trained in accordance with their position requirements; coordinate removal of waste containers; and maintain the Environmental Records binder and required documentation, including environmental permits compliance, Spanish Hazardous Waste Manifests, and close-out. Submit EC Qualifications to the Contracting Officer.

1.5.6 Employee Training Records

Prepare and maintain Employee Training Records throughout the term of the contract and maintain in the Environmental records binder. Provide Employee Training Records in the Environmental Records Binder. Ensure every employee completes a program of classroom instruction or on-the-job training that teaches them to perform their duties in a way that ensures compliance with FGS-SPAIN. Provide a Position Description for each employee, evaluating the employee's association with hazardous and regulated wastes. This Position Description will include training requirements for a Hazardous Waste Generator facility. Submit these Assembled Employee Training Records to the Contracting Officer at the conclusion of the project, unless otherwise directed.

Train personnel to meet FGS-SPAIN requirements. Conduct environmental protection/pollution control meetings for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting and avoiding pollution; familiarization with statutory and contractual pollution standards; installation and care of devices, vegetative covers, and instruments required for monitoring purposes to ensure adequate and continuous environmental protection/pollution control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of archaeological sites, artifacts, waters of Spain, and endangered species and their habitat that are known to be in the area. Provide copy of the Erosion and Sediment Control Inspector qualifications.

1.5.6.1 Pest Control Training

Train personnel in pest control. Conduct a pest control meeting for personnel prior to commencing construction activities. Conduct additional meetings for new personnel and when site conditions change. Include in the training and meeting agenda: methods of detecting a pest infestation; familiarization with statutory and contractual pest control standards; installation and care of devices and instruments, if required, for monitoring purposes to ensure adequate and continuous pest control; anticipated hazardous or toxic chemicals or wastes, and other regulated contaminants; recognition and protection of waters of the Host Nation, and endangered species and their habitat that are known to be in the area. Provide a Certificate of Competency for the personnel who will be conducting the pesticide application and management of pest control. All Pesticides to be used at the Installation must be approved by the NAVFAC Atlantic Entomologist/Professional Pest Management Consultant (PPMC).

1.5.7 Pesticide Delivery, Storage, and Handling

1.5.7.1 Delivery and Storage

Deliver pesticides to the site in the original, unopened containers bearing legible labels. Store pesticides according to manufacturer's instructions and under lock and key when unattended. Contractors must remove all pesticides from the installation at the end of the working day. Contractors are not permitted to store pesticides at the Installation after working hours.

1.5.7.2 Handling Requirements

Formulate, treat with, and dispose of pesticides and associated containers in accordance with label directions and use the clothing and personal protective equipment specified on the labeling for use during each phase of the application. Furnish SDSs for pesticide products.

1.5.8 Non-Compliance Notifications

The Contracting Officer will notify the Contractor in writing of any observed noncompliance with FGS-SPAIN and other elements of the Contractor's EPP. After receipt of such notice, inform the Contracting Officer of the proposed corrective action and take such action when approved by the Contracting Officer. The Contracting Officer may issue an order stopping all or part of the work until satisfactory corrective action has been taken. No time extensions will be granted or equitable adjustments allowed for any such suspensions. This is in addition to any other actions the Contracting Officer may take under the contract, or in accordance with the FAR.

1.6 LICENSES AND PERMITS

Obtain licenses and permits required for the construction of the project and in accordance with FAR 52.236-7. Notify the Government of all general use permitted equipment the Contractor plans to use on site. This paragraph supplements the Contractor's responsibility under FAR 52.236-7.

1.7 SOLID WASTE AND RECYCLING MANAGEMENT PLAN

Provide the Contracting Officer and NAVSTA Rota Environmental Division with written notification of the quantity of anticipated solid waste or debris that is anticipated or estimated to be generated by construction. Also, the anticipated quantity (weight) of materials to be recycled. Include in the report the locations where various types of waste will be disposed of or recycled. For instance, all scrap metal will be recycled at the NAVSTA Rota Recycling Center. Include letters of acceptance from the receiving location or as applicable; submit one copy of the receiving location license showing such entities approval of the disposal plan before transporting wastes off Government property.

1.7.1 Solid Waste Management Report

Monthly, submit a solid waste disposal and recycled materials report to the Contracting Officer and the NAVSTA Rota Environmental Division. For each waste, the report will state the classification (using the definitions provided in this section), amount, location, and name of the business receiving the solid waste.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 PROTECTION AND MANAGEMENT OF RESOURCES

3.1.1 Natural Resources

Minimize interference with, disturbance to, and damage to fish, wildlife,

and plants, including their habitats. Prior to the commencement of activities, consult with the Installation Environmental Division regarding rare species or sensitive habitats that need to be protected. The protection of rare, threatened, and endangered animal and plant species identified, including their habitats, is the Contractor's responsibility.

Preserve the natural resources within the project boundaries and outside the limits of permanent work. Restore to an equivalent or improved condition upon completion of work that is consistent with the requirements of the Installation Environmental Division or as otherwise specified. Confine construction activities to within the limits of the work indicated or specified.

3.1.2 Protection of Endangered Species

Known endangered or threatened species and Spanish protected species and their habitats must be protected.

The Protected Species Protocol for the Common Chameleon (March 2016) will be followed, as applicable. This includes submitting a plan for anticipated vegetation removal and site restoration and a survey of potentially suitable chameleon habitat in the project area. If chameleons are present in the project area they must be relocated by a specialist in advance of construction activities.

Preserve natural resources within the work area limits to the maximum extent possible. Restore natural resources to an equivalent or improved condition upon completion of work.

If a natural resource of interest is encountered during construction (e.g. chameleon or other sensitive species), the Contractor must stop all work until the resource is properly assessed by the Environmental Division and the Contracting Officer Representative has given the notice to resume work.

3.1.3 Vegetation

All planted vegetation must follow the NAVSTA Rota Landscaping Guidelines (LG). This guideline contains an approved list of vegetation and trees that may be planted on NAVSTA Rota.

Areas of Stone Pine and other Andalusian forest tree species must be avoided for new construction to the maximum extent possible. If trees must be removed, approval must be granted by the Contracting Officer. Trees that are approved for removal must be replaced at a 3:1 ratio at an area designated by the Contracting Officer. The replacement trees must be saplings (not seedlings) and must be irrigated for one year to ensure successful growth.

Do not damage trees. Do not fasten or attach ropes, cables, or guys to existing nearby trees or shrubs without the Contracting Officer's permission. With Contracting Officer approval, remove trees with 30 percent or more of their root systems destroyed as well as trees and other landscape features scarred or damaged by equipment operations. With Contracting Officer approval, replace damaged natural resources with equivalent trees and landscape features.

3.1.4 Surface Waters

Do not enter, disturb, destroy, or allow discharge of contaminants into

surface waters except as authorized herein. The protection of surface waters is the Contractor's responsibility. Authorization to enter specific waters identified does not relieve the Contractor from any obligation to protect other waters within, adjacent to, or in the vicinity of the construction site and associated boundaries.

3.1.4.1 Flow Ways

Do not alter water flows or otherwise significantly disturb the native habitat adjacent to the project and critical to the survival of fish and wildlife, except as specified.

3.1.4.2 Rivers and Streams

River and stream crossings must allow movement of materials or equipment without violating water pollution control standards of FGS-SPAIN. Construction of stream crossing structures must be in compliance with any required Spanish government permits.

The Contracting Officer's approval and appropriate permits are required before any equipment will be permitted to cross rivers and streams. In areas where frequent crossings are required, install temporary culverts or bridges. Obtain Contracting Officer's approval prior to installation. Remove temporary culverts or bridges upon completion of work, and repair the area to its original condition unless otherwise required by the Contracting Officer.

3.2 STORMWATER

3.2.1 Stormwater Pollution Prevention Plan (SWPPP)

Develop and implement a storm water pollution prevention plan (SWPPP) for activities where there is a potential for storm water pollution or erosion. Activities include: construction activities; hazardous material storage, outdoor material storage areas; outdoor painting/depainting operations; pesticide operations; power production; vehicle storage yards; aircraft ground support equipment maintenance; aircraft/runway deicing; Aircraft/vehicle fueling operations; aircraft vehicle maintenance and repair; aircraft/vehicle washing; bulk fuel storage areas; and corrosion control activities. Include the SWPPP as part of the Environmental Protection Plan (EPP) submittal.

3.2.2 Erosion and Sediment Control Measures

Implement measures to control effluent in stormwater discharges.

Control stormwater volume and velocity to minimize soil erosion in order to minimize pollutant discharges. Control stormwater discharges, including both peak flowrates and total stormwater volume, to minimize channel and streambank erosion and scour in the immediate vicinity of discharge points.

Provide and maintain natural buffers around waters of the HN. Direct stormwater to vegetated areas and maximize stormwater infiltration through use of low-impact development practices (identified by UFC 3-210-10) to reduce pollutant discharges, unless infeasible.

Minimize soil compaction. Minimizing soil compaction is not required where the intended function of a specific area of the site dictates that it be

compacted. Preserve topsoil, unless impractical. Preserving topsoil is not required where the intended function of a specific area of the site dictates that the topsoil be disturbed or removed.

Initialize soil stabilization whenever any clearing, grading, excavating, or other earth disturbing activities have permanently ceased on any portion of the site, or temporarily ceased on any portion of the site and will not resume for a period exceeding 14 calendar days. In arid, semi-arid, and drought-stricken areas where initiating vegetative stabilization measures immediately is infeasible, alternative stabilization measures must be employed as appropriate.

Minimize the exposure of building materials, building products, construction wastes, trash, landscape materials, fertilizers, pesticides, herbicides, detergents, sanitary waste, and other materials present on the site to precipitation and to stormwater. Minimization of exposure is not required in cases where the exposure to precipitation and to stormwater will not result in a discharge of pollutants, or where exposure of a specific material or product poses little risk of stormwater contamination (such as final products and materials intended for outdoor use).

Personnel who handle hazardous substances or perform activities that could contribute pollution in wet weather events must be trained in appropriate BMPs.

3.3 SURFACE AND GROUNDWATER

3.3.1 Cofferdams, Diversions, and Dewatering

Construction operations for dewatering, removal of cofferdams, tailrace excavation, and tunnel closure must be constantly controlled to maintain compliance with existing Spanish standards and designated uses of the surface water body. Do not discharge excavation ground water to the sanitary sewer, storm drains, or to surface waters without prior specific authorization in writing from the Installation Environmental Division. Discharge of hazardous substances will not be permitted under any circumstances. Use sediment control BMPs to prevent construction site runoff from directly entering any storm drain or surface waters.

If the construction dewatering is noted or suspected of being contaminated, it may only be released to the storm drain system if the discharge is treated and authorized by the Installation Environmental Division. Discharge of hazardous substances will not be permitted under any circumstances.

3.4 PROTECTION OF HISTORICAL AND ARCHAEOLOGICAL RESOURCES

Comply with Junta de Andalucia requirements for archaeological site protection. This may include having an archaeological monitor on site during earth moving activities. Contractor must coordinate all Spanish Navy and Junta de Andalucia authorization and permitting actions with Installation Environmental Division and NAVFAC Host Nation Liaison.

Construction of any type, that may alter the character of a historic and cultural resource or disturb its view, is prohibited without proper monitoring and documentation.

If an archeological item is uncovered during excavation, the Contractor must stop all work and contact the Contracting Officer immediately. Work

can continue once the archaeological resource has been evaluated and the Contracting Officer gives the direction to proceed.

3.5 WASTE MANAGEMENT AND DISPOSAL AND RECYCLING

Scrap metal, including wiring, shall remain the property of the U.S. Government and the Contractor shall transport all recovered scrap metal from the project site to the NAVSTA Rota Recycling Facility (near Tow Way). Other recyclable materials become the property of the Contractor upon demolition and must be taken by the Contractor off of the Navy Base to a local recycling center.

3.5.1 Solid Waste

All solid waste must be regularly removed from the project site to maintain a clean work area. Recycle solid waste at a recycling center to the greatest extent possible. Solid waste that cannot be recycled must be treated, stored, and disposed of properly at an authorized land-fill facility.

Report the weight (tonnage) of all materials disposed of as solid waste and of all materials recycled on a monthly basis. This report must be provided to the NAVSTA Rota's Environmental Division's Solid Waste/Recycling Manager. Document the weight of disposed materials in daily reports.

Pick up solid wastes, and place in covered containers that are regularly emptied. Do not prepare or cook food on the project site. Prevent contamination of the site or other areas when handling and disposing of wastes. At project completion, leave the areas clean. Employ segregation measures so that no hazardous or toxic waste will become co-mingled with non-hazardous solid waste. Transport solid waste off Government property and dispose of it in compliance with Chapter 7 of FGS-SPAIN for solid waste disposal. Verify that the selected transporters and disposal facilities have the necessary permits and licenses to operate. Segregate and separate treated wood components disposed at a lined landfill approved to accept this waste in accordance with Spanish regulations. Disposal must comply with Spanish regulations.

3.5.2 Hazardous Waste

No hazardous material or hazardous waste shall be brought onto the NAVSTA Rota property that does not directly relate to requirements for the performance of this contract. Manage hazardous material used in construction, including but not limited to, aerosol cans, waste paint, cleaning solvents, contaminated brushes, and used rags, in accordance with Chapters 5 and 6 of FGS-SPAIN. Do not dispose of hazardous waste on Government property. Do not discharge any waste to a sanitary sewer, storm drain, or to surface waters or conduct waste treatment or disposal on Government property without written approval of the Contracting Officer.

Comply with the Safety Data Sheets (SDSs) requirements per FGS-SPAIN.

All toxic or hazardous materials and hazardous wastes will be transported, stored, treated, and properly disposed of per FGS requirements and the Naval Station Rota Waste Analysis Plan. All hazardous materials and waste generated from the project will be characterized and properly stored. Hazardous waste must be stored outside the installation while awaiting disposition. Hazardous Material storage must be coordinated with the

NAVSTA Rota Safety Office and Environmental Division.

Storage and handling of hazardous materials will adhere to the Department of Defense Component policies, including Joint Service Publication on Storage and Handling of Hazardous Materials. Defense Logistics Agency Instruction 4145.11, Army Technical Manual 38-410, Naval Supply Publication 573, Air Force Joint Manual (AFJMAN) 23-209, and Marine Corps Order 4450.12A "Storage and Handling of Hazardous Materials" provide additional guidance on the storage and handling of hazardous materials.

Hazardous material dispensing areas must be properly maintained. Drums/containers must not be leaking. Drip pans/absorbent materials must be placed under containers, as necessary, to collect drips or spills. Container contents will be clearly marked. The dispensing area shall be designed and operated to ensure appropriate segregation of incompatible hazardous materials.

All containers of hazardous materials shall have manufacturer-generated labels or a Hazardous Chemical Warning Label in accordance with DoD Instruction 600.05 and have SDSs information available on site. These requirements apply throughout the life-cycle of these materials.

The use of materials listed in Table 5.3 of FGS-SPAIN is either prohibited or restricted in Spain, under specified conditions, with the exception of mission critical materials. The use of perfluorooctane sulfonates (PFOS)-based Fire-fighting foams are prohibited.

Completed Spanish Disposal Manifests of outgoing hazardous waste are required. They must specify the treatment and/or disposal method and location and must be provided to Contracting Officer.

3.5.2.1 Waste Storage/Satellite Accumulation/90 Day Storage Areas

Accumulate hazardous waste at satellite accumulation points and in compliance with Chapter 6 of FGS-SPAIN. Individual waste streams will be limited to 208 liters of accumulation (or 0.95 liter for acutely hazardous wastes). If the Contractor expects to generate hazardous waste at a rate and quantity that makes satellite accumulation impractical, the Contractor may request a temporary 90 day accumulation point be established. Submit a request in writing to the Contracting Officer and provide the following information (Attach Site Plan to the Request):

Contract Number	to be determined
Contractor	to be determined
Haz/Waste or Regulated Waste POC	to be determined
Phone Number	to be determined
Type of Waste	to be determined
Source of Waste	to be determined
Emergency POC	to be determined

Contract Number	to be determined
Phone Number	to be determined
Location of the Site	to be determined

Attach a Waste Determination form for the expected waste streams. Allow 10 working days for processing this request. Additional compliance requirements (e.g. training and contingency planning) that may be required are the responsibility of the Contractor. Barricade the designated area where waste is being stored and post a sign identifying as follows:

"DANGER - UNAUTHORIZED PERSONNEL KEEP OUT"

3.5.3 Universal Waste

Manage the following categories of universal waste in accordance with FGS-SPAIN and installation instructions:

- a. Batteries
- b. Lamps
- c. Mercury-containing equipment
- d. Pesticides

Mercury is prohibited in the construction of this facility, unless specified otherwise, and with the exception of mercury vapor lamps and fluorescent lamps. Dumping of mercury-containing materials and devices such as mercury vapor lamps, fluorescent lamps, and mercury switches, in rubbish containers is prohibited. Remove without breaking, pack to prevent breakage, and transport out of the activity in an unbroken condition for disposal or recycling as directed.

3.5.4 Electronics End-of-Life Management

Recycle or dispose of electronics waste, including, but not limited to, used electronic devices such computers, monitors, hard-copy devices, televisions, mobile devices, in accordance with FGS-SPAIN and installation instructions.

3.6 PETROLEUM, OIL, LUBRICANT (POL) STORAGE AND FUELING

POL products include flammable or combustible liquids, such as gasoline, diesel, lubricating oil, used engine oil, hydraulic oil, mineral oil, and cooking oil. Store POL products and fuel equipment and motor vehicles in a manner that affords the maximum protection against spills into the environment. Manage and store POL products in accordance with FGS-SPAIN. Use secondary containments, dikes, curbs, and other barriers, to prevent POL products from spilling and entering the ground, storm or sewer drains, stormwater ditches or canals, or Spanish waters. Describe in the EPP how POL tanks and containers will be stored, managed, and inspected and what protections will be provided. Storage of oil, including fuel, on the project site is not allowed. Fuel must be brought to the project site each day that work is performed.

3.6.1 Fuel Tanks

Petroleum products and lubricants required to sustain up to 30 days of construction activity may be kept on site. Storage and refilling practices

shall comply with FGS-SPAIN and Spanish Government requirements. Secondary containment shall be provided and be no less than 110 percent of the tank volume plus five inches of free-board. If a secondary berm is used for containment then the berm shall be impervious to oil for 72 hours and be constructed so that any discharge will not permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Drips pans are required and the tanks must be covered during inclement weather. Rain water collected in secondary containment that has been contaminated with POL must be treated and disposed of as hazardous waste.

3.6.2 Used Oil Management

Manage used oil generated on site in accordance with FGS-SPAIN. Determine if any used oil generated while onsite exhibits a characteristic of hazardous waste. Used oil containing 1,000 parts per million of solvents is considered a hazardous waste and disposed of at the Contractor's expense. Used oil mixed with a hazardous waste is also considered a hazardous waste. Dispose in accordance with paragraph HAZARDOUS WASTE DISPOSAL. Contractors can turn-in used oil that meets the recycling criteria to the Environmental Division for recycling.

3.6.3 Oil Storage Including Fuel Tanks

Provide secondary containment and overfill protection and flow shut off valve for oil storage tanks. A berm used to provide secondary containment must be of sufficient size and strength to contain the contents of the tanks plus 12 centimeters (5 inches) freeboard for precipitation. Construct the berm to be impervious to oil for 72 hours that no discharge will permeate, drain, infiltrate, or otherwise escape before cleanup occurs. Use drip pans during oil transfer operations; adequate absorbent material must be onsite to clean up any spills and prevent releases to the environment. Cover tanks and drip pans during inclement weather. Provide procedures and equipment to prevent overfilling of tanks. If tanks and containers with an aggregate aboveground capacity greater than 5,000 liter (1,320 gallons) will be used onsite (only containers with a capacity of 208 liter (55 gallons) or greater are counted), provide and implement a SPCC plan meeting the requirements of FGS-SPAIN and installation requirements. Do not bring underground storage tanks to the installation for Contractor use during a project. Submit the SPCC plan to the Contracting Officer for approval.

Proper labels are required on POL storage containers (Flammable, No Smoking, Capacity of tank, Type of POL etc.).

Generator (if inbuilt tank is not double walled and not equipped with overfill protection and shut off valve) and other fuel assemblies will be placed on a secondary containment and drainage of storm water from containment areas will be controlled by a valve that is locked closed when not in active use. Stormwater will be inspected for petroleum sheen before being drained from containment areas. If petroleum sheen is present it must be collected with sorbent materials prior to drainage, or treated using an oil-water separator. Sorbent material shall be disposed of in accordance with NAVSTA Rota Environmental Waste Management Procedures.

All flammable tanks shall be grounded properly to withstand the static current that may be generated during fuel transfer (FGS-SPAIN and UFC 3-460-03 address grounding and bonding of fuel oil storage tanks and fuel transfers).

Conduct the fueling and lubricating of equipment and motor vehicles in a manner that protects against spills and evaporation. Prevent oil or hazardous substances from entering the ground, drainage areas, or navigable waters. All fueling operations including fuel delivery on base shall follow NAVSTA Rota fuel management procedures.

3.7 RELEASES/SPILLS OF OIL, HAZARDOUS SUBSTANCES AND AQUEOUS FILM FORMING FOAM (AFFF)

Exercise due diligence to prevent, contain, and respond to spills of hazardous material, hazardous substances, hazardous waste, sewage, regulated gas, petroleum, lubrication oil, and other substances regulated by environmental law. Maintain spill cleanup equipment and materials at the work site. In the event of a spill, take prompt, effective action to stop, contain, curtail, or otherwise limit the amount, duration, and severity of the spill/release. In the event of any releases of oil and hazardous substances, chemicals, or gases; immediately (within 15 minutes) notify the NAVSTA Rota Fire Department, phone # 9-1-1 or Dispatch phone # 956-82-2911. If the Contractor's response is inadequate, the U.S. Navy may respond. If this should occur, the Contractor will be required to reimburse the government for spill response assistance and analysis. If the spill can be contained and cleaned up using available spill kit/supplies and trained manpower capabilities, complete response operations/clean up the spill, if it is safe to do so. Then report incidents to base Environmental Division.

Should the potential exist for fuel and other spills, a spill prevention, control, and response plan is required specific to this project.

For all in-water or above water projects, a thorough spill prevention, control, and response plan is required.

Conduct the fueling and lubricating of equipment and motor vehicles in a manner that protects against spills and evaporation. Manage all used oil generated on site in accordance with FGS-SPAIN and Spanish requirements. Determine if any used oil generated while on-site exhibits a characteristic of hazardous waste. Used oil containing 1000 parts per million of solvents will be considered a hazardous waste and disposed of at Contractor's expense. Used oil mixed with a hazardous waste will also be considered a hazardous waste.

Submit verbal and written notifications as required by Installation instructions and Spill Prevention Plan immediately. Spill response must be in accordance with Chapter 18 of FGS-SPAIN and the Installation Spill Prevention Plan. Contain and clean up these spills without cost to the Government.

The person discovering/causing the spill shall conduct the following immediate first responder actions as appropriate; but only if properly trained and they can be done safely:

- Shutdown all appropriate manifold storage tank and truck valves and/or pumps.
- Restrict all ignition sources and utilize spill control devices and equipment.
- Notify supervisor and people working nearby; and evacuate the immediate area to a safe distance upwind.
- Aid exposed personnel by assisting in safely removing personnel from immediate spill area, and requesting medical assistance for an

ambulance (as needed)

- Secure storm drains, if any with storm drain covers or diversion barriers.

While reporting provide as much of the following information as possible

- Name, telephone number, and identification of caller;
- Time and location of spill;
- Identity and quantity of spill materials;
- Origin and cause of spill;
- Description of spill (behavior of spill, affected areas);
- Anticipated movement of spill;
- Planned or initiated actions;
- Type of assistance required;
- Stay in the area, if it is safe to do so, until FRT arrives and direct them to the spill;
- Provide SDSs as needed.

Report any Aqueous Film Forming Foam (AFFF) intentional or unintentional releases at any location (regardless of any amount) to the Environmental Division.

3.8 WATER AND AIR QUALITY PROTECTION MEASURES

All precautionary measures practicable shall be taken to prevent and minimize any leakage and emissions of ozone-depleting substances.

To reduce air pollution emissions, minimize the number of vehicle trips and vehicle idling.

Protect all water supply aquifers (groundwater) and surface water sources from contamination.

Use only lead-free and aluminum free pipe, solder, flux, and fittings in the installation or repair of potable water distribution and plumbing. Use lead check swab test kit on pipes, solder, flux, and fittings to verify compliance.

Use Best Management Practices (BMPs) that will minimize or eliminate the possibility of pollution being introduced into the continental and coastal waters of Spain. Should a discharge occur, immediately contact Naval Station Rota Fire Department (phone # 956-82-2911) or Dispatch (phone # 9-1-1).

3.9 AIR RESOURCES

3.9.1 Oil or Dual-fuel Boilers and Furnaces

Provide product data and details for new, replacement, or relocated fuel fired boilers, heaters, or furnaces to the Installation Environmental Division through the Contracting Officer. Data to be reported include: equipment purpose (water heater, building heat, process), manufacturer, model number, serial number, fuel type (oil type, gas type) size (MMBTU heat input).

3.9.2 Burning

Burning is prohibited on the Government premises.

3.9.3 Class I and II ODS Prohibition

Certain reclaimed Class I and II ODS must be returned to the Government for appropriate management. Coordinate with the Installation Environmental Division to determine the appropriate management of Class I and II ODS.

3.9.4 Accidental Venting of Refrigerant

Accidental venting of a refrigerant is a release and must be reported immediately to the Contracting Officer. Any class I or class II ODS, HFC, and PFC refrigerant listed in FGS Tables 2.2 and 2.3 shall not be intentionally released in the course of maintaining, servicing, repairing, or disposing of appliances, industrial process refrigeration units, air conditioning units, or motor vehicle air conditioners.

All repairs, including leak repairs or services to appliances, industrial process refrigeration units, air conditioning units, or motor vehicle air conditioners, must be performed using commercially available refrigerant recovery/recycling equipment operated by trained personnel.

3.9.5 Dust Control

If Contractor vehicles transfer mud or dirt onto roads, they must remove the materials from the roads. Keep dust down at all times, including during nonworking periods. Sprinkle or treat, with dust suppressants, the soil at the site, haul roads, and other areas disturbed by operations. Avoid use of potable water for these purposes if practicable. Dry power brooming will not be permitted. Instead, use vacuuming, wet mopping, wet sweeping, or wet power brooming. Air blowing will be permitted only for cleaning nonparticulate debris such as steel reinforcing bars. Only wet cutting will be permitted for cutting concrete blocks, concrete, and bituminous concrete. Do not unnecessarily shake bags of cement, concrete mortar, or plaster.

3.9.5.1 Particulates

Dust particles, aerosols and gaseous by-products from construction activities, and processing and preparation of materials (such as from asphaltic batch plants) must be controlled at all times, including weekends, holidays, and hours when work is not in progress. Maintain excavations, stockpiles, haul roads, permanent and temporary access roads, plant sites, spoil areas, borrow areas, and other work areas within or outside the project boundaries free from particulates that would exceed FGS standards or that would cause a hazard or a nuisance. Sprinkling, chemical treatment of an approved type, baghouse, scrubbers, electrostatic precipitators, or other methods will be permitted to control particulates in the work area. Sprinkling, to be efficient, must be repeated to keep the disturbed area damp. Avoid use of potable water for these purposes if practicable. Provide sufficient, competent equipment available to accomplish these tasks. Perform particulate control as the work proceeds and whenever a particulate nuisance or hazard occurs.

3.9.5.2 Abrasive Blasting

Blasting operations cannot be performed without prior approval of the Installation Environmental Division. The use of silica sand is prohibited in sandblasting.

Provide tarpaulin drop cloths and windscreens to enclose abrasive blasting operations to confine and collect dust, abrasive agent, paint chips, and other debris. Perform work involving removal of hazardous material in accordance with Chapter 5 of FGS-SPAIN. Blasting residues will be treated and disposed of as hazardous waste in accordance with Chapter 6 of FGS-SPAIN.

3.9.6 Odors

Control odors from construction activities. The odors may not constitute a health hazard.

3.10 WASTEWATER

3.10.1 Treatment

Do not allow wastewater from construction activities, such as onsite material processing, concrete curing, foundation and concrete clean-up, water used in concrete trucks, and forms to enter water ways or to be discharged prior to being treated to remove pollutants. Dispose of the construction-related waste water off-Government property by collecting and placing it in a retention pond where suspended material can be settled out or the water can evaporate to separate pollutants from the water or other means as agreed to by the Contracting Officer. The site for the retention pond must be coordinated and approved with the Environmental Division and Contracting Officer. The residue left in the pond prior to completion of the project must be removed, tested, and disposed of off-Government property in accordance with Chapter 6 of FGS-SPAIN. Backfill the area to the original grade, top-soiled, and seeded or sodded. Coordinate with the Environmental Division on test requirements for the water in the retention pond for and have the results reviewed and approved by the Contracting Officer prior to being discharged or disposed of off-Government property.

3.10.2 Surface Discharge

For discharge of ground water obtain authorization from the Environmental Division prior to surface discharging.

3.10.3 Land Application

Water generated from the flushing of lines after disinfection or disinfection in conjunction with hydrostatic testing may be land-applied after de-chlorination or discharged into the sanitary sewer with prior approval by the Environmental Division.

3.11 PREVIOUSLY USED EQUIPMENT

Clean previously used construction equipment prior to bringing it onto the project site. Equipment must be free from soil residuals, egg deposits from plant pests, noxious weeds, and plant seeds.

3.12 CONTROL AND MANAGEMENT OF ASBESTOS-CONTAINING MATERIAL (ACM)

Manage and dispose of asbestos-containing waste in accordance with FGS-SPAIN. Manifest asbestos-containing waste and provide the Final Spanish Manifest to the Contracting Officer. Notifications to the Installation Air Program Manager are required before starting any asbestos work.

The Contractor needs to contact the Public Works Asbestos Program Manager and Environmental Division in identifying Asbestos Containing Material (ACM) locations and procedures to follow during any construction/maintenance to ensure ACM detected areas are managed properly. If asbestos is found at any stage of the project in confined areas or left unnoticed during the environmental survey - DO NOT disturb the area, stop work in the area of the questionable material or condition and notify both the Contracting Officer and Environmental Division immediately for further guidance. Sealant and adhesives used in the project must not contain asbestos. Submit a certificate proving the asbestos-free nature of products for materials that suspect asbestos-containing materials (Sealant, Adhesives, gasket etc.).

3.13 CONTROL AND MANAGEMENT OF LEAD-BASED PAINT (LBP)

Manage and dispose of lead-contaminated waste in accordance with FGS-SPAIN. Manifest any lead-contaminated waste and provide the manifest to the Contracting Officer.

3.14 CONTROL AND MANAGEMENT OF POLYCHLORINATED BIPHENYLS (PCBS)

Manage and dispose of PCB-contaminated waste in accordance with FGS-SPAIN.

3.15 CONTROL AND MANAGEMENT OF LIGHTING BALLAST AND LAMPS CONTAINING PCBS

Manage and dispose of contaminated waste in accordance with Chapter 6 of FGS-SPAIN.

3.16 MILITARY MUNITIONS

In the event military munitions, as defined in 40 CFR 260, are discovered or uncovered, immediately stop work in that area and immediately inform the Contracting Officer.

3.17 INADVERTENT DISCOVERY OF PETROLEUM-CONTAMINATED SOIL OR HAZARDOUS WASTES

The Government has performed studies with sampling and testing of soils and underground water focused on petroleum-contaminated soils along terrain close or adjacent to the pathway of the pipe. The studies conclude to the existence of some contamination in soils. Disposal of contaminated soils will require the preparation of documents to obtain the permits for disposal, such as, but not limited to, risk analysis per RD 9/2005. If petroleum-contaminated soil, groundwater or suspected hazardous waste is found during construction, immediately notify the Contracting Officer. Do not disturb this material until authorized by the Contracting Officer.

3.18 CHLORDANE

Evaluate excess soils and concrete foundation debris generated during the demolition of housing units or other wooden structures for the presence of chlordane or other pesticides prior to reuse or final disposal.

3.19 SOUND INTRUSION

Make the maximum use of low-noise emission products. Blasting or use of explosives are not permitted without written permission from the

Contracting Officer, and then only during the designated times. Confine pie driving from 8 a.m. and 4 p.m., Monday through Friday, exclusive of holidays, unless otherwise specified. Keep construction activities under surveillance and control to minimize environment damage by noise.

3.20 POST CONSTRUCTION CLEANUP

Clean up areas used for construction in accordance with Contract Clause: "Cleaning Up". Unless otherwise instructed in writing by the Contracting Officer, remove traces of temporary construction facilities such as haul roads, work area, structures, foundations of temporary structures, stockpiles of excess or waste materials, and other vestiges of construction prior to final acceptance of the work. Grade parking area and similar temporarily used areas to conform to surrounding contours.

3.21 LAYDOWN AREA

The Contractor laydown area will be subjected to periodic environmental inspections to ensure compliance with applicable environmental regulations. During demobilization and prior to returning the laydown area back to Government, a walk through site inspection shall be coordinated in advance with the Environmental Division to ensure the area is free from contamination/spills, etc. If required, remediation processes will be conducted at Contractor's expense.

3.22 DRINKING WATER

Identify and protect all above or below-ground drinking water distribution lines and wellhead equipment during all phases of construction. All aspects of drinking water systems in the installation, whether for repair and/or new systems, shall use certified lead-free and aluminum free materials, including pipe, solder, flux, appurtenances and all fixtures in accordance with 40 CFR 143. Use lead check swab test kit on pipes, solder, flux, and fittings to verify compliance.

- a. Submit results of lead check swap test kit to the Environmental Division for approval.
- b. Testing is required if material is made in another country to ensure it meets FGS-SPAIN Criteria.

Pipes, blow-offs, valves, fittings, pipe materials and products including, but not limited to, paints, linings, coatings, adhesives, and lubricants in direct contact with potable water should be certified to NSF/ANSI Standard 61. Prior to purchase, materials and products will be submitted to the NAVSTA Rota Public Works Department for approval.

Any connection to the existing water distribution system shall be coordinated with the NAVSTA Rota Public Works Department. Contractor shall ensure Cross Connection (CC) control inspections are done and Back Flow Prevention (BFP) devices are properly installed and certified prior to looping or installation of any new plumbing devices to the distribution line. All new pipelines shall be flushed in accordance with an approved flushing plan that follows the Requirements of American Water Works Association (AWWA) Standard 651 for Disinfecting Water Mains latest edition. Flushing plan shall be submitted to the installation water management board for review/approval.

Upon completion of project, Contractor shall coordinate with the NAVSTA

ROTA Environmental Division and Public Works Department to arrange for residual chlorine and lead swab (on fixtures, solder etc.) testing a minimum of 24 hours in advance.

Use energy/water conservation products (such as energy star, high efficiency - low flow flushing system etc.). Coordinate with the NAVSTA Rota Public Works Department, UEM Energy Manager. Water distributed onboard shall not be used for any construction purpose.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 58 00

PROJECT IDENTIFICATION

08/19, CHG 5: 08/22

PART 1 GENERAL

- 1.1 QUALITY CONTROL
- 1.2 PROJECT IDENTIFICATION SIGN

PART 2 PRODUCTS

PART 3 EXECUTION

ATTACHMENTS:

project sign detail

-- End of Section Table of Contents --

SECTION 01 58 00

PROJECT IDENTIFICATION
08/19, CHG 5: 08/22

PART 1 GENERAL

1.1 QUALITY CONTROL

1.2 PROJECT IDENTIFICATION SIGN

Prior to initiating any work on site, provide one project identification sign at the location designated. Construct the sign in accordance with project sign detail, which can be downloaded at: <http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>. Maintain sign throughout the life of the project. Upon completion of the project, remove the sign from the site.

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

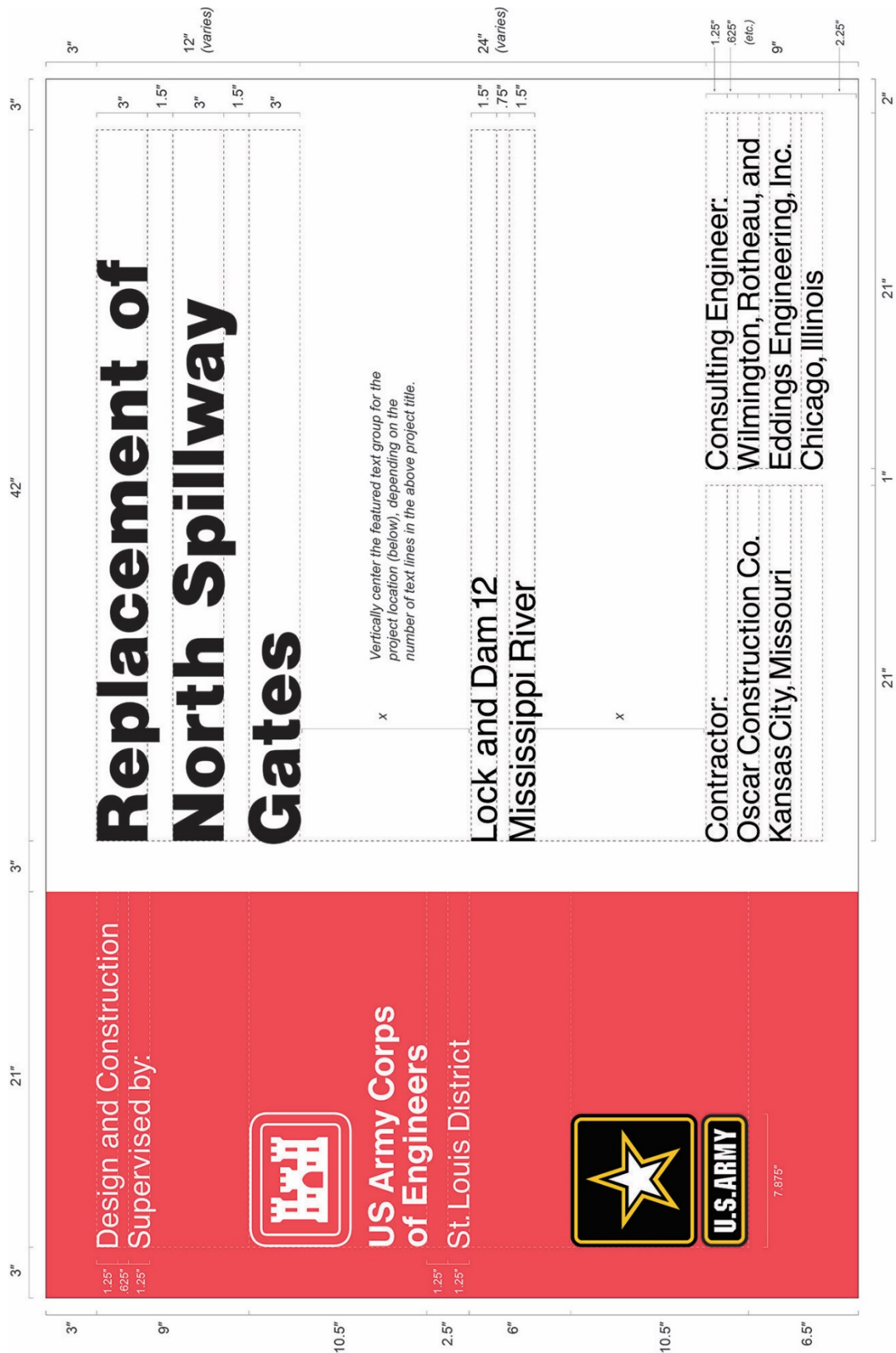
Not Used

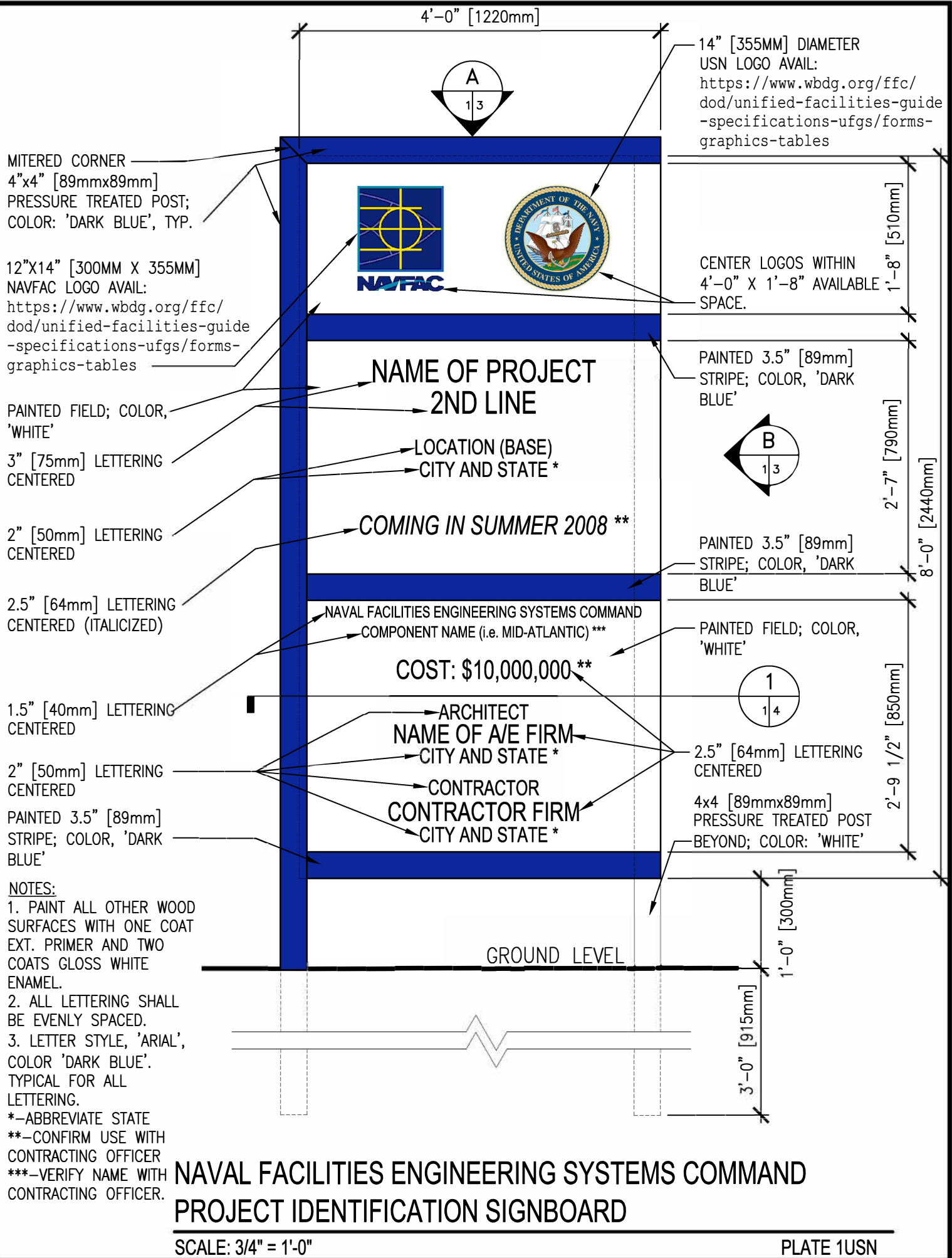
-- End of Section --

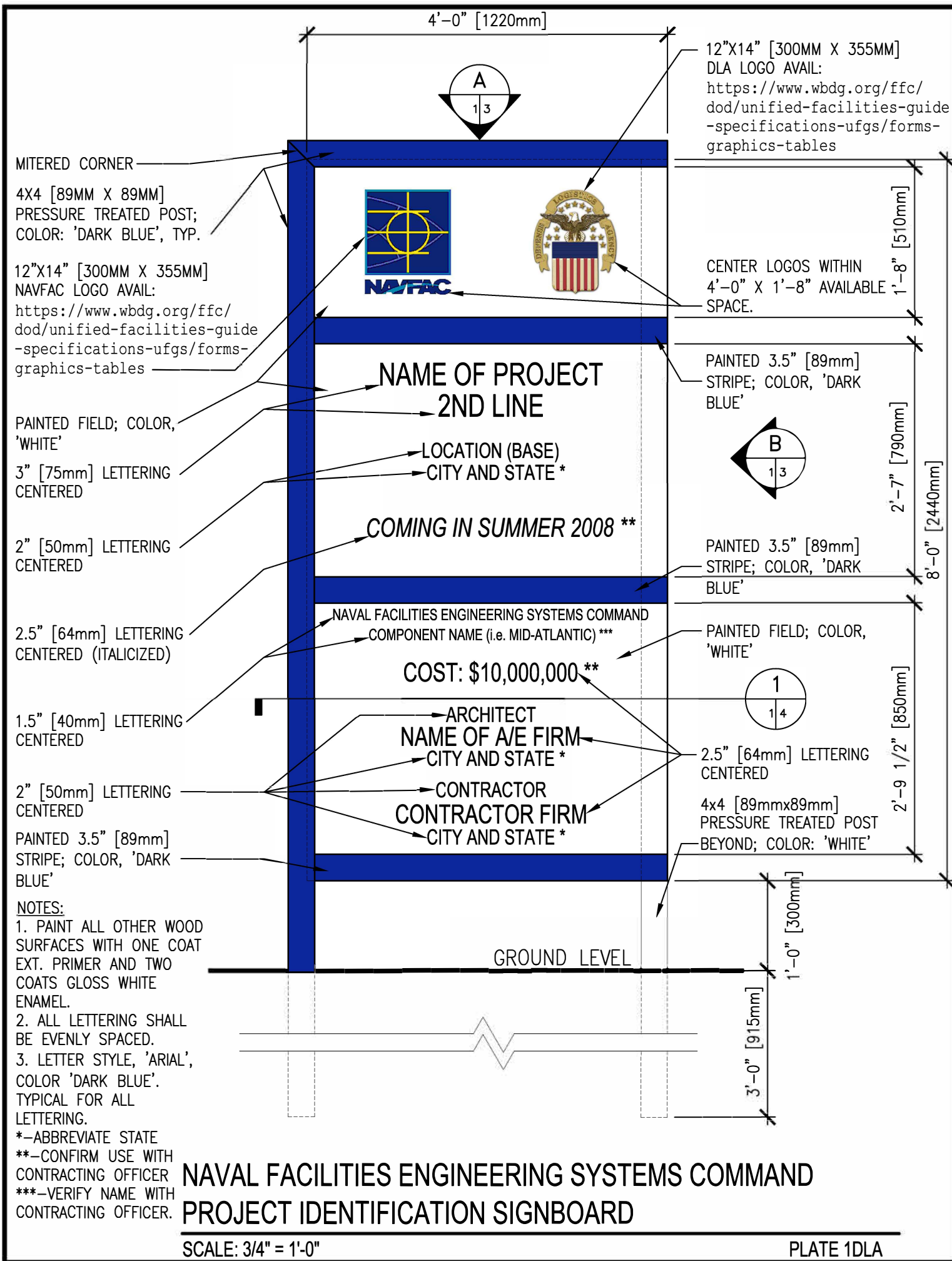
ATTACHMENT A: Example Graphic of Signage with Dimensions for MILCON Project

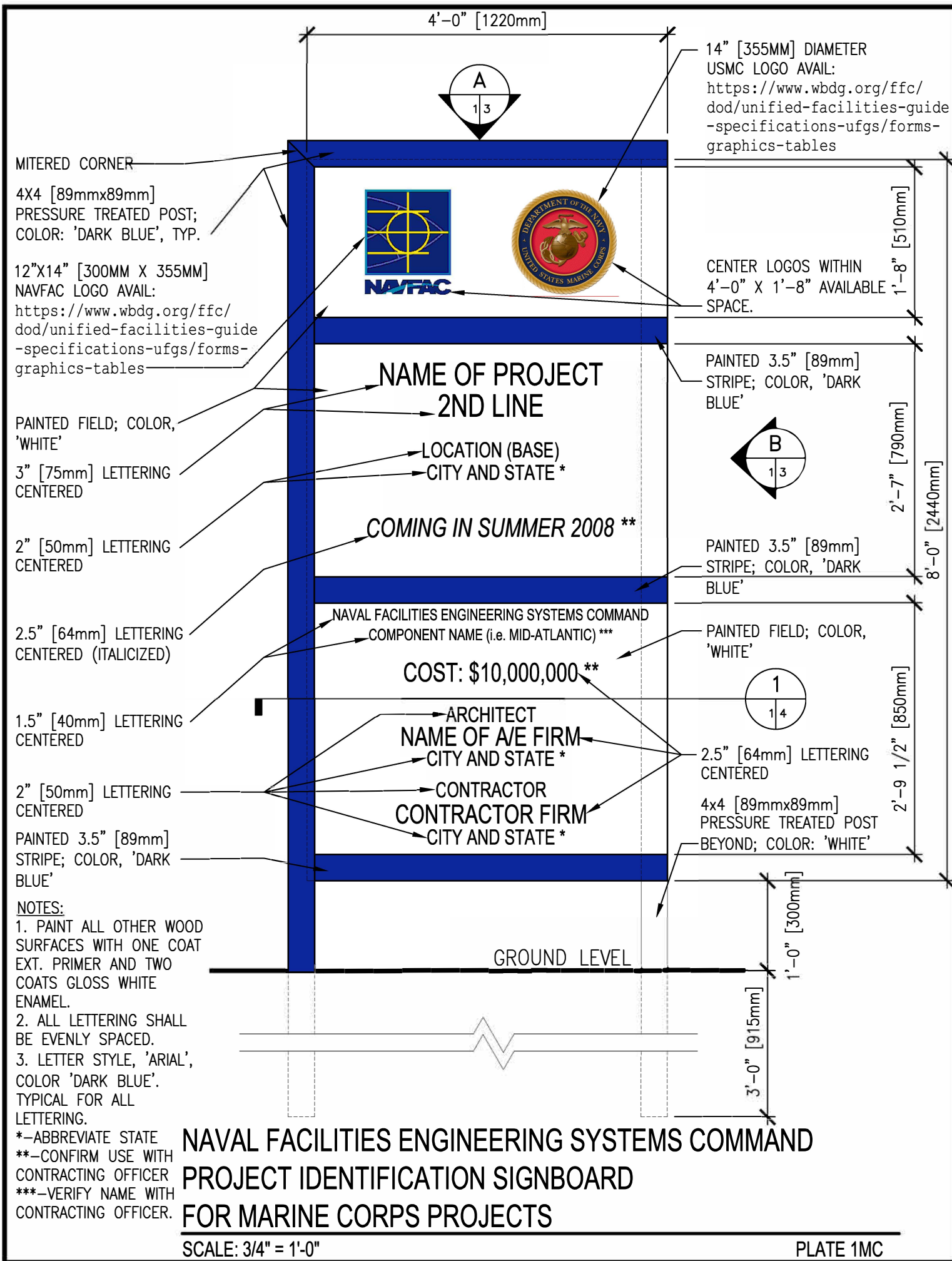


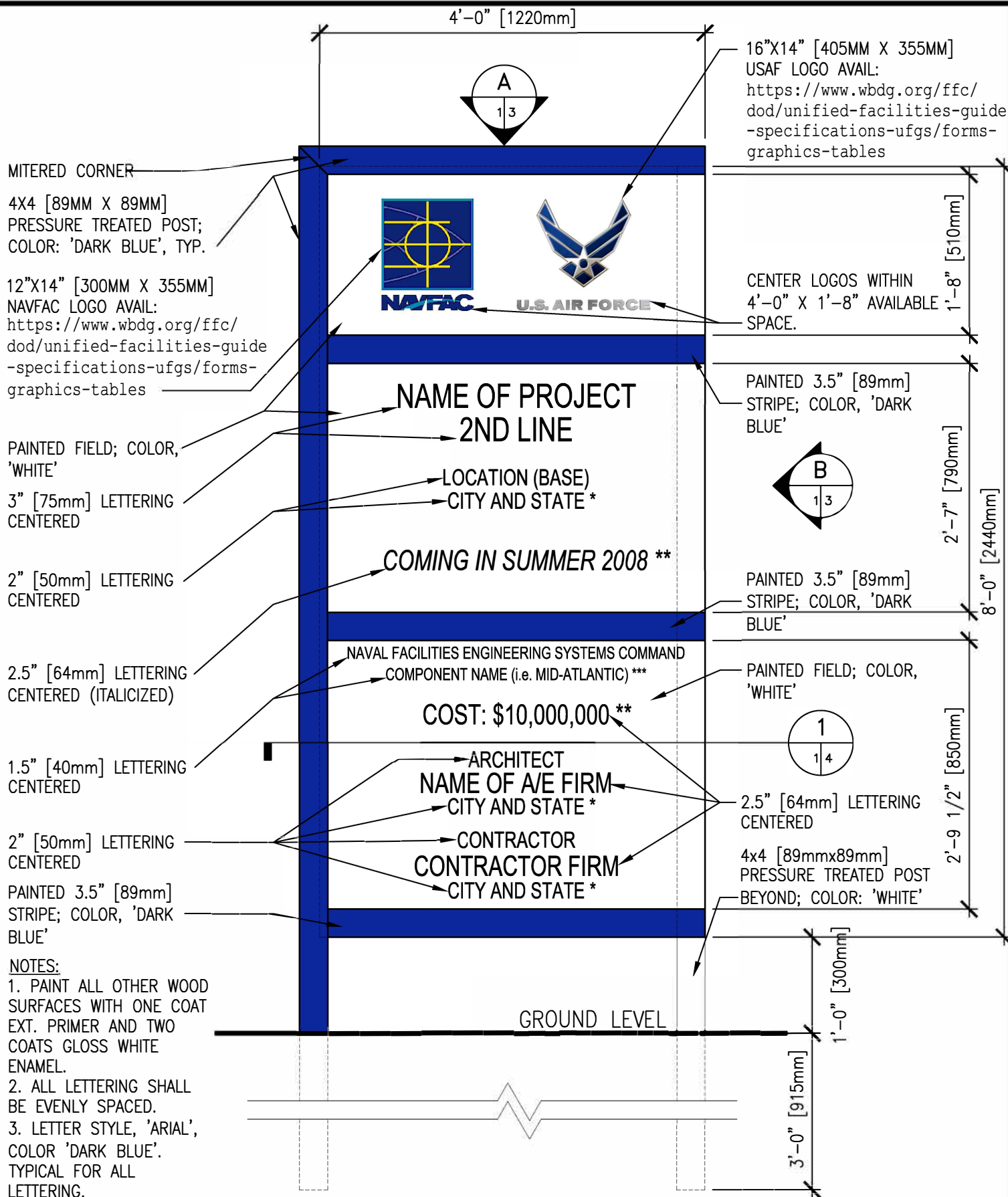
ATTACHMENT B: Example Graphic of Signage with Dimensions for Civil Works Project







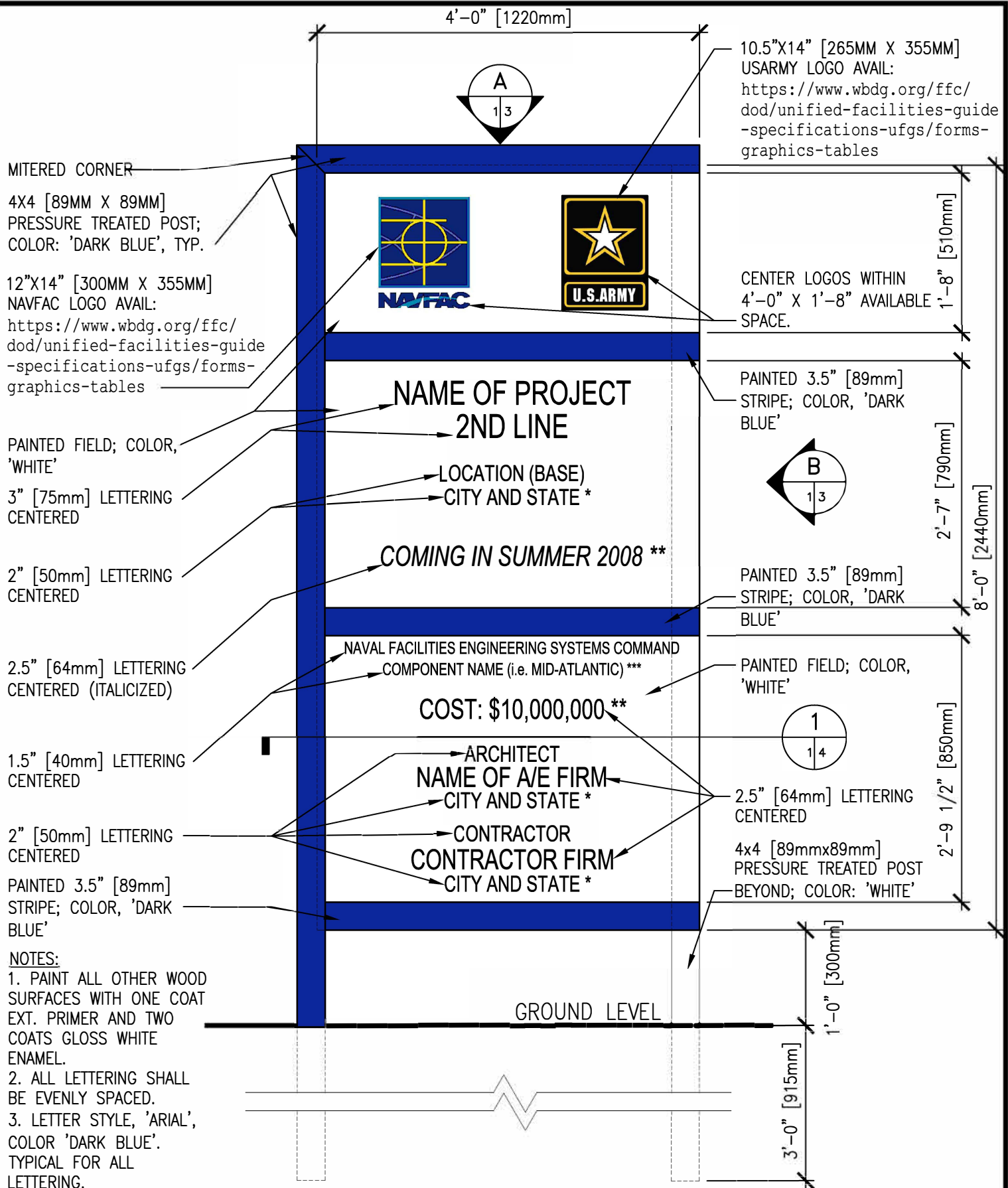




NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND PROJECT IDENTIFICATION SIGNBOARD

SCALE: 3/4" = 1'-0"

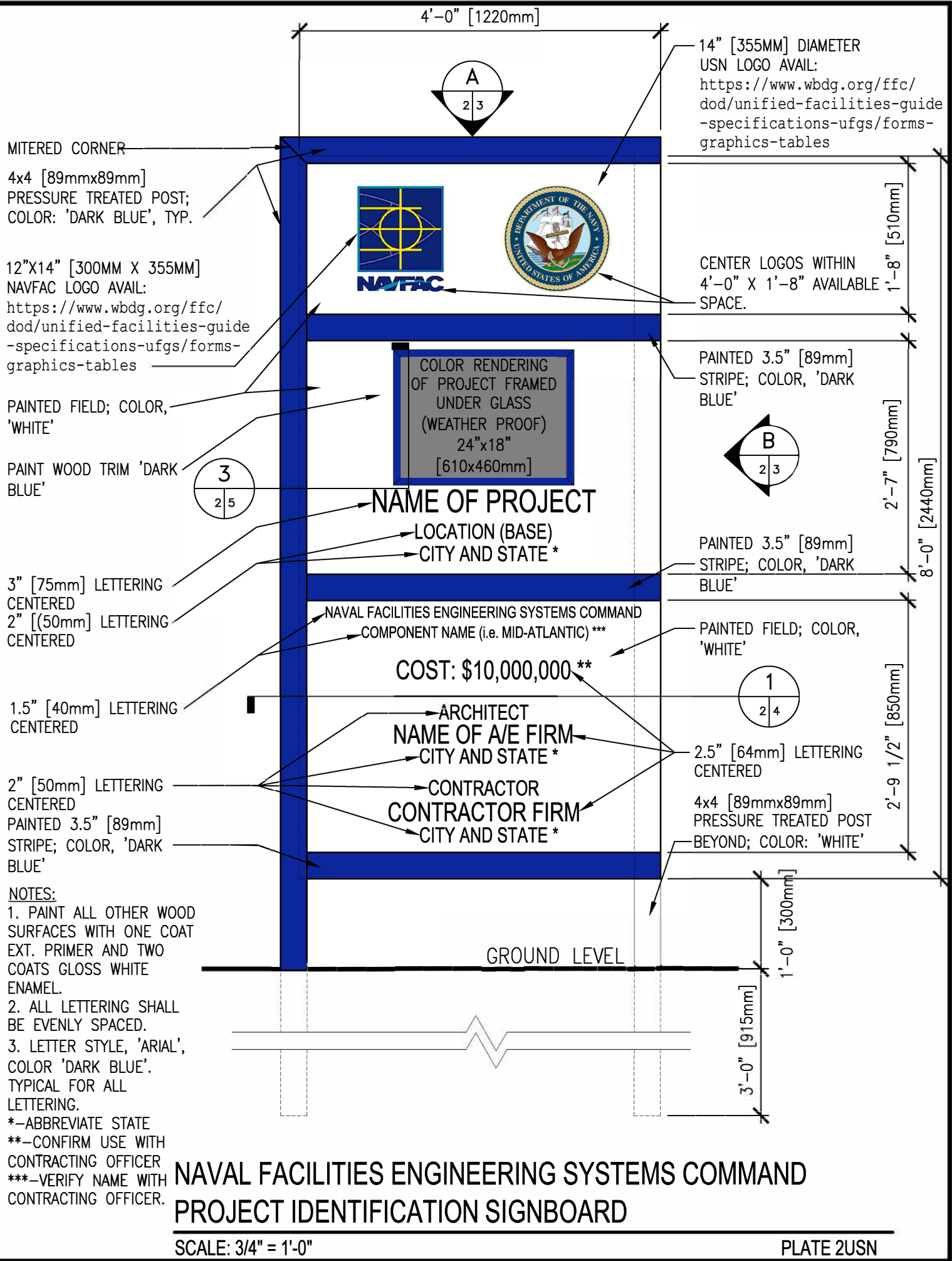
PLATE 1USAF

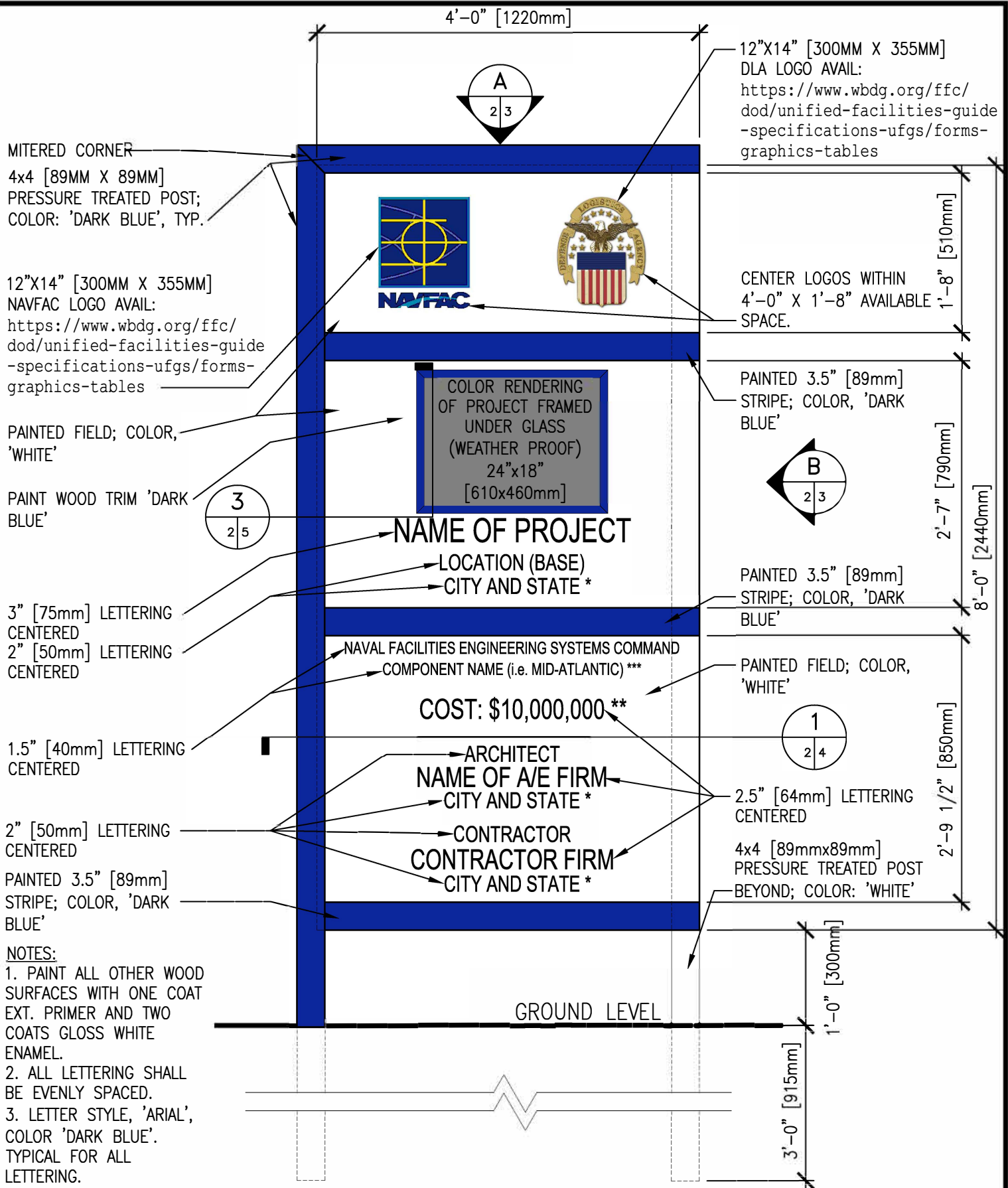


NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND PROJECT IDENTIFICATION SIGNBOARD

SCALE: 3/4" = 1'-0"

PLATE 1USARMY





NOTES:
1. PAINT ALL OTHER WOOD SURFACES WITH ONE COAT EXT. PRIMER AND TWO COATS GLOSS WHITE ENAMEL.

2. ALL LETTERING SHALL BE EVENLY SPACED.
3. LETTER STYLE, 'ARIAL', COLOR 'DARK BLUE'. TYPICAL FOR ALL LETTERING.

*-ABBREVIATE STATE

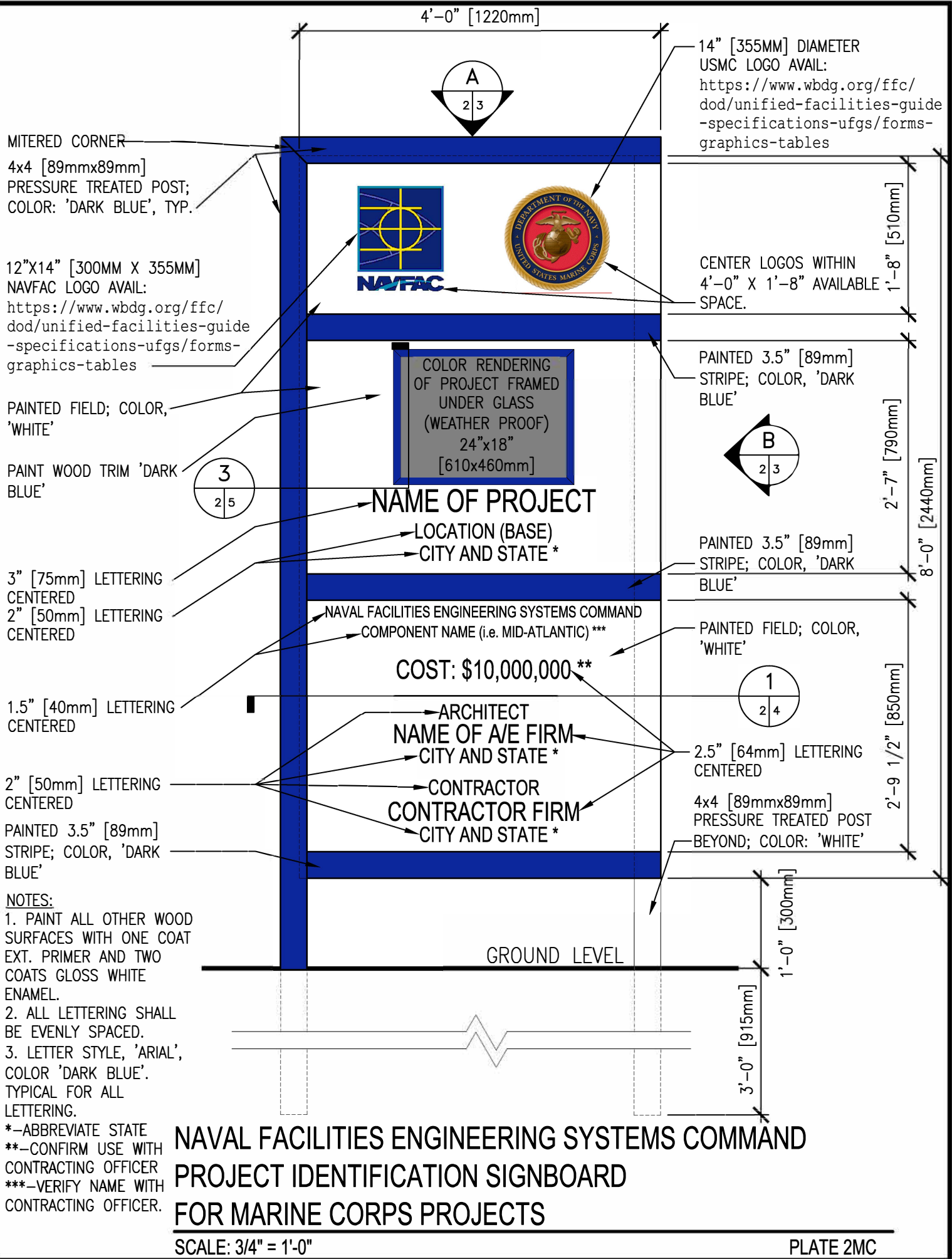
**-CONFIRM USE WITH CONTRACTING OFFICER

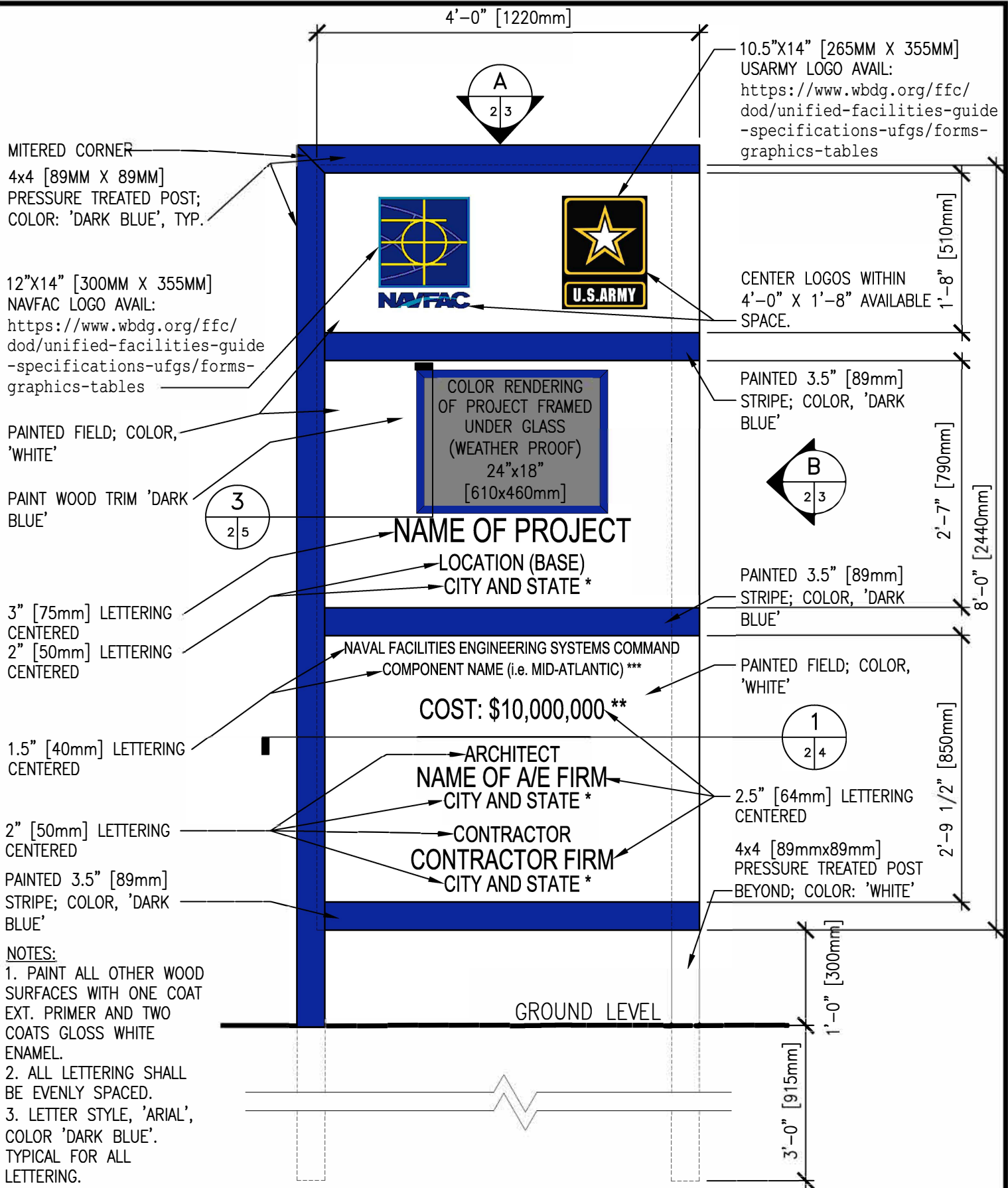
***-VERIFY NAME WITH CONTRACTING OFFICER.

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND PROJECT IDENTIFICATION SIGNBOARD

SCALE: 3/4" = 1'-0"

PLATE 2DLA

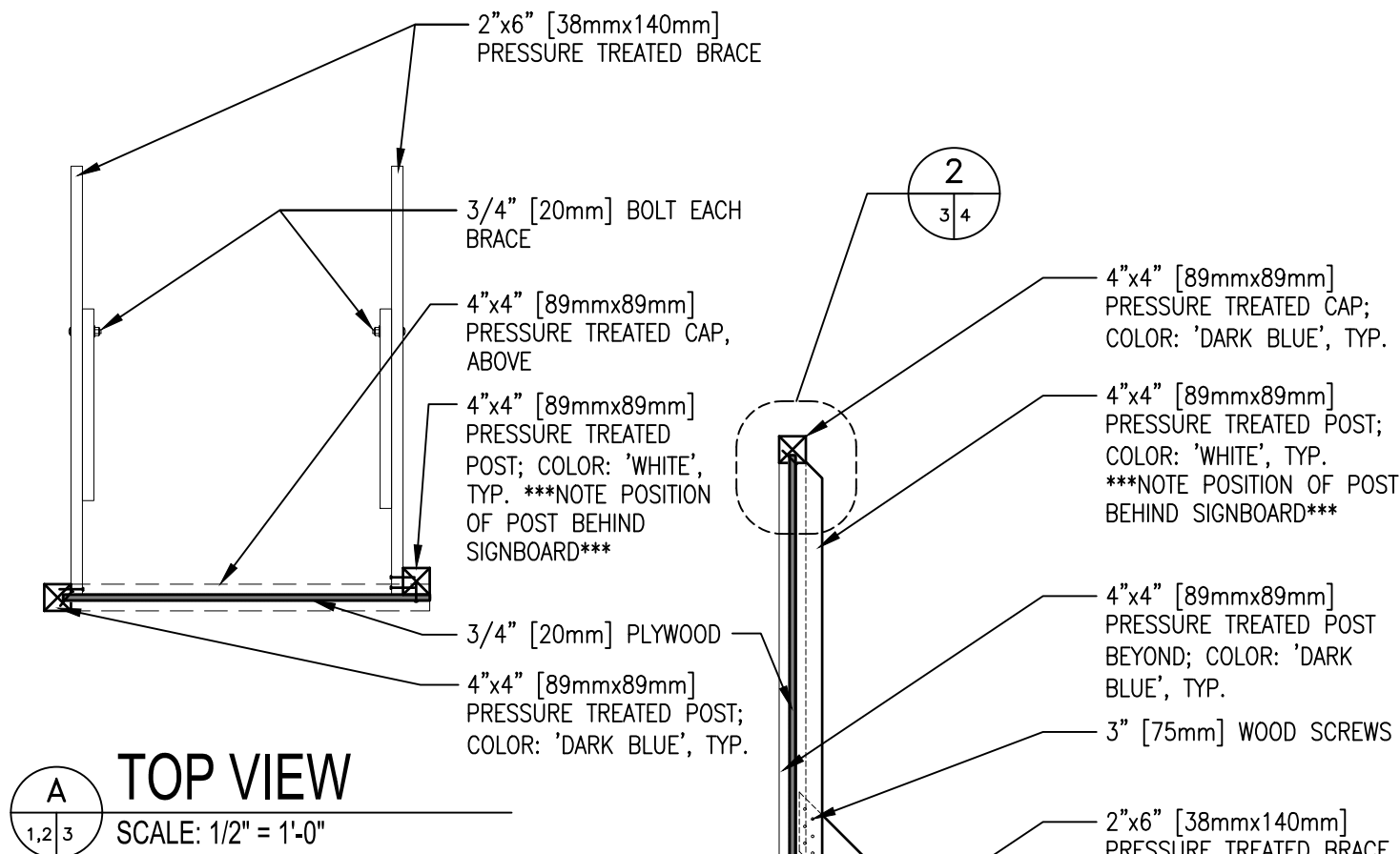




NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND PROJECT IDENTIFICATION SIGNBOARD

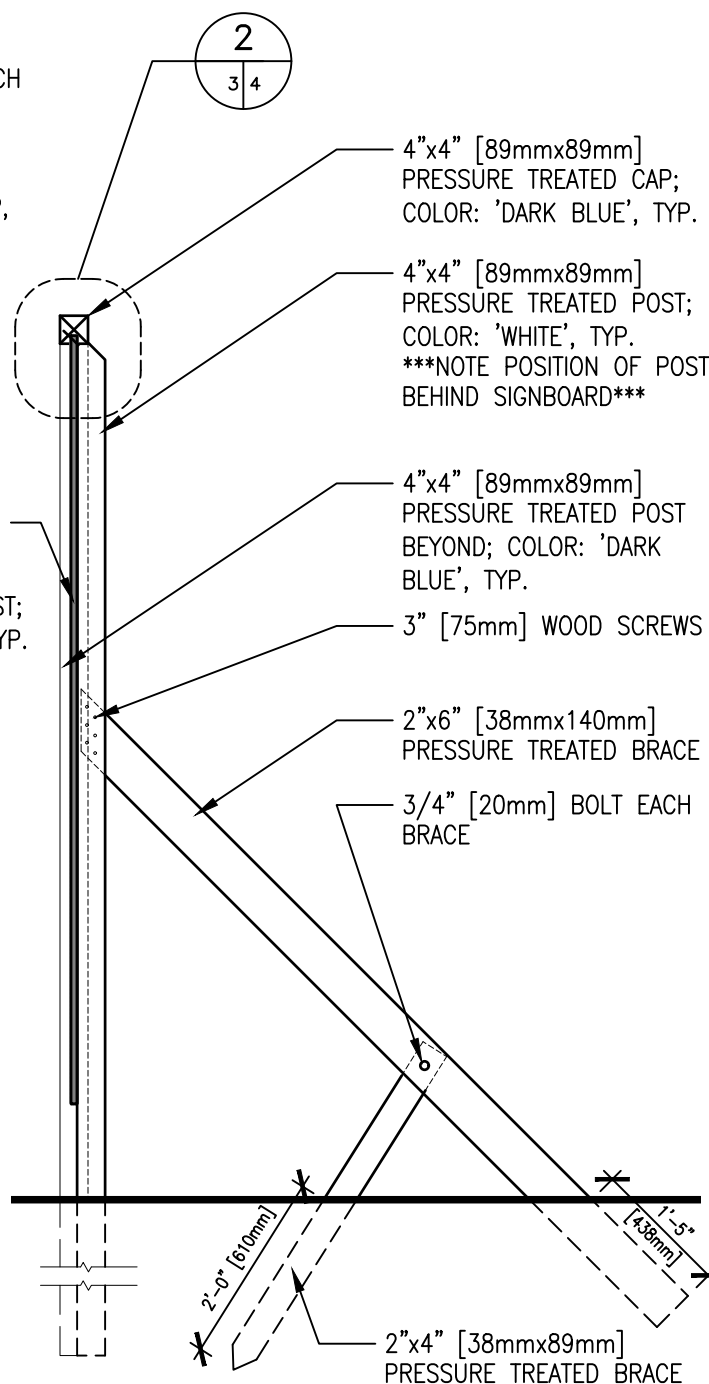
SCALE: 3/4" = 1'-0"

PLATE 2USARMY

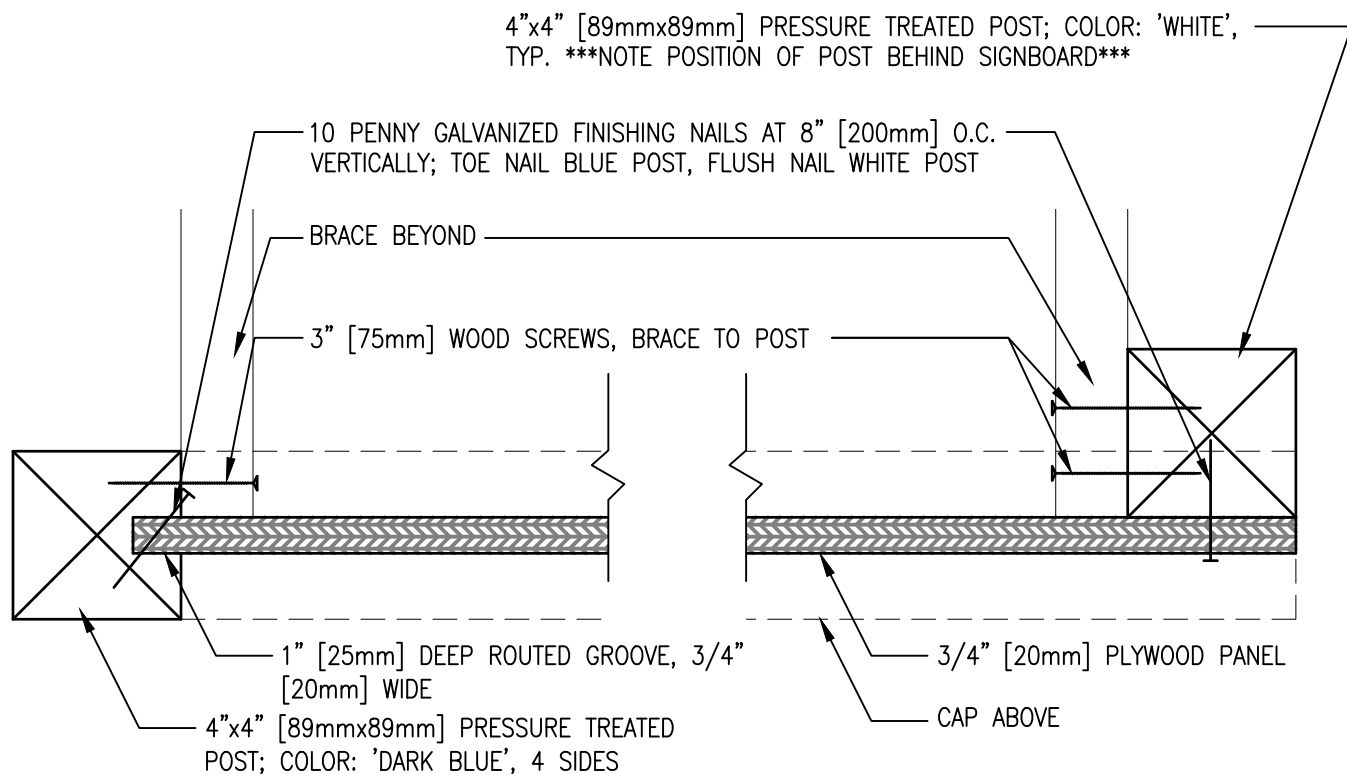


NOTES:

1. POSTS AND BRACES SHALL BE PRESSURE TREATED.
2. ALL FASTENERS SHALL BE ZINC COATED.
3. BRACING IS REQUIRED IN ALL SOIL CONDITIONS AND HIGH WIND ENVIRONMENTS.

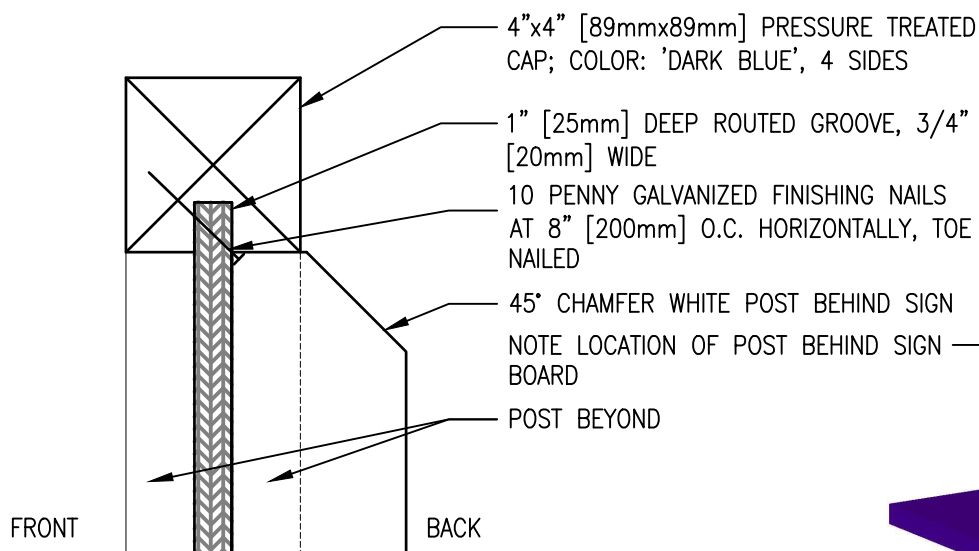


NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD
SUPPORT DETAILS



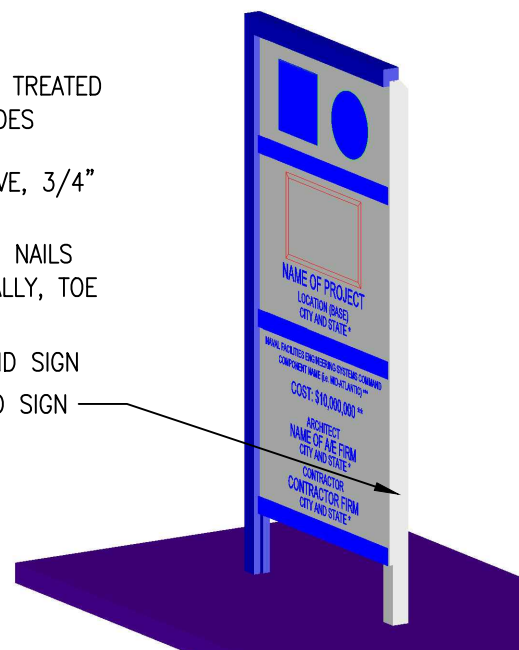
1 PLAN SECTION

SCALE: 3" = 1'-0"



2 SECTION AT TOP

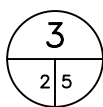
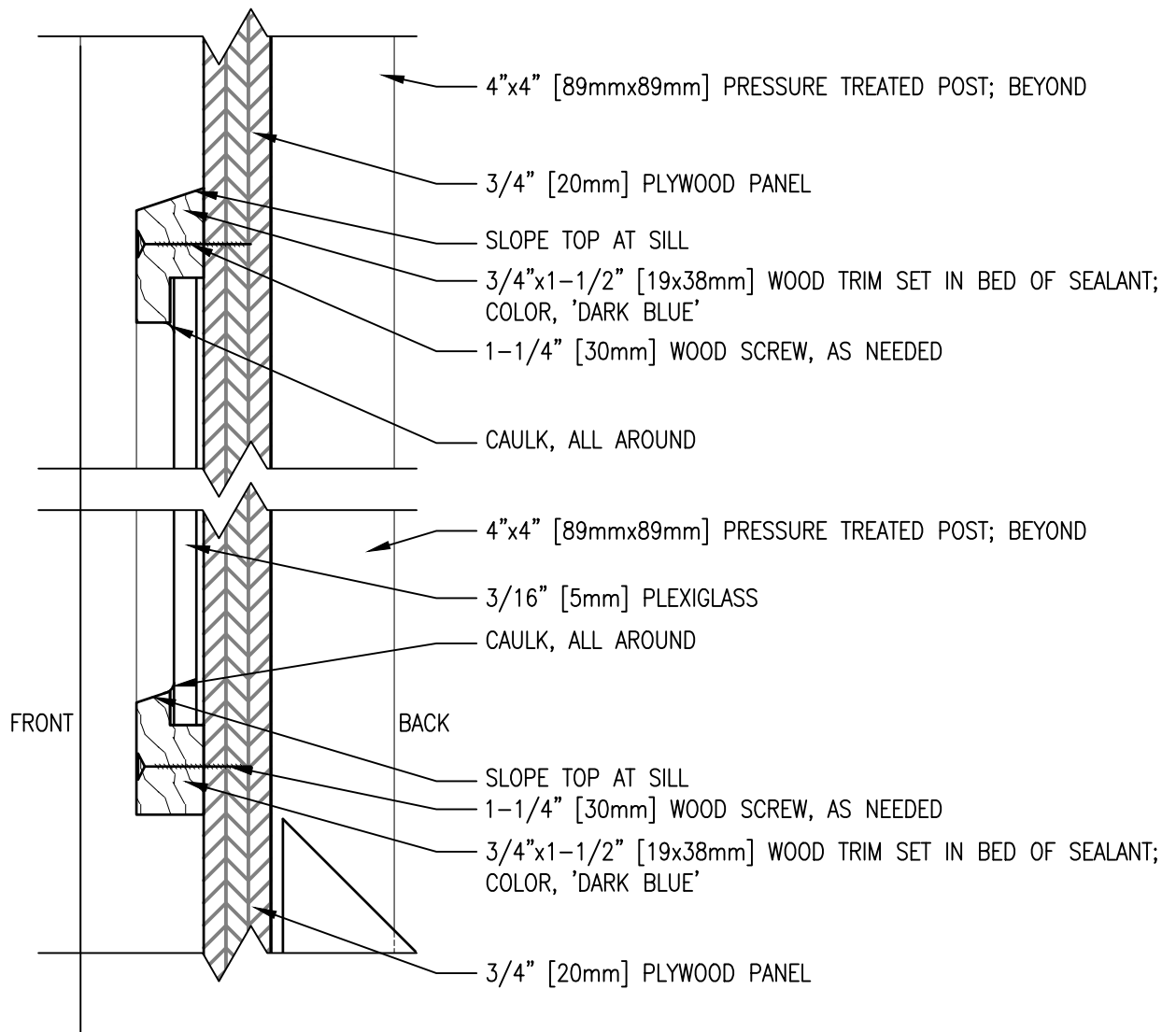
SCALE: 3" = 1'-0"



3 ISO VIEW

SCALE: NONE

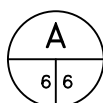
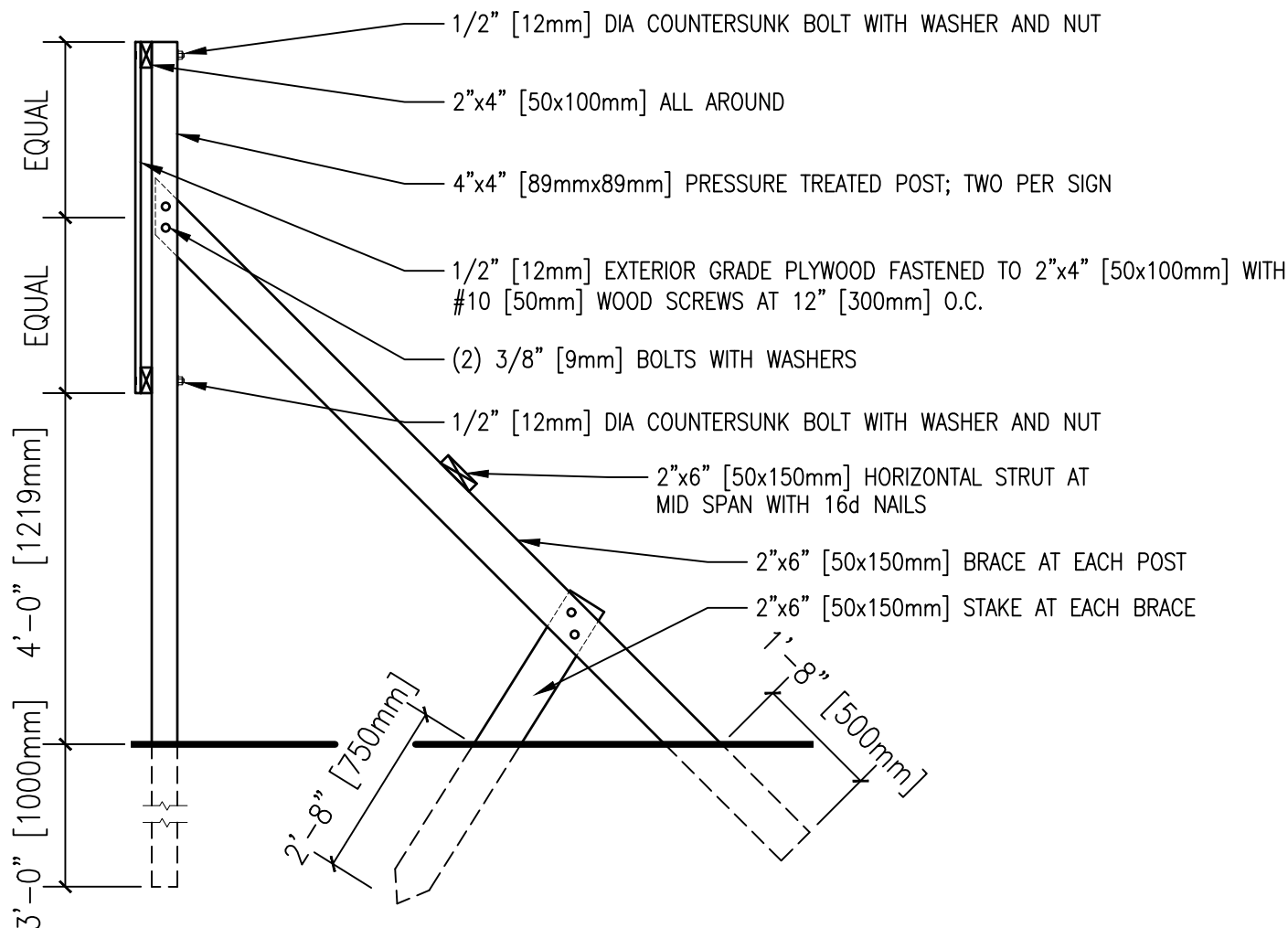
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD SECTIONS



SECTION AT RENDERING FRAME

SCALE: 6" = 1'-0"

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD SECTION



SECTION

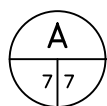
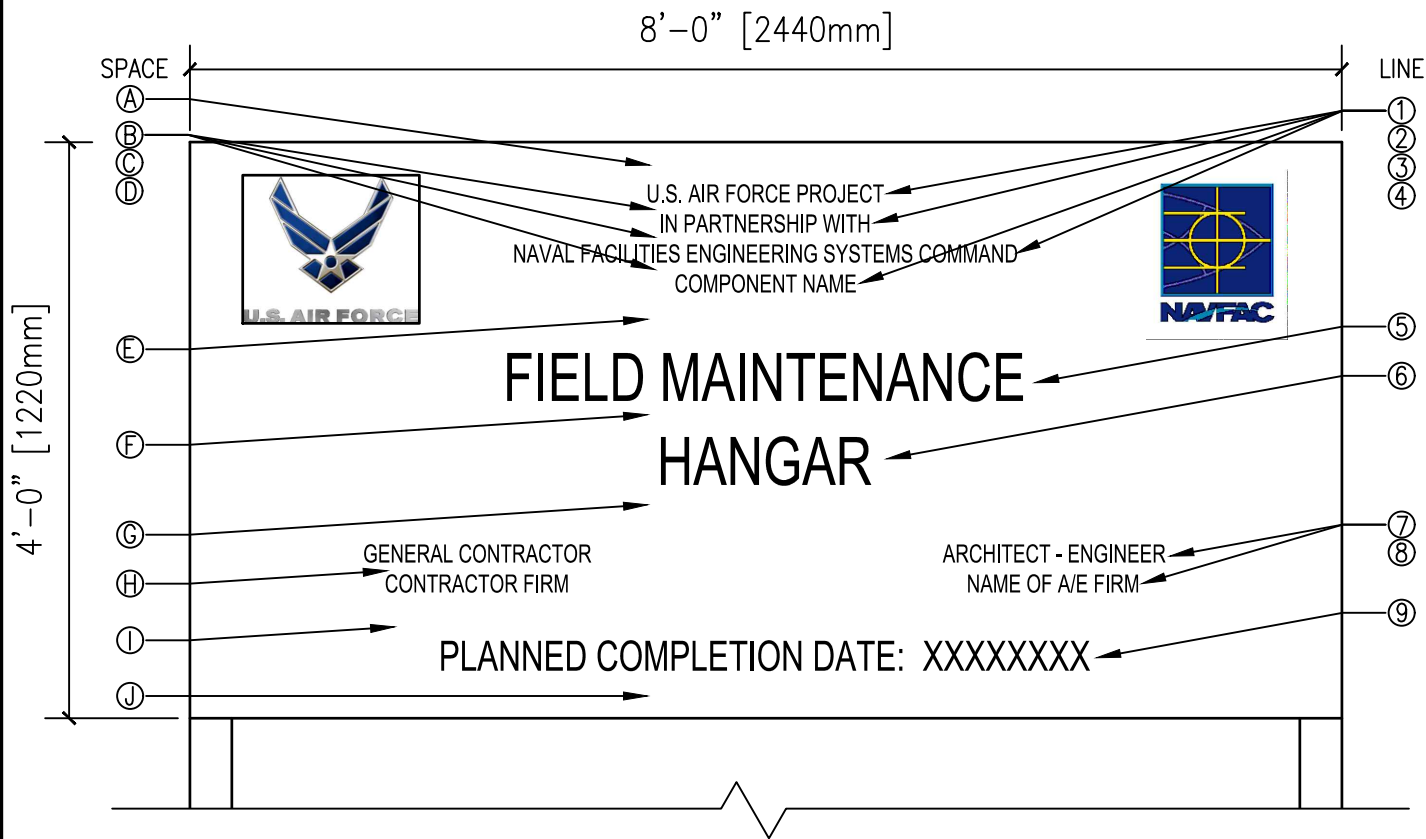
SCALE: 1/2" = 1'-0"

NOTES:

1. THE SIGN MUST BE CONSTRUCTED OF 4'x8' [1200mmx2400mm] 1/2" [12mm] GRADE A-C, EXTERIOR TYPE PLYWOOD MOUNTED ON A FRAME OF TREATED LUMBER.
2. THE HOST MAJCOM WILL PROVIDE SIGN COLORS AND DETAILS TO THE DESIGN MANAGER IN THE REQUIREMENTS AND MANAGEMENT PLAN (RAMP).
3. ALL LETTERING SHALL BE UPPERCASE HELVETICA MEDIUM.
4. PRESERVATIVE TREAT ALL WOOD SURFACES BURIED OR IN CONTACT WITH THE EARTH.
5. ALL FASTENERS SHALL BE GALVANIZED COATED.
6. LUMBER SHALL BE NON STRESS GRADE: $F_t = 8.5^5 \text{ KGS/M}^2 \text{ (1200PSI)}$
 $F_t = 10.5^5 \text{ KGS/M}^2 \text{ (1500PSI)}$

7. THE ENTIRE SIGNBOARD AND SUPPORTS SHALL BE GIVEN ONE COAT OF EXTERIOR ALKYD PRIMER AND TWO COATS OF EXTERIOR ALKYD ENAMEL PAINT. THE LETTERING AND SIGN WORK SHALL BE PERFORMED BY A SKILLED SIGN PAINTER USING PAINT KNOWN IN THE TRADE AS BULLETIN COLORS. WHERE PRESERVATIVE-TREATED LUMBER IS REQUIRED, UTILIZE ONLY CURED PRESSURE-TREATED WOOD WHICH HAS HAD THE CHEMICALS LEACHED FROM THE SURFACE OF THE WOOD PRIOR TO PAINTING.

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD SECTION
U.S. AIR FORCE PROJECTS



SIGNBOARD

SCALE: 3/4" = 1'-0"

SPACE	HEIGHT	LINE	DESCRIPTION	LETTER HEIGHT	STROKE
A	3.5"	①	U.S. AIR FORCE PROJECT	1.5"	3/16"
B	1"	②	IN PARTNERSHIP WITH	1.5"	3/16"
C	1"	③	NAVAL FACILITIES ENGINEERING COMMAND	1.5"	3/16"
D	1"	④	COMPONENT NAME	1.5"	3/16"
E	5"	⑤	PROJECT NAME	4"	1/2"
F	3"	⑥	PROJECT NAME CONT'D (IF REQUIRED)	4"	1/2"
G	5"	⑦	GENERAL CONTRACTOR/ A-E	1.5"	3/16"
H	1"	⑧	GENERAL CONTRACTOR/ A-E	1.5"	3/16"
I	4"	⑨	PLANNED COMPLETION DATE:	2.5"	1/4"
J	4"				

NOTES:

1. ALL LETTERING SHALL BE UPPERCASE HELVETICA MEDIUM.

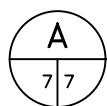
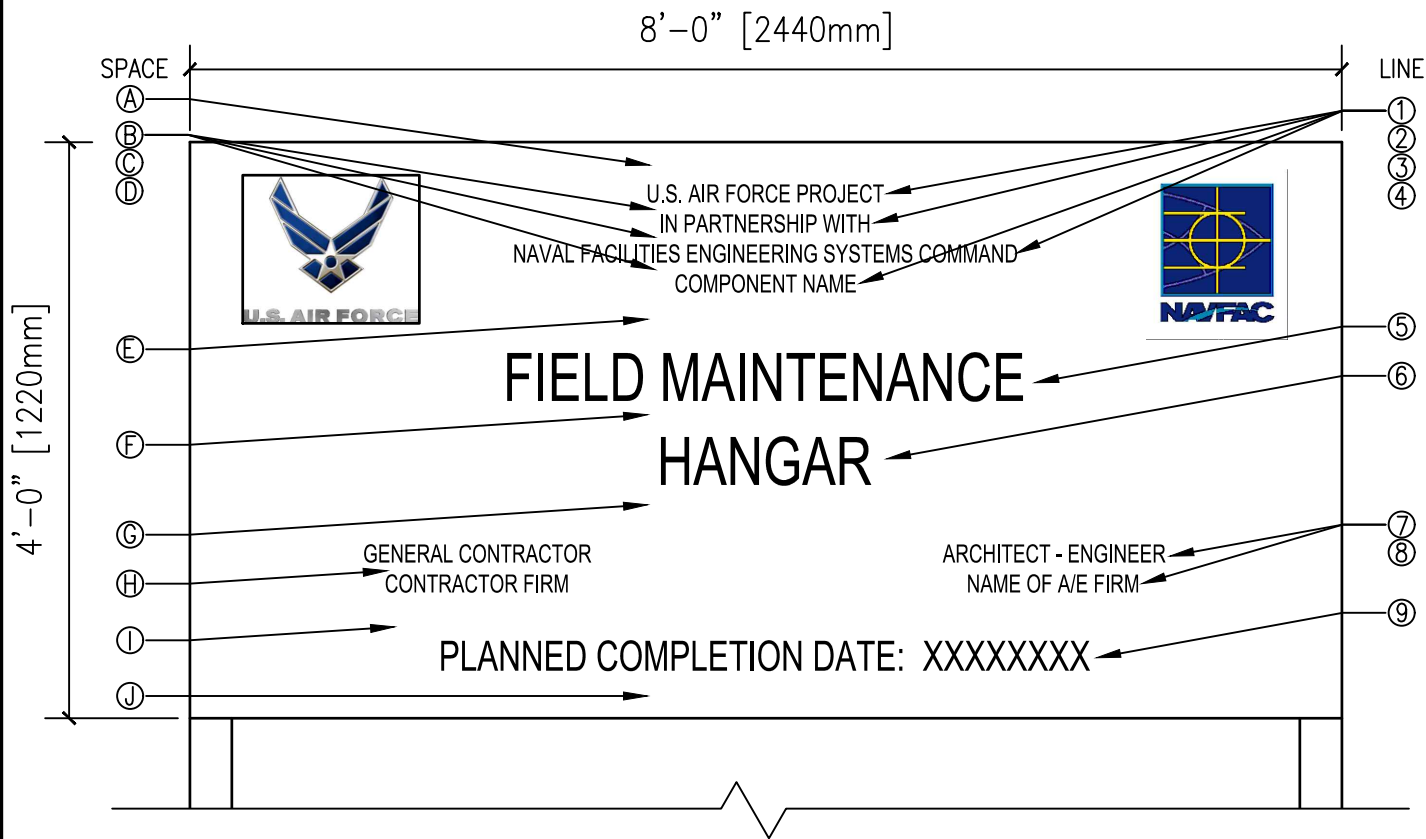
2. NAVFAC LOGO AVAILABLE AT:

<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>

3. AIR FORCE LOGO AVAILABLE AT:

<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD
U.S. AIR FORCE PROJECTS



SIGNBOARD

SCALE: 3/4" = 1'-0"

SPACE	HEIGHT	LINE	DESCRIPTION	LETTER HEIGHT	STROKE
A	3.5"	①	U.S. AIR FORCE PROJECT	1.5"	3/16"
B	1"	②	IN PARTNERSHIP WITH	1.5"	3/16"
C	1"	③	NAVAL FACILITIES ENGINEERING COMMAND	1.5"	3/16"
D	1"	④	COMPONENT NAME	1.5"	3/16"
E	5"	⑤	PROJECT NAME	4"	1/2"
F	3"	⑥	PROJECT NAME CONT'D (IF REQUIRED)	4"	1/2"
G	5"	⑦	GENERAL CONTRACTOR/ A-E	1.5"	3/16"
H	1"	⑧	GENERAL CONTRACTOR/ A-E	1.5"	3/16"
I	4"	⑨	PLANNED COMPLETION DATE:	2.5"	1/4"
J	4"				

NOTES:

1. ALL LETTERING SHALL BE UPPERCASE HELVETICA MEDIUM.

2. NAVFAC LOGO AVAILABLE AT:

<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>

3. AIR FORCE LOGO AVAILABLE AT:

<http://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/forms-graphics-tables>

NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
PROJECT IDENTIFICATION SIGNBOARD
U.S. AIR FORCE PROJECTS

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL

02/19, CHG 3: 11/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Co-mingle
 - 1.2.2 Construction Waste
 - 1.2.3 Demolition Debris/Waste
 - 1.2.4 Disposal
 - 1.2.5 Diversion
 - 1.2.6 Final Construction Waste Diversion Report
 - 1.2.7 Recycling
 - 1.2.8 Reuse
 - 1.2.9 Salvage
 - 1.2.10 Source Separation
- 1.3 CONSTRUCTION WASTE (INCLUDES DEMOLITION DEBRIS/WASTE)
- 1.4 CONSTRUCTION WASTE MANAGEMENT
 - 1.4.1 Implementation of Construction Waste Management Program
 - 1.4.2 Oversight
 - 1.4.3 Special Programs
 - 1.4.4 Special Instructions
 - 1.4.5 Waste Streams
- 1.5 SUBMITTALS
- 1.6 MEETINGS
- 1.7 CONSTRUCTION WASTE MANAGEMENT PLAN
- 1.8 RECORDS (DOCUMENTATION)
 - 1.8.1 General
 - 1.8.2 Accumulated
- 1.9 REPORTS
 - 1.9.1 General
- 1.10 FINAL CONSTRUCTION WASTE DIVERSION REPORT
- 1.11 COLLECTION
 - 1.11.1 Source Separation Method
 - 1.11.2 Co-Mingled Method
 - 1.11.3 Other Methods
- 1.12 DISPOSAL
 - 1.12.1 Reuse
 - 1.12.2 Recycle
 - 1.12.3 Compost
 - 1.12.4 Waste

PART 2 PRODUCTS

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 01 74 19

CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL
02/19, CHG 3: 11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

MINISTERIO DE LA PRESIDENCIA (MP)

RD 105/2008	(2008) Real Decreto 105/2008, de 1 de febrero, por el que se regula la producción y gestión de los residuos de construcción y demolición.
-------------	---

MINISTERIO DE MEDIO AMBIENTE (MA)

Orden MAM/304/2002	(2002) Orden MAM/304/2002, de 8 de febrero, por la que se publican las operaciones de valorización y eliminación de residuos y la lista europea de residuos.
--------------------	--

1.2 DEFINITIONS

1.2.1 Co-mingle

The practice of placing unrelated materials together in a single container, usually for benefits of convenience and speed.

1.2.2 Construction Waste

Waste generated by construction activities, such as scrap materials, damaged or spoiled materials, temporary and expendable construction materials, and other waste generated by the workforce during construction activities.

1.2.3 Demolition Debris/Waste

Waste generated from demolition activities, including minor incidental demolition waste materials generated as a result of Intentional dismantling of all or portions of a building, to include clearing of building contents that have been destroyed or damaged.

1.2.4 Disposal

Depositing waste in a solid waste disposal facility, usually a managed landfill or incinerator, regulated in the US under the Resource Conservation and Recovery Act (RCRA).

1.2.5 Diversion

The practice of diverting waste from disposal in a landfill or

incinerator, by means of eliminating or minimizing waste, or reuse of materials.

1.2.6 Final Construction Waste Diversion Report

A written assertion by a material recovery facility operator identifying constituent materials diverted from disposal, usually including summary tabulations of materials, weight in short-ton.

1.2.7 Recycling

The series of activities, including collection, separation, and processing, by which products or other materials are diverted from the solid waste stream for use in the form of raw materials in the manufacture of new products sold or distributed in commerce, or the reuse of such materials as substitutes for goods made of virgin materials, other than fuel.

1.2.8 Reuse

The use of a product or materials again for the same purpose, in its original form or with little enhancement or change.

1.2.9 Salvage

Usable, salable items derived from buildings undergoing demolition or deconstruction, parts from vehicles, machinery, other equipment, or other components.

1.2.10 Source Separation

The practice of administering and implementing a management strategy to identify and segregate unrelated waste at the first opportunity.

1.3 CONSTRUCTION WASTE (INCLUDES DEMOLITION DEBRIS/WASTE)

Divert a minimum of 70 percent by weight of the project construction waste and demolition debris/waste, except soils from earthwork activities, from the landfill or incinerator. Follow applicable industry standards in the management of waste. Apply sound environmental principles in the management of waste. (1) Practice efficient waste management when sizing, cutting, and installing products and materials and (2) use all reasonable means to divert construction waste and demolition debris/waste from landfills and incinerators and to facilitate the recycling or reuse of excess construction materials.

1.4 CONSTRUCTION WASTE MANAGEMENT

Implement a Construction Waste Management Program for the project, in compliance with RD 105/2008. Provide the services of an approved waste management firm (Gestor de Residuos). Take a pro-active, responsible role in the management of construction construction waste, recycling process, disposal of demolition debris/waste, and require all subcontractors, vendors, and suppliers to participate in the Construction Waste Management Program. Establish a process for clear tracking, and documentation of construction waste and demolition debris/waste.

1.4.1 Implementation of Construction Waste Management Program

Develop and document how the Construction Waste Management Program will be implemented in a Construction Waste Management Plan. Submit a Construction Waste Management Plan to the Contracting Officer for approval. Construction waste and demolition debris/waste materials include un-used construction materials not incorporated in the final work, as well as demolition debris/waste materials from demolition activities or deconstruction activities. In the management of waste, consider the availability of viable markets, the condition of materials, the ability to provide material in suitable condition and in a quantity acceptable to available markets, and time constraints imposed by internal project completion mandates.

1.4.2 Oversight

The Environmental Coordinator, as specified in Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS, is responsible for overseeing and documenting results from executing the Construction Waste Management Plan for the project.

1.4.3 Special Programs

Implement special programs involving rebates or similar incentives related to recycling of construction waste and demolition debris/waste materials. Retain revenue or savings from salvaged or recycling, unless otherwise directed. Ensure firms and facilities used for recycling, reuse, and disposal are permitted for the intended use to the extent required by federal, state, and local regulations.

1.4.4 Special Instructions

Provide on-site instruction of appropriate separation, handling, recycling, salvage, reuse, and return methods to be used by all parties at the appropriate stages of the projects. Designation of single source separating or commingling will be clearly marked on the containers.

1.4.5 Waste Streams

Delineate waste streams and characterization, including estimated material types and quantities of waste, in the Construction Waste Management Plan and Orden MAM/304/2002 . Manage all waste streams associated with the project. Typical waste streams are listed below. Include additional waste streams not listed:

- a. Land Clearing Debris
- b. Asphalt
- c. Concrete
- d. Metals (Includes, but is not limited to, banding, stud trim, ductwork, piping, rebar, roofing, other trim, steel, iron, galvanized, stainless steel, aluminum, copper, zinc, bronze.)
- e. Paper
- f. Plastics (PET, HDPE, PVC, LDPE, PP, PS, Other)
- g. Ceiling Tiles
- h. Insulation
- i. Beverage Containers

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Construction Waste Management Plan; G

SD-11 Closeout Submittals

Final Construction Waste Diversion Report; S

1.6 MEETINGS

Conduct Construction Waste Management meetings. After award of the Contract and prior to commencement of work, schedule and conduct a meeting with the Contracting Officer to discuss the proposed Construction Waste Management Plan and to develop a mutual understanding relative to the management of the Construction Waste Management Program and how waste diversion requirements will be met.

The requirements of this meeting may be fulfilled during the coordination and mutual Understanding meeting outlined in Section 01 45 00.00 20 QUALITY CONTROL. At a minimum, discuss and document waste management goals at following meetings:

- a. Preconstruction meeting.
- b. Regular Quality Control meetings.
- c. Work safety meeting (if applicable).

1.7 CONSTRUCTION WASTE MANAGEMENT PLAN

Submit Construction Waste Management Plan within 45 calendar days after contract award. Revise and resubmit Construction Waste Management Plan as necessary, in order for construction to begin.. Execute demolition or deconstruction activities in accordance with Section 02 41 00 DEMOLITION AND DECONSTRUCTION. Manage demolition debris/waste or deconstruction materials in accordance with the approved construction waste management plan.

An approved Construction Waste Management Plan will not relieve the Contractor of responsibility for compliance with applicable environmental regulations or meeting project cumulative waste diversion requirement. Ensure all subcontractors receive a copy of the approved Construction Waste Management Plan. The plan demonstrates how to meet the project waste diversion requirement. Also, include the following in the plan:

- a. Identify the names of individuals responsible for waste management and waste management tracking, along with roles and responsibilities on the project.
- b. Actions that will be taken to reduce solid waste generation, including coordination with subcontractors to ensure awareness and participation.
- c. Description of the regular meetings to be held to address waste management.

- d. Description of the specific approaches to be used in recycling/reuse of the various materials generated, including the areas on site and equipment to be used for processing, sorting, and temporary storage of materials.
- e. Name of landfill and incinerator to be used.
- f. Identification of local and regional re-use programs, including non-profit organizations such as schools, local housing agencies, and organization that accept used materials such as material exchange networks and resale stores. Include the name, location, phone number for each re-use facility identified, and provide a copy of the permit or license for each facility.
- g. List of specific materials, by type and quantity, that will be salvaged for resale, salvaged and reused on the current project, salvaged and stored for reuse on a future project, or recycled. Identify the recycling facilities by name, address, and phone number.
- h. Identification of materials that cannot be recycled or reused with an explanation or justification, to be approved by the Contracting Officer.
- i. Description of the means by which materials identified in item (g) above will be protected from contamination.
- j. Description of the means of transportation of the recyclable materials (whether materials will be site-separated and self-hauled to designated centers, or whether mixed materials will be collected by a waste hauler and removed from the site).
- k. Copy of training plan for subcontractors and other services to prevent contamination by co-mingling materials identified for diversion and waste materials.
- l. Facilities or subcontractors offering construction waste transport on-site or off-site must ensure that proper shipping orders, bill of lading, manifests, or other shipping documents containing waste diversion information meet requirements of the Spanish and Regional (Junta de Andalucia) legislation. Individuals signing manifests or other shipping documents should meet the minimum training requirements.
- m. List each supplier who deliver construction materials, in bulk, or package products in returnable containers or returnable packaging, or have take-back programs. List each program and the applicable material to actively monitor and track to assist in meeting waste diversion requirements on the project.
- n. Identify local jurisdiction requirements for waste management. Include local requirements and points of contact.

Distribute copies of the waste management plan to each subcontractor, Environmental Manager, and the Contracting Officer.

1.8 RECORDS (DOCUMENTATION)

1.8.1 General

Maintain records to document the types and quantities of waste generated and diverted through re-use, recycling and sale to third parties; through disposal to a landfill or incinerator facility. Provide explanations for materials not recycled, reused or sold. Collect and retain manifests, weight tickets, sales receipts, and invoices specifically identifying diverted project waste materials or disposed materials.

1.8.2 Accumulated

Maintain a running record of materials generated and diverted from landfill disposal, including accumulated diversion rates for the project. Make records available to the Contracting Officer during construction or incidental demolition activities. Provide a copy of the diversion records to the Contracting Officer upon completion of the construction, incidental demolitions or minor deconstruction activities.

1.9 REPORTS

1.9.1 General

Maintain current construction waste diversion information on site for periodic inspection by the Contracting Officer. Include in the quarterly reports, annual reports and final reports: the project name, contract information, information for waste generated, diverted and disposed of for the current reporting period and show cumulative totals for the project. Reports must identify quantities of waste by type and disposal method. Also include in each report, supporting documentation to include manifests, weigh tickets, receipts, and invoices specifically identifying the project and waste material type and weighted sum.

1.10 FINAL CONSTRUCTION WASTE DIVERSION REPORT

A Final Construction Waste Diversion Report is required at the end of the project. Provide Final Construction Waste Diversion Report 60 days prior to the Beneficial Occupancy Date (BOD).

1.11 COLLECTION

Collect, store, protect, and handle reusable and recyclable materials at the site in a manner which prevents contamination, and provides protection from the elements to preserve their usefulness and monetary value. Provide receptacles and storage areas designated specifically for recyclable and reusable materials and label them clearly and appropriately to prevent contamination from other waste materials. Keep receptacles or storage areas neat and clean.

Train subcontractors and other service providers to either separate waste streams or use the co-mingling method as described in the Construction Waste Management Plan. Handle hazardous waste and hazardous materials in accordance with applicable regulations and coordinate with Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS. Separate materials by one of the following methods described herein:

1.11.1 Source Separation Method

Separate waste products and materials that are recyclable from trash and sort as described below into appropriately marked separate containers and then transport to the respective recycling facility for further processing. Deliver materials in accordance with recycling or reuse facility requirements (e.g., free of dirt, adhesives, solvents, petroleum contamination, and other substances deleterious to the recycling process). Separate materials into the category types as defined in the Construction Waste Management Plan.

1.11.2 Co-Mingled Method

Place waste products and recyclable materials into a single container and then transport to an authorized recycling facility, which meets all applicable requirements to accept and dispose of recyclable materials in accordance with all applicable local, regional and Spanish regulations. The Co-mingled materials must be sorted and processed in accordance with the approved Construction Waste Management Plan.

1.11.3 Other Methods

Other methods proposed by the Contractor may be used when approved by the Contracting Officer.

1.12 DISPOSAL

Control accumulation of waste materials and trash. Recycle or dispose of collected materials off-site at intervals approved by the Contracting Officer and in compliance with waste management procedures as described in the waste management plan. Except as otherwise specified in other sections of the specifications, dispose of in accordance with the following:

1.12.1 Reuse

Give first consideration to reusing construction and demolition materials as a disposition strategy. Recover for reuse materials, products, and components as described in the approved Construction Waste Management Plan. Coordinate with the Contracting Officer to identify onsite reuse opportunities or material sales or donation available through Government resale or donation programs. Sale of recovered materials is not allowed on the Installation. Consider the use of surplus industrial supply broker services, who match entities with reusable or repurpose industrial materials with entities with need of such materials.

1.12.2 Recycle

Recycle non-hazardous construction and demolition/debris materials that are not suitable for reuse. Track rejection of contaminated recyclable materials by the recycling facility. Rejected recyclables materials will not be counted as a percentage of diversion calculation. Recycle all fluorescent lamps, HID lamps, mercury (Hg) -containing thermostats and ampoules, and PCBs-containing ballasts and electrical components as directed by the Contracting Officer. Do not crush lamps on site as this creates a hazardous waste stream with additional handling requirements.

1.12.3 Compost

Consider composting on site if a reasonable amount of compostable materials will be available and a utilization of compostable material can be determined and appropriately planned for. Compostable materials include plant materials, sawdust and certain food scraps. Composting as a strategy must be explicitly addressed in the Construction Waste Management Plan submitted for approval to ensure it is feasible.

1.12.4 Waste

Dispose by landfill or incineration only those waste materials with no practical use, economic benefit, or recycling opportunity.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

Not used.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 78 00

CLOSEOUT SUBMITTALS

05/19, CHG 1: 08/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 As-Built Drawings
- 1.3 SUBMITTALS
- 1.4 SPARE PARTS DATA
- 1.5 WARRANTY MANAGEMENT
 - 1.5.1 Warranty Management Plan
 - 1.5.2 Performance Bond
 - 1.5.3 Pre-Warranty Conference
 - 1.5.4 Warranty Tags

PART 2 PRODUCTS

- 2.1 CERTIFICATION OF EPA DESIGNATED ITEMS

PART 3 EXECUTION

- 3.1 AS-BUILT DRAWINGS
 - 3.1.1 Markup Guidelines
 - 3.1.2 As-Built Drawings Content
- 3.2 CLEANUP
- 3.3 REAL PROPERTY RECORD
 - 3.3.1 Interim DD FORM 1354
 - 3.3.2 Completed DD FORM 1354

-- End of Section Table of Contents --

SECTION 01 78 00

CLOSEOUT SUBMITTALS
05/19, CHG 1: 08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E1971 (2005; R 2011) Standard Guide for
Stewardship for the Cleaning of Commercial
and Institutional Buildings

GREEN SEAL (GS)

GS-37 (2017) Cleaning Products for Industrial
and Institutional Use

U.S. DEPARTMENT OF DEFENSE (DOD)

FC 1-300-09N (2014; with Change 4, 2018) Navy and
Marine Corps Design

UFC 1-300-08 (2009, with Change 2, 2011) Criteria for
Transfer and Acceptance of DoD Real
Property

1.2 DEFINITIONS

1.2.1 As-Built Drawings

As-built drawings are the marked-up drawings, maintained by the Contractor on-site, that depict actual conditions and deviations from the Contract Documents. These deviations and additions may result from coordination required by, but not limited to: contract modifications; official responses to submitted Requests for Information (RFI's); direction from the Contracting Officer; design that is the responsibility of the Contractor, and differing site conditions. Maintain the as-builts throughout construction as red-lined hard copies on site. These files serve as the basis for the creation of the record drawings.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Warranty Management Plan

Warranty Tags

Final Cleaning

Spare Parts Data

SD-08 Manufacturer's Instructions

Posted Instructions

SD-11 Closeout Submittals

As-Built Drawings; G

As-Built Record of Equipment and Materials

Certification of EPA Designated Items; G

Interim DD FORM 1354; G

Checklist for DD FORM 1354; G

1.4 SPARE PARTS DATA

Submit two copies of the Spare Parts Data list.

- a. Indicate manufacturer's name, part number, and stock level required for test and balance, pre-commissioning, maintenance and repair activities. List those items that may be standard to the normal maintenance of the system.

1.5 WARRANTY MANAGEMENT

1.5.1 Warranty Management Plan

Develop a warranty management plan which contains information relevant to FAR 52.246-21 Warranty of Construction. At least 30 days before the planned pre-warranty conference, submit one set of the warranty management plan. Include within the warranty management plan all required actions and documents to assure that the Government receives all warranties to which it is entitled. The plan narrative must contain sufficient detail to render it suitable for use by future maintenance and repair personnel, whether tradesmen, or of engineering background, not necessarily familiar with this contract. The term "status" as indicated below must include due date and whether item has been submitted or was accomplished. Submit warranty information, made available during the construction phase, to the Contracting Officer for approval prior to each monthly pay estimate. Assemble approved information in a binder and turn over to the Government upon acceptance of the work. The construction warranty period must begin on the date of project acceptance and continue for the full product warranty period. Conduct a joint 4 month and 9 month warranty inspection, measured from time of acceptance; with the Contractor, Contracting Officer and the Customer Representative. The warranty management plan must include, but is not limited to, the following:

- a. Roles and responsibilities of personnel associated with the warranty process, including points of contact and telephone numbers within the organizations of the Contractors, subcontractors, manufacturers or

suppliers involved.

- b. For each warranty, the name, address, telephone number, and e-mail of each of the guarantor's representatives nearest to the project location.
- c. A list and status of delivery of Certificates of Warranty for extended warranty items, including roofs, HVAC balancing, pumps, motors, transformers, and for commissioned systems, such as fire protection and alarm systems, sprinkler systems, and lightning protection systems.
- d. As-Built Record of Equipment and Materials list for each warranted equipment, item, feature of construction or system indicating:
 - (1) Name of item.
 - (2) Model and serial numbers.
 - (3) Location where installed.
 - (4) Name and phone numbers of manufacturers or suppliers.
 - (5) Names, addresses and telephone numbers of sources of spare parts.
 - (6) Warranties and terms of warranty. Include one-year overall warranty of construction, including the starting date of warranty of construction. Items which have warranties longer than one year must be indicated with separate warranty expiration dates.
 - (7) Cross-reference to warranty certificates as applicable.
 - (8) Starting point and duration of warranty period.
 - (9) Summary of maintenance procedures required to continue the warranty in force.
 - (10) Cross-reference to specific pertinent Operation and Maintenance manuals.
 - (11) Organization, names and phone numbers of persons to call for warranty service.
 - (12) Typical response time and repair time expected for various warranted equipment.
- e. The plans for attendance at the 4 and 9 month post-construction warranty inspections conducted by the Government.
- f. Procedure and status of tagging of equipment covered by warranties longer than one year.
- g. Copies of instructions to be posted near selected pieces of equipment where operation is critical for warranty or safety reasons.

1.5.2 Performance Bond

The Performance Bond must remain effective throughout the construction and warranty period.

- a. In the event the Contractor fails to commence and diligently pursue any construction warranty work required, the Contracting Officer will have the work performed by others, and after completion of the work, will charge the remaining construction warranty funds of expenses incurred by the Government while performing the work, including, but not limited to administrative expenses.
- b. In the event sufficient funds are not available to cover the construction warranty work performed by the Government at the Contractor's expense, the Contracting Officer will have the right to recoup expenses from the bonding company.

- c. Following oral or written notification of required construction warranty repair work, respond in a timely manner. Written verification will follow oral instructions. Failure to respond will be cause for the Contracting Officer to proceed against the Contractor.

1.5.3 Pre-Warranty Conference

Prior to contract completion, and at a time designated by the Contracting Officer, meet with the Contracting Officer to develop a mutual understanding with respect to the requirements of this section. At this meeting, establish and review communication procedures for Contractor notification of construction warranty defects, priorities with respect to the type of defect, reasonable time required for Contractor response, and other details deemed necessary by the Contracting Officer for the execution of the construction warranty. In connection with these requirements and at the time of the Contractor's quality control completion inspection, furnish the name, telephone number and address of a licensed and bonded company which is authorized to initiate and pursue construction warranty work action on behalf of the Contractor. This point of contact must be located within the local service area of the warranted construction, be continuously available, and be responsive to Government inquiry on warranty work action and status. This requirement does not relieve the Contractor of any of its responsibilities in connection with other portions of this provision.

1.5.4 Warranty Tags

At the time of installation, tag each warranted item with a durable, oil and water resistant tag approved by the Contracting Officer. Attach each tag with a copper wire and spray with a silicone waterproof coating. Also, submit two record copies of the warranty tags showing the layout and design. The date of acceptance and the QC signature must remain blank until the project is accepted for beneficial occupancy. Show the following information on the tag.

Type of product/material	
Model number	
Serial number	
Contract number	
Warranty period from/to	
Inspector's signature	
Construction Contractor	
Address	

Telephone number	
Warranty contact	
Address	
Telephone number	
Warranty response time priority code	
WARNING - PROJECT PERSONNEL TO PERFORM ONLY OPERATIONAL MAINTENANCE DURING THE WARRANTY PERIOD.	

PART 2 PRODUCTS

2.1 CERTIFICATION OF EPA DESIGNATED ITEMS

Submit the Certification of EPA Designated Items as required by FAR 52.223-9 Estimate of Percentage of Recovered Material Content for EPA Designated Items and FAR 52-223-17 Affirmative Procurement of EPA designated items in Service and Construction Contracts. Include on the certification form the following information: project name, project number, Contractor name, license number, Contractor address, and certification. The certification will read as follows and be signed and dated by the Contractor. "I hereby certify the information provided herein is accurate and that the requisition/procurement of all materials listed on this form comply with current EPA standards for recycled/recovered materials content. The following exemptions may apply to the non-procurement of recycled/recovered content materials:

- a. The product does not meet appropriate performance standards;
- b. The product is not available within a reasonable time frame;
- c. The product is not available competitively (from two or more sources);
- d. The product is only available at an unreasonable price (compared with a comparable non-recycled content product)."

Record each product used in the project that has a requirement or option of containing recycled content, noting total price, total value of post-industrial recycled content, total value of post-consumer recycled content, exemptions (a, b, c, or d, as indicated), and comments. Recycled content values may be determined by weight or volume percent, but must be consistent throughout.

PART 3 EXECUTION

3.1 AS-BUILT DRAWINGS

3.1.1 Markup Guidelines

Make comments and markup the drawings complete without reference to letters, memos, or materials that are not part of the As-Built drawing. Show what was changed, how it was changed, where item(s) were relocated

and change related details. These working as-built markup prints must be neat, legible and accurate as follows:

- a. Use base colors of red, green, and blue. Color code for changes as follows:
 - (1) Special (Blue) - Items requiring special information, coordination, or special detailing or detailing notes.
 - (2) Deletions (Red) - Over-strike deleted graphic items (lines), lettering in notes and leaders.
 - (3) Additions (Green) - Added items, lettering in notes and leaders.
- b. Provide a legend if colors other than the "base" colors of red, green, and blue are used.
- c. Add and denote any additional equipment or material facilities, service lines, incorporated under As-Built Revisions if not already shown in legend.
- d. Use frequent written explanations on markup drawings to describe changes. Do not totally rely on graphic means to convey the revision.
- e. Use legible lettering and precise and clear digital values when marking prints. Clarify ambiguities concerning the nature and application of change involved.
- f. Wherever a revision is made, also make changes to related section views, details, legend, profiles, plans and elevation views, schedules, notes and call out designations, and mark accordingly to avoid conflicting data on all other sheets.
- g. For deletions, cross out all features, data and captions that relate to that revision.
- h. For changes on small-scale drawings and in restricted areas, provide large-scale inserts, with leaders to the applicable location.
- i. Indicate one of the following when attaching a print or sketch to a markup print:
 - 1) Add an entire drawing to contract drawings
 - 2) Change the contract drawing to show changes on the drawing.
 - 3) Provided for reference only to further detail the initial design.
- j. Incorporate all shop and fabrication drawings into the markup drawings.

3.1.2 As-Built Drawings Content

Show on the as-built drawings, but not limited to, the following information:

- a. The actual location, kinds and sizes of all sub-surface utility lines. In order that the location of these lines and appurtenances may be determined in the event the surface openings or indicators become covered over or obscured, show by offset dimensions to two

permanently fixed surface features the end of each run including each change in direction on the record drawings. Locate valves, splice boxes and similar appurtenances by dimensioning along the utility run from a reference point. Also record the average depth below the surface of each run.

- b. The location and dimensions of any changes within the building structure.
- c. Layout and schematic drawings of electrical circuits and piping.
- d. Correct grade, elevations, cross section, or alignment of roads, earthwork, structures or utilities if any changes were made from contract plans.
- e. Changes in details of design or additional information obtained from working drawings specified to be prepared or furnished by the Contractor; including but not limited to shop drawings, fabrication, erection, installation plans and placing details, pipe sizes, insulation material, dimensions of equipment, and foundations.
- f. The topography, invert elevations and grades of drainage installed or affected as part of the project construction.
- g. Changes or Revisions which result from the final inspection.
- h. Where contract drawings or specifications present options, show only the option selected for construction on the working as-built markup drawings.
- i. If borrow material for this project is from sources on Government property, or if Government property is used as a spoil area, furnish a contour map of the final borrow pit/spoil area elevations.
- j. Systems designed or enhanced by the Contractor, such as HVAC controls, fire alarm, fire sprinkler, and irrigation systems.
- k. Changes in location of equipment and architectural features.
- l. Modifications and compliance with FC 1-300-09N procedures.
- m. Actual location of anchors, construction and control joints, etc., in concrete.
- n. Unusual or uncharted obstructions that are encountered in the contract work area during construction.
- o. Location, extent, thickness, and size of stone protection particularly where it will be normally submerged by water.

3.2 CLEANUP

Provide final cleaning in accordance with ASTM E1971 and submit two copies of the listing of completed final clean-up items. Leave premises "broom clean." Comply with GS-37 for general purpose cleaning and bathroom cleaning. Use only nonhazardous cleaning materials, including natural cleaning materials, in the final cleanup. Clean interior and exterior glass surfaces exposed to view; remove temporary labels, stains and foreign substances; polish transparent and glossy surfaces; vacuum

carpeted and soft surfaces. Clean equipment and fixtures to a sanitary condition. Replace filters of operating equipment and comply with the Indoor Air Quality (IAQ) Management Plan. Clean debris from roofs, gutters, downspouts and drainage systems. Sweep paved areas and rake clean landscaped areas. Remove waste and surplus materials, rubbish and construction facilities from the site. Recycle, salvage, and return construction and demolition waste from project in accordance with Section 01 57 19.01 21 TEMPORARY ENVIRONMENTAL CONTROLS, and 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL.

3.3 REAL PROPERTY RECORD

Refer to UFC 1-300-08 for instruction on completing the DD FORM 1354. Contact the Contracting Officer for any project specific information necessary to complete the DD FORM 1354.

3.3.1 Interim DD FORM 1354

Near the completion of Project, but a minimum of 60 days prior to final acceptance of the work, complete and submit an accounting of all installed property with Interim DD FORM 1354. Include any additional assets, improvements, and alterations from the Draft DD FORM 1354.

3.3.2 Completed DD FORM 1354

Submit the completed Checklist for DD FORM 1354 of Installed Equipment items. Attach this list to the updated DD FORM 1354.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA

08/15, CHG 2: 08/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 OPERATION AND MAINTENANCE DATA
 - 1.3.1 Package Quality
 - 1.3.2 Package Content
 - 1.3.3 Changes to Submittals
 - 1.3.4 Commissioning Authority Review and Approval
- 1.4 O&M DATABASE
- 1.5 OPERATION AND MAINTENANCE MANUAL FILE FORMAT
 - 1.5.1 Organization
 - 1.5.2 CD or DVD Label and Disk Holder or Case
- 1.6 TYPES OF INFORMATION REQUIRED IN O&M DATA PACKAGES
 - 1.6.1 Operating Instructions
 - 1.6.1.1 Safety Precautions and Hazards
 - 1.6.1.2 Operator Prestart
 - 1.6.1.3 Startup, Shutdown, and Post-Shutdown Procedures
 - 1.6.1.4 Normal Operations
 - 1.6.1.5 Emergency Operations
 - 1.6.1.6 Operator Service Requirements
 - 1.6.1.7 Environmental Conditions
 - 1.6.1.8 Operating Log
 - 1.6.1.9 Additional Requirements for HVAC Control Systems
 - 1.6.2 Preventive Maintenance
 - 1.6.2.1 Lubrication Data
 - 1.6.2.2 Preventive Maintenance Plan, Schedule, and Procedures
 - 1.6.2.3 Cleaning Recommendations
 - 1.6.3 Repair
 - 1.6.3.1 Troubleshooting Guides and Diagnostic Techniques
 - 1.6.3.2 Wiring Diagrams and Control Diagrams
 - 1.6.3.3 Repair Procedures
 - 1.6.3.4 Removal and Replacement Instructions
 - 1.6.3.5 Spare Parts and Supply Lists
 - 1.6.3.6 Repair Work-Hours
 - 1.6.4 Appendices
 - 1.6.4.1 Product Submittal Data
 - 1.6.4.2 Certificates
 - 1.6.4.3 Manufacturer's Instructions
 - 1.6.4.4 O&M Submittal Data
 - 1.6.4.5 Parts Identification
 - 1.6.4.6 Warranty Information
 - 1.6.4.7 Extended Warranty Information
 - 1.6.4.8 Personnel Training Requirements
 - 1.6.4.9 Testing Equipment and Special Tool Information
 - 1.6.4.10 Testing and Performance Data

- 1.6.4.11 Field Test Reports and Manufacturer's Field Reports
- 1.6.4.12 Contractor Information
- 1.7 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES
 - 1.7.1 Data Package 1
 - 1.7.2 Data Package 2
 - 1.7.3 Data Package 3

PART 2 PRODUCTS

PART 3 EXECUTION

- 3.1 TRAINING
 - 3.1.1 Training Plan
 - 3.1.2 Training Content
 - 3.1.3 Training Outline
 - 3.1.4 Training Video Recording
 - 3.1.5 Unresolved Questions from Attendees
 - 3.1.6 Validation of Training Completion
 - 3.1.7 Quality Control Coordination

-- End of Section Table of Contents --

SECTION 01 78 23

OPERATION AND MAINTENANCE DATA
08/15, CHG 2: 08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM E1971 (2005; R 2011) Standard Guide for
Stewardship for the Cleaning of Commercial
and Institutional Buildings

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-10 Operation and Maintenance Data

O&M Database; G

Training Plan; G

Training Outline; G

Training Content; G

SD-11 Closeout Submittals

Training Video Recording; G

Validation of Training Completion; G

1.3 OPERATION AND MAINTENANCE DATA

Submit Operation and Maintenance (O&M) Data for the provided equipment, product, or system, defining the importance of system interactions, troubleshooting, and long-term preventive operation and maintenance. Compile, prepare, and aggregate O&M data to include clarifying and updating the original sequences of operation to as-built conditions. Organize and present information in sufficient detail to clearly explain O&M requirements at the system, equipment, component, and subassembly level. Include an index preceding each submittal. Submit in accordance with this section and Section 01 33 00 SUBMITTAL PROCEDURES.

1.3.1 Package Quality

Documents must be fully legible. Operation and Maintenance data must be

consistent with the manufacturer's standard brochures, schematics, printed instructions, general operating procedures, and safety precautions.

1.3.2 Package Content

Provide data package content in accordance with paragraph SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES. Comply with the data package requirements specified in the individual technical sections, including the content of the packages and addressing each product, component, and system designated for data package submission, with any exceptions as approved by the contracting officer or COR..

1.3.3 Changes to Submittals

Provide manufacturer-originated changes or revisions to submitted data if a component of an item is so affected subsequent to acceptance of the O&M Data. Submit changes, additions, or revisions required by the Contracting Officer for final acceptance of submitted data within 30 calendar days of the notification of this change requirement.

1.3.4 Commissioning Authority Review and Approval

Submit the commissioned systems and equipment submittals to the Commissioning Authority (CxA) to review for completeness and applicability. Obtain validation from the CxA that the systems and equipment provided meet the requirements of the Contract documents and design intent, particularly as they relate to functionality, energy performance, water performance, maintainability, sustainability, system cost, indoor environmental quality, and local environmental impacts. The CxA communicates deficiencies to the Contracting Officer. Submit the O&M manuals to the Contracting Officer upon a successful review of the corrections, and with the CxA recommendation for approval and acceptance of these O&M manuals. This work is in addition to the normal review procedures for O&M data.

1.4 O&M DATABASE

Develop an editable, electronic spreadsheet based on the equipment in the Operation and Maintenance Manuals that contains the information required to start a preventive maintenance program. As a minimum, provide list of system equipment, location installed, warranty expiration date, manufacturer, model, and serial number.

1.5 OPERATION AND MAINTENANCE MANUAL FILE FORMAT

Assemble data packages into electronic Operation and Maintenance Manuals. Assemble each manual into a composite electronically indexed file using the most current version of Adobe Acrobat or similar software capable of producing PDF file format. Provide compact disks (CD) or data digital versatile disk (DVD) as appropriate, so that each one contains operation, maintenance and record files, project record documents, and training videos. Include a complete electronically linked operation and maintenance directory.

1.5.1 Organization

Bookmark Product and Drawing Information documents using the current version of CSI MasterFormat numbering system, and arrange submittals using the specification sections as a structure. Use CSI MasterFormat and UFGS

numbers along with descriptive bookmarked titles that explain the content of the information that is being bookmarked.

1.5.2 CD or DVD Label and Disk Holder or Case

Provide the following information on the disk label and disk holder or case:

- a. Project Title
- b. Activity and Location
- c. Construction Contract Number
- d. Prepared For: (Contracting Agency)
- e. Prepared By: (Name, title, phone number and email address)
- f. Include the disk content on the disk label
- g. Date
- h. Virus scanning program used

1.6 TYPES OF INFORMATION REQUIRED IN O&M DATA PACKAGES

The following are a detailed description of the data package items listed in paragraph SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES.

1.6.1 Operating Instructions

Provide specific instructions, procedures, and illustrations for the following phases of operation for the installed model and features of each system:

1.6.1.1 Safety Precautions and Hazards

List personnel hazards and equipment or product safety precautions for operating conditions. List all residual hazards identified in the Activity Hazard Analysis provided under Section 01 35 26 GOVERNMENT SAFETY REQUIREMENTS. Provide recommended safeguards for each identified hazard.

1.6.1.2 Operator Prestart

Provide procedures required to install, set up, and prepare each system for use.

1.6.1.3 Startup, Shutdown, and Post-Shutdown Procedures

Provide narrative description for Startup, Shutdown and Post-shutdown operating procedures including the control sequence for each procedure.

1.6.1.4 Normal Operations

Provide Control Diagrams with data to explain operation and control of systems and specific equipment. Provide narrative description of Normal Operating Procedures.

1.6.1.5 Emergency Operations

Provide Emergency Procedures for equipment malfunctions to permit a short period of continued operation or to shut down the equipment to prevent further damage to systems and equipment. Provide Emergency Shutdown Instructions for fire, explosion, spills, or other foreseeable contingencies. Provide guidance and procedures for emergency operation of utility systems including required valve positions, valve locations and zones or portions of systems controlled.

1.6.1.6 Operator Service Requirements

Provide instructions for services to be performed by the operator such as lubrication, adjustment, inspection, and recording gauge readings.

1.6.1.7 Environmental Conditions

Provide a list of Environmental Conditions (temperature, humidity, and other relevant data) that are best suited for the operation of each product, component or system. Describe conditions under which the item equipment should not be allowed to run.

1.6.1.8 Operating Log

Provide forms, sample logs, and instructions for maintaining necessary operating records.

1.6.1.9 Additional Requirements for HVAC Control Systems

Provide Data Package 5 and the following for control systems:

- a. Narrative description on how to perform and apply functions, features, modes, and other operations, including unoccupied operation, seasonal changeover, manual operation, and alarms. Include detailed technical manual for programming and customizing control loops and algorithms.
- b. Full as-built sequence of operations.
- c. Copies of checkout tests and calibrations performed by the Contractor (not Cx tests).
- d. Full points list. Provide a listing of rooms with the following information for each room:
 - (1) Floor
 - (2) Room number
 - (3) Room name
 - (4) Air handler unit ID
 - (5) Reference drawing number
 - (6) Air terminal unit tag ID
 - (7) Heating or cooling valve tag ID
 - (8) Minimum cfm

(9) Maximum cfm

- e. Full print out of all schedules and set points after testing and acceptance of the system.
- f. Full as-built print out of software program.
- g. Marking of system sensors and thermostats on the as-built floor plan and mechanical drawings with their control system designations.

1.6.2 Preventive Maintenance

Provide the following information for preventive and scheduled maintenance to minimize repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials.

1.6.2.1 Lubrication Data

Include the following preventive maintenance lubrication data, in addition to instructions for lubrication required under paragraph OPERATOR SERVICE REQUIREMENTS:

- a. A table showing recommended lubricants for specific temperature ranges and applications.
- b. Charts with a schematic diagram of the equipment showing lubrication points, recommended types and grades of lubricants, and capacities.
- c. A Lubrication Schedule showing service interval frequency.

1.6.2.2 Preventive Maintenance Plan, Schedule, and Procedures

Provide manufacturer's schedule for routine preventive maintenance, inspections, condition monitoring (predictive tests) and adjustments required to ensure proper and economical operation and to minimize repairs. Provide instructions stating when the systems should be retested. Provide manufacturer's projection of preventive maintenance work-hours on a daily, weekly, monthly, and annual basis including craft requirements by type of craft. For periodic calibrations, provide manufacturer's specified frequency and procedures for each separate operation.

- a. Define the anticipated time required to perform each of each test (work-hours), test apparatus, number of personnel identified by responsibility, and a testing validation procedure permitting the record operation capability requirements within the schedule. Provide a remarks column for the testing validation procedure referencing operating limits of time, pressure, temperature, volume, voltage, current, acceleration, velocity, alignment, calibration, adjustments, cleaning, or special system notes. Delineate procedures for preventive maintenance, inspection, adjustment, lubrication and cleaning necessary to minimize repairs.
- b. Repair requirements must inform operators how to check out, troubleshoot, repair, and replace components of the system. Include electrical and mechanical schematics and diagrams and diagnostic techniques necessary to enable operation and troubleshooting of the

system after acceptance.

1.6.2.3 Cleaning Recommendations

Provide environmentally preferable cleaning recommendations in accordance with ASTM E1971.

1.6.3 Repair

Provide manufacturer's recommended procedures and instructions for correcting problems and making repairs for the installed model and features of each system. Include potential environmental and indoor air quality impacts of recommended maintenance procedures and materials.

1.6.3.1 Troubleshooting Guides and Diagnostic Techniques

Provide step-by-step procedures to promptly isolate the cause of typical malfunctions. Describe clearly why the checkout is performed and what conditions are to be sought. Identify tests or inspections and test equipment required to determine whether parts and equipment may be reused or require replacement.

1.6.3.2 Wiring Diagrams and Control Diagrams

Provide point-to-point drawings of wiring and control circuits including factory-field interfaces. Provide a complete and accurate depiction of the actual job specific wiring and control work. On diagrams, number electrical and electronic wiring and pneumatic control tubing and the terminals for each type, identically to actual installation configuration and numbering.

1.6.3.3 Repair Procedures

Provide instructions and a list of tools required to repair or restore the product or equipment to proper condition or operating standards.

1.6.3.4 Removal and Replacement Instructions

Provide step-by-step procedures and a list of required tools and supplies for removal, replacement, disassembly, and assembly of components, assemblies, subassemblies, accessories, and attachments. Provide tolerances, dimensions, settings and adjustments required. Use a combination of text and illustrations.

1.6.3.5 Spare Parts and Supply Lists

Provide lists of spare parts and supplies required for repair to ensure continued service or operation without unreasonable delays. Special consideration is required for facilities at remote locations. List spare parts and supplies that have a long lead-time to obtain.

1.6.3.6 Repair Work-Hours

Provide manufacturer's projection of repair work-hours including requirements by type of craft. Identify, and tabulate separately, repair that requires the equipment manufacturer to complete or to participate.

1.6.4 Appendices

Provide information required below and information not specified in the preceding paragraphs but pertinent to the maintenance or operation of the product or equipment. Include the following:

1.6.4.1 Product Submittal Data

Provide a copy of SD-03 Product Data submittals documented with the required approval.

1.6.4.2 Certificates

Provide a copy of SD-07 Certificates submittals documented with the required approval.

1.6.4.3 Manufacturer's Instructions

Provide a copy of SD-08 Manufacturer's Instructions submittals documented with the required approval.

1.6.4.4 O&M Submittal Data

Provide a copy of SD-10 Operation and Maintenance Data submittals documented with the required approval.

1.6.4.5 Parts Identification

Provide identification and coverage for the parts of each component, assembly, subassembly, and accessory of the end items subject to replacement. Include special hardware requirements, such as requirement to use high-strength bolts and nuts. Identify parts by make, model, serial number, and source of supply to allow reordering without further identification. Provide clear and legible illustrations, drawings, and exploded views to enable easy identification of the items. When illustrations omit the part numbers and description, both the illustrations and separate listing must show the index, reference, or key number that will cross-reference the illustrated part to the listed part. Group the parts shown in the listings by components, assemblies, and subassemblies in accordance with the manufacturer's standard practice. Parts data may cover more than one model or series of equipment, components, assemblies, subassemblies, attachments, or accessories, such as typically shown in a master parts catalog.

1.6.4.6 Warranty Information

List and explain the various warranties and clearly identify the servicing and technical precautions prescribed by the manufacturers or contract documents in order to keep warranties in force. Include warranty information for primary components of the system. Provide copies of warranties required by Section 01 78 00 CLOSEOUT SUBMITTALS.

1.6.4.7 Extended Warranty Information

List all warranties for products, equipment, components, and sub-components whose duration exceeds one year. For each warranty listed, indicate the applicable specification section, duration, start date, end date, and the point of contact for warranty fulfillment. Also, list or reference the specific operation and maintenance procedures that must be

performed to keep the warranty valid. Provide copies of warranties required by Section 01 78 00 CLOSEOUT SUBMITTALS.

1.6.4.8 Personnel Training Requirements

Provide information available from the manufacturers that is needed for use in training designated personnel to properly operate and maintain the equipment and systems.

1.6.4.9 Testing Equipment and Special Tool Information

Include information on test equipment required to perform specified tests and on special tools needed for the operation, maintenance, and repair of components. Provide final set points.

1.6.4.10 Testing and Performance Data

Include completed prefunctional checklists, functional performance test forms, and monitoring reports. Include recommended schedule for retesting and blank test forms. Provide final set points.

1.6.4.11 Field Test Reports and Manufacturer's Field Reports

Provide a copy of Field Test Reports (SD-06) and Manufacturer's Field Reports (SD-09) submittals documented with the required approval.

1.6.4.12 Contractor Information

Provide a list that includes the name, address, and telephone number of the General Contractor and each Subcontractor who installed the product or equipment, or system. For each item, also provide the name address and telephone number of the manufacturer's representative and service organization that can provide replacements most convenient to the project site. Provide the name, address, and telephone number of the product, equipment, and system manufacturers.

1.7 SCHEDULE OF OPERATION AND MAINTENANCE DATA PACKAGES

Provide the O&M data packages specified in individual technical sections. The information required in each type of data package follows:

1.7.1 Data Package 1

- a. Safety precautions and hazards
- b. Cleaning recommendations
- c. Maintenance and repair procedures
- d. Warranty information
- e. Extended warranty information
- f. Contractor information
- g. Spare parts and supply list

1.7.2 Data Package 2

- a. Safety precautions and hazards
- b. Normal operations
- c. Environmental conditions
- d. Lubrication data
- e. Preventive maintenance plan, schedule, and procedures
- f. Cleaning recommendations
- g. Maintenance and repair procedures
- h. Removal and replacement instructions
- i. Spare parts and supply list
- j. Parts identification
- k. Warranty information
- l. Extended warranty information
- m. Contractor information

1.7.3 Data Package 3

- a. Safety precautions and hazards
- b. Operator prestart
- c. Startup, shutdown, and post-shutdown procedures
- d. Normal operations
- e. Emergency operations
- f. Environmental conditions
- g. Operating log
- h. Lubrication data
- i. Preventive maintenance plan, schedule, and procedures
- j. Cleaning recommendations
- k. Troubleshooting guides and diagnostic techniques
- l. Wiring diagrams and control diagrams
- m. Maintenance and repair procedures
- n. Removal and replacement instructions
- o. Spare parts and supply list

- p. Product submittal data
- q. O&M submittal data
- r. Parts identification
- s. Warranty information
- t. Extended warranty information
- u. Testing equipment and special tool information
- v. Testing and performance data
- w. Contractor information
- x. Field test reports

PART 2 PRODUCTS

Not Used

PART 3 EXECUTION

3.1 TRAINING

Prior to acceptance of the facility by the Contracting Officer for Beneficial Occupancy, provide comprehensive training for the systems and equipment specified in the technical specifications. The training must be targeted for the building maintenance personnel, and applicable building occupants. Instructors must be well-versed in the particular systems that they are presenting. Address aspects of the eOMSI Manual, as submitted in Section 01 78 24.00 20 FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI). Training must include classroom or field lectures based on the system operating requirements. The location of classroom training requires approval by the Contracting Officer.

3.1.1 Training Plan

Submit a written training plan to the Contracting Officer for approval at least 60 calendar days prior to the scheduled training. Training plan must be approved by the Quality Control Manager (QC) prior to forwarding to the Contracting Officer. Also, coordinate the training schedule with the Contracting Officer and QC. Include within the plan the following elements:

- a. Equipment included in training
- b. Intended audience
- c. Location of training
- d. Dates of training
- e. Objectives
- f. Outline of the information to be presented and subjects covered including description

- g. Start and finish times and duration of training on each subject
- h. Methods (e.g. classroom lecture, video, site walk-through, actual operational demonstrations, written handouts)
- i. Instructor names and instructor qualifications for each subject
- j. List of texts and other materials to be furnished by the Contractor that are required to support training
- k. Description of proposed software to be used for video recording of training sessions.

3.1.2 Training Content

The core of this training must be based on manufacturer's recommendations and the operation and maintenance information. The QC is responsible for overseeing and approving the content and adequacy of the training. Spend 95 percent of the instruction time during the presentation on the OPERATION AND MAINTENANCE DATA. Include the following for each system training presentation:

- a. Start-up, normal operation, shutdown, unoccupied operation, seasonal changeover, manual operation, controls set-up and programming, troubleshooting, and alarms.
- b. Relevant health and safety issues.
- c. Discussion of how the feature or system is environmentally responsive. Advise adjustments and optimizing methods for energy conservation.
- d. Design intent.
- e. Use of O&M Manual Files.
- f. Review of control drawings and schematics.
- g. Interactions with other systems.
- h. Special maintenance and replacement sources.
- i. Tenant interaction issues.

3.1.3 Training Outline

Provide the eOMSI Manual files as specified in Section 01 78 24.00 20, FACILITY ELECTRONIC OPERATION AND MAINTENANCE SUPPORT INFORMATION (eOMSI), and a written course outline listing the major and minor topics to be discussed by the instructor on each day of the course to each trainee in the course. Provide the course outline 14 calendar days prior to the training.

3.1.4 Training Video Recording

Record classroom training session(s) on video. Provide to the Contracting Officer two copies of the training session(s) in DVD video recording format. Capture within the recording, in video and audio, the

instructors' training presentations including question and answer periods with the attendees. The recording camera(s) must be attended by a person during the recording sessions to assure proper size of exhibits and projections during the recording are visible and readable when viewed as training.

3.1.5 Unresolved Questions from Attendees

If, at the end of the training course, there are questions from attendees that remain unresolved, the instructor must send the answers, in writing, to the Contracting Officer for transmittal to the attendees, and the training video must be modified to include the appropriate clarifications.

3.1.6 Validation of Training Completion

Ensure that each attendee at each training session signs a class roster daily to confirm Government participation in the training. At the completion of training, submit a signed validation letter that includes a sample record of training for reporting what systems were included in the training, who provided the training, when and where the training was performed, and copies of the signed class rosters. Provide two copies of the validation to the Contracting Officer, and one copy to the Operation and Maintenance Manual Preparer for inclusion into the Manual's documentation.

3.1.7 Quality Control Coordination

Coordinate this training with the QC in accordance with Section Section 01 45 00.00 20 QUALITY CONTROL.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 01 - GENERAL REQUIREMENTS

SECTION 01 78 24.00 10

FACILITY DATA REQUIREMENTS

05/18, CHG 1: 11/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS AND ABBREVIATIONS
 - 1.2.1 Assets
 - 1.2.2 Attributes
 - 1.2.3 Facility Data
 - 1.2.4 Facility Document Set (FDS)
 - 1.2.5 Facility Data Workbook (FDW)
 - 1.2.6 Facility Data Project Execution Plan (FDPxP)
- 1.3 UNITS OF MEASURE
- 1.4 SUBMITTALS
- 1.5 QUALITY CONTROL
 - 1.5.1 Facility Data Project Execution Plan (FDPxP)
 - 1.5.1.1 Front Matter
 - 1.5.1.2 Project Information
 - 1.5.1.3 Submittal Schedule
 - 1.5.1.4 Personnel
 - 1.5.1.5 Facility Data Workbook(s)
 - 1.5.1.6 Facility Document Set(s)
 - 1.5.1.7 Protocols
 - 1.5.2 Meetings
 - 1.5.2.1 Pre-Construction Meeting
 - 1.5.2.2 FDPxP Coordination Meeting
 - 1.5.2.3 Submittal Coordination Meeting
 - 1.5.3 Facility Turnover and Contract Closeout
 - 1.5.4 Facility Data Workbook Quality Requirements
 - 1.5.5 Facility Document Set Quality Requirements
 - 1.5.5.1 Document Files
 - 1.5.5.2 Photograph Files
 - 1.5.5.3 Drawing Files
 - 1.5.6 Facility Document Set Integrity Requirements
 - 1.5.6.1 File Protection
 - 1.5.6.2 Manufacturer-Specific Documents
- 1.6 DELIVERY, STORAGE, AND HANDLING
 - 1.6.1 Number of Copies
 - 1.6.2 Malicious Content
 - 1.6.3 Storage Media

PART 2 PRODUCTS

- 2.1 FACILITIES
- 2.2 FACILITY DATA WORKBOOK(S)
 - 2.2.1 Assets
 - 2.2.2 Attributes
- 2.3 FACILITY DOCUMENT SET

- 2.3.1 Organization
 - 2.3.1.1 O&M Data Hierarchy
 - 2.3.1.2 Record Drawings Hierarchy

PART 3 EXECUTION

- 3.1 CONSTRUCTION PROGRESS SUBMITTALS
- 3.2 CONSTRUCTION FINAL SUBMITTALS
- 3.3 FACILITY DATA WORKBOOK VERIFICATION

-- End of Section Table of Contents --

SECTION 01 78 24.00 10

FACILITY DATA REQUIREMENTS
05/18, CHG 1: 11/21

PART 1 GENERAL

This specification requires the collection, organization, and turnover of electronic Facility Data for specific assets designed and constructed as part of this contract. Provide a Facility Document Set (FDS) and Facility Data Workbook (FDW) as defined in this specification. See Sections 01 33 00 SUBMITTAL PROCEDURES, 01 78 00 CLOSEOUT SUBMITTALS, and 01 78 23 OPERATION AND MAINTENANCE DATA, for additional Facility Data delivery requirements.

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 19005-3	(2012) Document Management -- Electronic Document File Format for Long-Term Preservation -- Part 3: Use of ISO 32000-1 with Support for Embedded Files (PDF/A-3)
ISO 32000-1	(2008) Document Management -- Portable Document Format -- Part 1: PDF 1.7

1.2 DEFINITIONS AND ABBREVIATIONS

1.2.1 Assets

Assets are specific items of property or equipment.

1.2.2 Attributes

Attributes are individual pieces of Facility Data that describe facilities and their associated assets.

1.2.3 Facility Data

Information defined and collected in the Facility Data Workbook (FDW) and Facility Document Set (FDS).

1.2.4 Facility Document Set (FDS)

An electronically compiled and organized document containing the supporting documents and data used to populate the Facility Data Workbook during its respective phase of development.

- a. For construction-based deliverables, the FDS is comprised of the project Operation and Maintenance Data Packages and Government-Approved Record drawings.

1.2.5 Facility Data Workbook (FDW)

A pre-formatted spreadsheet template used to compile Asset, Attribute, Facility, and Space Data that the Government wishes to manage via electronic means. The FDW also contains all requirements associated with proper collection, organization, and turnover of the Facility Data.

1.2.6 Facility Data Project Execution Plan (FDPxP)

A document that describes the clear and organized plan for the collection, organization, and turnover of the Facility Data deliverables required by this specification.

1.3 UNITS OF MEASURE

Provide Facility Data deliverables utilizing the units of measure identified in the contract documents.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Facility Data Project Execution Plan(FDPxP)

SD-10 Operation and Maintenance Data

Facility Data Workbook, Construction Progress; G

Facility Document Set, Construction Progress; G

SD-11 Closeout Submittals

Facility Data Workbook, Construction Final; G

Facility Document Set, Construction Final; G

1.5 QUALITY CONTROL

1.5.1 Facility Data Project Execution Plan (FDPxP)

Provide the Government with a plan for the collection, organization, and turnover of the Facility Data deliverables to the Government. At a minimum, include the following items in the FDPxP:

1.5.1.1 Front Matter

Provide a Cover Page, Table of Contents, and Executive Summary/Objectives.

1.5.1.2 Project Information

List the Project Owner, Project Name, Project Location and address, Contract Type, Project Description, Project/Contract Number, Project Milestones.

1.5.1.3 Submittal Schedule

Identify delivery schedule for all deliverables in compliance with the submission requirements identified in this specification.

1.5.1.4 Personnel

Identify key personnel involved in the development of the Facility Data deliverables including Contractor and Government personnel.

1.5.1.5 Facility Data Workbook(s)

Identify Facility and Space Data as applicable at time of FDPxP submission. Individually list every asset group from the FDW Requirements that will require Facility Data collection. No attribute data is required at this time. Identify asset groups from the FDW Requirements that are not required within the scope of this Contract. Document the version of FDW to be used through the duration of the project.

1.5.1.6 Facility Document Set(s)

Define structure and format of the submittal. Provide a comprehensive outline of the final FDS to be delivered. Organize the outline with headings, titles, and descriptions such that the Government may ascertain that working documents comply with the formatting requirements defined by this specification.

1.5.1.7 Protocols

Detailed procedures:

- a. Facility Data documentation/collection process.
- b. Facility Document Set production/development process.
- c. Collaboration procedures including strategy, meetings, communication, and subcontractor/consultant involvement.
- d. Quality Control, including site verification of FDW, as applicable.
- e. File and folder naming structure.
- f. Hardware and software being used for collection and organization of Facility Data. Identify type, format, and anticipated organization of digital storage media to be provided as part of required deliverables. Include means and methods for checking deliverables for malicious content.

1.5.2 Meetings

To assure that Facility Data requirements are being met through the duration of the project, organize the following meetings and discuss the subsequent topics:

1.5.2.1 Pre-Construction Meeting

At a minimum, discuss the following:

- a. The requirement for Facility Data deliverables under this contract.

- b. Primary roles and responsibilities associated with the development and delivery of the Facility Data deliverables, and.
- c. Identify and agree upon a date and attendance list for the meetings described below:

1.5.2.2 FDPxP Coordination Meeting

- a. Facilitate a meeting following submission and Government review of the FDPxP. Include the Facility Data Preparer(s), Quality Control (QC) Manager, Government's Facility Data Proponent, Contracting Officer's Representative, and Directorate of Public Works (DPW) Base Civil Engineer (BCE) Facilities Management Specialist (FMS). Also include Government personnel required for obtaining security clearances and waivers for proper Facility Data collection in this meeting.
- b. The purpose of this meeting is to coordinate the efforts necessary by contract parties to ensure an accurate collection, preparation, quality control, and submittal of these deliverables.
- c. The FDPxP serves as the primary agenda for this meeting. At a minimum, discuss the following:
 - (1) Processes and methods of gathering facility data during construction. Discuss and obtain special permissions and waivers as necessary (such as photo waivers and data encryption);
 - (2) Contractor Quality Control practices and procedures;
 - (3) Corrective actions necessary for Government approval of FDPxP;
 - (4) Necessity for additional or recurring Facility Data Coordination Meetings outside of those required by this specification, as requested by the Contractor. Intent of these meetings would be to maintain regular contact between responsible parties of the Contractor and Government with regard to development of the facility data deliverables. Conduct status meetings with a frequency agreed upon at this meeting.

1.5.2.3 Submittal Coordination Meeting

- a. Facilitate a meeting following submission and Government review of each design or progress submittal of the Facility Data. Include the Facility Data Preparer(s), Quality Control (QC) Manager, Government's Facility Data Proponent, Contracting Officer's Representative, and Civil Engineer (BCE). Include Mechanical, Electrical, Plumbing, and Fire Protection subcontractors as applicable.
- b. The purpose of this meeting is to demonstrate ongoing compliance with the requirements identified in this specification.
- c. The applicable deliverables, along with Government remarks associated with review of these submittals serve as the primary guide and agenda for this meeting. At a minimum, discuss the following during this meeting:
 - (1) Review assets, applicable attributes, facility, and space data in FDW at time of submittal;

- (2) Demonstrate Quality Control and site verification procedures, as applicable, by Contractor QC;
- (3) Review contents and organization of FDS at time of submittal;
- (4) Discuss Government review comments and unresolved items preventing completion and Government approval of the Facility Data Workbook and Facility Document Set.

1.5.3 Facility Turnover and Contract Closeout

Include the Facility Document Set, Construction Final as a deliverable in Facility Turnover and Contract Closeout procedures as defined in 01 78 00 CLOSEOUT SUBMITTALS.

1.5.4 Facility Data Workbook Quality Requirements

For each submittal, ensure that the information contained in the FDW(s) reflects the minimum content requirements defined in the PART 3 EXECUTION portion of this section. Ensure that information provided as part of the FDW(s) conforms to the standards described below:

- a. Compile FDW(s) using approved spreadsheet templates. Do not alter the formatting or organizational layout of the templates. For this Contract, templates are available for download from the USACE CAD/BIM Technology Center website, site information provided in the PART 2 PRODUCTS portion of this section. Editable copies of the templates will be provided upon Contract Award.
- b. Instructions for the proper maintenance and completion of these FDWs are contained in the FDW Requirements contained within the FDW template.

1.5.5 Facility Document Set Quality Requirements

Ensure that information provided as part of each FDS conforms to the electronic and data formatting standards identified in 01 33 00 SUBMITTAL REQUIREMENTS and 01 78 23 OPERATION AND MAINTENANCE DATA.

1.5.5.1 Document Files

Utilize PDF file format in accordance with ISO 32000-1 and ISO 19005-3 for all document-based files. Provide files from original sources, text-searchable, and saved in "Standard" (uncompressed) resolution. Bookmark and label files as defined in the PART 2 PRODUCTS portion of this section.

1.5.5.2 Photograph Files

If photographs are required, utilize JPEG file format for all photograph and image files. Provide full-color photos with photo resolution of not less than 4 megapixels and not more than 12 megapixels.

Provide a copy of installation-specific letters or waivers allowing permission to take installed equipment photographs on this Contract. Waivers need not be attached to every photo, only one copy of each permission letter need be included in the Government deliverables.

1.5.5.3 Drawing Files

Provide all drawings required by this specification in full-size PDF format in accordance with ISO 32000-1 and ISO 19005-3. Produce PDF files from original sources, text-searchable, and saved in "Standard" (uncompressed) resolution whenever possible. Bookmark and label files as defined in the PART 2 PRODUCTS portion of this section.

Submission of scanned or photocopied drawing files is prohibited. Only vector-preserved PDF files are acceptable.

1.5.6 Facility Document Set Integrity Requirements

Ensure that information provided as part of each FDS conforms to the integrity standards identified below:

1.5.6.1 File Protection

Do not restrict data files, document files or photographic files from being printed, exported, modified or copied. Do not deliver files with restrictions such as expiration date and locks for access, viewing, archiving, or editing.

1.5.6.2 Manufacturer-Specific Documents

Provide text-searchable, vector-based document files from the manufacturer's online or electronic documentation. Color documents are preferred. Provide documents specific to the product(s) installed under this Contract. When possible, do not submit document files containing multiple product catalogs from the same manufacturer, or product data from multiple manufacturers in the same file. Provide documents directly from the manufacturer whenever possible. Do not provide scanned copies of hardcopy documents.

1.6 DELIVERY, STORAGE, AND HANDLING

Deliver facility data submittals in an organized and legible manner. Provide submittals adhering to the requirements of 01 33 00 SUBMITTAL REQUIREMENTS and 01 78 23 OPERATION AND MAINTENANCE DATA.

1.6.1 Number of Copies

Provide three identical copies of disks for approval; for each submittal and each facility required. Provide on approved electronic media (one copy per disk or set of disks) as defined below. Provide submittal files on electronic storage media in compliance with the quality requirements identified in this specification.

1.6.2 Malicious Content

Scan all files for malicious viruses using a commercially available scanning program that is routinely updated to identify and remove current virus threats.

1.6.3 Storage Media

Provide facility data on disk-based (DVD-R/RW) media. Deviations from the required storage media must be approved by the Government. Select and apply technology used for electronic data transmission to ensure that the

full Facility Data submittal for each facility is provided on one single disk, whenever possible. When separation of the submittal is required, first separate the FDS and the FDW onto separate media. Second, separate FDS into logical segments or components. Further divisions must be documented in the FDPxP and approved by the Government.

Provide Facility Data on disk-based (DVD-R/RW) media. Deviations from the required storage media must be approved by the Government. Select and apply technology used for electronic data transmission to ensure that the full Facility Data submittal for each facility is provided on one single disk, whenever possible. When separation of the submittal is required, first separate the FDS and the FDW onto separate media. Second, separate FDS into logical segments or components. Further divisions must be documented in the FDPxP and approved by the Government.

- a. Apply a label directly printed to storage media. Do not provide adhesive, paper-based labels. List the name of the facility, Project, Project location, Contract number, Designer of Record firm/Prime Contractor company's name, title of submission, and security classification (in accordance with the appropriate security classification labeling regulations) on the label. If multiple disks are provided, clearly document the contents of each disk on the label.
- b. Include the name and contact information of the individual who produced the final data disk to ensure that problems with the data or media can be easily resolved.
- c. When browsed on a computer, the disk must display the following folders and their associated content:
 - (1) Facility Data Workbook (containing 1 FDW per facility);
 - (2) Facility Document Set (containing 1 FDS per facility);
 - (3) FDPxP (containing 1 PxP per contract);
 - (4) Readme (Containing 1 TXT, PDF, or HTML file with general use information, organizational instructions, and basic preparer contact information. Include all information included on the storage media label).

PART 2 PRODUCTS

2.1 FACILITIES

Facilities that require individual (separate and complete) Facility Data deliverables as described in this specification are as follows:

- a. Tanks and related pump station

2.2 FACILITY DATA WORKBOOK(S)

Provide one compiled FDW for each facility identified above. Complete all portions of each FDW including facility, space, asset, and attribute data in compliance with the FDW Requirements. Download the current FDW template (.xlsm format) from the USACE CAD/BIM Technology Center website at <https://cadbimcenter.erdcdren.mil>.

2.2.1 Assets

- a. Compile an FDW that contains the tanks and related pump structure, maintainable and warrantable equipment (assets) associated with each

facility. This includes assets in contract scope and within the project extents. See 01 78 00 CLOSEOUT SUBMITTALS and 01 78 23 OPERATION AND MAINTENANCE DATA for related requirements. Assets include, but are not limited to, those types described in the "Required Assets" portion of the FDW template, those assets listed in the table below, and additional assets defined in the FDPxP. Itemized FDW asset entries (instance-based). Entries indicative of multiple assets (type-based) are not allowed.

SUPPLEMENTAL ASSET LIST (INCLUDE IN ADDITION TO "REQUIRED ASSETS" LISTED IN FDW TEMPLATE)	
Asset	Description/Example
Valves	All operable valves in facility excluding those installed in plumbing, HVAC, and fire protection systems.
Tanks and Vaults	Items such as fuel storage tanks expansion tanks, septic tanks, oil-water separators, grease separators

Document assets applicable to the scope of this project in the FDPxP.

- b. Sub-component assets that are an integral and functional part of another component (e.g. An electric motor that serves as part of an air-handling unit) need not be duplicated or listed separately as its own asset.
- c. Definitions, descriptions, and formatting requirements for these assets can be found in the FDW Requirements contained within the FDW template.
- d. If an asset type is not included in the scope of the Project, no Facility Data (assets or attributes) are to be included in the FDW (even as a placeholder) for that asset type.

2.2.2 Attributes

- a. Populate each individual asset with all required attributes defined in the "Required Attributes" portion of the FDW template.
- b. Definitions, descriptions, and formatting requirements for these attributes can be found in the FDW Requirements contained within the

FDW template.

- c. If an attribute is not applicable, populate that field with "N/A." Do not leave it blank.

2.3 FACILITY DOCUMENT SET

2.3.1 Organization

Organize the FDS in a hierarchical manner as follows. Use electronic bookmarks to create an easily navigable document. The first and primary hierarchical level must contain the following bookmarks:

- a. "Design Data" - See subordinate hierarchical requirements in the "DESIGN DATA HIERARCHY" paragraph.
- b. "O&M Data" - See subordinate hierarchical requirements in the "O&M DATA HIERARCHY" paragraph.
- c. "Record Drawings" - See subordinate hierarchical requirements in paragraph RECORD DRAWINGS HIERARCHY.

2.3.1.1 O&M Data Hierarchy

Under "O&M Data" provide all Government-Approved O&M Data Packages as defined in 01 78 23 OPERATION AND MAINTENANCE DATA and as required by technical specifications contained within this contract. Further organize this information under the following hierarchical levels:

- a. The contract specification and title under which the Data Package and the associated equipment or system references. (e.g. 26 23 00.00 40 - SWITCHBOARDS AND SWITCHGEAR)
- b. The Data Package Number as defined in 01 78 23 OPERATION AND MAINTENANCE DATA. (e.g. Data Package 2)

2.3.1.2 Record Drawings Hierarchy

Under "Record Drawings" provide an electronic copy of the Government-Approved record drawings, as specified in 01 78 00 CLOSEOUT SUBMITTALS, for the project in PDF format. Further group discipline sheets under the following hierarchical levels:

- a. The full discipline heading represented by the contents of the sheet and as shown in the Record Drawing Sheet Index. Organize these headings in the order that the drawings set is organized. (General, Civil, Structural, Architectural, Interiors, Plumbing, Mechanical, Electrical, Telecommunications)
- b. The Sheet ID and Sheet Name as found in the Record Drawing Sheet Index and in accordance with the AEC CAD Standard referenced in 01 78 00 CLOSEOUT SUBMITTALS. (e.g. G-001 - LEGEND; CS101 - SITE PLAN AREA 101; A-101 - OVERALL FIRST FLOOR PLAN; P-601 - FIRST FLOOR DWS WATER RISER DIAGRAM)

PART 3 EXECUTION

3.1 CONSTRUCTION PROGRESS SUBMITTALS

Submit the FDW and FDS construction progress submittals together. Meet the following completeness and formatting requirements listed below:

- a. Provide Facility Data Workbook, Construction Progress submittal(s) when all assets are identified, but not later than 60 days prior to Beneficial Occupancy Date (BOD) as identified in the Government-Approved construction schedule. Clearly identify assets or asset groups missing in the "variations" section of the ENG Form 4025 Transmittal Form provided with the submittal. Populate assets with front-loaded attribute data that is available at the time of asset input. See the FDW Requirements contained within the FDW template for a list of attributes to be completed for this submittal.
- b. Submit individual FDW templates for each facility identified in the "FACILITIES" paragraph. While FDWs are not required to be complete for this submittal, provide accurate and correctly formatted data according to the FDW Requirements.
- c. Submit a sample or working Facility Document Set, Construction Progress submittal containing "draft" or "example" documents that are organized in the manner defined by this specification. Draft or example documents need not be technically accurate or complete in their content, but defined and separated in a manner such that all organizational and formatting requirements defined by this specification may be evaluated.

3.2 CONSTRUCTION FINAL SUBMITTALS

Submit the FDW and FDS construction final submittals as they are completed. Coordinate the Facility Data Workbook, Construction Final submittal with data verification procedures as defined in the accepted FDPxP. Provide the Facility Document Set, Construction Final submittal only after Government acceptance of its individual components as defined by 01 78 00 CLOSEOUT SUBMITTALS and 01 78 23 OPERATION AND MAINTENANCE DATA.

3.3 FACILITY DATA WORKBOOK VERIFICATION

Verify the FDW through the quality control personnel and procedures as defined in the FDPxP. One-hundred percent accuracy of FDW information is required for Government acceptance of the Facility Data Workbook.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 41 00

DEMOLITION

08/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 PROJECT DESCRIPTION
 - 1.2.1 Definitions
 - 1.2.1.1 Demolition
 - 1.2.1.2 Demolition Plan
 - 1.2.2 Demolition Plan
 - 1.2.3 General Requirements
- 1.3 ITEMS TO REMAIN IN PLACE
 - 1.3.1 Existing Construction Limits and Protection
 - 1.3.2 Trees
 - 1.3.3 Utility Service
 - 1.3.4 Facilities
- 1.4 BURNING
- 1.5 AVAILABILITY OF WORK AREAS
- 1.6 SUBMITTALS
- 1.7 QUALITY ASSURANCE
 - 1.7.1 Dust and Debris Control
- 1.8 PROTECTION
 - 1.8.1 Traffic Control Signs
 - 1.8.2 Protection of Personnel
- 1.9 RELOCATIONS
- 1.10 EXISTING CONDITIONS

PART 2 PRODUCTS

- 2.1 FILL MATERIAL

PART 3 EXECUTION

- 3.1 EXISTING FACILITIES TO BE REMOVED
 - 3.1.1 Structures
 - 3.1.2 Utilities and Related Equipment
 - 3.1.2.1 General Requirements
 - 3.1.2.2 Disconnecting Existing Utilities
 - 3.1.3 Chain Link Fencing
 - 3.1.4 Paving and Slabs
 - 3.1.5 Concrete
 - 3.1.6 Structural Steel
 - 3.1.7 Miscellaneous Metal
 - 3.1.8 Patching
 - 3.1.9 Mechanical Equipment and Fixtures
 - 3.1.9.1 Piping
 - 3.1.10 Electrical Equipment and Fixtures
 - 3.1.10.1 Fixtures

- 3.1.10.2 Electrical Devices
- 3.1.10.3 Wiring Ducts or Troughs
- 3.1.10.4 Conduit and Miscellaneous Items
- 3.1.11 Items With Unique/Regulated Disposal Requirements
- 3.2 CONCURRENT EARTH-MOVING OPERATIONS
- 3.3 DISPOSITION OF MATERIAL
 - 3.3.1 Title to Materials
 - 3.3.2 Reuse of Materials and Equipment
 - 3.3.3 Salvaged Materials and Equipment
 - 3.3.4 Unsalvageable and Non-Recyclable Material
- 3.4 CLEANUP
- 3.5 DISPOSAL OF REMOVED MATERIALS
 - 3.5.1 Regulation of Removed Materials
 - 3.5.2 Burning on Government Property
 - 3.5.3 Removal from Government Property
- 3.6 REUSE OF SALVAGED ITEMS

-- End of Section Table of Contents --

SECTION 02 41 00

DEMOLITION
08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF SAFETY PROFESSIONALS (ASSP)

ASSP A10.6 (2006) Safety & Health Program
Requirements for Demolition Operations -
American National Standard for
Construction and Demolition Operations

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety -- Safety and Health
Requirements Manual

JUNTA DE ANDALUCIA (JA)

Decreto 73/2012 (2012) Decreto 73/2012, de 22 de marzo,
por el que se aprueba el Reglamento de
Residuos de Andalucía.

MINISTERIO DE MEDIO AMBIENTE (MA)

Orden MAM/304/2002 (2002) Orden MAM/304/2002, de 8 de
febrero, por la que se publican las
operaciones de valorización y eliminación
de residuos y la lista europea de residuos.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

40 CFR 61 National Emission Standards for Hazardous
Air Pollutants

1.2 PROJECT DESCRIPTION

1.2.1 Definitions

1.2.1.1 Demolition

Demolition is the process of tearing apart and removing any feature of a facility together with any related handling and disposal operations.

1.2.1.2 Demolition Plan

Demolition Plan is the planned steps and processes for managing demolition activities and identifying the required sequencing activities and disposal mechanisms.

1.2.2 Demolition Plan

Prepare a Demolition Plan and submit proposed salvage, demolition, and removal procedures for approval before work is started. Include in the plan procedures for careful removal and disposition of materials specified to be salvaged, coordination with other work in progress, a disconnection schedule of utility services, and a detailed description of methods and equipment to be used for each operation and of the sequence of operations. Comply with the requirements and methods of Decreto 73/2012 and Orden MAM/304/2002. Identify components and materials to be salvaged for reuse or recycling with reference to paragraph Existing Facilities to be Removed. Append tracking forms for all removed materials indicating type, quantities, condition, destination, and end use. Coordinate with Waste Management Plan in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Provide procedures for safe conduct of the work in accordance with EM 385-1-1. Plan must be approved by Contracting Officer prior to work beginning.

1.2.3 General Requirements

Do not begin demolition until authorization is received from the Contracting Officer. The work of this section is to be performed in a manner that maximizes the value derived from the salvage and recycling of materials. The work includes demolition, salvage of identified items and materials, and removal of resulting rubbish and debris. Remove rubbish and debris from Government property daily, unless otherwise directed. Store materials that cannot be removed daily in areas specified by the Contracting Officer. In the interest of occupational safety and health, perform the work in accordance with EM 385-1-1, Section 23, Demolition, and other applicable Sections.

1.3 ITEMS TO REMAIN IN PLACE

Comply with FAR 52.236-9 to protect existing vegetation, structures, equipment, utilities, and improvements. Coordinate the work of this section with all other work indicated. Construct and maintain shoring, bracing, and supports as required. Ensure that structural elements are not overloaded. Increase structural supports or add new supports as may be required as a result of any cutting, removal, or demolition work performed under this contract. Do not overload structural elements or pavements to remain. Provide new supports and reinforcement for existing construction weakened by demolition, or removal work. Repairs, reinforcement, or structural replacement require approval by the Contracting Officer prior to performing such work.

1.3.1 Existing Construction Limits and Protection

Do not disturb existing construction beyond the extent indicated or necessary for installation of new construction. Provide temporary shoring and bracing for support of building components to prevent settlement or other movement. Provide protective measures to control accumulation and migration of dust and dirt in all work areas. Remove dust, dirt, and debris from work areas daily.

1.3.2 Trees

Protect trees within the project site which might be damaged during demolition, and which are indicated to be left in place, by a 1.8 m high

fence. Erect and secure fence a minimum of 1.5 m from the trunk of individual trees or follow the outer perimeter of branches or clumps of trees. Replace any tree designated to remain that is damaged during the work under this contract with like-kind or as approved by the Contracting Officer.

1.3.3 Utility Service

Maintain existing utilities indicated to stay in service and protect against damage during demolition operations. Prior to start of work, utilities serving each area of alteration or removal will be shut off by the Government and disconnected and sealed by the Contractor.

1.3.4 Facilities

Protect electrical and mechanical services and utilities. Where removal of existing utilities and pavement is specified or indicated, provide approved barricades, temporary covering of exposed areas, and temporary services or connections for electrical and mechanical utilities. Floors, roofs, walls, columns, pilasters, and other structural components that are designed and constructed to stand without lateral support or shoring, and are determined to be in stable condition, must remain standing without additional bracing, shoring, or lateral support until demolished, unless directed otherwise by the Contracting Officer. Ensure that no elements determined to be unstable are left unsupported and place and secure bracing, shoring, or lateral supports as may be required as a result of any cutting, removal, or demolition work performed under this contract.

1.4 BURNING

The use of burning at the project site for the disposal of refuse and debris will not be permitted.

1.5 AVAILABILITY OF WORK AREAS

Areas in which the work is to be accomplished will be available at contract award.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Demolition Plan; G
Existing Conditions

1.7 QUALITY ASSURANCE

Notify the Contracting Officer in writing 10 working days prior to the commencement of work in accordance with 40 CFR 61, Subpart M. Comply with local and regional hauling and disposal regulations. In addition to the requirements of the "Contract Clauses," conform to the safety requirements contained in ASSP A10.6. Comply with the Environmental requirements specified. Use of explosives will not be permitted.

1.7.1 Dust and Debris Control

Prevent the spread of dust and debris and avoid the creation of a nuisance in the surrounding area. Do not use water if it results in hazardous or objectionable conditions such as, but not limited to, ice, flooding, or pollution.

1.8 PROTECTION

1.8.1 Traffic Control Signs

a. Where pedestrian and driver safety is endangered in the area of removal work, use traffic barricades with flashing lights. Anchor barricades in a manner to prevent displacement by wind. Notify the Contracting Officer prior to beginning such work.

1.8.2 Protection of Personnel

Before, during and after the demolition work continuously evaluate the condition of the site specific features being demolished and take immediate action to protect all personnel working in and around the project site. No area, section, or component of floors, roofs, walls, columns, pilasters, piping or other structural element will be allowed to be left standing without sufficient bracing, shoring, or lateral support to prevent collapse or failure while workmen remove debris or perform other work in the immediate area.

1.9 RELOCATIONS

Perform the removal and reinstallation of relocated items as indicated with workmen skilled in the trades involved. Repair or replace items to be relocated which are damaged by the Contractor with new undamaged items as approved by the Contracting Officer.

1.10 EXISTING CONDITIONS

Before beginning any demolition work, survey the site and examine the drawings and specifications to determine the extent of the work. Record existing conditions in the presence of the Contracting Officer showing the condition of structures and other facilities adjacent to areas of alteration or removal. Photographs or electronic images with a minimum resolution of 3072 x 2304 pixels, capable of a print resolution of 300 dpi, will be acceptable as a record of existing conditions. Include in the record the elevation of the top of foundations, piping and, possible conflicting electrical conduits, piping, etc. Include the location and extent of existing cracks and other damage and description of surface conditions that exist prior to starting work. It is the Contractor's responsibility to verify and document all required outages which will be required during the course of work, and to note these outages on the record document. Submit survey results to the Contracting Officer.

PART 2 PRODUCTS

2.1 FILL MATERIAL

a. Comply with excavating, backfilling, and compacting procedures for soils used as backfill material to fill voids, depressions or excavations resulting from demolition of piping and structures. Provide fill material consisting of waste products from demolition

until all waste appropriate for this purpose is consumed.

- b. Proposed fill material must be sampled and tested by an approved soil testing laboratory, as indicated in Section 31 23 00.00 20 EXCAVATION AND FILL:

PART 3 EXECUTION

3.1 EXISTING FACILITIES TO BE REMOVED

Dismantled and removed materials are to be separated, set aside, and prepared as specified, and stored or delivered to a collection point for reuse, remanufacture, recycling, or other disposal, as specified. Designate materials for reuse onsite whenever possible.

3.1.1 Structures

- a. Locate demolition equipment throughout the structure and remove materials so as to not impose excessive loads to supporting walls, floors, or framing.

3.1.2 Utilities and Related Equipment

3.1.2.1 General Requirements

Do not interrupt existing utilities serving occupied or used facilities, except when authorized in writing by the Contracting Officer. Do not interrupt existing utilities serving facilities occupied and used by the Government except when approved in writing and then only after temporary utility services have been approved and provided. Do not begin demolition work until all utility disconnections have been made. Shut off and cap utilities for future use, as indicated.

3.1.2.2 Disconnecting Existing Utilities

Remove existing utilities, as indicated and terminate in a manner conforming to the nationally recognized code covering the specific utility and approved by the Contracting Officer. When utility lines are encountered but are not indicated on the drawings, notify the Contracting Officer prior to further work in that area. Remove meters and related equipment and deliver to a location in accordance with instructions of the Contracting Officer.

3.1.3 Chain Link Fencing

Remove chain link fencing, gates and other related salvaged items scheduled for removal and transport to designated areas. Remove gates as whole units. Cut chain link fabric to 6 m lengths and store in rolls off the ground.

3.1.4 Paving and Slabs

Sawcut and remove concrete and asphaltic concrete paving and slabs including aggregate base as indicated. Provide neat sawcuts at limits of pavement removal as indicated. Move, grind and store pavement and slabs designated to be recycled and utilized in this project as directed by the Contracting Officer. Remove pavement and slabs not to be used in this project from the project site at Contractor's expense. If pavement remnant is 1 m or less, then remove and restore pavement.

3.1.5 Concrete

Saw concrete along straight lines to a depth of a minimum 50 mm. Make each cut in walls perpendicular to the face and in alignment with the cut in the opposite face. Break out the remainder of the concrete provided that the broken area is concealed in the finished work, and the remaining concrete is sound. At locations where the broken face cannot be concealed, grind smooth or saw cut entirely through the concrete. Sawcuts in existing concrete sidewalk, and roadsand parking areas must be at the nearest existing expansion joint or weakened plane joint and at full depth.

3.1.6 Structural Steel

Dismantle structural steel at field connections and in a manner that will prevent bending or damage. Salvage for recycle structural steel, steel joists, girders, angles, plates, columns and shapes. Do not use flame-cutting torches. Transport steel joists and girders as whole units and not dismantled. Transport structural steel shapes to a designated recycling facility in the area directed by the Contracting Officer (close to Tow Way Road) and stored off the ground, protected from the weather.

3.1.7 Miscellaneous Metal

Salvage shop-fabricated items such as access doors and frames, steel gratings, metal ladders, wire mesh partitions, metal railings, metal windows and similar items as whole units. Salvage light-gage and cold-formed metal framing, such as steel studs, metal gutters and similar items. Provide separate containers to collect scrap metal and transport to a scrap metal collection or recycling facility close to Tow Way Road, in accordance with the Waste Management Plan.

3.1.8 Patching

Where removals leave holes and damaged surfaces exposed in the finished work, patch and repair these holes and damaged surfaces to match adjacent finished surfaces, using on-site materials when available. Where new work is to be applied to existing surfaces, perform removals and patching in a manner to produce surfaces suitable for receiving new work. Make finished surfaces of patched area flush with the adjacent existing surface and match the existing adjacent surface as closely as possible to texture and finish. Provide patching as specified and indicated, and include the following:

- a. Concrete and Masonry: Completely fill holes and depressions, caused by previous physical damage or left as a result of removals with an approved masonry patching material, applied in accordance with the manufacturer's printed instructions.

3.1.9 Mechanical Equipment and Fixtures

3.1.9.1 Piping

Disconnect piping at unions, flanges and valves, and fittings as required to reduce the pipe into straight lengths for practical storage. Store salvaged piping according to size and type. If the piping that remains can become pressurized due to upstream valve failure, attach end caps, blind flanges, or other types of plugs or fittings with a pressure gage and bleed valve to the open end of the pipe to ensure positive leak

control. Carefully dismantle piping that previously contained gas, gasoline, oil, or other dangerous fluids, with precautions taken to prevent injury to persons and property. Store piping outdoors until all fumes and residues are removed. Box prefabricated supports, hangers, plates, valves, and specialty items according to size and type. Wrap sprinkler heads individually in plastic bags before boxing.

3.1.10 Electrical Equipment and Fixtures

Salvage motors, motor controllers, and operating and control equipment that are attached to the driven equipment. Salvage wiring systems and components. Box loose items and tag for identification. Disconnect primary, secondary, control, communication, and signal circuits at the point of attachment to their distribution system.

3.1.10.1 Fixtures

Remove and salvage electrical fixtures. Salvage unprotected glassware from the fixture and salvage separately. Salvage incandescent, mercury-vapor, and fluorescent lamps and fluorescent ballasts manufactured prior to 1978, boxed and tagged for identification, and protected from breakage.

3.1.10.2 Electrical Devices

Remove and salvage switches, switchgear, transformers, conductors including wire and nonmetallic sheathed and flexible armored cable, regulators, meters, instruments, plates, circuit breakers, panelboards, outlet boxes, and similar items. Box and tag these items for identification according to type and size.

3.1.10.3 Wiring Ducts or Troughs

Remove and salvage wiring ducts or troughs. Dismantle plug-in ducts and wiring troughs into unit lengths. Remove plug-in or disconnecting devices from the busway and store separately.

3.1.10.4 Conduit and Miscellaneous Items

Salvage conduit except where embedded in concrete or masonry. Consider corroded, bent, or damaged conduit as scrap metal. Sort straight and undamaged lengths of conduit according to size and type. Classify supports, knobs, tubes, cleats, and straps as debris to be removed and disposed.

3.1.11 Items With Unique/Regulated Disposal Requirements

Remove and dispose of items with unique or regulated disposal requirements in the manner dictated by law or in the most environmentally responsible manner.

3.2 CONCURRENT EARTH-MOVING OPERATIONS

Do not begin excavation, filling, and other earth-moving operations that are sequential to demolition work in areas occupied by structures to be demolished or deconstructed until all demolition in the area has been completed and debris removed. Fill holes, and other hazardous openings.

3.3 DISPOSITION OF MATERIAL

3.3.1 Title to Materials

Except for salvaged items, all materials and equipment removed and not reused or salvaged, become the property of the Contractor and must be removed from Government property. Materials approved for storage by the Contracting Officer must be removed before completion of the contract. Title to materials resulting from demolition, and materials and equipment to be removed, is vested in the Contractor upon approval by the Contracting Officer. The Government will not be responsible for the condition or loss of, or damage to, such property after contract award. Showing for sale or selling materials and equipment on site is prohibited.

3.3.2 Reuse of Materials and Equipment

Remove and store materials and equipment to be reused or relocated to prevent damage, and reinstall as the work progresses. Coordinate the re-use of materials and equipment with the re-use requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture re-use of materials in the diversion calculations for the project.

3.3.3 Salvaged Materials and Equipment

Remove materials and equipment that are indicated to be salvaged or are to be removed by the Contractor and that are to remain the property of the Government, and deliver to a storage site close to Tow Way Road.

- a. Salvage items and material to the maximum extent possible.
- b. Store all materials salvaged for the Contractor as approved by the Contracting Officer and remove from Government property before completion of the contract. Coordinate the salvaged materials with tracking requirements in accordance with Section 01 74 19 CONSTRUCTION WASTE MANAGEMENT AND DISPOSAL. Capture salvaged materials in the diversion calculations for the project.
- c. Remove salvaged items to remain the property of the Government in a manner to prevent damage, and packed or crated to protect the items from damage while in storage or during shipment. Items damaged during removal or storage must be repaired or replaced to match existing items. Properly identify the contents of containers.
- d. Remove the following items reserved as property of the using service prior to commencement of work under this contract: signs, concrete barriers, vinyl fencing, guardrail, chain link fence swing gate, and bus stop shelter.

3.3.4 Unsalvageable and Non-Recyclable Material

Dispose of unsalvageable and non-recyclable noncombustible materials per Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS. Dispose of unsalvageable and non-recyclable combustible material.

3.4 CLEANUP

Remove debris and rubbish from project site. Remove and transport the debris in a manner that prevents spillage on streets or adjacent areas.

Apply local regulations regarding hauling and disposal.

3.5 DISPOSAL OF REMOVED MATERIALS

3.5.1 Regulation of Removed Materials

Dispose of debris, rubbish, scrap, and other nonsalvageable materials resulting from removal operations with all applicable local and regional regulations as contractually specified.

3.5.2 Burning on Government Property

Burning of materials removed from demolished and deconstructed structures will not be permitted on Government property.

3.5.3 Removal from Government Property

Transport waste materials removed from demolished structures, except waste soil, from Government property for legal disposal. Dispose of waste soil per Section 01 57 19.01 21 ENVIRONMENTAL CONTROLS.

3.6 REUSE OF SALVAGED ITEMS

Recondition salvaged materials and equipment designated for reuse before installation. Replace items damaged during removal and salvage operations or restore them as necessary to usable condition.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 02 - EXISTING CONDITIONS

SECTION 02 61 13

EXCAVATION AND HANDLING OF CONTAMINATED MATERIAL

02/10, CHG 1: 02/21

PART 1 GENERAL

- 1.1 MEASUREMENT AND PAYMENT
 - 1.1.1 Measurement
 - 1.1.2 Payment
 - 1.1.2.1 Excavation and Transportation
 - 1.1.2.2 Backfilling
- 1.2 REFERENCES
- 1.3 DESCRIPTION OF WORK
 - 1.3.1 Scheduling
 - 1.3.2 Work Plan
 - 1.3.3 Other Submittal Requirements
- 1.4 SUBMITTALS
- 1.5 REGULATORY REQUIREMENTS
 - 1.5.1 Permits and Licenses
 - 1.5.2 Air Emissions

PART 2 PRODUCTS

- 2.1 SPILL RESPONSE MATERIALS
- 2.2 BACKFILL

PART 3 EXECUTION

- 3.1 SURVEYS
- 3.2 EXISTING STRUCTURES AND UTILITIES
- 3.3 CONTAMINATED MATERIAL REMOVAL
 - 3.3.1 Excavation
 - 3.3.2 Shoring
 - 3.3.3 Dewatering
- 3.4 CONFIRMATION SAMPLING AND ANALYSIS
- 3.5 CONTAMINATED MATERIAL STORAGE
 - 3.5.1 Stockpiles
 - 3.5.2 Roll-Off Units
 - 3.5.3 Liquid Storage
- 3.6 SAMPLING
 - 3.6.1 Sampling of Stored Material
- 3.7 SPILLS
- 3.8 BACKFILLING
 - 3.8.1 Confirmation Test Results
- 3.9 DISPOSAL REQUIREMENTS
- 3.10 CLOSURE REPORT

-- End of Section Table of Contents --

SECTION 02 61 13

EXCAVATION AND HANDLING OF CONTAMINATED MATERIAL
02/10, CHG 1: 02/21

PART 1 GENERAL

1.1 MEASUREMENT AND PAYMENT

1.1.1 Measurement

Measurement for excavation and onsite transportation shall be based on the actual number of cubic meters of contaminated material in-place prior to excavation. Determination of the volume of contaminated material excavated shall be based on cross-sectional volume determination reflecting the differential between the original elevations of the top of the contaminated material and the final elevations after removal of the contaminated material. Measurement for backfilling of excavated areas shall be based on in-place cubic meters of compacted fill.

1.1.2 Payment

1.1.2.1 Excavation and Transportation

Compensation for excavation and onsite transportation of contaminated material will be paid as a unit cost. This unit cost shall include any other items incidental to excavation and handling not defined as having a specific unit cost.

1.1.2.2 Backfilling

Compensation for backfill soil, transportation of backfill, backfill soil conditioning, backfilling, compaction, and geotechnical testing will be paid as a single unit cost.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE-EN ISO 14688-1	(2019) Geotechnical investigation and testing - Identification and classification of soil - Part 1: Identification and description (ISO 14688-1:2017)
--------------------	---

ASTM INTERNATIONAL (ASTM)

ASTM D2487	(2017; E 2020) Standard Practice for Classification of Soils for Engineering Purposes (Unified Soil Classification System)
ASTM D5434	(2012) Field Logging of Subsurface

Explorations of Soil and Rock

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1

(2014) Safety -- Safety and Health
Requirements Manual

JUNTA DE ANDALUCIA (JA)

Decreto 18/2015

(2015) Decreto 18/2015, de 27 de enero,
por el que se aprueba el reglamento que
regula el régimen aplicable a los suelos
contaminados.

MINISTERIO DE LA PRESIDENCIA (MP)

AAA/661/2013

(2013) Orden AAA/661/2013, de 18 de abril,
por la que se modifican los anexos I, II y
III del Real Decreto 1481/2001, de 27 de
diciembre, por el que se regula la
eliminación de residuos mediante depósito
en vertedero.

RD 9/2005

(2005) Real Decreto 9/2005, de 14 de
enero, por el que se establece la relación
de actividades potencialmente
contaminantes del suelo y los criterios y
estándares para la declaración de suelos
contaminados.

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1926

Safety and Health Regulations for
Construction

1.3 DESCRIPTION OF WORK

The work consists of excavation and temporary storage of approximately 250 cubic meters of contaminated material. Approximate locations of contaminated material are shown on the drawings. Characterization data on the nature and extent of the contaminated material is shown in Appendix A. Subsurface conditions are shown on the drawings. Submit a Work Plan in compliance with Spanish legislation RD 9/2005 and regional decree Decreto 18/2015, and as specified below. The requirements of those standards are captured in the "Plan de Prevención y Tratamiento de Suelos Contaminados en Instalaciones Militares" from the Spanish Ministry of Defense of March 2021. Provide a QRA and prepare the documentation to sample and analyze per AAA/661/2013 the soils potentially contaminated. The QRA and the execution of samples and laboratory test must be performed by a firm accredited by Junta de Andalucía for these works. Classify the contaminated soils as inert (Inerte), non-dangerous (No Peligroso) and dangerous (Peligroso). Quantify the volumes of each type of soil and identify landfills that are accredited for disposal of contaminated soil (vertidos de suelos contaminados en vertedero). Notify the Contracting Officer within 24 hours, and before excavation, if contaminated material is discovered that has not been previously identified or if other discrepancies between data provided and actual field conditions are discovered. Backfill material is not available onsite. Ground water is approximately 2 meters below pre-excavation ground surface.

1.3.1 Scheduling

Notify the Contracting Officer 10 calendar days prior to the start of excavation of contaminated material. The Contractor shall be responsible for contacting regulatory agencies in accordance with the applicable reporting requirements.

1.3.2 Work Plan

Submit a Work Plan within 30 calendar days after notice to proceed. No work at the site, with the exception of site inspections and surveys, shall be performed until the Work Plan is approved. Allow 30 calendar days in the schedule for the Government's review. No adjustment for time or money will be made if resubmittals of the Work Plan are required due to deficiencies in the plan. At a minimum, the Work Plan shall include:

- a. Schedule of activities.
- b. Method of excavation and equipment to be used.
- c. Shoring or side-wall slopes proposed.
- d. Dewatering plan.
- e. Storage methods and locations for liquid and solid contaminated material.
- f. Borrow sources and haul routes.
- g. Decontamination procedures.
- h. Spill contingency plan.
- i. Qualification and registration of the subcontractor supervising, sampling, testing, handling and preparing documentation for approval in front of the Regional Regulator (Junta de Andalucia).

1.3.3 Other Submittal Requirements

Submit separate cross-sections of each area before and after excavation and after backfilling, test results, and three copies of the Closure Report within 14 calendar days of work completion at the site.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Work Plan; G

SD-02 Shop Drawings

Surveys

SD-06 Test Reports

Closure Report; G

1.5 REGULATORY REQUIREMENTS

1.5.1 Permits and Licenses

Obtain required regional and local permits for excavation and storage of contaminated material. Permits shall be obtained at no additional cost to the Government. The contractor shall hire the services of a firm holding accreditation from Junta de Andalucía to control, supervise, perform samples and perform laboratory tests of contaminated soils for disposal in landfills. The study of contaminated soils and the final certification of remedial actions must be performed by a firm accredited by ENAC and recognized by Junta de Andalucía for contaminated soils inspection. The waste resulting from excavation of contaminated soils, technically called in Spain "residuo" has to be handled by an authorized waste manager ("Gestor de Residuos habilitado para el código LER de aplicación"), accredited for the code LER (Lista Europea de Residuos) of contaminated soils. All the personnel needed to complete these works shall be hired by the contractor without additional cost to the Government.

1.5.2 Air Emissions

Air emissions shall be monitored and controlled in accordance with local requirements.

PART 2 PRODUCTS

2.1 SPILL RESPONSE MATERIALS

Provide appropriate spill response materials including, but not limited to the following: containers, adsorbents, shovels, and personal protective equipment. Spill response materials shall be available at all times when contaminated materials/wastes are being handled or transported. Spill response materials shall be compatible with the type of materials and contaminants being handled.

2.2 BACKFILL

Backfill material shall be obtained from offsite sources approved by the Contracting Officer. Backfill shall be classified in accordance with ASTM D2487 or UNE-EN ISO 14688-1 as GW, GP, GM, SW, SP, or SM and shall be free from roots and other organic matter, trash, debris, snow, ice or frozen materials.

PART 3 EXECUTION

3.1 SURVEYS

Perform surveys immediately prior to and after excavation of contaminated material to determine the volume of contaminated material removed.

3.2 EXISTING STRUCTURES AND UTILITIES

No excavation shall be performed until site utilities have been field located. Take the necessary precautions to ensure no damage occurs to existing structures and utilities. Damage to existing structures and utilities resulting from the Contractor's operations shall be repaired at no additional cost to the Government. Utilities encountered that were not previously shown or otherwise located shall not be disturbed without approval from the Contracting Officer.

3.3 CONTAMINATED MATERIAL REMOVAL

3.3.1 Excavation

Areas of contamination shall be excavated to the depth and extent shown on the drawings and not more than 60 mm beyond the depth and extent shown on the drawings unless directed by the Contracting Officer. Excavation shall be performed in a manner that will limit spills and the potential for contaminated material to be mixed with uncontaminated material. An excavation log describing visible signs of contamination encountered shall be maintained for each area of excavation. Excavation logs shall be prepared in accordance with ASTM D5434.

3.3.2 Shoring

If workers must enter the excavation, it shall be evaluated, shored, sloped or braced as required by EM 385-1-1 and 29 CFR 1926 section 650.

3.3.3 Dewatering

Surface water shall be diverted to prevent entry into the excavation. Dewatering shall be limited to that necessary to assure adequate access, a safe excavation, prevent the spread of contamination, and to ensure that compaction requirements can be met. No dewatering shall be performed without prior approval of the Contracting Officer.

3.4 CONFIRMATION SAMPLING AND ANALYSIS

The Contracting Officer shall be present to inspect the removal of contaminated material from each site. After all material suspected of being contaminated has been removed, the excavation shall be examined for evidence of contamination. If the excavation appears to be free of contamination, field analysis shall be used to determine the presence of contamination using a real time vapor monitoring instrument or immunoassay field kits. Excavation of additional material shall be as directed by the Contracting Officer.

3.5 CONTAMINATED MATERIAL STORAGE

Material shall be placed in temporary storage immediately after excavation. The following paragraphs describe acceptable methods of material storage. Storage units shall be in good condition and constructed of materials that are compatible with the material or liquid to be stored. If multiple storage units are required, each unit shall be clearly labeled with an identification number and a written log shall be kept to track the source of contaminated material in each temporary storage unit.

3.5.1 Stockpiles

Stockpiles shall be constructed to isolate stored contaminated material from the environment. Stockpiles shall be constructed to include:

- a. A chemically resistant geomembrane liner free of holes and other damage. Non-reinforced geomembrane liners shall have a minimum thickness of 0.5 mm. Scrim reinforced geomembrane liners shall have a minimum weight of 20 kg/100 square m. The ground surface on which the geomembrane is to be placed shall be free of rocks greater than 13 mm in diameter and any other object which could damage the membrane.

- b. Geomembrane cover free of holes or other damage to prevent precipitation from entering the stockpile. Non-reinforced geomembrane covers shall have a minimum thickness of 0.25 mm. Scrim reinforced geomembrane covers shall have a minimum weight of 13 kg/100 square m. The cover material shall be extended over the berms and anchored or ballasted to prevent it from being removed or damaged by wind.
- c. Berms surrounding the stockpile, a minimum of 300 mm in height. Vehicle access points shall also be bermed.
- d. The liner system shall be sloped to allow collection of leachate. Storage and removal of liquid which collects in the stockpile, in accordance with paragraph Liquid Storage.

3.5.2 Roll-Off Units

Roll-off units used to temporarily store contaminated material shall be water tight. A cover shall be placed over the units to prevent precipitation from contacting the stored material. Liquid which collects inside the units shall be removed and stored in accordance with paragraph Liquid Storage.

3.5.3 Liquid Storage

Liquid collected from excavations and stockpiles shall be temporarily stored in 220 L barrels. Liquid storage containers shall be water-tight.

3.6 SAMPLING

3.6.1 Sampling of Stored Material

Samples of stored material shall be collected at a frequency based on applicable regional and local requirements. Samples shall be tested for parameters in which contamination is suspected.

Stored material with contaminant levels that exceed the action levels shall be treated offsite. Analyses for contaminated material to be taken to an offsite treatment facility shall conform to local, state, and regional criteria as well as to the requirements of the treatment facility. Documentation of all analyses performed shall be furnished to the Contracting Officer. Additional sampling and analyses to the extent required by the approved offsite treatment, storage or disposal (TSD) facility shall be the responsibility of the Contractor and shall be subject to approval by the Contracting Officer.

3.7 SPILLS

In the event of a spill or release of a hazardous substance. Immediate containment actions shall be taken to minimize the effect of any spill or leak. Cleanup shall be in accordance with applicable regional and local regulations. As directed by the Contracting Officer, additional sampling and testing shall be performed to verify spills have been cleaned up. Spill cleanup and testing shall be done at no additional cost to the Government.

3.8 BACKFILLING

3.8.1 Confirmation Test Results

Excavations shall be backfilled immediately after all contaminated materials have been removed and confirmation test results have been approved. Backfill shall be placed and compacted to the lines and grades shown on the drawings.

3.9 DISPOSAL REQUIREMENTS

Offsite disposal of contaminated material shall be in accordance with applicable regional and local requirements.

3.10 CLOSURE REPORT

Submit three copies of a Closure Report within 14 calendar days of completing work at the site. The report shall be labeled with the contract number, project name, location, date and name of general Contractor. The Closure Report shall include the following information as a minimum:

- a. A cover letter signed by a responsible company official certifying that all services involved have been performed in accordance with the terms and conditions of the contract documents and regulatory requirements.
- b. A narrative report including, but not limited to, the following:
 - (1) site conditions, ground water elevation, and cleanup criteria;
 - (2) excavation logs;
 - (3) field screening readings;
 - (4) quantity of materials removed from each area of contamination;
 - (5) quantity of water/product removed during dewatering;
 - (6) sampling locations and sampling methods;
 - (7) sample collection data such as time of collection and method of preservation;
 - (8) sample chain-of-custody forms; and
 - (9) source of backfill.
 - (10) landfill.
- c. Copies of all chemical and physical test results.
- d. Copies of all manifests and land disposal restriction notifications.
- e. Copies of all certifications of final disposal signed by the responsible disposal facility official.
- f. Waste profile sheets.
- g. Scale drawings showing limits of each excavation, limits of contamination, known underground utilities within 15 m of excavation, sample locations, and sample identification numbers. On-site stockpile, storage, treatment, loading, and disposal areas shall also be shown on the drawings.
- h. Progress Photographs. Color photographs shall be used to document progress of the work. A minimum of four views of the site showing the location of the area of contamination, entrance/exit road, and any

other notable site conditions shall be taken before work begins. After work has been started, activities at each work location shall be photographically recorded daily. Photographs shall be a minimum of 76.2 by 127.0 mm and shall include:

- (1) Soil removal and sampling.
- (2) Dewatering operations.
- (3) Unanticipated events such as spills and the discovery of additional contaminated material.
- (4) Contaminated material/water storage, handling, treatment, and transport.
- (5) Site or task-specific employee respiratory and personal protection.
- (6) Fill placement and grading.
- (7) Post-construction photographs. After completion of work at each site, take a minimum of four views of each excavation site.

A digital version of all photos shown in the report shall be included with the Closure Report. Photographs shall be a minimum of 76 by 127 mm and shall be mounted back-to-back in double face plastic sleeves punched to fit standard three ring binders. Each print shall have an information box attached. The box shall be typewritten and arranged as follows:

Project Name:	Direction of View:
Location:	Date/Time:
Photograph No.:	Description of View:

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 03 - CONCRETE

SECTION 03 30 00

CAST-IN-PLACE CONCRETE

02/19, CHG 4: 08/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
- 1.3 SUBMITTALS
- 1.4 MODIFICATION OF REFERENCES
- 1.5 DELIVERY, STORAGE, AND HANDLING
 - 1.5.1 Reinforcement
- 1.6 QUALITY ASSURANCE
 - 1.6.1 Design Data
 - 1.6.1.1 Formwork Calculations
 - 1.6.1.2 Concrete Mix Design
 - 1.6.2 Shop Drawings
 - 1.6.2.1 Formwork
 - 1.6.2.2 Reinforcing Steel
 - 1.6.3 Control Submittals
 - 1.6.3.1 Concrete Curing Plan
 - 1.6.3.2 Pumping Concrete
 - 1.6.3.3 VOC Content for form release agents, curing compounds, and concrete penetrating sealers
 - 1.6.3.4 Safety Data Sheets
 - 1.6.4 Test Reports
 - 1.6.4.1 Fly Ash and Pozzolan
 - 1.6.4.2 Slag Cement
 - 1.6.4.3 Aggregates
 - 1.6.5 Quality Control Plan
 - 1.6.6 Quality Control Personnel Certifications
 - 1.6.6.1 Quality Manager Qualifications
 - 1.6.6.2 Field Testing Technician and Testing Agency
 - 1.6.7 Laboratory Qualifications for Concrete Qualification Testing
 - 1.6.8 Laboratory Accreditation
- 1.7 ENVIRONMENTAL REQUIREMENTS
 - 1.7.1 Submittals for Environmental Performance
- 1.8 SUSTAINABLE DESIGN REQUIREMENTS
 - 1.8.1 Local/Regional Materials

PART 2 PRODUCTS

- 2.1 FORMWORK MATERIALS
 - 2.1.1 Wood Forms
 - 2.1.1.1 Concrete Form Plywood (Standard Rough)
 - 2.1.2 Steel Forms
- 2.2 FORMWORK ACCESSORIES
 - 2.2.1 Form Ties
 - 2.2.2 Waterstops
 - 2.2.2.1 Thermoplastic Elastomeric Rubber Waterstop

- 2.2.2.2 Hydrophilic Waterstop
- 2.2.3 Biodegradable Form Release Agent
- 2.2.4 Chamfer Materials
- 2.2.5 Construction and movement joints
- 2.2.6 Other Embedded items
- 2.3 CONCRETE MATERIALS
 - 2.3.1 Cementitious Materials
 - 2.3.1.1 Portland Cement
 - 2.3.1.2 Fly Ash
 - 2.3.1.3 Slag Cement
 - 2.3.1.4 Other Supplementary Cementitious Materials
 - 2.3.2 Water
 - 2.3.3 Aggregate
 - 2.3.3.1 Normal-Weight Aggregate
 - 2.3.3.2 Recycled Aggregate Materials
 - 2.3.4 Admixtures
- 2.4 MISCELLANEOUS MATERIALS
 - 2.4.1 Concrete Curing Materials
 - 2.4.2 Nonshrink Grout
 - 2.4.3 Expansion/Contraction Joint Filler
 - 2.4.4 Joint Sealants
 - 2.4.4.1 Horizontal Surfaces, 3 Percent Slope, Maximum
 - 2.4.4.2 Vertical Surfaces Greater Than 3 Percent Slope
- 2.5 CONCRETE MIX DESIGN
 - 2.5.1 Properties and Requirements
 - 2.5.2 Durability
 - 2.5.2.1 Alkali-Aggregate Reaction
 - 2.5.2.2 Concrete Temperature
 - 2.5.3 Ready-Mix Concrete
- 2.6 REINFORCEMENT
 - 2.6.1 Reinforcing Bars
 - 2.6.2 Welded Wire Reinforcement
 - 2.6.3 Reinforcing Bar Supports
 - 2.6.4 Dowels for Load Transfer in Floors

PART 3 EXECUTION

- 3.1 EXAMINATION
- 3.2 PREPARATION
 - 3.2.1 General
 - 3.2.2 Subgrade Under Foundations and Footings
 - 3.2.3 Subgrade Under Slabs on Ground
 - 3.2.4 Edge Forms and Screed Strips for Slabs
 - 3.2.5 Reinforcement and Other Embedded Items
- 3.3 FORMS
 - 3.3.1 Coating
 - 3.3.2 Reshoring
 - 3.3.3 Reuse
 - 3.3.4 Forms for Standard Rough Form Finish
 - 3.3.5 Forms for Standard Smooth Form Finish
 - 3.3.6 Form Ties
 - 3.3.7 Tolerances for Form Construction
 - 3.3.8 Removal of Forms and Supports
 - 3.3.9 Strength of Concrete Required for Removal of Formwork
- 3.4 WATERSTOP INSTALLATION AND SPLICES
 - 3.4.1 Thermoplastic Elastomeric Rubber Waterstop
 - 3.4.2 Hydrophilic Waterstop
- 3.5 PLACING REINFORCEMENT AND MISCELLANEOUS MATERIALS
 - 3.5.1 Reinforcement Supports

- 3.5.2 Splicing
- 3.5.3 Setting Miscellaneous Material
- 3.5.4 Fabrication
- 3.5.5 Placing Reinforcement
- 3.5.6 Spacing of Reinforcing Bars
- 3.5.7 Concrete Protection for Reinforcement
- 3.6 BATCHING, MEASURING, MIXING, AND TRANSPORTING CONCRETE
 - 3.6.1 Measuring
 - 3.6.2 Mixing
 - 3.6.3 Transporting
- 3.7 PLACING CONCRETE
 - 3.7.1 Pumping
 - 3.7.2 Cold Weather
 - 3.7.3 Hot Weather
 - 3.7.4 Bonding
- 3.8 WASTE MANAGEMENT
 - 3.8.1 Mixing Equipment
 - 3.8.2 Reinforcing Steel
 - 3.8.3 Other Waste
- 3.9 SURFACE FINISHES EXCEPT FLOOR, SLAB, AND PAVEMENT FINISHES
 - 3.9.1 Defects
 - 3.9.2 Not Against Forms (Top of Walls)
 - 3.9.3 Formed Surfaces
 - 3.9.3.1 Tolerances
 - 3.9.3.2 As-Cast Rough Form
 - 3.9.3.3 Standard Smooth Finish
 - 3.9.4 Smooth-Rubbed Finish
- 3.10 FLOOR, SLAB, AND PAVEMENT FINISHES AND MISCELLANEOUS CONSTRUCTION
 - 3.10.1 Finish
 - 3.10.1.1 Scratched
 - 3.10.1.2 Floated
 - 3.10.1.3 Broomed
 - 3.10.1.4 Pavement
- 3.11 JOINTS
 - 3.11.1 Construction Joints
 - 3.11.1.1 Construction Joints for Constructability Purposes
 - 3.11.2 Isolation Joints in Slabs on Ground
 - 3.11.3 Contraction Joints in Slabs on Ground
 - 3.11.4 Sealing Joints in Slabs on Ground
- 3.12 CURING AND PROTECTION
 - 3.12.1 Curing Periods
 - 3.12.2 Curing Formed Surfaces
 - 3.12.3 Curing Unformed Surfaces
 - 3.12.4 Temperature of Concrete During Curing
 - 3.12.5 Protection from Mechanical Injury
 - 3.12.6 Protection After Curing
- 3.13 FIELD QUALITY CONTROL
 - 3.13.1 Aggregate Testing
 - 3.13.1.1 Fine Aggregate
 - 3.13.1.2 Coarse Aggregate
 - 3.13.2 Concrete Sampling
 - 3.13.3 Concrete Testing
 - 3.13.3.1 Slump Tests
 - 3.13.3.2 Temperature Tests
 - 3.13.3.3 Compressive Strength Tests
 - 3.13.3.4 Strength of Concrete Structure
 - 3.13.3.5 Non-Conforming Materials
 - 3.13.3.6 Testing Concrete Structure for Strength
- 3.14 REPAIR, REHABILITATION AND REMOVAL

- 3.14.1 Crack Repair
- 3.14.2 Repair of Weak Surfaces
- 3.14.3 Failure of Quality Assurance Test Results

-- End of Section Table of Contents --

SECTION 03 30 00

CAST-IN-PLACE CONCRETE
02/19, CHG 4: 08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 36061	(2014) Steel welded fabric for structural use in concrete reinforcement. Steel welded fabric made out of steel bars B 500 S
UNE 36068	(2011) Ribbed bars of weldable steel for the reinforcement of concrete.
UNE 36092	(2014) Steel welded fabric for structural use in concrete reinforcement. Steel welded fabric made out of steel wires B 500 T
UNE 36097-1	(1981) Hot Rolled Plain Bars for the Reinforcement of Concrete. Characteristics.
UNE 36731	(1996) Plains Steel Wire For Welded Fabric And Lattice Girders.
UNE 41107	(1961) Prefabricated Products. Elastics and of Low Transversal Dilation for the Filling of Expansion Joints in Concrete Paving
UNE 146508	(2018) Test For Aggregates. Determination Of The Alkali-Silica And Alkali-Silicate Potential Reactivity Of Aggregates. Accelerated Mortar Bar Test.
UNE 146509	(2018) Determination Of The Potential Reactivity Of Aggregates. Concrete Prisms Method
UNE-EN 196-5	(2011) Methods of testing cement - Part 5: Pozzolanicity test for pozzolanic cement
UNE-EN 197-1	(2011) Cement - Part 1: Composition, specifications and conformity criteria for common cements.
UNE-EN 450-1	(2013) Fly ash for concrete - Part 1: Definition, specifications and conformity

criteria.

UNE-EN 932-1	(1997) Test For Geometrical Properties Of Aggregates. Part 1: Methods for Sampling.
UNE-EN 932-3	(A1:2004) Tests For General Properties Of Aggregates. Part 3: Procedure And Terminology For Simplified Petrographic Description.
UNE-EN 933-1	(2012) Tests for geometrical properties of aggregates - Part 1: Determination of particle size distribution - Sieving method
UNE-EN 933-3	(2012) Tests for geometrical properties of aggregates - Part 3: Determination of particle shape - Flakiness index
UNE-EN 933-4	(2008) Tests for geometrical properties of aggregates - Part 4: Determination of particle shape - Shape index
UNE-EN 934-2	(2010+A1:2012) Admixtures for concrete, mortar and grout. Part 2: Concrete admixtures. Definitions, requirements, conformity, marking and labeling.
UNE-EN 1367-2	(2010) Tests for thermal and weathering properties of aggregates - Part 2: Magnesium sulfate test
UNE-EN 10080	(2006) Steel for the reinforcement of concrete - Weldable reinforcing steel - General.
UNE-EN 12350-1	(2020) Testing fresh concrete - Part 1: Sampling and common apparatus
UNE-EN 12350-2	(2020) Testing fresh concrete - Part 2: Slump-test
UNE-EN 12390-2	(2020) Testing hardened concrete. Part 2: Making and curing specimens for strength tests.
UNE-EN 12390-3	(2020) Testing hardened concrete - Part 3: Compressive strength of test specimens.
UNE-EN 12504-1	(2020) Testing concrete in structures. Part 1: Cored specimens. Taking, examining and testing in compression.
UNE-EN 12620	(2003+A1:2009) Aggregates for concrete
UNE-EN 15167-1	(2008) Ground granulated blast furnace slag for use in concrete, mortar and grout - Part 1: Definitions, specifications and conformity criteria

UNE-EN ISO/IEC 17025 (2017) General requirements for the competence of testing and calibration laboratories (ISO/IEC 17025:2017)

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 117 (2010; Errata 2011) Specifications for Tolerances for Concrete Construction and Materials and Commentary

ACI 121R (2008) Guide for Concrete Construction Quality Systems in Conformance with ISO 9001

ACI 301 (2016) Specifications for Structural Concrete

ACI 302.1R (2015) Guide for Concrete Floor and Slab Construction

ACI 304R (2000; R 2009) Guide for Measuring, Mixing, Transporting, and Placing Concrete

ACI 305R (2020) Guide to Hot Weather Concreting

ASTM INTERNATIONAL (ASTM)

ASTM A934/A934M (2016) Standard Specification for Epoxy-Coated Prefabricated Steel Reinforcing Bars

ASTM C618 (2019) Standard Specification for Coal Fly Ash and Raw or Calcined Natural Pozzolan for Use in Concrete

ASTM C920 (2018) Standard Specification for Elastomeric Joint Sealants

ASTM C989/C989M (2018a) Standard Specification for Slag Cement for Use in Concrete and Mortars

ASTM C1107/C1107M (2020) Standard Specification for Packaged Dry, Hydraulic-Cement Grout (Nonshrink)

ASTM C1602/C1602M (2018) Standard Specification for Mixing Water Used in Production of Hydraulic Cement Concrete

ASTM D6690 (2015) Standard Specification for Joint and Crack Sealants, Hot Applied, for Concrete and Asphalt Pavements

MINISTERIO DE LA PRESIDENCIA (MP)

RD 470/2021 (2021) Real Decreto 470/2021, de 29 de junio, por el que se aprueba el Código Estructural.

MINISTERIO DE VIVIENDA (MV)

RD 410/2010

(2010) Requirements to Building Control
Entities and to Testing Laboratories for
building quality control.

U.S. GREEN BUILDING COUNCIL (USGBC)

LEED NC

(2013) Leadership in Energy and
Environmental Design(tm) New Construction
Rating System

1.2 DEFINITIONS

- a. "Cementitious material" as used herein must include all portland cement, pozzolan, fly ash, slag cement.
- b. "Exposed to public view" means situated so that it can be seen from eye level from a public location after completion of the building. A public location is accessible to persons not responsible for operation or maintenance of the building.
- c. "Chemical admixtures" are materials in the form of powder or fluids that are added to the concrete to give it certain characteristics not obtainable with plain concrete mixes.
- d. "Supplementary cementing materials" (SCM) include coal fly ash, slag cement, natural or calcined pozzolans, and ultra-fine coal ash when used in such proportions to replace the portland cement that result in improvement to sustainability and durability and reduced cost.
- e. "Design strength" (f'_c) is the specified compressive strength of concrete at time(s) specified in this section to meet structural design criteria.
- f. "Mass Concrete" is any concrete system that approaches a maximum temperature of 70 degrees C within the first 72 hours of placement. In addition, it includes all concrete elements with a section thickness of 1 meter or more regardless of temperature.
- g. "Mixture proportioning" is the process of designing concrete mixture proportions to enable it to meet the strength, service life and constructability requirements of the project while minimizing the initial and life-cycle cost.
- h. "Mixture proportions" are the masses or volumes of individual ingredients used to make a unit measure (cubic meter or cubic yard) of concrete.
- i. "Pozzolan" is a siliceous or siliceous and aluminous material, which in itself possesses little or no cementitious value but will, in finely divided form and in the presence of moisture, chemically react with calcium hydroxide at ordinary temperatures to form compounds possessing cementitious properties.
- j. "Workability (or consistence)" is the ability of a fresh (plastic) concrete mix to fill the form/mould properly with the desired work (vibration) and without reducing the concrete's quality. Workability depends on water content, chemical admixtures, aggregate (shape and

size distribution), cementitious content and age (level of hydration).

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

- Concrete Curing Plan
- Quality Control Plan; G
- Quality Control Personnel Certifications; G
- Quality Control Organizational Chart
- Laboratory Accreditation; G
- Form Removal Schedule; G

SD-02 Shop Drawings

- Formwork
- Reinforcing Steel; G

SD-03 Product Data

- Joint Sealants; (LEED NC)
- Joint Filler; (LEED NC)
- Formwork Materials
- Recycled Aggregate Materials; (LEED NC)
- Cementitious Materials; (LEED NC)
- Concrete Curing Materials
- Reinforcement; (LEED NC)
- Admixtures
- Waterstops
- Local/Regional Materials; (LEED NC)
- Biodegradable Form Release Agent
- Pumping Concrete
- Nonshrink Grout

SD-05 Design Data

- Concrete Mix Design; G
- Formwork Calculations

SD-06 Test Reports

- Concrete Mix Design; G
- Fly Ash
- Pozzolan
- Slag Cement
- Aggregates
- Compressive Strength Tests; G
- Slump Tests
- Water

SD-07 Certificates

- Reinforcing Bars
- VOC Content for Form Release Agents, Curing Compounds, and

Concrete Penetrating Sealers
Safety Data Sheets
Field Testing Technician and Testing Agency

SD-08 Manufacturer's Instructions

Curing Compound

1.4 MODIFICATION OF REFERENCES

Accomplish work in accordance with ACI publications and RD 470/2021, except as modified herein. Consider the advisory or recommended provisions to be mandatory. Interpret reference to the "Building Official," the "Structural Engineer," and the "Architect/Engineer" to mean the Contracting Officer.

1.5 DELIVERY, STORAGE, AND HANDLING

Follow ACI 301, ACI 304R and ASTM A934/A934M and RD 470/2021, Art. 14, 15, 49.2.3 and Anejo 4 requirements and recommendations. Do not deliver concrete until forms, reinforcement, embedded items, and chamfer strips are in place and ready for concrete placement. Do not store concrete curing compounds or sealers with materials that have a high capacity to adsorb volatile organic compound (VOC) emissions. Do not store concrete curing compounds or sealers in occupied spaces.

1.5.1 Reinforcement

RD 470/2021, Art. 49.7 and the following. Store reinforcement of different sizes and shapes in separate piles or racks raised above the ground to avoid excessive rusting. Protect from contaminants such as grease, oil, and dirt. Ensure bar sizes can be accurately identified after bundles are broken and tags removed.

1.6 QUALITY ASSURANCE

1.6.1 Design Data

1.6.1.1 Formwork Calculations

Include design calculations indicating arrangement of forms, sizes and grades of supports (lumber), panels, and related components. Furnish drawings and calculations of shoring and re-shoring methods proposed for retaining walls and horizontal slabs. Calculations must indicate concrete pressure with both live and dead loads, along with material types.

1.6.1.2 Concrete Mix Design

RD 470/2021, Art. 33.6 and the following. Sixty days minimum prior to concrete placement, submit a mix design for each strength and type of concrete. Submit a complete list of materials including type; brand; source and amount of cement, supplementary cementitious materials, and admixtures; and applicable reference specifications. Submit mill test and all other test for cement, supplementary cementitious materials, aggregates, and admixtures. Provide documentation of maximum nominal aggregate size, gradation analysis, percentage retained and passing sieve, and a graph of percentage retained versus sieve size. Provide mix proportion data using at least three different water-cementitious material ratios for each type of mixture, which produce a range of strength

encompassing those required for each type of concrete required. If source material changes, resubmit mix proportion data using revised source material. Provide only materials that have been proven by trial mix studies to meet the requirements of this specification, unless otherwise approved in writing by the Contracting Officer. Indicate clearly in the submittal where each mix design is used when more than one mix design is submitted. Resubmit data on concrete components if the qualities or source of components changes. Required average strength can be documented by field experience if field strength test data are available and represent a single group of at least 10 consecutive strength tests for one mixture, using materials and conditions similar to those expected for work, and encompassing a period of not less than 45 days. The average of field strength tests shall equal or exceed f_{cr}' . Changes in materials, conditions, and proportions within the test record shall not have been more closely restricted than those for the proposed work. Test records shall not be more than 24 months old. Obtain mix design approval from the contracting officer prior to concrete placement.

1.6.2 Shop Drawings

1.6.2.1 Formwork

Drawings showing details of formwork including, but not limited to; joints, supports, studding and shoring, and sequence of form and shoring removal. Indicate placement schedule, construction, location and method of forming control joints. Include locations of inserts, conduit, sleeves and other embedded items. Reproductions of contract drawings are unacceptable. Submit form removal schedule indicating element and minimum length of time for form removal.

Design, fabricate, erect, support, brace, and maintain formwork so that it is able to support, without failure, all vertical and lateral loads that may reasonably be anticipated to be applied to the formwork.

1.6.2.2 Reinforcing Steel

RD 470/2021, Figure 49.5.1.1 and Art. 49.5.1. Indicate bending diagrams, assembly diagrams, splicing and laps of bars, shapes, dimensions, and details of bar reinforcing, accessories, and concrete cover. Do not scale dimensions from structural drawings to determine lengths of reinforcing bars. Reproductions of contract drawings are unacceptable.

1.6.3 Control Submittals

1.6.3.1 Concrete Curing Plan

Submit proposed materials, methods and duration for curing concrete elements in accordance with RD 470/2021.

1.6.3.2 Pumping Concrete

Submit proposed materials and methods for pumping concrete. Submittal must include mix designs, pumping equipment including type of pump and size and material for pipe, and maximum length and height concrete is to be pumped.

1.6.3.3 VOC Content for form release agents, curing compounds, and concrete penetrating sealers

Submit certification for the form release agent, curing compounds, and concrete penetrating sealers that indicate the VOC content of each product.

1.6.3.4 Safety Data Sheets

Submit Safety Data Sheets (SDS) for all materials that are regulated for hazardous health effects. SDS must be readily accessible during each work shift to employees when they are at the construction site.

1.6.4 Test Reports

1.6.4.1 Fly Ash and Pozzolan

Submit test results in accordance with ASTM C618 or alternatively UNE-EN 196-5 and UNE-EN 450-1 for fly ash and pozzolan. Submit test results performed within 6 months of submittal date.

1.6.4.2 Slag Cement

Submit test results in accordance with ASTM C989/C989M or UNE-EN 15167-1 for slag cement. Submit test results performed within 6 months of submittal date.

1.6.4.3 Aggregates

Submit test results in accordance with UNE-EN 1367-2, UNE-EN 12620, UNE 146508 and UNE 146509, as required in the paragraph titled ALKALI-AGGREGATE REACTION.

1.6.5 Quality Control Plan

Develop and submit for approval a concrete quality control program in accordance with the guidelines of ACI 121R or alternatively RD 470/2021, Art. 17 through 23 and Art. 63 and as specified herein. The plan must include approved laboratories. Provide direct oversight for the concrete qualification program inclusive of associated sampling and testing. All quality control reports must be provided to the Contracting Officer, Quality Manager and Concrete Supplier. Maintain a copy of the reports at project site.

1.6.6 Quality Control Personnel Certifications

The Contractor must submit for approval the responsibilities of the various quality control personnel, including the names and qualifications of the individuals in those positions and a quality control organizational chart defining the quality control hierarchy and the responsibility of the various positions. Quality control personnel must be employed by the Contractor.

Submit certification for the following:

- a. Personnel responsible for inspection of concrete operations.
- b. Lead Foreman or Journeyman of the Concrete Placing, Finishing, and Curing Crews.
- c. Field Testing Technicians

1.6.6.1 Quality Manager Qualifications

The quality manager must hold a current license as a professional engineer in Spain, with experience on at least five similar projects. Evidence of extraordinary proven experience may be considered by the Contracting Officer as sufficient to act as the Quality Manager.

1.6.6.2 Field Testing Technician and Testing Agency

Submit data on qualifications of proposed testing agency and technicians for approval by the Contracting Officer prior to performing testing on concrete.

- a. Work on concrete under this contract must be performed by a professional Engineer, Ingeniero de Caminos, Canales y Puertos, Ingeniero Industrial or Ingeniero de la Edificación.
- b. Testing agencies that perform testing services on reinforcing steel must meet the requirements of UNE-EN ISO/IEC 17025.
- c. Testing agencies that perform testing services on concrete materials must meet the requirements of UNE-EN ISO/IEC 17025.

1.6.7 Laboratory Qualifications for Concrete Qualification Testing

The concrete testing laboratory must have the necessary equipment and experience to accomplish required testing. The laboratory must meet the requirements of UNE-EN ISO/IEC 17025 and be accredited by ENAC and comply with RD 410/2010.

1.6.8 Laboratory Accreditation

Laboratory and testing facilities must be provided by and at the expense of the Contractor. The laboratories performing the tests must be accredited by ENAC and comply with RD 410/2010, including UNE-EN 12390-2 and UNE 146508. The accreditation must be current and must include the required test methods, as specified. Furthermore, the testing must comply with the following requirements:

- a. Aggregate Testing and Mix Proportioning: Aggregate testing and mixture proportioning studies must be performed by an accredited laboratory and under the direction of a licensed professional engineer in Spain who is competent in concrete materials and must sign all reports and designs.
- b. Acceptance Testing: Furnish all materials, labor, and facilities required for molding, curing, testing, and protecting test specimens at the site and in the laboratory. Furnish and maintain boxes or other facilities suitable for storing and curing the specimens at the site while in the mold within the temperature range stipulated by UNE-EN 12390-2.
- c. Contractor Quality Control: All sampling and testing must be performed by an approved, onsite, independent, accredited laboratory.

1.7 ENVIRONMENTAL REQUIREMENTS

RD 470/2021, Art. 6 and Anejo 2 and the following. Provide space ventilation according to material manufacturer recommendations, at a

minimum, during and following installation of concrete curing compound and sealer.

1.7.1 Submittals for Environmental Performance

- a. Provide data indication the percentage of post-industrial pozzolan (fly ash, slag cement) cement substitution as a percentage of the full product composite by weight.
- b. Provide data indicating the percentage of post-industrial and post-consumer recycled content aggregate.
- c. Provide product data indicating the percentage of post-consumer recycled steel content in each type of steel reinforcement as a percentage of the full product composite by weight.
- d. Provide product data stating the location where all products were manufactured
- e. For projects using reusable formwork, provide data showing how formwork is reused.
- f. Provide SDS product information data showing that form release agents meet any environmental performance goals such as using vegetable and soy based products.
- g. Provide SDS product information data showing that concrete adhesives meet any environmental performance goals including low emitting, low volatile organic compound products.

1.8 SUSTAINABLE DESIGN REQUIREMENTS

1.8.1 Local/Regional Materials

Use materials or products extracted, harvested, or recovered, as well as manufactured, within a 805 kilometer radius from the project site, if available from a minimum of three sources.

PART 2 PRODUCTS

2.1 FORMWORK MATERIALS

- a. Form-facing material in contact with concrete must be lumber, plywood or. Submit product information on proposed form-facing materials if different from that specified herein.
- b. Design formwork, shores, reshores, and backshores to support loads transmitted to them and to comply with applicable building code requirements.
- c. Design formwork to withstand pressure resulting from placement and vibration of concrete and to maintain specified tolerances.
- d. Design formwork to accommodate waterstop materials in joints at locations indicated in Contract Documents.
- e. Provide temporary openings in formwork if needed to facilitate cleaning and inspection.

- f. Design formwork joints to inhibit leakage of mortar.
- g. Limit deflection of facing materials for concrete surfaces exposed to view to 1/400 of center-to-center spacing of facing supports.
- h. Submit product information on proposed form-facing materials if different from that specified herein.
- i. Submit shop drawings for formwork, shoring, reshoring, and backshoring. Shop drawings must be signed and sealed by a licensed design engineer.
- j. Submit design calculations for formwork, shoring, reshoring, and backshoring. Design calculations must be signed and sealed by a licensed design engineer.
- k. Submit procedure for reshoring and backshoring, including drawings signed and sealed by a licensed design engineer. Include on shop drawings the formwork removal procedure and magnitude of construction loads used for design of reshoring or backshoring system. Indicate in procedure the magnitude of live and dead loads assumed for required capacity of the structure at time of reshoring or backshoring.
- l. Submit manufacturer's product data on form liner proposed for use with each formed surface.

2.1.1.1 Wood Forms

Provide lumber that is square edged or tongue-and-groove boards, free of raised grain, knotholes, or other surface defects. Provide plywood panels or hardboard for smooth form lining.

2.1.1.1.1 Concrete Form Plywood (Standard Rough)

Provide plywood concrete form not less than 16 mm thick.

2.1.2 Steel Forms

Provide steel form surfaces that do not contain irregularities, dents, or sags.

2.2 FORMWORK ACCESSORIES

- a. Use commercially manufactured formwork accessories, including ties and hangers.
- b. Form ties and accessories must not reduce the effective cover of the reinforcement.

2.2.1 Form Ties

- a. Use form ties with ends or end fasteners that can be removed without damage to concrete.
- b. Use form ties with integral water barrier plates or other acceptable positive water barriers in below-grade walls.
- c. The breakback distance for ferrous ties must be at least 50 mm.

- d. Submit manufacturer's data sheet on form ties.

2.2.2 Waterstops

Submit manufacturer's data sheet on waterstop materials and splices.

2.2.2.1 Thermoplastic Elastomeric Rubber Waterstop

Thermoplastic elastomeric rubber waterstops must be jet fuel resistant.

2.2.2.2 Hydrophilic Waterstop

Swellable strip type compound of polymer modified chloroprene rubber that swells upon contact with water: Tensile strength 2.9 MPa minimum; ultimate elongation 600 percent minimum. Hardness must be 50 minimum on the type A durometer and the volumetric expansion ratio in distilled water at 20 degrees C must be 3 to 1 minimum.

2.2.3 Biodegradable Form Release Agent

- a. Provide form release agent that is colorless, biodegradable, and water-based.
- b. Provide product that does not bond with, stain, or adversely affect concrete surfaces and does not impair subsequent treatments of concrete surfaces.
- c. Provide form release agent that reduces formwork moisture absorption, and does not contain diesel fuel, petroleum-based lubricating oils, waxes, or kerosene. Submit documentation indicating type of biobased material in product and biobased content. Indicate relative dollar value of biobased content products to total dollar value of products included in project.
- d. Submit manufacturer's product data on formwork release agent for use on each form-facing material.

2.2.4 Chamfer Materials

Use lumber materials with dimensions of 19 x 19 mm.

2.2.5 Construction and movement joints

- a. Submit details and locations of construction joints in accordance with the requirements herein.
- b. Locate construction joints within middle one-third of spans of slabs, beams, and girders.
- c. Locate construction joints in walls and columns at underside of slabs, beams, or girders and at tops of footings or slabs.
- d. Make construction joints perpendicular to main reinforcement.
- e. Provide movement joints where indicated in Contract Documents or in accepted alternate locations.
- f. Submit location and detail of movement joints if different from those indicated in Contract Documents.

- g. Submit manufacturer's data sheet on expansion joint materials.
- h. Provide keyways where indicated in Contract Documents.

2.2.6 Other Embedded items

Use sleeves, inserts, anchors, and other embedded items of material and design indicated in Contract Documents.

2.3 CONCRETE MATERIALS

2.3.1 Cementitious Materials

2.3.1.1 Portland Cement

- a. Unless otherwise specified, provide cement that conforms to RD 470/2021, Art. 28 and Table 28, and to UNE-EN 197-1, CEM I or CEM II, 42.5, Types BS, BP or BV.
- b. Use one brand and type of cement for formed concrete having exposed-to-view finished surfaces.
- c. Supplier must certify that no hazardous waste is used in the fuel mix or raw materials.
- d. Submit information along with evidence demonstrating compliance with referenced standards. Submittals must include types of cementitious materials, manufacturing locations, shipping locations, and certificates showing compliance.
- e. Cementitious materials must be stored and kept dry and free from contaminants.

2.3.1.2 Fly Ash

- a. RD 470/2021, Art. 32.1 and UNE-EN 450-1, Class F, except that the maximum allowable loss on ignition must not exceed 6 percent.
- b. If fly ash is used it shall range from 15 to 35 percent by weight of cementitious material, provided the fly ash does not reduce the amount of cement in the concrete mix below the minimum requirements of local building codes. Where the use of fly ash cannot meet the minimum level, it shall not be used. Report the chemical analysis of the fly ash in accordance with UNE-EN 197-1.

2.3.1.3 Slag Cement

UNE-EN 15167-1, Grade 100.

2.3.1.4 Other Supplementary Cementitious Materials

Natural pozzolan must be raw or calcined and conform to UNE-EN 450-1 and UNE-EN 196-5, Class N, including the optional requirement for uniformity.

Ultra Fine Fly Ash (UFFA) and Ultra Fine Pozzolan (UFP) must conform to UNE-EN 196-5 Class F or N, and the following additional requirements:

- a. The strength activity index at 28 days of age must be at least 95

percent of the control specimens.

- b. The average particle size must not exceed 6 microns.
- c. The sum of $\text{SiO}_2 + \text{Al}_2\text{O}_3 + \text{Fe}_2\text{O}_3$ must be greater than 77 percent.

2.3.2 Water

- a. Water must comply with the requirements of RD 470/2021, Art. 29.
- b. Minimize the amount of water in the mix. Improve workability by adjusting the grading of the aggregate and using admixture rather than by adding water.
- c. Water must be potable; free from injurious amounts of oils, acids, alkalis, salts, organic materials, or other substances deleterious to concrete.
- d. Protect mixing water and ice from contamination during storage and delivery.
- e. Submit test report showing water complies with ASTM C1602/C1602M.

2.3.3 Aggregate

2.3.3.1 Normal-Weight Aggregate

- a. Aggregates must conform to RD 470/2021, Art. 30 and to UNE-EN 12620. Comply with the following articles from RD 470/2021: for max. size, Art. 30.3; for grading, Art. 30.4; for fines content, Art. 30.4.1; for coarse aggregate shape, Art. 30.5; for physical requirements, Art. 30.6; and for chemical requirements, Art. 30.7.
- b. Aggregates used in concrete must be obtained from the same sources and have the same size range as aggregates used in concrete represented by submitted field test records or used in trial mixtures.
- c. Store and handle aggregate in a manner that will avoid segregation and prevents contamination by other materials or other sizes of aggregates. Store aggregates in locations that will permit them to drain freely. Do not use aggregates that contain frozen lumps.
- d. Submit types, pit or quarry locations, producers' names, aggregate supplier statement of compliance with UNE-EN 932-1, UNE-EN 932-3, UNE-EN 933-3 and UNE-EN 933-4.

2.3.3.2 Recycled Aggregate Materials

Use recycled aggregate, depending on local availability and conforming to requirements of the mix design. Recycled aggregate to include: recovered concrete or recovered stone that meets the aggregate requirements specified. Submit recycled material request with the aggregate certification submittals and do not use until approved by the Contracting Officer.

2.3.4 Admixtures

RD 470/2021, Art. 31 and the following.

- a. Chemical admixtures must conform to UNE-EN 934-2.
- b. Air-entraining admixtures must conform to UNE-EN 934-2.
- c. Chemical admixtures for use in producing flowing concrete must conform to UNE-EN 934-2.
- d. Do not use calcium chloride admixtures.
- e. Use an ASR-inhibiting admixture for concrete containing aggregate susceptible to ASR.
- f. Admixtures used in concrete must be the same as those used in the concrete represented by submitted field test records or used in trial mixtures.
- g. Protect stored admixtures against contamination, evaporation, or damage.
- h. To ensure uniform distribution of constituents, provide agitating equipment for admixtures used in the form of suspensions or unstable solutions. Protect liquid admixtures from temperature changes that would adversely affect their characteristics.
- i. Submit types, brand names, producers' names, manufacturer's technical data sheets, and certificates showing compliance with standards required herein.

2.4 MISCELLANEOUS MATERIALS

2.4.1 Concrete Curing Materials

Provide concrete curing material in accordance with RD 470/2021, Art. 52.5. Submit product data for concrete curing compounds. Submit manufacturer's instructions for placement of curing compound.

2.4.2 Nonshrink Grout

Nonshrink grout in accordance with ASTM C1107/C1107M.

2.4.3 Expansion/Contraction Joint Filler

UNE 41107. Material must be 13 mm thick, unless otherwise indicated.

2.4.4 Joint Sealants

Submit manufacturer's product data, indicating VOC content.

2.4.4.1 Horizontal Surfaces, 3 Percent Slope, Maximum

ASTM D6690 or ASTM C920, Type M, Class 25, Use T, fuel resistant.

2.4.4.2 Vertical Surfaces Greater Than 3 Percent Slope

ASTM C920, Type M, Grade NS, Class 25, Use NT. No sag, fuel resistant.

2.5 CONCRETE MIX DESIGN

2.5.1 Properties and Requirements

- a. Use materials and material combinations listed in this section and the contract documents.
- b. Cementitious material content must be adequate for concrete to satisfy the specified requirements for strength, w/cm, durability, and finishability described in this section and the contract documents.

The minimum cementitious material content for concrete used in floors must meet the following requirements:

Nominal maximum size of aggregate, mm	Minimum cementitious material content, kg per cubic meter
25	310

- c. Selected target slump must meet the requirements this section, the contract documents, and must not exceed 230 mm. Concrete must not show visible signs of segregation.
- d. The target slump must be enforced for the duration of the project. Determine the slump by UNE-EN 12350-2. Slump tolerances must meet the requirements of RD 470/2021, Art. 33.5 and Tables 51.2.4 and 57.5.2.2..
- e. The nominal maximum size of coarse aggregate for a mixture must not exceed three-fourths of the minimum clear spacing between reinforcement, one-fifth of the narrowest dimension between sides of forms, or one-third of the thickness of slabs or toppings.
- f. Concrete for slabs to receive a hard-troweled finish must not contain an air-entraining admixture or have a total air content greater than 3 percent.
- g. Concrete properties and requirements for each portion of the structure are specified in the table below. Refer to the paragraph titled DURABILITY for more details on exposure categories and their requirements.

	Minimum $f'c$ MPa	Miscellaneous Requirements
All concrete	30 at 28 days	Max. slump: 15 cm Nominal maximum aggregate size must be 25 mm

2.5.2 Durability

2.5.2.1 Alkali-Aggregate Reaction

RD 470/2021, Art. 30.7.5 and the following. Do not use any aggregate susceptible to alkali-carbonate reaction (ACR). Use one of the three options below for qualifying concrete mixtures to reduce the potential of alkali-silica reaction (ASR):

- For each aggregate used in concrete, the expansion result determined in accordance with UNE 146509 must not exceed 0.04 percent at one year.
- For each aggregate used in concrete, the expansion result of the aggregate and cementitious materials combination determined in accordance with UNE 146509 must not exceed 0.10 percent at an age of 16 days.
- Alkali content in concrete (LBA) must not exceed 2.4 kg per cubic meter for moderately reactive aggregate or 1.8 kg per cubic meter for highly reactive aggregate. Reactivity must be determined by testing in accordance with UNE 146509. Alkali content is calculated as follows:

$$LBA = (\text{cement content, kg per cubic meter}) \times (\text{equivalent alkali content of portland cement in percent}/100 \text{ percent})$$

2.5.2.2 Concrete Temperature

The temperature of concrete as delivered must not exceed 35°C.

2.5.3 Ready-Mix Concrete

Provide concrete that meets the requirements of RD 470/2021 Art. 33, 43.2 and Anejo 4.

Ready-mixed concrete manufacturer must provide duplicate delivery tickets with each load of concrete delivered. Provide delivery tickets with the following information in addition to that required by RD 470/2021 Anejo 4:

- Type and brand cement
- Cement and supplementary cementitious materials content in kilogram per cubic meter of concrete
- Maximum size of aggregate
- Amount and brand name of admixtures
- Total water content expressed by water cementitious material ratio

2.6 REINFORCEMENT

- Bend reinforcement cold. Fabricate reinforcement in accordance with

fabricating tolerances of RD 470/2021, Art. 34, 49 and 59.2, and Anejo 14 Art. 5.

- b. Submit manufacturer's certified test report for reinforcement.
- c. Submit placing drawings showing fabrication dimensions and placement locations of reinforcement and reinforcement supports. Placing drawings must indicate locations of splices, lengths of lap splices, and details of mechanical and welded splices.
- d. Submit request with locations and details of splices not indicated in Contract Documents.
- e. Submit request to place column dowels without using templates.
- f. Submit request and procedure to field-bend or straighten reinforcing bars partially embedded in concrete.
- g. Submit request for field cutting, including location and type of bar to be cut and reason field cutting is required.

2.6.1 Reinforcing Bars

- a. Reinforcing bars must be deformed, except load-transfer dowels, and welded wire reinforcement, which may be plain.
- b. UNE-EN 10080 or UNE 36068, B-500-T or B-500-S.
- c. Reinforcing bars may contain post-consumer or post-industrial recycled content.
- d. Submit mill certificates for reinforcing bars.

2.6.2 Welded Wire Reinforcement

- a. Use welded wire reinforcement specified in Contract Documents and conforming to one or more of the specifications given herein.
- b. Plain welded wire reinforcement must conform to RD 470/2021, Art. 35.2.1, to UNE 36731 and to UNE 36092, with welded intersections spaced no greater than 300 mm apart in direction of principal reinforcement.
- c. Deformed welded wire reinforcement must conform to UNE 36061, with welded intersections spaced no greater than 400 mm apart in direction of principal reinforcement.

2.6.3 Reinforcing Bar Supports

- a. Provide reinforcement support types within structure as required by Contract Documents.
- b. Legs of supports in contact with formwork must be hot-dip galvanized, or plastic coated after fabrication, or stainless-steel bar supports.

2.6.4 Dowels for Load Transfer in Floors

Provide greased dowels for load transfer in floors of the type, design, weight, and dimensions indicated. Provide dowel bars that are

plain-billet steel conforming to UNE 36097-1, 235 Mpa minimum yield stress.

PART 3 EXECUTION

3.1 EXAMINATION

- a. Do not begin installation until substrates have been properly constructed; verify that substrates are level. Comply with RD 470/2021, Art. 48.1.
- b. If substrate preparation is the responsibility of another installer, notify Contracting Officer of unsatisfactory preparation before processing.
- c. Check field dimensions before beginning installation. If dimensions vary too much from design dimensions for proper installation, notify Contracting Officer and wait for instructions before beginning installation.

3.2 PREPARATION

Determine quantity of concrete needed and minimize the production of excess concrete. Designate locations or uses for potential excess concrete before the concrete is poured.

3.2.1 General

Comply with RD 470/2021, Art. 48.2 and the following.

- a. Surfaces against which concrete is to be placed must be free of debris, loose material, standing water, and other deleterious substances before start of concrete placing.
- b. Remove standing water without washing over freshly deposited concrete. Divert flow of water through side drains provided for such purpose.

3.2.2 Subgrade Under Foundations and Footings

- a. When subgrade material is semi-porous and dry, sprinkle subgrade surface with water as required to eliminate suction at the time concrete is deposited.

3.2.3 Subgrade Under Slabs on Ground

- a. Before construction of slabs on ground, have underground work on pipes and conduits completed and approved.
- b. Previously constructed subgrade or fill must be cleaned of foreign materials
- c. Finish surface of capillary water barrier under interior slabs on ground must not show deviation in excess of 6.4 mm when tested with a 3000 mm straightedge parallel with and at right angles to building lines.
- d. Finished surface of subgrade or fill under exterior slabs on ground must not be more than 6.10 mm above or 30.50 mm below elevation indicated.

3.2.4 Edge Forms and Screed Strips for Slabs

- a. Set edge forms or bulkheads and intermediate screed strips for slabs to obtain indicated elevations and contours in finished slab surface and must be strong enough to support vibrating bridge screeds or roller pipe screeds if nature of specified slab finish requires use of such equipment.
- b. Align concrete surface to elevation of screed strips by use of strike-off templates or approved compacting-type screeds.

3.2.5 Reinforcement and Other Embedded Items

- a. Secure reinforcement, joint materials, and other embedded materials in position, inspected, and approved before start of concrete placing.
- b. When concrete is placed, reinforcement must be free of materials deleterious to bond. Reinforcement with rust, mill scale, or a combination of both will be considered satisfactory, provided minimum nominal dimensions, nominal weight, and minimum average height of deformations of a hand-wire-brushed test specimen are not less than applicable ASTM specification requirements.

3.3 FORMS

- a. Provide forms, shoring, and scaffolding for concrete placement. Set forms mortar-tight and true to line and grade.
- b. Chamfer above grade exposed joints, edges, and external corners of concrete 19 mm unless indicated otherwise. Place chamfer strips in corners of formwork to produce beveled edges on permanently exposed surfaces.
- c. Provide formwork with clean-out openings to permit inspection and removal of debris.
- d. Inspect formwork and remove foreign material before concrete is placed.
- e. At construction joints, lap form-facing materials over the concrete of previous placement. Ensure formwork is placed against hardened concrete so offsets at construction joints conform to specified tolerances.
- f. Provide positive means of adjustment (such as wedges or jacks) of shores and struts. Do not make adjustments in formwork after concrete has reached initial setting. Brace formwork to resist lateral deflection and lateral instability.
- g. Fasten form wedges in place after final adjustment of forms and before concrete placement.
- h. Provide anchoring and bracing to control upward and lateral movement of formwork system.
- i. Construct formwork for openings to facilitate removal and to produce opening dimensions as specified and within tolerances.
- j. Provide runways for moving equipment. Support runways directly on formwork or structural members. Do not support runways on

reinforcement. Loading applied by runways must not exceed capacity of formwork or structural members.

- k. Position and support expansion joint materials, waterstops, and other embedded items to prevent displacement. Fill voids in sleeves, inserts, and anchor slots temporarily with removable material to prevent concrete entry into voids.
- l. Clean surfaces of formwork and embedded materials of mortar, grout, and foreign materials before concrete placement.

3.3.1 Coating

- a. Cover formwork surfaces with an acceptable material that inhibits bond with concrete.
- b. If formwork release agent is used, apply to formwork surfaces in accordance with manufacturer's recommendations before placing reinforcement. Remove excess release agent on formwork prior to concrete placement.
- c. Do not allow formwork release agent to contact reinforcement or hardened concrete against which fresh concrete is to be placed.

3.3.2 Reshoring

- a. Do not allow structural members to be loaded with combined dead and construction loads in excess of loads indicated in the accepted procedure.
- b. Install and remove reshores or backshores in accordance with accepted procedure.

3.3.3 Reuse

- a. Reuse forms providing the structural integrity of concrete and the aesthetics of exposed concrete are not compromised.
- b. Wood forms must not be clogged with paste and must be capable of absorbing high water-cementitious material ratio paste.
- c. Remove leaked mortar from formwork joints before reuse.

3.3.4 Forms for Standard Rough Form Finish

Provide formwork in accordance with ACI 301 Section 5 with a surface finish, SF-1.0, for formed surfaces that are to be concealed by other construction.

3.3.5 Forms for Standard Smooth Form Finish

Provide formwork in accordance with ACI 301 Section 5 with a surface finish, SF-3.0, for formed surfaces that are exposed to view.

3.3.6 Form Ties

- a. After ends or end fasteners of form ties have been removed, repair tie holes in accordance with ACI 301 Section 5 requirements.

3.3.7 Tolerances for Form Construction

- a. Construct formwork so concrete surfaces conform to tolerances in ACI 117.
- b. Position and secure sleeves, inserts, anchors, and other embedded items such that embedded items are positioned within tolerances.
- c. To maintain specified elevation and thickness within tolerances, install formwork to compensate for deflection and anticipated settlement in formwork during concrete placement. Set formwork and intermediate screed strips for slabs to produce designated elevation, camber, and contour of finished surface before formwork removal. If specified finish requires use of vibrating screeds or roller pipe screeds, ensure that edge forms and screed strips are strong enough to support such equipment.

3.3.8 Removal of Forms and Supports

- a. If vertical formed surfaces require finishing, remove forms as soon as removal operations will not damage concrete.
- b. Remove top forms on sloping surfaces of concrete as soon as removal will not allow concrete to sag. Perform repairs and finishing operations required. If forms are removed before end of specified curing period, provide curing and protection.
- c. Do not damage concrete during removal of vertical formwork for columns, walls, and sides of beams. Perform needed repair and finishing operations required on vertical surfaces. If forms are removed before end of specified curing period, provide curing and protection.
- d. Leave formwork and shoring in place to support construction loads and weight of concrete in beams, slabs, and other structural members until in-place required strength of concrete is reached.
- e. Form-facing material and horizontal facing support members may be removed before in-place concrete reaches specified compressive strength if shores and other supports are designed to allow facing removal without deflection of supported slab or member.

3.3.9 Strength of Concrete Required for Removal of Formwork

If removal of formwork, reshoring, or backshoring is based on concrete reaching a specified in-place strength, mold and field-cure cylinders in accordance with UNE-EN 12390-2. Test cylinders in accordance with UNE-EN 12390-3.

3.4 WATERSTOP INSTALLATION AND SPLICES

- a. Provide waterstops in construction joints as indicated.
- b. Install formwork to accommodate waterstop materials. Locate waterstops in joints where indicated in Contract Documents. Minimize number of splices in waterstop. Splice waterstops in accordance with manufacturer's written instructions. Install factory-manufactured premolded mitered corners.

- c. Install waterstops to form a continuous diaphragm in each joint. Make adequate provisions to support and protect waterstops during progress of work. Protect waterstops protruding from joints from damage.

3.4.1 Thermoplastic Elastomeric Rubber Waterstop

Fittings must be shop made using a machine specifically designed to mechanically weld the waterstop. A portable power saw must be used to miter or straight cut the ends to be joined to ensure good alignment and contact between joined surfaces. Maintain continuity of the characteristic features of the cross section of the waterstop (for example ribs, tabular center axis, and protrusions) across the splice.

3.4.2 Hydrophilic Waterstop

Miter cut ends to be joined with sharp knife or shears. The ends must be adhered with adhesive.

3.5 PLACING REINFORCEMENT AND MISCELLANEOUS MATERIALS

- a. Unless otherwise specified, placing reinforcement and miscellaneous materials must be in accordance to and RD 470/2021 Art. 49. Comply with Art. 49.4.1 for spacing and Art. 49.4.3 for general requirements of reinforcement and Art. 49.8 for placing. Provide bars, welded wire reinforcement, wire ties, supports, and other devices necessary to install and secure reinforcement.
- b. Reinforcement must not have rust, scale, oil, grease, clay, or foreign substances that would reduce the bond. Rusting of reinforcement is a basis of rejection if the effective cross-sectional area or the nominal weight per unit length has been reduced. Remove loose rust prior to placing steel. Tack welding is prohibited.
- c. Nonprestressed cast-in-place concrete members must have concrete cover for reinforcement as indicated on drawings.

3.5.1 Reinforcement Supports

Provide reinforcement support in accordance with RD 470/2021, Art. 49.4.3.1 and 49.8.2.

3.5.2 Splicing

As indicated in the Contract Documents. For splices not indicated follow RD 470/2021 Art. 49.5.1 and 49.5.2.2. Do not splice at points of maximum stress. Overlap welded wire reinforcement the spacing of the cross wires, plus 50 mm.

3.5.3 Setting Miscellaneous Material

Place and secure anchors and bolts, pipe sleeves, conduits, and other such items in position before concrete placement and support against displacement. Plumb anchor bolts and check location and elevation. Temporarily fill voids in sleeves with readily removable material to prevent the entry of concrete.

3.5.4 Fabrication

Shop fabricate reinforcing bars to conform to shapes and dimensions

indicated for reinforcement, and as follows:

- a. Provide fabrication tolerances that are in accordance with ACI 117 and RD 470/2021 Art. 34.2 and 49.2.4.
- b. Provide hooks and bends that are in accordance with the Contract Documents and RD 470/2021, Art. 49.3.4 and Figure 49.5.1.1.

Reinforcement must be bent cold to shapes as indicated. Bending must be done in the shop. Rebending of a reinforcing bar that has been bent incorrectly is not permitted. Bending must be in accordance with standard approved practice and by approved machine methods.

Deliver reinforcing bars bundled, tagged, and marked. Tags must be metal with bar size, length, mark, and other information pressed in by machine. Marks must correspond with those used on the placing drawings.

Do not use reinforcement that has any of the following defects:

- a. Bar lengths, depths, and bends beyond specified fabrication tolerances
- b. Bends or kinks not indicated on drawings or approved shop drawings
- c. Bars with reduced cross-section due to rusting or other cause

Replace defective reinforcement with new reinforcement having required shape, form, and cross-section area.

3.5.5 Placing Reinforcement

Place reinforcement in accordance with ACI 301 and RD 470/2021 Art. 49.8.

For slabs on grade (over earth or over capillary water barrier) and for footing reinforcement, support bars or welded wire reinforcement on precast concrete blocks, spaced at intervals required by size of reinforcement, to keep reinforcement the minimum height specified above the underside of slab or footing.

For slabs other than on grade, supports for which any portion is less than 25 mm from concrete surfaces that are exposed to view or to be painted must be of precast concrete units, plastic-coated steel, or stainless steel protected bar supports. Precast concrete units must be wedge shaped, not larger than 90 by 90 mm, and of thickness equal to that indicated for concrete protection of reinforcement. Provide precast units that have cast-in galvanized tie wire hooked for anchorage and blend with concrete surfaces after finishing is completed.

Provide reinforcement that is supported and secured together to prevent displacement by construction loads or by placing of wet concrete, and as follows:

- a. Provide supports for reinforcing bars that are sufficient in number and have sufficient strength to carry the reinforcement they support. Do not use supports to support runways for concrete conveying equipment and similar construction loads.
- b. Equip supports on ground and similar surfaces with sand-plates.
- c. Support welded wire reinforcement as required for reinforcing bars.
- d. Secure reinforcements to supports by means of tie wire. Wire must be

black, soft iron wire, not less than 1.6 mm.

- e. Reinforcement must be accurately placed, securely tied at intersections, and held in position during placing of concrete by spacers, chairs, or other approved supports. Point wire-tie ends away from the form. Unless otherwise indicated, numbers, type, and spacing of supports must conform to the Contract Documents.
- f. Bending of reinforcing bars partially embedded in concrete is permitted only as specified in the Contract Documents.

3.5.6 Spacing of Reinforcing Bars

- a. Spacing must be as indicated in the Contract Documents.
- b. Reinforcing bars may be relocated to avoid interference with other reinforcement, or with conduit, pipe, or other embedded items. If any reinforcing bar is moved a distance exceeding one bar diameter or specified placing tolerance, resulting rearrangement of reinforcement is subject to preapproval by the Contracting Officer.

3.5.7 Concrete Protection for Reinforcement

Additional concrete protection must be in accordance with the Contract Documents.

3.6 BATCHING, MEASURING, MIXING, AND TRANSPORTING CONCRETE

In accordance with RD 470/2021 Art. 51.2 and 51.3, except as modified herein. Batching equipment must be such that the concrete ingredients are consistently measured within the following tolerances: 1 percent for cement and water, 2 percent for aggregate, and 3 percent for admixtures. Furnish mandatory batch ticket information for each load of ready mix concrete.

3.6.1 Measuring

Make measurements at intervals as specified in paragraphs SAMPLING and TESTING.

3.6.2 Mixing

- a. Mix concrete in accordance with RD 470/2021 Art. 51.2.3 and 51.2.4.
- b. Machine mix concrete. Begin mixing within 30 minutes after the cement has been added to the aggregates. Place concrete within 90 minutes of either addition of mixing water to cement and aggregates or addition of cement to aggregates if the concrete temperature is less than 29 degrees C.
- c. Place concrete within 60 minutes if the concrete temperature is greater than 29 degrees C except as follows: if set retarding admixture is used and slump requirements can be met, limit for placing concrete may remain at 90 minutes. Additional water may be added, provided that both the specified maximum slump and submitted water-cementitious material ratio are not exceeded and the required concrete strength is still met. When additional water is added, an additional 30 revolutions of the mixer at mixing speed is required.

- d. Dissolve admixtures in the mixing water and mix in the drum to uniformly distribute the admixture throughout the batch. Do not reconstitute concrete that has begun to solidify.

3.6.3 Transporting

Transport concrete from the mixer to the forms as rapidly as practicable. Prevent segregation or loss of ingredients. Clean transporting equipment thoroughly before each batch. Do not use aluminum pipe or chutes. Remove concrete which has segregated in transporting and dispose of as directed.

3.7 PLACING CONCRETE

Place concrete in accordance with RD 470/2021 Art. 52.1 and 52.2. Concrete shall be placed within 15 minutes of discharge into non-agitating equipment.

3.7.1 Pumping

Pumping must not result in separation or loss of materials nor cause interruptions sufficient to permit loss of plasticity between successive increments. Loss of slump in pumping equipment must not exceed 50 mm at discharge/placement. Do not convey concrete through pipe made of aluminum or aluminum alloy. Avoid rapid changes in pipe sizes. Limit maximum size of coarse aggregate to 33 percent of the diameter of the pipe. Limit maximum size of well-rounded aggregate to 40 percent of the pipe diameter. Take samples for testing at both the point of delivery to the pump and at the discharge end.

3.7.2 Cold Weather

Cold weather concrete must meet the requirements of RD 470/2021 Art. 52.3.1 unless otherwise specified. Do not allow concrete temperature to decrease below 10 degrees C. Obtain approval prior to placing concrete when the ambient temperature is below 4 degrees C or when concrete is likely to be subjected to freezing temperatures within 24 hours. Cover concrete and provide sufficient heat to maintain 10 degrees C minimum adjacent to both the formwork and the structure while curing. Limit the rate of cooling to 3 degrees C in any one hour and 10 degrees C per 24 hours after heat application.

3.7.3 Hot Weather

Hot weather concrete must meet the requirements of RD 470/2021 Art. 52.3.2 unless otherwise specified. Maintain required concrete temperature using Figure 4.2 in ACI 305R to prevent the evaporation rate from exceeding 1 kg per square meter of exposed concrete per hour. Cool ingredients before mixing or use other suitable means to control concrete temperature and prevent rapid drying of newly placed concrete. Shade the fresh concrete as soon as possible after placing. Start curing when the surface of the fresh concrete is sufficiently hard to permit curing without damage. Provide water hoses, pipes, spraying equipment, and water hauling equipment, where job site is remote to water source, to maintain a moist concrete surface throughout the curing period. Provide burlap cover or other suitable, permeable material with fog spray or continuous wetting of the concrete when weather conditions prevent the use of either liquid membrane curing compound or impervious sheets. For vertical surfaces, protect forms from direct sunlight and add water to top of structure once concrete is set.

3.7.4 Bonding

Surfaces of set concrete at joints, must be roughened and cleaned of laitance, coatings, loose particles, and foreign matter. Roughen surfaces in a manner that exposes the aggregate uniformly and does not leave laitance, loosened particles of aggregate, nor damaged concrete at the surface.

Obtain bonding of fresh concrete that has set as follows:

- a. At joints between footings and walls or columns, between walls or columns and the beams or slabs they support, and elsewhere unless otherwise specified; roughened and cleaned surface of set concrete must be dampened, but not saturated, immediately prior to placing of fresh concrete.
- b. At joints in exposed-to-view work; at vertical joints in walls; at joints near midpoint of span in girders, beams, supported slabs, other structural members; in work designed to contain liquids; the roughened and cleaned surface of set concrete must be dampened but not saturated and covered with a cement grout coating.
- c. Provide cement grout that consists of equal parts of portland cement and fine aggregate by weight with not more than 22.5 liters of water per sack of cement. Apply cement grout with a stiff broom or brush to a minimum thickness of 1.6 mm. Deposit fresh concrete before cement grout has attained its initial set.

3.8 WASTE MANAGEMENT

Provide as specified in the Waste Management Plan and as follows.

3.8.1 Mixing Equipment

Before concrete pours, designate Contractor-owned site meeting environmental standards for cleaning out concrete mixing trucks. Minimize water used to wash equipment.

3.8.2 Reinforcing Steel

Collect reinforcing steel and place in designated area for recycling.

3.8.3 Other Waste

Identify concrete manufacturer's or supplier's policy for collection or return of construction waste, unused material, deconstruction waste, and/or packaging material.

3.9 SURFACE FINISHES EXCEPT FLOOR, SLAB, AND PAVEMENT FINISHES

3.9.1 Defects

Repair surface defects in accordance with ACI 301 Section 5 and RD 470/2021 Art. 53.3.

3.9.2 Not Against Forms (Top of Walls)

Surfaces not otherwise specified must be finished with wood floats to even

surfaces. Finish must match adjacent finishes.

3.9.3 Formed Surfaces

3.9.3.1 Tolerances

Tolerances in accordance with ACI 117 and RD 470/2021 Art. 53.3 and Anejo 14, and as indicated.

3.9.3.2 As-Cast Rough Form

Provide for surfaces not exposed to public view a surface finish SF-1.0. Patch holes and defects.

3.9.3.3 Standard Smooth Finish

Provide for surfaces exposed to public view a surface finish SF-3.0. Patch holes and defects.

3.9.4 Smooth-Rubbed Finish

Provide a smooth-rubbed finish per ACI 301 Section 5 in the locations exposed to view.

3.10 FLOOR, SLAB, AND PAVEMENT FINISHES AND MISCELLANEOUS CONSTRUCTION

In accordance with ACI 301 and ACI 302.1R and RD 470/2021 Art. 53.3, unless otherwise specified. Slope floors uniformly to drains where drains are provided. Where straightedge measurements are specified, Contractor must provide straightedge.

3.10.1 Finish

Place, consolidate, and immediately strike off concrete to obtain proper contour, grade, and elevation before bleedwater appears. Permit concrete to attain a set sufficient for floating and supporting the weight of the finisher and equipment. If bleedwater is present prior to floating the surface, drag the excess water off or remove by absorption with porous materials. Do not use dry cement to absorb bleedwater. Grate tampers ("jitterbugs") shall not be used.

3.10.1.1 Scratched

Use for surfaces intended to receive bonded applied cementitious applications. Finish concrete in accordance with ACI 301 Section 5 for a scratched finish.

3.10.1.2 Floated

Use for exterior slabs where not otherwise specified. Finish concrete in accordance with ACI 301 Section 5 for a floated finish.

3.10.1.3 Broomed

Use on surfaces of exterior walks, platforms, patios, and ramps, unless otherwise indicated. Finish concrete in accordance with ACI 301 Section 5 for a broomed finish.

3.10.1.4 Pavement

Screed the concrete with a template advanced with a combined longitudinal and crosswise motion. Maintain a slight surplus of concrete ahead of the template. After screeding, float the concrete longitudinally. Use a straightedge to check slope and flatness; correct and refloat as necessary. Obtain final finish by a burlap drag. Drag a strip of clean, wet burlap from 900 to 3000 mm wide and 600 mm longer than the pavement width across the slab. Produce a fine, granular, sandy textured surface without disfiguring marks. Round edges and joints with an edger having a radius of 3 mm.

3.11 JOINTS

RD 470/2021, Art. 52.4 and the following:

3.11.1 Construction Joints

Make and locate joints not indicated so as not to impair strength and appearance of the structure, as approved. Joints must be perpendicular to main reinforcement. Reinforcement must be continued and developed across construction joints. Locate construction joints as follows:

3.11.1.1 Construction Joints for Constructability Purposes

- a. In walls, at top of footing; at top of slabs on ground; at top and bottom of door and window openings or where required to conform to architectural details; and at underside of deepest beam or girder framing into wall.
- b. In columns or piers, at top of footing; at top of slabs on ground; and at underside of deepest beam or girder framing into column or pier.
- c. Near midpoint of spans for supported slabs, beams, and girders unless a beam intersects a girder at the center, in which case construction joints in girder must offset a distance equal to twice the width of the beam. Make transfer of shear through construction joint by use of inclined reinforcement.

Provide keyways at least 40 mm deep in construction joints in walls and slabs and between walls and footings; approved bulkheads may be used for slabs.

3.11.2 Isolation Joints in Slabs on Ground

- a. Provide joints at points of contact between slabs on ground and vertical surfaces, such as column pedestals, foundation walls, grade beams, and elsewhere as indicated.
- b. Fill joints with premolded joint filler strips 13 mm thick, extending full slab depth. Install filler strips at proper level below finish floor elevation with a slightly tapered, dress-and-oiled wood strip temporarily secured to top of filler strip to form a groove not less than 19 mm in depth where joint is sealed with sealing compound and not less than 6 mm in depth where joint sealing is not required. Remove wood strip after concrete has set. Contractor must clean groove of foreign matter and loose particles after surface has dried.

3.11.3 Contraction Joints in Slabs on Ground

- a. Provide joints to form panels as indicated.
- b. Under and on exact line of each control joint, cut 50 percent of welded wire reinforcement before placing concrete.
- c. Sawcut contraction joints into slab on ground.

3.11.4 Sealing Joints in Slabs on Ground

- a. Contraction and control joints which are to receive finish flooring material must be sealed with joint sealing compound after concrete curing period. Slightly underfill groove with joint sealing compound to prevent extrusion of compound. Remove excess material as soon after sealing as possible.
- b. Sealed groove must be left ready to receive filling material that is provided as part of finish floor covering work.

3.12 CURING AND PROTECTION

Curing and protection in accordance with RD 470/2021, unless otherwise specified. Begin curing immediately following form removal. Avoid damage to concrete from vibration created by blasting, pile driving, movement of equipment in the vicinity, disturbance of formwork or protruding reinforcement, and any other activity resulting in ground vibrations. Protect concrete from injurious action by sun, rain, flowing water, frost, mechanical injury, tire marks, and oil stains. Do not allow concrete to dry out from time of placement until the expiration of the specified curing period. Do not use membrane-forming compound on surfaces where appearance would be objectionable, on any surface to be painted, where coverings are to be bonded to the concrete, or on concrete to which other concrete is to be bonded. If forms are removed prior to the expiration of the curing period, provide another curing procedure specified herein for the remaining portion of the curing period. Provide moist curing for those areas receiving liquid chemical sealer, hardener, or epoxy coating. Allow curing compound/sealer installations to cure prior to the installation of materials that adsorb VOCs.

3.12.1 Curing Periods

ACI 301 Section 5 and RD 470/2021 Art. 52.5, except 10 days for retaining walls, pavement or chimneys. Begin curing immediately after placement. Protect concrete from premature drying, excessively hot temperatures, and mechanical injury; and maintain minimal moisture loss at a relatively constant temperature for the period necessary for hydration of the cement and hardening of the concrete. The materials and methods of curing are subject to approval by the Contracting Officer.

3.12.2 Curing Formed Surfaces

Accomplish curing of formed surfaces, including undersurfaces of girders, beams, supported slabs, and other similar surfaces by moist curing with forms in place for full curing period or until forms are removed. If forms are removed before end of curing period, accomplish final curing of formed surfaces by any of the curing methods specified above, as applicable.

3.12.3 Curing Unformed Surfaces

- a. Accomplish initial curing of unformed surfaces, such as monolithic slabs, floor topping, and other flat surfaces, by membrane curing.
- b. Accomplish final curing of unformed surfaces by any of curing methods specified, as applicable.
- c. Accomplish final curing of concrete surfaces to receive liquid floor hardener of finish flooring by moisture-retaining cover curing.

3.12.4 Temperature of Concrete During Curing

When temperature of atmosphere is 5 degrees C and below, maintain temperature of concrete at not less than 13 degrees C throughout concrete curing period or 7 degrees C when the curing period is measured by maturity. When necessary, make arrangements before start of concrete placing for heating, covering, insulation, or housing as required to maintain specified temperature and moisture conditions for concrete during curing period.

When the temperature of atmosphere is 27 degrees C and above or during other climatic conditions which cause too rapid drying of concrete, make arrangements before start of concrete placing for installation of wind breaks, of shading, and for fog spraying, wet sprinkling, or moisture-retaining covering of light color as required to protect concrete during curing period.

Changes in temperature of concrete must be uniform and not exceed 3 degrees C in any one hour nor 27 degrees C in any 24-hour period.

3.12.5 Protection from Mechanical Injury

During curing period, protect concrete from damaging mechanical disturbances, particularly load stresses, heavy shock, and excessive vibration and from damage caused by rain or running water.

3.12.6 Protection After Curing

Protect finished concrete surfaces from damage by construction operations.

3.13 FIELD QUALITY CONTROL

3.13.1 Aggregate Testing

3.13.1.1 Fine Aggregate

At least once during each shift when the concrete plant is operating, there shall be one sieve analysis and fineness modulus determination in accordance with UNE-EN 933-1 for the fine aggregate or for each fine aggregate if it is batched in more than one size or classification. The location at which samples are taken may be selected by the Contractor as the most advantageous for control. However, the Contractor is responsible for delivering fine aggregate to the mixer within specification limits. When the amount passing on any sieve is outside the specification limits, the fine aggregate shall be immediately resampled and retested. If there is another failure on any sieve, the fact shall be immediately reported to the Contracting Officer, concreting shall be stopped, and immediate steps taken to correct the grading.

3.13.1.2 Coarse Aggregate

At least once during each shift in which the concrete plant is operating, there shall be a sieve analysis in accordance with UNE-EN 933-1 for each size of coarse aggregate. The location at which samples are taken may be selected by the Contractor as the most advantageous for production control. However, the Contractor shall be responsible for delivering the aggregate to the mixer within specification limits. A test record of samples of aggregate taken at the same locations shall show the results of the current test as well as the average results of the five most recent tests including the current test. The Contractor may adopt limits for control coarser than the specification limits for samples taken other than as delivered to the mixer to allow for degradation during handling. When the amount passing any sieve is outside the specification limits, the coarse aggregate shall be immediately resampled and retested. If the second sample fails on any sieve, that fact shall be reported to the Contracting Officer. Where two consecutive averages of 5 tests are outside specification limits, the operation shall be considered out of control and reported to the Contracting Officer. Concreting shall be stopped and immediate steps shall be taken to correct the grading.

3.13.2 Concrete Sampling

UNE-EN 12350-1. Collect samples of fresh concrete to perform tests specified. UNE-EN 12390-2 for making test specimens.

3.13.3 Concrete Testing

3.13.3.1 Slump Tests

UNE-EN 12350-2. Take concrete samples during concrete placement/discharge. The maximum slump may be increased as specified with the addition of an approved admixture provided that the water-cementitious material ratio is not exceeded. Perform tests at commencement of concrete placement, when test cylinders are made, and for each batch (minimum) or every 16 cubic meters (maximum) of concrete.

3.13.3.2 Temperature Tests

Test the concrete delivered and the concrete in the forms. Perform tests in hot or cold weather conditions (below 10 degrees C and above 27 degrees C) for each batch (minimum) or every 16 cubic meters (maximum) of concrete, until the specified temperature is obtained, and whenever test cylinders and slump tests are made.

3.13.3.3 Compressive Strength Tests

UNE-EN 12390-3. Make six 150 mm by 300 mm test cylinders for each set of tests in accordance with UNE-EN 12390-2 applicable requirements of RD 470/2021, Art. 57.5.3, 57.5.4 and 57.5.5. Take precautions to prevent evaporation and loss of water from the specimen. Test two cylinders at 7 days, two cylinders at 28 days, and hold two cylinder in reserve. Take samples for strength tests of concrete placed each day not less than once a day, nor less than once for each 75 cubic meters of concrete for the first 380 cubic meters, then every 380 cubic meters thereafter, nor less than once for each 500 square meters of surface area for slabs or walls. For the entire project, take no less than five sets of samples and perform strength tests for each mix design of concrete placed. Each strength test

result must be the average of two cylinders from the same concrete sample tested at 28 days. Concrete compressive tests must meet the requirements of this section and the Contract Document. Retest locations represented by erratic core strengths. Where retest does not meet concrete compressive strength requirements submit a mitigation or remediation plan for review and approval by the contracting officer. Repair core holes with nonshrink grout. Match color and finish of adjacent concrete.

3.13.3.4 Strength of Concrete Structure

The strength of the concrete structure will be considered to be deficient if any of the following conditions are identified:

- a. Failure to meet compressive strength tests as evaluated.
- b. Reinforcement not conforming to requirements specified.
- c. Concrete which differs from required dimensions or location in such a manner as to reduce strength.
- d. Concrete curing and protection of concrete against extremes of temperature during curing, not conforming to requirements specified.
- e. Concrete subjected to damaging mechanical disturbances, particularly load stresses, heavy shock, and excessive vibration.
- f. Poor workmanship likely to result in deficient strength.

Where the strength of the concrete structure is considered deficient submit a mitigation or remediation plan for review and approval by the contracting officer.

3.13.3.5 Non-Conforming Materials

Factors that indicate that there are non-conforming materials include (but not limited to) excessive compressive strength, inadequate compressive strength, excessive slump, excessive voids and honeycombing, concrete delivery records that indicate excessive time between mixing and placement, or excessive water was added to the mixture during delivery and placement. Any of these indicators alone are sufficient reason for the Contracting Officer to request additional sampling and testing.

Investigations into non-conforming materials must be conducted at the Contractor's expense. The Contractor must be responsible for the investigation and must make written recommendations to adequately mitigate or remediate the non-conforming material. The Contracting Officer may accept, accept with reduced payment, require mitigation, or require removal and replacement of non-conforming material at no additional cost to the Government.

3.13.3.6 Testing Concrete Structure for Strength

When there is evidence that strength of concrete structure in place does not meet specification requirements or there are non-conforming materials, make cores drilled from hardened concrete for compressive strength determination in accordance with UNE-EN 12504-1, and as follows:

- a. Take at least three representative cores from each member or area of concrete-in-place that is considered potentially deficient. Location

of cores will be determined by the Contracting Officer.

- b. Test cores after moisture conditioning in accordance with UNE-EN 12504-1 if concrete they represent is more than superficially wet under service.
- c. Air dry cores, (16 to 27 degrees C with relative humidity less than 60 percent) for 7 days before test and test dry if concrete they represent is dry under service conditions.
- d. Strength of cores from each member or area are considered satisfactory if their average is equal to or greater than 85 percent of the 28-day design compressive strength of the class of concrete.

Fill core holes solid with patching mortar and finished to match adjacent concrete surfaces.

Correct concrete work that is found inadequate by core tests in a manner approved by the Contracting Officer.

3.14 REPAIR, REHABILITATION AND REMOVAL

Before the Contracting Officer accepts the structure the Contractor must inspect the structure for cracks, damage and substandard concrete placements that may adversely affect the service life of the structure. A report documenting these defects must be prepared which includes recommendations for repair, removal or remediation must be submitted to the Contracting Officer for approval before any corrective work is accomplished.

3.14.1 Crack Repair

Prior to final acceptance, all cracks in excess of 0.50 mm wide must be documented and repaired. The proposed method and materials to repair the cracks must be submitted to the Contracting Officer for approval. The proposal must address the amount of movement expected in the crack due to temperature changes and loading.

3.14.2 Repair of Weak Surfaces

Weak surfaces are defined as mortar-rich, rain-damaged, uncured, or containing exposed voids or deleterious materials. Concrete surfaces with weak surfaces less than 6 mm thick must be diamond ground to remove the weak surface. Surfaces containing weak surfaces greater than 6 mm thick must be removed and replaced or mitigated in a manner acceptable to the Contracting Officer.

3.14.3 Failure of Quality Assurance Test Results

Proposed mitigation efforts by the Contractor must be approved by the Contracting Officer prior to proceeding.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 05 - METALS

SECTION 05 50 13

MISCELLANEOUS METAL FABRICATIONS

05/17, CHG 1: 08/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALIFICATION OF WELDERS
- 1.4 DELIVERY, STORAGE, AND PROTECTION
- 1.5 MISCELLANEOUS REQUIREMENTS
 - 1.5.1 Fabrication Drawings
 - 1.5.2 Installation Drawings

PART 2 PRODUCTS

- 2.1 MATERIALS
 - 2.1.1 Structural Carbon Steel
 - 2.1.2 Structural Tubing
 - 2.1.3 Steel Pipe
 - 2.1.4 Fittings for Steel Pipe
 - 2.1.5 Gratings
 - 2.1.6 Floor Plates, Patterned
 - 2.1.7 Anchor Bolts
 - 2.1.7.1 Expansion Anchors and Adhesive Anchors
 - 2.1.7.2 Lag Screws and Bolts
 - 2.1.7.3 Toggle Bolts
 - 2.1.7.4 Bolts, Nuts, Studs and Rivets
 - 2.1.7.5 Screws
 - 2.1.7.6 Washers
 - 2.1.7.7 Welded Headed Shear Studs
- 2.2 FABRICATION FINISHES
 - 2.2.1 Galvanizing
 - 2.2.2 Galvanize
 - 2.2.3 Repair of Zinc-Coated Surfaces
 - 2.2.4 Shop Cleaning and Painting
 - 2.2.4.1 Surface Preparation
 - 2.2.4.2 Pretreatment, Priming and Painting
 - 2.2.5 Nonferrous Metal Surfaces
- 2.3 FLOOR GRATINGS
- 2.4 BOLLARDS
- 2.5 MISCELLANEOUS PLATES AND SHAPES

PART 3 EXECUTION

- 3.1 GENERAL INSTALLATION REQUIREMENTS
- 3.2 WORKMANSHIP
- 3.3 ANCHORAGE, FASTENINGS, AND CONNECTIONS
- 3.4 BUILT-IN WORK
- 3.5 WELDING

- 3.6 DISSIMILAR METALS
- 3.7 PREPARATION
 - 3.7.1 Material Coatings and Surfaces
 - 3.7.2 Environmental Conditions
- 3.8 INSTALLATION OF BOLLARDS

-- End of Section Table of Contents --

SECTION 05 50 13

MISCELLANEOUS METAL FABRICATIONS
05/17, CHG 1: 08/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 104231	(1999) Waterproofing. Bituminous And Bituminous Modified Materials. Asphalt Emulsions
UNE-EN 1562	(2020) Founding - Malleable cast irons
UNE-EN 1993-1-1	(2013) Eurocode 3: Design of steel structures - Part 1-1: General rules and rules for buildings
UNE-EN 10025-2	(2020) Hot rolled products of structural steels - Part 2: Technical delivery conditions for non-alloy structural steels
UNE-EN 10025-3	(2006) Hot rolled products of structural steels - Part 3: Technical delivery conditions for normalized/normalized rolled weldable fine grain structural steels
UNE-EN 10083-1	(2008) Steels for quenching and tempering - Part 1: General technical delivery conditions.
UNE-EN 10162	(2005) Cold formed steel sections - Technical Supply Conditions. Dimensional, tolerances and on Cross-section.
UNE-EN 10216-1	(2014) Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 1: Non-alloy steel tubes with specified room temperature properties
UNE-EN 10217-1	(2019) Welded steel tubes for pressure purposes - Technical delivery conditions - Part 1: Non-alloy steel tubes with specified room temperature properties
UNE-EN 10219-1	(2007) Cold formed welded structural hollow sections of non-alloy and fine grain steels - Part 1: Technical delivery conditions

UNE-EN 10363	(2016) Continuously hot-rolled patterned steel strip and plate/sheet cut from wide strip - Tolerances on dimensions and shape
UNE-EN ISO 898-1	(2015) Mechanical properties of fasteners made of carbon steel and alloy steel - Part 1: Bolts, screws and studs with specified property classes - Coarse thread and fine pitch thread (ISO 898-1:2013)
UNE-EN ISO 898-2	(2013) Mechanical properties of fasteners made of carbon steel and alloy steel - Part 2: Nuts with specified property classes - Coarse thread and fine pitch thread (ISO 898-2:2012)
UNE-EN ISO 1461	(2010) Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods (ISO1461:1999)
UNE-EN ISO 1482	(2011) Slotted countersunk (flat) head tapping screws (ISO 1482:2011)
UNE-EN ISO 9606-1	(2017) Qualification testing of welders - Fusion welding - Part 1: Steels (ISO 9606-1:2012 including Cor 1:2012)
UNE-EN ISO 9606-2	(2005) Qualification test of welders-Fusion welding-Part 2: Aluminium and aluminium alloys (ISO 9606-2:2004)
UNE-EN ISO 10673	(2010) Plain washers for screw and washer assemblies. Small, normal and large series. Product grade A (ISO 10673:2009)

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318	(2014; Errata 1-2 2014; Errata 3-5 2015; Errata 6 2016; Errata 7-9 2017) Building Code Requirements for Structural Concrete (ACI 318-14) and Commentary (ACI 318R-14)
---------	---

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

AISC 303	(2016) Code of Standard Practice for Steel Buildings and Bridges
----------	--

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B18.2.1	(2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)
ASME B18.6.2	(2020) Square Head Set Screws and Slotted Headless Set Screws (Inch Series)
ASME B18.6.3	(2013; R 2017) Machine Screws, Tapping Screws, and Machine Drive Screws (Inch

Series)

ASME B18.21.1	(2009; R 2016) Washers: Helical Spring-Lock, Tooth Lock, and Plain Washers (Inch Series)
ASME B18.21.2M	(1999; R 2014) Lock Washers (Metric Series)
ASME B18.22M	(1981; R 2017) Metric Plain Washers

AMERICAN WELDING SOCIETY (AWS)

AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel
----------------	---

ASTM INTERNATIONAL (ASTM)

ASTM A53/A53M	(2020) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless
ASTM A108	(2013) Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished
ASTM A780/A780M	(2020) Standard Practice for Repair of Damaged and Uncoated Areas of Hot-Dip Galvanized Coatings
ASTM C1513	(2018) Standard Specification for Steel Tapping Screws for Cold-Formed Steel Framing Connections

MASTER PAINTERS INSTITUTE (MPI)

MPI 79	(2016) Primer, Alkyd, Anti-Corrosive for Metal
--------	--

NATIONAL ASSOCIATION OF ARCHITECTURAL METAL MANUFACTURERS (NAAMM)

NAAMM MBG 531	(2017) Metal Bar Grating Manual
---------------	---------------------------------

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 3	(2018) Power Tool Cleaning
SSPC SP 6/NACE No.3	(2007) Commercial Blast Cleaning

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Floor Gratings, Installation Drawings; G
Bollards; G
Embedded Angles and Plates, Installation Drawings; G

SD-03 Product Data

Floor Gratings; G

SD-07 Certificates

Certified Mill Test Reports for Chemistry and Mechanical
Properties; G

1.3 QUALIFICATION OF WELDERS

Qualify welders in accordance with AWS D1.1/D1.1M or in accordance with UNE-EN ISO 9606-1 for steel and UNE-EN ISO 9606-2 for aluminum. Use procedures, materials, and equipment of the type required for the work.

1.4 DELIVERY, STORAGE, AND PROTECTION

Protect from corrosion, deformation, and other types of damage. Store items in an enclosed area free from contact with soil and weather. Remove and replace damaged items with new items.

1.5 MISCELLANEOUS REQUIREMENTS

1.5.1 Fabrication Drawings

Submit fabrication drawings showing layout(s), connections to structural system, and anchoring details as specified in AISC 303 and in accordance with UNE-EN 1993-1-1.

1.5.2 Installation Drawings

Submit templates, erection, and installation drawings indicating thickness, type, grade, class of metal, and dimensions. Show construction details, reinforcement, anchorage, and installation in relation to the building construction.

PART 2 PRODUCTS

2.1 MATERIALS

Provide exposed fastenings of compatible materials (avoid contact of dissimilar metals). Coordinate color and finish with the material to which fastenings are applied. Submit the manufacturer's certified mill reports which clearly show the applicable ASTM mechanical and chemical requirements together with the actual test results for the supplied materials.

2.1.1 Structural Carbon Steel

Provide in accordance with UNE-EN 10025-2 and UNE-EN 10025-3.

2.1.2 Structural Tubing

Provide in accordance with UNE-EN 10219-1.

2.1.3 Steel Pipe

Provide in accordance with UNE-EN 10216-1 or UNE-EN 10217-1. Minimum

tensile strength 330 Mpa.

2.1.4 Fittings for Steel Pipe

Provide standard malleable iron fittings in accordance with UNE-EN 1562.

2.1.5 Gratings

a. Provide metal bar type grating in accordance with NAAMM MBG 531.

2.1.6 Floor Plates, Patterned

Provide floor plate in accordance with UNE-EN 10363. Provide steel plate not less than 1.9 mm.

2.1.7 Anchor Bolts

Provide in accordance with UNE-EN ISO 898-1. Where exposed, provide anchor bolts of the same material, color, and finish as the metal to which they are applied.

2.1.7.1 Expansion Anchors and Adhesive Anchors

Provide expansion anchors and adhesive anchors as indicated in the Contract Documents to the minimum embedment indicated.

- a. Provide minimum allowable pullout value as required by design. Calculate pullout capacity according to ACI 318.
- b. Provide minimum allowable shear value as required by design. Calculate shear capacity according to ACI 318.

2.1.7.2 Lag Screws and Bolts

Provide in accordance with ASME B18.2.1, type and grade best suited for the purpose.

2.1.7.3 Toggle Bolts

Provide in accordance with ASME B18.2.1.

2.1.7.4 Bolts, Nuts, Studs and Rivets

Provide in accordance with UNE-EN ISO 898-1 and UNE-EN ISO 898-2.

2.1.7.5 Screws

Provide in accordance with ASME B18.2.1, ASME B18.6.2, ASME B18.6.3 and ASTM C1513 or UNE-EN ISO 1482.

2.1.7.6 Washers

Provide plain washers in accordance with ASME B18.22M, ASME B18.21.1 or UNE-EN ISO 10673. Provide beveled washers for American Standard beams and channels, square or rectangular, tapered in thickness, and smooth. Provide lock washers in accordance with ASME B18.21.2M, ASME B18.21.1.

2.1.7.7 Welded Headed Shear Studs

Provide in accordance with ASTM A108 or UNE-EN 10162 or UNE-EN 10083-1.

2.2 FABRICATION FINISHES

2.2.1 Galvanizing

Hot-dip galvanize items specified to be zinc-coated, after fabrication where practicable. Provide galvanizing in accordance with UNE-EN ISO 1461, Z275 G90.

2.2.2 Galvanize

Anchor bolts, grating fasteners, washers, and parts or devices necessary for proper installation, unless indicated otherwise.

2.2.3 Repair of Zinc-Coated Surfaces

Repair damaged surfaces with galvanizing repair method and paint in accordance with ASTM A780/A780M or UNE-EN ISO 1461, or by application of stick or thick paste material specifically designed for repair of galvanizing, as approved by Contracting Officer. Clean areas to be repaired and remove slag from welds. Heat, with a torch, surfaces to which stick or paste material will be applied. Heat to a temperature sufficient to melt the metals in the stick or paste. Spread molten material uniformly over surfaces to be coated and wipe off excess material.

2.2.4 Shop Cleaning and Painting

2.2.4.1 Surface Preparation

Blast clean surfaces in accordance with SSPC SP 6/NACE No.3. Surfaces that will be exposed in spaces above ceiling or in attic spaces, crawl spaces, furred spaces, and chases may be cleaned in accordance with SSPC SP 3 in lieu of being blast cleaned. Wash cleaned surfaces which become contaminated with rust, dirt, oil, grease, or other contaminants with solvents until thoroughly clean. Steel to be embedded in concrete must be free of dirt and grease prior to embed. Do not paint or galvanize bearing surfaces, including contact surfaces within slip critical joints. Shop coat these surfaces with rust prevention.

2.2.4.2 Pretreatment, Priming and Painting

Apply pre-treatment, primer, and paint in accordance with manufacturer's printed instructions. On surfaces concealed in the finished construction or not accessible for finish painting, apply an additional prime coat to a minimum dry film thickness of 0.03 mm. Tint additional prime coat with a small amount of tinting pigment.

2.2.5 Nonferrous Metal Surfaces

Protect by plating, anodic, or organic coatings.

2.3 FLOOR GRATINGS

- a. NAAMM MBG 531, band ends of gratings with bars of the same or greater thickness than the metal used for grating. Weld banding bars to bearing bars or channels at least every fourth bar or channel and in

every corner. Tack weld intervening bars or channels. Band diagonal or round cuts by welding bars of the same or greater thickness as the grating and in accordance with the manufacturer's standard for trim unless otherwise indicated.

- b. Anchor gratings to structural members with bolts, toggle bolts, or expansion shields and bolts.
- c. Provide slip resistant surface finishes.

2.4 BOLLARDS

Provide 200 mm prime coated standard weight steel pipe in accordance with ASTM A53/A53M or UNE-EN 10216-1. Anchor posts in concrete as indicated and fill solidly with concrete with minimum compressive strength of 20 MPa.

2.5 MISCELLANEOUS PLATES AND SHAPES

Provide items that do not form a part of the structural framework, such as miscellaneous mountings and frames. Provide with connections and fasteners or welds.

Provide angles and plates in accordance with UNE-EN 10025-2 and UNE-EN 10025-3, for embedment as indicated. Galvanize embedded items exposed to the elements in accordance with UNE-EN ISO 1461.

PART 3 EXECUTION

3.1 GENERAL INSTALLATION REQUIREMENTS

Install items at locations indicated in accordance with manufacturer's instructions. Verify all field dimensions prior to fabrication. Include materials and parts necessary to complete each assembly, whether indicated or not. Miss-alignment and miss-sizing of holes for fasteners is cause for rejection. Conceal fastenings where practicable. Joints exposed to weather must be watertight.

3.2 WORKMANSHIP

Provide miscellaneous metalwork that is true and accurate in shape, size, and profile. Make angles and lines continuous and straight. Make curves consistent, smooth and unfaceted. Provide continuous welding along the entire area of contact except where tack welding is permitted. Do not tack weld exposed connections. Unless otherwise indicated and approved, provide a smooth finish on exposed surfaces. Provide countersunk rivets where exposed. Provide coped and mitered corner joints aligned flush and without gaps.

3.3 ANCHORAGE, FASTENINGS, AND CONNECTIONS

Provide anchorage as necessary, whether indicated or not, for fastening miscellaneous metal items securely in place. Include slotted inserts, expansion shields, powder-driven fasteners, toggle bolts (when approved for concrete), headed shear studs, machine and carriage bolts for steel. Provide non-ferrous attachments for non-ferrous metal. Provide exposed fastenings of compatible materials (avoid contact of dissimilar metals), that generally match in color and finish the surfaces to which they are applied. Conceal fastenings where practicable. Provide all fasteners flush with the surfaces they fasten, unless indicated otherwise.

3.4 BUILT-IN WORK

Where necessary and not otherwise indicated, form built-in metal work for anchorage with concrete. Provide built-in metal work in ample time for securing in place as the work progresses.

3.5 WELDING

Perform welding, welding inspection, and corrective welding in accordance with AWS D1.1/D1.1M or UNE-EN ISO 9606-1. Use continuous welds on all exposed connections. Grind visible welds smooth in the finished installation.

3.6 DISSIMILAR METALS

Where dissimilar metals are in contact, protect surfaces with a coating in accordance with MPI 79 to prevent galvanic or corrosive action. Where aluminum is in contact with concrete, plaster, mortar, masonry, wood, or absorptive materials subject to wetting, protect in accordance with UNE 104231, asphalt-base emulsion. Clean surfaces with metal shavings from installation at the end of each work day.

3.7 PREPARATION

3.7.1 Material Coatings and Surfaces

Remove rust preventive coating just prior to field erection, using a remover approved by the metal manufacturer. Surfaces, when assembled, must be free of rust, grease, dirt and other foreign matter.

3.7.2 Environmental Conditions

Do not clean or paint surfaces when damp or exposed to foggy or rainy weather, when metallic surface temperature is less than minus 15 degrees C above the dew point of the surrounding air, or when surface temperature is below 7 degrees C or over 35 degrees C, unless approved by the Contracting Officer. Metal surfaces to be painted must be dry for a minimum of 48 hours prior to the application of primer or paint.

3.8 INSTALLATION OF BOLLARDS

Set bollards/pipe guards vertically in concrete piers. Fill hollow cores with concrete having a compressive strength of 30 MPa.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 09 - FINISHES

SECTION 09 97 13.27

HIGH PERFORMANCE COATING FOR STEEL STRUCTURES

02/21, CHG 1: 08/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Contract Errors, Omissions, and Other Discrepancies
 - 1.4.2 Corrective Action (CA)
 - 1.4.2.1 Corrective Action Procedures
 - 1.4.2.2 Implement Corrective Action
 - 1.4.3 Coating Work Plan
 - 1.4.4 Design Data
 - 1.4.4.1 Containment System
 - 1.4.5 Test Reports
 - 1.4.5.1 Joint Sealant Qualification Test Reports
 - 1.4.5.2 Coatings Qualification Test Reports
 - 1.4.5.3 Metallic Abrasive Qualification Test Reports
 - 1.4.5.4 Recycled Metallic Abrasive Field Test Reports (Daily and Weekly)
 - 1.4.6 Qualifications
 - 1.4.6.1 Qualifications of Certified Industrial Hygienist (CIH)
 - 1.4.6.2 Qualifications of Certified Protective Coatings Specialist (PCS)
 - 1.4.6.3 Qualifications of Coating Inspection Company
 - 1.4.6.4 Qualifications of QC Specialist Coating Inspector
 - 1.4.6.5 Qualifications Of Individuals Performing Abrasive Blasting
 - 1.4.6.6 Qualifications of Testing Laboratory for Coatings
 - 1.4.6.7 Qualifications of Testing Laboratory for Abrasive
 - 1.4.6.8 Qualifications of Coating Contractors
 - 1.4.6.9 Joint Sealant Materials
 - 1.4.6.10 Coating Materials
 - 1.4.6.11 Coating System Component Compatibility
 - 1.4.6.12 Non-metallic Abrasive
 - 1.4.6.13 Metallic Abrasive
 - 1.4.7 Protective Coating Specialist (PCS)
 - 1.4.8 Pre-Application Meeting
- 1.5 PRODUCT DATA
 - 1.5.1 Coating System Instructions
- 1.6 DELIVERY AND STORAGE
- 1.7 COATING HAZARDS
- 1.8 JOB SITE REFERENCES

PART 2 PRODUCTS

- 2.1 COATING SYSTEM
 - 2.1.1 Zinc-Rich Epoxy Primer Coat

- 2.1.2 Epoxy Intermediate Coat
- 2.1.3 Polyurethane Topcoat
- 2.2 COLOR IDENTIFICATION OF FUEL HANDLING AND STORAGE FACILITIES
- 2.3 COATING SAMPLE COLLECTION AND SHIPPING KIT
- 2.4 ABRASIVE SAMPLE COLLECTION AND SHIPPING KIT
- 2.5 TEST KITS
 - 2.5.1 Test Kit for Measuring Chloride, Sulfate and Nitrate Ions on Steel and Coated Surfaces
 - 2.5.2 Test Kit for Identifying Amine Blush on Epoxy Surfaces
- 2.6 ABRASIVE
 - 2.6.1 Non-metallic Abrasive
 - 2.6.2 Metallic Abrasive
 - 2.6.2.1 New and Remanufactured Steel Grit
 - 2.6.2.2 Recycled Steel Grit
- 2.7 WHITE ALUMINUM OXIDE NON-SKID GRIT

PART 3 EXECUTION

- 3.1 COATING AND ABRASIVE SAMPLE COLLECTION AND TESTING
 - 3.1.1 Coating Sample Collection
 - 3.1.2 Abrasive Sample Collection
 - 3.1.3 Coating Sample Test Reports
 - 3.1.4 Abrasive Sample Test Reports
- 3.2 SURFACES TO BE COATED
- 3.3 LIGHTING
- 3.4 ENVIRONMENTAL CONDITIONS
 - 3.4.1 Containment
- 3.5 SURFACE PREPARATION
 - 3.5.1 Abrasive Blasting Equipment
 - 3.5.2 Operational Evaluation of Abrasive
 - 3.5.3 Surface Standard
 - 3.5.4 Pre-Preparation Testing for Surface Contamination
 - 3.5.4.1 Pre-Preparation Testing for Oil and Grease Contamination
 - 3.5.4.2 Pre-Preparation Testing for Soluble Salts Contamination
 - 3.5.5 Abrasive Blasting
 - 3.5.6 Disposal of Used Abrasive
 - 3.5.7 Pre-Application Testing For Surface Contamination
 - 3.5.7.1 Pre-Application Testing for Oil and Grease Contamination
 - 3.5.7.2 Pre-Application Testing for Soluble Salts Contamination
 - 3.5.7.3 Pre-Application Testing for Surface Cleanliness
- 3.6 MIXING AND APPLICATION OF SEALANT AND COATING SYSTEM
 - 3.6.1 Preparation of Sealant and Coating Materials for Application
 - 3.6.1.1 Mixing Topcoat Material
 - 3.6.1.2 Pot Life
 - 3.6.1.3 Application Conditions and Recoat Windows
 - 3.6.2 Amine Blush Testing of Epoxy Coat Prior to Overcoating
 - 3.6.3 Application of Coating System and Joint Sealant
 - 3.6.3.1 Application of Primer
 - 3.6.3.2 Application of Intermediate Coat
 - 3.6.3.3 Non-skid for Stairs and Top
 - 3.6.3.4 Application of Topcoat
 - 3.6.3.5 Application of Joint Sealant
 - 3.6.3.6 Procedure for Holiday and Spot Repairs of Newly Applied Coating
 - 3.6.3.7 Structure Occupancy After Coating Application
- 3.7 FIELD QUALITY CONTROL
 - 3.7.1 Coating Inspector
 - 3.7.2 Field Inspection
 - 3.7.2.1 Inspection Requirements

- 3.7.2.2 Inspection Report Forms
- 3.7.2.3 Daily Inspection Reports
- 3.7.2.4 Inspection Logbook
- 3.7.2.5 Inspection Equipment
- 3.8 FINAL CLEANUP

-- End of Section Table of Contents --

SECTION 09 97 13.27

HIGH PERFORMANCE COATING FOR STEEL STRUCTURES
02/21, CHG 1: 08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM C920	(2018) Standard Specification for Elastomeric Joint Sealants
ASTM D1200	(2010; R 2014) Viscosity by Ford Viscosity Cup
ASTM D1640/D1640M	(2014) Standard Test Methods for Drying, Curing, or Film Formation of Organic Coatings
ASTM D3276	(2015; E 2016) Standard Guide for Painting Inspectors (Metal Substrates)
ASTM D3925	(2002; R 2015) Sampling Liquid Paints and Related Pigmented Coatings
ASTM D4285	(1983; R 2018) Indicating Oil or Water in Compressed Air
ASTM D7127	(2017) Standard Test Method for Measurement of Surface Roughness of Abrasive Blast Cleaned Metal Surfaces using a Portable Stylus Instrument
ASTM E11	(2022) Standard Specification for Woven Wire Test Sieve Cloth and Test Sieves

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 9001	(2015) Quality Management Systems- Requirements
----------	---

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC 7/NACE No.4	(2007) Brush-Off Blast Cleaning
SSPC AB 2	(2015; E 2016) Cleanliness of Recycled Ferrous Metallic Abrasive
SSPC AB 3	(2003; E 2004) Ferrous Metallic Abrasive
SSPC Guide 6	(2015) Guide for Containing Surface

Preparation Debris Generated During Paint
Removal Operations

SSPC Guide 12	(1998; E 2004) Guide for Illumination of Industrial Painting Projects
SSPC PA 1	(2016) Shop, Field, and Maintenance Coating of Metals
SSPC PA 2	(2015; E 2018) Procedure for Determining Conformance to Dry Coating Thickness Requirements
SSPC QP 1	(2019) Standard Procedure for Evaluating the Qualifications of Industrial/Marine Painting Contractors (Field Application to Complex Industrial Steel Structures and Other Metal Components)
SSPC QP 5	(2012) Standard Procedure for Evaluating the Qualifications of Coating and Lining Inspection Companies
SSPC QS 1	(2015) Standard Procedure for Evaluating a Contractor's Advanced Quality Management System
SSPC SP 1	(2015) Solvent Cleaning
SSPC SP 10/NACE No. 2	(2015) Near-White Blast Cleaning
SSPC SP COM	(2016; E 2017) Surface Preparation Commentary for Steel and Concrete Substrates
SSPC VIS 1	(2002; E 2004) Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE AMS-STD-595A	(2017) Colors used in Government Procurement
------------------	--

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-A-22262	(1993; Rev B; Am 1 1994; Am 2 1996; Notice 1 2021) Abrasive Blasting Media Ship Hull Blast Cleaning
MIL-DTL-24441	(2009; Rev D; Notice 1 2021) Paint, Epoxy-Polyamide, General Specification for
MIL-DTL-24441/19	(2009; Rev C) Paint, Epoxy-Polyamide, Zinc Primer, Formula 159, Type III
MIL-DTL-24441/31	(2009; Rev B; Notice 1 2021) Paint, Epoxy-Polyamide, White, Formula 152, Type IV

MIL-PRF-85285	(2022; Rev F) Topcoat, Aircraft and Support Equipment
MIL-STD-161	(2005; Rev G; Notice 1 2010) Identification Methods for Bulk Petroleum Products Including Hydrocarbon Missile Fuels

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.134	Respiratory Protection
29 CFR 1910.1000	Air Contaminants
29 CFR 1926.59	Hazard Communication

1.2 DEFINITIONS

Definitions are provided throughout this Section, generally in the paragraph where used, and denoted by capital letters.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-05, Design Data

Containment System

SD-06 Test Reports

Joint Sealant Qualification Test Reports
Coatings Qualification Test Reports
Metallic Abrasive Qualification Test Reports
Coating Sample Test Reports
Abrasive Sample Test Reports
Inspection Report Forms
Daily Inspection Reports
Recycled Metallic Abrasive Field Test Reports (Daily and Weekly)

SD-07 Certificates

Contract Errors, Omissions, and Other Discrepancies
Corrective Action Procedures
Coating Work Plan
Qualifications of Certified Industrial Hygienist (CIH)
Qualifications Of Individuals Performing Abrasive Blasting
Qualifications of Certified Protective Coatings Specialist (PCS)
Qualifications of Coating Inspection Company
Qualifications of QC Specialist Coating Inspector
Qualifications of Testing Laboratory for Coatings
Qualifications of Testing Laboratory for Abrasive
Qualifications of Coating Contractors
Joint Sealant Materials
Coating Materials

Coating System Component Compatibility
Non-metallic Abrasive
Metallic Abrasive

SD-08 Manufacturer's Instructions

Coating System Instructions

SD-11 Closeout Submittals

Disposal of Used Abrasive
Inspection Logbook; G

1.4 QUALITY ASSURANCE

1.4.1 Contract Errors, Omissions, and Other Discrepancies

Submit all errors, omissions, and other discrepancies in contract documents the Contracting Officer within 30 days of contract award for all work covered in this Section, other than the work that will not be uncovered until a later date. All such discrepancies shall be addressed and resolved, and the Coating Work Plan modified, prior to beginning the Initial and Follow-Up phases of work. Discrepancies that become apparent only after work is uncovered shall be identified at the earliest discoverable time and submitted for resolution. Schedule time (Float) should be built into the project schedule at those points where old work is to be uncovered or where access is not available during the first 30 days after award, to allow for resolution of contract discrepancies.

1.4.2 Corrective Action (CA)

CA shall be included in the Quality Control Plan.

1.4.2.1 Corrective Action Procedures

Develop procedures for determining the root cause of each non-compliance, developing a plan to eliminate the root cause so that the non-compliance does not recur, and following up to ensure that the root cause was eliminated. Develop Corrective Action Request (CAR) forms for initiating CA, and for tracking and documenting each step.

1.4.2.2 Implement Corrective Action

The Contractor shall take action to identify and eliminate the root cause of each non-compliance so as to prevent recurrence. These procedures shall apply to non-compliance in the work, and to non-compliance in the QC System. Corrective actions shall be appropriate to the effects of the non-compliance encountered. Each CAR shall be serialized, tracked in a Log to completion and acceptance by the Contracting Officer, and retained in project records. The Corrective Action Log, showing status of each CAR, shall be submitted to the Contracting Officer monthly. A CAR may be initiated by either the Contractor or the Contracting Officer. The Contracting Officer must approve each CAR at the root cause identification stage, the plan for elimination stage, and the close out stage after verification that the root cause has been eliminated.

1.4.3 Coating Work Plan

This work plan shall be considered as part of the Quality Control Plan.

Provide procedures for reviewing contract documents immediately after award to identify errors, omissions, and discrepancies so that any such issues can be resolved prior to project planning and development of detailed procedures.

Provide procedures for all phases of coating operations, including planned work, rework, repair, inspection, and documentation. Address mobilization and setup, surface preparation, coating application, coating initial cure, tracking and correction of noncompliant work, and demobilization. Coordinate work processes with health and safety plans and confined space entry plans. For each process, provide procedures that include appropriate work instructions, material and equipment requirements, personnel qualifications, controls, and process verification procedures. Provide procedures for inspecting work to verify and document compliance with contract requirements, including inspection forms and checklists, and acceptance and rejection criteria.

Provide procedures for determining the existing surface profile under paint, and procedures for ensuring that the profile is not increased beyond the maximum profile specified herein.

Provide procedures for correcting noncompliant work. Detailed procedures are required in advance to avoid delays in meeting overcoat windows as well as to avoid delays in production. Provide procedures for repairing defects in the coating film, such as runs, drips, sags, holidays, overspray, as well as how to handle correct coating thickness noncompliance, any other areas of repair or rework that might be adversely affected by delays in preparing and approving new procedures.

If a procedure is based on a proposed or approved request for deviation, the deviation shall be referenced. Changes to procedures shall be noted by submittal number and date approved, clearly delineating old requirements and new requirements, so that the records provide a continuous log of requirements and procedures.

1.4.4 Design Data

1.4.4.1 Containment System

Submit complete design drawings and calculations for the scaffolding and containment system, including an analysis of the loads which will be added to the structure by the containment system and waste materials. A registered engineer shall approve calculations and scaffold system design.

1.4.5 Test Reports

1.4.5.1 Joint Sealant Qualification Test Reports

Submit test results from independent laboratory of representative samples of joint sealant material. Samples must have been tested within the last three years. Submit results as required in paragraph QUALITY ASSURANCE PROVISIONS of ASTM C920. Note that testing in accordance with QUALITY ASSURANCE PROVISIONS is a pre-qualification requirement.

1.4.5.2 Coatings Qualification Test Reports

Submit test results from independent laboratory of representative samples of each coating material. U.S. Department of Defense laboratories are considered to be independent laboratories for purposes of compliance with

"QUALIFICATION INSPECTION" requirements herein. Samples must have been tested within the last three years. Submit results for epoxy materials as required in paragraph QUALIFICATION INSPECTION of MIL-DTL-24441, and as revised by paragraph COATING SYSTEM herein. Submit results for polyurethane materials as required in paragraph QUALIFICATION INSPECTION of MIL-PRF-85285, and as revised by paragraph COATING SYSTEM herein. Note that requirement for QUALIFICATION INSPECTION is a pre-qualification requirement, and involves the same testing required for listing in the Qualified Products List of the respective material. See appropriate Military Specification for specific test requirements.

1.4.5.3 Metallic Abrasive Qualification Test Reports

Submit results for abrasive as required in paragraph 4 REQUIREMENTS of SSPC AB 3. Submit test results from independent laboratory of representative samples of each abrasive to be used on the jobsite. Samples must have been tested within the last three years. Note that this testing is for the purpose of prequalifying the abrasive.

1.4.5.4 Recycled Metallic Abrasive Field Test Reports (Daily and Weekly)

Submit test results from independent laboratory of daily and weekly Quality Control testing required by SSPC AB 2, as modified in paragraph ABRASIVE.

1.4.6 Qualifications

1.4.6.1 Qualifications of Certified Industrial Hygienist (CIH)

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party CIH. Submit documentation that hygienist is certified by the American Board of Industrial Hygiene in comprehensive practice, including certification number and date of certification/recertification. Provide evidence of experience with hazards involved in industrial coating application work.

1.4.6.2 Qualifications of Certified Protective Coatings Specialist (PCS)

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party PCS. Submit documentation that specialist is certified by SSPC: The Society for Protective Coatings (SSPC) as a PCS, including certification number and date of certification/recertification. If the PCS is employed by the same coating inspection company to which the coating inspector is employed, this does not violate the independent third-party requirements. The PCS shall remain certified during the entire project, and the Contracting Officer shall be notified of any change in certification status within 10 days of the change. The PCS shall not be the designated coating inspector.

1.4.6.3 Qualifications of Coating Inspection Company

Submit documentation that the coating inspection company that will be performing all coating inspection functions is certified by SSPC to the requirements of SSPC QP 5 prior to contract award, and shall remain certified while accomplishing any coating inspection functions. The coating inspection company must remain so certified for the duration of the project. If a coating inspection company's certification expires, the firm will not be allowed to perform any inspection functions, and all surface preparation and coating application work must stop, until the

certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in coating inspection company certification status.

1.4.6.4 Qualifications of QC Specialist Coating Inspector

Submit documentation that each coating inspector is employed, and qualified to SSPC QP 5, Level III, by the selected coating inspection company. Each inspector shall remain employed by the coating inspection company while performing any coating inspection functions.

1.4.6.5 Qualifications Of Individuals Performing Abrasive Blasting

Submit name, address, and telephone number of each person that will be performing abrasive blasting. Submit documentation that each blaster is qualified by SSPC to the SSPC C-7 Dry Abrasive Blaster Qualification Program. Each blaster shall remain qualified during the entire period of abrasive blasting, and the Contracting Officer shall be notified of any change in qualification status.

1.4.6.6 Qualifications of Testing Laboratory for Coatings

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party laboratory selected to perform testing of coating samples for compliance with specification requirements. Submit documentation that laboratory is regularly engaged in testing of paint samples for conformance with specifications, and that persons performing analyses are qualified.

1.4.6.7 Qualifications of Testing Laboratory for Abrasive

Submit name, address, telephone number, FAX number, and e-mail address of the independent third party laboratory selected to perform testing of abrasive for compliance with specification requirements. Submit documentation that laboratory has experience in testing samples of abrasive for conformance with specifications, and that persons performing analyses are qualified.

1.4.6.8 Qualifications of Coating Contractors

All Contractors and Subcontractors that perform surface preparation or coating application shall be certified to either ISO 9001 or SSPC QP 1 and SSPC QS 1 prior to contract award, and shall remain certified while accomplishing any surface preparation or coating application. The painting Contractors and painting Subcontractors must remain so certified for the duration of the project. If a Contractor's or Subcontractor's certification expires, the firm will not be allowed to perform any work until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in Contractor certification status.

1.4.6.9 Joint Sealant Materials

Provide manufacturer's certification of conformance to contract requirements.

1.4.6.10 Coating Materials

Provide manufacturer's certification of conformance to contract requirements.

1.4.6.11 Coating System Component Compatibility

Provide certification from each manufacturer of components of the coating system, epoxy primer, epoxy intermediate, and polyurethane topcoat, that the supplied coating material is suitable for use in the specified coating system. Each manufacturer shall identify the specific products, including manufacturer's name, which their product may be used with. The certification shall provide the name of the manufacturer that will provide technical support for the entire system. When all coating materials are manufactured by one manufacturer, this certification is not required.

1.4.6.12 Non-metallic Abrasive

Provide manufacturer's certification that the materials are currently approved by the Naval Sea Systems Command and listed on the Qualified Products Lists (QPL) for the specified materials.

1.4.6.13 Metallic Abrasive

Provide manufacturer's certification of conformance to contract requirements and provide copies of test results.

1.4.7 Protective Coating Specialist (PCS)

The PCS shall be considered a QC Specialist and shall report to the QC Manager, as specified in Section 01 45 00.00 20 QUALITY CONTROL. The PCS shall approve all submittals prior to submission to the QC Manager for approval or submission to the government for approval.

1.4.8 Pre-Application Meeting

After approval of submittals but prior to the initiation of coating work, Contractor representatives, including at a minimum, project superintendent and QC manager, paint foreman, coating inspector, and PCS shall have a pre-application coating preparatory meeting. This meeting shall be in addition to the pre-construction conference. Specific items addressed shall include: corrective action requirements and procedures, coating work plan, safety plan, coordination with other Sections, inspection standards, inspection requirements and tools, test procedures, environmental control system, safety plan, and test logs. Notify Contracting Officer at least ten days prior to meeting.

1.5 PRODUCT DATA

1.5.1 Coating System Instructions

Submit manufacturer's printed instructions including detailed mixing and application procedures, number and types of coats required, minimum and maximum application temperatures, and curing procedures. Include Safety Data Sheets (SDS) for materials to be used at the job site in accordance with 29 CFR 1926.59.

1.6 DELIVERY AND STORAGE

Ship, store, and handle materials in accordance with SSPC PA 1, and as modified in this Section. Maintain temperature in storage spaces between 5 and 29 degrees C, and air temperature more than 3 degrees C above the dew-point at all times. Inspect materials for damage prior to use and return non-compliant materials to manufacturer. Remove materials with expired shelf life from government property immediately and notify the Contracting Officer.

If materials are approaching shelf life expiration and an extension is desired, samples may be sent to the manufacturer, along with complete records of storage conditions, with a request for shelf life extension. If the manufacturer finds the samples and storage data suitable for shelf life extension, the manufacturer may issue an extension, referencing the product evaluation and the review of storage records. Products may not be extended longer than allowed in the product specification.

1.7 COATING HAZARDS

Ensure that employees are trained in all aspects of the safety plan. Specified coatings may have potential health hazards if ingested or improperly handled. The coating manufacturer's written safety precautions shall be followed throughout mixing, application, and curing of the coatings. During all cleaning, cleanup, surface preparation, and paint application phases, ensure that employees are protected from toxic and hazardous chemical agents which exceed concentrations in 29 CFR 1910.1000. Comply with respiratory protection requirements in 29 CFR 1910.134. The CIH shall approve work procedures and personal protective equipment.

1.8 JOB SITE REFERENCES

Make available to the Contracting Officer at least one copy each of ASTM D3276, ASTM D3925, ASTM D4285, ASTM D7127, SSPC SP COM, SSPC SP 1, SSPC 7/NACE No.4, SSPC SP 10/NACE No. 2, SSPC PA 1, SSPC PA 2, SSPC Guide 6, SSPC VIS 1, SSPC QP 1, SSPC QS 1, and an SSPC Certified Contractor Evaluation Form at the job site.

PART 2 PRODUCTS

2.1 COATING SYSTEM

Alternate systems or products will not be considered. Provide a complete system (primer, top coat) material from one supplier.

The Military specification epoxy and polyurethane products specified in this Section do not require approval for listing on the QPL prior to contract award, as indicated in paragraph 3.2 of MIL-DTL-24441 and paragraph 3.1 of MIL-PRF-85285. Testing of products by an independent laboratory to the QUALIFICATION INSPECTION requirements of MIL-DTL-24441 and MIL-PRF-85285 prior to contract award is required. See specific submittal requirements in paragraph QUALITY ASSURANCE.

2.1.1 Zinc-Rich Epoxy Primer Coat

Epoxy polyamide, MIL-DTL-24441/19 (Formula 159, Type III).

2.1.2 Epoxy Intermediate Coat

Epoxy polyamide, MIL-DTL-24441/31 (Formula 152, Type IV, White (Tinted)). Tint to approximately SAE AMS-STD-595A color number 27778 parchment using pigment dispersions prepared for epoxy paint tinting. Manufacturer shall tint material and appropriately label. All other requirements of this Military Specification apply.

2.1.3 Polyurethane Topcoat

Polyurethane coating topcoat of MIL-PRF-85285, Type II, White SAE AMS-STD-595A color number 17925.

Modify paragraph 3.6.4 of MIL-PRF-85285, Viscosity and Pot Life, as follows:

The viscosity of the admixed coating, when tested in accordance with ASTM D1200 through a No. 4 Ford cup, shall be as follows:

Time from mix (minimum)	Maximum time through a No. 4 Ford cup
Initially	30 seconds
2 hours	60 seconds
4 hours	No gel

Modify paragraph 3.7.1 of MIL-PRF-85285, Drying Time, as follows:

When applied by spray techniques and when tested in accordance with ASTM D1640/D1640M, the coating shall be set-to-touch within four hours and dry-hard within eight hours (see 4.6 and table I).

2.2 COLOR IDENTIFICATION OF FUEL HANDLING AND STORAGE FACILITIES

Piping, conduit, and tank identification shall be in accordance with MIL-STD-161. Mark direction of fluids in accordance with MIL-STD-161. The NATO symbol for JP-8 is F-34.

2.3 COATING SAMPLE COLLECTION AND SHIPPING KIT

Provide a kit that contains one liter can for the base of each coating material, an appropriately sized can for each activator, dipping cups for each component to be sampled, a shipping box sized for the samples to to be shipped, and packing material. Mark cans for the appropriate component. Provide shipping documents, including either pre-paid shipping or a shipper number that can be used by the QC Manager to arrange pickup, addressed to the approved coating testing laboratory.

2.4 ABRASIVE SAMPLE COLLECTION AND SHIPPING KIT

Provide a kit that contains one suitable plastic bag or container for each sample to be collected. Mark containers for the appropriate component. Provide shipping documents, including either pre-paid shipping or a shipper number that can be used by the QC Manager to arrange pickup, addressed to the approved coating testing laboratory.

2.5 TEST KITS

2.5.1 Test Kit for Measuring Chloride, Sulfate and Nitrate Ions on Steel and Coated Surfaces

Provide test kits that meet all of the following requirements:

- a. Contains all materials, supplies, tools, and instructions for field testing and on-site quantitative evaluation of chloride, sulfate, and nitrate ions;
- b. Extract solution is acidic, factory pre-measured, pre-packaged, and of uniform concentration;
- c. Components and solutions are mercury free and environmentally friendly;
- d. Contains new materials and solutions for each test extraction;
- e. Contains an extraction test container (vessel, sleeve, cell) that creates a sealed, encapsulated environment during salt ion extraction;
- f. Contains a test extract container suitable for testing the following steel surfaces: horizontal (up/down configuration), vertical, flat, curved, smooth, pitted, and rough;
- g. All salt ion concentrations are directly measured in micrograms per square centimeter.

2.5.2 Test Kit for Identifying Amine Blush on Epoxy Surfaces

Provide test kits that meet all of the following requirements:

- a. Is a completely self-contained field test kit with all materials, supplies, tools, and instructions to perform tests and indicate the presence of unreacted amines;
- b. Uses an identifiable, consistent, uniform, pre-packaged, factory pre-measured indicating solution;
- c. Contains no mercury or lead and is environmentally friendly;
- d. Contains a solution of an unreacted amine for the purpose of "self checking" the indicator solution;

2.6 ABRASIVE

The referenced abrasive specifications have maximum limits for soluble salts contamination, however, this maximum level of contamination does not guarantee that contamination will not be transferred to the steel surface during abrasive blasting. Other factors such as on-site handling and recycling can allow contamination of abrasive. Contractors are cautioned to verify that the chosen abrasive, along with work and storage processes, allow the final surface cleanliness requirements to be achieved. Successful testing of chlorides in abrasive does not negate the final acceptance testing of steel surfaces.

2.6.1 Non-metallic Abrasive

Conform to MIL-A-22262, Type I (Inorganic materials) except that the gross

gamma radioactivity shall not exceed 5 picocuries per gram. Abrasive shall be approved by the Naval Sea Systems Command and listed on the appropriate Qualified Products List (QPL) for the specified materials. Use sampling procedures and testing frequencies as prescribed in MIL-A-22262. Use abrasive that is specifically selected and graded to provide a sharp, angular profile to the specified depth. Do not use ungraded abrasive. Make adjustments to processes or abrasive gradation to achieve specified surface profile. Recycled non-metallic abrasive shall meet all requirements of the specification each time that it is placed in the blast pot.

2.6.2 Metallic Abrasive

2.6.2.1 New and Remanufactured Steel Grit

Conform to the chemical and physical properties of SSPC AB 3 Class 1 (Steel) only. Class 2 (Iron) abrasive shall not be used.

To develop a suitable work mix from new steel abrasive, a minimum of 200 - 400 recycles is required, therefore, it is advantageous for a Contractor to use remanufactured steel grit or grit reclaimed from a previous project. Such grit shall be considered to conform if it can be traced to new grit conforming to SSPC AB 3 Class 1 and it meets all cleanliness requirements of SSPC AB 3 Class 1 when brought to the current jobsite. Submit one representative sample of this work mix to the laboratory for testing, along with samples of new material. Acceptance and use of this work mix shall not be used to justify any deviation from surface preparation requirements.

2.6.2.2 Recycled Steel Grit

Conform to the chemical and physical properties of SSPC AB 2

2.7 WHITE ALUMINUM OXIDE NON-SKID GRIT

Size #60, dust free (washed and dry), minimum 99 percent pure, having the following sieve analysis when tested in accordance with ASTM E11 using a 1000 gram sample:

Sieve #	Percent Retained
40	0
50	15-40
60	60-85

PART 3 EXECUTION

Perform all work, rework, and repair in accordance with approved procedures in the Coating Work Plan.

3.1 COATING AND ABRASIVE SAMPLE COLLECTION AND TESTING

Sample and test materials delivered to the jobsite. Notify Contracting Officer three days in advance of sampling. The QC Manager and either the PCS or coating inspector shall witness all sampling.

3.1.1 Coating Sample Collection

Provide a sample collection kit as required in paragraph COATING SAMPLE COLLECTION AND SHIPPING KIT. From each lot, obtain a one liter sample of each base material, and proportional samples of each activator based on mix ratio, by random selection from sealed containers in accordance with ASTM D3925. Prior to sampling, mix contents of each sealed container to ensure uniformity. As an alternative to collecting small samples from kits, entire kits may be randomly selected and shipped to laboratory, observing all requirements for witnessing and traceability. For purposes of quality conformance inspection, a lot is defined as that quantity of materials from a single, uniform batch produced and offered for delivery at one time. A batch is defined as that quantity of material processed by the manufacturer at one time and identified by number on the label. Identify samples by designated name, specification number, batch number, project contract number, sample date, intended use, and quantity involved. The QC manager will take possession of the packaged samples, contact the shipping company to arrange for pickup, and relinquish the samples only to the shipping representative for shipment to the approved laboratory for testing as required by the paragraph COATING SAMPLE TEST REPORTS.

3.1.2 Abrasive Sample Collection

Provide a sample collection kit as required in paragraph ABRASIVE SAMPLE COLLECTION AND SHIPPING KIT. For purposes of quality conformance inspection, a lot shall consist of all abrasive materials of the same type from a single, uniform batch produced and offered for delivery at one time. Obtain samples of each abrasive lot using the sampling techniques and schedule of MIL-A-22262. The addition of any substance to a batch shall constitute a new lot. Identify samples by designated name, specification number, lot number, project contract number, sample date, intended use, and quantity involved. The QC manager will take possession of the packaged samples, contact the shipping company to arrange for pickup, and relinquish the samples only to the shipping representative for shipment to the approved laboratory for testing as required by the paragraph ABRASIVE SAMPLE TEST REPORTS.

3.1.3 Coating Sample Test Reports

Submit test results for each lot of coating material delivered to the jobsite. Test samples of primer, intermediate, and topcoat materials for compliance with requirements of Table I. Reject entire lot represented by samples that fail one or more tests, select new lots, and test samples.

3.1.4 Abrasive Sample Test Reports

Submit test results for each lot of abrasive delivered to the jobsite. Test samples of metallic abrasive to the requirements of paragraph REQUIREMENTS of SSPC AB 3, except paragraph 4.1.5 DURABILITY. Test samples of non-metallic abrasive as required in paragraph QUALITY CONFORMANCE INSPECTION of MIL-A-22262. Reject entire lot represented by samples that fail one or more tests, select new lots, and test samples.

3.2 SURFACES TO BE COATED

Coat exterior surfaces of fuel piping.

3.3 LIGHTING

Provide lighting for all work areas as prescribed in SSPC Guide 12.

3.4 ENVIRONMENTAL CONDITIONS

3.4.1 Containment

Design and provide a containment system for the capture, containment, collection, storage and disposal of the waste materials generated by the work under this Section, to meet the requirements of SSPC Guide 6, Class 3. Vapor concentrations shall be kept at or below 10 percent of Lower Explosive Limit (LEL) at all times. Containment may be designed as fixed containment for complete structure or portable containment for sections of structure, however, containment shall remain in any one place from beginning of abrasive blasting through initial cure of coating. Waste materials covered by this paragraph shall not include any material or residue from removal of coatings containing lead, chromium, cadmium, PCB, or any other hazardous material.

It is the Contractors responsibility to insure the feasibility and workability of the containment system. The Contractor shall perform his operations and work schedule in a manner as to minimize leakage of the containment system. The containment system shall be properly maintained and shall not deviate from the approved drawings. If the containment system fails to function satisfactorily, the Contractor shall suspend all operations, except those required to minimize adverse impact on the environment or government property. Operations shall not resume until modifications have been made to correct the cause of the failure.

3.5 SURFACE PREPARATION

3.5.1 Abrasive Blasting Equipment

Use abrasive blasting equipment of conventional air, force-feed, or pressure type. Maintain a minimum pressure of 650 kPa at nozzle. Confirm that air supply for abrasive blasting is free of oil and moisture when tested in accordance with ASTM D4285. Test air quality at each startup, but in no case less often than every five operating hours.

3.5.2 Operational Evaluation of Abrasive

Test abrasive for salt contamination and oil contamination as required by the appropriate abrasive specification daily at startup and every five operating hours thereafter.

3.5.3 Surface Standard

Inspect surfaces to be coated, and select plate with similar properties and surface characteristics for use as a surface standard. Blast clean one or more 300 mm square steel panels as specified in paragraph SURFACE PREPARATION. Record blast nozzle type and size, air pressure at nozzle and compressor, distance of nozzle from panel, and angle of blast to establish procedures for blast cleaning. Measure surface profile in accordance with ASTM D7127. When the surface standard complies with all specified requirements, seal with a clearcoat protectant. Use the surface standard for comparison to abrasive blasted surfaces throughout the course of work.

3.5.4 Pre-Preparation Testing for Surface Contamination

Perform testing, abrasive blasting, and testing in the prescribed order.

3.5.4.1 Pre-Preparation Testing for Oil and Grease Contamination

Inspect all surfaces for oil and/or grease contamination using two or more of the following inspection techniques: 1) Visual inspection, 2) WATER BREAK TEST, 3) CLOTH RUB TEST. Reject oil and/or grease contaminated surfaces, clean using a water based pH neutral degreaser in accordance with SSPC SP 1, and recheck for contamination until surfaces are free of oil and grease.

WATER BREAK TEST - Spray atomized mist of distilled water onto surface, and observe for water beading. If water "wets" surface rather than beading up, surface can be considered free of oil or grease contamination. Beading of water (water forms droplets) is evidence of oil or grease contamination.

CLOTH RUB TEST - Rub a clean, white, lint free, cotton cloth onto surface and observe for discoloration. To confirm oil or grease contamination in lightly stained areas, a non-staining solvent may be used to aid in oil or grease extraction. Any visible discoloration is evidence of oil or grease contamination.

3.5.4.2 Pre-Preparation Testing for Soluble Salts Contamination

Test surfaces for soluble salts, and wash as required, prior to abrasive blasting. Soluble salt testing is also required in paragraph PRE-APPLICATION TESTING FOR SOLUBLE SALTS CONTAMINATION as a final acceptance test of prepared surfaces after abrasive blasting, and successful completion of this phase does not negate that requirement. This phase is recommended since pre-preparation testing and washing are generally more advantageous than attempting to remove soluble salt contamination after abrasive blasting. Effective removal of soluble salts will require removal of any barrier to the steel surface, including rust. This procedure may necessitate combinations of wet abrasive blasting, high pressure water rinsing, and cleaning using a solution of water washing and soluble salts remover. The soluble salts remover shall be acidic, biodegradable, nontoxic, noncorrosive, and after application, will not interfere with primer adhesion. Delays between testing and preparation, or testing and coating application, may allow for the formation of new contamination. Use potable water, or potable water modified with soluble salt remover, for all washing or wet abrasive blasting. Test methods and equipment used in this phase are selected at the Contractor's discretion.

3.5.5 Abrasive Blasting

Abrasive blast steel surfaces to near-white metal in accordance with SSPC SP 10/NACE No. 2. Prepared surfaces shall conform to SSPC VIS 1 and shall match the prepared test-panels. Provide a 50 to 75 micron surface profile. Reject profile greater than 75 microns, discontinue abrasive blasting, and modify processes and materials to provide the specified profile. Measure surface profile in accordance with ASTM D7127, using Rmax as the measure of profile height. Record all measurements required in this standard. Measure profile at rate of three test areas for the first 100 square meters plus one test area for each additional 100 square meters or part thereof. When surfaces are reblasted for any reason, retest profile as specified. Following abrasive blasting, remove dust and

debris by vacuum cleaning. Do not attempt to wipe surface clean.

3.5.6 Disposal of Used Abrasive

Dispose of used abrasive off Government property in accordance with Federal, State, and Local mandated regulations.

3.5.7 Pre-Application Testing For Surface Contamination

3.5.7.1 Pre-Application Testing for Oil and Grease Contamination

Ensure surfaces are free of contamination as described in paragraph PRE-PREPARATION TESTING FOR OIL AND GREASE CONTAMINATION, except that only questionable areas need be checked for beading of water misted onto surface.

3.5.7.2 Pre-Application Testing for Soluble Salts Contamination

Test surfaces for chloride contamination using the Test Kit described in TEST KIT FOR MEASURING CHLORIDE, SULFATE AND NITRATE IONS ON STEEL AND COATED SURFACES. Test all surfaces at rate of three tests for the first 100 square meters plus one test for each additional 200 square meters or part thereof. One or more readings greater than 3 micrograms per square centimeter of chlorides or 10 micrograms per square centimeter of sulfates or 5 micrograms per square centimeter of nitrates is evidence of soluble salt contamination. Reject contaminated surfaces, wash as discussed in paragraph PRE-PREPARATION TESTING FOR SOLUBLE SALTS CONTAMINATION, allow to dry, and re-test until all required tests show allowable results. Reblast tested and cleaned areas as required. Label all test tubes and retain for test verification.

3.5.7.3 Pre-Application Testing for Surface Cleanliness

Apply coatings to dust free surfaces. To test surfaces, apply strip of clear adhesive tape to surface and rub onto surface with finger. When removed, the tape should show little or no dust, blast abrasive, or other contaminant. Reject contaminated surfaces and retest. Test surfaces at rate of three tests for the first 100 square meters plus one test for each additional 100 square meters or part thereof. Provide two additional tests for each failed test or questionable test. Attach test tapes to Daily Inspection Reports.

3.6 MIXING AND APPLICATION OF SEALANT AND COATING SYSTEM

3.6.1 Preparation of Sealant and Coating Materials for Application

Each of the sealant, primer, intermediate, and topcoat materials is a two-component material supplied in separate containers.

3.6.1.1 Mixing Topcoat Material

Do not mix partial kits, or alter mix ratios. Mix polyurethane coating materials in same temperature conditions specified in paragraph DELIVERY AND STORAGE. The polyurethane coating material is moisture sensitive and any introduction of moisture or water into the material during mixing or application will shorten usable pot life. Use a mixer that does not create a vortex. Do not add solvent without specific written recommendation from the manufacturer. No induction time is required, only thorough agitation of the mixed material.

3.6.1.2 Pot Life

Apply mixed products within stated pot life for each product. Stop applying when material becomes difficult to apply in a smooth, uniform wet film. Add all required solvent at time of mixing. Do not add solvent to extend pot life. Pot life is based on standard conditions at 21 degrees C and 50 percent relative humidity. For every 10 degrees C rise in temperature, pot life is reduced by approximately half, and for every 10 degrees C drop it is approximately doubled. Usable pot life depends on the temperature of the material at the time of mixing and the sustained temperature at the time of application. Other factors such as the shape of the container and volume of mixed material may also affect pot life. Precooling or exterior icing of components for at least 24 hours to a minimum of 10 degrees C in hot climates will extend pot life. High humidity at time of mixing and application shortens pot life of the Polyurethane topcoat material. Following are approximate pot life times:

Sealant	As specified by manufacturer
Epoxy primer and intermediate materials	4 hours
Polyurethane topcoat materials	2 hours.

3.6.1.3 Application Conditions and Recoat Windows

The application condition requirements for the coating system are very time and temperature sensitive, and are intended to avoid the delamination problems frequently found on industrial structures. Plan coating application to ensure that specified temperature, humidity, and condensation conditions are met. If conditions do not allow for orderly application of sealant, primer, stripe coat, intermediate coat and topcoat, use appropriate means of controlling air and surface temperatures, as required. Partial or total enclosures, insulation, heating or cooling, or other appropriate measures may be required to control conditions to allow for orderly application of all required coats.

Maintain air and steel surface temperature between 16 and 38 degrees C during application and the first four hours of cure for epoxy coats and the first eight hours of cure for polyurethane coats. Maintain steel surface temperature more than 3 degrees C above the dew-point of the ambient air for the same period.

Use Table entitled "RECOAT WINDOWS" to determine appropriate recoat windows for each coat after the initial coat. Apply each coat during appropriate RECOAT WINDOW of preceding coat. If a RECOAT WINDOW is missed, the minimum and maximum primer and intermediate coat thickness may be adjusted to accommodate a FILL COAT, however, requirements for total epoxy coating thickness and total coating thickness will not be modified. Missing more than one RECOAT WINDOW may require complete removal of coating if maximum total coating thickness requirements cannot be achieved.

If coating is not applied during RECOAT WINDOW, or if surface temperature exceeds 49 degrees C between applications, provide GLOSS REMOVAL, apply next coat within 24 hours. If next planned coat is topcoat, apply FILL COAT if required to fill sanding marks. Sanding marks from GLOSS REMOVAL of intermediate coat reflecting through topcoat will be considered as noncompliant. Apply FILL COAT within 24 hours of GLOSS REMOVAL, then apply topcoat within RECOAT WINDOW of FILL COAT.

RECOAT WINDOWS						
<u>EPOXY OVER EPOXY</u>						
Temperature degrees C	16-21	22-27	28-32	33-38	39-43	44-49
RECOAT WINDOW (Hrs.)	24-72	18-60	16-48	12-36	8-18	4-6
<u>POLYURETHANE OVER EPOXY</u>						
Temperature degrees C	16-21	22-27	28-32	33-38	39-43	44-49
RECOAT WINDOW (Hrs.)	24-96	24-72	16-48	12-36	10-24	8-16
<u>POLYURETHANE OVER POLYURETHANE</u>						
Temperature degrees C	16-21	22-27	28-32	33-38	39-43	44-49
RECOAT WINDOW (Hrs.)	8-48	6-48	4-36	3-24	2-12	1-2

The temperature ranges shown in the table above are for determining recoat windows. Choose recoat window based on the highest surface temperature that was sustained for one or more hours between coats. This applies to the entire time between coats. Measure and record air and surface temperatures on hourly basis to determine appropriate recoat windows. If surface temperature goes above 38 degrees C, measure and record temperatures every half hour.

FILL COAT - Where indicated, apply coat of intermediate coat epoxy, at 50 to 75 microns DFT, then apply next specified full coat within recoat window of FILL COAT. A FILL COAT may be used to adjust coating thickness to comply with requirements or to fill sanding marks in intermediate coat.

GLOSS REMOVAL - Where required, hand sand in a linear fashion to remove gloss using 120-200 grit wet/dry sandpaper, followed by solvent wiping with a clean rag soaked with denatured alcohol to remove all dust. GLOSS REMOVAL of primer coat is to scarify surface and shall consist of removal of approximately 25 microns of coating. If steel is exposed during GLOSS REMOVAL, repair in accordance with paragraph PROCEDURE FOR HOLIDAY AND SPOT REPAIRS OF NEWLY APPLIED COATING. GLOSS REMOVAL of intermediate coat may include removal of up to 75 microns of coating to avoid excess thickness, prior to application of FILL COAT.

3.6.2 Amine Blush Testing of Epoxy Coat Prior to Overcoating

Test epoxy surfaces prior to application of roof joint sealant, epoxy coat, or polyurethane topcoat for amine blush contamination using the Test Kit described in paragraph TEST KIT FOR IDENTIFYING AMINE BLUSH ON EPOXY SURFACES. Test all surfaces at rate of three tests for the first 100 square meters plus one test for each additional 200 square meters or part thereof. Remove any identified contamination using an approved procedure.

3.6.3 Application of Coating System and Joint Sealant

Apply coatings in accordance with SSPC PA 1 and as specified herein. Apply coatings to surfaces that meet all stated surface preparation requirements.

After application of primer coat and prior to application of each subsequent coat, perform testing prescribed in paragraph PRE-APPLICATION TESTING FOR SURFACE CONTAMINATION, as necessary, to ensure minimal intercoat contamination. This testing may be reduced to one half of the prescribed rate for bare steel if the testing indicates no contamination when sampling is evenly distributed over surfaces being tested. If contamination is found between coats, revert to the specified testing rate. Generally, oil and grease contamination and soluble salts contamination are not encountered if subsequent coats are applied within specified recoat windows and unusual atmospheric events do not occur. Such atmospheric events as a coastal storm blowing onshore can bring unusual chloride contamination. Concern for intercoat contamination should be continually prevalent, and spot testing should be accomplished to verify satisfactory conditions. Where visual examination or spot testing indicates contamination, perform sufficient testing to verify non-contamination, or to define extent of contamination for appropriate treatment.

Apply each coat in a consistent wet film, at 90 degrees to previous coat. Ensure that primer and intermediate coat "cold joints" are no less than 150 mm from welds. Apply stripe coat by brush. For convenience, stripe coat material may be delivered by spray if followed immediately with brush-out and approved procedures include appropriate controls on thickness. Apply all other coats by spray application. Use appropriate controls to prevent airborne coating fog from drifting beyond three meters from the structure perimeter. Cover or protect all surfaces that will not be coated. The cleanliness, temperature, recoat windows, and airborne paint containment requirements may necessitate the use of enclosures, portable shelters, or other appropriate controls.

Apply coatings at the following specified thickness:

Coat	Minimum DFT (Microns)	Maximum DFT (Microns)
Primer	75	125
Intermediate	75	125
Top	50	75
Total system	200	325

3.6.3.1 Application of Primer

Apply primer coat, maintaining paint supply container height within 1 meter of the paint nozzle for applying zinc primer. Maintain constant agitation of paint pot to ensure that zinc does not settle in container.

3.6.3.2 Application of Intermediate Coat

Apply intermediate coat within RECOAT WINDOW of primer coat.

3.6.3.3 Non-skid for Stairs and Top

Where non-skid is required, apply a second intermediate coat, and immediately follow with application of non-skid grit, broadcast at the rate of 2 pounds per 100 square feet, and backroll. Apply topcoat as specified.

3.6.3.4 Application of Topcoat

Make all required repairs to primer and intermediate coats as specified in paragraph entitled "Procedure for Holiday and Spot Repairs of Newly Applied Coating" prior to applying topcoat. Apply topcoat within RECOAT WINDOW of intermediate coat. The polyurethane topcoat may require multiple passes to achieve desired aesthetics and required thickness. Consult manufacturer for thinning and application procedures for anticipated temperature, humidity, and wind conditions. Touch-up blemishes and defects within recoat window of polyurethane topcoat. Retain sample of polyurethane topcoat, from the same batch used to coat structure, to make touch-ups that might be required later.

3.6.3.5 Application of Joint Sealant

Apply joint sealant to back-to-back steel joints that are less than 3/8 inches wide and are not seal welded. Apply sealant to top and bottom, or each side, of narrow joints. Apply sealant within 48 hours of application of the topcoat, and touch-up with topcoat after appropriate cure of the sealant.

3.6.3.6 Procedure for Holiday and Spot Repairs of Newly Applied Coating

Repair coating film defects at the earliest practicable time, preferably before application of the succeeding coat. Observe all requirements for soluble salts contamination, cleanliness between coats, and application conditions. Prepare defective area in accordance with SSPC SP 10/NACE No. 2, and feather coating as required to leave 100 mm of each succeeding coat feathered and abraded. Protect adjacent areas from damage and overspray. Remove dust and solvent wipe the prepared area plus

an additional 100 mm beyond the prepared area with clean denatured alcohol. Apply each coat within RECOAT WINDOW of preceeding coat. Within four hours of preparation, apply zinc-rich primer to prepared steel and feather onto prepared primer. Apply intermediate coat to primed area and feather to prepared intermediate area. Apply topcoat to intermediate coat and feather to prepared topcoat. Apply each repair coat to approximate thickness of surrounding coating system.

3.6.3.7 Structure Occupancy After Coating Application

Use clean canvas or other approved shoe covers when walking on coated surfaces, regardless of curing time allowed. For heavily trafficked areas, provide cushioned mats for additional protection.

3.7 FIELD QUALITY CONTROL

For marking of tank surfaces, use chalk for marking bare steel, and water based markers for marking coated surfaces, and remove marks prior to coating. Do not use any wax or grease based markers, or any other markers that leave a residue or stain.

3.7.1 Coating Inspector

The coating inspector shall be considered a QC Specialist and shall report to the QC Manager, as specified in Section 01 45 00.00 20 QUALITY CONTROL. The Coating Inspector shall be present during all pre-preparation testing, surface preparation, coating application, initial cure of the coating system, during all coating repair work, and during completion activities as specified in Section 01 45 00.00 20 QUALITY CONTROL. The Coating Inspector shall provide complete documentation of conditions and occurrences on the job site, and be aware of conditions and occurrences that are potentially detrimental to the coating system. The requirements for inspection listed in this Section are in addition to the QC inspection and reporting requirements specified in Section 01 45 00.00 20 QUALITY CONTROL.

3.7.2 Field Inspection

3.7.2.1 Inspection Requirements

Perform field inspection in accordance with ASTM D3276 and the approved Coating Work Plan. Document Contractor's compliance with the approved Coating Work Plan.

Provide all tools and instruments required to perform the required testing, as well as any tools or instruments that the inspector considers necessary to perform the required inspections and tests. Document each inspection and test, including required hold points and other required inspections and tests, as well as those inspections and tests deemed prudent from on-site evaluation to document a particular process or condition, as follows:

- a. Location or area;
- b. Purpose (required or special);
- c. Method;
- d. Criteria for evaluation;
- e. Results;
- f. Determination of compliance;
- g. List of required rework;

h. Observations.

Collect and record Environmental Conditions as described in ASTM D3276 on a 24 hour basis, as follows:

- a. During surface preparation, every two hours or when changes occur;
- b. During coating application and the first four days of initial cure, every hour, or when changes occur;
- c. Note location, time, and temperature of the highest and lowest surface temperatures each day;
- d. Use a non-contact thermometer to locate temperature extremes, then verify with contact thermometers.

Document all equipment used in inspections and testing, including manufacturer, model number, serial number, last calibration date and future calibration date, and results of on-site calibration performed.

Document Contractors compliance with the approved Coating Work Plan.

3.7.2.2 Inspection Report Forms

Develop project-specific report forms as required to report measurements, test results, and observations being complete and conforming to contract requirements. This includes all direct requirements of the contract documents and indirect requirements of referenced documents. Show acceptance criteria with each requirement and indication of conformity of each inspected item. The data may be in any format, but must be legible and presented so that entered data can be quickly compared to the appropriate requirement.

3.7.2.3 Daily Inspection Reports

Submit one copy of daily inspection report completed each day when performing work under this Section, to the Contracting Officer. Note all non-compliance issues, and all issues that were reported for rework in accordance with QC procedures of Section 01 45 00.00 20 QUALITY CONTROL. Each report shall be signed by the coating inspector and the QC Manager. Submit report within 24 hours of date recorded on the report.

3.7.2.4 Inspection Logbook

A continuous record of all activity related to this Section shall be maintained in an Inspection Logbook on a daily basis. The logbook shall be hard or spiral bound with consecutively numbered pages, and shall be used to record all information provided in the Daily Inspection Reports, as well as other pertinent observations and information. The Coating Inspector's Logbook that is sold by NACE is satisfactory. Submit the original Inspection Logbook to the Contracting Officer upon completion of the project and prior to final payment.

3.7.2.5 Inspection Equipment

All equipment shall be in good condition, operational within its design range, and calibrated as required by the specified standard for use of each device.

3.8 FINAL CLEANUP

Following completion of the work, remove debris, equipment, and materials from the site. Remove temporary connections to Government or Contractor furnished water and electrical services. Restore existing facilities in and around the work areas to their original condition.

TABLE 1						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ia - Zinc-rich Epoxy Primer Coat MIL-DTL-24441/19 Formula 159						
Test	Component A		Component B		Mixed	
	Min.	Max.	Min.	Max.	Min.	Max.
Pigment content, percent (zinc dust)	---	---	81.5	85.5	---	---
Volatiles, percent	42.8	44.3	8.0	8.4	---	---
Non-volatile vehicle percent	53.7	57.7	8.3	8.7	---	---
Weight, Kilograms/liter	0.87	1.01	3.30	3.40	2.80	2.91
Weight, Pounds/gallon	7.3	8.4	27.5	28.4	23.4	24.4
Flashpoint, Degrees C	35.6	---	37.8	---	---	---
Flashpoint, Degrees F	96	---	100	---	---	---
Consistency, grams	---	---	250	500	150	300
Set to touch time, hours at 23 degrees C, 73 degrees F	---	---	---	---	---	2
Dry hard time, hours at 23 degrees C, 73 degrees F	---	---	---	---	---	8
Pot life, hours at 23 degrees C, 73 degrees F	---	---	---	---	4	---
Sag resistance, Micrometers	---	---	---	---	300	---
Sag resistance, Mils	---	---	---	---	12	---

TABLE 1						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ia - Zinc-rich Epoxy Primer Coat MIL-DTL-24441/19 Formula 159						
Test	Component A		Component B		Mixed	
	Min.	Max.	Min.	Max.	Min.	Max.
VOC, Grams/liter	---	---	---	---	---	304
VOC, Pounds/gallon	---	---	---	---	---	2.5
NOTES: Test methods as specified in MIL-DTL-24441.						

TABLE 1						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ib. - Epoxy Intermediate Coat MIL-DTL-24441/31 Formula 152 Type IV (White (Tinted))						
Test	Component A		Component B		Mixed	
	Min.	Max.	Min.	Max.	Min.	Max.
Pigment content, percent	44.0	49.0	33.0	38.0	---	---
Volatiles, percent	29.0	35.0	16.0	21.0	---	---
Non-volatile vehicle percent	17.5	23.5	44.0	49.0	---	---
Coarse particles, percent	---	0.3	---	0.3	---	---
Consistency, grams	180	320	300	470	180	245
Weight, Kilograms/liter	1.39	1.45	1.29	1.35	1.34	1.4
Weight, Pounds/gallon	11.6	12.1	10.8	11.3	11.2	11.7

TABLE 1						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ib. - Epoxy Intermediate Coat MIL-DTL-24441/31 Formula 152 Type IV (White (Tinted))						
Test	Component A		Component B		Mixed	
	Min.	Max.	Min.	Max.	Min.	Max.
Set to touch time, hours at 23 degrees C, 73 degrees F	---	---	---	---	---	3
Dry hard time, hours at 23 degrees C, 73 degrees F	---	---	---	---	---	8
Fineness of grind, Hegman	4	---	4	---	---	---
Flashpoint, Degrees C	35.5	---	37.8	---	---	---
Flashpoint, Degrees F	96	---	100	---	---	---
Titanium dioxide, percent of pigment	91	---	---	---	---	---
Pot life, hours at 23 degrees C, 73 degrees F	---	---	---	---	4	---
Sag resistance, Micrometers	---	---	---	---	300	---
Sag resistance, Mils	---	---	---	---	12	---
Color of dry film to approximate color of SAE AMS-STD-595A color 27778	---	---	---	---	---	Conform
Contrast ratio, at 75 micrometers, 3 mils DFT	---	---	---	---	.098	---
Gloss, 60 degree specular	---	---	---	---	35	---
VOC, Grams/liter	---	---	---	---	---	340
VOC, Pounds/gallon	---	---	---	---	---	2.8

TABLE 1						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ib. - Epoxy Intermediate Coat MIL-DTL-24441/31 Formula 152 Type IV (White (Tinted))						
Test	Component A		Component B		Mixed	
	Min.	Max.	Min.	Max.	Min.	Max.
GENERAL NOTES: Test methods as specified in MIL-DTL-24441. Where "Conform" is indicated, refer to specific requirements of MIL-DTL-24441/31.						

TABLE I						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ic - Polyurethane TopcoatMIL-PRF-85285 Type II (White and Colors)						
Test	Component A		Component B		Mixed	
	Min.	Max.	Min.	Max.	Min.	Max.
Moisture content, percent	---	2	---	---	---	---
Course particles, percent	---	---	---	---	---	.5
Viscosity	---	---	---	---	---	See Note 1
Fineness of grind, Hegman	---	---	---	---	7	---
Drying to touch (See Note 2)	---	---	---	---	---	4
Dry-hard (See Note 2)	---	---	---	---	---	8
VOC, grams per liter	---	---	---	---	---	340
Color	---	---	---	---	delta E+-1.0	
Gloss 60 degree specular gloss						

TABLE I						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ic - Polyurethane TopcoatMIL-PRF-85285 Type II (White and Colors)						
<u>Test</u>	<u>Component A</u>		<u>Component B</u>		<u>Mixed</u>	
	<u>Min.</u>	<u>Max.</u>	<u>Min.</u>	<u>Max.</u>	<u>Min.</u>	<u>Max.</u>
Gloss	---	---	---	---	---	90
Semi-gloss	---	---	---	---	15	45
Opacity	---	---	---	---	0.95	---
Flexibility	---	---	---	---	---	Conform
Fluid resistance	---	---	---	---	---	Conform
Heat resistance (cure)	---	---	---	---	---	Conform
Solvent resistance (cure)	---	---	---	---	---	Conform
Condition in container	---	---	---	---	---	Conform
Odor	---	---	---	---	---	Conform
Lead percent	---	---	---	---	---	0.06
Cadmium percent	---	---	---	---	---	0.06
Chromium percent	---	---	---	---	---	0.00
NOTES: (1) Modify paragraph 3.6.4 Viscosity and Pot Life, of MIL-PRF-85285 as follows: The viscosity of the admixed coating, when tested in accordance with ASTM D1200 through a No. 4 Ford cup, shall be as follows:						
Time from mix (minimum)			Maximum time through a No. 4 Ford Cup			
Initially			30 seconds			
2 hours			60 seconds			
4 hours			No gel			

TABLE I						
COATING QUALITY CONFORMANCE INSPECTION REQUIREMENTS						
Table Ic - Polyurethane TopcoatMIL-PRF-85285 Type II (White and Colors)						
<u>Test</u>	<u>Component A</u>		<u>Component B</u>		<u>Mixed</u>	
	<u>Min.</u>	<u>Max.</u>	<u>Min.</u>	<u>Max.</u>	<u>Min.</u>	<u>Max.</u>
(2) Modify paragraph 3.7.1 Drying Time, of MIL-PRF-85285. When applied by spray techniques and when tested in accordance with ASTM D1640/D1640M, the coating shall be set-to-touch within four hours and dry-hard within eight hours (see 4.6 and table I).						
GENERAL NOTES: Test methods as specified in MIL-PRF-85285, except those marked with "*". Where "Conform" is indicated, refer to specific requirements of MIL-PRF-85285.						

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 25 - INTEGRATED AUTOMATION

SECTION 25 05 11

CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS

05/21

PART 1 GENERAL

- 1.1 RELATED REQUIREMENTS
- 1.2 DEFINITIONS
 - 1.2.1 Administrator Account
 - 1.2.2 Computer
 - 1.2.3 Controller
 - 1.2.4 Mission Space
 - 1.2.5 Network
 - 1.2.6 Network Connected
 - 1.2.6.1 Wireless Network Connected
 - 1.2.7 Network Media
 - 1.2.8 User Account Support Levels
 - 1.2.8.1 FULLY Supported
 - 1.2.8.2 MINIMALLY Supported
 - 1.2.8.3 NOT Supported
 - 1.2.9 Manual Local Input
 - 1.2.10 Card Reader
 - 1.2.11 User Interface
 - 1.2.11.1 Local User Interface
 - 1.2.11.2 Remote User Interface
 - 1.2.11.3 Types of User Interface (by capability)
 - 1.2.11.3.1 Read-Only User Interface
 - 1.2.11.3.2 Limited User Interface
 - 1.2.11.3.3 Full User Interface
 - 1.2.11.3.4 View-Only User Interface
 - 1.2.11.4 Other User Interface Terminology
 - 1.2.11.4.1 Writable User Interface
 - 1.2.11.4.2 Privileged User Interface
 - 1.2.12 Wireless Network
 - 1.2.13 Wired Broadcast Network
- 1.3 ADMINISTRATIVE REQUIREMENTS
 - 1.3.1 Points of Contact
 - 1.3.2 Coordination
- 1.4 SUBMITTALS
- 1.5 QUALITY CONTROL
- 1.6 CYBERSECURITY DOCUMENTATION
 - 1.6.1 Control System Inventory Report
 - 1.6.2 Cybersecurity Riser Diagram
 - 1.6.3 Control System Cybersecurity Documentation
 - 1.6.3.1 Default Requirements for Control System Devices
- 1.7 CYBERSECURITY DURING CONSTRUCTION
 - 1.7.1 Contractor Computer Equipment
 - 1.7.1.1 Operating System
 - 1.7.1.2 Anti-Malware Software
 - 1.7.1.3 Passwords and Passphrases

- 1.7.1.4 User-Based Authentication
- 1.7.1.5 Demonstration of Compliance
- 1.7.1.6 Contractor Computer Cybersecurity Compliance Statements
- 1.7.2 Temporary IP Networks
 - 1.7.2.1 Network Boundaries and Connections
- 1.7.3 Government Access to Network
- 1.7.4 Temporary Wireless IP Networks
- 1.7.5 Passwords and Passphrases
- 1.7.6 Contractor Temporary Network Cybersecurity Compliance Statements
- 1.8 CYBERSECURITY DURING WARRANTY PERIOD

PART 2 PRODUCTS

PART 3 EXECUTION

- 3.1 CYBERSECURITY HARDENING AND CONFIGURATION GUIDES
- 3.2 NETWORK REQUIREMENTS
 - 3.2.1 Wireless and Wired Broadcast Communication for Systems Other than Fire Protection Systems
 - 3.2.1.1 Wireless and Wired Broadcast IP Communications
 - 3.2.2 Non-IP Control Networks
- 3.3 ACCESS CONTROL REQUIREMENTS
 - 3.3.1 User Accounts
 - 3.3.1.1 Controllers
 - 3.3.1.1.1 Default Requirements for Other Control Systems
 - 3.3.2 Enclosures
- 3.4 USER IDENTIFICATION AND AUTHENTICATION
 - 3.4.1 User Identification and Authentication for All System Types
 - 3.4.2 Password-Based Authentication Requirements
 - 3.4.2.1 Passwords for Devices Minimally Supporting Accounts
 - 3.4.2.2 Password Configuration and Reporting
- 3.5 SYSTEM AND COMMUNICATION PROTECTION
 - 3.5.1 Collaborative Computing
 - 3.5.2 Denial of Service Protection
- 3.6 CONTROL SYSTEM CYBERSECURITY TESTING
 - 3.6.1 Control System Cybersecurity Testing Procedures
 - 3.6.2 Control System Cybersecurity Testing Execution
 - 3.6.3 Control System Cybersecurity Testing Report
- 3.7 FIELD QUALITY CONTROL, CYBERSECURITY VALIDATION SUPPORT

-- End of Section Table of Contents --

SECTION 25 05 11

CYBERSECURITY FOR FACILITY-RELATED CONTROL SYSTEMS 05/21

PART 1 GENERAL

Many subparts in this Section contain text in curly braces ("{" and "}") indicating which cybersecurity control and control correlation identifier (CCI) the requirements of the subpart relate to. The text inside these curly braces is for Government reference only and enables coordination of the requirements of this Section with the RMF process throughout the design and construction process. Text in curly braces are not contractor requirements.

This Section refers to Security Requirements Guide (SRGs) and Security Technical Implementation Guide (STIGs). STIGs and SRGs are available online at the Information Assurance Support Environment (IASE) website at <https://public.cyber.mil/stigs/downloads/> and an SRG/STIG Applicability Guide and Collection Tool is available at <https://public.cyber.mil/stigs/SCAP/>. Not all control system components have applicable STIGs or SRGs. The "Control Systems SRG" does not apply to work performed under this Section; all requirements within this section to apply applicable SRGs DO NOT include the "Control Systems SRG".

1.1 RELATED REQUIREMENTS

This section does not contain sufficient requirements to procure a control system and must be used in conjunction with other Sections which specify control systems. This Section adds cybersecurity requirements to the control systems specified in other Sections, and as these requirements are conditioned on the control system being provided, there may be requirements in this Section that will not apply to this project. All Sections containing facility-related control systems or control system components are related to the requirements of this Section. Review all specification sections to determine related requirements.

In cases where a requirement is specified in both this Section and in another Section, the more stringent requirement must be met. In cases where a requirement in this Section conflicts with the requirements of another Section such that both requirements cannot be met at the same time, request direction from the Representative to determine which requirement applies to the project.

1.2 DEFINITIONS

1.2.1 Administrator Account

An administrator account is an account with full permissions to a device, application, or operating system, including the ability to create and modify other user accounts.

Note that the operating system Administrator Account may be different than

Administrator Accounts for applications hosted on that operating system. Also, most controllers will not have any support for accounts and will therefore not have an 'Administrator Account'.

1.2.2 Computer

A computer is one of the following:

- a. a device running a non-embedded desktop or server version of Microsoft Windows
- b. a device running a non-embedded version of MacOS
- c. a device running a non-embedded version of Linux
- d. a device running a version or derivative of the Android Operating System, where Android is considered separate from Linux
- e. a device running a version of Apple iOS

Unless otherwise indicated or clear from context use of the word "device" in this Section includes computers.

1.2.3 Controller

A device other than a computer or Ethernet switch.

1.2.4 Mission Space

A device or media is in mission space if physical access to the device or media is controlled by the organization served by the device. For example, a VAV box controller in a suspended ceiling is in mission space if the VAV box serves that room; an electrical switchgear in an electrical room or an AHU in a mechanical room or on a rooftop may still be considered to be in mission space if the organization (mission) served by that switchgear or AHU controls access to the electrical room, mechanical room or rooftop.

1.2.5 Network

A network is a group of two or more devices that can communicate using a network protocol. Network protocols must provide a method for addressing devices on the network; a communication method that does not provide an addressing scheme is not a networked form of communication. Devices that communicate using a method of communication that does not support device addressing are not using a network.

1.2.6 Network Connected

A component is network connected (or "connected to a network") only when the device has a network transceiver which is directly connected to the network and implements the network protocol. A device lacking a network transceiver (and accompanying protocol implementation) can never be considered network connected. Note that (unlike many IT definitions of "Network Connected") a device connected to a non-IP network is still considered network connected (an IP connection or IP address is not required for a device to be network connected).

1.2.6.1 Wireless Network Connected

Any device that supports wireless network communication is network connected to a wireless network, regardless of whether the device is communicating using wireless. Unless physically disabled, devices with wireless transceivers support wireless, it is not sufficient to disable the wireless in software.

1.2.7 Network Media

The thing that provides the communication channel between the devices on a network. Typically wire, but might include wireless, fiber optic, or even power line (some network protocols allow sending network signals over power wiring).

1.2.8 User Account Support Levels

The support for user accounts is categorized in this Section as one of three levels:

1.2.8.1 FULLY Supported

Fully supported user accounts are not used for this project. Device supports configurable individual accounts. Accounts can be created, deleted, modified, etc. Privileges can be assigned to accounts. These devices support user-based (as opposed to role-based) authentication.

1.2.8.2 MINIMALLY Supported

Device supports a small, fixed number of accounts (perhaps only one). Accounts cannot be modified. A device with only a "User" and an "Administrator" account would fit this category. Similarly, a device with two PINs for logon - one for restricted and one for unrestricted rights would fit here (in other words, the accounts do not have to be the traditional "username and password" structure). These devices typically only support role-based authentication.

Examples of devices which MINIMALLY support accounts are a) a variable frequency drive with a single account which requires a PIN for access to configuration; and b) a room lighting control touchpad interface that has a single account.

1.2.8.3 NOT Supported

Device does not support any Access Enforcement therefore the whole concept of "account" is meaningless.

1.2.9 Manual Local Input

Manual Local Inputs are system analog or binary inputs that are adjustable by a person but are, by intrinsic hardware design, very limited in potential capabilities. Manual Local Inputs do not have touch screens or full keyboards, but may have a few buttons or dials to allow input. Manual Local Inputs do not have full graphic screens or dot-matrix displays, but may have simple lights (LEDs) or 7-segment displays. Manual Local Inputs do not have any sort of menu structure, each button has a single well-defined function.

Examples of Manual Local Inputs are H-O-A switches, simple thermostats,

and disconnect switches.

1.2.10 Card Reader

A card reader is an input/output device whose primary function is to assist in two-factor authentication. A card reader must have an interface to read data from a card and may be able to write data to a card. A card reader may have a means (such as buttons, keypad, touchscreen, etc.) for a user to input a PIN or password, as well as a limited display.

1.2.11 User Interface

A User Interface (UI) is something other than a Manual Local Input or Card Reader that allows a person to interact with the system or device. Note that while a Card Reader is not by itself a User Interface, a User Interface may contain a Card Reader in order for it to authenticate its user. Within control systems, there are a wide range of User Interfaces.

Two important distinctions are 1) whether the user interface is Local or Remote, and 2) the effective capabilities of the User Interface to alter data, which is the "privilege" of the user interface (where effective privilege available to a specific user at a specific user interface is the combination of the greatest privilege offered by the user interface and the specific account the user is logged into).

1.2.11.1 Local User Interface

A Local User Interface is a user interface where the physical hardware the user interacts with (keyboard, buttons, display, etc.) is physically part of the device being affected. All of the relevant characteristics of the user interface are embodied within a single device.

Note that a Local UI may be able to access data in a different device, Local versus Remote in this context refers to the user interface itself; the capability to access data in a different device is covered under "Full User Interface".

1.2.11.2 Remote User Interface

Remote User Interfaces are not utilized for this project. A Remote User Interface implements a Client/Server model where the physical hardware the user interacts with (Client) is physically distinct from the device being affected (Server). Most or all of the security and functionality characteristics of the user interface are defined by the Server, not the Client. The Client and Server communicate via a network connection. A common example of a remote user interface is a web-based interface where the browser (client) is generally on different hardware than the web server (server). A Remote UI remains a Remote UI even if the user happens to be at a Client on the same hardware as the Server. What is important is that a) the Client may be on different hardware than the Server and b) the majority of the security and functional characteristics of the interface are defined at the Server.

Note that this definition of "remote" is consistent with that generally used in the control industry but is not aligned with the NIST 800-53 definition of "Remote", which refers to "outside the system". The term "Remote" here better aligns with the NIST 800-53 definition of "Network" (remote from within the system) Access.

1.2.11.3 Types of User Interface (by capability)

User interfaces are also categorized by their capabilities as being Read Only, Limited, or Full.

1.2.11.3.1 Read-Only User Interface

A Read Only User Interface (also referred to as a View-Only User Interface) is a user interface that only allows for reading data, it does not allow (have the capability to) modify data. A Read Only User Interface may be either Local or Remote. A User Interface that is configured to be Read Only (by some other means than the interface itself, such as using configuration software on a laptop) is a Read-Only Interface. Note a Read Only User Interface may have buttons (or touch screen, etc.) allowing the user to navigate through the presentation of data.

Examples of a Read Only User Interfaces are a) a publicly viewable "energy dashboard" showing weather data and energy usage within a building and b) digital wayfinding signage.

1.2.11.3.2 Limited User Interface

A Limited User Interface is a user interface that - by design - can only alter information local to the user interface. Note that the determination of "alter" includes only direct interactions, it explicitly excludes interactions that might occur as secondary effects. For example, an interface changing the flow setpoint in a pump controller is a direct interaction, the subsequent change in flow (as well as any subsequent downstream changes in valve position) are not direct interactions.

Two examples of LIMITED UIs are: a) a variable speed drive has a Limited Local User Interface which allows the user to change properties within the drive, but does not allow affecting things outside the drive; and b) a typical home WiFi Router has a Limited Remote User Interface which allows configuration of the Router, but does not allow direct interaction with other devices.

1.2.11.3.3 Full User Interface

A Full User Interface can alter information in devices outside the device with the user interface. For example, a typical Local Display Panel is a Full Local User Interface while a browser-based front end is a Full Remote User Interface.

1.2.11.3.4 View-Only User Interface

See Read-Only User Interface

1.2.11.4 Other User Interface Terminology

In addition to defining whether a user interface is a Hardware Limited, Read-Only, Limited or Full, and whether it is Local or Remote, user interfaces are classified by whether they are writable or privileged.

1.2.11.4.1 Writable User Interface

Any User Interface that is not Read-Only is Writable. (Limited User Interfaces and Full User Interfaces are both writable user interfaces (as

they are capable of changing a value)).

1.2.11.4.2 Privileged User Interface

A Privileged UI is a UI that has sufficient capabilities or functionality that it requires specific cybersecurity measures to be put in place to limit its unauthorized use. Ultimately, whether a specific user interface is considered a Privileged User Interface must be determined by usage. Unless otherwise specified, user interfaces can be determined to be privileged or not using the following:

- a. Read-Only User Interfaces are not privileged user interfaces.
- b. Except as specified above, a Limited User Interface must be determined to be privileged or not based on the specific capabilities and use case of the user interface. In general however, user interfaces that do not offer significant capabilities above and beyond those available at that location via other means (e.g. such as a disconnect switch, breaker, or hand-off-auto switch, or physical attack) are not privileged.

1.2.12 Wireless Network

Any network that communicates without using wires or fiber optics as the communication media. Wireless networks include: WiFi, Bluetooth, ZigBee, cellular, satellite, 900 MHz radio, 2.4 GHz, free space optical, point-to-point laser, and IR.

1.2.13 Wired Broadcast Network

Wired Broadcast Networks are any network, such as powerline carrier networks and modem (wired telephony), that use wire-based technologies where there is not a clearly defined boundary for signal propagation.

1.3 ADMINISTRATIVE REQUIREMENTS

1.3.1 Points of Contact

Coordinate with the following Points of Contact as indicated in this Section and as required. Not all projects will require coordination with all Points of Contact. When coordination is required and no Point of Contact is indicated, coordinate with The Contracting Office Representative (COR).

- a. Password Point of Contact: The Contracting Office Representative (COR)

1.3.2 Coordination

Coordinate the execution of this Section with the execution of all other Sections related to control systems as indicated in the paragraph RELATED REQUIREMENTS. Items that must be considered when coordinating project efforts include but are not limited to:

- a. If requesting permission for wireless or wired broadcast communication, the Wireless and Wired Broadcast Communication Request submittal must be approved prior to control system device selection and installation.
- b. Passwords must be coordinated with the indicated contact for the

project site.

- c. Contractor Computer Cybersecurity Compliance Statements must be provided for each contractor using contractor owned computers.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contractor Computer Cybersecurity Compliance Statements; G

Contractor Temporary Network Cybersecurity Compliance Statements; G

SD-02 Shop Drawings

Cybersecurity Riser Diagram; G

SD-03 Product Data

Control System Cybersecurity Documentation; G

SD-06 Test Reports

Control System Cybersecurity Testing Procedures; G

Control System Cybersecurity Testing Report; G

SD-11 Closeout Submittals

Password Change Summary Report; G

Enclosure Keys; G

Control System Inventory Report; G

1.5 QUALITY CONTROL

1.6 CYBERSECURITY DOCUMENTATION

{For Government Reference Only: This subpart (and its subparts) relates to PL-7; CCI-003071}

1.6.1 Control System Inventory Report

{For Government Reference Only: This subpart (and its subparts) relates to CM-8(a), SI-17, IA-3; CCI-000389, CCI-000392, CCI-000398, CCI-002773, CCI-002774, CCI-002775, CCI-000777, CCI-000778, CCI-001958}

Provide a Control System Inventory report using the Inventory Spreadsheet listed under this Section at <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-25-05-11> documenting all devices, including networked devices, network infrastructure devices, non-networked devices, input devices (e.g.

sensors) and output devices (e.g. actuators). For each device provide all applicable information for which there is a field on the spreadsheet in accordance with the instructions on the spreadsheet.

In addition to the requirements of Section 01 33 00 SUBMITTAL PROCEDURES, provide the Control System Inventory Report as an editable Microsoft Excel file.

1.6.2 Cybersecurity Riser Diagram

{For Government Reference Only: This subpart (and its subparts) relates to PL-2(a), PL-8; CCI-003051, CCI-003053, CCI-003072, CCI-003073, CCI-003075}

Provide a cybersecurity riser diagram of the complete control system including all network and device hardware. If the control system specifications require a riser diagram submittal, provide a copy of that submittal as the cybersecurity riser diagram. Otherwise, provide a riser diagram in one-line format.

1.6.3 Control System Cybersecurity Documentation

{For Government Reference Only: This subpart (and its subparts) relates to SA-5 (a),(b),(c); CCIs: CCI-003124, CCI-003125, CCI-003126, CCI-003127, CCI-003128, CCI-003129, CCI-003130, CCI-003131}

Provide a Control System Cybersecurity Documentation submittal containing the indicated information for each device and software application.

1.6.3.1 Default Requirements for Control System Devices

For control system devices where Control System Cybersecurity Documentation requirements are not otherwise indicated in this Section, provide:

- a. Documentation that describes secure configuration of the device {For Government Reference Only: relates to CCI-003124}
- b. Documentation that describes secure installation of the device {For Government Reference Only: relates to CCI-003125}
- c. Documentation that describes secure operation of the device {For Government Reference Only: relates to CCI-003124}
- d. Documentation that describes effective use and maintenance of security functions or mechanisms for the device {For Government Reference Only: relates to CCI-003127}
- e. Documentation that describes known vulnerabilities regarding configuration and use of administrative (i.e. privileged) functions for the device {For Government Reference Only: relates to CCI-003128}
- f. Documentation that describes user-accessible security functions or mechanisms in the device and how to effectively use those security functions or mechanisms {For Government Reference Only: relates to CCI-003129}
- g. Documentation that describes methods for user interaction which enables individuals to use the device in a more secure manner {For

Government Reference Only: relates to CCI-003130}

- h. Documentation that describes user responsibilities in maintaining the security of the device {For Government Reference Only: relates to CCI-003131}

1.7 CYBERSECURITY DURING CONSTRUCTION

{For Government Reference Only: This subpart (and its subparts) relates to AC-18, SA-3; CCI-000258}

In addition to the control system cybersecurity requirements indicated in this section, meet following requirement throughout the construction process.

1.7.1 Contractor Computer Equipment

Contractor owned computers may be used for construction. Contractor computers connected to the control system, control system network, or a control system component at any point during construction must meet the following requirements:

1.7.1.1 Operating System

The operating system must be an operating system currently supported by the manufacturer of the operating system. The operating system must be current on security patches and operating system manufacturer required updates.

1.7.1.2 Anti-Malware Software

The computer must run anti-malware software from a reputable software manufacturer. Anti-malware software must be a version currently supported by the software manufacturer, must be current on all patches and updates, and must use the latest definitions file. Computers used on this project must be scanned using the installed software at least once per day.

1.7.1.3 Passwords and Passphrases

The passwords and passphrases for computers, applications, and web-based applications supporting passwords must be changed from their default values. Passwords must be a minimum of eight characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.7.1.4 User-Based Authentication

Each user must have a unique account; sharing of a single account between multiple users is prohibited.

1.7.1.5 Demonstration of Compliance

The Government has the right to require demonstration of computer compliance with these requirements at any time during the project.

1.7.1.6 Contractor Computer Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Computer Cybersecurity Compliance Statements for each company using contractor

owned computers. Contractor Computer Cybersecurity Compliance Statements must use the template published at <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-25-05-11>. Each Statement must be signed by a cybersecurity representative for the relevant company.

1.7.2 Temporary IP Networks

Temporary contractor-installed IP networks may be used during construction. When used, temporary contractor-installed IP networks connected to the control system, control system network, or a control system component at any point during construction must meet the following requirements:

1.7.2.1 Network Boundaries and Connections

The network must not extend outside the project site and must not connect to any IP network other than those specifically provided or furnished for this project. Any and all access to the network from outside the project site is prohibited.

1.7.3 Government Access to Network

Government personnel must be allowed to have complete and immediate access to the network at any time in order to verify compliance with this specification.

1.7.4 Temporary Wireless IP Networks

No temporary wireless IP networks allowed.

1.7.5 Passwords and Passphrases

The passwords and passphrases for all network devices and network access must be changed from their default values. Passwords must be a minimum 8 characters with a minimum of one uppercase letter, one lowercase letter, one number and one special character.

1.7.6 Contractor Temporary Network Cybersecurity Compliance Statements

Provide a single submittal containing completed Contractor Temporary Network Cybersecurity Compliance Statements for each company implementing a temporary IP network. Contractor Temporary Network Cybersecurity Compliance Statements must use the template published at <https://www.wbdg.org/ffc/dod/unified-facilities-guide-specifications-ufgs/ufgs-25-05-11>. Each Statement must be signed by a cybersecurity representative for the relevant company. If no temporary IP networks will be used, provide a single copy of the Statement indicating this.

1.8 CYBERSECURITY DURING WARRANTY PERIOD

All work performed on the control system after acceptance must be performed using Government Furnished Equipment or equipment specifically and individually approved by the Government.

PART 2 PRODUCTS

All products used on this project must meet the indicated requirements, but not all products specified here will be required by every project.

PART 3 EXECUTION

3.1 CYBERSECURITY HARDENING AND CONFIGURATION GUIDES

Install, configure, and harden all hardware and software furnished on this project in accordance with manufacturer provided documentation, procedures, or methods for secure configuration or installation. Do not implement specific hardening actions if that action would conflict with required functionality or another requirement of this Section.

3.2 NETWORK REQUIREMENTS

3.2.1 Wireless and Wired Broadcast Communication for Systems Other than Fire Protection Systems

3.2.1.1 Wireless and Wired Broadcast IP Communications

Do not install wireless or wired broadcast IP networks, including: do not install a wireless access point; do not install or configure an ad-hoc wireless network; do not install or configure a WiFi Direct communication.

3.2.2 Non-IP Control Networks

When control system specifications require particular communication protocols, use only those communication protocols and only as specified. Do not implement any other communication protocol.

When control system specifications do not indicate requirements for communication protocols, use only those protocols required for operation of the system as specified.

3.3 ACCESS CONTROL REQUIREMENTS

3.3.1 User Accounts

{For Government Reference Only: This subpart (and its subparts) relate to AC-2(a), AC-3, AC-6(1), AC-6(10), AC-6(2), AC-6(9), CM-11(2), and IA-2; CCI-002110, CCI-000213, CCI-001558, CCI-002221, CCI-002222, CCI-002223, CCI-002235, CCI-000039, CCI-001419, CCI-002234, CCI-001812, and CCI-000764.

For MODERATE Impact systems, this subpart (and its subparts) also relate to AC-2 (2), AC-2(3), AC-2(4), AC-6(1), and CM-5(1); CCI-001361, CCI-000017, CCI-000217, CCI-000018, CCI-001403, CCI-001404, CCI-001405, CCI-002130, CCI-001683, CCI-001684, CCI-001685, CCI-001686, CCI-002132, CCI-001558, CCI-002221, CCI-002222, CCI-002223, CCI-001813.}

Any user interface supporting user accounts (either FULLY or MINIMALLY) must limit access according to specified limitations for each account.

All user interfaces FULLY supporting accounts must implement user-based authentication where each account is uniquely assigned to a specific user. User interfaces FULLY supporting accounts must implement at least three (3) levels of user account privilege including: 1) an account with read-only permissions 2) an account with full permissions including account creation and modification and 3) an account with greater permissions than read-only but without account creation and modification.

3.3.1.1 Controllers

For user interfaces provided by controllers, provide access control in accordance with the User Interface Requirements table for the applicable control system and user interface type.

- a. For table entries of "NA": NA means Not Applicable, there are no interfaces in this category.
- b. For table entries of "None Required": The user interface is not required to support user accounts.
- c. For table entries of "MINIMALLY": The user interface must at least MINIMALLY support user accounts.
- d. For table entries of "FULLY": The user interface must at FULLY support user accounts.
- e. For table entries of "KEY": The user interface must have physical security in the form of either a key lock on the interface itself or be furnished inside a locked enclosure. Where this is required for a read only interface, this lock must prevent viewing of data on the interface; for other interfaces, this lock must prevent using the interface to alter data.
- f. For table entries of "Physical Security": For Local FULL interfaces, the interface must be located inside mission space. For Local Limited (not FULL) interfaces, the user interface must either a) be located within mission space or b) be protected by physical security at least as good as the control devices (and equipment controlled by the control devices) affected by the interface. For purposes of this requirement, 'affected' includes controllers with data that can be directly altered by the interface, as well as mechanical and/or electrical equipment directly controlled by those controllers, but does not include other interactions.
- g. Entries of the form "X and Y" must meet both the requirement indicated for X and the requirement indicated for Y. For example, an entry of "MINIMALLY and Physical Security" indicates the user interface must both MINIMALLY support accounts and have physical security.
- h. Entries of the form "X or Y" must meet either the requirement indicated for X or the requirement indicated for Y.

3.3.1.1.1 Default Requirements for Other Control Systems

For control system devices where User Interface Requirements are not otherwise indicated in this Section, use the Default User Interface Requirements tables.

Default User Interface Requirements for LOW Impact Control Systems	
<u>User Interface Type</u>	<u>Access Control Requirement</u>
Local Read Only (see note 1)	None Required

Default User Interface Requirements for LOW Impact Control Systems	
<u>User Interface Type</u>	<u>Access Control Requirement</u>
Local Limited, Non-privileged	None Required
Local Limited, Privileged	Physical Security
Local Full	MINIMALLY
Remote Read Only	None Required
Remote Limited, Non-Privileged	MINIMALLY
Remote Limited, Privileged AND Remote Full (see note 2)	FULLY
Notes: 1)Local Read Only User Interfaces are always Non-Privileged 2)Remote Full User Interfaces are always Privileged	

3.3.2 Enclosures

Prior to final acceptance of the system, lock all lockable enclosures. Submit an Enclosure Keys submittal with all copies of keys for all enclosures and a key inventory list documenting all keys. Label each key with the matching enclosure identifier.

3.4 USER IDENTIFICATION AND AUTHENTICATION

{For Government Reference Only: This subpart (and its subparts) relates to IA-2, IA-2(1), IA-2(12), IA-5 IA-5(b), IA-5(c), IA-5(e), IA-5(g), IA-5(1), IA-5(11); CCI-000764, CCI-000765, CCI-001953, CCI-001954, CCI-001544, CCI-001989, CCI-000182, CCI-001610, CCI-000192, CCI-000193, CCI-000194, CCI-000205, CCI-001619, CCI-001611, CCI-001612, CCI-001613, CCI-001614, CCI-000195, CCI-001615, CCI-000196, CCI-000197, CCI-000199, CCI-000198, CCI-001616, CCI-001617, CCI-000200, CCI-001618, CCI-002041, CCI-002002, CCI-002003. For MODERATE Impact systems, this subpart also relates to AC-6 (1), AC-6(10), AC-6(2), AC-6(9)IA-2(4), IA-5(13); CCI-001558, CCI-002221, CCI-002222, CCI-002223, CCI-002235, CCI-000039, CCI-001419, CCI-002234, CCI-000768, CCI-002007.}

This subpart indicates requirements for specific methods of identification and authentication for users and user accounts. Where these requirements conflict apply the following order of precedence: 1) If present, Device Specific Requirements take precedence over any other requirements; and then 2) multifactor authentication requirements take precedence over password requirements.

3.4.1 User Identification and Authentication for All System Types

Unless otherwise indicated, all user interfaces supporting accounts (either FULLY or MINIMALLY) must implement Identification and Authorization via passwords.

3.4.2 Password-Based Authentication Requirements

3.4.2.1 Passwords for Devices Minimally Supporting Accounts

Devices MINIMALLY supporting accounts must support passwords with a minimum length of four characters.

3.4.2.2 Password Configuration and Reporting

For all devices with a password, coordinate the changing of passwords with the project site following testing of the system but prior to turnover to the Government. Coordinate with Password Point of Contact to determine appropriate project site personnel to complete password changes. Accompany identified personnel to each device with a password and instruct personnel on the process of changing password. Record the time, date and personnel present when each device's password is changed and submit a Password Change Summary Report documenting this information.

Provide the Password Summary Report electronically in both PDF and Microsoft Excel.

3.5 SYSTEM AND COMMUNICATION PROTECTION

3.5.1 Collaborative Computing

{For Government Reference Only: This subpart relates to SC-15(a), SC-15(b); CCI-001150, CCI-001152.}

Without explicit approval from the project site, control systems must not use collaborative computing technologies.

3.5.2 Denial of Service Protection

{For Government Reference Only: This subpart relates to SC-5, SC-39, SC-7(a); CCI-001093, CCI-002385, CCI-002386, CCI-002430, CCI-001097. For MODERATE Impact systems, this subpart also relates to SC-2; CCI-001082.}

To the greatest extent practical, implement control logic without reliance on the network. Except when required to meet the requirements of the control system Section (where the requirement can only be met using computer hardware), do not implement control logic in computers.

3.6 CONTROL SYSTEM CYBERSECURITY TESTING

{For Government Reference Only: For MODERATE Impact systems, this subpart (and its subparts) relates to SA-11(a), SA-11(b), SA-11(c), SA-11(d), SA-11(e); CCI-003171, CCI-003172, CCI-003173, CCI-003174, CCI-003175, CCI-003176, CCI-003177, CCI-003178.}

3.6.1 Control System Cybersecurity Testing Procedures

Prepare Control System Cybersecurity Testing Procedures explaining step-by-step, the actions and expected results that will demonstrate that the control system meets the requirements of this Section.

Submit 4 copies of the Control System Cybersecurity Testing Procedures. The Control System Cybersecurity Testing Procedures may be submitted as a Technical Data Package.

3.6.2 Control System Cybersecurity Testing Execution

Using the Control System Cybersecurity Testing Procedures verify that the control system meets the requirements of this Section. UNLESS GOVERNMENT WITNESSING OF A TEST IS SPECIFICALLY WAIVED BY THE GOVERNMENT, PERFORM ALL TESTS WITH A GOVERNMENT WITNESS. If testing reveals deficiencies in the system, correct the deficiency and retest until successful.

3.6.3 Control System Cybersecurity Testing Report

Prepare and submit a Control System Cybersecurity Testing Report documenting all tests performed and their results. Include all tests in the Control System Cybersecurity Testing Procedures and any additional tests performed during testing. Document test failures and repairs conducted with the test results.

Submit four copies of the Control System Cybersecurity Testing Report. The Control System Cybersecurity Testing Report may be submitted as a Technical Data Package.

3.7 FIELD QUALITY CONTROL, CYBERSECURITY VALIDATION SUPPORT

In addition to testing and testing support required by other Sections, provide a minimum of 8 hours of technical support for cybersecurity testing of control systems to support the DoD Risk Management Framework process Cybersecurity assessment of the control system. This support is independent of (and in addition to) the Control System Cybersecurity Testing specified in this section.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 08 00

APPARATUS INSPECTION AND TESTING

11/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 RELATED REQUIREMENTS
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Qualifications
 - 1.4.2 Acceptance Tests and Inspections Reports
 - 1.4.3 Acceptance Test and Inspections Procedure

PART 2 PRODUCTS

PART 3 EXECUTION

- 3.1 ACCEPTANCE TESTS AND INSPECTIONS
- 3.2 SYSTEM ACCEPTANCE
- 3.3 PLACING EQUIPMENT IN SERVICE

-- End of Section Table of Contents --

SECTION 26 08 00

APPARATUS INSPECTION AND TESTING
11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS	(2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems
----------	--

1.2 RELATED REQUIREMENTS

Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM applies to this section with additions and modifications specified herein.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-06 Test Reports

Acceptance Tests and Inspections; G

SD-07 Certificates

Qualifications of Organization, and Lead Engineering Technician; G

Acceptance Test and Inspections Procedure; G

1.4 QUALITY ASSURANCE

1.4.1 Qualifications

Contractor shall engage the services of a qualified testing organization to provide inspection, testing, calibration, and adjustment of the electrical distribution system and generation equipment listed in paragraph entitled "Acceptance Tests and Inspections" herein. Organization shall be independent of the supplier, manufacturer, and installer of the equipment. The organization shall be a first tier subcontractor. No work required by this section of the specification shall be performed by a second tier subcontractor.

- a. Submit name and qualifications of organization. Organization shall have been regularly engaged in the testing of electrical materials, devices, installations, and systems for a minimum of 5 years. The

organization shall have a calibration program, and test instruments used shall be calibrated in accordance with NETA ATS.

- b. Submit name and qualifications of the lead engineering technician performing the required testing services. Include a list of three comparable jobs performed by the technician with specific names and telephone numbers for reference. Testing, inspection, calibration, and adjustments shall be performed by an engineering technician, certified by NETA (Level III) or the National Institute for Certification in Engineering Technologies (NICET) with a minimum of 5 years' experience inspecting, testing, and calibrating electrical distribution and generation equipment, systems, and devices.

1.4.2 Acceptance Tests and Inspections Reports

Submit certified copies of inspection reports and test reports. Reports shall include certification of compliance with specified requirements, identify deficiencies, and recommend corrective action when appropriate. Type and neatly bind test reports to form a part of the final record. Submit test reports documenting the results of each test not more than 10 days after test is completed.

1.4.3 Acceptance Test and Inspections Procedure

Submit test procedure reports for each item of equipment to be field tested at least 45 days prior to planned testing date. Do not perform testing until after test procedure has been approved.

PART 2 PRODUCTS

Not used.

PART 3 EXECUTION

3.1 ACCEPTANCE TESTS AND INSPECTIONS

Testing organization shall perform acceptance tests and inspections. Test methods, procedures, and test values shall be performed and evaluated in accordance with NETA ATS, the manufacturer's recommendations, and paragraph entitled "Field Quality Control" of each applicable specification section. Tests identified as optional in NETA ATS are not required unless otherwise specified. Equipment shall be placed in service only after completion of required tests and evaluation of the test results have been completed. Contractor shall supply to the testing organization complete sets of shop drawings, settings of adjustable devices, and other information necessary for an accurate test and inspection of the system prior to the performance of any final testing. Contracting Officer shall be notified at least 14 days in advance of when tests will be conducted by the testing organization. Perform acceptance tests and inspections on applicable equipment and systems specified in the following sections:

- a. Functional engine shutdown tests, vibration base-line test, and load bank test shall not be performed by the testing organization. These tests shall be performed by the start-up engineer.
- b. Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION. Low voltage cables and grounding systems only.

3.2 SYSTEM ACCEPTANCE

Final acceptance of the system is contingent upon satisfactory completion of acceptance tests and inspections.

3.3 PLACING EQUIPMENT IN SERVICE

A representative of the approved testing organization shall be present when equipment tested by the organization is initially energized and placed in service.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 20 00

INTERIOR DISTRIBUTION SYSTEM

08/19, CHG 3: 11/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Regulatory Requirements
 - 1.4.2 Standard Products
 - 1.4.2.1 Alternative Qualifications
 - 1.4.2.2 Material and Equipment Manufacturing Date
- 1.5 MAINTENANCE
 - 1.5.1 Electrical Systems
- 1.6 WARRANTY

PART 2 PRODUCTS

- 2.1 MATERIALS AND EQUIPMENT
- 2.2 CONDUIT AND FITTINGS
 - 2.2.1 Rigid Metallic Conduit
 - 2.2.1.1 Rigid, Threaded Zinc-Coated Steel Conduit
 - 2.2.2 Rigid Nonmetallic Conduit
 - 2.2.3 Fittings for Metal Conduit
 - 2.2.3.1 Fittings for Rigid Metal Conduit
 - 2.2.4 Fittings for Rigid Nonmetallic Conduit
- 2.3 OUTLET BOXES AND COVERS
- 2.4 CABINETS, JUNCTION BOXES, AND PULL BOXES
- 2.5 WIRES AND CABLES
 - 2.5.1 Conductors
 - 2.5.1.1 Minimum Conductor Sizes
 - 2.5.2 Color Coding
 - 2.5.2.1 Ground and Neutral Conductors
 - 2.5.2.2 Ungrounded Conductors
 - 2.5.3 Insulation
 - 2.5.4 Bonding Conductors
- 2.6 SPLICES AND TERMINATION COMPONENTS
- 2.7 PANELBOARDS
 - 2.7.1 Circuit Breakers
- 2.8 MOTOR SHORT-CIRCUIT PROTECTOR (MSCP)
- 2.9 MOTORS
 - 2.9.1 High Efficiency Single-Phase Motors
 - 2.9.2 Premium Efficiency Polyphase and Single-Phase Motors
 - 2.9.3 Motor Sizes
 - 2.9.4 Wiring and Conduit
- 2.10 MOTOR CONTROLLERS
 - 2.10.1 Control Wiring
 - 2.10.2 Control Circuit Terminal Blocks

- 2.10.2.1 Types of Terminal Blocks
- 2.10.3 Control Circuits
- 2.10.4 Enclosures for Motor Controllers
- 2.10.5 Pushbutton Stations
- 2.10.6 Pilot and Indicating Lights
- 2.10.7 Reduced-Voltage Controllers
- 2.11 LOCKOUT REQUIREMENTS
- 2.12 GROUNDING AND BONDING EQUIPMENT
 - 2.12.1 Ground Rods
 - 2.12.2 Ground Bus
 - 2.12.3 Primary Bonding Busbar
- 2.13 HAZARDOUS LOCATIONS
- 2.14 MANUFACTURER'S NAMEPLATE
- 2.15 FIELD FABRICATED NAMEPLATES
- 2.16 WARNING SIGNS
- 2.17 FACTORY APPLIED FINISH
- 2.18 SOURCE QUALITY CONTROL
 - 2.18.1 Transformer Factory Tests
- 2.19 COORDINATED POWER SYSTEM PROTECTION

PART 3 EXECUTION

- 3.1 INSTALLATION
 - 3.1.1 Underground Service
 - 3.1.2 Hazardous Locations
 - 3.1.3 Wiring Methods
 - 3.1.3.1 Pull Wire
 - 3.1.4 Conduit Installation
 - 3.1.4.1 Restrictions Applicable to EMT
 - 3.1.4.2 Restrictions Applicable to Flexible Conduit
 - 3.1.4.3 Underground Conduit
 - 3.1.4.4 Conduit Through Floor Slabs
 - 3.1.4.5 Stub-Ups
 - 3.1.4.6 Conduit Support
 - 3.1.4.7 Directional Changes in Conduit Runs
 - 3.1.4.8 Locknuts and Bushings
 - 3.1.4.9 Flexible Connections
 - 3.1.5 Boxes, Outlets, and Supports
 - 3.1.5.1 Boxes
 - 3.1.5.2 Pull Boxes
 - 3.1.6 Mounting Heights
 - 3.1.7 Conductor Identification
 - 3.1.7.1 Marking Strips
 - 3.1.8 Splices
 - 3.1.9 Covers and Device Plates
 - 3.1.10 Grounding and Bonding
 - 3.1.10.1 Ground Rods
 - 3.1.10.2 Grounding Connections
 - 3.1.10.3 Ground Bus
 - 3.1.10.4 Resistance
 - 3.1.11 Equipment Connections
 - 3.1.12 Repair of Existing Work
 - 3.1.12.1 Workmanship
 - 3.1.12.2 Removal of Existing Electrical Distribution System
 - 3.1.12.3 Continuation of Service
- 3.2 FIELD FABRICATED NAMEPLATE MOUNTING
- 3.3 WARNING SIGN MOUNTING
- 3.4 FIELD APPLIED PAINTING
- 3.5 FIELD QUALITY CONTROL

- 3.5.1 Devices Subject to Manual Operation
- 3.5.2 600-Volt Wiring Test
- 3.5.3 Ground-Fault Receptacle Test
- 3.5.4 Grounding System Test
- 3.5.5 Phase Rotation Test

-- End of Section Table of Contents --

SECTION 26 20 00

INTERIOR DISTRIBUTION SYSTEM
08/19, CHG 3: 11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM B1	(2013) Standard Specification for Hard-Drawn Copper Wire
ASTM B8	(2011; R 2017) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft
ASTM D709	(2017) Standard Specification for Laminated Thermosetting Materials

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 81	(2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System
IEEE 100	(2000; Archived) The Authoritative Dictionary of IEEE Standards Terms
IEEE C2	(2017; Errata 1-2 2017; INT 1 2017) National Electrical Safety Code

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

NETA ATS	(2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems
----------	--

NATIONAL ELECTRICAL CONTRACTORS ASSOCIATION (NECA)

NECA NEIS 1	(2015) Standard for Good Workmanship in Electrical Construction
-------------	---

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C80.1	(2020) American National Standard for Electrical Rigid Steel Conduit (ERSC)
NEMA 250	(2020) Enclosures for Electrical Equipment (1000 Volts Maximum)
NEMA ICS 1	(2000; R 2020) Standard for Industrial

Control and Systems: General Requirements

NEMA ICS 2	(2000; R 2020) Industrial Control and Systems Controllers, Contactors, and Overload Relays Rated 600 V
NEMA ICS 4	(2015) Application Guideline for Terminal Blocks
NEMA ICS 6	(1993; R 2016) Industrial Control and Systems: Enclosures
NEMA MG 1	(2018) Motors and Generators
NEMA MG 10	(2017) Energy Management Guide for Selection and Use of Fixed Frequency Medium AC Squirrel-Cage Polyphase Induction Motors
NEMA MG 11	(1977; R 2012) Energy Management Guide for Selection and Use of Single Phase Motors
NEMA ST 20	(2014) Dry-Type Transformers for General Applications
NEMA TC 2	(2020) Standard for Electrical Polyvinyl Chloride (PVC) Conduit
NEMA TC 3	(2021) Polyvinyl Chloride (PVC) Fittings for Use With Rigid PVC Conduit and Tubing
NEMA Z535.4	(2011; R 2017) Product Safety Signs and Labels

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70	(2020; ERTA 20-1 2020; ERTA 20-2 2020; ERTA 20-3 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4; TIA 20-5; TIA 20-6; TIA 20-7; TIA 20-8; TIA 20-9; TIA 20-10; TIA 20-11; TIA 20-12; TIA 20-13; TIA 20-14; TIA 20-15; TIA 20-16; ERTA 20-4 2022) National Electrical Code
NFPA 70E	(2021) Standard for Electrical Safety in the Workplace
NFPA 780	(2023) Standard for the Installation of Lightning Protection Systems

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

TIA-607	(2019d) Generic Telecommunications Bonding and Grounding (Earthing) for Customer Premises
---------	---

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.147	The Control of Hazardous Energy (Lock
-----------------	---------------------------------------

Out/Tag Out)

29 CFR 1910.303

Electrical, General

UNDERWRITERS LABORATORIES (UL)

UL 6	(2007; Reprint Sep 2019) UL Standard for Safety Electrical Rigid Metal Conduit-Steel
UL 44	(2018; Reprint May 2021) UL Standard for Safety Thermoset-Insulated Wires and Cables
UL 50	(2015) UL Standard for Safety Enclosures for Electrical Equipment, Non-Environmental Considerations
UL 83	(2017; Reprint Mar 2020) UL Standard for Safety Thermoplastic-Insulated Wires and Cables
UL 467	(2013; Reprint Jun 2017) UL Standard for Safety Grounding and Bonding Equipment
UL 486A-486B	(2018) UL Standard for Safety Wire Connectors
UL 486C	(2018; Reprint May 2021) UL Standard for Safety Splicing Wire Connectors
UL 489	(2016; Rev 2019) UL Standard for Safety Molded-Case Circuit Breakers, Molded-Case Switches and Circuit-Breaker Enclosures
UL 506	(2017; Reprint Jan 2022) UL Standard for Safety Specialty Transformers
UL 508	(2018; Reprint Jul 2021) UL Standard for Safety Industrial Control Equipment
UL 510	(2020) UL Standard for Safety Polyvinyl Chloride, Polyethylene and Rubber Insulating Tape
UL 514A	(2013; Reprint Aug 2017) UL Standard for Safety Metallic Outlet Boxes
UL 514B	(2012; Reprint May 2020) Conduit, Tubing and Cable Fittings
UL 514C	(2014; Reprint Feb 2020) UL Standard for Safety Nonmetallic Outlet Boxes, Flush-Device Boxes, and Covers
UL 651	(2011; Reprint May 2022) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings
UL 674	(2011; Reprint Dec 2020) UL Standard for Safety Electric Motors and Generators for

Use in Hazardous (Classified) Locations

UL 943	(2016; Reprint Feb 2018) UL Standard for Safety Ground-Fault Circuit-Interrupters
UL 984	(1996; Reprint Sep 2005) Hermetic Refrigerant Motor-Compressors
UL 1063	(2017; Reprint Jun 2020) UL Standard for Safety Machine-Tool Wires and Cables
UL 1203	(2013; Reprint Apr 2022) UL Standard for Safety Explosion-Proof and Dust-Ignition-Proof Electrical Equipment for Use in Hazardous (Classified) Locations

1.2 DEFINITIONS

Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE 100.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Panelboards; G

Marking Strips Drawings; G

SD-03 Product Data

Circuit Breakers; G

Motor Controllers; G

Primary Bonding Busbar; G

SD-06 Test Reports

600-volt Wiring Test; G

Grounding System Test; G

Ground-fault Receptacle Test; G

SD-09 Manufacturer's Field Reports

Transformer Factory Tests

SD-10 Operation and Maintenance Data

Electrical Systems, Data Package 5; G

1.4 QUALITY ASSURANCE

1.4.1 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Provide equipment, materials, installation, and workmanship in accordance with NFPA 70 unless more stringent requirements are specified or indicated. NECA NEIS 1 shall be considered the minimum standard for workmanship.

1.4.2 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship and:

- a. Have been in satisfactory commercial or industrial use for 2 years prior to bid opening including applications of equipment and materials under similar circumstances and of similar size.
- b. Have been on sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period.
- c. Where two or more items of the same class of equipment are required, provide products of a single manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.4.2.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.4.2.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable.

1.5 MAINTENANCE

1.5.1 Electrical Systems

Submit operation and maintenance data in accordance with Section 01 78 23, OPERATION AND MAINTENANCE DATA and as specified herein. Submit operation and maintenance manuals for electrical systems that provide basic data relating to the design, operation, and maintenance of the electrical distribution system for the building. Include the following:

- a. Single line diagram of the "as-built" building electrical system.
- b. Schematic diagram of electrical control system (other than HVAC, covered elsewhere).

- c. Manufacturers' operating and maintenance manuals on active electrical equipment.

1.6 WARRANTY

Provide equipment items supported by service organizations that are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

PART 2 PRODUCTS

2.1 MATERIALS AND EQUIPMENT

As a minimum, meet requirements of UL, where UL standards are established for those items, and requirements of NFPA 70 for all materials, equipment, and devices.

2.2 CONDUIT AND FITTINGS

Conform to the following:

2.2.1 Rigid Metallic Conduit

2.2.1.1 Rigid, Threaded Zinc-Coated Steel Conduit

ANSI C80.1, UL 6.

2.2.2 Rigid Nonmetallic Conduit

PVC Type EPC-40, and EPC-80 in accordance with NEMA TC 2, UL 651.

2.2.3 Fittings for Metal Conduit

UL 514B. Ferrous fittings: cadmium- or zinc-coated in accordance with UL 514B.

2.2.3.1 Fittings for Rigid Metal Conduit

Threaded-type. Split couplings unacceptable.

2.2.4 Fittings for Rigid Nonmetallic Conduit

NEMA TC 3 for PVC, and UL 514B.

2.3 OUTLET BOXES AND COVERS

UL 514A, cadmium- or zinc-coated, if ferrous metal. UL 514C, if nonmetallic.

2.4 CABINETS, JUNCTION BOXES, AND PULL BOXES

UL 50; volume greater than 1640 mL, NEMA Type 1 enclosure; sheet steel, hot-dip, zinc-coated. Where exposed to wet, damp, or corrosive environments, NEMA Type 4X.

2.5 WIRES AND CABLES

Provide wires and cables in accordance applicable requirements of NFPA 70

and UL for type of insulation, jacket, and conductor specified or indicated. Do not use wires and cables manufactured more than 12 months prior to date of delivery to site.

2.5.1 Conductors

Provide the following:

- a. Conductor sizes and capacities shown are based on copper, unless indicated otherwise.
- b. Conductors No. 8 AWG and larger diameter: stranded.
- c. Conductors No. 10 AWG and smaller diameter: solid.
- d. Conductors for remote control, alarm, and signal circuits, classes 1, 2, and 3: stranded unless specifically indicated otherwise.
- e. All conductors: copper.

2.5.1.1 Minimum Conductor Sizes

Provide minimum conductor size in accordance with the following:

- a. Branch circuits: No. 12 AWG.
- b. Class 1 remote-control and signal circuits: No. 14 AWG.
- c. Class 2 low-energy, remote-control and signal circuits: No. 16 AWG.
- d. Class 3 low-energy, remote-control, alarm and signal circuits: No. 22 AWG.

2.5.2 Color Coding

Provide color coding for service, feeder, branch, control, and signaling circuit conductors.

2.5.2.1 Ground and Neutral Conductors

Provide color coding of ground and neutral conductors as follows:

- a. Grounding conductors: Green.
- b. Neutral conductors: White.
- c. Exception, where neutrals of more than one system are installed in same raceway or box, other neutrals color coding: white with a different colored (not green) stripe for each.

2.5.2.2 Ungrounded Conductors

Provide color coding of ungrounded conductors in different voltage systems as follows:

- a. 208/120 volt, three-phase
 - (1) Phase A - black

(2) Phase B - red

(3) Phase C - blue

b. 480/277 volt, three-phase

(1) Phase A - brown

(2) Phase B - orange

(3) Phase C - yellow

c. 120/240 volt, single phase: Black and red

2.5.3 Insulation

Unless specified or indicated otherwise or required by NFPA 70, provide power and lighting wires rated for 600-volts, Type THWN/THHN conforming to UL 83 or Type XHHW conforming to UL 44, except that grounding wire may be type TW conforming to UL 83; remote-control and signal circuits: Type TW or TF, conforming to UL 83. Where equipment or devices require 90-degree Centigrade (C) conductors, provide only conductors with 90-degree C insulation or better.

2.5.4 Bonding Conductors

ASTM B1, solid bare copper wire for sizes No. 8 AWG and smaller diameter; ASTM B8, Class B, stranded bare copper wire for sizes No. 6 AWG and larger diameter.

2.6 SPLICES AND TERMINATION COMPONENTS

UL 486A-486B for wire connectors and UL 510 for insulating tapes. Connectors for No. 10 AWG and smaller diameter wires: insulated, pressure-type in accordance with UL 486A-486B or UL 486C (twist-on splicing connector). Provide solderless terminal lugs on stranded conductors.

2.7 PANELBOARDS

2.7.1 Circuit Breakers

UL 489, thermal magnetic-type having a minimum short-circuit current rating equal to the short-circuit current rating of the panelboard in which the circuit breaker will be mounted. Breaker terminals: UL listed as suitable for type of conductor provided. Series rated circuit breakers and plug-in circuit breakers are unacceptable.

2.8 MOTOR SHORT-CIRCUIT PROTECTOR (MSCP)

Motor short-circuit protectors, also called motor circuit protectors (MCPs): UL 508 and UL 489, and provided as shown. Provide MSCPs that consist of an adjustable instantaneous trip circuit breaker used only in conjunction with a combination motor controller which provides coordinated motor branch-circuit overload and short-circuit protection. Rate MSCPs in accordance with the requirements of NFPA 70.

2.9 MOTORS

Provide motors in accordance with the following:

- a. NEMA MG 1.
- b. Hermetic-type sealed motor compressors: Also comply with UL 984.
- c. Provide the size in terms of hp or full-load current and other characteristics, of each motor as indicated or specified.
- d. Determine specific motor characteristics to ensure provision of correctly sized starters and overload heaters.
- e. Rate motors for operation on 208-volt, 3-phase circuits with a terminal voltage rating of 200 volts, and those for operation on 480-volt, 3-phase circuits with a terminal voltage rating of 460 volts.
- f. Use motors designed to operate at full capacity with voltage variation of plus or minus 10 percent of motor voltage rating.
- g. Unless otherwise indicated, use continuous duty type motors if rated 746 Watts and above.
- h. Where fuse protection is specifically recommended by the equipment manufacturer, provide fused switches in lieu of non-fused switches indicated.

2.9.1 High Efficiency Single-Phase Motors

Single-phase fractional-horsepower alternating-current motors: high efficiency types are not acceptable. In exception, for special purpose motors and motor-driven equipment with a minimum seasonal or overall efficiency rating, such as a SEER rating, provide equipment with motor to meet the overall system rating indicated.

2.9.2 Premium Efficiency Polyphase and Single-Phase Motors

Select polyphase and continuous-duty single phase motors based on high efficiency characteristics relative to typical characteristics and applications as listed in NEMA MG 10 and NEMA MG 11. In addition, continuous rated, polyphase squirrel-cage medium induction motors must meet the requirements for premium efficiency electric motors in accordance with NEMA MG 1, including the NEMA full load efficiency ratings. In exception, for motor-driven equipment with a minimum seasonal or overall efficiency rating, such as a SEER rating, provide equipment with motor to meet the overall system rating indicated.

2.9.3 Motor Sizes

Provide size for duty to be performed, not exceeding the full-load nameplate current rating when driven equipment is operated at specified capacity under most severe conditions likely to be encountered. When motor size provided differs from size indicated or specified, make adjustments to wiring, disconnect devices, and branch circuit protection to accommodate equipment actually provided. Provide controllers for motors rated 1-hp and above with electronic phase-voltage monitors designed to protect motors from phase-loss, undervoltage, and

overvoltage. Provide protection for motors from immediate restart by a time adjustable restart relay.

2.9.4 Wiring and Conduit

Provide internal wiring for components of packaged equipment as an integral part of the equipment. Provide power wiring and conduit for field-installed equipment and motor control equipment forming part of motor control centers or switchgear assemblies, the conduit and wiring connecting such centers, assemblies, or other power sources to equipment as specified herein. Power wiring and conduit: conform to the requirements specified herein. Control wiring: provided under, and conform to, the requirements of the section specifying the associated equipment.

2.10 MOTOR CONTROLLERS

Provide motor controllers in accordance with the following:

- a. UL 508, NEMA ICS 1, and NEMA ICS 2.
- b. Provide controllers with thermal overload protection in each phase, and one spare normally open auxiliary contact, and one spare normally closed auxiliary contact.
- c. Provide controllers for motors rated 1-hp and above with electronic phase-voltage monitors designed to protect motors from phase-loss, undervoltage, and overvoltage.
- d. Provide protection for motors from immediate restart by a time adjustable restart relay.
- e. When used with pressure, float, or similar automatic-type or maintained-contact switch, provide a hand/off/automatic selector switch with the controller.
- f. Connections to selector switch: wired such that only normal automatic regulatory control devices are bypassed when switch is in "hand" position.
- g. Safety control devices, such as low and high pressure cutouts, high temperature cutouts, and motor overload protective devices: connected in motor control circuit in "hand" and "automatic" positions.
- h. Control circuit connections to hand/off/automatic selector switch or to more than one automatic regulatory control device: made in accordance with indicated or manufacturer's approved wiring diagram.
- i. Provide selector switch with the means for locking in any position.
- j. Provide a disconnecting means, capable of being locked in the open position, for the motor that is located in sight from the motor location and the driven machinery location. As an alternative, provide a motor controller disconnect, capable of being locked in the open position, to serve as the disconnecting means for the motor if it is in sight from the motor location and the driven machinery location.
- k. Overload protective devices: provide adequate protection to motor windings; be thermal inverse-time-limit type; and include manual

reset-type pushbutton on outside of motor controller case.

- l. Cover of combination motor controller and manual switch or circuit breaker: interlocked with operating handle of switch or circuit breaker so that cover cannot be opened unless handle of switch or circuit breaker is in "off" position.

2.10.1 Control Wiring

Provide control wiring in accordance with the following:

- a. All control wire: stranded tinned copper switchboard wire with 600-volt flame-retardant insulation Type SIS meeting UL 44, or Type MTW meeting UL 1063, and passing the VW-1 flame tests included in those standards.
- b. Hinge wire: Class K stranding.
- c. Current transformer secondary leads: not smaller than No. 10 AWG.
- d. Control wire minimum size: No. 14 AWG.
- e. Power wiring for 480-volt circuits and below: the same type as control wiring with No. 12 AWG minimum size.
- f. Provide wiring and terminal arrangement on the terminal blocks to permit the individual conductors of each external cable to be terminated on adjacent terminal points.

2.10.2 Control Circuit Terminal Blocks

Provide control circuit terminal blocks in accordance with the following:

- a. NEMA ICS 4.
- b. Control circuit terminal blocks for control wiring: molded or fabricated type with barriers, rated not less than 600 volts.
- c. Provide terminals with removable binding, fillister or washer head screw type, or of the stud type with contact and locking nuts.
- d. Terminals: not less than No. 10 in size with sufficient length and space for connecting at least two indented terminals for 10 AWG conductors to each terminal.
- e. Terminal arrangement: subject to the approval of the Contracting Officer with not less than four spare terminals or 10 percent, whichever is greater, provided on each block or group of blocks.
- f. Modular, pull apart, terminal blocks are acceptable provided they are of the channel or rail-mounted type.
- g. Submit data showing that any proposed alternate will accommodate the specified number of wires, are of adequate current-carrying capacity, and are constructed to assure positive contact between current-carrying parts.

2.10.2.1 Types of Terminal Blocks

- a. Short-Circuiting Type: Short-circuiting type terminal blocks: furnished for all current transformer secondary leads with provision for shorting together all leads from each current transformer without first opening any circuit. Terminal blocks: comply with the requirements of paragraph CONTROL CIRCUIT TERMINAL BLOCKS above.
- b. Load Type: Load terminal blocks rated not less than 600 volts and of adequate capacity: provided for the conductors for NEMA Size 3 and smaller motor controllers and for other power circuits, except those for feeder tap units. Provide terminals of either the stud type with contact nuts and locking nuts or of the removable screw type, having length and space for at least two indented terminals of the size required on the conductors to be terminated. For conductors rated more than 50 amperes, provide screws with hexagonal heads. Conducting parts between connected terminals must have adequate contact surface and cross-section to operate without overheating. Provide each connected terminal with the circuit designation or wire number placed on or near the terminal in permanent contrasting color.

2.10.3 Control Circuits

Control circuits: maximum voltage of 120 volts derived from control transformer in same enclosure. Transformers: conform to UL 506, as applicable. Transformers, other than transformers in bridge circuits: provide primaries wound for voltage available and secondaries wound for correct control circuit voltage. Size transformers so that 80 percent of rated capacity equals connected load. Provide disconnect switch on primary side. Provide fuses in each ungrounded primary feeder. Provide one fused secondary lead with the other lead grounded.

2.10.4 Enclosures for Motor Controllers

NEMA ICS 6.

2.10.5 Pushbutton Stations

Provide with "start/stop" momentary contacts having one normally open and one normally closed set of contacts, and red lights to indicate when motor is running. Stations: heavy duty, oil-tight design.

2.10.6 Pilot and Indicating Lights

Provide LED cluster lamps.

2.10.7 Reduced-Voltage Controllers

Provide for polyphase motors 20 horsepower and larger. Reduced-voltage starters: single-step, closed transition solid state-type, or as indicated, with an adjustable time interval between application of reduced and full voltages to motors. Provide with integral bypass contactor. Provide with 300 percent start-up current limitation.

2.11 LOCKOUT REQUIREMENTS

Provide circuit breakers, disconnecting means, and other devices that are electrical energy-isolating capable of being locked out for machines and other equipment to prevent unexpected startup or release of stored energy

in accordance with 29 CFR 1910.147, NFPA 70E and 29 CFR 1910.303. Comply with requirements of Division 23, "Mechanical" for mechanical isolation of machines and other equipment.

2.12 GROUNDING AND BONDING EQUIPMENT

2.12.1 Ground Rods

UL 467. Ground rods: cone pointed copper-clad steel, with minimum diameter of 19 mm and minimum length of 3050 mm.

2.12.2 Ground Bus

Copper ground bus: provided in the electrical equipment rooms as indicated.

2.12.3 Primary Bonding Busbar

Provide corrosion-resistant grounding busbar suitable for indoor installation in accordance with TIA-607. Busbars: plated for reduced contact resistance. If not plated, clean the busbar prior to fastening the conductors to the busbar and apply an anti-oxidant to the contact area to control corrosion and reduce contact resistance. The Primary bonding busbar (PBB): sized in accordance with the immediate application requirements and with consideration of future growth. Provide Primary bonding busbars with the following:

- a. Predrilled copper busbar provided with holes for use with standard sized lugs,
- b. Minimum dimensions of 6 mm thick by 100 mm wide for the PBB with length as indicated;
- c. Listed by a nationally recognized testing laboratory.

2.13 HAZARDOUS LOCATIONS

Electrical materials, equipment, and devices for installation in hazardous locations, as defined by NFPA 70: specifically approved by Underwriters' Laboratories, Inc., or Factory Mutual for particular "Class," "Division," and "Group" of hazardous locations involved. Boundaries and classifications of hazardous locations: as indicated. Equipment in hazardous locations: comply with UL 1203 for electrical equipment and industrial controls and UL 674 for motors.

2.14 MANUFACTURER'S NAMEPLATE

Provide on each item of equipment a nameplate bearing the manufacturer's name, address, model number, and serial number securely affixed in a conspicuous place; the nameplate of the distributing agent will not be acceptable.

2.15 FIELD FABRICATED NAMEPLATES

Provide field fabricated nameplates in accordance with the following:

- a. ASTM D709.
- b. Provide laminated plastic nameplates for each equipment enclosure,

relay, switch, and device; as specified or as indicated on the drawings.

- c. Each nameplate inscription: identify the function and, when applicable, the position.
- d. Nameplates: melamine plastic, 3 mm thick, white with black center core.
- e. Provide red laminated plastic label with white center core where indicated.
- f. Surface: matte finish. Corners: square. Accurately align lettering and engrave into the core.
- g. Minimum size of nameplates: 25 by 65 mm.
- h. Lettering size and style: a minimum of 6.35 mm high normal block style.

2.16 WARNING SIGNS

Provide warning signs for flash protection in accordance with NFPA 70E and NEMA Z535.4 for switchboards, panelboards, industrial control panels, and motor control centers that are in other than dwelling occupancies and are likely to require examination, adjustment, servicing, or maintenance while energized. Provide field installed signs to warn qualified persons of potential electric arc flash hazards when warning signs are not provided by the manufacturer. Provide marking that is clearly visible to qualified persons before examination, adjustment, servicing, or maintenance of the equipment.

2.17 FACTORY APPLIED FINISH

Provide factory-applied finish on electrical equipment in accordance with the following:

- a. NEMA 250 corrosion-resistance test and the additional requirements as specified herein.
- b. Interior and exterior steel surfaces of equipment enclosures: thoroughly cleaned followed by a rust-inhibitive phosphatizing or equivalent treatment prior to painting.
- c. Exterior surfaces: free from holes, seams, dents, weld marks, loose scale or other imperfections.
- d. Interior surfaces: receive not less than one coat of corrosion-resisting paint in accordance with the manufacturer's standard practice.
- e. Exterior surfaces: primed, filled where necessary, and given not less than two coats baked enamel with semigloss finish.
- f. Equipment located indoors: ANSI Light Gray, and equipment located outdoors: ANSI Dark Gray.
- g. Provide manufacturer's coatings for touch-up work and as specified in paragraph FIELD APPLIED PAINTING.

2.18 SOURCE QUALITY CONTROL

2.18.1 Transformer Factory Tests

Submittal: include routine NEMA ST 20 transformer test results on each transformer and also provide the results of NEMA "design" and "prototype" tests that were made on transformers electrically and mechanically equal to those specified.

2.19 COORDINATED POWER SYSTEM PROTECTION

Prepare analyses as specified in Section 26 28 01.00 10 COORDINATED POWER SYSTEM PROTECTION.

PART 3 EXECUTION

3.1 INSTALLATION

Electrical installations, including weatherproof and hazardous locations and ducts, plenums and other air-handling spaces: conform to requirements of NFPA 70 and IEEE C2 and to requirements specified herein.

3.1.1 Underground Service

Underground service conductors and associated conduit: continuous from service entrance equipment to outdoor power system connection.

3.1.2 Hazardous Locations

Perform work in hazardous locations, as defined by NFPA 70, in strict accordance with NFPA 70 for particular "Class," "Division," and "Group" of hazardous locations involved. Provide conduit and cable seals where required by NFPA 70. Provide conduit with tapered threads.

3.1.3 Wiring Methods

Provide insulated conductors installed in rigid steel conduit - or EMT, except where specifically indicated or specified otherwise or required by NFPA 70 to be installed otherwise. Grounding conductor: separate from electrical system neutral conductor. Provide insulated green equipment grounding conductor for circuit(s) installed in conduit and raceways. Minimum conduit size: 20 mm in diameter for low voltage lighting and power circuits.

3.1.3.1 Pull Wire

Install pull wires in empty conduits. Pull wire: plastic having minimum 890-N force tensile strength. Leave minimum 915 mm of slack at each end of pull wire.

3.1.4 Conduit Installation

Unless indicated otherwise, conceal conduit under floor slabs and within finished walls, ceilings, and floors. Install conduit parallel with or at right angles to ceilings, walls, and structural members where located above accessible ceilings and where conduit will be visible after completion of project. Run conduits under floor slab as if exposed.

3.1.4.1 Restrictions Applicable to EMT

- a. Do not install underground.
- b. Do not encase in concrete, mortar, grout, or other cementitious materials.
- c. Do not use in areas subject to physical damage including but not limited to equipment rooms where moving or replacing equipment could physically damage the EMT.
- d. Do not use in hazardous areas.
- e. Do not use outdoors.

3.1.4.2 Restrictions Applicable to Flexible Conduit

Use only as specified in paragraph FLEXIBLE CONNECTIONS.

3.1.4.3 Underground Conduit

Refer to Section 33 71 02 UNDERGROUND ELECTRICAL DISTRIBUTION.

3.1.4.4 Conduit Through Floor Slabs

Where conduits rise through floor slabs, do not allow curved portion of bends to be visible above finished slab. Where conduit rises through slab-on grade, seal all electrical penetrations to address radon mitigation and prevent infiltration of air, insects, and vermin.

3.1.4.5 Stub-Ups

Provide conduits stubbed up through concrete floor for connection to free-standing equipment with adjustable top or coupling threaded inside for plugs, set flush with finished floor. Extend conductors to equipment in rigid steel conduit, except that flexible metal conduit may be used 150 mm above floor. Where no equipment connections are made, install screwdriver-operated threaded flush plugs in conduit end.

3.1.4.6 Conduit Support

Support conduit by pipe straps, wall brackets, slotted metallic channel, threaded rod conduit hangers, or ceiling trapeze. Plastic cable ties are not acceptable. Fasten by wood screws to wood; by toggle bolts on hollow masonry units; by concrete inserts or expansion bolts on concrete or brick; and by machine screws, welded threaded studs, or spring-tension clamps on steel work. Threaded C-clamps may be used on rigid steel conduit only. Do not weld conduits or pipe straps to steel structures. Do not exceed one-fourth proof test load for load applied to fasteners. Provide vibration resistant and shock-resistant fasteners attached to concrete ceiling. Do not cut main reinforcing bars for any holes cut to depth of more than 40 mm in reinforced concrete beams or to depth of more than 20 mm in concrete joints. Fill unused holes. In partitions of light steel construction, use sheet metal screws. In suspended-ceiling construction, run conduit above ceiling. Do not support conduit by ceiling support system. Conduit and box systems: supported independently of both (a) tie wires supporting ceiling grid system, and (b) ceiling grid system into which ceiling panels are placed. Do not share supporting means between electrical raceways and mechanical piping or ducts.

Coordinate installation with above-ceiling mechanical systems to assure maximum accessibility to all systems. Spring-steel fasteners may be used for lighting branch circuit conduit supports in suspended ceilings in dry locations. Where conduit crosses building expansion joints, provide suitable watertight expansion fitting that maintains conduit electrical continuity by bonding jumpers or other means. For conduits greater than 63 mm inside diameter, provide supports to resist forces of 0.5 times the equipment weight in any direction and 1.5 times the equipment weight in the downward direction.

3.1.4.7 Directional Changes in Conduit Runs

Make changes in direction of runs with symmetrical bends or cast-metal fittings. Make field-made bends and offsets with hickey or conduit-bending machine. Do not install crushed or deformed conduits. Avoid trapped conduits. Prevent plaster, dirt, or trash from lodging in conduits, boxes, fittings, and equipment during construction. Free clogged conduits of obstructions.

3.1.4.8 Locknuts and Bushings

Fasten conduits to sheet metal boxes and cabinets with two locknuts where required by NFPA 70, where insulated bushings are used, and where bushings cannot be brought into firm contact with the box; otherwise, use at least minimum single locknut and bushing. Provide locknuts with sharp edges for digging into wall of metal enclosures. Install bushings on ends of conduits, and provide insulating type where required by NFPA 70.

3.1.4.9 Flexible Connections

Provide flexible steel conduit between 915 and 1830 mm in length for recessed and semirecessed lighting fixtures; for equipment subject to vibration, noise transmission, or movement; and for motors. Install flexible conduit to allow 20 percent slack. Minimum flexible steel conduit size: 16 mm diameter. Provide liquid tight flexible conduit in wet and damp locations for equipment subject to vibration, noise transmission, movement or motors. Provide separate ground conductor across flexible connections. Plastic cable ties are not acceptable as a support method.

3.1.5 Boxes, Outlets, and Supports

Provide boxes in wiring and raceway systems wherever required for pulling of wires, making connections, and mounting of devices or fixtures. Boxes for metallic raceways: cast-metal, hub-type when located in wet locations, when surface mounted on outside of exterior surfaces, when surface mounted on interior walls exposed up to 2135 mm above floors and walkways, or when installed in hazardous areas and when specifically indicated. Boxes in other locations: sheet steel. Provide each box with volume required by NFPA 70 for number of conductors enclosed in box. Boxes for mounting lighting fixtures: minimum 100 mm square, or octagonal, except that smaller boxes may be installed as required by fixture configurations, as approved. Provide gaskets for cast-metal boxes installed in wet locations and boxes installed flush with outside of exterior surfaces. Provide separate boxes for flush or recessed fixtures when required by fixture terminal operating temperature; provide readily removable fixtures for access to boxes unless ceiling access panels are provided. Support boxes and pendants for surface-mounted fixtures on suspended ceilings independently of ceiling supports. Fasten boxes and

supports with wood screws on wood, with bolts and expansion shields on concrete or brick, with toggle bolts on hollow masonry units, and with machine screws or welded studs on steel. In open overhead spaces, cast boxes threaded to raceways need not be separately supported except where used for fixture support; support sheet metal boxes directly from building structure or by bar hangers. Where bar hangers are used, attach bar to raceways on opposite sides of box, and support raceway with approved-type fastener maximum 610 mm from box. When penetrating reinforced concrete members, avoid cutting reinforcing steel.

3.1.5.1 Boxes

Boxes for use with raceway systems: minimum 40 mm deep, except where shallower boxes required by structural conditions are approved. Boxes for other than lighting fixture outlets: minimum 100 mm square, except that 100 by 50 mm boxes may be used where only one raceway enters outlet. Mount outlet boxes flush in finished walls.

3.1.5.2 Pull Boxes

Construct of at least minimum size required by NFPA 70 of galvanized sheet steel, except where cast-metal boxes are required in locations specified herein. Provide boxes with screw-fastened covers. Where several feeders pass through common pull box, tag feeders to indicate clearly electrical characteristics, circuit number, and panel designation.

3.1.6 Mounting Heights

Mount panelboards, circuit breakers, motor controller and disconnecting switches so height of center of grip of the operating handle of the switch or circuit breaker at its highest position is maximum 2007 mm above floor or working platform or as allowed in Section 404.8 per NFPA 70. Mount lighting switches 1220 mm above finished floor. Mount receptacles and telecommunications outlets 460 mm above finished floor, unless otherwise indicated. Wall-mounted telecommunications outlets: mounted at height 1525 mm above finished floor. Mount other devices as indicated. Measure mounting heights of wiring devices and outlets in non-hazardous areas to center of device or outlet. Measure mounting heights of receptacle outlet boxes in the hazardous areas to the bottom of the outlet box.

3.1.7 Conductor Identification

Provide conductor identification within each enclosure where tap, splice, or termination is made. For conductors No. 6 AWG and smaller diameter, provide color coding by factory-applied, color-impregnated insulation. For conductors No. 4 AWG and larger diameter, provide color coding by plastic-coated, self-sticking markers; colored nylon cable ties and plates; or heat shrink-type sleeves. Identify control circuit terminations in accordance with manufacturer's recommendations.

3.1.7.1 Marking Strips

Provide marking strips for identification of power distribution, control, data, and communications cables in accordance with the following:

- a. Provide white or other light-colored plastic marking strips, fastened by screws to each terminal block, for wire designations.
- b. Use permanent ink for the wire numbers.

- c. Provide reversible marking strips to permit marking both sides, or provide two marking strips with each block.
- d. Size marking strips to accommodate the two sets of wire numbers.
- e. Assign a device designation in accordance with NEMA ICS 1 to each device to which a connection is made. Mark each device terminal to which a connection is made with a distinct terminal marking corresponding to the wire designation used on the Contractor's schematic and connection diagrams.
- f. The wire (terminal point) designations used on the Contractor's wiring diagrams and printed on terminal block marking strips may be according to the Contractor's standard practice; however, provide additional wire and cable designations for identification of remote (external) circuits for the Government's wire designations.
- g. Prints of the marking strips drawings submitted for approval will be so marked and returned to the Contractor for addition of the designations to the terminal strips and tracings, along with any rearrangement of points required.

3.1.8 Splices

Make splices in accessible locations. Make splices in conductors No. 10 AWG and smaller diameter with insulated, pressure-type connector. Make splices in conductors No. 8 AWG and larger diameter with solderless connector, and cover with insulation material equivalent to conductor insulation.

3.1.9 Covers and Device Plates

Install with edges in continuous contact with finished wall surfaces without use of mats or similar devices. Plaster fillings are not permitted. Install plates with alignment tolerance of 0.58 mm. Use of sectional-type device plates are not permitted. Provide gasket for plates installed in wet locations.

3.1.10 Grounding and Bonding

Provide in accordance with NFPA 70 and NFPA 780. Ground exposed, non-current-carrying metallic parts of electrical equipment, access flooring support system, metallic raceway systems, grounding conductor in metallic and nonmetallic raceways, telecommunications system grounds, and neutral conductor of wiring systems. Make ground connection at main service equipment, and extend grounding conductor to point of entrance of metallic water service. Make connection to water pipe by suitable ground clamp or lug connection to plugged tee. If flanged pipes are encountered, make connection with lug bolted to street side of flanged connection. Supplement metallic water service grounding system with additional made electrode in compliance with NFPA 70. Make ground connection to driven ground rods on exterior of building. Bond additional driven rods together with a minimum of 4/0 AWG soft bare copper wire buried to a depth of at least 300 mm. Interconnect all grounding media in or on the structure to provide a common ground potential. This includes lightning protection, electrical service, telecommunications system grounds, as well as underground metallic piping systems. Use main size lightning conductors for interconnecting these grounding systems to the lightning protection

system. In addition to the requirements specified herein, provide telecommunications grounding in accordance with TIA-607.

3.1.10.1 Ground Rods

Provide ground rods and measure the resistance to ground using the fall-of-potential method described in IEEE 81. Do not exceed 25 ohms under normally dry conditions for the maximum resistance of a driven ground. If this resistance cannot be obtained with a single rod, provide additional rods, spaced 3000 mm on center. If the resultant resistance exceeds 25 ohms measured not less than 48 hours after rainfall, notify the Contracting Officer who will decide on the number of ground rods to add.

3.1.10.2 Grounding Connections

Make grounding connections which are buried or otherwise normally inaccessible, excepting specifically those connections for which access for periodic testing is required, by exothermic weld or high compression connector.

- a. Make exothermic welds strictly in accordance with the weld manufacturer's written recommendations. Welds which are "puffed up" or which show convex surfaces indicating improper cleaning are not acceptable. Mechanical connectors are not required at exothermic welds.
- b. Make high compression connections using a hydraulic or electric compression tool to provide the correct circumferential pressure. Provide tools and dies as recommended by the manufacturer. Use an embossing die code or other standard method to provide visible indication that a connector has been adequately compressed on the ground wire.

3.1.10.3 Ground Bus

Provide a copper ground bus as indicated. Noncurrent-carrying metal parts of transformer neutrals and other electrical equipment: effectively grounded by bonding to the ground bus. Bond the ground bus to both the entrance ground, and to a ground rod or rods as specified above having the upper ends terminating approximately 100 mm above the floor. Make connections and splices of the brazed, welded, bolted, or pressure-connector type, except use pressure connectors or bolted connections for connections to removable equipment. For raised floor equipment rooms in computer and data processing centers, provide a minimum of four, one at each corner, ground buses connected to the building grounding system. Use bolted connections in lieu of thermoweld, so they can be changed as required by additions and alterations.

3.1.10.4 Resistance

Maximum resistance-to-ground of grounding system: do not exceed 5 ohms under dry conditions. Where resistance obtained exceeds 5 ohms, contact Contracting Officer for further instructions.

3.1.11 Equipment Connections

Provide power wiring for the connection of motors and control equipment under this section of the specification. Except as otherwise specifically noted or specified, automatic control wiring, control devices, and

protective devices within the control circuitry are not included in this section of the specifications and are provided under the section specifying the associated equipment.

3.1.12 Repair of Existing Work

Perform repair of existing work, demolition, and modification of existing electrical distribution systems as follows:

3.1.12.1 Workmanship

Lay out work in advance. Exercise care where cutting, channeling, chasing, or drilling of floors, walls, partitions, ceilings, or other surfaces is necessary for proper installation, support, or anchorage of conduit, raceways, or other electrical work. Repair damage to buildings, piping, and equipment using skilled craftsmen of trades involved.

3.1.12.2 Removal of Existing Electrical Distribution System

Removal of existing electrical distribution system equipment includes equipment's associated wiring, including conductors, cables, exposed conduit, surface metal raceways, boxes, and fittings, back to equipment's power source as indicated.

3.1.12.3 Continuation of Service

Maintain continuity of existing circuits of equipment to remain. Maintain existing circuits of equipment energized. Restore circuits wiring and power which are to remain but were disturbed during demolition back to original condition.

3.2 FIELD FABRICATED NAMEPLATE MOUNTING

Provide number, location, and letter designation of nameplates as indicated. Fasten nameplates to the device with a minimum of two sheet-metal screws or two rivets.

3.3 WARNING SIGN MOUNTING

Provide the number of signs required to be readable from each accessible side. Space the signs in accordance with NFPA 70E.

3.4 FIELD APPLIED PAINTING

Paint electrical equipment as required to match finish of adjacent surfaces or to meet the indicated or specified safety criteria.

3.5 FIELD QUALITY CONTROL

Furnish test equipment and personnel and submit written copies of test results. Give Contracting Officer 5 working days notice prior to each test. Where applicable, test electrical equipment in accordance with NETA ATS.

3.5.1 Devices Subject to Manual Operation

Operate each device subject to manual operation at least five times, demonstrating satisfactory operation each time.

3.5.2 600-Volt Wiring Test

Test wiring rated 600 volt and less to verify that no short circuits or accidental grounds exist. Perform insulation resistance tests on wiring No. 6 AWG and larger diameter using instrument which applies voltage of 1,000 volts DC for 600 volt rated wiring and 500 volts DC for 300 volt rated wiring per NETA ATS to provide direct reading of resistance. All existing wiring to be reused must also be tested.

3.5.3 Ground-Fault Receptacle Test

Test ground-fault receptacles with a "load" (such as a plug in light) to verify that the "line" and "load" leads are not reversed. Press the TEST button and then the RESET button to verify by LED status that the device is a self-test model as specified in UL 943.

3.5.4 Grounding System Test

Test grounding system to ensure continuity, and that resistance to ground is not excessive. Test each ground rod for resistance to ground before making connections to rod; tie grounding system together and test for resistance to ground. Make resistance measurements in dry weather, not earlier than 48 hours after rainfall. Submit written results of each test to Contracting Officer, and indicate location of rods as well as resistance and soil conditions at time measurements were made.

3.5.5 Phase Rotation Test

Perform phase rotation test to ensure proper rotation of service power prior to operation of new or reinstalled equipment using a phase rotation meter. Follow the meter manual directions performing the test.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 28 01.00 10

COORDINATED POWER SYSTEM PROTECTION

08/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SYSTEM DESCRIPTION
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 System Coordinator

PART 2 PRODUCTS

- 2.1 COORDINATED POWER SYSTEM PROTECTION
 - 2.1.1 Scope of Analyses
 - 2.1.2 Determination of Facts
 - 2.1.3 Single Line Diagram
 - 2.1.4 Fault Current Analysis
 - 2.1.4.1 Method
 - 2.1.4.2 Data
 - 2.1.4.3 Fault Current Availability
 - 2.1.5 Coordination Study
 - 2.1.6 Study report

PART 3 EXECUTION

-- End of Section Table of Contents --

SECTION 26 28 01.00 10

COORDINATED POWER SYSTEM PROTECTION
08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 242 (2001; Errata 2003) Recommended Practice
for Protection and Coordination of
Industrial and Commercial Power Systems -
Buff Book

IEEE 399 (1997) Brown Book IEEE Recommended
Practice for Power Systems Analysis

1.2 SYSTEM DESCRIPTION

The power system covered by this specification consists of: electrical panels and motors.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Fault Current Analysis

Protective Device Coordination Study

System Coordinator

1.4 QUALITY ASSURANCE

1.4.1 System Coordinator

System coordination, recommended ratings and settings of protective devices, and design analysis must be accomplished by a registered professional electrical power engineer with a minimum of 3 years of current experience in the coordination of electrical power systems. Submit verification of experience and license number, of a registered Professional Engineer as specified above. Provide experience data consisting of at least five references for work of a magnitude comparable to this contract, including points of contact, addresses and telephone numbers.

PART 2 PRODUCTS

2.1 COORDINATED POWER SYSTEM PROTECTION

Prepare analyses to demonstrate that the equipment selected and system constructed meet the contract requirements for ratings, coordination, and protection. Include a load flow analysis, a fault current analysis, and a protective device coordination study. Submit the study along with protective device equipment submittals. No time extensions or similar contact modifications will be granted for work arising out of the requirements for this study. Approval of protective devices proposed will be based on recommendations of this study. The Government is not responsible for any changes to equipment, device ratings, settings, or additional labor for installation of equipment or devices ordered and/or procured prior to approval of the study. The studies must be performed by a registered professional engineer with demonstrated experience in power system coordination in the last 3 years. Provide a list of references complete with points of contact, addresses and telephone numbers. The selection of the engineer is subject to the approval of the Contracting Officer.

2.1.1 Scope of Analyses

The fault current analysis and protective device coordination study must begin at: the nearest upstream device in the existing source system and extend through the downstream devices at the load end.

2.1.2 Determination of Facts

Determine and document the time-current characteristics, features, and nameplate data for each existing protective device. Coordinate with the commercial power company for fault current availability at the site.

2.1.3 Single Line Diagram

Prepare a single line diagram to show the electrical system buses, devices, transformation points, and all sources of fault current (including generator and motor contributions). A fault-impedance diagram or a computer analysis diagram may be provided. Each bus, device or transformation point must have a unique identifier. If a fault-impedance diagram is provided, show impedance data. Show location of switches, breakers, and circuit interrupting devices on the diagram together with available fault data, and the device interrupting rating.

2.1.4 Fault Current Analysis

2.1.4.1 Method

Perform the fault current analysis in accordance with methods described in IEEE 242, and IEEE 399.

2.1.4.2 Data

Utilize actual data in fault calculations. Bus characteristics and transformer impedance must be those proposed. Document data in the report.

2.1.4.3 Fault Current Availability

Provide balanced three-phase fault, bolted line-to-line fault, and

line-to-ground fault current values at each voltage transformation point and at each power distribution bus. Show the maximum and minimum values of fault available at each location in tabular form on the diagram or in the report.

2.1.1.5 Coordination Study

Demonstrate that the maximum possible degree of selectivity has been obtained between devices specified, consistent with protection of equipment and conductors from damage from overloads and fault conditions. Include a description of the coordination of the protective devices in this project. Provide a written narrative describing: which devices may operate in the event of a fault at each bus; the logic used to arrive at device ratings and settings; situations where system coordination is not achievable due to device limitations (an analysis of any device curves which overlap); coordination between upstream and downstream devices; and relay settings. Provide recommendations to improve or enhance system reliability, and detail where such changes would involve additions or modifications to the contract and cost damages (addition or reduction). Provide composite coordination plots on log-log graph paper.

2.1.1.6 Study report

- a. Include a narrative describing: the analyses performed; the bases and methods used; and the desired method of coordinated protection of the power system.
- b. Include descriptive and technical data for existing devices and new protective devices proposed. Include manufacturers published data, nameplate data, and definition of the fixed or adjustable features of the existing or new protective devices.
- c. Document utility company data including system voltages, fault MVA, system X/R ratio, time-current characteristic curves, current transformer ratios, and relay device numbers and settings; and existing power system data including time-current characteristic curves and protective device ratings and settings.
- d. The report must contain fully coordinated composite time-current characteristics curves for each bus in the system, as required to ensure coordinated power system protection between protective devices or equipment. Include recommended ratings and settings of all protective devices in tabulated form.
- e. Provide the calculation performed for the analyses, including computer analysis programs utilized. Provide the name of the software package, developer, and version number.

PART 3 EXECUTION

Not Used

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 26 - ELECTRICAL

SECTION 26 42 17

IMPRESSED CURRENT CATHODIC PROTECTION (ICCP) SYSTEM

05/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Cathodic Protection
 - 1.2.2 Corrosion
 - 1.2.3 Alternating Current (AC) Corrosion
 - 1.2.4 AC Interference
 - 1.2.5 Uniform Attack
 - 1.2.6 Galvanic or Two-Metal Corrosion
 - 1.2.7 Crevice Corrosion
 - 1.2.8 Pitting Corrosion
 - 1.2.9 Intergranular Corrosion
 - 1.2.10 Selective Leaching
 - 1.2.11 Erosion Corrosion
 - 1.2.12 Stress-Corrosion Cracking
 - 1.2.13 Exothermic Welding
 - 1.2.14 Error-Free
- 1.3 ADMINISTRATIVE REQUIREMENTS
- 1.4 SUBMITTALS
 - 1.4.1 Material and Equipment Manufacturer Data
- 1.5 MAINTENANCE MATERIAL SUBMITTALS
 - 1.5.1 Spare Parts
 - 1.5.2 Extra Materials
- 1.6 QUALITY CONTROL
 - 1.6.1 Regulatory Requirements
 - 1.6.2 Qualifications
 - 1.6.3 Services of Corrosion Expert
- 1.7 DELIVERY, STORAGE AND HANDLING
- 1.8 PROJECT/SITE CONDITIONS
 - 1.8.1 Environmental Requirements
 - 1.8.2 Existing Conditions
- 1.9 WARRANTY

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
 - 2.1.1 Corrosion Control System Description
 - 2.1.2 Design Requirements
 - 2.1.2.1 Electrical Isolators
 - 2.1.2.2 Anode and Bond Wires
 - 2.1.2.3 Surge Protection
 - 2.1.2.4 Non-metallic Pipe System
 - 2.1.2.5 Coatings
 - 2.1.2.6 Drawings
 - 2.1.2.7 Summary of Services Required

- 2.1.2.8 Contractor's Modifications
- 2.1.2.9 Tests of Components
- 2.1.2.10 Electrical Potential Measurements
- 2.1.2.11 Typical Metallic Components on Non-metallic Systems
 - 2.1.2.11.1 Pipe Under Concrete Slab
- 2.1.2.12 Metallic Component Coating
- 2.1.2.13 Location of Test Stations
- 2.1.2.14 Electrical Isolation of Structures
 - 2.1.2.14.1 Isolation Joint Testing
 - 2.1.2.14.2 Underground Structure Coating
 - 2.1.2.14.3 Field Joints
 - 2.1.2.14.4 Inspection of Pipe Coatings
 - 2.1.2.14.5 Protective Coating for Aboveground Piping System
 - 2.1.2.14.6 Ferrous Surfaces
- 2.1.3 Performance Requirements
 - 2.1.3.1 Criteria of Cathodic Protection
 - 2.1.3.1.1 Steel
- 2.2 EQUIPMENT
 - 2.2.1 Remote Monitoring
 - 2.2.2 Rectifiers
- 2.3 COMPONENTS
 - 2.3.1 Test Stations
 - 2.3.1.1 Flush Mounted
 - 2.3.1.2 Post Top Mounted
 - 2.3.2 Shunts for Test Stations and Junction Boxes
 - 2.3.3 Junction Box Enclosures
 - 2.3.3.1 Nameplates
 - 2.3.4 Terminal Boards
- 2.4 MATERIALS
 - 2.4.1 Wire and Cable
 - 2.4.1.1 Structure Cable
 - 2.4.1.2 Test Wires
 - 2.4.1.3 Joint and Continuity Bond Cables
 - 2.4.1.4 Polyethylene Insulation
 - 2.4.1.4.1 HMWPE
 - 2.4.1.4.2 High Density Polyethylene (HDPE)
 - 2.4.1.5 Rectifier DC Negative (Structure) Cable(s)
 - 2.4.2 Cable and Wire Identification Tags
 - 2.4.3 Permanent Reference Electrodes
- 2.5 ACCESSORIES
 - 2.5.1 Wire Connectors
 - 2.5.2 Electrical Tape
 - 2.5.3 Cable Marker Tape
 - 2.5.4 Insulating Tape
 - 2.5.5 Underground Splices
 - 2.5.5.1 Cast-Type Splice
 - 2.5.6 Buried Cable Warning and Identification Tape
 - 2.5.7 Conduit
 - 2.5.8 Protective Coating
 - 2.5.8.1 Pipeline Metallic Components
 - 2.5.8.2 Field Joints
 - 2.5.9 Isolation Flange Sets/Kits
 - 2.5.10 Electrical Isolation of Structures
 - 2.5.10.1 Gaskets
 - 2.5.10.2 Isolation Washers and Sleeves
 - 2.5.11 Electrically Isolating Pipe Joints
 - 2.5.11.1 Threaded Fittings
 - 2.5.12 Electrically Conductive Couplings
 - 2.5.13 Stray Current Measurements

- 2.5.14 Cast-In-Place Concrete
- 2.5.15 Steel Flanges and Bolting
 - 2.5.15.1 Steel Flanges
 - 2.5.15.2 Bolting

PART 3 EXECUTION

- 3.1 SAFETY PRECAUTIONS AND HAZARDOUS LOCATIONS
- 3.2 INSTALLATION
 - 3.2.1 Cable Marker Tape
- 3.3 MISCELLANEOUS INSTALLATION
 - 3.3.1 Test Stations
 - 3.3.2 Permanent Reference Electrodes
 - 3.3.2.1 Permanent Reference Electrode Verification
 - 3.3.3 Bonding Boxes
 - 3.3.4 Joint Bonds
 - 3.3.5 Wire Connections
 - 3.3.5.1 Wire-To-Structure Connections
 - 3.3.5.2 Cable Protection
 - 3.3.5.3 Wire Splicing
 - 3.3.6 Steel Surfaces
 - 3.3.7 Electrical Continuity
 - 3.3.8 Electrical Isolation of Structures
 - 3.3.8.1 Locations of Electrical Isolation
 - 3.3.9 Ferrous Valves
 - 3.3.10 Brass or Bronze Valves
- 3.4 INSTALLATION DETAILS
 - 3.4.1 Underground Pipe Joint Bonds
 - 3.4.2 Test Stations and Permanent Reference Electrodes
 - 3.4.3 Permanent Reference Electrode Calibration and Installation
 - 3.4.3.1 Field Drawings
- 3.5 ELECTRICAL ISOLATION OF STRUCTURES
 - 3.5.1 Isolation Fittings
 - 3.5.2 Joint Bonds
- 3.6 FIELD QUALITY CONTROL
 - 3.6.1 Tests and Measurements
 - 3.6.1.1 Native Potentials
 - 3.6.1.2 Protected Potentials
 - 3.6.1.3 Isolation Testing
 - 3.6.1.4 Isolation Tester
 - 3.6.1.5 Anode Output
 - 3.6.1.6 Reference Electrode Potential Measurements
 - 3.6.1.7 Casing Tests
 - 3.6.1.8 Holiday Test
 - 3.6.1.9 Existing Rectifier Testing
 - 3.6.1.10 Stray Current Measurements
 - 3.6.1.11 Induced AC Testing
 - 3.6.1.12 Interference Tests
 - 3.6.1.13 Initial Cathodic Protection System Field Testing
 - 3.6.1.14 Government Field Testing
 - 3.6.1.15 One-Year-Warranty-Period-Testing
 - 3.6.1.16 Final Acceptance Field Testing
- 3.7 CLOSEOUT ACTIVITIES
 - 3.7.1 Reconditioning of Surfaces
 - 3.7.1.1 Concrete
 - 3.7.1.2 Restoration of Sod
 - 3.7.1.3 Restoration of Pavement
 - 3.7.1.4 Cleanup
 - 3.7.2 Training

3.7.2.1 Instruction to Government Personnel

-- End of Section Table of Contents --

SECTION 26 42 17

IMPRESSED CURRENT CATHODIC PROTECTION (ICCP) SYSTEM
05/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1	(2003; R 2018) Unified Inch Screw Threads (UN and UNR Thread Form)
ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.21	(2021) Nonmetallic Flat Gaskets for Pipe Flanges
ASME B18.2.1	(2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)
ASME B18.2.2	(2022) Nuts for General Applications: Machine Screw Nuts, Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)

ASTM INTERNATIONAL (ASTM)

ASTM A194/A194M	(2022) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both
ASTM A307	(2021) Standard Specification for Carbon Steel Bolts, Studs, and Threaded Rod 60 000 PSI Tensile Strength
ASTM B3	(2013) Standard Specification for Soft or Annealed Copper Wire
ASTM B8	(2011; R 2017) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft
ASTM D709	(2017) Standard Specification for Laminated Thermosetting Materials
ASTM D1248	(2016) Standard Specification for Polyethylene Plastics Extrusion Materials for Wire and Cable
ASTM D2028/D2028M	(2015) Cutback Asphalt (Rapid-Curing Type)

ASTM D3381/D3381M (2018) Standard Specification for
Viscosity-Graded Asphalt Binder for Use in
Pavement Construction

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 15589-2 (2012) Petroleum, Petrochemical and
Natural Gas Industries - Cathodic
Protection of Pipeline Transportation
Systems - Part 2: Offshore Pipelines

NACE INTERNATIONAL (NACE)

NACE SP0106 (2018) Control of Internal Corrosion in
Steel Pipelines and Piping Systems

NACE SP0176 (2007) Corrosion Control of Submerged
Areas of Permanently Installed Steel
Offshore Structures Associated with
Petroleum Production

NACE SP0177 (2019) Mitigation of Alternating Current
and Lightning Effects on Metallic
Structures and Corrosion Control Systems

NACE SP0186 (1986; R 2007) Application of Cathodic
Protection for External Surfaces of Steel
Well Casings

NACE SP0188 (1999; R 2006) Discontinuity (Holiday)
Testing of New Protective Coatings on
Conductive Substrates

NACE SP0193 (2016) Application of Cathodic Protection
to Control External Corrosion of Carbon
Steel On-Grade Storage Tank Bottoms

NACE SP0196 (2020) Galvanic Anode Cathodic Protection
of Internal Submerged Surfaces of Steel
Water Storage Tanks

NACE SP0200 (2014) Standard Practice Steel-Cased
Pipeline Practices

NACE SP0285 (2021) External Corrosion Control of
Underground Storage Tank Systems by
Cathodic Protection

NACE SP0286 (1997; R 2007) Standard Practice
Electrical Isolation of Cathodically
Protected Pipelines

NACE SP0388 (2018) Impressed Current Cathodic
Protection of Internal Submerged Surfaces
of Carbon Steel Water Storage Tanks - Item
No. 21040

NACE SP0607 (2007) Petroleum and Natural Gas

Industries – Cathodic Protection of
Pipeline Transportation Systems – Part 2:
Offshore Pipelines

NACE SP21424

(2018) Alternating Current Corrosion on
Cathodically Protected Pipelines: Risk
Assessment, Mitigation and Monitoring

NACE TPC 11

(2008) A Guide to the Organization of
Underground Corrosion-Control Coordinating
Committees

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI C119.1

(2016) Electric Connectors - Sealed
Insulated Underground Connector Systems
Rated 600 Volts

NEMA ICS 6

(1993; R 2016) Industrial Control and
Systems: Enclosures

NEMA RN 1

(2005; R 2013) Polyvinyl-Chloride (PVC)
Externally Coated Galvanized Rigid Steel
Conduit and Intermediate Metal Conduit

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 70

(2020; ERTA 20-1 2020; ERTA 20-2 2020;
ERTA 20-3 2020; TIA 20-1; TIA 20-2; TIA
20-3; TIA 20-4; TIA 20-5; TIA 20-6; TIA
20-7; TIA 20-8; TIA 20-9; TIA 20-10; TIA
20-11; TIA 20-12; TIA 20-13; TIA 20-14;
TIA 20-15; TIA 20-16; ERTA 20-4 2022)
National Electrical Code

UNDERWRITERS LABORATORIES (UL)

UL 6

(2007; Reprint Sep 2019) UL Standard for
Safety Electrical Rigid Metal Conduit-Steel

UL 486A-486B

(2018; Reprint May 2021) UL Standard for
Safety Wire Connectors

UL 510

(2020) UL Standard for Safety Polyvinyl
Chloride, Polyethylene and Rubber
Insulating Tape

UL 514A

(2013; Reprint Aug 2017) UL Standard for
Safety Metallic Outlet Boxes

UL 514B

(2012; Reprint May 2020) Conduit, Tubing
and Cable Fittings

1.2 DEFINITIONS

It is convenient to classify corrosion by the forms in which it manifests itself, the basis for this classification being the appearance of the corroded metal. Each form can be identified by visual observation, although, in some cases, magnification is required. Valuable information

for the solution of a corrosion problem can often be obtained through careful observation of the corroded test specimens or failed equipment. Examination before cleaning is particularly desirable. Cathodic Protection is a method used to control corrosion.

1.2.1 Cathodic Protection

Cathodic Protection (CP) is an electrochemical (half electrical and half chemical) method used to control corrosion of buried or submerged metallic structures. It prevents corrosion by making the protected structure a cathode by installing a more anodic metal (sacrificial or galvanic) anode or a metallic (Impressed Current) anode connected to a Direct Current (DC) power source. When the proper amount of current is applied to the structure, it becomes a cathode. Since all corrosion occurs at the anode, the structure no longer corrodes. The electrons move in the metallic path (electrical). Reduction (chemical) reactions occur at the surface of the cathode resulting in a hydrogen coating and more alkaline environment. Oxidation (chemical) reactions occur at the surface of the anode resulting in corrosion and a more acidic environment. After a CP system is installed and adjusted to provide adequate protection, the hydrogen coats the defects in the coating and polarizes in the negative direction (to a copper/copper sulfate reference electrode) over time the current and potentials remain relatively stable; changes in currents or potentials indicate a problem. An error-free measurement of negative 850 millivolts DC or more negative to the copper/copper-sulfate reference electrode proves the structure is a cathode and corrosion has been mitigated.

1.2.2 Corrosion

It is convenient to classify corrosion by the forms in which it manifests itself, the basis for this classification being the appearance of the corroded metal. Each form can be identified by visual observation, although, in some cases, magnification is required. Valuable information for the solution of a corrosion problem can often be obtained through careful observation of the corroded test specimens or failed equipment. Examination before cleaning is particularly desirable. Some of the eight forms of corrosion are unique, but all of them are more or less interrelated.

The eight forms of corrosion are: (1) Uniform Attack, (2) Galvanic or Two-Metal Corrosion, (3) Crevice Corrosion, (4) Pitting Corrosion, (5) Intergranular Corrosion, (6) Selective Leaching, (7) Erosion Corrosion, and (8) Stress Corrosion Cracking. This listing is arbitrary but covers practically all corrosion failures and problems. The forms are not listed in any particular order of importance. Below, the eight forms of corrosion are discussed in terms of their characteristics, mechanisms, and preventive measures. Hydrogen damage, although not a form of corrosion, often occurs indirectly as a result of corrosive attack and is, therefore, included in this discussion.

1.2.3 Alternating Current (AC) Corrosion

AC corrosion occurs when there is a source of AC current, typically from a high-voltage overhead AC (OHAC) power-line, when there is a low soil resistivity (typically less than 5,000 ohm-cm) and there is a very small coating holiday. The AC corrosion pits typically have a tubercle of corrosion product at the pit. AC interference study modeling software can determine the mitigation solution to solve this problem. Typically, AC Corrosion mitigation is done in conjunction with high AC potentials and

fault current mitigation.

1.2.4 AC Interference

AC interference occurs when a pipeline parallels a high-voltage overhead AC (OHAC) power-line. An interference study is required when this situation occurs as AC interference can cause high AC potentials along the pipeline (safety), can cause a fault condition between the pipeline and power-line and could cause AC corrosion to occur. The pipeline coating when exposed can have blisters/bubbles caused by the excessive AC. The interference study will use modeling software to determine what combination of interference may be occurring (if any) and provide the mitigation solution to solve the problem.

1.2.5 Uniform Attack

Uniform attack is the most common form of corrosion. It is normally characterized by a chemical or electrochemical reaction that proceeds uniformly over the entire exposed surface or over a large area. The metal becomes thinner and eventually fails. For example, a piece of steel or zinc immersed in dilute sulfuric acid normally dissolves at a uniform rate over its entire surface. A sheet iron roof shows essentially the same degree of rusting over its entire outside surface.

Uniform attack, or general overall corrosion, represents the greatest destruction of metal on a tonnage basis. This form of corrosion, however, is not of great concern from a technical standpoint, because the life of equipment can be accurately estimated on the basis of comparatively simple tests. Merely immersing specimens in the fluid involved is often sufficient. Uniform attack can be prevented or reduced by (1) materials, such as coatings, that reduce contact between metal and electrolytes, (2) inhibitors, or (3) cathodic protection.

1.2.6 Galvanic or Two-Metal Corrosion

A potential difference usually exists between two dissimilar-metals when they are immersed in a corrosive or conductive solution. If these metals are placed in contact (or otherwise electrically connected), this potential difference produces electron flow between them. Corrosion of the less corrosion-resistant metal is usually increased, and attack of the more resistant material is decreased, compared to the behavior of these metals when they are not in contact. The less resistant metal becomes anodic and the more resistant metal becomes cathodic. Usually the cathode or cathodic metal corrodes very little or not at all in this type of couple. Because of the electric currents and dissimilar-metals involved, this form of corrosion is called galvanic, bi-metallic or two-metal, corrosion. Galvanic corrosion is restricted to electrochemical corrosion caused by dissimilar-metal effects. It is electrochemical corrosion, but this document must restrict the term galvanic to dissimilar-metal effects for purposes of clarity.

1.2.7 Crevice Corrosion

Intense localized corrosion frequently occurs within crevices and other shielded areas on metal surfaces exposed to corrosives. This type of attack is usually associated with small volumes of stagnant solution caused by holes, gasket surfaces, lap joints, surface deposits, and crevices under bolt and rivet heads. As a result, this form of corrosion is called crevice corrosion or, sometimes, deposit or gasket corrosion.

1.2.8 Pitting Corrosion

Pitting is a form of extremely localized attack that results in holes in the metal. These holes may be small or large in diameter, but in most cases they are relatively small. Pits are sometimes isolated or so close together that they look like a rough surface. Generally a pit may be described as a cavity or hole with the surface diameter about the same as or less than the depth. Pitting is one of the most destructive and insidious forms of corrosion. It causes equipment to fail because of perforation with only a small percent weight loss of the entire structure. It is often difficult to detect pits because of their small size and because the pits are often covered with corrosion products. In addition, it is difficult to measure quantitatively and compare the extent of pitting because of the varying depths and numbers of pits that may occur under identical conditions. Pitting is also difficult to predict by laboratory tests. Sometimes the pits require a long time (several months or a year) to show up in actual service. Pitting is particularly vicious because it is a localized and intense form of corrosion, and failures often occur with extreme suddenness.

1.2.9 Intergranular Corrosion

Grain boundary effects are of little or no consequence in most applications or uses of metals. If a metal corrodes, uniform attack results since grain boundaries are usually only slightly more reactive than the matrix. However, under certain conditions, grain interfaces are very reactive and intergranular corrosion results. Localized attack at and adjacent to grain boundaries, with relatively little corrosion of the grains, is intergranular corrosion. The alloy disintegrates (grains fall out) or loses its strength. Intergranular corrosion can be caused by impurities at the grain boundaries, enrichment of one of the alloying elements, or depletion of one of these elements in the grain-boundary areas. Small amounts of iron in aluminum, wherein the solubility of iron is low, have been shown to segregate in the grain boundaries and cause intergranular corrosion. It has been shown that, based on surface tension considerations, the zinc content of a brass is higher at the grain boundaries. Depletion of chromium in the grain-boundary regions results in intergranular corrosion of stainless steels.

1.2.10 Selective Leaching

Selective leaching is the removal of one element from a solid alloy by corrosion processes. The most common example is the selective removal of zinc in brass alloys (dezincification). Similar processes occur in other alloy systems in which aluminum, iron, cobalt, chromium, and other elements are removed. Selective leaching is the general term to describe these processes, and its use precludes the creation of terms such as de-aluminumification, de-cobaltification. Parting is a metallurgical term that is sometimes applied, but selective leaching is preferred.

1.2.11 Erosion Corrosion

Erosion corrosion is the acceleration or increase in rate of deterioration or attack on a metal because of relative movement between a corrosive fluid and the metal surface. Generally, this movement is quite rapid, and mechanical wear effects or abrasion are involved. Metal is removed from the surface as dissolved ions, or it forms solid corrosion products, which are mechanically swept from the metal surface. Sometimes, movement of the

environment decreases corrosion, particularly when localized attack occurs under stagnant conditions; this is not erosion corrosion because deterioration is not increased. Erosion corrosion is characterized in appearance by grooves, gullies, waves, rounded holes, and valleys and usually exhibits a directional pattern. In many cases, failures because of erosion corrosion occur in a relatively short time, and they are unexpected largely because evaluation corrosion tests were run under static conditions or because the erosion effects were not considered.

1.2.12 Stress-Corrosion Cracking

Stress-corrosion cracking refers to cracking caused by the simultaneous presence of tensile stress and a specific corrosive medium. Many investigators have classified all cracking failures occurring in corrosive media as stress-corrosion cracking, including failures due to hydrogen embrittlement. However, these two types of cracking failures respond differently to environmental variables. To illustrate, CP is an effective method for preventing stress-corrosion cracking; however, hydrogen-embrittlement may be caused when excessive current is applied, especially on stainless steel. Hence, the importance of considering stress-corrosion cracking and hydrogen embrittlement as separate phenomena is obvious. During stress-corrosion cracking, the metal or alloy is virtually unattacked over most of its surface, while fine cracks progress through it. This cracking phenomenon has serious consequences, since it can occur at stresses within the range of typical design stress.

1.2.13 Exothermic Welding

Exothermic welding is used in CP to connect a copper wire to a metallic structure, usually steel or cast-iron. It is a pyrotechnic composition of copper oxide, aluminum powder and magnesium powder. The magnesium powder is ignited with a spark gun or electronic ignition equipment. The aluminum powder serves as fuel, and melts the copper oxide, which bonds the wire to the structure. Although not explosive, it can create brief bursts of heat and high temperature in a small area.

1.2.14 Error-Free

Potential measurement error due to a voltage drop caused by current flowing through a resistor (the electrolyte) between the reference electrode and the protected structure.

1.3 ADMINISTRATIVE REQUIREMENTS

After award of the contract, but prior to commencement of any work at the site, meet with the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Develop a mutual understanding relative to the administration of the value engineering, the safety program, preparation of the schedule of prices or earned value report. Review shop drawings, other submittals, scheduling programming, execution of the work, and clear expectations of the "Interim Department of Defense (DD) Form 1354" submittal. Major subcontractors who will engage in the work must also attend.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance

with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Preconstruction Survey

SD-02 Shop Drawings

Drawings; G, AE

Test stations

Joint bonds

Contractor's Modifications; G, AE

SD-03 Product Data

Qualifications

Equipment; G, AE

Test stations

Isolation flange kits

Wires

Cable and wire

Shunts

Permanent reference electrodes; G, AE

Spare Parts

SD-06 Test Reports

Tests and Measurements; G, AE

Contractor's Modifications; G, AE

SD-10 Operation and Maintenance Data

Cathodic Protection System; G, AE

Training Course; G, AE

Cathodic Protection System, Data Package 5; G, AE

SD-11, Closeout Submittals

Initial Cathodic Protection System Field Testing; G, AE

One Year Warranty Period Cathodic Protection System Field Test
Report; G, AE

Final Cathodic Protection System Field Test Report; G, AE

1.4.1 Material and Equipment Manufacturer Data

DATE	ISSUE NO.	REQUEST DATE	REQUESTED BY	REQUEST REF. NO.
MANUFACTURER NAME				
DESCRIPTION OF EQUIPMENT				

1.5 MAINTENANCE MATERIAL SUBMITTALS

1.5.1 Spare Parts

After approval of shop drawings, furnish spare parts data for each different item of material and equipment specified. The data must include a complete list of parts, special tools, and supplies, with current unit prices and source of supply.

After approval of shop drawings, furnish revised spare parts for any changes made from original submittal. In addition, supply information for material and equipment replacement for all other components of the complete system, including cables, splice kits and connectors, corrosion test stations, and any other components not listed above. Furnish one reference electrode on a hand reel with 120 meters of conductor, one digital voltmeter that can be used in the maintenance of this CP system. Demonstrate use of furnished equipment in actual tests during the training course. Provide a description of equipment of the pipe-to-soil protected structure and foreign structures at electrical isolation between the utility supplier and the facility piping.

1.5.2 Extra Materials

Furnish one reference electrode on a hand reel with 120 meters of conductor, one high-input-impedance digital multimeter that can be used in the maintenance of this CP system. Demonstrate use of furnished equipment in actual tests during the training course. Provide a description of equipment of the pipe-to-soil protected structure and foreign structures at electrical isolation between the utility supplier and the facility piping. Include a description of the equipment and measurement of the pipe-to-soil potentials, and soil condition.

1.6 QUALITY CONTROL

1.6.1 Regulatory Requirements

Obtain the services of a corrosion expert to supervise, inspect, and test the installation and performance of the CP system. The term "corrosion expert" refers to a person, who by thorough knowledge of the physical sciences and the principles of engineering and mathematics, acquired by professional education and related practical experience, is qualified to engage in the practice of corrosion control of buried or submerged metallic structures.

1.6.2 Qualifications

The corrosion expert must be accredited or certified by NACE International, as a CP-4 CP Specialist or be a NACE International certified Corrosion Specialist or a registered professional engineer who has certification or licensing that includes education and experience in CP of the type of CP system being installed. The corrosion expert must have not less than 10 years of experience in the type of CP for buried or submerged metallic structures under this contract. Submit evidence of qualifications of the corrosion expert including their name and qualifications certified in writing to the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager prior to the start of construction. Certification must be submitted giving the name of the firm, the number of years of experience, and a list of not less than five of the firm's installations, three or more years old, that have been tested and found satisfactory.

1.6.3 Services of Corrosion Expert

The "corrosion expert" must make a minimum of three visits to the project site. The first of these visits will include obtaining soil resistivity data, acknowledging the type of pipeline coatings to be used and reporting to the contractor if Impressed Current Cathodic Protection (ICCP) or Galvanic Anode Cathodic Protection (GACP) is required. Once the submittals are approved and the materials delivered, the "corrosion expert" will revisit the site to verify the materials meet submittal requirements, ensure the contractor understands installation practices and that the contractor is capable and qualified to complete the installation.

The "corrosion expert" will be available (but not necessarily be onsite the entire time) during the installation of the CP system to answer questions, approve any changes or additions required during construction, or to provide recommendations as required. The third visit is to complete the training and demonstrations to applicable personnel on proper testing and maintenance techniques and to complete testing the installed CP systems to ensure it has been installed properly and meets adequate CP criteria. An additional visit is required if the One-Year-Warranty-Period-Testing is required.

1.7 DELIVERY, STORAGE AND HANDLING

Storage area for corrosion material will be designated by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. If material are not stored in a building, tarps or similar protection must be used to protect material from inclement weather. Resack and add backfill to packaged anodes that are damaged as a result of improper handling or exposure to rain.

1.8 PROJECT/SITE CONDITIONS

1.8.1 Environmental Requirements

1.8.2 Existing Conditions

Prior to start of any onsite construction activities, perform a Preconstruction Survey of the project site with the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager, and take photographs showing existing environmental conditions in and adjacent to the site. Submit a report for the record. Include in the report a plan describing the features requiring protection under the provisions of the Contract Clauses, which are not specifically identified on the drawings as environmental features requiring protection along with the condition of trees, shrubs and grassed areas immediately adjacent to the site of work and adjacent to the contractor's assigned storage area and access route(s), as applicable. The contractor and the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager will sign this survey report upon mutual agreement regarding its accuracy and completeness. Protect those environmental features included in the survey report and any indicated on the drawings, regardless of interference that their preservation may cause to the work under the Contract.

1.9 WARRANTY

Provide equipment items that are supported by service organizations which are reasonably convenient to the equipment installation in order to render satisfactory service to the equipment on a regular and emergency basis during the warranty period of the contract.

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

2.1.1 Corrosion Control System Description

A corrosion control system consists of several systems which work together to mitigate corrosion on buried or submerged metallic structures. Failure to comply with the requirements of any one of these systems may result in inadequate corrosion control and premature failure of the structure being protected. Each system's guide specifications must be included in the design and installation of a complete corrosion control system and must be adhered to in the design and execution of the corrosion control of a structure being protected. Determination of Need for CP must be made by government requirements and policy directives.

- a. Construction Design Requirements (CDR) for protected structures are found in the UFGS for the structure being protected. For water storage tanks refer to NACE SP0388, and NACE SP0196, underground storage tanks NACE SP0285, aboveground storage NACE SP0193, fuel storage piping Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT), offshore pipelines NACE SP0607 and ISO 15589-2, offshore structures and NACE SP0176, pipeline casings structures (railroad, highway and water crossing) NACE SP0200 and for well casings NACE SP0186.
- b. Coating Systems (CS) are a critical factor in performance of an ICCP system. All coatings, including coatings in structure guide

specifications and Green Seal (GS) coatings, must be compatible with the structure and the CP system, and have high disbondment capabilities. A high resistance to cathodic disbondment is critical for long term service life of coatings on buried or submerged metallic structures under CP. For discontinuity (Holiday) testing of new protective coatings on conductive substrates refer to NACE SP0188.

- c. Mechanical Damage Systems (MDS) such as bedding and rock control barriers normally included in Structure GS may be required by design for some locations. Electrical Isolation is required for all galvanic anode CP systems. For an Electrical Isolation System (EIS) refer to NACE SP0286.
- d. An Electrical Continuity System (ECS) of the protected structure is critical to the operation of the CP system. The types of joints such as bonding and couplings are normally included in Structure GS, this is particularly important to nonwelded pipelines to allow sufficient CP current to conduct to the entire structure.
- e. Stray Current (Interference) Systems must be considered in design, monitored during construction, and interference testing must be completed during the final testing. Design must consider all other cathodic protection systems which may affect other systems or systems which may affect the project, including foreign systems. All foreign systems must be contacted for information and notification and any joint testing which may be required. Corrosion Coordinating Committees may exist. Reference NACE TPC 11.
- f. Pipelines that parallel overhead high-voltage AC transmission power systems are subject to induced AC. Induced AC has several potential adverse impacts on the safety of personnel and pipeline integrity. Assuming that these conditions exist, there are several measures that can be taken to mitigate the induced AC present on a pipeline. These induced AC mitigation strategies are detailed in various international standards including NACE SP0177 and NACE SP21424.
- g. An Impressed Current CP Monitoring System is a solution for remote monitoring (and optionally controlling) different kinds of CP installations. Normally provides for measuring AC voltage input, DC voltage and current output. Optionally may be used to include potentials with permanent reference cells. Remote monitoring systems may be set to provide notifications of high and low alarms on any of the monitored inputs/outputs. Automated recording of all measurements may be scheduled and provide a historical database. Remote monitoring is widely used for underground pipelines used in oil and gas distribution systems, remote facilities such as missile launch facilities, for resistance bonds and remote test stations. These Monitoring Systems are detailed in various NACE international technical papers.
- h. When a project is connecting to an existing infrastructure with CP the design must be compatible with the existing structure(s) CP system. Existing structures may have Impressed Current CP (ICCP) Anode Systems using Remote Anode Systems, Deep Anode Systems, Linear Anode Systems, or Distributed Anode Systems. Existing structures may also have Galvanic Cathodic Protection (GACP) Systems which may be distributed or remote. Existing structures might not have CP. They may use alternative methods of corrosion mitigation instead of CP such as Inhibitor System/Internal Corrosion Control. For control of internal

corrosion in steel pipelines and piping systems refer to NACE SP0106. Due to the limited voltage and current of galvanic anodes the protected structure must be coated and isolated from other structures.

- i. A highly dielectric bonded coating is required to attain adequate CP. Unbonded coatings block the protective current from the pipeline or structure and must not be used with CP. Failure to isolate other metallic structures will result in loss of protection. Isolation from other metallic structures must be maintained.
- j. Continuity of the structure with low resistance is crucial to proper operation of a galvanic anode system. All joints must be continuous or be bonded to both sides of the joint.
- k. A conductive electrolyte is required to allow current flow from the galvanic anodes. Use of galvanic anode systems are normally restricted to electrolytes with resistivities below 30,000 ohm-cm. Small well-coated structures such as coated valves, tees and elbows have very high resistance to earth in high resistance soils. Galvanic anodes in electrolytes over 30,000 ohms also have very high resistance to earth. High circuit resistance with the low voltage of galvanic anodes will not allow sufficient current to meet instant off or depolarization criteria. Additional anodes under these conditions will not noticeably increase the current applied to the structure. Reference SP0169 for criteria in high resistance electrolytes.

2.1.2 Design Requirements

2.1.2.1 Electrical Isolators

Isolators are required to isolate the pipes from any other structure. Provide isolators at all locations where the indicated pipes contact any other metallic structure. Provide locations and detailed drawings of required installations. Include any requirements for lightning protection, test stations, surge protection, or other requirements and include locations and details in design drawings.

2.1.2.2 Anode and Bond Wires

For each CP system, bond metallic components and structures that are not electrically continuous by installing bond wires between the various structures. Bonding of existing buried structures may also be required to preclude detrimental stray current effects and safety hazards. Return stray current to its source without damaging structures intercepting the stray current. Provide electrical isolation of underground facilities in accordance with acceptable industry practice. All tests must be witnessed by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager.

2.1.2.3 Surge Protection

Install sealed weatherproof lightning and surge arrestor devices across isolation flanges or fittings installed in underground piping as indicated on the drawings. Provide gapless, self-healing, solid state type arrestor. Provide number 4 AWG copper lead wires with High Molecular Weight Polyethylene (HMWPE) insulation. Lightning arrestors are not required for isolation flanges on metallic components used on non-metallic piping systems.

2.1.2.4 Non-metallic Pipe System

In the event pipe other than metallic pipe is approved and used in lieu of metallic pipe, protect all metallic components of this pipe system with CP. Submit detailed drawings of CP for each component to the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager for approval within 45 days after date of receipt of notice to proceed, and before commencement of any work.

2.1.2.5 Coatings

Provide coatings for metallic components as required for metallic fittings. Complete and test protective coating on each metallic component (such as valves, hydrants and fittings). Provide coating as required for underground metallic pipe. Each test must be witnessed by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Select, apply, and inspect coatings as specified in these specifications. The use of non-metallic pipe does not change other requirements of the specifications. Submit any deviations due to the use of non-metallic pipe for approval.

2.1.2.6 Drawings

Submit an electronic copy of detail drawings consisting of a complete list of equipment and material including manufacturer's descriptive and technical literature, catalog cuts, results of system design calculations including soil-resistivity, installation instructions and certified test data showing location of anodes and stating the maximum recommended anode current output density. Include in the detail drawings complete wiring and schematic diagrams, isolation fittings, test stations, permanent reference electrodes and bonding and any other details required to demonstrate that the system has been coordinated and will function properly as a unit. Reference locations to two permanent facilities or mark points. Provide one electronic PDF copy and digital photos of the completed installation.

2.1.2.7 Summary of Services Required

Include the following scope of services:

- a. Close-interval potential surveys,
- b. CP Installation System,
- c. System testing,
- d. Interference testing,
- e. Training,
- f. Operating and maintenance manual,
- g. Isolation testing and bonding testing,
- h. Coating and holiday testing.

2.1.2.8 Contractor's Modifications

The specified system is based on a complete system with Impressed Current

in ICCP anodes. The contractor may modify the CP system after review of the project, site verification, and analysis, if the proposed modifications include the anodes specified and will provide better overall system performance.

- a. Submit an electronic copy of detail drawings showing proposed changes in location, scope of performance indicating any variations from, additions to, or clarifications of contract drawings. Show proposed changes in conduit size, wire size, mounting details, wiring diagram, method for electrically-isolating the structure, and any other pertinent information for proper installation and performance of the system. Reference locations to two permanent facilities or mark points. Include in the detail drawings complete wiring and schematic diagrams, permanent reference electrodes and bonding and any other details required to demonstrate that the system has been coordinated and will function properly as a unit. The modifications must be fully described and must be approved by the Contracting Officer. Provide one electronic PDF copy and digital photos of the completed installation.
- b. The proposed system must achieve performance requirements found in section 2.1.3 Performance Requirements, meet the criteria of CP and be verified by testing requirements in section 3.6.1 Tests and Measurements.

2.1.2.9 Tests of Components

Perform a minimum of four tests at each metallic component in the piping system. Two measurements must be made directly over the anodes and the other two tests must be over the outer edge of the component, but at the farthest point from the anodes. Provide a field drawing showing the component, the structure, all components of the CP system and their relationship to each other. Also provide a narrative describing how the CP system will work and the testing at each component. Components requiring CP must include but not be limited to the following:

- a. Pipes under the floor slab or foundations.
- b. Post Indicator Valve (PIV).
- c. Shutoff valves.
- d. Metallic pipes extended from aboveground locations.
- e. Connectors or change-of-direction devices.
- f. Metallic pipe components or sections.
- g. Backflow preventers.
- h. Casings.

2.1.2.10 Electrical Potential Measurements

Make all potential tests at a minimum of 3 m intervals witnessed by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Provide submittals identifying test locations on separate drawings, showing all metal to be protected and all CP equipment. Distinguish and identify test points equipment and protected metal.

2.1.2.11 Typical Metallic Components on Non-metallic Systems

2.1.2.11.1 Pipe Under Concrete Slab

Pipe under concrete slab must have 2 permanent reference electrodes located under the slab. Locate one permanent reference electrode where the pipe enters the concrete slab. Route all conductors to a test station.

2.1.2.12 Metallic Component Coating

Coatings for metallic components will be required for metallic fittings as indicated. These metallic fittings will include fire hydrants, tees, elbows, and valves. Coatings must be selected, applied, and inspected as specified in the coating specifications referenced and be compatible with the structure being protected. All coatings must be in accordance with all applicable Federal, State, and local regulations. Unbonded coatings must not be used with CP.

2.1.2.13 Location of Test Stations

Provide test stations of the type and location shown and curb box or post mount. Provide buried isolation joints with test wire connections brought to a test station. Reference all test stations with GPS coordinates. Unless otherwise shown, locate other test stations as follows:

- a. At 300 m intervals or less.
- b. Where the pipe or conduit crosses any other metal pipe.

2.1.2.14 Electrical Isolation of Structures

As a minimum, provide isolating flanges or unions at the following locations:

- a. Connection of new metallic piping or components to existing piping.
- b. Pressure piping under floor slab to a building.

Provide isolation at metallic connection of all lines to existing system and where connecting to a building. Additionally, provide isolation between fuel line; and foreign pipes that cross the new lines within 3.05 m. Install isolation fittings, including isolating flanges and couplings, aboveground or in a concrete pit.

2.1.2.14.1 Isolation Joint Testing

An isolator checker or insulation tester will be used for isolation or insulating joint (flange or dielectric) electrical testing.

2.1.2.14.2 Underground Structure Coating

This coating specification takes precedence over any other project specification and drawing notes, whether stated or implied, and also applies to the pipeline or tank supplier. Variance in coating quality is not allowed by the contractor or Base Construction Representative without the written consent of the designer. All underground metallic pipelines and tanks to be cathodically protected must have a high quality factory-applied coating. This includes all carbon steel, cast-iron and

ductile-iron pipelines or vessels. Select, apply, and inspect coatings as specified. If non-metallic pipelines are installed, coat all metallic fittings on pipe sections in accordance with this specification section.

2.1.2.14.3 Field Joints

Coat all field joints with materials compatible with the pipeline coating compound. Apply the field joint coating material to an equal thickness as the pipeline coating. Do not use unbonded coatings for these buried metallic components. This includes the elimination of all unbonded polymer wraps or tubes. Once the pipeline or vessel is set in the trench, conduct an inspection of the coating. This inspection must include electrical holiday detection. Repair any damaged areas of the coating. The Contracting Officer or the Contracting Officer's Representative, Technical Expert or Project Manager must be asked to witness inspection of the coating and testing using a holiday detector.

2.1.2.14.4 Inspection of Pipe Coatings

Any damage to the protective coating during transit and handling must be repaired before installation. After field coating has been applied, inspect the entire pipe using an electric holiday detector in accordance with NACE SP0188 using a full-ring, spring-type coil electrode. The holiday detector must be equipped with a bell, buzzer, or other type of audible signal which sounds when a holiday is detected. Upon detection, immediately repair all holidays in the protective coating. Occasional checks of holiday detector, operation will be made by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager to determine suitability of the detector. Provide all labor, materials, and equipment necessary for conducting the inspection.

2.1.2.14.5 Protective Coating for Aboveground Piping System

Provide finish painting as follows:

2.1.2.14.6 Ferrous Surfaces

Touch-up shop-primed surfaces with ferrous metal primer. Solvent-clean surfaces that have not been shop-primed. Surfaces that contain loose rust, loose mil scale, and other foreign substances must be mechanically-cleaned by power wire-brushing and primed with ferrous metal primer. Finish primed surface with two coats of exterior oil paint and vinyl paint.

2.1.3 Performance Requirements

The design must allow for synchronized interruption of all applied current.

2.1.3.1 Criteria of Cathodic Protection

Criteria for adequate CP must be identified by the designer or the contractor's corrosion engineer and approved by the government corrosion engineer. The method of voltage drop consideration must also be identified by the contractor's corrosion engineer and approved by the government. The 100 mV shift criteria are not applicable to bi-metallic structures.

- a. Determination of the on and polarized (instant off) potentials must be made with the protective current applied to the structure for a

minimum of 4 days. Polarized potentials must be determined by interrupting all the current being applied to the structure or coupon.

- b. The potential measurements for the native measurement and the polarized potential must be made with the reference electrode at the same exact location. The polarization decay measurements must also be made with the reference electrode at the same exact location as the polarization potential.
- c. The polarization decay measurements will be the difference between the polarized potential and a voltage measurement made 24 hours after the interruption of protective current.

2.1.3.1.1 Steel

A negative polarized potential of 0.85 volts (850 millivolts) or more negative. The voltage must not be more negative than a negative polarized potential of 1.200 volts (1200 millivolts).

2.2 EQUIPMENT

2.2.1 Remote Monitoring

Remote monitoring equipment must be designed, manufactured and procured specifically for CP use and should be provided as per design and drawings to monitor potential (requires permanent reference electrode) test station(s) shunts and must match or be compatible with previously installed remote monitoring equipment in use at the installation. Rectifier monitoring must include DC Current output(using rectifier shunt), DC Voltage Output, AC voltage input.

2.2.2 Rectifiers

Existing rectifier will be utilized.

2.3 COMPONENTS

2.3.1 Test Stations

2.3.1.1 Flush Mounted

NEMA ICS 6. Metallic or non-metallic with terminal board, 5 terminal posts and lockable lid. A non-metallic enclosure must be molded of glass filled polycarbonate and urethane coated or Acrylonitrile Butadiene Styrene (ABS) plastic. The unit must be of standard design, manufactured for use as a CP test station, complete with cover, terminal board, shunts, and brass or Type 316 stainless steel hardware. The terminal board must be removable for easy access to wires. Provide traffic valve box capable of withstanding H-20 traffic loads. The cover must have a cast in legend "CP TEST."

2.3.1.2 Post Top Mounted

NEMA ICS 6. Metallic or non-metallic with terminal board, 5 terminal posts and lockable lid. A non-metallic enclosure must be high impact strength molded plastic. The unit must be of standard design, manufactured for use as a CP test station, complete with cover, terminal board, shunts, and brass or Type 316 stainless steel hardware. The terminal board must be removable for easy access to wires. The test

station must be mounted atop 1830 mm long polyethylene conduit with anchor. Terminal connections will be permanently tagged to identify each termination of conductors (e.g. identify the conductors connected to the protected structure, anodes, and reference electrodes).

2.3.2 Shunts for Test Stations and Junction Boxes

0.01 ohm, 8 ampere, accuracy plus or minus one percent, polycarbonate circuit board type, color coded for value recognition yellow for 0.01 ohm shunt with nickel-plated brass posts and standard 6.35 mm 1/4 inch holes on 2.54 cm 1 inch centers to fit test stations and terminal boards, accuracy plus or minus one percent, manganin wire type.

2.3.3 Junction Box Enclosures

NEMA ICS 6, Type 3R enclosure with clamped cover and padlocked hasp. Enclosure must be of galvanized steel construction with terminal board. Knockout for conduit must be the size and location as per design drawings.

2.3.3.1 Nameplates

Provide nameplate in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM and ASTM D709. Provide laminated plastic nameplates for each enclosure as specified or as indicated on the drawings. Each nameplate inscription must identify the function. Nameplates will be melamine plastic, 3 mm thick, white with black center core. Surface will be matte finish. Corners will be square. Accurately align lettering and engrave into the core. Minimum size of nameplates must be 635 mm by 65 mm. Lettering must be a minimum of 6.35 mm high normal block style.

2.3.4 Terminal Boards

Provide terminal boards for test stations made of phenolic plastic 6 mm thick with dimensions as indicated. Insulated terminal boards must have the required number of terminals (one terminal required for each conductor). Install solderless copper lugs and copper buss bars, shunts, and variable resistors on the terminal board as indicated. Test station terminal connections will be permanently tagged to identify each termination of conductors (e.g. identify the conductors connected to the protected structure, anodes, reference electrodes and coupons).

2.4 MATERIALS

2.4.1 Wire and Cable

2.4.1.1 Structure Cable

Structure bonding connecting wire must be No. 2 AWG stranded copper wire with type CP high molecular weight polyethylene (HMWPE) insulation, 600 volt rating. Copper conductors conforming to ASTM B3 and ASTM B8.

2.4.1.2 Test Wires

Test wires must be No. 8 AWG stranded copper wire with NFPA 70 Type HMWPE TW insulation. Copper conductors conforming to ASTM B3 and ASTM B8.

2.4.1.3 Joint and Continuity Bond Cables

Provide bonds across joints or any electrically discontinuous connections

in the piping, and other pipes and structures with other than welded or threaded joints included in this CP system. Unless otherwise specified, bonds between structures and across joints in pipe with other than welded or threaded joints must be with No. 4 AWG stranded copper cable with HMWPE insulation. Bonds between structures must contain sufficient slack for any anticipated movement between structures. Bonds across pipe joints must contain a minimum of 100 mm of slack to allow for pipe movement and soil stress. Bonds must be attached by exothermic welding. Exothermic weld areas must be insulated with coating compound and approved by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Continuity bonds must be installed as necessary to reduce stray current interference. Additional joint bonding must be done where determined during construction or testing or as directed. Joint bonding must include excavation and backfilling. There must be a minimum of 2 continuity bonds between each structure and other than welded or threaded joints. Electrical continuity must be tested across joints with other than welded or threaded joints and across metallic portions of sewage lift stations and water booster stations. Copper conductors conforming to ASTM B3 and ASTM B8.

2.4.1.4 Polyethylene Insulation

Polyethylene insulation must comply with the requirements of ASTM D1248 and of the following types, classes, and grades:

2.4.1.4.1 HMWPE

HMWP must be Type I, Class C, Grade E5.

2.4.1.4.2 High Density Polyethylene (HDPE)

HDPE must be Type III, Class C, Grade E3.

2.4.1.5 Rectifier DC Negative (Structure) Cable(s)

ASTM D1248 HMWPE insulation, stranded copper conductors, gauge (AWG) as indicated.

2.4.2 Cable and Wire Identification Tags

Laminated plastic material with black letters on a yellow background. Print letters and numbers a minimum of 5 mm in height. Provide identifier legend in accordance with the cable schedule.

2.4.3 Permanent Reference Electrodes

Permanent reference electrodes must be copper/copper-sulfate specifically manufactured for underground use, 32 mm diameter, by 203 mm long, plastic tube with an ion trap to minimize contamination of the electrode. Must never need recharging, maintenance, or recalibration. Must have impregnated membrane which keeps electrode electrolytes from drying out or getting the reference electrode electrolyte contaminated. Must have ion trap to prevent reference electrode damage from hydrogen sulfide or excess chloride ions. The electrode will be prepackaged by the manufacturer with a backfill material as recommended by the manufacturer. Provide electrodes with No. 12 AWG, RHW cable of sufficient length to extend to the test station without splicing. Reference electrodes will have a minimum 20-year life, stability of plus or minus 5 millivolts under 3 microamp load. The manufacturer must calibrate the PRE to 316 mV plus or

minus 10mV referenced to a standard hydrogen electrode (SHE) and provide a calibration certificate detailing the results of the calibration. Procedures for evaluating the accuracy annually must be included in the Operation and Maintenance Manual.

2.5 ACCESSORIES

2.5.1 Wire Connectors

Safety Standard for Wire Connectors must conform to UL 486A-486B.

2.5.2 Electrical Tape

Pressure-sensitive vinyl plastic electrical tape must conform to UL 510.

2.5.3 Cable Marker Tape

Traceable marker tape must be manufactured for the purpose and clearly labeled "Cathodic Protection Cable Buried Below".

2.5.4 Insulating Tape

Pressure-sensitive vinyl plastic electrical tape and rubber insulated tape must conform to UL 510.

2.5.5 Underground Splices

Provide splices with a compression connector on the conductors, and insulation and waterproofing using one of the following methods which are suitable for continuous submersion in water and comply with ANSI C119.1.

2.5.5.1 Cast-Type Splice

Provide cast-type splice insulation by means of molded casting process employing a thermosetting epoxy resin insulating material applied by a gravity poured method or pressure injected method. Provide component materials of the resin insulation in a packaged form ready for convenient mixing without removing from the package.

2.5.6 Buried Cable Warning and Identification Tape

Polyethylene tape, manufactured for warning and identification of buried cable and conduit. Tape must be 75 mm wide, Yellow in color and read "CAUTION CATHODIC PROTECTION CABLE BURIED BELOW". Color and lettering must be permanent and unaffected by moisture or other substances in backfill materials.

2.5.7 Conduit

UL 6, rigid galvanized steel, Outlet boxes: UL 514A and fittings UL 514B, threaded hubs. Metallic conduit and fittings to be PVC coated in accordance with NEMA RN 1, Type A40.

2.5.8 Protective Coating

Except as otherwise specified, protective coating for underground metallic components including pipe and fittings must be applied mechanically in a factory or field plant specially equipped for the purpose. Valves and fittings that cannot be coated mechanically must have the protective

coating applied by hand, preferably at the plant. Joints must be coated by hand. Hand coating must produce a coating equal in thickness to the coating applied mechanically. Piping and components installed in valve boxes or manholes must also receive the specified protective coating.

2.5.8.1 Pipeline Metallic Components

Underground metallic pipelines and structures must have a good quality factory-applied coating. This includes carbon steel, cast iron and ductile-iron pipelines or vessels. If non-metallic pipelines are installed, metallic fittings or pipe sections must be coated as follows.

- a. The nominal thickness of the metallic pipe joint or other component coating must be 0.6 mm, plus or minus 5 percent.
- b. Pipe and joint coating for factory-applied or field-repair material must be applied as recommended by the manufacturer and must be one of the following:
 - (1) Fusion bonded epoxy.
 - (2) Continuously-extruded polyethylene and adhesive coating system.
 - (3) Polyvinyl chloride pressure-sensitive adhesive tape.
 - (4) High-density polyethylene/bituminous rubber compound tape.
 - (5) Butyl rubber tape.
 - (6) Coal tar epoxy.

2.5.8.2 Field Joints

Coat field joints with material compatible with the existing pipeline coating compound. Apply the joint coating material to an equal thickness as the pipeline coating. Unbonded coatings must not be used on buried metallic piping. This prohibition includes unbonded wraps or tubes.

2.5.9 Isolation Flange Sets/Kits

Provide full-faced gaskets, isolating sleeves and washers, and steel washers. Provide isolation flange kits rated for operation at the rated pressure and temperature. Bolts must be replaced with similar type longer bolts if less than two threads extend beyond the end of both steel washers.

2.5.10 Electrical Isolation of Structures

Isolating fittings, including isolating flanges and couplings, must be installed above ground or in a concrete hand hole. As a minimum, isolating flanges or unions must be provided at the following locations:

- a. Connection of new piping to existing pipes.
- b. Pressure piping under floor slab to a building.
- c. Additionally, isolation must be provided between new pipe lines and foreign pipes that cross the new lines within 3 m.

2.5.10.1 Gaskets

ASME B16.21. Laminated phenolic material for operation at 2,000 KPa, 65 degrees C. Asbestos materials must not be used.

2.5.10.2 Isolation Washers and Sleeves

Two sets 3 mm laminated phenolic for operation at 65 degrees C. Isolating washers must fit within the bolt facing on the flange over the outside of the fabric reinforced phenolic sleeve.

2.5.11 Electrically Isolating Pipe Joints

Electrically isolating pipe joints for above- or below-ground use must be nonflexible flanged type electrically isolating pipe joints to be field assembled.

2.5.11.1 Threaded Fittings

Threaded type electrically isolating pipe joints must have molded plastic screw threads and be used above ground only. Machined plastic screw threads must not be used.

2.5.12 Electrically Conductive Couplings

Electrically conductive couplings must be of a type that is in regular factory production.

2.5.13 Stray Current Measurements

Perform stray current measurements as indicated. Alternate methods may be used when approved. The stray current test report must indicate location of test, type of pipes tested, method of testing.

2.5.14 Cast-In-Place Concrete

Concrete must be 20 MPa minimum ultimate 28-day compressive strength with 25 mm minimum aggregate conforming to Section 03 30 00 CAST-IN-PLACE CONCRETE.

2.5.15 Steel Flanges and Bolting

2.5.15.1 Steel Flanges

ASME B16.5, 1335 N.

2.5.15.2 Bolting

ASTM A307, Grade B for bolts; ASTM A194/A194M, Grade 2 for nuts. Dimensions: ASME B18.2.1 for bolts, ASME B18.2.2 for nuts. Threads: ASME B1.1, Class 2A fit for bolts, Class 2B fit for nuts. Bolts must extend completely through the nuts and may have reduced shanks of a diameter not less than the diameter at the roof of threads.

PART 3 EXECUTION

3.1 SAFETY PRECAUTIONS AND HAZARDOUS LOCATIONS

Any personnel performing operations that will generate heat, sparks, or

flame in hazardous locations must first perform adequate safety precautions. A trained responsible person must ensure the area is safe to perform the operation. Required actions include ensuring adequate ventilation before work starts, air monitoring, and a fire watch must be provided and remain for 30 minutes after the operation is completed. A minimum of a 20 pound ABC type fire extinguisher must be available and must be inspected before each use. Equipment being used must be inspected and used in accordance with manufacturer recommendations. Combustibles that are in the work area(s) must be moved or if they cannot be moved, be covered with fire retardant welding blankets. When performing exothermic welding, properly sized charges and inspection of the structure condition must be accomplished to ensure a safe operation.

3.2 INSTALLATION

3.2.1 Cable Marker Tape

Locate traceable marker tape in the same trench above CP cables including structure leads, test station leads, bonding cables.

3.3 MISCELLANEOUS INSTALLATION

3.3.1 Test Stations

Provide test stations complete with an insulated terminal block having the indicated number of terminals; provided with a lockable cover and have a cast-in legend, "C.P. Test" and complete with an insulated terminal block having the required number of terminals. (One terminal required for each conductor). Provide sufficient test stations to monitor underground isolation points. Test-bond stations (potential measurement and stray current control) must be provided to monitor pipe-to-soil potential of proposed underground pipes or existing underground metallic structures which may conduct stray current from the new CP system. The location of the test-bond stations must ensure that the pipe-to-soil potential of metallic pipe not designated to be protected is not made less negative by the energization of the CP system. Test station terminal connections and the terminal conductor must be permanently tagged to identify each termination of the conductors (e.g. identify the conductors connected to the protected structures). Conductors must be permanently identified in the station by means of plastic or metal tags, or plastic sleeves to indicate termination. Each conductor must be color coded in accordance with the drawings. The station test facility, including permanent copper/copper-sulfate reference electrodes and test returns must be installed as indicated. Pavement inserts must be non-metallic and must allow copper/copper-sulfate reference electrode to contact the electrolyte beneath the pavement surface. Abbreviations must not be used. Welding of electrical connections must be as follows: Exothermic welds must be as specified. Use and selection of these materials and welding equipment must be in accordance with the manufacturer's recommendations.

3.3.2 Permanent Reference Electrodes

3.3.2.1 Permanent Reference Electrode Verification

Verify permanent reference electrodes against a calibrated portable electrode in the presence of the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager before installation. Verify in a non-metallic container of water. Permanent electrode must measure a reference potential agreeing with that measured

by the portable electrode within plus or minus 0.010 volts when the sensing windows of the two electrodes being compared are not more than 2 mm apart but not touching. Remove permanent reference electrodes not within this potential range from the construction site by the end of the day and replace at the contractor's expense. The testing provision applies to replacement permanent reference electrodes as well.

3.3.3 Bonding Boxes

Provide structure bonding boxes in locations as indicated.

3.3.4 Joint Bonds

Provide joint bonds on metallic pipe to and across buried flexible couplings, mechanical joints, flanged joints except at places where isolation joints are specified and joints not welded or threaded to provide electrical continuity. Connect bond wire(s) to the structure(s) by use of exothermic weld kit(s). Clean the structure surface by scraping, filing or wire brushing to produce a clean, bright surface. Check and verify adherence of the bond to the substrate for mechanical integrity by striking the weld with a 908 gram hammer. Cover connections with an electrically insulating coating which is compatible with the existing coating on the structure.

3.3.5 Wire Connections

3.3.5.1 Wire-To-Structure Connections

Lead wire to structure connections must be by exothermic welding process. Weld charges made specifically for use on cast iron must be used on cast iron pipe. A backfill shield filled with a pipeline mastic sealant or material compatible with the coating must be placed over the weld connection and must cover the exposed metal adequately. If other methods of connection are required, they must be approved by the corrosion expert and the Contract Officer or the Contracting Officer's Representative.

3.3.5.2 Cable Protection

Negative cable to the pipe to be protected must be buried a minimum depth of 750 mm except where above ground construction utilizing conduit is used.

3.3.5.3 Wire Splicing

Connecting wire splicing is not allowed.

3.3.6 Steel Surfaces

Connections to metallic structure must be made by exothermic weld methods as manufactured by an approved manufacturer for the type of structure. If other methods of connection are required, such as brazing, mechanical, electric-arc welding and other types of welded connections to ferrous pipe and structures, they must be approved by the corrosion expert and the Contracting Officer or the Contracting Officer's Representative before use.

3.3.7 Electrical Continuity

Underground pipe must be electrically continuous except at places where electrically isolating joints are specified. Pipe joined by means other than welding must meet the following electrical continuity requirements:

- a. Mechanical joints that are not factory designed to provide electrical continuity must be bonded by installing a metallic bond across the joint. The bonding connections must be made by the exothermic welding process.
- b. Mechanical joints designed to provide electrical continuity may be used.

3.3.8 Electrical Isolation of Structures

Perform electrical isolation of structures as follows:

3.3.8.1 Locations of Electrical Isolation

- a. Structural reinforcing steel
- b. Metallic piping
- c. Metallic conduits
- d. Fuel-storage tanks
- e. Gasoline-storage tanks

Provide electrical isolation at each building entrance, and at other locations as indicated.

3.3.9 Ferrous Valves

Dissimilar ferrous valves in a buried ferrous pipeline, including the pipe, must be coated with an underground type dielectric coating for a minimum distance of 10 diameters on each side of the valve.

3.3.10 Brass or Bronze Valves

Brass or bronze valves must not be used in a buried ferrous pipeline.

3.4 INSTALLATION DETAILS

3.4.1 Underground Pipe Joint Bonds

Underground pipe having other than welded or threaded coupling joints must be made electrically continuous by means of a bonding connection installed across the joint.

3.4.2 Test Stations and Permanent Reference Electrodes

Test stations will be of the type and location shown and will be curb box or post mounted. Provide buried isolation joints with test wire connections brought to a test station. Reference all test stations with GPS coordinates. Unless otherwise shown, locate other test stations and permanent reference electrodes as indicated and as follows:

- a. At 305 meters intervals.
- b. At all isolation joints.
- c. At both ends of casings.

- d. Where the pipe crosses any other metal pipes.
- e. Where the pipe connects to an existing piping system.
- f. Where the pipe connects to a dissimilar metal pipe.

Do not fill the bottom of the test station with concrete unless otherwise specified. Do not place rubbish, scrap or other debris into the test station.

3.4.3 Permanent Reference Electrode Calibration and Installation

Calibrate permanent reference electrodes against a portable electrode in the presence of the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager before installation. Calibrate in a test tank containing water with the same composition as the tank to be protected. Permanent electrode must measure a reference potential agreeing with that measured by the portable electrode within plus or minus 0.005 volt when the sensing windows of the two electrodes being compared are not more than 2 mm apart but not touching. Remove permanent reference electrodes not within this potential range from the construction site by the end of the day and replace at the contractor's expense. The testing provision applies to replacement reference electrodes as well.

3.4.3.1 Field Drawings

Complete a field drawing of lead wire and any other pertinent details, test station location and diagram, Underground Pipe Joint bond locations and details, Permanent Reference Electrode locations and Location of all Electrical Isolations. Submit copy with daily report to the government.

3.5 ELECTRICAL ISOLATION OF STRUCTURES

3.5.1 Isolation Fittings

Isolating fittings, including isolating flange kits, dielectric unions and couplings, must be installed aboveground, or within manholes, wherever possible. Where isolating joints must be covered with soil, they must be fitted with a proper joint cover specifically manufactured for covering the particular joint, and the space within the cover filled with hot coal-tar enamel. Isolating fittings in lines entering buildings must be located at least 305 mm above grade of floor level, when possible. Isolating joints must be provided with over voltage protection (OVP) devices to protect against over-voltage surges or approved surge protection devices. The cells must provide a low resistance across isolating joint without excessive loss of cathodic current.

3.5.2 Joint Bonds

Provide joint bonds on metallic pipe to and across buried flexible couplings, mechanical joints, flanged joints except at places where isolation joints are specified and joints not welded or threaded to provide electrical continuity. Connect bond wire(s) to the structure(s) by use of exothermic weld kit(s). Clean the structure surface by scraping, filing or wire brushing to produce a clean, bright surface. Check and verify adherence of the bond to the substrate for mechanical integrity by striking the weld with a 908 gram hammer. Cover connections

with an electrically insulating coating which is compatible with the existing coating on the structure.

3.6 FIELD QUALITY CONTROL

Field tests must be witnessed by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager or their designated representative. Advise the Contracting Officer or Contracting Officer's Representative 14 days prior to performing each field test. Quality control for the cathodic protection system must consist of the following:

- a. Initial field testing by the contractor upon construction.
- b. Government Field Testing after contractor initial field test report submission.
- c. Warranty period field testing by the contractor.
- d. Final field testing by the contractor after one year of service.

3.6.1 Tests and Measurements

3.6.1.1 Native Potentials

Notify the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager a minimum of 14 working days prior to each test. Base potential tests: At least 24 hours after installation of structure to be protected. Perform measurements at test stations and other locations suitable for test purposes, at intervals not exceeding 300 meters with readings at each end point and the midpoints as a minimum. The locations of these measurements must be identical to the locations specified for potential measurements with anodes connected. Use the same measuring equipment that is specified for measuring protected potential measurements.

3.6.1.2 Protected Potentials

Systems must be tested and inspected by the contractor's corrosion engineer in the presence of the Contracting Officer or the Contracting Officer's Representative, Technical Project Manager corrosion expert or an approved representative. Notify the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager a minimum of five working days prior to each test. At least one week after native potential testing and connection to the structure, measure protected structure-to-electrolyte potentials. The locations of these measurements must be identical to the locations specified for native potential measurements. Use the same measuring equipment that is specified for measuring protected potential measurements. Record test data, including date, time, and locations of testing and submit report to the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Contractor must correct and retest, at the contractor's expense, deficiencies in the materials and installation observed by these tests and inspections.

3.6.1.3 Isolation Testing

Before the cathodic protection system is connected to the pipe, an isolation test must be made at each isolating joint or fitting. This test

will demonstrate that no metallic contact, or short circuit exists between the two isolated sections of the pipe. Any isolating fittings installed and found to be defective must be reported to the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager.

3.6.1.4 Isolation Tester

An isolation tester designed and manufactured for use in CP, using the continuity check circuit, must be used for all isolating joint (flange) electrical testing. Testing must conform to the manufacturer's operating instructions. Test must be witnessed by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. An isolating joint that is good will read full scale on the meter. If an isolating joint is shorted, the meter pointer will be deflected or near zero on the meter scale. Location of the fault will be determined from the instructions, and the joint must be repaired.

3.6.1.5 Anode Output

As the cable bonds are connected to the pipe, current output will be measured with an approved clamp-on milliammeter, calibrated shunt with a suitable millivoltmeter or multimeter, or a low resistance ammeter. (Of the three methods, the low-resistance ammeter is the least desirable and most inaccurate. The clamp-on milliammeter is the most accurate.) The values obtained and the date, time, and location must be recorded.

3.6.1.6 Reference Electrode Potential Measurements

Upon completion of the installation and with the entire CP system in operation, electrode potential measurements must be made using a copper/copper sulfate reference electrode and a potentiometer-voltmeter, or a direct-current voltmeter having an internal resistance (sensitivity) of not less than 10 megohms per volt and a full scale of 10 volts. The locations of these measurements must be identical to the locations used for baseline potentials. The values obtained and the date, time, and locations of measurements must be recorded. No less than eight (8) measurements will be made over any length of line or component. Additional measurements will be made at the direction of the cathodic protection design engineer.

3.6.1.7 Casing Tests

Before final acceptance of the installation, the electrical isolation of carrier pipe from casings must be tested and any short circuits corrected.

3.6.1.8 Holiday Test

Any damage to the protective coating during transit and handling must be repaired before installation. After field-coating has been applied, the entire pipe must be inspected by an electric holiday detector with Impressed Current in accordance with NACE SP0188 using a full-ring, spring-type coil electrode or wet sponge test. The holiday detector will be equipped with a bell, buzzer, or other type of audible signal which sounds when a holiday is detected. Holidays in the protective covering must be repaired upon detection. Occasional checks of holiday detector potential will be made by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager to determine suitability of the detector. Labor, materials, and equipment

necessary for conducting the inspection must be furnished by the contractor. The coating system must be inspected for holes, voids, cracks, and other damage during installation.

3.6.1.9 Existing Rectifier Testing

The purpose of the existing rectifier operational inspection is to determine the serviceability of all components required to impress current to the anodes of the Impressed Current system. The inspection must be thorough to ensure dependable current until the next inspection.

- a. Before energizing the rectifier, visually check all rectifier components, shunt box components, safety switches, circuit breakers, and other system power components.
- b. Tighten all accessible connections and check temperature of all the components.
- c. Ensure all permanent reference electrode caps are removed and placed inside the rectifier cabinet for verification of removal and for use during testing.
- d. Startup testing of the rectifier must include voltage and current testing at all tap settings up to the level of protection or maximum of the rectifier rated current, whichever is the lowest. Do not apply excessive current to the structure. For automatic rectifiers, record each tap setting (if available) before switching to automatic potential control.
- e. Using a dependable hand-held meter, measure the output voltage and current, and calibrate the rectifier meters, if present. For rectifiers with more than one circuit, measure the output voltage and current for each circuit using a dependable hand-held meter, and calibrate the rectifier meters, if present.
- f. For systems with permanent reference electrodes, using a calibrated reference electrode, measure the potential difference to each installed permanent reference electrode by placing both electrodes together in the electrolyte with CP current off (may be tested before installation). If the difference is more than 10 mV, replace the permanent reference electrode. For rectifiers with potential voltmeters, using a dependable hand-held meter, measure the potentials for each voltmeter, and calibrate that rectifier meter.
- g. Calculate the CP system circuit resistance of each circuit by dividing the rectifier DC voltage output of each circuit by the rectifier DC ampere output for that circuit.
- h. Calculate the rectifier efficiency.

3.6.1.10 Stray Current Measurements

Before final acceptance of the installation, stray current tests must be performed on any foreign pipes and other metallic structures in close proximity to the installed anodes. A full report of the tests giving all details must be made.

3.6.1.11 Induced AC Testing

Before final acceptance of the installation, induced AC voltage tests must be performed on the pipes near high AC voltage infrastructure and where crossing above ground and underground AC transmission systems. A full report of these tests must be included in the final testing reports with all details and data taken. The touch potential of any testing over 5 volts must be reported to the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Any touch potential over 10 Volts must be mitigated by effective mitigation techniques.

3.6.1.12 Interference Tests

Before final acceptance of the installation, interference tests will be made with respect to any foreign structures in cooperation with the owner of the foreign structures. A full report of the tests giving all details must be made. Stray current measurements must be performed at all isolating locations and at locations where the new pipeline crosses foreign metallic pipes; results of stray current measurements must also be submitted for approval. The method of measurements and locations of measurements must be submitted for approval. As a minimum, stray current measurements must be performed at the following locations:

- a. Connection points of new structures to existing structures.
- b. Crossing points of new pipeline with other existing metallic pipelines.

3.6.1.13 Initial Cathodic Protection System Field Testing

Initial field testing must be completed by the contractor upon completion of construction. Field testing must be witnessed by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager or their designated representative. Advise the Contracting Officer or Contracting Officer's Representative 14 days prior to performing each field test. Field testing must include native and protected potentials, anode current and rectifier testing.

The contractor must submit an initial field test report of the CP system. All structure-to-electrolyte measurements, including native potentials, protected potentials, anode current testing, rectifier testing, and other required testing must be recorded on applicable forms. Identification of test locations, test station and anode test stations will coordinate with the as-built drawings and be provided on system drawings included in the report. The contractor must locate, correct, and report to the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager any short circuits encountered during the checkout of the installed CP system.

3.6.1.14 Government Field Testing

The government corrosion engineer must review the contractor's initial field testing report. Approximately four weeks after receipt of the contractor's initial test report, the system will be tested and inspected in the contractor's presence by the government corrosion engineer. The contractor must correct, at the contractor's expense, materials and installations observed by these tests and inspections not to be in conformance with the plans and specifications. The contractor will pay for all retesting done by the government engineer made necessary by the

correction of deficiencies.

3.6.1.15 One-Year-Warranty-Period-Testing

The contractor must inspect, test, and adjust the CP system monthly for one year, 12 interim inspections total, to ensure its continued conformance with the criteria outlined below. The performance period for these tests will commence upon the completion of all CP work, including changes required to correct deficiencies identified during initial testing, and preliminary acceptance of the CP system by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. Copies of the One Year Warranty Period Cathodic Protection System Field Test Report, including field data, and certified by the contractor's corrosion engineer must be submitted to the Contracting Officer or Contracting Officer's Representative, the activity, and the geographic EFD corrosion engineer.

3.6.1.16 Final Acceptance Field Testing

Conduct final field testing of the CP system utilizing the same procedures specified under, "Initial Cathodic Protection System Field Testing". The contractor will inspect, test, and adjust the CP system after one year of operation to ensure its continued conformance with the criteria outlined below. The performance period for these tests will commence upon preliminary acceptance for the CP system by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager. One electronic copy of the Final Cathodic Protection System Field Test Report, certified by the contractor's corrosion engineer must be submitted to the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager and the geographic EFD corrosion engineer for approval, and as an attachment to the Operation and Maintenance Manual in accordance with Section 01 78 23 OPERATION AND MAINTENANCE DATA. The government corrosion engineer must review the contractor's final field testing report.

3.7 CLOSEOUT ACTIVITIES

3.7.1 Reconditioning of Surfaces

3.7.1.1 Concrete

Concrete must be 20 MPa minimum ultimate 28-day compressive strength with 25 mm minimum aggregate conforming to Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.7.1.2 Restoration of Sod

Restore unpaved surfaces disturbed during the installation of cables to their original elevation and condition. In areas where grass cover exists, it is possible that sod can be carefully removed, watered, and stored during construction operations, and replaced after the operations are completed since it is estimated that no section of pipeline must remain uncovered for more than two (2) days. Where the surface is disturbed in a newly seeded area, re-seed the area with the same quality and formula of seed as that used in the original seeding. Seeding must be done as directed, in all unsurfaced locations where sod and topsoil could not be preserved and replaced. The use of sod in lieu of seeding will require approval by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager.

3.7.1.3 Restoration of Pavement

Repair pavement, sidewalks, curbs, and gutters where existing surfaces are removed or disturbed for construction. Saw cut pavement edges. Graded aggregate base course must have a maximum aggregate size of 40 millimeters. Prime base course with liquid asphalt, ASTM D2028/D2028M, Grade RC-70 prior to paving. Match base course thickness to existing but must not be less than 150 millimeters. Asphalt aggregate size must be 15 mm, asphalt cement must conform to ASTM D3381/D3381M, Grade AR-2000. Match asphalt concrete thickness to existing but must not be less than 50 millimeters. Repair Portland cement concrete pavement, sidewalks, curbs, and gutters using 20.67 MPa concrete conforming to Section 03 30 00 CAST-IN-PLACE CONCRETE. Match existing pavement, sidewalk, curb, and gutter thicknesses.

3.7.1.4 Cleanup

The contractor is responsible for cleanup of the construction site. All paper bags, wire clippings, must be disposed of as directed. Paper bags, wire clippings and other waste will not be put in bell holes or anodes excavation.

3.7.2 Training

3.7.2.1 Instruction to Government Personnel

During the warranty testing or at a time designated by the Contracting Officer or the Contracting Officer's Representative, Technical Expert and Project Manager, make available the services of a technician regularly employed or authorized by the manufacturer of the CP system for instructing government personnel in the proper operation, maintenance, safety, and emergency procedures of the CP system. The period of instruction must be not less than two hours and not more than two 8-hour working days. Conduct the training at the jobsite or at another location mutually satisfactory to the government and the contractor. The field instructions will cover all of the items contained in the Operation and Maintenance Manual.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 31 - EARTHWORK

SECTION 31 23 00.00 20

EXCAVATION AND FILL

02/11, CHG 2: 08/15

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Degree of Compaction
 - 1.2.2 Hard Materials
 - 1.2.3 Rock
- 1.3 SUBMITTALS
- 1.4 DELIVERY, STORAGE, AND HANDLING
- 1.5 CRITERIA FOR BIDDING
- 1.6 REQUIREMENTS FOR OFF SITE SOIL
- 1.7 QUALITY ASSURANCE
 - 1.7.1 Shoring and Sheet piling Plan
 - 1.7.2 Dewatering Work Plan
 - 1.7.3 Utilities

PART 2 PRODUCTS

- 2.1 SOIL MATERIALS
 - 2.1.1 Satisfactory Materials
 - 2.1.2 Unsatisfactory Materials
 - 2.1.3 Cohesionless and Cohesive Materials
 - 2.1.4 Common Fill
 - 2.1.5 Backfill and Fill Material
 - 2.1.6 Select Material
 - 2.1.7 Topsoil
- 2.2 UTILITY BEDDING MATERIAL
 - 2.2.1 Fuel Line Bedding
- 2.3 BORROW
- 2.4 BURIED WARNING AND IDENTIFICATION TAPE
 - 2.4.1 Warning Tape for Metallic Piping
 - 2.4.2 Detectable Warning Tape for Non-Metallic Piping
- 2.5 DETECTION WIRE FOR NON-METALLIC PIPING

PART 3 EXECUTION

- 3.1 PROTECTION
 - 3.1.1 Shoring and Sheet piling
 - 3.1.2 Drainage and Dewatering
 - 3.1.2.1 Drainage
 - 3.1.2.2 Dewatering
 - 3.1.3 Underground Utilities
 - 3.1.4 Machinery and Equipment
- 3.2 SURFACE PREPARATION
 - 3.2.1 Clearing and Grubbing
 - 3.2.2 Stripping

- 3.2.3 Unsuitable Material
- 3.3 EXCAVATION
 - 3.3.1 Structures With Spread Footings
 - 3.3.2 Pipe Trenches
 - 3.3.3 Hard Material and Rock Excavation
 - 3.3.4 Excavated Materials
 - 3.3.5 Final Grade of Surfaces to Support Concrete
- 3.4 SUBGRADE PREPARATION
 - 3.4.1 Proof Rolling
- 3.5 FILLING AND BACKFILLING
 - 3.5.1 Common Fill Placement
 - 3.5.2 Backfill and Fill Material Placement
 - 3.5.3 Select Material Placement
 - 3.5.4 Backfill and Fill Material Placement Over Pipes and at Walls
 - 3.5.5 Trench Backfilling
- 3.6 BORROW
- 3.7 BURIED WARNING AND IDENTIFICATION TAPE
- 3.8 BURIED DETECTION WIRE
- 3.9 COMPACTION
 - 3.9.1 General Site
 - 3.9.2 Structures, Spread Footings, and Concrete Slabs
 - 3.9.3 Adjacent Area
 - 3.9.4 Paved Areas
- 3.10 FINISH OPERATIONS
 - 3.10.1 Grading
 - 3.10.2 Topsoil and Seed
 - 3.10.3 Protection of Surfaces
- 3.11 DISPOSITION OF SURPLUS MATERIAL
- 3.12 FIELD QUALITY CONTROL
 - 3.12.1 Sampling
 - 3.12.2 Testing
 - 3.12.2.1 Fill and Backfill Material Testing
 - 3.12.2.2 Select Material Testing
 - 3.12.2.3 Density Tests

-- End of Section Table of Contents --

SECTION 31 23 00.00 20

EXCAVATION AND FILL
02/11, CHG 2: 08/15

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 103501	(1994) Geotechnic. Compaction Test. Modified Proctor.
UNE 103503	(1995) Determination "In Situ" Of Density Of Soil By The Sand Method.
UNE 103900	(2013) In situ determination of density and moisture content of soil and granular materials by nuclear methods: low depths.
UNE-EN 933-1	(2012) Tests for geometrical properties of aggregates - Part 1: Determination of particle size distribution - Sieving method
UNE-EN 933-2	(1996) Test For Geometrical Properties Of Aggregates. Part 2: Determination Of Particle Size Distribution. Test Sieves, Nominal Size Of Apertures.
UNE-EN 933-10	(2010) Tests for geometrical properties of aggregates - Part 10: Assessment of fines - Grading of filler aggregates (air jet sieving)
UNE-EN ISO 14688-1	(2019) Geotechnical investigation and testing - Identification and classification of soil - Part 1: Identification and description (ISO 14688-1:2017)
UNE-EN ISO 17892-4	(2019) Geotechnical investigation and testing - Laboratory testing of soil - Part 4: Determination of particle size distribution (ISO 17892-4:2016)
UNE-EN ISO 17892-12	(2019) Geotechnical investigation and testing - Laboratory testing of soil - Part 12: Determination of liquid and plastic limits (ISO 17892-12:2018).

AMERICAN WATER WORKS ASSOCIATION (AWWA)

AWWA C600 (2017) Installation of Ductile-Iron Mains
and Their Appurtenances

ASTM INTERNATIONAL (ASTM)

ASTM D1140 (2017) Standard Test Methods for
Determining the Amount of Material Finer
than 75- μ m (No. 200) Sieve in Soils by
Washing

ASTM D2321 (2020) Standard Practice for Underground
Installation of Thermoplastic Pipe for
Sewers and Other Gravity-Flow Applications

ASTM D2487 (2017; E 2020) Standard Practice for
Classification of Soils for Engineering
Purposes (Unified Soil Classification
System)

U.S. ARMY CORPS OF ENGINEERS (USACE)

EM 385-1-1 (2014) Safety and Health Requirements
Manual

1.2 DEFINITIONS

1.2.1 Degree of Compaction

Degree of compaction is expressed as a percentage of the maximum density obtained by the test procedure presented in UNE 103501, for general soil types, abbreviated as percent laboratory maximum density.

1.2.2 Hard Materials

Weathered rock, dense consolidated deposits, or conglomerate materials which are not included in the definition of "rock" but which usually require the use of heavy excavation equipment, ripper teeth, or jack hammers for removal.

1.2.3 Rock

Solid homogeneous interlocking crystalline material with firmly cemented, laminated, or foliated masses or conglomerate deposits, neither of which can be removed without systematic drilling, drilling and the use of expansion jacks or feather wedges, or the use of backhoe-mounted pneumatic hole punchers or rock breakers; also large boulders, buried masonry, or concrete other than pavement exceeding 0.75 cubic meter in volume. Removal of hard material will not be considered rock excavation because of intermittent drilling that is performed merely to increase production.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Shoring and Sheet piling Plan
Dewatering work plan

Submit 15 days prior to starting work.

SD-06 Test Reports

Borrow Site Testing; G
Fill and backfill test
Select material test
Density tests

Copies of all laboratory and field test reports within 24 hours of the completion of the test.

1.4 DELIVERY, STORAGE, AND HANDLING

Perform in a manner to prevent contamination or segregation of materials.

1.5 CRITERIA FOR BIDDING

Base bids on the following criteria:

- a. Surface elevations are as indicated.
- b. Pipes or other artificial obstructions, except those indicated, will not be encountered.
- c. Ground water elevations indicated by the boring log were those existing at the time subsurface investigations were made and do not necessarily represent ground water elevation at the time of construction.
- d. Material character is indicated by the boring logs.
- e. Suitable backfill and bedding material in the quantities required is not available at the project site.
- f. Blasting will not be permitted. Remove material in an approved manner.

1.6 REQUIREMENTS FOR OFF SITE SOIL

Soil and sand brought in from off site for use as backfill or bedding shall be tested for petroleum hydrocarbons, BTEX, PCBs and HW characteristics (including toxicity, ignitability, corrosivity, and reactivity). Backfill shall not contain concentrations of these analytes above the appropriate local environmental agency criteria, and shall pass the tests for HW characteristics. Determine petroleum hydrocarbon concentrations by using appropriate local protocols. Perform HW characteristic tests for ignitability, corrosivity, and reactivity in accordance with accepted standard methods. Perform PCB testing in accordance with accepted standard methods for sampling and analysis of bulk solid samples. Provide borrow site testing for petroleum hydrocarbons and BTEX from a grab sample of material from the area most likely to be contaminated at the borrow site (as indicated by visual or olfactory evidence), with at least one test from each borrow site. For each borrow site, provide borrow site testing for HW characteristics from

a composite sample of material, collected in accordance with standard soil sampling techniques. Do not bring material onsite until tests results have been received and approved by the Contracting Officer.

1.7 QUALITY ASSURANCE

1.7.1 Shoring and Sheet Piling Plan

Submit drawings and calculations, certified by a registered professional engineer, describing the methods for shoring and sheet piling of excavations. Drawings shall include material sizes and types, arrangement of members, and the sequence and method of installation and removal. Calculations shall include data and references used.

The Contractor is required to retain a Professional Geotechnical Engineer to provide technical assistance and/or inspection of excavations, soil/groundwater conditions, and other issues as needed to address construction concerns.

1.7.2 Dewatering Work Plan

Submit procedures for accomplishing dewatering work.

1.7.3 Utilities

Movement of construction machinery and equipment over pipes and utilities during construction shall be at the Contractor's risk. Excavation made with power-driven equipment is not permitted within 600 mm of known Government-owned utility or subsurface construction. For work immediately adjacent to an existing utility or for excavations exposing a utility or other buried obstruction, excavate by hand. Start hand excavation on each side of and parallel to the indicated obstruction and continue until the obstruction is uncovered or until clearance for the new grade is assured. Support uncovered lines or other existing work affected by the contract excavation until approval for backfill is granted by the Contracting Officer. Report damage to utility lines or subsurface construction immediately to the Contracting Officer.

PART 2 PRODUCTS

2.1 SOIL MATERIALS

2.1.1 Satisfactory Materials

Any materials classified by ASTM D2487 or UNE-EN ISO 14688-1 as GW, GP, GM, GP-GM, GW-GM, GC, GP-GC, GM-GC, SW, SP, SM, SW-SM, SC, SW-SC, and natural graded-aggregate free of debris, roots, wood, scrap material, vegetation, refuse, soft unsound particles, and deleterious, or objectionable materials. Unless specified otherwise, the maximum particle diameter shall be one-half the lift thickness at the intended location.

2.1.2 Unsatisfactory Materials

Materials which do not comply with the requirements for satisfactory materials. Unsatisfactory materials also include man-made fills, trash, refuse, or backfills from previous construction. Unsatisfactory material also includes material classified as satisfactory which contains root and other organic matter, and stones larger than 75 mm. The Contracting Officer shall be notified of any contaminated materials.

2.1.3 Cohesionless and Cohesive Materials

Cohesionless materials include materials classified in ASTM D2487 or UNE-EN ISO 14688-1 as GW, GP, SW, and SP. Cohesive materials include materials classified as GC, SC, ML, CL, MH, and CH. Materials classified as GM, GP-GM, GW-GM, SW-SM, SP-SM, and SM shall be identified as cohesionless only when the fines are nonplastic (plasticity index equals zero). Materials classified as GM and SM will be identified as cohesive only when the fines have a plasticity index greater than zero.

2.1.4 Common Fill

Approved, unclassified soil material with the characteristics required to compact to the soil density specified for the intended location.

2.1.5 Backfill and Fill Material

ASTM D2487 or UNE-EN ISO 14688-1, classification GW, GP, GM, GC, SW, SP, SM, SC with a maximum UNE-EN ISO 17892-12 liquid limit of 35 and maximum plasticity index of 12, and a maximum of 25 percent by weight passing ASTM D1140, 75 micrometers sieve or UNE-EN 933-2, 63 micrometer sieve.

2.1.6 Select Material

Provide materials classified as GW, GP, SW, or SP, by ASTM D2487 or UNE-EN ISO 14688-1 where indicated.

2.1.7 Topsoil

Provide as specified in Section 32 92 19 SEEDING.

2.2 UTILITY BEDDING MATERIAL

Except as specified otherwise in the individual piping section, provide bedding for buried piping in accordance with AWWA C600, Type 4, except as specified herein. Backfill to top of pipe shall be compacted to 90 percent of UNE 103501 maximum density. Plastic piping shall have bedding to 300 mm above the crown of the pipe. Provide ASTM D2321 materials as follows:

- a. Class I: Angular, 6 to 40 mm, graded stone, including a number of fill materials that have regional significance such as coral, slag, cinders, crushed stone, and crushed shells.
- b. Class II: Coarse sands and gravels with maximum particle size of 40 mm, including various graded sands and gravels containing small percentages of fines, generally granular and noncohesive, either wet or dry. Soil Types GW, GP, SW, and SP are included in this class.

2.2.1 Fuel Line Bedding

Clean, coarse-grained sand classified as SW or SP by ASTM D2487 or UNE-EN ISO 14688-1 for bedding as indicated.

2.3 BORROW

Obtain borrow materials required, in excess of those furnished from excavations, from sources outside of Government property.

2.4 BURIED WARNING AND IDENTIFICATION TAPE

Polyethylene plastic and metallic core or metallic-faced, acid- and alkali-resistant, polyethylene plastic warning tape manufactured specifically for warning and identification of buried utility lines. Provide tape on rolls, 75 mm minimum width, color coded as specified below for the intended utility with warning and identification imprinted in bold black letters continuously over the entire tape length. Warning and identification to read, "CAUTION, BURIED (intended service) LINE BELOW" or similar wording. Color and printing shall be permanent, unaffected by moisture or soil.

Warning Tape Color Codes	
Red:	Electric
Yellow:	Gas, Fuel Oil; Dangerous Materials
Orange:	Telephone and Other Communications
Blue:	Potable Water Systems
Green:	Sewer Systems
White:	Steam Systems
Gray:	Compressed Air
Purple:	Non Potable, Reclaimed Water, Irrigation and Slurry lines

2.4.1 Warning Tape for Metallic Piping

Acid and alkali-resistant polyethylene plastic tape conforming to the width, color, and printing requirements specified above. Minimum thickness of tape shall be 0.08 mm. Tape shall have a minimum strength of 10.3 MPa lengthwise, and 8.6 MPa crosswise, with a maximum 350 percent elongation.

2.4.2 Detectable Warning Tape for Non-Metallic Piping

Polyethylene plastic tape conforming to the width, color, and printing requirements specified above. Minimum thickness of the tape shall be 0.10 mm. Tape shall have a minimum strength of 10.3 MPa lengthwise and 8.6 MPa crosswise. Tape shall be manufactured with integral wires, foil backing, or other means of enabling detection by a metal detector when tape is buried up to 920 mm deep. Encase metallic element of the tape in a protective jacket or provide with other means of corrosion protection.

2.5 DETECTION WIRE FOR NON-METALLIC PIPING

Detection wire shall be insulated single strand, solid copper with a minimum of 4 sq. mm section minimum.

PART 3 EXECUTION

3.1 PROTECTION

3.1.1 Shoring and Sheeting

Provide shoring bracing, trench boxes, and sheeting where indicated. In addition to Section 25 A and B of EM 385-1-1, include provisions in the shoring and sheeting plan that will accomplish the following:

- a. Prevent undermining of pavements, foundations, thrust blocks, and slabs.
- b. Prevent slippage or movement in banks or slopes adjacent to the excavation.
- c. Allow for the abandonment of shoring and sheeting materials in place in critical areas as the work is completed. In these areas, backfill the excavation to within 900 mm of the finished grade and remove the remaining exposed portion of the shoring before completing the backfill.

3.1.2 Drainage and Dewatering

Provide for the collection and disposal of surface and subsurface water encountered during construction.

3.1.2.1 Drainage

So that construction operations progress successfully, completely drain construction site during periods of construction to keep soil materials sufficiently dry. The Contractor shall establish/construct storm drainage features (ponds/basins) at the earliest stages of site development, and throughout construction grade the construction area to provide positive surface water runoff away from the construction activity and/or provide temporary ditches, dikes, swales, and other drainage features and equipment as required to maintain dry soils, preventing erosion and undermining of foundations. When unsuitable working platforms for equipment operation and unsuitable soil support for subsequent construction features develop, remove unsuitable material and provide new soil material as specified herein. It is the responsibility of the Contractor to assess the soil and ground water conditions presented by the plans and specifications and to employ necessary measures to permit construction to proceed. Excavated slopes and backfill surfaces shall be protected to prevent erosion and sloughing. Excavation shall be performed so that the site, the area immediately surrounding the site, and the area affecting operations at the site shall be continually and effectively drained.

3.1.2.2 Dewatering

Groundwater flowing toward or into excavations shall be controlled to prevent sloughing of excavation slopes and walls, boils, uplift and heave in the excavation and to eliminate interference with orderly progress of construction. French drains, sumps, ditches or trenches will not be permitted within 0.9 m of the foundation of any structure, except with specific written approval, and after specific contractual provisions for restoration of the foundation area have been made. Control measures shall be taken by the time the excavation reaches the water level in order to

maintain the integrity of the in situ material. While the excavation is open, the water level shall be maintained continuously, at least 0.6 m below the working level.

Operate dewatering system continuously until construction work below existing water levels is complete. Submit performance records weekly. Measure and record performance of dewatering system at same time each day by use of observation wells or piezometers installed in conjunction with the dewatering system.

3.1.3 Underground Utilities

Location of the existing utilities indicated is approximate. The Contractor shall physically verify the location and elevation of the existing utilities indicated prior to starting construction. The Contractor shall contact the Public Works Department for assistance in locating existing utilities.

3.1.4 Machinery and Equipment

Movement of construction machinery and equipment over pipes during construction shall be at the Contractor's risk. Repair, or remove and provide new pipe for existing or newly installed pipe that has been displaced or damaged.

3.2 SURFACE PREPARATION

3.2.1 Clearing and Grubbing

Unless indicated otherwise, remove trees, stumps, logs, shrubs, brush and vegetation and other items that would interfere with construction operations within the clearing limits. Remove stumps entirely. Grub out matted roots and roots over 50 mm in diameter to at least 460 mm below existing surface.

3.2.2 Stripping

Strip suitable soil from the site where excavation or grading is indicated and stockpile separately from other excavated material. Material unsuitable for use as topsoil shall be stockpiled and used for backfilling. Locate topsoil so that the material can be used readily for the finished grading. Where sufficient existing topsoil conforming to the material requirements is not available on site, provide borrow materials suitable for use as topsoil. Protect topsoil and keep in segregated piles until needed.

3.2.3 Unsuitable Material

Remove vegetation, debris, decayed vegetable matter, sod, mulch, and rubbish underneath paved areas or concrete slabs.

3.3 EXCAVATION

Excavate to contours, elevation, and dimensions indicated. Reuse excavated materials that meet the specified requirements for the material type required at the intended location. Keep excavations free from water. Excavate soil disturbed or weakened by Contractor's operations, soils softened or made unsuitable for subsequent construction due to exposure to weather. Excavations below indicated depths will not be permitted except

to remove unsatisfactory material. Unsatisfactory material encountered below the grades shown shall be removed as directed by the Contracting Officer. Refill with satisfactory material and compact to 95 percent of UNE 103501 maximum density. Unless specified otherwise, refill excavations cut below indicated depth with satisfactory material and compact to 95 percent of UNE 103501 maximum density. Satisfactory material removed below the depths indicated, without specific direction of the Contracting Officer, shall be replaced with satisfactory materials to the indicated excavation grade; except as specified for spread footings. Determination of elevations and measurements of approved overdepth excavation of unsatisfactory material below grades indicated shall be done under the direction of the Contracting Officer.

3.3.1 Structures With Spread Footings

Ensure that footing subgrades have been inspected and approved by the Contracting Officer prior to concrete placement. Fill over excavations with concrete or select, well graded, tightly compacted granular material during foundation placement.

3.3.2 Pipe Trenches

Excavate to the dimension indicated. Grade bottom of trenches to provide uniform support for each section of pipe after pipe bedding placement. Tamp if necessary to provide a firm pipe bed. Recesses shall be excavated to accommodate bells and joints so that pipe will be uniformly supported for the entire length. Rock, where encountered, shall be excavated to a depth of at least 150 mm below the bottom of the pipe.

3.3.3 Hard Material and Rock Excavation

Remove hard material and rock to elevations indicated in a manner that will leave foundation material in an unshattered and solid condition. Roughen level surfaces and cut sloped surfaces into benches for bond with concrete. Protect shale from conditions causing decomposition along joints or cleavage planes and other types of erosion. Removal of hard material and rock beyond lines and grades indicated will not be grounds for a claim for additional payment unless previously authorized by the Contracting Officer. Excavation of the material claimed as rock shall not be performed until the material has been cross sectioned by the Contractor and approved by the Contracting Officer. Common excavation shall consist of all excavation not classified as rock excavation. Rock was not found in the geotechnical exploration for this project.

3.3.4 Excavated Materials

Satisfactory excavated material required for fill or backfill shall be placed in the proper section of the permanent work required or shall be separately stockpiled if it cannot be readily placed. Satisfactory material in excess of that required for the permanent work and all unsatisfactory material shall be disposed of as specified in Paragraph "DISPOSITION OF SURPLUS MATERIAL."

3.3.5 Final Grade of Surfaces to Support Concrete

Excavation to final grade shall not be made until just before concrete is to be placed. Only excavation methods that will leave the foundation rock in a solid and unshattered condition shall be used. Approximately level surfaces shall be roughened, and sloped surfaces shall be cut as indicated

into rough steps or benches to provide a satisfactory bond. Shales shall be protected from slaking and all surfaces shall be protected from erosion resulting from ponding or flow of water.

3.4 SUBGRADE PREPARATION

Unsatisfactory material in surfaces to receive fill or in excavated areas shall be removed and replaced with satisfactory materials as directed by the Contracting Officer. The surface shall be scarified to a depth of 150 mm before the fill is started. Sloped surfaces steeper than 1 vertical to 4 horizontal shall be plowed, stepped, benched, or broken up so that the fill material will bond with the existing material. When subgrades are less than the specified density, the ground surface shall be broken up to a minimum depth of 150 mm, pulverized, and compacted to the specified density. When the subgrade is part fill and part excavation or natural ground, the excavated or natural ground portion shall be scarified to a depth of 300 mm and compacted as specified for the adjacent fill. Material shall not be placed on surfaces that are muddy. Compaction shall be accomplished by sheepsfoot rollers, pneumatic-tired rollers, steel-wheeled rollers, or other approved equipment well suited to the soil being compacted. Material shall be moistened or aerated as necessary to provide the moisture content that will readily facilitate obtaining the specified compaction with the equipment used. Minimum subgrade density shall be as specified herein.

3.4.1 Proof Rolling

Proof rolling shall be done on an exposed subgrade free of surface water (wet conditions resulting from rainfall) which would promote degradation of an otherwise acceptable subgrade. Proof roll the existing subgrade of the trench with six passes of a dump truck loaded with 6 cubic meters of soil. Operate the truck in a systematic manner to ensure the number of passes over all areas, and at speeds between 4 to 5.5 kilometers per hour. Notify the Contracting Officer a minimum of 3 days prior to proof rolling. Proof rolling shall be performed in the presence of the Contracting Officer. Rutting or pumping of material shall be undercut as directed by the Contracting Officer to a depth of 300 mm and replaced with fill and backfill material.

3.5 FILLING AND BACKFILLING

Fill and backfill to contours, elevations, and dimensions indicated. Compact each lift before placing overlaying lift.

3.5.1 Common Fill Placement

Provide for general site. Use satisfactory materials. Place in 150 mm lifts. Compact areas not accessible to rollers or compactors with mechanical hand tampers. Aerate material excessively moistened by rain to a satisfactory moisture content. Finish to a smooth surface by blading, rolling with a smooth roller, or both.

3.5.2 Backfill and Fill Material Placement

Provide for paved areas and under concrete slabs, except where select material is provided. Place in 150 mm lifts. Do not place over wet areas. Place backfill material adjacent to structures as the structural elements are completed and accepted. Backfill against concrete only when approved. Place and compact material to avoid loading upon or against the

structure.

3.5.3 Select Material Placement

Provide under structures not pile supported. Place in 150 mm lifts. Do not place over wet areas. Backfill adjacent to structures shall be placed as structural elements are completed and accepted. Backfill against concrete only when approved. Place and compact material to avoid loading upon or against structure.

3.5.4 Backfill and Fill Material Placement Over Pipes and at Walls

Backfilling shall not begin until construction below finish grade has been approved, underground utilities systems have been inspected, tested and approved, forms removed, and the excavation cleaned of trash and debris. Backfill shall be brought to indicated finish grade and shall include backfill for outside underground fuel tanks. Where pipe is coated or wrapped for protection against corrosion, the backfill material up to an elevation 600 mm above sewer lines and 300 mm above other utility lines shall be free from stones larger than 25 mm in any dimension. Heavy equipment for spreading and compacting backfill shall not be operated closer to foundation or retaining walls than a distance equal to the height of backfill above the top of footing; the area remaining shall be compacted in layers not more than 100 mm in compacted thickness with power-driven hand tampers suitable for the material being compacted. Backfill shall be placed carefully around pipes or tanks to avoid damage to coatings, wrappings, or tanks. Backfill shall not be placed against foundation walls prior to 7 days after completion of the walls. As far as practicable, backfill shall be brought up evenly on each side of the wall and sloped to drain away from the wall.

3.5.5 Trench Backfilling

Backfill as rapidly as construction, testing, and acceptance of work permits. Place and compact backfill under structures and paved areas in 150 mm lifts to top of trench and in 150 mm lifts to 300 mm over pipe outside structures and paved areas.

3.6 BORROW

Where satisfactory materials are not available in sufficient quantity from required excavations, approved borrow materials shall be obtained as specified herein.

3.7 BURIED WARNING AND IDENTIFICATION TAPE

Provide buried utility lines with utility identification tape. Bury tape 300 mm below finished grade; under pavements and slabs, bury tape 150 mm below top of subgrade.

3.8 BURIED DETECTION WIRE

Bury detection wire directly above non-metallic piping at a distance not to exceed 300 mm above the top of pipe. The wire shall extend continuously and unbroken, from manhole to manhole. The ends of the wire shall terminate inside the manholes at each end of the pipe, with a minimum of 0.9 m of wire, coiled, remaining accessible in each manhole. The wire shall remain insulated over its entire length. The wire shall enter manholes between the top of the corbel and the frame, and extend up

through the chimney seal between the frame and the chimney seal. For force mains, the wire shall terminate in the valve pit at the pump station end of the pipe.

3.9 COMPACTION

Determine in-place density of existing subgrade; if required density exists, no compaction of existing subgrade will be required.

3.9.1 General Site

Compact underneath areas designated for vegetation and areas outside the 1.5 meter line of the paved area or structure to 85 percent of UNE 103501.

3.9.2 Structures, Spread Footings, and Concrete Slabs

Compact top 300 mm of subgrades to 95 percent of UNE 103501. Compact select material to 95 percent of UNE 103501.

3.9.3 Adjacent Area

Compact areas within 1.5 m of structures to 90 percent of UNE 103501.

3.9.4 Paved Areas

Compact top 300 mm of subgrades to 95 percent of UNE 103501. Compact fill and backfill materials to 95 percent of UNE 103501.

3.10 FINISH OPERATIONS

3.10.1 Grading

Finish grades as indicated within 30 mm. Grade areas to drain water away from structures. Maintain areas free of trash and debris. For existing grades that will remain but which were disturbed by Contractor's operations, grade as directed.

3.10.2 Topsoil and Seed

Provide as specified in Section 32 92 19 SEEDING.

3.10.3 Protection of Surfaces

Protect newly backfilled, graded, and topsoiled areas from traffic, erosion, and settlements that may occur. Repair or reestablish damaged grades, elevations, or slopes.

3.11 DISPOSITION OF SURPLUS MATERIAL

Remove from Government property surplus or other soil material not required or suitable for filling or backfilling, and brush, refuse, stumps, roots, and timber.

3.12 FIELD QUALITY CONTROL

3.12.1 Sampling

Take the number and size of samples required to perform the following tests.

3.12.2 Testing

Perform one of each of the following tests for each material used. Provide additional tests for each source change.

3.12.2.1 Fill and Backfill Material Testing

Test fill and backfill material in accordance with UNE-EN 933-1 for conformance to UNE-EN ISO 14688-1 and UNE-EN ISO 17892-4 for gradation limits; UNE-EN 933-10 for material finer than the 63 micrometers sieve; UNE-EN ISO 17892-12 for liquid limit and for plastic limit; UNE 103501 for moisture density relations, as applicable.

3.12.2.2 Select Material Testing

Test select material in accordance with UNE-EN 933-1 for conformance to UNE-EN ISO 14688-1 and UNE-EN ISO 17892-4 for gradation limits; UNE-EN 933-10 for material finer than the 635 micrometers sieve; UNE 103501 for moisture density relations, as applicable.

3.12.2.3 Density Tests

Test density in accordance with UNE 103503, or UNE 103900. When UNE 103900 density tests are used, verify density test results by performing an UNE 103503 density test at a location already UNE 103900 tested as specified herein. Perform an UNE 103503 density test at the start of the job, and for every 10 UNE 103900 density tests thereafter. Test each lift at randomly selected locations every 200 square meters of existing grade in fills for structures and concrete slabs, and every 250 square meters for other fill areas and every 200 square meters of subgrade in cut. Include density test results in daily report.

Bedding and backfill in trenches: One test per 15 meters in each lift.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 11 23

AGGREGATE BASE COURSE

05/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Aggregate Base Course
 - 1.2.2 Degree of Compaction
- 1.3 SUBMITTALS
- 1.4 QUALITY ASSURANCE
 - 1.4.1 Sampling
 - 1.4.2 Tests
 - 1.4.2.1 Gradation Analysis
 - 1.4.2.2 Liquid Limit and Plasticity Index
 - 1.4.2.3 Moisture-Density Determinations
 - 1.4.2.4 Field Density Tests
 - 1.4.2.5 Wear Test
 - 1.4.2.6 Flat and Elongated Pieces
 - 1.4.2.7 Fractured Faces
 - 1.4.2.8 Weight of Slag
- 1.5 ENVIRONMENTAL REQUIREMENTS
- 1.6 ACCEPTANCE
 - 1.6.1 Tolerances
 - 1.6.2 Test Section

PART 2 PRODUCTS

- 2.1 AGGREGATES
 - 2.1.1 Coarse Aggregate
 - 2.1.1.1 Aggregate Base Course
 - 2.1.2 Fine Aggregate
 - 2.1.2.1 Aggregate Base Course
 - 2.1.3 Gradation Requirements
- 2.2 LIQUID LIMIT AND PLASTICITY INDEX
- 2.3 TESTS, INSPECTIONS, AND VERIFICATIONS
 - 2.3.1 Initial Tests
 - 2.3.2 Approval of Material
- 2.4 EQUIPMENT, TOOLS, AND MACHINES

PART 3 EXECUTION

- 3.1 GENERAL REQUIREMENTS
- 3.2 OPERATION OF AGGREGATE SOURCES
- 3.3 STOCKPILING MATERIAL
- 3.4 PREPARATION OF UNDERLYING COURSE OR SUBGRADE
- 3.5 GRADE CONTROL
- 3.6 MIXING AND PLACING MATERIALS
 - 3.6.1 Mixing

- 3.6.2 Placing
- 3.7 LAYER THICKNESS
- 3.8 COMPACTION
- 3.9 EDGES OF BASE COURSE
- 3.10 FINISHING
- 3.11 SMOOTHNESS TEST
- 3.12 FIELD QUALITY CONTROL
 - 3.12.1 In-Place Tests
 - 3.12.2 Approval of Material
- 3.13 TRAFFIC
- 3.14 MAINTENANCE
- 3.15 DISPOSAL OF UNSATISFACTORY MATERIALS

-- End of Section Table of Contents --

SECTION 32 11 23

AGGREGATE BASE COURSE
05/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 103501	(1994) Geotechnic. Compaction Test. Modified Proctor.
UNE 103503	(1995) Determination "In Situ" Of Density Of Soil By The Sand Method.
UNE 103900	(2013) In situ determination of density and moisture content of soil and granular materials by nuclear methods: low depths.
UNE-EN 932-1	(1997) Test For Geometrical Properties Of Aggregates. Part 1: Methods for Sampling.
UNE-EN 933-1	(2012) Tests for geometrical properties of aggregates - Part 1: Determination of particle size distribution - Sieving method
UNE-EN 933-2	(1996) Test For Geometrical Properties Of Aggregates. Part 2: Determination Of Particle Size Distribution. Test Sieves, Nominal Size Of Apertures.
UNE-EN 933-3	(2012) Tests for geometrical properties of aggregates - Part 3: Determination of particle shape - Flakiness index
UNE-EN 933-5	(1999) Test For Geometrical Properties Of Aggregates. Part 5: Determination Of Percentage Of Crushed And Broken Surfaces In Coarse Aggregate Particles.
UNE-EN 1097-2	(2010) Tests for mechanical and physical properties of aggregates - Part 2: Methods for the determination of resistance to fragmentation
UNE-EN 1097-3	(1999) Tests For Mechanical And Physical Properties Of Aggregates. Part 3: Determination Of Loose Bulk Density And Voids.
UNE-EN ISO 14688-1	(2019) Geotechnical investigation and

testing - Identification and
classification of soil - Part 1:
Identification and description (ISO
14688-1:2017)

UNE-EN ISO 17892-12

(2019) Geotechnical investigation and
testing - Laboratory testing of soil -
Part 12: Determination of liquid and
plastic limits (ISO 17892-12:2018).

CENTRO DE ESTUDIOS Y EXPERIMENTACION DE OBRAS PUBLICAS (CEDEX)

NLT-153

(1992) Densidad relativa y absorción de
áridos gruesos

NLT-156

(1994) Densidad aparente de los áridos.

ASTM INTERNATIONAL (ASTM)

ASTM C127

(2015) Standard Test Method for Density,
Relative Density (Specific Gravity), and
Absorption of Coarse Aggregate

ASTM D3665

(2012; R 2017) Standard Practice for
Random Sampling of Construction Materials

ASTM D4791

(2019) Flat Particles, Elongated
Particles, or Flat and Elongated Particles
in Coarse Aggregate

ASTM D5821

(2013; R 2017) Standard Test Method for
Determining the Percentage of Fractured
Particles in Coarse Aggregate

ASTM D6938

(2017a) Standard Test Method for In-Place
Density and Water Content of Soil and
Soil-Aggregate by Nuclear Methods (Shallow
Depth)

MINISTERIO DE FOMENTO (MF)

FOM/2523/2014

(2014) Orden de 12 de diciembre, por la
que se actualizan determinados artículos
del pliego de prescripciones técnicas
generales para obras de carreteras y
puentes, relativos a materiales básicos, a
firmes y pavimentos, y a señalización,
balizamiento y sistemas de contención de
vehículos.

1.2 DEFINITIONS

For the purposes of this specification, the following definitions apply.

1.2.1 Aggregate Base Course

Aggregate base course (ABC) is well graded, durable aggregate uniformly
moistened and mechanically stabilized by compaction.

1.2.2 Degree of Compaction

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum laboratory dry density obtained by the test procedure presented in UNE 103501 abbreviated as a percent of laboratory maximum dry density. Since UNE 103501 applies only to soils that have 30 percent or less by weight of their particles retained on the 19.0 mm sieve, express the degree of compaction for material having more than 30 percent by weight of their particles retained on the 19.0 mm sieve as a percentage of the laboratory maximum dry density in accordance with UNE 103501.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Plant, Equipment, and Tools; G

SD-06 Test Reports

Initial Tests; G
In-Place Tests; G

1.4 QUALITY ASSURANCE

Perform sampling and testing using a laboratory approved in accordance with Section 01 45 00.00 20 QUALITY CONTROL. Do not start work requiring testing until the testing laboratory has been inspected and approved. Test the materials to establish compliance with the specified requirements and perform testing at the specified frequency. Furnish copies of test results within 24 hours of completion of the tests.

1.4.1 Sampling

Take samples for laboratory testing in conformance with UNE-EN 932-1.

1.4.2 Tests

1.4.2.1 Gradation Analysis

Perform gradation analysis in conformance with UNE-EN 933-1 using sieves conforming to UNE-EN 933-2.

1.4.2.2 Liquid Limit and Plasticity Index

Determine liquid limit and plasticity index in accordance with UNE-EN ISO 17892-12.

1.4.2.3 Moisture-Density Determinations

Determine the laboratory maximum dry density and optimum moisture content in accordance with paragraph DEGREE OF COMPACTION.

1.4.2.4 Field Density Tests

Measure field density in accordance with UNE 103503, or UNE 103900. For the method presented in UNE 103900 check the calibration curves and adjust them, if necessary, using only the sand cone method as described in Annex A2 of ASTM D6938 or UNE 103900. Use ASTM D6938 or UNE 103900 to determine the moisture content of the soil. Check the calibration curves furnished with the moisture gauges along with density calibration checks as described in ASTM D6938. Make the calibration checks of both the density and moisture gauges using the prepared containers of material method, on each different type of material being tested at the beginning of a job and at intervals as directed. Submit calibration curves and related test results prior to using the device or equipment being calibrated.

1.4.2.5 Wear Test

Perform wear tests on ABC course material in conformance with UNE-EN 1097-2.

1.4.2.6 Flat and Elongated Pieces

Determine flat and elongated pieces on ABC course material in conformance with ASTM D4791, Method A.

1.4.2.7 Fractured Faces

Perform fractured faces test on ABC coarse aggregate in conformance with UNE-EN 933-5.

1.4.2.8 Weight of Slag

Determine weight per cubic meter of slag in accordance with UNE-EN 1097-3 on the ABC course material.

1.5 ENVIRONMENTAL REQUIREMENTS

Correct completed areas damaged by freezing, rainfall, or other weather conditions to meet specified requirements.

1.6 ACCEPTANCE

1.6.1 Tolerances

Acceptance of ABC is based on compliance with the tolerances presented in Table 1. Remove any materials found to be non-compliant and replace with compliant material or rework, as directed, to meet the requirements of this specification.

TABLE 1	
Measurement	Tolerance
Grade	Plus 6 mm, Minus 13 mm
Smoothness	Plus/Minus 10 mm

TABLE 1	
Individual Test Total Thickness	Plus/Minus 13 mm 1/2 inch
Average Job Thickness	Plus/Minus 6 mm
Compaction	Minimum 100 percent

1.6.2 Test Section

A test section is not required.

PART 2 PRODUCTS

2.1 AGGREGATES

Provide ABC aggregates compliant with FOM/2523/2014, Chapter 10, consisting of clean, sound, durable particles of crushed stone, crushed gravel, angular sand, or other approved material. Provide ABC that is free of lumps of clay, organic matter, and other objectionable materials or coatings. The portion retained on the 4.75 mm sieve is known as coarse aggregate; that portion passing the 4.75 mm sieve is known as fine aggregate. When the coarse and fine aggregate is supplied from more than one source, provide aggregate from each source that meets the specified requirements.

2.1.1 Coarse Aggregate

Provide coarse aggregates with angular particles of uniform density. Separately stockpile coarse aggregate supplied from more than one source.

- a. Crushed Gravel: Provide crushed gravel that has been manufactured by crushing gravels and that meets all the requirements specified below.
- b. Crushed Stone: Provide crushed stone consisting of freshly mined quarry rock, meeting all the requirements specified below.

2.1.1.1 Aggregate Base Course

Limit the percentage of loss of ABC coarse aggregate to a maximum of 50 percent when tested in accordance with UNE-EN 1097-2. Provide aggregate that contains a maximum of 30 percent flat and elongated particles when tested in accordance with UNE-EN 933-3, Method A. A flat particle is one having a ratio of width to thickness greater than 3; an elongated particle is one having a ratio of length to width greater than 3. In the portion retained on each sieve specified, provide crushed aggregates containing a minimum of 50 percent by weight of crushed pieces having two or more freshly fractured faces determined in accordance with ASTM D5821. When two fractures are contiguous, the angle between planes of the fractures is required to be a minimum of 30 degrees in order to count as two fractured faces. Manufacture crushed gravel from gravel particles 50 percent of which, by weight, are retained on the maximum size sieve listed in TABLE 2.

2.1.2 Fine Aggregate

Provide fine aggregates consisting of angular particles of uniform density.

2.1.2.1 Aggregate Base Course

Provide ABC fine aggregate that consists of screenings, angular sand, crushed recycled concrete fines, or other finely divided mineral matter processed or naturally combined with the coarse aggregate.

2.1.3 Gradation Requirements

Select a gradation from Table 2 and apply the specified gradation requirements to the completed base course. Provide aggregates that are continuously well graded within the limits specified in TABLE 2.

TABLE 2. GRADATION OF AGGREGATES			
Percentage By Weight Passing Square-Mesh Sieve			
Sieve Designation	No. 1	No. 2	No.3
50 mm	100	---	---
37.5 mm	70-100	100	---
25.0 mm	45-80	60-100	100
12.5 mm	30-60	30-65	40-70
4.75 mm	20-50	20-50	20-50
2.0 mm	15-40	15-40	15-40
0.425 mm	5-25	5-25	5-25
0.075 mm	0-8	0-8	0-8

NOTE 1: The values are based on aggregates of uniform specific gravity. If materials from different sources are used for the coarse and fine aggregates, test the materials in accordance with ASTM C127 or NLT-153 and NLT-156 to determine their specific gravities. Correct the percentages passing the various sieves as directed if the specific gravities vary by more than 10 percent.

NOTE 2: Gradations containing more than 30 percent retained on the 37.5 mm sieve can produce inconsistent compacted density values when tested in accordance with paragraph DEGREE OF COMPACTION.

2.2 LIQUID LIMIT AND PLASTICITY INDEX

Apply liquid limit and plasticity index requirements to the completed course and to any component that is blended to meet the required gradation. Limit the portion of any component or of the completed course passing the 0.425 mm sieve to be either nonplastic or have a maximum liquid limit of 25 and a maximum plasticity index of 5.

2.3 TESTS, INSPECTIONS, AND VERIFICATIONS

2.3.1 Initial Tests

Perform one of each of the following initial tests on the proposed material prior to commencing construction to demonstrate that the proposed material meets all specified requirements when furnished. Complete this testing for each source if materials from more than one source are proposed. Submit certified copies of test results for approval a minimum of 30 days before material is required for the work.

- a. Gradation Analysis.
- b. Liquid limit and plasticity index.
- c. Moisture-density relationship.
- d. Wear.
- e. Flat and Elongated Pieces.
- f. Fractured Faces.

2.3.2 Approval of Material

Tentative approval of material will be based on initial test results.

2.4 EQUIPMENT, TOOLS, AND MACHINES

All plant, equipment, and tools used in the performance of the work are subject to approval by the Government before the work is started. Maintain all plant, equipment, and tools in satisfactory working condition at all times. Submit a list of proposed equipment, including descriptive data. Use equipment capable of minimizing segregation, producing the required compaction, meeting grade controls, thickness control, and smoothness requirements as set forth herein.

PART 3 EXECUTION

3.1 GENERAL REQUIREMENTS

When the ABC is constructed in more than one lift, clean the previously constructed lift of loose and foreign matter by sweeping with power sweepers or power brooms. Use hand brooms in areas where power cleaning is not practicable. Provide adequate drainage during the entire period of construction to prevent water from collecting or standing on the working area.

3.2 OPERATION OF AGGREGATE SOURCES

Condition aggregate sources on private lands in accordance with local laws or authorities.

3.3 STOCKPILING MATERIAL

Clear and level storage sites prior to stockpiling of material. Stockpile all materials, including approved material available from excavation and grading, in the manner and at the locations designated. Stockpile aggregates on the cleared and leveled areas designated to prevent segregation. Stockpile materials obtained from different sources separately.

3.4 PREPARATION OF UNDERLYING COURSE OR SUBGRADE

Clean the underlying course or subgrade of all foreign substances prior to constructing the base course(s). Do not construct base course(s) on underlying course or subgrade that is muddy. Construct the surface of the underlying course or subgrade to meet specified compaction and surface tolerances. Correct ruts or soft yielding spots in the underlying courses, areas having inadequate compaction, and deviations of the surface from the specified requirements set forth herein by loosening and removing soft or unsatisfactory material and adding approved material, reshaping to line and grade, and recompacting to specified density requirements. For cohesionless underlying courses or subgrades containing sands or gravels, as defined in UNE-EN ISO 14688-1, stabilize the surface prior to placement of the base course(s). Stabilize by mixing ABC into the underlying course and compacting by approved methods. Consider the stabilized material as part of the underlying course and meet all requirements of the underlying course. Do not allow traffic or other operations to disturb the finished underlying course and maintain in a compliant condition until the base course is placed.

3.5 GRADE CONTROL

Provide a finished and completed base course conforming to the lines, grades, and cross sections shown. Place line and grade stakes as necessary for control.

3.6 MIXING AND PLACING MATERIALS

3.6.1 Mixing

Mix the coarse and fine aggregates in a stationary plant. Make adjustments in mixing procedures or in equipment to obtain true grades, to minimize segregation or degradation, to obtain the required water content, and to produce a satisfactory base course meeting all requirements of this specification.

3.6.2 Placing

Place the mixed material on the prepared subgrade or subbase in lifts of uniform thickness with an approved spreader. Place the lifts so that when compacted they are true to the grades or levels required with the least possible surface disturbance. Where the base course is placed in more than one lift, clean the previously constructed lift of loose and foreign matter by sweeping with power sweepers, power brooms, or hand brooms. Make adjustments in placing procedures or equipment to obtain true grades, to minimize segregation and degradation, to adjust the water content, and to produce an acceptable base course.

3.7 LAYER THICKNESS

Compact the completed base course to the thickness indicated. Limit individual compacted lifts to a maximum thickness of 150 mm and a minimum thickness of 75 mm. Compact the base course(s) to a total thickness that is within the tolerances of paragraph ACCEPTANCE of the thickness indicated. Where the measured thickness is more than 13 mm deficient, correct such areas by scarifying, adding new material of proper gradation, reblading, and recompacting as directed. Where the measured thickness is more than 13 mm thicker than indicated, the course will be considered as conforming to the specified thickness requirements. However, the

requirements for wearing course thickness and plan grade are still applicable. The average job thickness will be the average of all thickness measurements taken for the job and within the tolerances of paragraph ACCEPTANCE of the thickness indicated.

3.8 COMPACTION

Compact each lift of the base course, as specified, with approved compaction equipment. For cohesive soils, maintain water content during the compaction procedure to within plus or minus 2 percent of the optimum water content determined from laboratory tests as specified and for cohesionless soils, maintain the water content to facilitate compaction without bulking. Begin rolling at the outside edge of the surface and proceed to the center, overlapping on successive trips at least one-half the width of the roller. Slightly vary the length of alternate trips of the roller. Adjust speed of the roller as needed so that displacement of the aggregate does not occur. Compact mixture with hand-operated power tampers in all places not accessible to the rollers. Continue compaction until each lift is compacted through the full depth to meet the compaction requirements of Table 1. Make such adjustments in compacting or finishing procedures to obtain true grades, to minimize segregation and degradation, to reduce or increase water content, and to produce a compliant base course. Remove any materials found to be non-compliant and replace with compliant material or rework, as directed, to meet the requirements of this specification.

3.9 EDGES OF BASE COURSE

Place the base course(s) so that the completed section is a minimum of 600 mm wider, on all sides, than the next lift that will be placed above it. Place approved material along the outer edges of the base course in sufficient quantity to compact to the thickness of the course being constructed. When the course is being constructed in two or more lifts, simultaneously roll and compact at least a 600 mm width of this shoulder material with the rolling and compacting of each lift of the base course.

3.10 FINISHING

Finish the surface of the top lift of base course after final compaction by cutting any overbuild to grade and rolling with a steel-wheeled roller. Do not add thin lifts of material to the top lift of base course to meet grade. If the elevation of the top lift of base course exceeds the tolerances of paragraph ACCEPTANCE, scarify the top lift to a depth of at least 75 mm and blend new material in and compact to bring to grade. Make adjustments to rolling and finishing procedures to minimize segregation and degradation, obtain grades, maintain moisture content, and produce an acceptable base course. If the surface become rough, corrugated, uneven in texture, or traffic marked prior to completion, scarify the non-compliant portion and rework and recompact it or replace as directed.

3.11 SMOOTHNESS TEST

Construct the top lift so that the surface shows no deviations exceeding the tolerances of paragraph ACCEPTANCE when tested with a 3.66 meter straightedge. Test the entire area in both a longitudinal and a transverse direction on parallel lines. Perform the transverse lines at a maximum spacing of 4.5 m or less apart, as directed. Perform the longitudinal lines at the centerline of each placement lane, regardless of

whether multiple lanes are allowed to be paved at the same time, and at the 1/8th point in from each side of the lane. Hold the straightedge in contact with the surface and moved ahead one-half the length of the straightedge for each successive measurement. Determine the amount of surface irregularity by placing the freestanding (unleveled) straightedge on the pavement surface and measuring the maximum gap between the straightedge and the pavement surface. Determine measurements along the entire length of the straight edge. Correct deviations exceeding this amount by removing material and replacing with new material, or by reworking existing material and compacting it to meet these specifications.

3.12 FIELD QUALITY CONTROL

3.12.1 In-Place Tests

Perform each of the following in-place tests on samples taken from the placed and compacted ABC. Determine sample locations using random sampling in accordance with ASTM D3665. Take samples and test at the rates indicated.

- a. Perform density tests on every lift of material placed and at a frequency of one set of tests for every 250 square meters, or portion thereof, of completed area. Gradations containing more than 30 percent retained on the 37.5 mm sieve can produce inconsistent compacted density values when tested in accordance with paragraph DEGREE OF COMPACTION.
- b. Perform gradation analysis on every lift of material placed and at a frequency of one sieve analysis for every 500 square meters, or portion thereof, of material placed.
- c. Perform liquid limit and plasticity index tests at the same frequency as the sieve analysis.
- d. Measure the thickness of the base course at intervals providing at least one measurement for each 500 square meters of base course or part thereof. Measure the thickness using test holes, at least 75 mm in diameter through the base course.

3.12.2 Approval of Material

Final approval of the materials will be based on tests for gradation, liquid limit, and plasticity index performed on samples taken from the completed and fully compacted course(s).

3.13 TRAFFIC

Completed portions of the base course can be opened to limited traffic, provided there is no marring or distorting of the surface by the traffic. Do not allow heavy equipment on the completed base course except when necessary for construction. When it is necessary for heavy equipment to travel on the completed base course, protect the area against marring or damage to the completed work. Repair damage to meet these specifications.

3.14 MAINTENANCE

Maintain the base course in a satisfactory condition until the full pavement section is completed and accepted. Immediately repair any defects and repeat repairs as often as necessary to keep the area intact.

Retest any base course that was not paved over prior to the onset of winter to verify that it still complies with the requirements of this specification. Rework or replace any area of base course that is damaged as necessary to comply with this specification.

3.15 DISPOSAL OF UNSATISFACTORY MATERIALS

Dispose of any unsuitable materials that have been removed outside the limits of Government-controlled land. No additional payments will be made for materials that have to be replaced.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 12 13

BITUMINOUS TACK AND PRIME COATS

05/17

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY ASSURANCE
- 1.4 DELIVERY, STORAGE, AND HANDLING
- 1.5 EQUIPMENT, TOOLS AND MACHINES
 - 1.5.1 General Requirements
 - 1.5.2 Bituminous Distributor
 - 1.5.3 Heating Equipment for Storage Tanks
 - 1.5.4 Power Brooms and Power Blowers
- 1.6 ENVIRONMENTAL REQUIREMENTS

PART 2 PRODUCTS

- 2.1 PRIME COAT
 - 2.1.1 Emulsified Asphalt
- 2.2 TACK COAT
 - 2.2.1 Emulsified Asphalt

PART 3 EXECUTION

- 3.1 PREPARATION OF SURFACE
- 3.2 APPLICATION RATE
 - 3.2.1 Tack Coat
 - 3.2.2 Prime Coat
- 3.3 APPLICATION TEMPERATURE
 - 3.3.1 Viscosity Relationship
 - 3.3.2 Temperature Ranges
- 3.4 APPLICATION
 - 3.4.1 General
 - 3.4.2 Prime Coat
 - 3.4.3 Tack Coat
- 3.5 CURING PERIOD
- 3.6 FIELD QUALITY CONTROL
- 3.7 SAMPLING AND TESTING
 - 3.7.1 Sampling
 - 3.7.2 Calibration Test
 - 3.7.3 Trial Applications
 - 3.7.3.1 Tack Coat Trial Application Rate
 - 3.7.3.2 Prime Coat Trial Application Rate
 - 3.7.4 Sampling and Testing During Construction
- 3.8 TRAFFIC CONTROLS

-- End of Section Table of Contents --

SECTION 32 12 13

BITUMINOUS TACK AND PRIME COATS
05/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE-EN 13808 (2013) Bitumen and bituminous binders - Framework for specifying cationic bituminous emulsions

CENTRO DE ESTUDIOS Y EXPERIMENTACION DE OBRAS PUBLICAS (CEDEX)

NLT-121 (1999) Toma de Muestras de los Materiales Bituminosos

NLT-138 (1999) Viscosidad Saybolt de las emulsiones bituminosas

NLT-348 (1990) Toma de Muestras de Mezclas Bituminosas para Pavimentacion.

MINISTERIO DE FOMENTO (MF)

FOM/891/2004 (2004) Orden Por La Que Se Actualizan Determinados Articulos Del Pliego De Prescripciones Tecnicas Generales Para Obras De Carreteras Y Puentes, Relativos A Firmes Y Pavimentos

FOM/2523/2014 (2014) Orden de 12 de diciembre, por la que se actualizan determinados artículos del pliego de prescripciones técnicas generales para obras de carreteras y puentes, relativos a materiales básicos, a firmes y pavimentos, y a señalización, balizamiento y sistemas de contención de vehículos.

MF PG3 (2014) Pliego de Prescripciones Tecnicas Generales para Obras de Carreteras y Puentes

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO T 102 (2009; R 2013) Standard Method of Test for Spot Test of Asphaltic Materials

ASTM INTERNATIONAL (ASTM)

ASTM D2995

(1999; R 2009) Determining Application
Rate of Bituminous Distributors

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Prime Coat; G
Tack Coat; G

SD-06 Test Reports

Sampling and Testing

1.3 QUALITY ASSURANCE

Certificates of compliance for asphalt materials delivered will be obtained and checked to ensure that specification requirements are met. Tack coat materials will not be diluted. Prime coat materials when emulsions are used can be diluted on site with potable water up to 1 part emulsion to 1 part water.

1.4 DELIVERY, STORAGE, AND HANDLING

Inspect the materials delivered to the site for contamination and damage. Unload and store the materials with a minimum of handling.

1.5 EQUIPMENT, TOOLS AND MACHINES

1.5.1 General Requirements

Equipment, tools and machines used in the work are subject to approval. Maintain in a satisfactory working condition at all times. Calibrate equipment such as asphalt distributors, scales, batching equipment, spreaders and similar equipment within 12 months of their use. If the calibration expires during project, recalibrate the equipment before work can continue.

1.5.2 Bituminous Distributor

Provide a self propelled distributor with pneumatic tires of such size and number to prevent rutting, shoving or otherwise damaging the surface being sprayed. Calibrate the distributor in accordance with ASTM D2995. Design and equip the distributor to spray the bituminous material in a uniform coverage at the specified temperature, at readily determined and controlled total liquid rates from 0.14 to 4.5 L/square meter, with a pressure range of 172.4 to 517.1 kPa and with an allowable variation from the specified rate of not more than plus or minus 5 percent, and at variable widths. Include with the distributor equipment a separate power unit for the bitumen pump, full-circulation spray bars, tachometer, pressure gauges, volume-measuring devices, adequate heaters for heating of materials to the proper application temperature, a thermometer for reading

the temperature of tank contents, and a hand hose attachment suitable for applying bituminous material manually to areas inaccessible to the distributor. The distributor will be capable of circulating and agitating the bituminous material during the heating process.

1.5.3 Heating Equipment for Storage Tanks

Use steam, electric, or hot oil heaters for heating the bituminous material. Provide steam heaters consisting of steam coils and equipment for producing steam, so designed that the steam cannot come in contact with the bituminous material. Fix an armored thermometer to the tank with a temperature range from 4.4 to 204.4 degrees C so that the temperature of the bituminous material may be determined at all times.

1.5.4 Power Brooms and Power Blowers

Use power brooms and power blowers suitable for cleaning the surfaces to which the bituminous coat is to be applied.

1.6 ENVIRONMENTAL REQUIREMENTS

Apply bituminous coat only when the surface to receive the bituminous coat is dry. A limited amount of moisture (approximately 0.14 liter/square meter) can be sprayed on the surface of unbound material when prime coat is used to improve coverage and penetration of asphalt material. Apply bituminous coat only when the atmospheric temperature in the shade is 10 degrees C or above and when the temperature has not been below 2 degrees C for the 12 hours prior to application, unless otherwise directed.

PART 2 PRODUCTS

2.1 PRIME COAT

Provide asphalt conforming to one of the following grades:

2.1.1 Emulsified Asphalt

Provide emulsified asphalt conforming to MF PG3 Article 214 and UNE-EN 13808 C60BF4. Asphalt emulsion can be diluted up to 1 part water to 1 part emulsion for prime coat use. Do not dilute asphalt emulsion for tack coat use.

2.2 TACK COAT

2.2.1 Emulsified Asphalt

Provide emulsified asphalt conforming to MF PG3 Article 214 and UNE-EN 13808 types C60B3 ADH or C60B3 TER for hot weather and C60B2 TER for cold weather. For prime coats the emulsified asphalt can be diluted with up to 1 part emulsion to 1 part water. No dilution is allowed for tack coat applications. The base asphalt used to manufacture the emulsion is required to show a negative spot when tested in accordance with AASHTO T 102 using standard naphtha.

PART 3 EXECUTION

3.1 PREPARATION OF SURFACE

Immediately before applying the bituminous coat, remove all loose

material, dirt, clay, or other objectionable material from the surface to be treated by means of a power broom or blower supplemented with hand brooms. Apply treatment only when the surface is dry and clean.

3.2 APPLICATION RATE

The exact quantities within the range specified, which may be varied to suit field conditions, will be determined by the Contracting Officer.

3.2.1 Tack Coat

Apply bituminous material for the tack coat in quantities of not less than 0.14 L nor more than 0.46 L/square meter of residual asphalt onto the pavement surface as approved by the Contracting Officer. Do not dilute asphalt emulsion when used as a tack coat.

3.2.2 Prime Coat

Apply bituminous material for the prime coat in quantities of not less than 0.23 L nor more than 0.54 L/square meter of residual asphalt for asphalt emulsion up to a 1 to 1 dilution rate or for residual asphalt for cutback asphalt.

3.3 APPLICATION TEMPERATURE

3.3.1 Viscosity Relationship

Apply asphalt at a temperature that will provide a viscosity between 10 and 60 seconds, Saybolt Furol, or between 20 and 120 square mm/sec, kinematic. Furnish the temperature viscosity relation to the Contracting Officer.

3.3.2 Temperature Ranges

The viscosity requirements determine the application temperature to be used. Apply the emulsion so that the viscosity is within 10 and 40 SSF per NLT-138. Comply with FOM/891/2004 and FOM/2523/2014, Art. 531.5.2

3.4 APPLICATION

3.4.1 General

Following preparation and subsequent inspection of the surface, apply the bituminous prime or tack coat with the bituminous distributor at the specified rate with uniform distribution over the surface to be treated. Properly treat all areas and spots, not capable of being sprayed with the distributor, with the hand spray. Until the succeeding layer of pavement is placed, maintain the surface by protecting the surface against damage and by repairing deficient areas at no additional cost to the Government. If required, spread clean dry sand to effectively blot up any excess bituminous material. No smoking, fires, or flames other than those from the heaters that are a part of the equipment are permitted within 8 meters of heating, distributing, and transferring operations of cutback materials. Prevent all traffic, except for paving equipment used in constructing the surfacing, from using the underlying material, whether primed or not, until the surfacing is completed. The bituminous coat requirements are described herein.

3.4.2 Prime Coat

Apply a prime coat at locations shown on the Drawings. The type of liquid asphalt and application rate will be as specified herein. Protect the underlying layer from any damage (water, traffic, etc.) until the surfacing is placed. If the Contractor places the surfacing within seven days, the choice of protection measures or actions to be taken is at the Contractor's option. Repair (recompact or replace) damage to the underlying material caused by lack of, or inadequate, protection by approved methods at no additional cost to the Government. If the Contractor opts to use the prime coat, apply as soon as possible after consolidation of the underlying material. Apply the bituminous material uniformly over the surface to be treated at a pressure range of 172.4 to 517.1 kPa; the rate will be as specified above in paragraph APPLICATION RATE. To obtain uniform application of the prime coat on the surface treated at the junction of previous and subsequent applications, spread building paper on the surface for a sufficient distance back from the ends of each application to start and stop the prime coat on the paper and to ensure that all sprayers will operate at full force on the surface to be treated. Immediately after application remove and destroy the building paper.

3.4.3 Tack Coat

A tack coat should be applied to every bound surface (asphalt or concrete pavement) that is being overlaid with asphalt mixture and at transverse and longitudinal joints. Apply the tack coat when the surface to be treated is clean and dry. Immediately following the preparation of the surface for treatment, apply the bituminous material by means of the bituminous distributor, within the limits of temperature specified herein and at a rate as specified above in paragraph APPLICATION RATE. Apply the bituminous material so that uniform distribution is obtained over the entire surface to be treated. Treat lightly coated areas and spots missed by the distributor by spraying with a hand wand or using other approved method. Following the application of bituminous material, allow the surface to cure without being disturbed for period of time necessary to permit setting of the tack coat. Apply the bituminous tack coat only as far in advance of the placing of the overlying layer as required for that day's operation. Maintain and protect the treated surface from damage until the succeeding course of pavement is placed.

3.5 CURING PERIOD

Following application of the bituminous material and prior to application of the succeeding layer of asphalt mixture allow the bituminous coat to cure and water or volatiles to evaporate prior to overlaying. Maintain the tacked surface in good condition until the succeeding layer of pavement is placed, by protecting the surface against damage and by repairing and recoating deficient areas. Allow the prime coat to cure without being disturbed for a period of at least 48 hours or longer, as may be necessary to attain penetration into the treated course. Furnish and spread enough sand to effectively blot up excess bituminous material.

3.6 FIELD QUALITY CONTROL

Obtain certificates of compliance for all asphalt material delivered to the project. Obtain samples of the bituminous material under the supervision of the Contracting Officer. The sample may be retained and tested by the Government at no cost to the Contractor.

3.7 SAMPLING AND TESTING

Furnish certified copies of the manufacturer's test reports indicating temperature viscosity relationship for cutback asphalt or asphalt cement, compliance with applicable specified requirements, not less than 5 days before the material is required in the work.

3.7.1 Sampling

Unless otherwise specified, sample bituminous material in accordance with NLT-121 and NLT-348.

3.7.2 Calibration Test

Furnish all equipment, materials, and labor necessary to calibrate the bituminous distributor. Calibrate using the approved job material and prior to applying the bituminous coat material to the prepared surface. Calibrate the bituminous distributor in accordance with MF PG3 Art. 531.4.1 and 531.8.3.

3.7.3 Trial Applications

Before applying the spray application of tack or prime coat, apply three lengths of at least 30 meters for the full width of the distributor bar to evaluate the amount of bituminous material that can be satisfactorily applied.

3.7.3.1 Tack Coat Trial Application Rate

Unless otherwise authorized, apply the trial application rate of bituminous tack coat materials in the amount of 0.23 L/square meter. Make other trial applications using various amounts of material as may be deemed necessary.

3.7.3.2 Prime Coat Trial Application Rate

Unless otherwise authorized, apply the trial application rate of bituminous materials in the amount of 0.66 L/square meter. Make other trial applications using various amounts of material as may be deemed necessary.

3.7.4 Sampling and Testing During Construction

Perform quality control sampling and testing as required in paragraph FIELD QUALITY CONTROL.

3.8 TRAFFIC CONTROLS

Keep traffic off surfaces freshly treated with bituminous material. Provide sufficient warning signs and barricades so that traffic will not travel over freshly treated surfaces.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 12 16.16

ROAD-MIX ASPHALT PAVING

11/20

PART 1 GENERAL

- 1.1 PERCENT PAYMENT
 - 1.1.1 Method of Measurement
 - 1.1.2 Basis of Payment
 - 1.1.3 Lot Pay Factor
 - 1.1.4 Payment Adjustment for Laboratory Air Voids
 - 1.1.4.1 Pay Factor Example for Laboratory Air Voids
 - 1.1.5 Payment Adjustment for In-place Densities
 - 1.1.5.1 Pay Factor Example for In-place Density
 - 1.1.6 Payment Adjustment for Plan Grade
- 1.2 PAYMENT
 - 1.2.1 Method of Measurement
 - 1.2.2 Basis of Payment
- 1.3 REFERENCES
- 1.4 SUBMITTALS
- 1.5 ACCEPTANCE
 - 1.5.1 Acceptability of Work
 - 1.5.2 Acceptance Requirements
 - 1.5.3 Pavement Lots
 - 1.5.4 Sublot Sampling
 - 1.5.5 Additional Sampling and Testing
 - 1.5.6 Theoretical Maximum Density (TMD)
 - 1.5.7 Laboratory Air Voids
 - 1.5.7.1 Tolerance
 - 1.5.7.2 Calculating Laboratory Air Voids
 - 1.5.8 In-place Density
 - 1.5.8.1 Tolerance
 - 1.5.9 Surface Smoothness
 - 1.5.9.1 Smoothness Requirements
 - 1.5.9.1.1 Straightedge Testing
 - 1.5.9.2 Testing Method
 - 1.5.9.2.1 Straightedge Testing
 - 1.5.10 Plan Grade
 - 1.5.11 Laboratory Accreditation and Validation
- 1.6 ENVIRONMENTAL REQUIREMENTS

PART 2 PRODUCTS

- 2.1 SYSTEM DESCRIPTION
 - 2.1.1 Asphalt Mixing Plant
 - 2.1.1.1 Truck Scales
 - 2.1.1.2 Inspection of Plant
 - 2.1.1.3 Storage bins
 - 2.1.2 Hauling Equipment
 - 2.1.3 Asphalt Pavers

- 2.1.3.1 Receiving Hopper
- 2.1.4 Rollers
- 2.1.5 Diamond Grinding
- 2.2 AGGREGATES
 - 2.2.1 Coarse Aggregate
 - 2.2.2 Fine Aggregate
 - 2.2.3 Mineral Filler
 - 2.2.4 Aggregate Gradation
- 2.3 ASPHALT CEMENT BINDER
- 2.4 MIX DESIGN
 - 2.4.1 JMF Requirements
 - 2.4.2 Adjustments to JMF
- 2.5 RECYCLED HOT MIX ASPHALT
 - 2.5.1 RAP Aggregates and Asphalt Cement
 - 2.5.2 RAP Mix

PART 3 EXECUTION

- 3.1 CONTRACTOR QUALITY CONTROL
 - 3.1.1 General Quality Control Requirements
 - 3.1.2 Testing Laboratory
 - 3.1.3 Quality Control Testing
 - 3.1.3.1 Asphalt Content
 - 3.1.3.2 Aggregate Properties
 - 3.1.3.3 Moisture Content of Aggregate
 - 3.1.3.4 Moisture Content of Asphalt Mixture
 - 3.1.3.5 Temperatures
 - 3.1.3.6 VMA, Marshall Stability, and Flow
 - 3.1.3.7 In-Place Density
 - 3.1.3.8 Additional Testing
 - 3.1.3.9 QC Monitoring
 - 3.1.4 Sampling
- 3.2 PREPARATION OF ASPHALT BINDER MATERIAL
- 3.3 PREPARATION OF MINERAL AGGREGATE
- 3.4 PREPARATION OF ASPHALT MIXTURE
- 3.5 PREPARATION OF THE UNDERLYING SURFACE
- 3.6 TRANSPORTING AND PLACING
 - 3.6.1 Transporting
 - 3.6.2 Placing
- 3.7 COMPACTION OF MIXTURE
 - 3.7.1 General
 - 3.7.2 Segregation
- 3.8 JOINTS
 - 3.8.1 Transverse Joints
 - 3.8.2 Longitudinal Joints

-- End of Section Table of Contents --

SECTION 32 12 16.16

ROAD-MIX ASPHALT PAVING
11/20

PART 1 GENERAL

1.1 PERCENT PAYMENT

1.1.1 Method of Measurement

Measurement of the quantity of hot-mix asphalt pavement, per tonne placed and accepted, will be made for the purposes of assessing the pay factors stipulated below.

1.1.2 Basis of Payment

The measured quantity of hot-mixed asphalt pavement will be paid for and included in the lump sum contract price. If less than 100 percent payment is due based on the pay factors stipulated in paragraph PERCENT PAYMENT, a unit price of 400 U.S. dollars per metric ton (tonne) will be used for purposes of calculating the payment reduction.

1.1.3 Lot Pay Factor

The lot pay factor is determined by taking the lowest computed pay factor based on either laboratory air voids, in-place density, smoothness, or grade (each discussed below). Remove and replace lots when the lowest computed pay factor requires rejection. At the end of the project calculate the average pay factor for all lots. If this average lot pay factor exceeds 95.0 percent and no individual lot has a pay factor less than 75.0 percent, then the percent payment for the entire project will be 100 percent of the unit bid price. If the average lot pay factor is less than 95.0 percent, then each lot will be paid for at the unit price multiplied by the lot's pay factor. For any lots which are less than 2,000 tonnes, a weighted lot pay factor will be used to calculate the average lot pay factor. When work on a lot is required to be terminated before all four sublots are completed, the results from the completed sublots will be analyzed to determine the percent payment for the lot following the same procedures and requirements for full lots but with fewer or more test results as determined in paragraph PAVEMENT LOTS.

1.1.4 Payment Adjustment for Laboratory Air Voids

Laboratory air void calculations for each lot will use the average theoretical maximum density values obtained for the lot. Determine the average TMD in accordance with paragraph THEORETICAL MAXIMUM DENSITY (TMD). The mean absolute deviation of the laboratory air void contents (one from each subplot) from the JMF air void content will be evaluated as shown in the example below and a pay factor will be determined from Table 1. When 0 percent payment is determined, remove and replace the rejected lot at least 100 mm into the cold (existing) lane adjacent to the longitudinal joint.

Table 1. Pay Factor Based on Laboratory Air Voids	
Mean Absolute Deviation of Lab Air Voids from JMF	Pay Factor, percent
0.60 or less	100
0.61 - 0.80	98
0.81 - 1.00	95
1.01 - 1.20	90
Above 1.20	reject (0)

1.1.4.1 Pay Factor Example for Laboratory Air Voids

An example of the computation of mean absolute deviation for laboratory air voids is as follows: Assume that the laboratory air voids are determined from 4 sublots where one set of laboratory compacted specimens is from a single subplot. The laboratory air voids for the 4 sublots are determined to be 3.5, 3.0, 4.0, and 3.7. Assume that the target air voids from the JMF is 4.0. The mean absolute deviation is then:

$$\text{Mean Absolute Deviation} = (|3.5 - 4.0| + |3.0 - 4.0| + |4.0 - 4.0| + |3.7 - 4.0|)/4$$

$$\text{Mean Absolute Deviation} = (0.5 + 1.0 + 0.0 + 0.3)/4 = (1.8)/4 = 0.45$$

The mean absolute deviation for laboratory air voids is determined to be 0.45. It can be seen from Table 1 that the lot's pay factor based on laboratory air voids is 100 percent.

1.1.5 Payment Adjustment for In-place Densities

The average in-place mat and joint densities are expressed as a percentage of the average theoretical maximum density (TMD) for the lot. Determine the average TMD in accordance with paragraph THEORETICAL MAXIMUM DENSITY (TMD). The average in-place mat density and joint density for a lot are determined and compared with Table 2 to calculate a single pay factor per lot. Use the following process to determine the single pay factor for in-place density:

- a. Step 1: Determine the pay factors for mat density and joint density using Table 2.
- b. Step 2: Determine ratio of joint area to mat area. The area associated with the joint is considered to be 3 m wide times the length of completed longitudinal construction joint in the lot. This joint area will not exceed the total lot size. The length of joint to be considered will be that length where a new lane has been placed against an adjacent lane of asphalt pavement, either an adjacent freshly paved lane or one paved at any time previously.
- c. Step 3: Compute the weighted pay factor for the joint using the formula in the example below.

- d. Step 4: Compare weighted pay factor for joint density to pay factor for mat density and select the smaller. This selected pay factor is the pay factor based on density for the lot.

When 0 percent payment is determined for mat density, remove and replace the rejected lot at least 100 mm into the cold (existing) lane adjacent to the longitudinal joint. When 0 percent payment is determined for joint density, remove and replace the rejected longitudinal joint with a 3 m wide paving lane that is centered over the joint.

Table 2. Pay Factor Based on In-place Density		
Average Mat Density (4 Cores) (Percent of TMD)	Pay Factor, Percent	Average Joint Density (4 Cores) (Percent of TMD)
93.0 - 96.0	100.0	91.5 or above
92.9	100.0	91.4
92.8 or 96.1	99.9	91.3
92.7	99.8	91.2
92.6 or 96.2	99.6	91.1
92.5	99.4	91.0
92.4 or 96.3	99.1	90.9
92.3	98.7	90.8
92.2 or 96.4	98.3	90.7
92.1	97.8	90.6
92.0 or 96.5	97.3	90.5
91.9	96.3	90.4
91.8 or 96.6	94.1	90.3
91.7	92.2	90.2
91.6 or 96.7	90.3	90.1
91.5	87.9	90.0
91.4 or 96.8	85.7	89.9
91.3	83.3	89.8
91.2 or 96.9	80.6	89.7
91.1	78.0	89.6
91.0 or 97.0	75.0	89.5
below 91.0, above 97.0	0.0 (reject)	below 89.5

1.1.5.1 Pay Factor Example for In-place Density

An example of the computation of a pay factor based on in-place density, is as follows: Assume the following test results for field density made on the lot: (1) Average mat density = 92.2 percent (of lab TMD). (2) Average joint density = 90.5 percent (of lab TMD). (3) Total area of lot = 2,800 square meters. (4) Length of completed longitudinal construction

joint = 610 meters.

- a. Step 1: Determine pay factor based on mat density and on joint density, using Table 2:

Mat density of 92.2 percent = 98.3 pay factor.

Joint density of 90.5 percent = 97.3 pay factor.

- b. Step 2: Determine ratio of joint area to mat area. Multiply the length of completed longitudinal construction joint by the specified 3 meter width and divide by the mat area (total paved area in the lot).

Ratio = Ratio of joint area to mat area

Ratio = 610 meters/2,800 square meters

Ratio = 0.6667

- c. Step 3: Weighted pay factor (wpf) for joint is determined as indicated below:

$wpf = \text{joint pay factor} + (100 - \text{joint pay factor}) \times (1 - \text{ratio})$

$wpf = 97.3 + (100 - 97.3) \times (1 - 0.6667) = 98.2 \text{ percent}$

- d. Step 4: Compare weighted pay factor for joint density to pay factor for mat density and select the smaller:

Pay factor for mat density: 98.3 percent.

Weighted pay factor for joint density: 98.2 percent

Selected pay factor: 98.2 percent

1.1.6 Payment Adjustment for Plan Grade

When more than 5 percent of all measurements made within a lot are outside the 15 mm tolerance, the pay factor based on grade for that lot will be 95 percent. For individual locations where the grade exceeds 22.5 mm tolerance, remove the surface lift full depth and replace the lift with asphalt pavement to meet specification requirements at no additional cost to the Government. High spots can be diamond ground as an alternative to remove and replace in order to meet grade requirements for the lot and at individual locations.

1.2 PAYMENT

1.2.1 Method of Measurement

Measurement of the quantity of hot-mix asphalt pavement per lot will be made for the purposes of assessing acceptance stipulated in paragraph ACCEPTANCE.

1.2.2 Basis of Payment

The measured quantity of hot-mixed asphalt pavement will be paid for and included in the lump sum contract price.

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 146403	(2018) Determination of clay lumps and crumbly particles in aggregate for the making of mortars and concretes.
UNE-EN 58	(2012) Bitumen and bituminous binders - Sampling bituminous binders
UNE-EN 932-1	(1997) Test For Geometrical Properties Of Aggregates. Part 1: Methods for Sampling.
UNE-EN 933-1	(2012) Tests for geometrical properties of aggregates - Part 1: Determination of particle size distribution - Sieving method
UNE-EN 933-2	(1996) Test For Geometrical Properties Of Aggregates. Part 2: Determination Of Particle Size Distribution. Test Sieves, Nominal Size Of Apertures.
UNE-EN 933-4	(2008) Tests for geometrical properties of aggregates - Part 4: Determination of particle shape - Shape index
UNE-EN 933-5	(1999) Test For Geometrical Properties Of Aggregates. Part 5: Determination Of Percentage Of Crushed And Broken Surfaces In Coarse Aggregate Particles.
UNE-EN 933-8	(2012+A1:2015) Tests for geometrical properties of aggregates - Part 8: Assessment of fines - Sand equivalent test
UNE-EN 1097-2	(2010) Tests for mechanical and physical properties of aggregates - Part 2: Methods for the determination of resistance to fragmentation
UNE-EN 1097-3	(1999) Tests For Mechanical And Physical Properties Of Aggregates. Part 3: Determination Of Loose Bulk Density And Voids.
UNE-EN 1097-5	(2009) Tests for mechanical and physical properties of aggregates - Part 5: Determination of the water content by drying in a ventilated oven
UNE-EN 1367-2	(2010) Tests for thermal and weathering properties of aggregates - Part 2: Magnesium sulfate test

UNE-EN 12697-5	(2020) Bituminous mixtures - Test methods - Part 5: Determination of the maximum density
UNE-EN 12697-10	(2019) Bituminous mixtures - Test methods - Part 10: Compactability
UNE-EN 12697-31	(2020) Bituminous mixtures - Test methods - Part 31: Specimen preparation by gyratory compactor
UNE-EN 12697-34	(2013) Bituminous mixtures - Test methods - Part 34: Marshall test
UNE-EN 13108-20	(2007) Bituminous mixtures - Material specifications - Part 20: Type Testing
UNE-EN 13108-21	(2007) Bituminous mixtures - Material specifications - Part 21: Factory Production Control
UNE-EN ISO/IEC 17025	(2017) General requirements for the competence of testing and calibration laboratories (ISO/IEC 17025:2017)

CENTRO DE ESTUDIOS Y EXPERIMENTACION DE OBRAS PUBLICAS (CEDEX)

NLT-121	(1999) Toma de Muestras de los Materiales Bituminosos
NLT-156	(1994) Densidad aparente de los áridos.
NLT-159	(1986) Resistencia a la Deformacion Plastica de Mezclas Butuminosas Empleando el Aparato Marshall
NLT-164	(1990) Contenido de ligante en mezclas bituminosas.
NLT-168	(1990) Densidad y Huecos en Mezclas Bituminosas Compactadas
NLT-176	(1992) Densidad aparente del polvo mineral en tolueno.
NLT-314	(1992) Toma de muestras testigos en pavimento
NLT-348	(1990) Toma de Muestras de Mezclas Bituminosas para Pavimentacion.
NLT-384	(2000) Contenido de ligante en mezclas bituminosas por el método de combustión.

MINISTERIO DE FOMENTO (MF)

FOM/510/2018	WARNING: Text in tags exceeds the maximum length of 300 characters
--------------	---

FOM/2523/2014	(2014) Orden de 12 de diciembre, por la que se actualizan determinados artículos del pliego de prescripciones técnicas generales para obras de carreteras y puentes, relativos a materiales básicos, a firmes y pavimentos, y a señalización, balizamiento y sistemas de contención de vehículos.
MF PG3	(2014) Pliego de Prescripciones Técnicas Generales para Obras de Carreteras y Puentes
OC 3/2019	(2019) Orden Circular Oc 3/2019 Sobre Mezclas Bituminosas Tipo SMA
RD 305/2011	(2011) Reglamento (UE) nº 305/2011 del Parlamento Europeo y del Consejo, de 9 de marzo de 2011, por el que se establecen condiciones armonizadas para la comercialización de productos de construcción y se deroga la Directiva 89/106/CEE del Consejo.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO T 304	(2011; R 2015) Standard Method of Test for Uncompacted Void Content of Fine Aggregate
AASHTO T 329	(2015) Standard Test Method for Moisture Content of Hot Mix Asphalt (HMA) by Oven Method

ASPHALT INSTITUTE (AI)

AI MS-2	(2015) Asphalt Mix Design Methods
---------	-----------------------------------

ASTM INTERNATIONAL (ASTM)

ASTM C29/C29M	(2017a) Standard Test Method for Bulk Density ("Unit Weight") and Voids in Aggregate
ASTM C117	(2017) Standard Test Method for Materials Finer than 75-um (No. 200) Sieve in Mineral Aggregates by Washing
ASTM C127	(2015) Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Coarse Aggregate
ASTM C128	(2015) Standard Test Method for Density, Relative Density (Specific Gravity), and Absorption of Fine Aggregate
ASTM C136/C136M	(2019) Standard Test Method for Sieve Analysis of Fine and Coarse Aggregates

ASTM C566	(2013) Standard Test Method for Total Evaporable Moisture Content of Aggregate by Drying
ASTM D75/D75M	(2019) Standard Practice for Sampling Aggregates
ASTM D242/D242M	(2009; R 2014) Mineral Filler for Bituminous Paving Mixtures
ASTM D979/D979M	(2015) Sampling Bituminous Paving Mixtures
ASTM D2041/D2041M	(2011) Theoretical Maximum Specific Gravity and Density of Bituminous Paving Mixtures
ASTM D2172/D2172M	(2017; E 2018) Standard Test Methods for Quantitative Extraction of Asphalt Binder from Asphalt Mixtures
ASTM D2726/D2726M	(2019) Standard Test Method for Bulk Specific Gravity and Density of Non-Absorptive Compacted Bituminous Mixtures
ASTM D3203/D3203M	(2017) Standard Test Method for Percent Air Voids in Compacted Asphalt Mixtures
ASTM D3665	(2012; R 2017) Standard Practice for Random Sampling of Construction Materials
ASTM D4791	(2019) Flat Particles, Elongated Particles, or Flat and Elongated Particles in Coarse Aggregate
ASTM D4867/D4867M	(2009; R 2014) Effect of Moisture on Asphalt Concrete Paving Mixtures
ASTM D5361/D5361M	(2016) Standard Practice for Sampling Compacted Asphalt Mixtures for Laboratory Testing
ASTM D5444	(2015) Mechanical Size Analysis of Extracted Aggregate
ASTM D5821	(2013; R 2017) Standard Test Method for Determining the Percentage of Fractured Particles in Coarse Aggregate
ASTM D6307	(2019) Standard Test Method for Asphalt Content of Asphalt Mixture by Ignition Method
ASTM D6925	(2014) Standard Test Method for Preparation and Determination of the Relative Density of Hot Mix Asphalt (HMA) Specimens by Means of the Superpave Gyratory Compactor

ASTM D6926	(2020) Standard Practice for Preparation of Asphalt Mixture Specimens Using Marshall Apparatus
ASTM D6927	(2015) Standard Test Method for Marshall Stability and Flow of Bituminous Mixtures

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Placement Plan; G

SD-03 Product Data

Diamond Grinding Plan; G
Mix Design; G
Contractor Quality Control; G

SD-04 Samples

Aggregates
Asphalt Cement Binder

SD-06 Test Reports

Aggregates; G
QC Monitoring

SD-07 Certificates

Asphalt Cement Binder; G
Laboratory Accreditation and Validation

1.5 ACCEPTANCE

1.5.1 Acceptability of Work

Acquire the services of an independent commercial laboratory to perform acceptance testing. Acceptance of the plant produced mix and in-place requirements will be on a lot to lot basis. The materials and the pavement itself will be accepted on the basis of production testing. The Government may make check tests from split samples to validate the results of the production testing. Testing performed by the Government does not reduce the required testing of the independent commercial laboratory. Split samples will be taken for Government testing to reduce the variability between the independent commercial laboratory and the Government's test results. When the difference between the independent commercial laboratory and the Government's test results for split samples exceed the acceptable range of two results for multilaboratory precision for the appropriate test method (i.e. ASTM) then at least one of the laboratories is determined to be in error. An evaluation of procedures and equipment in both laboratories will be made to determine the cause(s) for the differences. Develop steps to correct procedures and equipment to

bring multilaboratory precision to within acceptable limits.

1.5.2 Acceptance Requirements

Provide all sampling and testing required for acceptance and payment adjustment. Where appropriate, adjustments in percent payment acceptance for individual lots of asphalt pavement will be made based on laboratory air voids, in-place density, smoothness, and grade in accordance with the following paragraphs. Surface smoothness and grade determinations will be made on the lot as a whole. Exceptions or adjustments to this will be made in situations where the mix within one lot is placed as part of both the intermediate and surface courses, thus smoothness and grade measurements for the entire lot cannot be made.

1.5.3 Pavement Lots

A standard lot for all requirements is equal to one day's production or 2,000 tonnes, whichever is smaller. Divide each lot into four equal sublots in order to evaluate laboratory air voids and in-place density. When operational conditions cause a lot to be terminated before the specified four sublots have been completed, use the following procedure to adjust the lot size and number of tests for the lot. Where three sublots have been completed, they constitute a lot. Where one or two sublots have been completed, incorporate them into the next lot and the total number of sublots (i.e. 5 or 6 sublots) is used for acceptance criteria. Include partial lots at the end of asphalt production into the previous lot. Complete and report all theoretical maximum density, laboratory air voids, and in-place density testing within 24 hours after construction of each lot.

1.5.4 Sublot Sampling

Take one mixture sample for each sublot in accordance with ASTM D979/D979M or NLT-121 and NLT-348 and UNE-EN 58 from a random truck or another location for determining theoretical maximum density, laboratory air voids, any additional testing the Government desires, and Contractor Quality Control. All samples will be selected randomly, using commonly recognized methods of assuring randomness conforming to ASTM D3665 and employing tables of random numbers or computer programs and executed by registered laboratories in accordance with UNE-EN ISO/IEC 17025.

1.5.5 Additional Sampling and Testing

The Government reserves the right to direct additional samples and tests for any area which appears to deviate from the specification requirements. The cost of any additional testing will be paid for by the Government. Testing in these areas will be treated as a separate lot. Payment Acceptance will be made for the quantity of asphalt pavement represented by these tests in accordance with the provisions of this section.

1.5.6 Theoretical Maximum Density (TMD)

Measure theoretical maximum density one time for each sublot in accordance with ASTM D2041/D2041M or UNE-EN 12697-5 for purposes of calculating laboratory air voids and determining in-place density. The average TMD for each lot will be determined as the average TMD of the random sublot samples. When the TMD on both sides of a longitudinal joint is different, the average of these two TMD values will be used as the TMD needed to

calculate the percent joint density.

1.5.7 Laboratory Air Voids

Prepare one set of laboratory compacted specimens for each subplot in accordance with ASTM D6926 or UNE-EN 12697-34 using the hand-held hammer for the Marshall Method. Prepare one set of laboratory compacted specimens for each subplot in accordance with ASTM D6925 or UNE-EN 12697-10 and UNE-EN 12697-31, in which case an angle of 0.82° is acceptable, in lieu of an angle of 1.16° as requested by ASTM, using the Superpave gyratory compactor. Provide three test specimens prepared from the same sample for each set of laboratory compacted specimens. Compact the specimens within 2 hours of the time the mixture was loaded into trucks at the asphalt plant. Do not reheat samples prior to compaction. Provide insulated containers as necessary to maintain the sample temperature. Measure the bulk density of laboratory compacted specimens in accordance with ASTM D2726/D2726M or NLT-168. Determine laboratory air voids from one set (three laboratory compacted specimens) for each subplot sample in accordance with ASTM D3203/D3203M or NLT-168.

1.5.7.1 Tolerance

Provide laboratory air voids with a mean absolute deviation of 1.00 percent or less from the JMF for each lot. Remove and replace lots that do not meet the laboratory air voids requirement at least 100 mm into the cold (existing) lane adjacent to the longitudinal joint, at no additional cost to the Government. The mean absolute deviation of the laboratory air void contents from the JMF air void content will be evaluated as shown in the example below.

1.5.7.2 Calculating Laboratory Air Voids

Laboratory air void calculations for each lot will use the average theoretical maximum density values obtained for the lot. Determine the average TMD in accordance with paragraph THEORETICAL MAXIMUM DENSITY (TMD). The mean absolute deviation of the laboratory air void contents (one from each subplot) from the JMF air void content will be evaluated as in the following example:

Assume that the laboratory air voids are determined from 4 sublots where one set of laboratory compacted specimens is from a single subplot. The laboratory air voids for the 4 sublots are determined to be 3.5, 3.0, 4.0, and 3.7. Assume that the target air voids from the JMF is 4.0. The mean absolute deviation is then:

$$\text{Mean Absolute Deviation} = (|3.5 - 4.0| + |3.0 - 4.0| + |4.0 - 4.0| + |3.7 - 4.0|)/4$$

$$\text{Mean Absolute Deviation} = (0.5 + 1.0 + 0.0 + 0.3)/4 = (1.8)/4 = 0.45$$

The mean absolute deviation for laboratory air voids is determined to be 0.45. It can be seen that 0.45 is less than 1.00 percent. The lot is acceptable for laboratory air voids.

1.5.8 In-place Density

Obtain one random 100 mm or 150 mm diameter core from the mat and joint of each subplot in accordance with ASTM D5361/D5361M or NLT-314 for determining in-place density. Cut samples neatly with a diamond core

drill bit. Obtain random cores that are the full thickness of the layer being placed. Select core locations randomly using the procedures contained in ASTM D3665 or NLT-314, par. 3 Procedures 1 or 2. Locate cores for mat density no closer than 300 mm from a transverse or longitudinal joint including the pavement edge. Center all cores for joint density on the joint. Discard samples that are clearly defective as a result of sampling and take an additional random core. When the random core is less than 25 mm thick, it will not be included in the analysis. In this case, obtain another random core sample. Clean and tack coat dry core holes before filling with asphalt mixture. Fill all core holes with asphalt mixture and compact using a standard Marshall hammer to the density specified. Provide all tools, labor, and materials for cutting samples, cleaning, and filling the cored pavement. Measure in-place density in accordance with ASTM D2726/D2726M or NLT-168 using each core obtained from the mat and joint.

1.5.8.1 Tolerance

Provide a minimum in-place mat density of 93.0 percent and a minimum in-place joint density of 90.0 percent for each lot. The average in-place mat and joint densities are expressed as a percentage of the average theoretical maximum density (TMD) for the lot. Determine the average TMD in accordance with paragraph THEORETICAL MAXIMUM DENSITY (TMD). Remove and replace lots that do not meet the in-place mat density requirement at least 100 mm into the cold (existing) lane adjacent to the longitudinal joint, at no additional cost to the Government. Remove and replace the longitudinal joint when the lot does not meet the in-place joint density, at no additional cost to the Government. Use a 3 m wide paving lane that is centered over the joint.

1.5.9 Surface Smoothness

Use a straightedge and profilograph for measuring surface smoothness. Use the profilograph method for all longitudinal testing, except for paving lanes less than 400 meters in length. Use the straightedge method for transverse testing, for longitudinal testing where the length of each pavement lane is less than 400 meters, and at the ends of the paving limits for the project. Smoothness requirements do not apply over crowns or grade breaks. Maintain detailed notes of the testing results and provide a copy to the Government immediately after each day's testing.

1.5.9.1 Smoothness Requirements

1.5.9.1.1 Straightedge Testing

Provide finished surfaces of the pavements with no abrupt change of 6 mm or more when checked with an approved 4 m straightedge. Remove and replace surface lift lots when the surface smoothness exceeds 9 mm, at no additional cost to the Government. High spots can be diamond ground as an alternative to remove and replace in order to meet surface smoothness requirements at individual locations.

1.5.9.2 Testing Method

After the final rolling, but not later than 24 hours after placement, test the surface of the pavement in each entire lot in a manner to reveal surface irregularities exceeding the tolerances specified above. If any pavement areas are diamond ground, retest these areas immediately after diamond grinding. The maximum area allowed to be corrected by diamond

grinding is 10 percent of the total area of the lot. Check a number of random locations along with any observed suspicious locations primarily at transverse and longitudinal joints with the straightedge.

1.5.9.2.1 Straightedge Testing

Use the straightedge to measure abrupt changes in surface smoothness. Hold the straightedge in contact with the pavement surface and measure the maximum distance between the straightedge and the pavement surface. Determine the amount of surface irregularity by placing the freestanding (unleveled) straightedge on the pavement surface and allowing it to rest upon the two highest spots covered by its length, and measuring the maximum gap between the straightedge and the pavement surface in the area between these two high points.

1.5.10 Plan Grade

Provide a final wearing surface of pavement conforming to the elevations and cross sections shown and not vary more than 15 mm from the plan grade established and approved at site of work. Within 5 working days after completion of a particular lot incorporating the final wearing course, test the final wearing surface of the pavement for conformance with specified plan grade requirements. Match finished surfaces at juncture with other pavements with finished surfaces of abutting pavements. Deviation from the plan elevation will not be permitted in areas of pavements where closer conformance with planned elevation is required for the proper functioning of drainage and other appurtenant structures involved. For roads, the grade will be determined by running lines of levels along the centerline at intervals of 7.6 m or less longitudinally to determine the elevation of the completed pavement surface. Measure transverse grades at appropriate intervals. For parking lots, the grade will be determined by running lines of levels at intervals of 7.6 m or less longitudinally and transversely to determine the elevation of the completed pavement surface. Diamond grinding can be used to remove high spots to meet grade requirements. Skin patching for correcting low areas or planing or milling for correcting high areas will not be permitted. Maintain detailed notes of the results of the testing and provide a copy to the Government immediately after each day's testing. Remove and replace surface lift lots when individual locations exceed the 15 mm tolerance, at no additional cost to the Government. High spots can be diamond ground as an alternative to remove and replace in order to meet plan grade requirements at individual locations.

1.5.11 Laboratory Accreditation and Validation

Provide laboratories used to develop the Job Mix Formula (JMF), perform acceptance testing, and Contractor Quality Control testing that meet the requirements of UNE-EN ISO/IEC 17025 and are ENAC accredited. Provide laboratories with a masonry saw having a diamond blade for trimming pavement cores and samples. Perform all required test methods by an accredited laboratory. Schedule and provide payment for laboratory inspections. Additional payment or a time extension due to failure to acquire the required laboratory accreditation is not allowed. Submit a certificate of compliance signed by the manager of the laboratory stating that it meets these requirements to the Government prior to the start of construction. At a minimum, include the following certifications:

- a. Qualifications of personnel; laboratory manager, supervising technician, and testing technicians.

- b. A listing of equipment to be used in developing the job mix.
- c. A copy of the laboratory's quality control system.

1.6 ENVIRONMENTAL REQUIREMENTS

Do not place the asphalt mixture upon a wet surface or when the surface temperature of the underlying course is less than specified in Table 3. The temperature requirements may be waived by the Government, if requested; however, meet all other requirements including compaction.

Table 3. Surface Temperature Limitations of Underlying Course	
Mat Thickness, mm	Degrees C
75 or greater	4
Less than 75	7

PART 2 PRODUCTS

2.1 SYSTEM DESCRIPTION

Perform the work consisting of pavement courses composed of mineral aggregate and asphalt material heated and mixed in a central mixing plant and placed on a prepared course. Provide asphalt pavement designed and constructed in accordance with this section conforming to the lines, grades, thicknesses, and typical cross sections shown on the drawings. Construct each course to the depth, section, or elevation required by the drawings and rolled, finished, and approved before the placement of the next course. Submit proposed Placement Plan indicating lane widths and longitudinal joints for each course or lift.

2.1.1 Asphalt Mixing Plant

RD 305/2011. Provide plants used for the preparation of asphalt mixture conforming to the requirements of UNE-EN 13108-20 and UNE-EN 13108-21 with the following changes:

2.1.1.1 Truck Scales

Weigh the asphalt mixture on approved scales, or on certified public scales at no additional expense to the Government. Inspect and seal scales at least annually by an approved calibration laboratory.

2.1.1.2 Inspection of Plant

Provide access to the Government at all times, to all areas of the plant for checking adequacy of equipment; inspecting operation of the plant; verifying weights, proportions, and material properties; checking the temperatures maintained in the preparation of the mixtures and for taking samples. Provide assistance as requested, for the Government to procure any desired samples.

2.1.1.3 Storage bins

The asphalt mixture can be stored in non-insulated storage bins for a period of time not exceeding 3 hours. The asphalt mixture can be stored in insulated storage bins for a period of time not exceeding 8 hours.

Provide the mix drawn from bins that meets the same requirements as mix loaded directly into trucks.

2.1.2 Hauling Equipment

Provide trucks used for hauling asphalt mixture that have tight, clean, and smooth metal beds. To prevent the mixture from adhering to them, lightly coat the truck beds with a minimum amount of paraffin oil, lime solution, or other approved material. Do not use petroleum based products as a release agent. Provide each truck with a suitable cover to protect the mixture from adverse weather, contamination, and loss of material during hauling. When necessary due to long haul distance and cold weather, provide insulated truck beds with covers (tarps) that are securely fastened.

2.1.3 Asphalt Pavers

Provide mechanical spreading and finishing equipment consisting of a self-powered paver, capable of spreading and finishing the mixture to the specified line, grade, and cross section. Provide paver screed capable of laying a uniform mixture to meet the specified thickness, smoothness, and grade without physical or temperature segregation, the full width of the material being placed. Provide a paver with a vibrating screed to be used during all placement.

2.1.3.1 Receiving Hopper

Provide paver with a receiving hopper of sufficient capacity to permit a uniform spreading operation and a distribution system to place the mixture uniformly in front of the screed without segregation. Provide a screed that effectively produces a finished surface of the required evenness and texture without tearing, shoving, or gouging the mixture.

2.1.4 Rollers

Provide rollers in good condition and operate at slow speeds to avoid displacement of the asphalt mixture. Provide sufficient number, type, and weight of rollers to compact the mixture to the required density while it is still in a workable condition. Do not use equipment which causes excessive crushing of the aggregate.

2.1.5 Diamond Grinding

Those performing diamond grinding are required to have a minimum of three years experience in diamond grinding. In areas not meeting the specified limits for surface smoothness and plan grade, reduce high areas to attain the required smoothness and grade, except as depth is limited below. Reduce high areas by diamond grinding the asphalt pavement with approved equipment. Perform diamond grinding by sawing with saw blades impregnated with an industrial diamond abrasive. Assemble the saw blades in a cutting head mounted on a machine designed specifically for diamond grinding that produces the required texture and smoothness level without damage to the asphalt pavement or joint faces. Provide diamond grinding equipment with saw blades that are 3 mm wide, a minimum of 60 blades per 300 mm of cutting head width, and capable of cutting a path a minimum of 0.9 m wide. Diamond grinding equipment that causes raveling, fracturing of aggregate, or disturbance to the underlying material will not be allowed. The maximum area corrected by diamond grinding the surface of the asphalt pavement is 10 percent of the total area of any lot. The maximum depth of

diamond grinding is 12 mm. Provide diamond grinding machine equipped to flush and vacuum the pavement surface. Dispose of all debris from diamond grinding operations off Government property. Prior to diamond grinding, submit a Diamond Grinding Plan for review and approval. At a minimum, include the daily reports for the deficient areas, the location and extent of deficiencies, corrective actions, and equipment. Remove and replace all pavement areas requiring plan grade or surface smoothness corrections in excess of the limits specified.

Prior to production diamond grinding operations, perform a test section at the approved location, consisting of a minimum of two adjacent passes with a minimum length of 12 m to allow evaluation of the finish and transition between adjacent passes. Production diamond grinding operations cannot be performed prior to approval.

2.2 AGGREGATES

Notify the Government at least 7 days before sampling aggregates. Obtain samples in accordance with ASTM D75/D75M or UNE-EN 932-1 that are representative of the materials to be used for the project. Provide aggregates consisting of crushed stone, crushed gravel, crushed slag, screenings, natural sand, and mineral filler as required. The portion of material retained on the 4.75 mm sieve is coarse aggregate. The portion of material passing the 4.75 mm sieve and retained on the 0.075 mm sieve is fine aggregate. The portion passing the 0.075 mm sieve is defined as mineral filler. Submit sufficient materials to produce 90 kg of blended mixture for mix design verification. Submit all aggregate test results and samples to the Government at least 14 days prior to start of construction. Perform job aggregate testing no earlier than 6 months before contract award.

2.2.1 Coarse Aggregate

Provide coarse aggregate consisting of sound, tough, durable particles, free from films of material that would prevent thorough coating and bonding with the asphalt material and free from organic matter and other deleterious substances. Provide coarse aggregate particles meeting the following requirements:

- a. The percentage of loss not greater than 40 percent after 500 revolutions when tested in accordance with UNE-EN 1097-2.
- b. The sodium sulfate soundness loss not exceeding 12 percent, or the magnesium sulfate soundness loss not exceeding 18 percent after five cycles when tested in accordance with UNE-EN 1367-2.
- c. At least 75 percent by weight of coarse aggregate containing two or more fractured faces when tested in accordance with UNE-EN 933-5 with fractured faces produced by crushing.
- d. The particle shape essentially cubical and the aggregate containing not more than 10 percent, by weight, of flat and elongated particles (5:1 ratio of length to thickness) when tested in accordance with ASTM D4791, Method B or UNE-EN 933-4.
- e. Slag consisting of air-cooled, blast furnace slag with a compacted weight of not less than 1200 kg/cubic meter when tested in accordance with ASTM C29/C29M or NLT-176.

- f. Clay lumps and friable particles not exceeding 0.3 percent, by weight, when tested in accordance with UNE 146403.

2.2.2 Fine Aggregate

Provide fine aggregate consisting of clean, sound, tough, durable particles. Provide aggregate particles that are free from coatings of clay, silt, or any objectionable material, contain no clay balls, and meet the following requirements:

- a. Quantity of natural sand (noncrushed material) added to the aggregate blend not exceeding 15 percent by weight of total aggregate.
- b. Individual fine aggregate sources with a sand equivalent value greater than 45 when tested in accordance with UNE-EN 933-8.
- c. Fine aggregate portion of the blended aggregate with an uncompacted void content greater than 45.0 percent when tested in accordance with AASHTO T 304 Method A.
- d. Clay lumps and friable particles not exceeding 0.3 percent, by weight, when tested in accordance with UNE 146403.

2.2.3 Mineral Filler

Provide mineral filler consisting of a nonplastic material meeting the requirements of ASTM D242/D242M or compliant with FOM/2523/2014, Table 542.7 for material passing sieve 63 microns and with density between 0.50 and 0.90.

2.2.4 Aggregate Gradation

Provide a combined aggregate gradation that conforms to gradations specified in Table 4, when tested in accordance with ASTM C136/C136M and UNE-EN 933-1, and does not vary from the low limit on one sieve to the high limit on the adjacent sieve or vice versa, but grades uniformly from coarse to fine. Provide a JMF within the specification limits; however, the gradation can exceed the limits when the allowable deviation from the JMF shown in Tables 4 and 5 are applied.

Table 4. Aggregate Gradations	
Sieve Size, mm	Gradation Percent Passing by Mass
25.0	---
19.0	100
12.5	90-100
9.5	69-89
4.75	53-73
2.36	38-60
1.18	26-48
0.60	18-38
0.30	11-27

Table 4. Aggregate Gradations	
Sieve Size, mm	Gradation Percent Passing by Mass
0.15	6-18
0.075	3-6

2.3 ASPHALT CEMENT BINDER

Provide asphalt cement binder that conforms to Art. 211 and 542 of MF PG3 Performance Grade BC 35/50. Provide test data indicating grade certification by the supplier at the time of delivery of each load to the mix plant. Submit copies of these certifications to the Government. The supplier is defined as the last source of any modification to the binder. The Government may sample and test the binder at the mix plant at any time before or during mix production.

2.4 MIX DESIGN

Develop the mix design in accordance with FOM/2523/2014, Art. 542 and as modified by FOM/510/2018, Type AC-22 SURF 35/50D and the following. Perform Job Mix formula (JMF) and aggregates testing no earlier than 6 months before contract award. Provide asphalt mixture composed of well-graded aggregate, mineral filler if required, and asphalt material. Provide aggregate fractions sized, handled in separate size groups, and combined in such proportions that the resulting mixture meets the grading requirements. Do not produce asphalt pavement for payment acceptance until a JMF has been approved. Prepare samples at various asphalt contents and compacted in accordance with ASTM D6925 or UNE-EN 12697-10 and UNE-EN 12697-31. Use laboratory compaction temperatures for Polymer Modified Asphalts as recommended by the asphalt binder manufacturer. Determine the Tensile Strength Ratio (TSR) of the composite mixture in accordance with ASTM D4867/D4867M. Compact the TSR specimens to an air void content of 7 percent plus or minus 1 percent. If the Tensile Strength Ratio (TSR) of the composite mixture is less than 75, reject the aggregates or treat the asphalt mixture with an anti-stripping agent. Add a sufficient amount of anti-stripping agent to produce a TSR of not less than 75. If an antistrip agent is required, provide it at no additional cost to the Government. Provide sufficient materials to produce 90 kg of blended mixture to the Government for verification of mix design at least 14 days prior to construction of test section.

2.4.1 JMF Requirements

Submit the proposed JMF in writing, for approval, at least 14 days prior to the start of the test section including, as a minimum:

- Percent passing each sieve size.
- Percent of asphalt cement.
- Percent of each aggregate and mineral filler to be used.
- Asphalt performance grade or penetration grade.
- Number of blows of hammer per side of molded specimen or number of Superpave gyratory compactor gyrations.
- Laboratory mixing temperature.
- Laboratory compaction temperature.

- h. Temperature-viscosity relationship of the asphalt cement
- i. Plot of the combined gradation on the 0.45 power gradation chart, stating the nominal maximum size.
- j. Graphical plots and summary tabulation of Marshall stability, flow, air voids, voids in the mineral aggregate, and unit weight versus asphalt content as shown in AI MS-2. Include summary tabulation that includes individual specimen data for each specimen tested.
- k. Specific gravity and absorption of each aggregate.
- l. Percent natural sand.
- m. Percent particles with two or more fractured faces (in coarse aggregate).
- n. Fine aggregate angularity.
- o. Percent flat or elongated particles in coarse aggregate.
- p. Tensile Strength Ratio and wet/dry specimen test results.
- q. Antistrip agent (if required).
- r. List of all modifiers.
- s. Percentage and properties (asphalt content, aggregate gradation, and aggregate properties) of RAP in accordance with paragraph RECYCLED ASPHALT PAVEMENT, if RAP is used.

Table 5. Mix Design Criteria			
Test Property	Marshall (50 Blows)	Marshall (75 Blows)	AC-22 SURF 35/50D
Stability, N, minimum	4450 ⁽¹⁾	8000 ⁽¹⁾	8000
Air voids, percent	3-5	3-5	4-7
Minimum Percent Voids in Mineral Aggregate (VMA) ⁽²⁾			
Gradation	14.0	14.0	14.0
TSR, minimum percent	75	75	80
(1) This is a minimum requirement. Provide significantly higher average during construction to ensure compliance with the specifications.			
(2) Calculate VMA in accordance with AI MS-2, based on ASTM C127 and ASTM C128 bulk specific gravity for the aggregate.			

2.4.2 Adjustments to JMF

The JMF for each mixture is in effect until a new formula is approved in writing by the Government. Should a change in sources of any materials be made, perform a new mix design and a new JMF approved before the new material is used. Make minor adjustments within the specification limits to the JMF to optimize mix volumetric properties. Adjustments to the original JMF are limited to plus or minus 4 percent on the 4.75 mm and coarser sieves; plus or minus 3 percent on the 2.36 mm to 0.30 mm sieves; and plus or minus 1 percent on the 0.15 mm sieve and 0.075 mm sieve. Asphalt content adjustments are limited to plus or minus 0.40 from the original JMF. If adjustments are needed that exceed these limits, develop a new mix design.

2.5 RECYCLED HOT MIX ASPHALT

Provide recycled asphalt mixture consisting of reclaimed asphalt pavement (RAP), coarse aggregate, fine aggregate, mineral filler, and asphalt cement. Provide RAP of a consistent gradation, asphalt content, and properties. Maintain RAP stockpiles free from contamination including coal-tar sealers. Limit the maximum RAP chunk size to 50 mm when feeding RAP into the plant. The individual aggregates in a RAP chunk are not to exceed the maximum size aggregate of the gradation specified in Table 4. Design the recycled asphalt mixture using procedures contained in AI MS-2. Provide RAP job mix that meets the requirements of paragraph MIX DESIGN. Limit the amount of RAP so the asphalt binder from the RAP does not exceed 30 percent of the total asphalt content.

2.5.1 RAP Aggregates and Asphalt Cement

Provide a blend of aggregates used in the recycled mix that meet the requirements of paragraph AGGREGATES. Establish the percentage of asphalt binder in the RAP for the mixture design according to ASTM D2172/D2172M or ASTM D6307 or NLT-164 using the appropriate dust correction procedure.

2.5.2 RAP Mix

Select the virgin asphalt binder as described below:

- a. For 0 to 20 percent recycled binder content - no change in virgin binder selection.
- b. For 20+ percent to 30 percent recycled binder content - select virgin binder one grade softer than normal.

PART 3 EXECUTION

MF PG3, FOM/2523/2014, Art. 542 and OC 3/2019.

3.1 CONTRACTOR QUALITY CONTROL

3.1.1 General Quality Control Requirements

Submit the Quality Control Plan. Do not produce hot-mix asphalt for payment acceptance until the quality control plan has been approved. In the quality control plan, address all elements which affect the quality of the pavement including, but not limited to:

- a. Mix Design and unique JMF identification code
- b. Aggregate Grading
- c. Quality of Materials
- d. Stockpile Management and procedures to prevent contamination
- e. Proportioning
- f. Mixing and Transportation
- g. Mixture Volumetrics
- h. Moisture Content of Mixtures
- i. Placing and Compaction
- j. Joints
- k. Surface Smoothness
- l. Truck bed release agent
- m. Correlation of mechanical hammer to hand hammer. Determine the number of blows of the mechanical hammer required to provide the same density of the JMF as provided by the hand hammer. Use the average of three

specimens per trial blow application.

3.1.2 Testing Laboratory

Provide a fully equipped asphalt laboratory located at the plant or job site that is equipped with heating and air conditioning units to maintain a temperature of 24 plus or minus 2.3 degrees C. Provide laboratory facilities that are kept clean and all equipment maintained in proper working condition. Provide the Government with unrestricted access to inspect the laboratory facility, to witness quality control activities, and to perform any check testing desired. The Government will advise in writing of any noted deficiencies concerning the laboratory facility, equipment, supplies, or testing personnel and procedures. When the deficiencies are serious enough to adversely affect test results, immediately suspend the incorporation of the materials into the work. Incorporation of the materials into the work will not be permitted to resume until the deficiencies are corrected.

3.1.3 Quality Control Testing

Perform all quality control tests applicable to these specifications and as set forth in the Quality Control Program. Use the independent commercial laboratory that meets the requirements of UNE-EN 13108-21 for acceptance testing in paragraph ACCEPTANCE. Use in-house capabilities or the independent commercial laboratory for quality control testing. Required elements of the testing program include, but are not limited to tests for the control of asphalt content, aggregate gradation, aggregate moisture, moisture in the asphalt mixture, temperatures, VMA, Marshall stability, flow, and in-place density. Develop a Quality Control Testing Plan as part of the Quality Control Program.

3.1.3.1 Asphalt Content

Comply with FOM/2523/2014, Table 542.10 and as modified by FOM/510/2018. Determine asphalt content a minimum of twice per lot (a lot is defined in paragraph PAVEMENT LOTS) using the ignition method in accordance with ASTM D6307 or NLT-384. Use the extraction method in accordance with ASTM D2172/D2172M or NLT-164 if the correction factor for the ignition method in ASTM D6307 is greater than 1.0. The asphalt content for the lot will be determined by averaging the test results.

3.1.3.2 Aggregate Properties

Determine aggregate gradations a minimum of twice per lot from mechanical analysis of extracted aggregate in accordance with ASTM D5444, ASTM C136/C136M, and ASTM C117 or UNE-EN 933-1 and UNE-EN 933-2. Determine the specific gravity of each aggregate size grouping for each 18,000 tonnes in accordance with ASTM C127 or ASTM C128 or NLT-156 and UNE-EN 1097-3. Determine fractured faces for gravel sources for each 18,000 tonnes in accordance with ASTM D5821. Determine the uncompacted void content of natural sand, manufactured sand, and blended aggregate for each 18,000 tonnes in accordance with AASHTO T 304 Method A.

3.1.3.3 Moisture Content of Aggregate

Determine the moisture content of aggregate used for production a minimum of once per lot in accordance with ASTM C566 or UNE-EN 1097-5.

3.1.3.4 Moisture Content of Asphalt Mixture

Determine the moisture content of the asphalt mixture at least once per lot in accordance with AASHTO T 329.

3.1.3.5 Temperatures

Check temperatures at least four times per lot, at necessary locations to determine the temperature at the dryer, the asphalt cement binder in the storage tank, the asphalt mixture at the plant, and the asphalt mixture at the job site.

3.1.3.6 VMA, Marshall Stability, and Flow

Obtain mixture samples at least four times per lot. Calculate the VMA of each specimen in accordance with AI MS-2 based on ASTM C127 and ASTM C128 bulk specific gravity for the aggregate, as well as the Marshall stability and flow, as described in ASTM D6927 or NLT-159. Provide VMA within the limits of Table 5.

3.1.3.7 In-Place Density

Conduct any necessary testing to ensure the specified density is achieved. A nuclear gauge or other non-destructive testing device can be used to monitor pavement density.

3.1.3.8 Additional Testing

Perform any additional testing deemed necessary to control the process.

3.1.3.9 QC Monitoring

Submit all QC test results to the Government on a daily basis as the tests are performed. The Government reserves the right to monitor any of the Contractor's quality control testing and to perform duplicate testing as a check to the Contractor's quality control testing.

3.1.4 Sampling

When directed by the Government, sample and test any material which appears to not meet specification requirements unless such material is voluntarily removed and replaced or deficiencies corrected. Perform all sampling in accordance with standard procedures specified.

3.2 PREPARATION OF ASPHALT BINDER MATERIAL

Heat the asphalt cement material while avoiding local overheating. Provide a continuous supply of the asphalt material to the mixer at a uniform temperature. Maintain the temperature of the asphalt delivered to the mixer to provide a suitable viscosity for adequate coating of the aggregate particles. For hot-mix, do not heat unmodified asphalt to a temperature exceeding 160 degrees C when added to the aggregate. Do not heat modified asphalt to a temperature exceeding 175 degrees C when added to the aggregate.

3.3 PREPARATION OF MINERAL AGGREGATE

Heat and dry the aggregate prior to mixing. Provide a rate of heating and a maximum temperature that does not damage the aggregates. Do not heat

the aggregate to a temperature exceeding 175 degrees C when the asphalt binder is added. Maintain the temperature no lower than is required to obtain complete coating and uniform distribution on the aggregate particles and to provide a mixture of satisfactory workability.

3.4 PREPARATION OF ASPHALT MIXTURE

Weigh or meter the aggregates and the asphalt cement and introduce into the mixer the amount specified by the JMF. Mix the combined materials until the aggregate obtains a uniform coating of asphalt binder and is thoroughly distributed throughout the mixture. The moisture content of all asphalt mixture upon discharge from the plant is not to exceed 0.5 percent by total weight of mixture as measured by AASHTO T 329.

3.5 PREPARATION OF THE UNDERLYING SURFACE

Immediately before placing the asphalt mixture, clean the underlying course of dust and debris. Apply a prime coat or tack coat in accordance with Section 32 12 13 BITUMINOUS TACK AND PRIME COATS.

3.6 TRANSPORTING AND PLACING

3.6.1 Transporting

Transport asphalt mixture from the mixing plant to the site in clean, tight vehicles. Schedule deliveries so that placing and compacting of mixture is uniform with minimum stopping and starting of the paver. Provide adequate artificial lighting for night placements. Hauling over freshly placed material will not be permitted until the material has been compacted as specified, and allowed to cool to 60 degrees C.

3.6.2 Placing

Place the mix in lifts of adequate thickness and compact at a temperature suitable for obtaining density, surface smoothness, and other specified requirements. Upon arrival, place the mixture to the full width by an asphalt paver; strike off in a uniform layer of such depth that, when the work is completed, the required thickness is obtained and the surface conforms to the grade and contour indicated. Do not broadcast waste mixture onto the mat or recycle into the paver hopper. Collect waste mixture and dispose off site. Regulate the speed of the paver to eliminate pulling and tearing of the asphalt mat. Begin placement of the mixture along the centerline of a crowned section or on the high side of areas with a one-way slope. Place the mixture in consecutive adjacent strips having a minimum width of 3 m. Offset the longitudinal joint in one course from the longitudinal joint in the course immediately below by at least 300 mm; however, locate the joint in the surface course at the centerline of the pavement. Offset transverse joints in one course by at least 3 m from transverse joints in the previous course. Offset transverse joints in adjacent lanes a minimum of 3 m. On isolated areas where irregularities or unavoidable obstacles make the use of mechanical spreading and finishing equipment impractical, the mixture can be spread and luted by hand tools.

3.7 COMPACTION OF MIXTURE

3.7.1 General

- a. After placing, thoroughly and uniformly compact the mixture by

rolling. Compact the surface as soon as possible without causing displacement, cracking, or shoving. Determine the sequence of rolling operations and the type of rollers used with the exception that application of more than three passes with a vibratory roller in the vibrating mode is prohibited. Maintain the speed of the roller, at all times, sufficiently slow to avoid displacement of the asphalt mixture and to be effective in compaction. Correct at once any displacement occurring as a result of reversing the direction of the roller, or from any other cause.

- b. Furnish sufficient rollers to handle the output of the plant. Continue rolling until the surface is of uniform texture, true to grade and cross section, and the required field density is obtained. To prevent adhesion of the mixture to the roller, keep the wheels properly moistened, but excessive water is not permitted. In areas not accessible to the roller, thoroughly compact the mixture with hand tampers or small compactors. Remove the full depth of any mixture that becomes loose and broken, mixed with dirt, contains check-cracking, or is in any way defective. Replace with fresh asphalt mixture and immediately compact to conform to the surrounding area. Perform this work at no expense to the Government. Skin patching is not allowed.

3.7.2 Segregation

The Government can sample and test any material that looks deficient. When the in-place material appears to be segregated, the Government has the option to sample the material and have it tested and compared to the in-place density requirements in Table 2 paragraph ACCEPTANCE. If the material fails to meet these specification requirements, remove and replace the extent of the segregated material the full depth of the layer of asphalt mixture at no additional cost to the Government. When segregation occurs in the mat, take appropriate action to correct the process so that additional segregation does not occur.

3.8 JOINTS

Construct joints to ensure a continuous bond between the courses and to obtain the required density. Provide all joints with the same texture as other sections of the course and meet the requirements for smoothness and grade.

3.8.1 Transverse Joints

Do not pass the roller over the unprotected end of the freshly laid mixture, except when necessary to form a transverse joint. When necessary to form a transverse joint, construct by means of placing a bulkhead or by tapering the course. Utilize a dry saw cut on the transverse joint full depth and width on a straight line to expose a vertical face prior to placing the adjacent lane. Remove the cutback material from the project. In both methods, provide a light tack coat of asphalt material to all contact surfaces before placing any fresh mixture against the joint.

3.8.2 Longitudinal Joints

Provide a joint that meets density and smoothness requirements for joints and has uniform texture. Cut back longitudinal joints which are irregular, damaged, uncompacted, cold (less than 80 degrees C at the time of placing adjacent lanes), or otherwise defective, a maximum of 75 mm

from the top of the course with a cutting wheel to expose a clean, sound, near vertical surface for the full depth of the course. Remove all cutback material from the project. Provide a light tack coat of asphalt material to all contact surfaces prior to placing any fresh mixture against the joint.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 15 00

AGGREGATE SURFACING

05/17

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEGREE OF COMPACTION
- 1.3 SUBMITTALS
- 1.4 EQUIPMENT, TOOLS, AND MACHINES
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Sampling
 - 1.5.2 Testing
 - 1.5.2.1 Sieve Analysis
 - 1.5.2.2 Liquid Limit and Plasticity Index
 - 1.5.2.3 Moisture-Density Determinations
 - 1.5.2.4 Field Density Tests
 - 1.5.2.5 Wear Test

PART 2 PRODUCTS

- 2.1 AGGREGATES
 - 2.1.1 Coarse Aggregates
 - 2.1.2 Fine Aggregates
 - 2.1.3 Gradation Requirements
- 2.2 LIQUID LIMIT AND PLASTICITY INDEX
- 2.3 TESTS, INSPECTIONS, AND VERIFICATIONS
 - 2.3.1 Initial Tests
 - 2.3.2 Approval of Material

PART 3 EXECUTION

- 3.1 STOCKPILING MATERIAL
- 3.2 PREPARATION OF UNDERLYING SUBGRADE
- 3.3 GRADE CONTROL
- 3.4 MIXING AND PLACING MATERIALS
- 3.5 LAYER THICKNESS
- 3.6 COMPACTION
- 3.7 EDGES OF AGGREGATE SURFACE COURSE
- 3.8 SMOOTHNESS TEST
- 3.9 FIELD QUALITY CONTROL
 - 3.9.1 In-Place Tests
 - 3.9.2 Approval of Material
- 3.10 MAINTENANCE

-- End of Section Table of Contents --

SECTION 32 15 00

AGGREGATE SURFACING
05/17

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 103501	(1994) Geotechnic. Compaction Test. Modified Proctor.
UNE 103503	(1995) Determination "In Situ" Of Density Of Soil By The Sand Method.
UNE 103900	(2013) In situ determination of density and moisture content of soil and granular materials by nuclear methods: low depths.
UNE-EN 932-1	(1997) Test For Geometrical Properties Of Aggregates. Part 1: Methods for Sampling.
UNE-EN 933-1	(2012) Tests for geometrical properties of aggregates - Part 1: Determination of particle size distribution - Sieving method
UNE-EN 933-2	(1996) Test For Geometrical Properties Of Aggregates. Part 2: Determination Of Particle Size Distribution. Test Sieves, Nominal Size Of Apertures.
UNE-EN 1097-2	(2010) Tests for mechanical and physical properties of aggregates - Part 2: Methods for the determination of resistance to fragmentation
UNE-EN ISO 17892-12	(2019) Geotechnical investigation and testing - Laboratory testing of soil - Part 12: Determination of liquid and plastic limits (ISO 17892-12:2018).

ASTM INTERNATIONAL (ASTM)

ASTM D1556/D1556M	(2015; E 2016) Standard Test Method for Density and Unit Weight of Soil in Place by Sand-Cone Method
-------------------	--

CENTRO DE ESTUDIOS Y EXPERIMENTACION DE OBRAS PUBLICAS (CEDEX)

NLT-149	(1991) Resistencia al Desgaste de los Aridos por medio de la Maquina de Los
---------	---

Angeles

1.2 DEGREE OF COMPACTION

Degree of compaction required, except as noted in the second sentence, is expressed as a percentage of the maximum laboratory dry density obtained by the test procedure presented in UNE 103501 abbreviated as a percent of laboratory maximum dry density.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Plant, Equipment, and Tools; G

SD-06 Test Reports

Initial Tests; G
In-Place Tests; G

1.4 EQUIPMENT, TOOLS, AND MACHINES

All plant, equipment, and tools used in the performance of the work will be subject to approval by the Contracting Officer before the work is started. Maintain all plant, equipment, and tools in satisfactory working condition at all times. Submit a list of proposed equipment, including descriptive data. Provide adequate equipment having the capability of minimizing segregation, producing the required compaction, meeting grade controls, thickness control, and smoothness requirements as set forth herein.

1.5 QUALITY ASSURANCE

Sampling and testing are the responsibility of the Contractor. Perform sampling and testing using a laboratory approved in accordance with Section 01 45 00.00 20 QUALITY CONTROL. Work requiring testing will not be permitted until the testing laboratory has been inspected and approved. Test the materials to establish compliance with the specified requirements and perform testing at the specified frequency. The Contracting Officer may specify the time and location of the tests. Furnish copies of test results to the Contracting Officer within 24 hours of completion of the tests.

1.5.1 Sampling

Take samples for laboratory testing in conformance with UNE-EN 932-1. When deemed necessary, the sampling will be observed by the Contracting Officer.

1.5.2 Testing

1.5.2.1 Sieve Analysis

Perform sieve analysis in conformance with UNE-EN 933-1 using sieves

conforming to UNE-EN 933-2.

1.5.2.2 Liquid Limit and Plasticity Index

Determine liquid limit and plasticity index in accordance with UNE-EN ISO 17892-12.

1.5.2.3 Moisture-Density Determinations

Determine the laboratory maximum dry density and optimum moisture content in accordance with paragraph DEGREE OF COMPACTION.

1.5.2.4 Field Density Tests

Measure field density in accordance with UNE 103503 or UNE 103900. For the method presented in ASTM D1556/D1556M use the base plate as shown in the drawing. For the method presented in UNE 103900 check the calibration curves and adjust them, if necessary, using only the sand cone method. Tests performed in accordance with UNE 103900 result in a wet unit weight of soil and UNE 103900 will be used to determine the moisture content of the soil. Also check the calibration curves furnished with the moisture gauges along with density calibration checks as described in UNE 103900. Make the calibration checks of both the density and moisture gauges using the prepared containers of material. Submit calibration curves and related test results prior to using the device or equipment being calibrated.

1.5.2.5 Wear Test

Perform wear tests on aggregate surface course material in conformance with UNE-EN 1097-2 or NLT-149.

PART 2 PRODUCTS

2.1 AGGREGATES

Provide aggregates consisting of clean, sound, durable particles of natural gravel, crushed gravel, crushed stone, sand, slag, soil, or other approved materials processed and blended or naturally combined. Provide aggregates free from lumps and balls of clay, organic matter, objectionable coatings, and other foreign materials. The Contractor is responsible for obtaining materials that meet the specification and can be used to meet the grade and smoothness requirements specified herein after all compaction operations have been completed.

2.1.1 Coarse Aggregates

The material retained on the 5 mm sieve is known as coarse aggregate. Use only coarse aggregates that are reasonably uniform in density and quality. Use only coarse aggregate having a percentage of wear not exceeding 50 percent after 500 revolutions as determined by UNE-EN 1097-2 or NLT-149. The amount of flat and/or elongated particles must not exceed 20 percent. A flat particle is one having a ratio of width to thickness greater than three; an elongated particle is one having a ratio of length to width greater than three. When the coarse aggregate is supplied from more than one source, aggregate from each source must meet the requirements set forth herein.

2.1.2 Fine Aggregates

The material passing the 5 mm sieve is known as fine aggregate. Fine aggregate consists of screenings, sand, soil, or other finely divided mineral matter that is processed or naturally combined with the coarse aggregate.

2.1.3 Gradation Requirements

Gradation requirements specified in TABLE I apply to the completed aggregate surface. It is the responsibility of the Contractor to obtain materials that will meet the gradation requirements after mixing, placing, compacting, and other operations. TABLE I shows permissible gradings for granular material used in aggregate surface roads. Use sieves conforming to UNE-EN 933-2.

TABLE I. GRADATION FOR AGGREGATE SURFACE COURSES Percentage by Weight Passing Square-Mesh Sieve		
Sieve Designation (mm)	No. 1	No. 2
20	100	100
12.50	50-85	60-100
4.00	35-65	50-85
2.00	25-50	40-70
0.50	15-30	24-45
0.063	8-15	8-15

2.2 LIQUID LIMIT AND PLASTICITY INDEX

The portion of the completed aggregate surface course passing the 0.425 mm sieve must have a maximum liquid limit of 35 and a plasticity index of 4 to 9.

2.3 TESTS, INSPECTIONS, AND VERIFICATIONS

2.3.1 Initial Tests

Perform one of each of the following tests, on the proposed material prior to commencing construction, to demonstrate that the proposed material meets all specified requirements when furnished. Complete this testing for each source if materials from more than one source are proposed.

- Sieve Analysis.
- Liquid limit and plasticity index.
- Moisture-density relationship.
- Wear.

Submit certified copies of test results for approval not less than 30 days before material is required for the work.

2.3.2 Approval of Material

Tentative approval of material will be based on initial test results.

PART 3 EXECUTION

3.1 STOCKPILING MATERIAL

Prior to stockpiling the material, clear and level the storage sites. Stockpile all materials, including approved material available from excavation and grading, in the manner and at the locations designated. Stockpile aggregates in such a manner that will prevent segregation. Stockpile aggregates and binders obtained from different sources separately.

3.2 PREPARATION OF UNDERLYING SUBGRADE

Clean the subgrade and shoulders of all foreign substances. Correct ruts or soft yielding spots in the subgrade, areas having inadequate compaction and deviations of the surface from the requirements set forth herein by loosening and removing soft or unsatisfactory material and by adding approved material, reshaping to line and grade and recompacting to density requirements specified in Section 31 23 00.00 20 EXCAVATION AND FILL. Do not allow traffic or other operations to disturb the completed subgrade and maintain in a satisfactory condition until the surface course is placed.

3.3 GRADE CONTROL

During construction, maintain the lines and grades including crown and cross slope indicated for the aggregate surface course by means of line and grade stakes placed by the Contractor in accordance with the SPECIAL CONTRACT REQUIREMENTS.

3.4 MIXING AND PLACING MATERIALS

Mix and place the materials to obtain uniformity of the material and a uniform optimum water content for compaction. Make adjustments in mixing, placing procedures, or in equipment to obtain the true grades, to minimize segregation and degradation, to obtain the desired water content, and to ensure a satisfactory surface course.

3.5 LAYER THICKNESS

Place the aggregate material on the subgrade in layers of uniform thickness. Compact the completed aggregate surface course to the thickness indicated. No individual layer may be thicker than 150 mm nor be thinner than 75 mm in compacted thickness. Compact the aggregate surface course to a total thickness that is within 13 mm of the thickness indicated. Where the measured thickness is more than 13 mm deficient, correct such areas by scarifying, adding new material of proper gradation, reblading, and recompacting as directed. Where the measured thickness is more than 13 mm thicker than indicated, the course will be considered as conforming to the specified thickness requirements. The average job thickness will be the average of all thickness measurements taken for the job and must be within 6 mm of the thickness indicated. Measure the total thickness of the aggregate surface course at intervals of one measurement for each 500 square meters of surface course. Measure total

thickness using 75 mm diameter test holes penetrating the aggregate surface course.

3.6 COMPACTION

Degree of compaction is a percentage of the maximum density obtained by the test procedure presented in UNE 103501 abbreviated herein as percent laboratory maximum density. Compact each layer of the aggregate surface course with approved compaction equipment, as required in the following paragraphs. Maintain the water content during the compaction procedure at optimum or at the percentage specified by the Contracting Officer. Compact the mixture with mechanical tampers in locations not accessible to rollers. Continue compaction until each layer through the full depth is compacted to at least 100 percent of laboratory maximum density. Remove any materials that are found to be unsatisfactory and replace them with satisfactory material or rework them to produce a satisfactory material.

3.7 EDGES OF AGGREGATE SURFACE COURSE

Place approved material along the edges of the aggregate surface course in such quantity as to compact to the thickness of the course being constructed.

3.8 SMOOTHNESS TEST

Construct each layer so that the surface shows no deviations in excess of 10 mm when tested with a 3 m straightedge applied both parallel with and at right angles to the centerline of the area to be surfaced. Correct deviations exceeding this amount by removing material, replacing with new material, or reworking existing material and compacting, as directed.

3.9 FIELD QUALITY CONTROL

3.9.1 In-Place Tests

Perform each of the following tests on samples taken from the placed and compacted aggregate surface course. Take samples and test at the rates indicated.

- a. Perform density tests on every lift of material placed and at a frequency of one set of tests for every 250 square meters, or portion thereof, of completed area.
- b. Perform sieve analysis on every lift of material placed and at a frequency of one sieve analysis for every 500 square meters, or portion thereof, of material placed.
- c. Perform liquid limit and plasticity index tests at the same frequency as the sieve analysis.
- d. Measure the thickness of the aggregate surface course at intervals providing at least one measurement for each 500 square meters of base course or part thereof. Measure the thickness using test holes, at least 75 mm in diameter through the aggregate surface course.

3.9.2 Approval of Material

Final approval of the materials will be based on tests for gradation, liquid limit, and plasticity index performed on samples taken from the

completed and full compacted aggregate surface course.

3.10 MAINTENANCE

Maintain the aggregate surface course in a condition that will meet all specification requirements until accepted.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 16 19

CONCRETE CURBS AND SIDEWALKS

05/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 EQUIPMENT, TOOLS, AND MACHINES
 - 1.3.1 General Requirements
- 1.4 ENVIRONMENTAL REQUIREMENTS
 - 1.4.1 Placing During Cold Weather
 - 1.4.2 Placing During Warm Weather

PART 2 PRODUCTS

- 2.1 CONCRETE
 - 2.1.1 Air Content
 - 2.1.2 Slump
 - 2.1.3 Precast Concrete Curb
- 2.2 CONCRETE CURING MATERIALS
 - 2.2.1 Impervious Sheet Materials
 - 2.2.2 Burlap
 - 2.2.3 White Pigmented Membrane-Forming Curing Compound
- 2.3 CONCRETE PROTECTION MATERIALS
- 2.4 JOINT FILLER STRIPS
 - 2.4.1 Contraction Joint Filler for Curb
 - 2.4.2 Expansion Joint Filler, Premolded
- 2.5 JOINT SEALANTS
- 2.6 FORM WORK
 - 2.6.1 Wood Forms
 - 2.6.2 Steel Forms
 - 2.6.3 Sidewalk Forms
- 2.7 DETECTABLE WARNING SYSTEM

PART 3 EXECUTION

- 3.1 SUBGRADE PREPARATION
 - 3.1.1 Sidewalk Subgrade
 - 3.1.2 Curb Subgrade
 - 3.1.3 Maintenance of Subgrade
- 3.2 FORM SETTING
 - 3.2.1 Sidewalks
- 3.3 SIDEWALK CONCRETE PLACEMENT AND FINISHING
 - 3.3.1 Formed Sidewalks
 - 3.3.2 Concrete Finishing
 - 3.3.3 Edge and Joint Finishing
 - 3.3.4 Surface and Thickness Tolerances
- 3.4 CURB CONCRETE PLACEMENT AND FINISHING
 - 3.4.1 Concrete Finishing

- 3.4.2 Joint Finishing
- 3.4.3 Surface and Thickness Tolerances
- 3.4.4 Installation of Precast Concrete Curbs
- 3.5 SIDEWALK JOINTS
 - 3.5.1 Sidewalk Contraction Joints
 - 3.5.2 Sidewalk Expansion Joints
- 3.6 CURING AND PROTECTION
 - 3.6.1 General Requirements
 - 3.6.1.1 Mat Method
 - 3.6.1.2 Impervious Sheeting Method
 - 3.6.1.3 Membrane Curing Method
 - 3.6.2 Backfilling
 - 3.6.3 Protection
 - 3.6.4 Protective Coating
 - 3.6.4.1 Application
 - 3.6.4.2 Precautions
- 3.7 FIELD QUALITY CONTROL
 - 3.7.1 General Requirements
 - 3.7.2 Concrete Testing
 - 3.7.2.1 Strength Testing
 - 3.7.2.2 Air Content
 - 3.7.2.3 Slump Test
 - 3.7.3 Thickness Evaluation
 - 3.7.4 Surface Evaluation
- 3.8 SURFACE DEFICIENCIES AND CORRECTIONS
 - 3.8.1 Thickness Deficiency
 - 3.8.2 High Areas
 - 3.8.3 Appearance
- 3.9 DETECTABLE WARNING SYSTEM

-- End of Section Table of Contents --

SECTION 32 16 19

CONCRETE CURBS AND SIDEWALKS
05/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 41107	(1961) Prefabricated Products. Elastics and of Low Transversal Dilation for the Filling of Expansion Joints in Concrete Paving
UNE-EN 12350-1	(2020) Testing fresh concrete - Part 1: Sampling and common apparatus
UNE-EN 12350-2	(2020) Testing fresh concrete - Part 2: Slump-test
UNE-EN 12350-7	(2010) Testing fresh concrete - Part 7: Air content - Pressure methods.
UNE-EN 12390-1	(2013) Testing hardened concrete. Part 1: Shape, dimensions and other requirements for specimens and moulds.
UNE-EN 12390-2	(2020) Testing hardened concrete. Part 2: Making and curing specimens for strength tests.
UNE-EN 13163	(2013+A2:2017) Thermal insulation products for buildings - Factory made expanded polystyrene (EPS) products - Specification

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO M 182	(2005; R 2017) Standard Specification for Burlap Cloth Made from Jute or Kenaf and Cotton Mats
--------------	--

ASTM INTERNATIONAL (ASTM)

ASTM C920	(2018) Standard Specification for Elastomeric Joint Sealants
ASTM D1752	(2018) Standard Specification for Preformed Sponge Rubber, Cork and Recycled PVC Expansion Joint Fillers for Concrete Paving and Structural Construction

INTERNATIONAL CODE COUNCIL (ICC)

ICC A117.1 (2017) Standard And Commentary Accessible
and Usable Buildings and Facilities

MINISTERIO DE FOMENTO (MF)

FOM/2523/2014 (2014) Orden de 12 de diciembre, por la
que se actualizan determinados artículos
del pliego de prescripciones técnicas
generales para obras de carreteras y
puentes, relativos a materiales básicos, a
firmes y pavimentos, y a señalización,
balizamiento y sistemas de contención de
vehículos.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Concrete

SD-04 Samples

Concrete Curbs

SD-06 Test Reports

Field Quality Control

1.3 EQUIPMENT, TOOLS, AND MACHINES

1.3.1 General Requirements

Plant, equipment, machines, and tools used in the work will be subject to approval and must be maintained in a satisfactory working condition at all times. Use equipment capable of producing the required product, meeting grade controls, thickness control and smoothness requirements as specified. Discontinue using equipment that produces unsatisfactory results. Allow the Contracting Officer access at all times to the plant and equipment to ensure proper operation and compliance with specifications.

1.4 ENVIRONMENTAL REQUIREMENTS

1.4.1 Placing During Cold Weather

Do not place concrete when the air temperature reaches 5 degrees C and is falling, or is already below that point. Provide covering or other means as needed to maintain the concrete at a temperature of at least 10 degrees C for not less than 72 hours after placing, for the remainder of the curing period.

1.4.2 Placing During Warm Weather

The temperature of the concrete as placed must not exceed 30 degrees C except where an approved retarder is used. Cool the mixing water and aggregates as necessary to maintain a satisfactory placing temperature. The placing temperature must not exceed 35 degrees C at any time.

PART 2 PRODUCTS

2.1 CONCRETE

Provide concrete conforming to the applicable requirements of Section 03 30 00 CAST-IN-PLACE CONCRETE except as otherwise specified. Size of aggregate must not exceed 37.5 mm. Concrete for sidewalk and precast curb must have a minimum compressive strength of 24 MPa at 28 days. Concrete for the concrete foundation of the precast curb must have a minimum compressive strength of 20 MPa at 28 days. Submit copies of certified delivery tickets for all concrete used in the construction.

2.1.1 Air Content

Use concrete mixtures that have an air content by volume of concrete of 5 to 7 percent, based on measurements made immediately after discharge from the mixer.

2.1.2 Slump

Use concrete with a slump of 75 mm plus or minus 25 mm for hand placed concrete as determined in accordance with UNE-EN 12350-2.

2.1.3 Precast Concrete Curb

Provide precast concrete curbs as detailed in the drawings from an approved manufacturer. Hydraulically press precast sections in approved molds under conditions of controlled temperatures and humidity. Water or steam cure sections until the concrete attains 100 percent of the specified strength. Provide precast elements uniform in color, free from cracks, flaws or other defects, have a clean finish with smooth surfaces and have well defined arises. Provide precast elements free of segregation, honeycombing or broken corners will not be and without remedial measures. Sample and test precast concrete elements in accordance with paragraph CONCRETE TESTING. Provide concrete materials in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE except otherwise specified. Provide precast concrete curbs comprised of homogenous elements 60 cm or 90 cm long where laid in straight lines, but may be reduced to 20 cm long to be laid on curves.

2.2 CONCRETE CURING MATERIALS

2.2.1 Impervious Sheet Materials

Use impervious sheet materials, type optional, except that polyethylene film, if used, must be white opaque.

2.2.2 Burlap

Use burlap conforming to AASHTO M 182.

2.2.3 White Pigmented Membrane-Forming Curing Compound

Use white pigmented membrane-forming curing compound conforming to FOM/2523/2014, Art. 550.2.9.

2.3 CONCRETE PROTECTION MATERIALS

Use concrete protection materials consisting of a linseed oil mixture of equal parts, by volume, of linseed oil and either mineral spirits, naphtha, or turpentine. At the option of the Contractor, commercially prepared linseed oil mixtures, formulated specifically for application to concrete to provide protection against the action of deicing chemicals may be used, except that emulsified mixtures are not acceptable.

2.4 JOINT FILLER STRIPS

2.4.1 Contraction Joint Filler for Curb

Use hard-pressed fiberboard contraction joint filler for curb.

2.4.2 Expansion Joint Filler, Premolded

Unless otherwise indicated, use 13 mm thick premolded expansion joint filler conforming to UNE 41107 and UNE-EN 13163.

2.5 JOINT SEALANTS

Use cold-applied joint sealant conforming to ASTM C920.

2.6 FORM WORK

Design and construct form work to ensure that the finished concrete will conform accurately to the indicated dimensions, lines, and elevations, and within the tolerances specified. Use wood or steel forms that are straight and of sufficient strength to resist springing during depositing and consolidating concrete.

2.6.1 Wood Forms

Use forms that are surfaced plank, 50 mm nominal thickness, straight and free from warp, twist, loose knots, splits or other defects. Use forms with a nominal length of 3 m. Radius bends may be formed with 19 mm boards, laminated to the required thickness.

2.6.2 Steel Forms

Use channel-formed sections with a flat top surface and welded braces at each end and at not less than two intermediate points. Use forms with interlocking and self-aligning ends. Provide flexible forms for radius forming, corner forms, form spreaders, and fillers as needed. Use forms with a nominal length of 3 m and that have a minimum of 3 welded stake pockets per form. Use stake pins consisting of solid steel rods with chamfered heads and pointed tips designed for use with steel forms.

2.6.3 Sidewalk Forms

Use sidewalk forms that are of a height equal to the full depth of the finished sidewalk.

2.7 DETECTABLE WARNING SYSTEM

Detectable Warning Systems shown on the Contract plans are to meet requirements of ICC A117.1 - Section 705.

PART 3 EXECUTION

3.1 SUBGRADE PREPARATION

Construct subgrade to the specified grade and cross section prior to concrete placement.

3.1.1 Sidewalk Subgrade

Place and compact the subgrade in accordance with Section 31 23 00.00 20 EXCAVATION AND FILL. Test the subgrade for grade and cross section with a template extending the full width of the sidewalk and supported between side forms.

3.1.2 Curb Subgrade

Place and compact the subgrade in accordance with Section 31 23 00.00 20 EXCAVATION AND FILL. Test the subgrade for grade and cross section by means of a template extending the full width of the curb. Use subgrade materials equal in bearing quality to the subgrade under the adjacent pavement.

3.1.3 Maintenance of Subgrade

Maintain subgrade in a smooth, compacted condition in conformity with the required section and established grade until the concrete is placed. The subgrade must be in a moist condition when concrete is placed. Prepare and protect subgrade so that it is free from frost when the concrete is deposited.

3.2 FORM SETTING

Set forms to the indicated alignment, grade and dimensions. Hold forms rigidly in place by a minimum of 3 stakes per form placed at intervals not to exceed 1.2 m. Use additional stakes and braces at corners, deep sections, and radius bends, as required. Use clamps, spreaders, and braces where required to ensure rigidity in the forms. Remove forms in a manner that will not injure the concrete. Do not use bars or heavy tools against the concrete when removing the forms. Promptly and satisfactorily repair concrete found to be defective after form removal. Clean forms and coat with form oil each time before concrete is placed. Wood forms may, instead, be thoroughly wetted with water before concrete is placed, except that with probable freezing temperatures, oiling is mandatory.

3.2.1 Sidewalks

Set forms for sidewalks with the upper edge true to line and grade with an allowable tolerance of 3 mm in any 3 m long section. After forms are set, grade and alignment must be checked with a 3 m straightedge. Sidewalks must have a transverse slope of maximum 20 mm per meter. Unless otherwise indicated, construct sidewalks that are located adjacent to curbs with the low side adjacent to the curb. Do not remove side forms less than 12 hours after finishing has been completed.

3.3 SIDEWALK CONCRETE PLACEMENT AND FINISHING

3.3.1 Formed Sidewalks

Place concrete in the forms in one layer. When consolidated and finished, the sidewalks must be of the thickness indicated. Use a strike-off guided by side forms after concrete has been placed in the forms to bring the surface to proper section to be compacted. Consolidate concrete by tamping and spading or with an approved vibrator. Finish the surface to grade with a strike off.

3.3.2 Concrete Finishing

After straightedging, when most of the water sheen has disappeared, and just before the concrete hardens, finish the surface with a wood or magnesium float or darby to a smooth and uniformly fine granular or sandy texture free of waves, irregularities, or tool marks. Produce a scored surface by brooming with a fiber-bristle brush in a direction transverse to that of the traffic, followed by edging.

3.3.3 Edge and Joint Finishing

Finish all slab edges, including those at formed joints, with an edger having a radius of 3 mm. Edge transverse joints before brooming. Eliminate the flat surface left by the surface face of the edger with brooming. Clean and solidly fill corners and edges which have crumbled and areas which lack sufficient mortar for proper finishing with a properly proportioned mortar mixture and then finish.

3.3.4 Surface and Thickness Tolerances

Finished surfaces must not vary more than 8 mm from the testing edge of a 3 m straightedge. Permissible deficiency in section thickness will be up to 6 mm.

3.4 CURB CONCRETE PLACEMENT AND FINISHING

3.4.1 Concrete Finishing

Float and finish exposed surfaces with a smooth wood float until true to grade and section and uniform in texture. Brush floated surfaces with a fine-hair brush using longitudinal strokes. Round the edges of the top of the curb with an edging tool to a radius of 13 mm. Immediately after removing the front curb form, rub the face of the curb with a wood or concrete rubbing block and water until blemishes, form marks, and tool marks have been removed. Brush the front curb surface, while still wet, in the same manner as the curb top.

3.4.2 Joint Finishing

Finish curb edges at formed joints as indicated.

3.4.3 Surface and Thickness Tolerances

Finished surfaces must not vary more than 6 mm from the testing edge of a 3 m straightedge. Permissible deficiency in section thickness will be up to 6 mm.

3.4.4 Installation of Precast Concrete Curbs

Place precast concrete curbs where indicated on the drawings, or as directed by the Contracting Officer, and prior to constructing pavement on roads or parking areas. Do not place precast curbs until after the subbase course of the pavement has been placed, compacted and approved by the Contracting Officer's Representation unless otherwise specifically directed in writing. Submit samples of the various types of concrete curbs for approval prior to commencing the work. Place precast concrete curb sections on a setting bed of mortar consisting of one part cement to two parts sand with constant width joints not exceeding 0.5 cm. Place the just damp mortar by ramming it in place with a hammer and cold chisel or piece of wood when required. Do not use mortar after it has partially set. Do not re-temper or use of any material from any previous mortar batch to mix a new batch of mortar. Prepared mortar only in a sufficient amount to be used within a short period of time and of such a quantity that a partial set does not occur before all the batch is used in the work.

3.5 SIDEWALK JOINTS

Construct sidewalk joints to divide the surface into rectangular areas. Space transverse contraction joints at a distance equal to the sidewalk width or 1.5 m on centers, whichever is less, and continuous across the slab. Construct longitudinal contraction joints along the centerline of all sidewalks 3 m or more in width. Construct transverse expansion joints at sidewalk returns and opposite expansion joints in adjoining curbs. Where the sidewalk is not in contact with the curb, install transverse expansion joints as indicated. Form expansion joints around structures and features which project through or into the sidewalk pavement, using joint filler of the type, thickness, and width indicated. Expansion joints are not required between sidewalks and curb that abut the sidewalk longitudinally.

3.5.1 Sidewalk Contraction Joints

Form contraction joints in the fresh concrete by cutting a groove in the top portion of the slab to a depth of at least one-fourth of the sidewalk slab thickness. Unless otherwise approved or indicated, either use a jointer to cut the groove or saw a groove in the hardened concrete with a power-driven saw. Construct sawed joints by sawing a groove in the concrete with a 3 mm blade. Provide an ample supply of saw blades on the jobsite before concrete placement is started. Provide at least one standby sawing unit in good working order at the jobsite at all times during the sawing operations.

3.5.2 Sidewalk Expansion Joints

Form expansion joints using 13 mm joint filler strips. Joint filler in expansion joints surrounding structures and features within the sidewalk may consist of preformed filler material conforming to ASTM D1752 or building paper. Hold joint filler in place with steel pins or other devices to prevent warping of the filler during floating and finishing. Immediately after finishing operations are completed, round joint edges using an edging tool having a radius of 3 mm. Remove any concrete over the joint filler. At the end of the curing period, clean the top of expansion joints and fill with cold-applied joint sealant. Use joint sealant that is gray or stone in color. Thoroughly clean the joint opening before the sealing material is placed. Do not spill sealing material on exposed surfaces of the concrete. Apply joint sealing

material only when the concrete at the joint is surface dry and atmospheric and concrete temperatures are above 10 degrees C. Immediately remove any excess material on exposed surfaces of the concrete and clean the concrete surfaces.

3.6 CURING AND PROTECTION

3.6.1 General Requirements

Protect concrete against loss of moisture and rapid temperature changes for at least 7 days from the beginning of the curing operation. Protect unhardened concrete from rain and flowing water. All equipment needed for adequate curing and protection of the concrete must be on hand and ready for use before actual concrete placement begins. Protect concrete as necessary to prevent cracking of the pavement due to temperature changes during the curing period.

3.6.1.1 Mat Method

Cover the entire exposed surface with two or more layers of burlap. Overlap mats at least 150 mm. Thoroughly wet the mat with water prior to placing on concrete surface and keep the mat continuously in a saturated condition and in intimate contact with concrete for not less than 7 days.

3.6.1.2 Impervious Sheeting Method

Wet the entire exposed surface with a fine spray of water and then cover with impervious sheeting material. Lay sheets directly on the concrete surface with the light-colored side up and overlapped 300 mm when a continuous sheet is not used. Use sheeting that is not less than 450 mm wider than the concrete surface to be cured. Secure sheeting using heavy wood planks or a bank of moist earth placed along edges and laps in the sheets. Satisfactorily repair or replace sheets that are torn or otherwise damaged during curing. Sheeting must remain on the concrete surface to be cured for not less than 7 days.

3.6.1.3 Membrane Curing Method

Apply a uniform coating of white-pigmented membrane-curing compound to the entire exposed surface of the concrete as soon after finishing as the free water has disappeared from the finished surface. Coat formed surfaces immediately after the forms are removed and in no case longer than 1 hour after the removal of forms. Do not allow concrete surface to dry before application of the membrane. If drying has occurred, moisten the surface of the concrete with a fine spray of water and apply the curing compound as soon as the free water disappears. Apply curing compound in two coats by hand-operated pressure sprayers at a coverage of approximately 5 square meters/L for the total of both coats. Apply the second coat in a direction approximately at right angles to the direction of application of the first coat. The compound must form a uniform, continuous, coherent film that will not check, crack, or peel and must be free from pinholes or other imperfections. If pinholes, abrasion, or other discontinuities exist, apply an additional coat to the affected areas within 30 minutes. Respray concrete surfaces that are subjected to heavy rainfall within 3 hours after the curing compound has been applied by the method and at the coverage specified above. Respray areas where the curing compound is damaged by subsequent construction operations within the curing period. Take precautions necessary to ensure that the concrete is properly cured at sawed joints, and that no curing compound enters the joints. Tightly

seal the top of the joint opening and the joint groove at exposed edges before the concrete in the region of the joint is resprayed with curing compound. Use a method used for sealing the joint groove that prevents loss of moisture from the joint during the entire specified curing period. Provide approved standby facilities for curing concrete pavement at a location accessible to the jobsite for use in the event of mechanical failure of the spraying equipment or other conditions that might prevent correct application of the membrane-curing compound at the proper time. Adequately protect concrete surfaces to which membrane-curing compounds have been applied during the entire curing period from pedestrian and vehicular traffic, except as required for joint-sawing operations and surface tests, and from other possible damage to the continuity of the membrane.

3.6.2 Backfilling

After curing, remove debris and backfill, grade, and compact the area adjoining the concrete to conform to the surrounding area in accordance with lines and grades indicated.

3.6.3 Protection

Protect completed concrete from damage until accepted. Repair damaged concrete and clean concrete discolored during construction. Remove and reconstruct concrete that is damaged for the entire length between regularly scheduled joints. Refinishing the damaged portion will not be acceptable. Dispose of removed material as directed.

3.6.4 Protective Coating

Apply a protective coating of linseed oil mixture to the exposed-to-view concrete surface after the curing period, if concrete will be exposed to de-icing chemicals within 6 weeks after placement. Moist cure concrete to receive a protective coating.

3.6.4.1 Application

Complete curing and backfilling operation prior to applying two coats of protective coating. Concrete must be surface dry and clean before each application. Spray apply at a rate of not more than 11 square meters/L for first application and not more than 15.5 square meters/L for second application, except that the number of applications and coverage for each application for commercially prepared mixture must be in accordance with the manufacturer's instructions. Protect coated surfaces from vehicular and pedestrian traffic until dry.

3.6.4.2 Precautions

Do not heat protective coating by direct application of flame or electrical heaters and protect the coating from exposure to open flame, sparks, and fire adjacent to open containers or applicators. Do not apply material at ambient or material temperatures lower than 10 degrees C.

3.7 FIELD QUALITY CONTROL

Submit copies of all test reports within 24 hours of completion of the test.

3.7.1 General Requirements

Perform the inspection and tests described and meet the specified requirements for inspection details and frequency of testing. Based upon the results of these inspections and tests, take the action and submit reports as required below, and additional tests to ensure that the requirements of these specifications are met.

3.7.2 Concrete Testing

3.7.2.1 Strength Testing

Take concrete samples in accordance with UNE-EN 12350-1 and UNE-EN 12390-1 not less than once a day nor less than once for every 190 cubic meters of concrete placed. Mold cylinders in accordance with UNE-EN 12390-2 for strength testing by an approved laboratory. Each strength test result must be the average of 2 test cylinders from the same concrete sample tested at 28 days, unless otherwise specified or approved. Concrete specified on the basis of compressive strength will be considered satisfactory if the averages of all sets of three consecutive strength test results equal or exceed the specified strength, and no individual strength test result falls below the specified strength by more than 4 MPa.

3.7.2.2 Air Content

Determine air content in accordance with UNE-EN 12350-7. Make two tests for air content on randomly selected batches of each class of concrete placed during each shift. Make additional tests when excessive variation in concrete workability is reported by the placing foreman or the Government inspector. Notify the placing foreman if results are out of tolerance. The placing foreman must take appropriate action to have the air content corrected at the plant. Additional tests for air content will be performed on each truckload of material until such time as the air content is within the tolerance specified.

3.7.2.3 Slump Test

Perform two slump tests on randomly selected batches of each class of concrete for every 190 cubic meters, or fraction thereof, of concrete placed during each shift. Perform additional tests when excessive variation in the workability of the concrete is noted or when excessive crumbling or slumping is noted along the edges of slip-formed concrete.

3.7.3 Thickness Evaluation

Determine the anticipated thickness of the concrete prior to placement by passing a template through the formed section or by measuring the depth of opening of the extrusion template of the curb forming machine.

3.7.4 Surface Evaluation

Provide finished surfaces for each category of the completed work that are uniform in color and free of blemishes and form or tool marks.

3.8 SURFACE DEFICIENCIES AND CORRECTIONS

3.8.1 Thickness Deficiency

When measurements indicate that the completed concrete section is

deficient in thickness by more than 6 mm the deficient section will be removed, between regularly scheduled joints, and replaced.

3.8.2 High Areas

In areas not meeting surface smoothness and plan grade requirements, reduce high areas either by rubbing the freshly finished concrete with carborundum brick and water when the concrete is less than 36 hours old or by grinding the hardened concrete with an approved surface grinding machine after the concrete is 36 hours old or more. The area corrected by grinding the surface of the hardened concrete must not exceed 5 percent of the area of any integral slab, and the depth of grinding must not exceed 6 mm. Remove and replace pavement areas requiring grade or surface smoothness corrections in excess of the limits specified.

3.8.3 Appearance

Exposed surfaces of the finished work will be inspected by the Contracting Officer and deficiencies in appearance will be identified. Remove and replace areas which exhibit excessive cracking, discoloration, form marks, or tool marks or which are otherwise inconsistent with the overall appearances of the work.

3.9 DETECTABLE WARNING SYSTEM

Install Detectable Warning Systems required by Contract plans in accordance with ICC A117.1, Section 705, and by manufacturers' installation instructions.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 17 23

PAVEMENT MARKINGS

08/16, CHG 5: 11/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 QUALITY ASSURANCE
 - 1.3.1 Regulatory Requirements
 - 1.3.2 Qualifications
- 1.4 DELIVERY AND STORAGE
- 1.5 PROJECT/SITE CONDITIONS
 - 1.5.1 Environmental Requirements
 - 1.5.1.1 Weather Limitations for Application
 - 1.5.1.2 Weather Limitations for Removal of Pavement Markings on Roads and Automotive Parking Areas
 - 1.5.2 Traffic Controls
 - 1.5.3 Lighting

PART 2 PRODUCTS

- 2.1 EQUIPMENT
 - 2.1.1 Surface Preparation and Paint Removal
 - 2.1.1.1 Surface Preparation Equipment for Roads and Automotive Parking Areas
 - 2.1.1.1.1 Waterblasting Equipment
 - 2.1.1.1.2 Grinding or Scarifying Equipment
 - 2.1.2 Application Equipment
 - 2.1.2.1 Paint Application Equipment
 - 2.1.2.1.1 Hand-Operated, Push-Type Machines
 - 2.1.2.1.2 Self-Propelled or Mobile-Drawn Spraying Machines
 - 2.1.2.1.2.1 Road Marking
 - 2.1.2.1.2.2 Hand Application
 - 2.1.2.2 Reflective Media Dispenser
- 2.2 MATERIALS
 - 2.2.1 Waterborne Paint
 - 2.2.2 Reflective Media
 - 2.2.2.1 Reflective Media for Roads

PART 3 EXECUTION

- 3.1 EXAMINATION
 - 3.1.1 Testing for Moisture
 - 3.1.2 Surface Preparation Demonstration
 - 3.1.3 Test Stripe Demonstration
 - 3.1.4 Application Rate Demonstration
 - 3.1.5 Level of Performance Demonstration
- 3.2 EXTERIOR SURFACE PREPARATION
 - 3.2.1 Early Painting of Asphalt Pavements

3.3 APPLICATION

3.3.1 Paint

3.3.1.1 Waterborne Paint

3.3.1.1.1 Roads

3.3.2 Cleanup and Waste Disposal

3.4 FIELD QUALITY CONTROL

3.4.1 Sampling and Testing

3.4.2 Material Inspection

3.4.3 Dimensional Tolerances

3.4.4 Reflective Media and Coating Application Verification

-- End of Section Table of Contents --

SECTION 32 17 23

PAVEMENT MARKINGS
08/16, CHG 5: 11/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to in the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE 48076	(1992) Paints and varnishes. Determination of apparent viscosity or consistency. Method krebs-stormer.
UNE 48083	(1992) Paints And Varnishes. Conservation and Stability In Containers
UNE 135202	(2010) Road signalization equipment. Horizontal signalisation. Paints and cold plastic. Drying time "no pick-up".
UNE 135213	(2018) Road signalization equipment. Horizontal signalization. Paints. Contrast ratio of the dried film.
UNE-EN 1436	(2018) Road marking materials - Road marking performance for road users and test methods
UNE-EN 1871	(2000) Road Marking Materials. Physical Properties.
UNE-EN 12802	(2012) Road marking materials - Laboratory methods for identification.
UNE-EN ISO 11664-2	(2011) Colorimetry - Part 2: CIE standard illuminants (ISO 11664-2:2007)

CENTRO DE ESTUDIOS Y EXPERIMENTACION DE OBRAS PUBLICAS.
LABORATORIO CENTRAL DE ESTRUCTURAS Y MATERIALES (MELC)

MELC 12.30	(1990) Determination of percentage of default glass microspheres in road signaling.
MELC 12.31	(1990) Refraction measurement of glass microspheres for road signaling.
MELC 12.32	(1990) Sample taking of glass microspheres for road signaling.

MINISTERIO DE FOMENTO (MF)

FOM/2523/2014 (2014) Orden de 12 de diciembre, por la que se actualizan determinados artículos del pliego de prescripciones técnicas generales para obras de carreteras y puentes, relativos a materiales básicos, a firmes y pavimentos, y a señalización, balizamiento y sistemas de contención de vehículos.

MF PG3 (2014) Pliego de Prescripciones Tecnicas Generales para Obras de Carreteras y Puentes

Norma 8.2-IC (1987) Marcas Viales.

UNA NORMA ESPAÑOLA (UNE)

UNE-EN 1423 (2013) Road marking materials - Drop on materials - Glass beads, antiskid aggregates and mixtures of the two

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS (AASHTO)

AASHTO M 247 (2013) Standard Specification for Glass Beads Used in Pavement Markings

MASTER PAINTERS INSTITUTE (MPI)

MPI 97 (2012) Traffic Marking Paint, Latex

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Surface Preparation Equipment List; G
Application Equipment List; G
Exterior Surface Preparation
Safety Data Sheets; G
Reflective media for roads; G
Waterborne Paint; G

SD-06 Test Reports

Reflective Media for Roads; G
Waterborne Paint; G

SD-07 Certificates

Qualifications; G
Reflective Media for Roads
Waterborne Paint

Volatile Organic Compound, (VOC); G

SD-08 Manufacturer's Instructions

Waterborne Paint; G

1.3 QUALITY ASSURANCE

1.3.1 Regulatory Requirements

Submit certificate stating that the proposed pavement marking paint meets the Volatile Organic Compound, (VOC) regulations of the local Air Pollution Control District having jurisdiction over the geographical area in which the project is located. Submit Safety Data Sheets for each product.

1.3.2 Qualifications

Submit documentation certifying that pertinent personnel are qualified for equipment operation and handling of applicable chemicals. The documentation should include experience on five projects of similar size and scope with references for all personnel.

1.4 DELIVERY AND STORAGE

Deliver paint materials, thermoplastic compound materials, and reflective media in original sealed containers that plainly show the designated name, specification number, batch number, color, date of manufacture, manufacturer's directions, and name of manufacturer.

Provide storage facilities at the job site, only in areas approved by the Contracting Officer, for maintaining materials at temperatures recommended by the manufacturer. Make available paint stored at the project site or segregated at the source for sampling not less than 30 days prior to date of required approval for use to allow sufficient time for testing. Notify the Contracting Officer when paint is available for sampling.

1.5 PROJECT/SITE CONDITIONS

1.5.1 Environmental Requirements

1.5.1.1 Weather Limitations for Application

Apply pavement markings to clean, dry surfaces, and unless otherwise approved, only when the air and pavement surface temperature is at least 3 degrees C above the dew point and the air and pavement temperatures are within the limits recommended by the pavement marking manufacturer. Allow pavement surfaces to dry after water has been used for cleaning or rainfall has occurred prior to striping or marking. Test the pavement surface for moisture before beginning work each day and after cleaning. Do not commence marking until the pavement is sufficiently dry and the pavement condition has been approved by the Contracting Officer. Employ the "plastic wrap method" to test the pavement for moisture as specified in paragraph TESTING FOR MOISTURE.

1.5.1.2 Weather Limitations for Removal of Pavement Markings on Roads and Automotive Parking Areas

Cease operation during thunderstorms, or during rainfall, except for

waterblasting and removal of previously applied chemicals. Cease waterblasting where surface water accumulation alters the effectiveness of material removal.

1.5.2 Traffic Controls

Place warning signs near the beginning of the worksite and well ahead of the worksite for alerting approaching traffic from both directions. Place small markers along newly painted lines or freshly placed raised markers to control traffic and prevent damage to newly painted surfaces or displacement of raised pavement markers. Mark painting equipment with large warning signs indicating slow-moving painting equipment in operation.

When traffic must be rerouted or controlled to accomplish the work, provide necessary warning signs, flag persons, and related equipment for the safe passage of vehicles.

1.5.3 Lighting

When night operations are necessary, provide all necessary lighting and equipment. The Government reserves the right to accept or reject night work on the day following night activities by the Contractor.

PART 2 PRODUCTS

2.1 EQUIPMENT

2.1.1 Surface Preparation and Paint Removal

2.1.1.1 Surface Preparation Equipment for Roads and Automotive Parking Areas

Submit a surface preparation equipment list by serial number, type, model, and manufacturer. Include descriptive data indicating area of coverage per pass, pressure adjustment range, tank and flow capacities, and safety precautions required for the equipment operation. Mobile equipment must allow for removal of markings without damaging the pavement surface or joint sealant. Maintain machines, tools, and equipment used in the performance of the work in satisfactory operating condition.

2.1.1.1.1 Waterblasting Equipment

Use mobile waterblasting equipment capable of producing a pressurized stream of water that effectively removes paint from the pavement surface without significantly damaging the pavement. Provide equipment, tools, and machinery which are safe and in good working order at all times.

2.1.1.1.2 Grinding or Scarifying Equipment

Use equipment capable of removing surface contaminants, paint build-up, or extraneous markings from the pavement surface without leaving any residue. Clean the surface by hydro blast to remove surface contaminants and ash after a weed torch is used to remove paint.

2.1.2 Application Equipment

Submit application equipment list appropriate for the material(s) to be used. Include manufacturer's descriptive data and certification for the planned use that indicates area of coverage per pass, pressure adjustment

range, tank and flow capacities, and all safety precautions required for operating and maintaining the equipment. Provide and maintain machines, tools, and equipment used in the performance of the work in satisfactory operating condition, or remove them from the work site. Provide mobile and maneuverable application equipment to the extent that straight lines can be followed and normal curves can be made in a true arc.

2.1.2.1 Paint Application Equipment

2.1.2.1.1 Hand-Operated, Push-Type Machines

Provide hand-operated push-type applicator machine of a type commonly used for application of water based paint or two-component, chemically curing paint, thermoplastic, or preformed tape, to pavement surfaces for small marking projects, such as legends and cross-walks, automotive parking areas, or surface painted signs. Provide applicator machine equipped with the necessary tanks and spraying nozzles capable of applying paint uniformly at coverage specified. Hand operated spray guns may be used in areas where push-type machines cannot be used.

2.1.2.1.2 Self-Propelled or Mobile-Drawn Spraying Machines

Provide self-propelled or mobile-drawn spraying machine with suitable arrangements of atomizing nozzles and controls to obtain the specified results. Provide machine having a speed during application capable of applying the stripe widths indicated at the paint coverage rate specified herein and of even uniform thickness with clear-cut edges.

2.1.2.1.2.1 Road Marking

Provide equipment used for marking roads capable of placing the prescribed number of lines at a single pass as solid lines, intermittent lines, or a combination of solid and intermittent lines using a maximum of three different colors of paint as specified.

2.1.2.1.2.2 Hand Application

Provide spray guns for hand application of paint in areas where the mobile paint applicator cannot be used.

2.1.2.2 Reflective Media Dispenser

Attach the dispenser for applying the reflective media to the paint or thermoplastic dispenser and designed to operate automatically and simultaneously with the applicator through the same control mechanism. The bead applicator must be capable of adjustment and designed to provide uniform flow of reflective media over the full length and width of the stripe at the rate of coverage specified in paragraph APPLICATION.

2.2 MATERIALS

Use waterborne paint for roads. Use non-reflectorized waterborne paint for automotive parking areas. The maximum allowable VOC content of pavement markings is 150 grams per liter. Color of markings are indicated on the drawings and must conform to Norma 8.2-IC and Art. 700 of MF PG3 for roads and automotive parking areas. Provide materials conforming to the requirements specified herein.

2.2.1 Waterborne Paint

MPI 97 and the following. UNE-EN 1436 and UNE-EN 1871, 100% acrylic waterborne paint, color as indicated. Paints with content of mercury, lead, hexavalent chromium, toluene, chlorinated solvents, chlorine derivatives or ethylene-based glycol shall not be authorized. The paints shall have the following characteristics:

- Consistency: Krebs Units. UNE 48076. Min. 84 UK +/- 10 UK.
- Daylight diffuse illuminance factor (Beta). UNE-EN ISO 11664-2. Beta > 0.85 minimum for white paint, drying time UNE 135202 ≤ 30 min.
- Contrast value with adjacent pavement: UNE 135213, $rc \geq 0.90$.
- Accelerated package stability: UNE 48083. Variation in consistency: less than 5 Krebs Units.
- Minimum content of solids: UNE-EN 12802. Minimum value: 75%. Maximum deviation versus declared value: +/- 2.5%.
- Accelerated aging: UNE-EN 1871. Maximum deviation of illuminance: Beta ≤ 0.05, Class BR2.
- Bleeding of paint for markings on bituminous pavement. UNE-EN 1871. Min. value 0.95.
- Alkali resistance for marking on concrete pavement. UNE-EN 1871. Without superficial damages.

2.2.2 Reflective Media

2.2.2.1 Reflective Media for Roads

Post mixed microspheres per UNE-EN 1423 and MELC 12.30, MELC 12.31 and MELC 12.32. The size of beads will be between 125 and 600 microns. The spheres will have a crash resistance of 193 N minimum. The beads will have 2% maximum of spheres with a ratio between maximum diameter and minimum diameter less than 0.9. Additionally, the beads will comply with the following requirements:

- Roundness: MELC 12.30, 80% minimum of true sphere.
- Index of refraction: MELC 12.31, Class A, 1.5 minimum.
- Specific weight: between 2.30 and 2.50.
- Gradation: MELC 12.32.
Sieve UNE:
#800: pass 100%
#630: pass 95% to 100%
#500: pass 90% to 100%
#320: pass 30% to 70%
#125: pass 0% to 5%
- Chemical resistance: the beads shall not develop haze or dulling when immersed in sulfuric acid for 5 minutes and rinsed with distilled water; when immersed in calcium chloride for 3 hours and rinsed with distilled water; when immersed in distilled water and 0.1 N hydrochloric acid for 2 hours and rinsed with distilled water. The beads will not darken when immersed in a dilution of 50% sodium sulfide for 1 hour and rinsed with distilled water.

PART 3 EXECUTION

FOM/2523/2014, Art. 700.6 through Art. 700.10.

3.1 EXAMINATION

3.1.1 Testing for Moisture

Test the pavement surface for moisture before beginning pavement marking after each period of rainfall, fog, high humidity, or cleaning, or when the ambient temperature has fallen below the dew point. Do not commence marking until the pavement is sufficiently dry and the pavement condition has been approved by the Contracting Officer or authorized representative.

Employ the "plastic wrap method" to test the pavement for moisture as follows: Cover the pavement with a 300 mm by 300 mm section of clear plastic wrap and seal the edges with tape. After 15 minutes, examine the plastic wrap for any visible moisture accumulation inside the plastic. Do not begin marking operations until the test can be performed with no visible moisture accumulation inside the plastic wrap. Re-test surfaces when work has been stopped due to rain.

3.1.2 Surface Preparation Demonstration

Prior to surface preparation, demonstrate the proposed procedures and equipment. Prepare areas large enough to determine cleanliness, adhesion of remaining coating and rate of cleaning. Approved demonstration area establishes the standard for the remainder of the work.

3.1.3 Test Stripe Demonstration

Prior to paint application, demonstrate test stripe application within the work area using the proposed materials and equipment. Apply separate test stripes in each of the line widths and configurations required herein using the proposed equipment. Make the test stripes long enough to determine the proper speed and operating pressures for the vehicle(s) and machinery, but not less than 15 m long.

3.1.4 Application Rate Demonstration

During the Test Stripe Demonstration, demonstrate compliance with the application rates specified herein. Document the equipment speed and operating pressures required to meet the specified rates in each configuration of the equipment and provide a copy of the documentation to the Contracting Officer prior to proceeding with the work.

3.1.5 Level of Performance Demonstration

The Contracting Officer will be present at the application demonstrations to observe the results obtained and to validate the operating parameters of the vehicle(s) and equipment. If accepted by the Contracting Officer, the test stripe is the measure of performance required for this project. Do not proceed with the work until the demonstration results are satisfactory to the Contracting Officer.

3.2 EXTERIOR SURFACE PREPARATION

Allow new pavement surfaces to cure for a period of not less than 30 days before application of marking materials. Thoroughly clean surfaces to be marked before application of the paint. Remove dust, dirt, and other granular surface deposits by sweeping, blowing with compressed air, rinsing with water, or a combination of these methods as required. Remove rubber deposits, existing paint markings, residual curing compounds, and

other coatings adhering to the pavement by water blasting.

3.2.1 Early Painting of Asphalt Pavements

For asphalt pavement systems requiring painting application at less than 30 days, apply the paint and beads at half the normal application rate, followed by a second application at the normal rate after 30 days.

3.3 APPLICATION

Apply pavement markings to dry pavements only.

3.3.1 Paint

Apply paint with approved equipment at rate of coverage specified herein. Provide guidelines and templates as necessary to control paint application. Take special precautions in marking numbers, letters, and symbols. Manually paint numbers, letters, and symbols. Sharply outline all edges of markings. The maximum drying time requirements of the paint specifications will be strictly enforced, to prevent undue softening of bitumen, and pickup, displacement, or discoloration by tires of traffic. If there is a deficiency in drying of the markings, painting operations must cease until the cause of the slow drying is determined and corrected.

3.3.1.1 Waterborne Paint

3.3.1.1.1 Roads

Apply paint at a rate of 2.6 plus or minus 0.1 square meter per liter. Apply AASHTO M 247 Type 1 beads at a rate of 834 plus or minus 60 grams of glass spheres per liter.

3.3.2 Cleanup and Waste Disposal

Keep the worksite clean and free of debris and waste from the removal and application operations. Dispose of debris at approved sites.

3.4 FIELD QUALITY CONTROL

3.4.1 Sampling and Testing

As soon as the paint and reflective media are available for sampling, obtain by random selection from the sealed containers, two quart samples of each batch in the presence of the Contracting Officer. Accomplish adequate mixing prior to sampling to ensure a uniform, representative sample. A batch is defined as that quantity of material processed by the manufacturer at one time and identified by number on the label. Clearly identify samples by designated name, specification number, batch number, project contract number, intended use, and quantity involved.

At the discretion of the Contracting Officer, samples provided may be tested by the Government for verification.

3.4.2 Material Inspection

Examine material at the job site to determine that it is the material referenced in the report of test results or certificate of compliance. Provide test results substantiating conformance to the specified requirements with each certificate of compliance.

3.4.3 Dimensional Tolerances

Apply all markings in the standard dimensions provide in the drawings. New markings may deviate a maximum of 10 percent larger than the standard dimension. The maximum deviation allowed when painting over an old marking is up to 20 percent larger than the standard dimensions.

3.4.4 Reflective Media and Coating Application Verification

Use a wet film thickness gauge to measure the application of wet paint. Use a microscope or magnifying glass to evaluate the embedment of glass beads in the paint. Verify the glass bead embedment with approximately 50 percent of the individual bead spheres embedded and 50 percent of the individual bead spheres exposed.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 31 13.53

HIGH-SECURITY FENCES AND GATES

11/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SUBMITTALS
- 1.3 DELIVERY, STORAGE, AND HANDLING

PART 2 PRODUCTS

- 2.1 COMPONENTS
 - 2.1.1 Chain Link Fence Fabric
 - 2.1.1.1 General
 - 2.1.2 Posts
 - 2.1.2.1 Metal Posts for Chain Link Fence
 - 2.1.2.2 Accessories
 - 2.1.3 Chain Link Braces and Rails
 - 2.1.4 Chain Link Gates
 - 2.1.4.1 Gate Assembly
 - 2.1.4.2 Gate Leaves
 - 2.1.4.3 Gate Hardware and Accessories
 - 2.1.5 Padlocks
- 2.2 MATERIALS
 - 2.2.1 Wire
 - 2.2.1.1 Wire Ties
 - 2.2.1.2 Barbed Wire
 - 2.2.1.3 Tension Wire
 - 2.2.2 Concrete

PART 3 EXECUTION

- 3.1 PREPARATION
 - 3.1.1 Line and Grade
 - 3.1.2 Excavation
- 3.2 INSTALLATION
 - 3.2.1 Installation Drawings
 - 3.2.2 Security Fencing
 - 3.2.3 Posts
 - 3.2.3.1 Earth and Bedrock
 - 3.2.4 Rails
 - 3.2.5 Fabric
 - 3.2.6 Supporting Arms
 - 3.2.7 Gate Installation
 - 3.2.8 Grounding
- 3.3 CLOSEOUT ACTIVITIES
 - 3.3.1 Cleanup

-- End of Section Table of Contents --

SECTION 32 31 13.53

HIGH-SECURITY FENCES AND GATES
11/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE-EN 10210-1	(2007) Hot finished structural hollow sections of non-alloy and fine grain steels - Part 1: Technical delivery conditions
UNE-EN 10216-1	(2014) Seamless steel tubes for pressure purposes - Technical delivery conditions - Part 1: Non-alloy steel tubes with specified room temperature properties
UNE-EN 10244-2	(2010) Steel wire and wire products - Non-ferrous metallic coatings on steel wire - Part 2: Zinc or zinc alloy coatings
UNE-EN ISO 1461	(2010) Hot dip galvanized coatings on fabricated iron and steel articles. Specifications and test methods (ISO1461:1999)

ASTM INTERNATIONAL (ASTM)

ASTM A121	(2022) Standard Specification for Metallic-Coated Carbon Steel Barbed Wire
ASTM A153/A153M	(2016a) Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware
ASTM A491	(2011; R 2017) Standard Specification for Aluminum-Coated Steel Chain-Link Fence Fabric
ASTM F567	(2014a; R 2019) Standard Practice for Installation of Chain Link Fence
ASTM F626	(2014; R 2019) Standard Specification for Fence Fittings
ASTM F883	(2013; R 2022) Standard Performance Specification for Padlocks
ASTM F900	(2011; R 2017) Standard Specification for

Industrial and Commercial Swing Gates

ASTM F1043 (2018) Standard Specification for Strength and Protective Coatings on Steel Industrial Fence Framework

ASTM F1083 (2018) Standard Specification for Pipe, Steel, Hot-Dipped Zinc Coated (Galvanized) Welded, for Fence Structures

MINISTERIO DE LA PRESIDENCIA (MP)

RD 470/2021 (2021) Real Decreto 470/2021, de 29 de junio, por el que se aprueba el Código Estructural.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Fence Installation Drawings; G

SD-03 Product Data

Fabric
Posts
Post Caps
Chain Link Braces
Rails
Tension Wire
Barbed Wire
Barbed Wire Supporting Arms
Latches
Hinges
Stops
Keepers
Padlocks
Wire Ties

SD-07 Certificates

Chain Link Fence
Fabric
Barbed Wire
Gate Hardware and Accessories
Concrete

SD-10 Operation and Maintenance Data

Operating and maintenance instructions

1.3 DELIVERY, STORAGE, AND HANDLING

Deliver materials to site in an undamaged condition. Store materials

elevated off of the ground to protect against oxidation caused by ground contact.

PART 2 PRODUCTS

2.1 COMPONENTS

2.1.1 Chain Link Fence Fabric

2.1.1.1 General

Provide ASTM A491, aluminum-coated steel wire or UNE-EN 10244-2 zinc coated steel wire with minimum coating of 610 g of zinc per square meter of coated surface. Fabricate fence fabric of 3.7 mm diameter wire woven in 50 mm diamond mesh. Provide twisted and barbed fabric on selvage with tension wire and knuckled on the selvage with rails. Set fence heights to match existing. Approximate heights are shown on drawings.

2.1.2 Posts

2.1.2.1 Metal Posts for Chain Link Fence

Provide posts conforming to ASTM F1083, zinc-coated. Group IA Regular Strength steel pipe or alternatively conforming to UNE-EN 10210-1 or UNE-EN 10216-1, coated per UNE-EN ISO 1461. Provide sizes as shown on the drawings. Use line posts and terminal (corner, gate, and pull) posts of the same designation throughout the fence. Provide gate post for the gate type specified subject to the limitation specified in ASTM F900.

2.1.2.2 Accessories

- a. Provide accessories conforming to ASTM F626. Coat ferrous accessories with zinc coating.
- b. Provide truss rods (with turnbuckles or other means of adjustment) for each terminal post.
- c. Provide barbed wire supporting arms of the 45 degree outward angle 3-strand arm type and of the design required for the post furnished. Secure arms as shown on drawings.
- d. Furnish post caps in accordance with manufacturer's standard accessories.
- e. Provide 9 gauge tie wire for attaching fabric to rails, braces, and posts and match the material and coating of the fence fabric. Use tie wires for attaching fabric to tension wire on high security fences made from 1.6 mm stainless steel. Provide double loop tie wires 165 mm in length. Provide miscellaneous hardware coatings which conform to ASTM A153/A153M or UNE-EN ISO 1461 unless modified.

2.1.3 Chain Link Braces and Rails

ASTM F1083 or alternatively conforming to UNE-EN 10210-1 or UNE-EN 10216-1, zinc-coated, Group IA Regular Strength, steel pipe, Nominal Pipe Size (NPS) 1-1/4. Use braces and rails that are Group IA Regular Strength, steel pipe, size NPS 1-1/4 and zinc coated, minimum thickness, 0.25 mm conforming to the requirements of ASTM F1043.

2.1.4 Chain Link Gates

2.1.4.1 Gate Assembly

Provide gate assembly conforming to ASTM F900 of the type and swing shown. Provide gate frames conforming to strength and coating requirements of ASTM F1083 for Group IA Regular Strength, steel pipe, nominal pipe size (NPS) 1-1/2. Use gate fabric that matches the specified chain link fabric.

2.1.4.2 Gate Leaves

For gate leaves, more than 2.44 m wide, provide either intermediate members and diagonal truss rods or tubular members as necessary to provide rigid construction, free from sag or twist. For gate leaves less than 2.44 m wide, provide truss rods or intermediate braces. Provide intermediate braces on all gate frames with an electro-mechanical lock. Attach fabric to the gate frame by method standard with the manufacturer. Welding is not an acceptable method for attaching fabric to gate frames.

2.1.4.3 Gate Hardware and Accessories

Submit manufacturer's catalog data. Furnish and install latches, hinges, stops, keepers, and other hardware items as required for the operation of the gate. All items are required to match the material characteristics of the fence system being installed. Provide hinge with stainless steel pin. Arrange latches for padlocking so that the padlock will be accessible from both sides of the gate. Provide stops for holding the gates in the open position. For high security applications, extend each end member of gate frames sufficiently above the top member to carry three strands of barbed wire in horizontal alignment with barbed wire strands on the fence. Coating for steel latches, stops, hinges, keepers, and accessories, must be galvanized.

2.1.5 Padlocks

Provide padlocks conforming to ASTM F883, Type P01, Option A. Size 44 mm.

2.2 MATERIALS

2.2.1 Wire

2.2.1.1 Wire Ties

Submit samples as specified. Provide wire ties constructed of the same material and finish as the fencing fabric.

2.2.1.2 Barbed Wire

Provide barbed wire conforming to ASTM A121 aluminum-coated, Type A, with 12.5 gauge wire with 14 gauge, round, 4-point barbs spaced no more than 125 mm apart.

2.2.1.3 Tension Wire

Provide metallic coated steel marcelled tension wire (No. 7-gauge). Provide aluminum-coated (aluminized) steel wire with coating that weighs not less than 122 gram per square meter.

2.2.2 Concrete

Conforming to RD 470/2021 and Section 03 30 00 CAST-IN-PLACE CONCRETE, using 19 mm maximum size aggregate, and having minimum compressive strength of 21 MPa at 28 days. Use grout consisting of one part portland cement to three parts clean, well-graded sand and the minimum amount of water to produce a workable mix.

PART 3 EXECUTION

3.1 PREPARATION

Perform complete installation conforming to ASTM F567.

3.1.1 Line and Grade

Install fence to the lines and grades indicated. Clear the area on either side of the fence line to the extent indicated. Space line posts equidistant at intervals not exceeding 3 m. Set terminal (corner, gate, and pull) posts whenever abrupt changes in vertical and horizontal alignment are encountered. Provide continuous fabric between terminal posts; however, ensure runs between terminal posts do not exceed 152.4 m. Repair any damage to galvanized surfaces, including welding, with paint containing zinc dust.

3.1.2 Excavation

Excavate holes to depths indicated. Clear all post holes of loose material and spread waste material where directed. Eliminate ground surface irregularities along the fence line to the extent necessary to maintain a 50 mm clearance between the bottom of the fabric and finish grade.

3.2 INSTALLATION

3.2.1 Installation Drawings

Submit complete Fence Installation Drawings for review and approval by the Contracting Officer prior to shipment. Submit drawing details that include, but are not limited to the following information: Fence Installation Drawings, Location of gate, corner, end, and pull posts, Gate Assembly, and Gate Hardware and Accessories. Install fence system per approved drawings.

3.2.2 Security Fencing

Install new security fencing, remove existing security fencing, and perform related work to provide continuous security for facility. Schedule and fully coordinate work with Contracting Officer.

3.2.3 Posts

3.2.3.1 Earth and Bedrock

- a. Set posts plumb and in alignment. Except where solid rock is encountered, set posts in concrete to the depth indicated on the drawings. Where solid rock is encountered with no overburden, set posts to a minimum depth of 457 mm in rock. Where solid rock is covered with an overburden of soil or loose rock, set posts to the

minimum depth indicated on the drawing unless a penetration of 457 mm in solid rock is achieved before reaching the indicated depth, in which case terminate depth of penetration. Grout all portions of posts set in rock.

- b. Set portions of posts not set in rock in concrete from the rock to ground level. Set posts in holes not less than the diameter shown on the drawings. Make diameters of holes in solid rock at least 25 mm greater than the largest cross section of the post. Thoroughly consolidate concrete and grout around each post, free of voids and finished to form a dome. Allow concrete and grout to cure for 72 hours prior to attachment of any item to the posts. Group II line posts may be mechanically driven, for temporary fence construction only, if rock is not encountered. Set driven posts to a minimum depth of 914 mm and protect with drive caps when setting.
- c. Test fence post rigidity by applying a 222.4 newtons force on the post, perpendicular to the fabric, at 1.52 m above ground. Ensure post movement measured at the point where the force is applied is less than or equal to 19 mm from the relaxed position. Test every tenth post for rigidity. When a post fails this test, make further tests on the next four posts on either side of the failed post. Remove, replace, and retest all failed parts at the Contractor's expense.

3.2.4 Rails

Bolt rail to double rail ends and securely fasten double rail ends to the posts. Peen bolts to prevent easy removal. Install rail before chain link fabric.

3.2.5 Fabric

- a. Set fabric height as shown on drawings.
- b. Install chain link fabric on the non-secure side of the post. Attach fabric to terminal posts with stretcher bars and tension bands. Space bands at approximately 381 mm intervals. Install fabric and pull taut to provide a smooth and uniform appearance free from sag, without permanently distorting the fabric diamond or reducing the fabric height. Fasten fabric to line posts at approximately 381 mm intervals and fastened to all rails and tension wires at approximately 305 mm intervals.
- c. Cut fabric by untwisting and removing pickets. Accomplish splicing by weaving a single picket into the ends of the rolls to be joined. Install the bottom of the fabric 50 mm plus or minus 13 mm above the ground.
- d. After the fabric installation is complete, exercise the fabric by applying a 222 newton push-pull force at the center of the fabric between posts; use a 133 newton pull at the center of the panel to ensure fabric deflection of not more than 63.5 mm when pulling fabric from the post side of the fence. Every second fence panel is required to meet this requirement. Resecure and retest all failed panels at the Contractor's expense.

3.2.6 Supporting Arms

Install barbed wire supporting arms and barbed wire as indicated on the

drawings and as recommended by the manufacturer. Anchor supporting arms to the posts in a manner to prevent easy removal with hand tools. Pull barbed wire taut and attach to the arms with clips or other means that will prevent easy removal.

3.2.7 Gate Installation

- a. Install gates at the locations shown. Mount gates to swing as indicated. Install latches, stops, and keepers as required.
- b. Attach padlocks to gates or gate posts with chains. Weld or otherwise secure hinge pins, and hardware assembly to prevent removal.
- c. Submit 6 copies of operating and maintenance instructions. Outline the step-by-step procedures required for system startup, operation, and shutdown. Include the manufacturer's name, model number, service manual, parts list, and brief description of all equipment and their basic operating features. Include in the maintenance instructions routine maintenance procedures, possible breakdowns and repairs, and troubleshooting guide. Also include the general gate layout, equipment layout and simplified wiring and control diagrams of the system as installed.

3.2.8 Grounding

- a. Ground fencing as indicated on drawings and specified.
- b. Ground fences crossed by overhead powerlines in excess of 600 volts and ground all electrical equipment attached to the fence.
- c. Ground fences on each side of all gates, at each corner, at the closest approach to each building located within 15 m of the fence, and where the fence alignment changes more than 15 degrees. Ensure grounding locations are located no more than 198 m apart. Bond each gate panel with a flexible bond strap to its gate post. Ground fences crossed by powerlines of 600 volts or more at or near the point of crossing and at distances not exceeding 45 m on each side of crossing.
- d. Provide ground conductor consisting of 3.3 mm solid copper wire. Use grounding electrodes that measures 19 mm by 3.05 m long and are a solid copper rod. Drive electrodes into the earth so that the top of the electrode is at least 152 mm below the grade. Where driving is impracticable, bury electrodes a minimum of 305 mm deep and radially from the fence. Install the top of the electrode to be less than 610 mm or more than 2.4 m from the fence. Clamp ground conductor to the fence and electrodes with bronze grounding clamps to create electrical continuity between fence posts, fence fabric, and ground rods. Measure total resistance of the fence to ground and ensure it is not greater than 25 ohms.

3.3 CLOSEOUT ACTIVITIES

3.3.1 Cleanup

Remove waste fencing materials and other debris from the work site each workday.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 32 - EXTERIOR IMPROVEMENTS

SECTION 32 92 19

SEEDING

08/17, CHG 1: 08/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
 - 1.2.1 Stand of Turf
- 1.3 RELATED REQUIREMENTS
- 1.4 SUBMITTALS
- 1.5 DELIVERY, STORAGE, AND HANDLING
 - 1.5.1 Delivery
 - 1.5.1.1 Seed Protection
 - 1.5.1.2 Fertilizer Gypsum Sulfur and Iron Delivery
 - 1.5.2 Storage
 - 1.5.2.1 Seed and Fertilizer Gypsum Sulfur and Iron Storage
 - 1.5.2.2 Topsoil
 - 1.5.2.3 Handling
- 1.6 TIME RESTRICTIONS AND PLANTING CONDITIONS
 - 1.6.1 Restrictions
- 1.7 TIME LIMITATIONS
 - 1.7.1 Seed

PART 2 PRODUCTS

- 2.1 SEED
 - 2.1.1 Classification
 - 2.1.2 Seed Mix
 - 2.1.3 Seed Mixture by Weight
- 2.2 TOPSOIL
 - 2.2.1 On-Site Topsoil
 - 2.2.2 Off-Site Topsoil
 - 2.2.3 Composition
- 2.3 SOIL CONDITIONERS
 - 2.3.1 Aluminum Sulfate
 - 2.3.2 Sulfur
 - 2.3.3 Iron
 - 2.3.4 Sand
 - 2.3.5 Perlite
 - 2.3.6 Composted Derivatives
 - 2.3.6.1 Particle Size
 - 2.3.6.2 Nitrogen Content
 - 2.3.7 Gypsum
- 2.4 FERTILIZER
 - 2.4.1 Granular Fertilizer
 - 2.4.2 Hydroseeding Fertilizer
- 2.5 MULCH
 - 2.5.1 Wood Cellulose Fiber Mulch
- 2.6 WATER

2.7 EROSION CONTROL MATERIALS

- 2.7.1 Erosion Control Blanket
- 2.7.2 Erosion Control Fabric
- 2.7.3 Erosion Control Net
- 2.7.4 Hydrophilic Colloids
- 2.7.5 Erosion Control Material Anchors

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 EXTENT OF WORK

- 3.1.1.1 Topsoil
- 3.1.1.2 Soil Conditioner Application Rates
- 3.1.1.3 Fertilizer Application Rates

3.2 SEEDING

- 3.2.1 Seed Application Seasons and Conditions
- 3.2.2 Seed Application Method
 - 3.2.2.1 Hydroseeding
- 3.2.3 Mulching
 - 3.2.3.1 Mulch
 - 3.2.3.2 Non-Asphaltic Tackifier
- 3.2.4 Erosion Control Material
- 3.2.5 Watering

3.3 PROTECTION OF TURF AREAS

3.4 RENOVATION OF EXISTING TURF AREA

- 3.4.1 Overseeding

3.5 RESTORATION

-- End of Section Table of Contents --

SECTION 32 92 19

SEEDING

08/17, CHG 1: 08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AENOR - UNA NORMA ESPAÑOLA (UNE)

UNE-ISO 10390 (2012) Soil quality. Determination of pH.

ASTM INTERNATIONAL (ASTM)

ASTM D4972 (2018) Standard Test Methods for pH of Soils

U.S. DEPARTMENT OF AGRICULTURE (USDA)

DOA SSIR 42 (1996) Soil Survey Investigation Report No. 42, Soil Survey Laboratory Methods Manual, Version 3.0

1.2 DEFINITIONS

1.2.1 Stand of Turf

95 percent ground cover of the established species.

1.3 RELATED REQUIREMENTS

Section 31 23 00.00 20 EXCAVATION AND FILL applies to this section for pesticide use and plant establishment requirements, with additions and modifications herein.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-03 Product Data

Wood Cellulose Fiber Mulch
Fertilizer

Include physical characteristics, and recommendations.

SD-06 Test Reports

Topsoil Composition Tests (reports and recommendations).

SD-07 Certificates

Seed

SD-08 Manufacturer's Instructions

Erosion Control Materials

1.5 DELIVERY, STORAGE, AND HANDLING

1.5.1 Delivery

1.5.1.1 Seed Protection

Protect from drying out and from contamination during delivery, on-site storage, and handling.

1.5.1.2 Fertilizer Gypsum Sulfur and Iron Delivery

Deliver to the site in original, unopened containers bearing manufacturer's chemical analysis, name, trade name, trademark, and indication of conformance to local and federal laws. Instead of containers, fertilizer, gypsum, sulphur, and iron may be furnished in bulk with certificate indicating the above information.

1.5.2 Storage

1.5.2.1 Seed and Fertilizer Gypsum Sulfur and Iron Storage

Store in cool, dry locations away from contaminants.

1.5.2.2 Topsoil

Prior to stockpiling topsoil, treat growing vegetation with application of appropriate specified non-selective herbicide. Clear and grub existing vegetation three to four weeks prior to stockpiling topsoil.

1.5.2.3 Handling

Do not drop or dump materials from vehicles.

1.6 TIME RESTRICTIONS AND PLANTING CONDITIONS

1.6.1 Restrictions

Do not plant when the ground is muddy or when air temperature exceeds 32 degrees Celsius.

1.7 TIME LIMITATIONS

1.7.1 Seed

Apply seed within twenty four hours after seed bed preparation.

PART 2 PRODUCTS

2.1 SEED

2.1.1 Classification

Provide certified seed of the latest season's crop delivered in original sealed packages, bearing producer's guaranteed analysis for percentages of mixtures, purity, germination, weedseed content, and inert material. Wet, moldy, or otherwise damaged seed will be rejected. Field mixes will be acceptable when field mix is performed on site in the presence of the Contracting Officer.

2.1.2 Seed Mix

Botanical Name	Minimum Percent Germination and Hard Seed	Maximum Percent Weed Seed
Cynodon Dactylon	40 percent	60 percent
Festuca Rubra	15 percent	20 percent
Festuca Arundinacea	15 percent	20 percent
Medicago Sativa	5 percent	10 percent
Agropyron cristatum	5 percent	10 percent
Onobrychis viciifolia	5 percent	10 percent

2.1.3 Seed Mixture by Weight

Proportion seed mixtures by weight. Provide 15-20 g of seed per square meter.

2.2 TOPSOIL

2.2.1 On-Site Topsoil

Surface soil stripped and stockpiled on site and modified as necessary to meet the requirements specified for topsoil in paragraph COMPOSITION. When available topsoil must be existing surface soil stripped and

stockpiled on-site in accordance with Section 31 23 00.00 20 EXCAVATION AND FILL.

2.2.2 Off-Site Topsoil

Conform to requirements specified in paragraph COMPOSITION. Additional topsoil must be furnished by the Contractor.

2.2.3 Composition

Containing from 5 to 10 percent organic matter as determined by the topsoil composition tests of the Organic Carbon, 6A, Chemical Analysis Method described in DOA SSIR 42. Maximum particle size, 19 mm, with maximum 3 percent retained on 6 mm screen. The pH must be tested in accordance with ASTM D4972 or UNE-ISO 10390. Topsoil must be free of sticks, stones, roots, and other debris and objectionable materials. Other components must conform to the following limits:

Silt	25-50 percent
Clay	10-30 percent
Sand	20-35 percent
pH	5.5 to 7.0
Soluble Salts	600 ppm maximum

2.3 SOIL CONDITIONERS

Add conditioners to topsoil as required to bring into compliance with "composition" standard for topsoil as specified herein.

2.3.1 Aluminum Sulfate

Commercial grade.

2.3.2 Sulfur

100 percent elemental

2.3.3 Iron

100 percent elemental

2.3.4 Sand

Clean and free of materials harmful to plants.

2.3.5 Perlite

Horticultural grade.

2.3.6 Composted Derivatives

Ground bark, nitrolized sawdust, humus or other green wood waste material free of stones, sticks, and soil stabilized with nitrogen and having the following properties:

2.3.6.1 Particle Size

Minimum percent by weight passing:

4.75 mm screen	95
2.36 mm screen	80

2.3.6.2 Nitrogen Content

Minimum percent based on dry weight:

Fir Sawdust	0.7
Fir or Pine Bark	1.0

2.3.7 Gypsum

Coarsely ground gypsum comprised of calcium sulfate dihydrate 80 percent, calcium 18 percent, sulfur 14 percent; minimum 96 percent passing through 850 micrometers, 100 percent passing thru 970 micrometers screen.

2.4 FERTILIZER

2.4.1 Granular Fertilizer

Organic, granular controlled release fertilizer containing the following minimum percentages, by weight, of plant food nutrients:

- 10 percent available nitrogen
- 10 percent available phosphorus
- 10 percent available potassium

2.4.2 Hydroseeding Fertilizer

Controlled release fertilizer, to use with hydroseeding and composed of pills coated with plastic resin to provide a continuous release of nutrients for at least 6 months and containing the following minimum percentages, by weight, of plant food nutrients.

- 10 percent available nitrogen
- 10 percent available phosphorus
- 10 percent available potassium

2.5 MULCH

Mulch must be free from noxious weeds, mold, and other deleterious materials.

2.5.1 Wood Cellulose Fiber Mulch

Use recovered materials of either paper-based (100 percent post-consumer content) or wood-based (100 percent total recovered content) hydraulic mulch. Processed to contain no growth or germination-inhibiting factors and dyed an appropriate color to facilitate visual metering of materials application. Use with hydraulic application of grass seed and fertilizer.

2.6 WATER

Source of water must be approved by Contracting Officer and of suitable

quality for irrigation, containing no elements toxic to plant life.

2.7 EROSION CONTROL MATERIALS

Erosion control material must conform to the following:

2.7.1 Erosion Control Blanket

100 percent agricultural straw stitched with a degradable nettings, designed to degrade within 12 months.

2.7.2 Erosion Control Fabric

Fabric must be knitted construction of polypropylene yarn with uniform mesh openings 19 to 25 mm square with strips of biodegradable paper. Filler paper strips must have a minimum life of 6 months.

2.7.3 Erosion Control Net

Net must be heavy, twisted jute mesh, weighing approximately 605 grams per meter and 1200 mm wide with mesh openings of approximately 25 mm square.

2.7.4 Hydrophilic Colloids

Hydrophilic colloids must be physiologically harmless to plant and animal life without phytotoxic agents. Colloids must be naturally occurring, silicate powder based, and must form a water insoluble membrane after curing. Colloids must resist mold growth.

2.7.5 Erosion Control Material Anchors

Erosion control anchors must be as recommended by the manufacturer.

PART 3 EXECUTION

3.1 PREPARATION

3.1.1 EXTENT OF WORK

Provide soil preparation prior to planting (including soil conditioners as required), fertilizing, seeding, and surface topdressing of all newly graded finished earth surfaces, unless indicated otherwise, and at all areas inside or outside the limits of construction that are disturbed by the Contractor's operations.

3.1.1.1 Topsoil

Provide 102 mm of off-site topsoil or existing soil to meet indicated finish grade. After areas have been brought to indicated finish grade, incorporate fertilizer, pH adjusters, soil conditioners into soil a minimum depth of 100 mm by disking, harrowing, tilling or other method approved by the Contracting Officer. Remove debris and stones larger than 19 mm in any dimension remaining on the surface after finish grading. Correct irregularities in finish surfaces to eliminate depressions. Protect finished topsoil areas from damage by vehicular or pedestrian traffic.

3.1.1.2 Soil Conditioner Application Rates

Apply soil conditioners at rates as determined by laboratory soil analysis of the soils at the job site.

3.1.1.3 Fertilizer Application Rates

Apply fertilizer at rates as determined by laboratory soil analysis of the soils at the job site.

3.2 SEEDING

3.2.1 Seed Application Seasons and Conditions

Immediately before seeding, restore soil to proper grade. Do not seed when ground is muddy or in an unsatisfactory condition for seeding. If special conditions exist that may warrant a variance in the above seeding dates or conditions, submit a written request to the Contracting Officer stating the special conditions and proposed variance. Apply seed within twenty four hours after seedbed preparation. Sow seed by approved sowing equipment.

3.2.2 Seed Application Method

Seeding method must be hydroseeding.

3.2.2.1 Hydroseeding

First, mix water and fiber. Wood cellulose fiber, paper fiber, or recycled paper must be applied as part of the hydroseeding operation. Fiber must be added at 11.2 kg per 100 square meter. Then add and mix seed and fertilizer to produce a homogeneous slurry. Seed must be mixed to ensure broadcasting at the rate of 15-20 g/sm. When hydraulically sprayed on the ground, material must form a blotter like cover impregnated uniformly with grass seed. Spread with one application with no second application of mulch.

3.2.3 Mulching

3.2.3.1 Mulch

Mulch must be spread uniformly at the rate of 0.75 metric tons per hectare. Mulch must be spread by hand, blower-type mulch spreader, or other approved method. Mulching must be started on the windward side of relatively flat areas or on the upper part of steep slopes, and continued uniformly until the area is covered. Sunlight must not be completely excluded from penetrating to the ground surface.

3.2.3.2 Non-Asphaltic Tackifier

Hydrophilic colloid must be applied at the rate recommended by the manufacturer, using hydraulic equipment suitable for thoroughly mixing with water. A uniform mixture must be applied over the area.

3.2.4 Erosion Control Material

Install in accordance with manufacturer's instructions, where indicated or as directed by the Contracting Officer.

3.2.5 Watering

Start watering areas seeded as required by temperature and wind conditions. Apply water at a rate sufficient to insure thorough wetting of soil to a depth of 50 mm without run off. During the germination process, seed is to be kept actively growing and not allowed to dry out.

3.3 PROTECTION OF TURF AREAS

Immediately after turfing, protect area against traffic and other use.

3.4 RENOVATION OF EXISTING TURF AREA

3.4.1 Overseeding

Apply seed in accordance with and at rates indicated in applicable portions of paragraph SEED APPLICATION METHOD.

3.5 RESTORATION

Restore to original condition existing turf areas which have been damaged during turf installation operations at the Contractor's expense. Keep clean at all times at least one paved pedestrian access route and one paved vehicular access route to each building. Clean other paving when work in adjacent areas is complete.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 08 55

FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT)

08/19

PART 1 GENERAL

- 1.1 SUMMARY/APPLICABILITY
- 1.2 REFERENCES
- 1.3 ADMINISTRATIVE REQUIREMENTS
 - 1.3.1 Plans
 - 1.3.2 Copies of API Publication
 - 1.3.3 Existing System Cleanliness Confirmation Plan
 - 1.3.4 Fuel Testing Laboratory Qualifications
 - 1.3.5 System Start-up Plan
 - 1.3.6 Performance Testing Plan
 - 1.3.7 Fuel Provisioning Plan
 - 1.3.8 Certification (Ready for Start-Up and Performance Testing)
- 1.4 SUBMITTALS
- 1.5 CLOSEOUT SUBMITTALS
 - 1.5.1 Final Reports
- 1.6 QUALITY ASSURANCE
 - 1.6.1 Contractor Start-Up Qualifications
 - 1.6.2 Certification of Entire System
 - 1.6.3 Service Headquarters Definition
- 1.7 DISPOSAL OF WASTE MATERIALS

PART 2 PRODUCTS

- 2.1 GOVERNMENT-FURNISHED MATERIAL AND EQUIPMENT
 - 2.1.1 Fuel
- 2.2 CONTRACTOR-FURNISHED MATERIAL AND EQUIPMENT
 - 2.2.1 Contractor-furnished
 - 2.2.2 Design Conditions

PART 3 EXECUTION

- 3.1 SEQUENCE OF EVENTS
- 3.2 PRELIMINARY REQUIREMENTS
 - 3.2.1 Safety
 - 3.2.2 Electrical Preparations
 - 3.2.3 Piping System
 - 3.2.3.1 Pier Piping Systems
 - 3.2.4 Existing System Cleanliness Confirmation
- 3.3 PREPARATIONS FOR FLUSHING
 - 3.3.1 Protection of System Components
 - 3.3.2 Strainers
 - 3.3.3 Water Draw-off
- 3.4 INITIAL FUEL RECEIPT INTO STORAGE TANK
 - 3.4.1 General
 - 3.4.2 Storage Tanks

- 3.4.3 Components
 - 3.4.4 Fuel Quality
 - 3.4.5 Fuel Receipt
 - 3.4.5.1 Fuel Receipt from Pier
 - 3.4.6 Initial Low Point Flush
 - 3.4.7 Piping Hydrostatic Tests
 - 3.4.8 Piping Soak Tests
 - 3.5 FLUSHING
 - 3.5.1 Flushing Requirements
 - 3.5.2 Fueling System Piping
 - 3.5.2.1 Receipt Pipeline
 - 3.5.2.2 Pier Piping
 - 3.5.3 Piping Flushing Checklist
 - 3.6 PERFORMANCE TESTS
 - 3.6.1 Final Performance Test
 - 3.6.2 Satisfactory Performance
 - 3.6.3 Performance Testing Plan
- End of Section Table of Contents --

SECTION 33 08 55

FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT)
08/19

PART 1 GENERAL

1.1 SUMMARY/APPLICABILITY

This specification defines the requirements and procedures for startup and performance testing of all equipment, components, control systems, devices, and associated appurtenances which are used for the receipt, storage, transfer and issue of petroleum fuel products for non-hydrant fuel facility systems. It covers requirements for safety, Government scheduling and coordination, device testing, existing and new system cleanliness, system flushing, demonstration of indicated and specified system performance, and final acceptance and reporting. The types of fuel systems covered include:

- a. installation pipelines and interterminal pipelines

It does not cover private vehicle fueling stations such as exchange service stations, nor aircraft direct refueling hydrant systems.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API RP 1595	(2012; R 2019; 2nd Ed) Design, Construction, Operation, Maintenance, and Inspection of Aviation Pre-Airfield Storage Terminals
-------------	--

JEFATURA DEL ESTADO (JE)

Ley 22/2011	(2011) Ley 22/2011, de 28 de julio, de residuos y suelos contaminados
-------------	---

JUNTA DE ANDALUCIA (JA)

Decreto 293/2009	(2009) Decreto 293/2009, de 7 de julio, por el que se aprueba el reglamento que regula las normas para la accesibilidad en las infraestructuras, el urbanismo, la edificación y el transporte en Andalucía.
------------------	---

MINISTERIO DE LA PRESIDENCIA (MP)

Directiva 2008/98/CE	(1998) Directiva 2008/98/CE del Parlamento Europeo y del Consejo, de 19 de noviembre de 2008, sobre los residuos y por la que se derogan determinadas Directivas.
----------------------	---

U.S. ARMY CORPS OF ENGINEERS (USACE)

ER 1110-1-8167

(2016) Engineering and Design -- Petroleum,
Oil, and Lubricants Mandatory Center Of
Expertise

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-3004-1

(2018) Quality Assurance for Bulk Fuels,
Lubricants and Related Products

1.3 ADMINISTRATIVE REQUIREMENTS

1.3.1 Plans

The Contractor must submit detailed written plans for implementation of the System Start-Up Plan, the Existing System Cleanliness Confirmation Plan, the Fuel Provisioning Plan, and the Performance Testing Plan. The plans must specify a detailed plan incorporating in a sequential manner all work specified in PART 3 EXECUTION of this specification section. Plan elements must include:

- a. Personnel. List of Contractor's personnel by trade, list of key personnel, list of safety equipment, list of miscellaneous equipment such as two-way radios, and personnel transportation vehicles.
- b. Fluid Used for Hydrotesting, Cleaning Pigging, Verification Pigging, Flushing, and Cleaning. Identify what fluid will be used for each operation in each system section.
- c. Fuel. See the Fuel Provisioning Plan.
- d. Storage-Into. Identify where the contents of the piping will be pushed into including the capacity required, how the storage vessel will be prepared for receiving the piping content, how it will be emptied and how it will be cleaned after the contents are removed.
- e. Storage-From. Identify where the fluid that will be drained or pushed into the piping will come from including the capacity required, the number, size and type of the storage vessel, how the storage vessel is filled, how it will be emptied, and how it will be cleaned after the contents are removed.
- f. Piping. Notwithstanding the requirements described elsewhere, every pipe section and individual component in the system covered by this specification section must be filled, vented, hydrostatically tested, flushed, cleaned, and tested for fuel quality. It must also be able to be isolated and drained if found that it needs repair. The Plans must detail how this is to be done.
- g. Equipment Tests. Detailed procedures and schedules for each system component to perform all system tests under each operating scenario in accordance with paragraph EQUIPMENT TESTS.
- h. Performance Testing. Detailed procedures and schedules for each system component to perform all system tests under each operating scenario in accordance with paragraph PERFORMANCE TESTING.
- i. Pigging Plans. See Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).

- j. Hydrotesting Plans. See the Work Plan in Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- k. Schedule. Schedules must be generated with listing dates and durations of all system start-up activities as well as regular coordination and safety meetings and dates of key events for Government participation.
- l. Contingency plans. Information on spill and fire contingencies, along with the required Government Fire and Safety Office involvement and approvals.
- m. Coordination with Installation. Description of how Contractor must implement system start-up in coordination with ongoing operations at the Installation. Plan must incorporate all phasing and work restriction requirements of the Contract documents.

1.3.2 Copies of API Publication

Provide four copies of API RP 1595 to the Contracting Officer.

1.3.3 Existing System Cleanliness Confirmation Plan

Submit a detailed written plan prepared by the Contractor for the Contractor to verify and confirm the cleanliness of the existing system before connecting it to the new system or extension. The cleanliness confirmation test of the existing system (pre-cleanliness test) must be per MIL-STD-3004-1. Submit the plan 90 (OCONUS) days prior to making the connection to the existing system. The Contractor are responsible for implementing the Existing System Cleanliness Confirmation Plan in coordination with ongoing base operations.

1.3.4 Fuel Testing Laboratory Qualifications

Contractor must submit the qualifications of the DoD Fuel Testing Laboratory which will be used to test the fuel samples.

1.3.5 System Start-up Plan

Submit a detailed written plan prepared by the Contractor for implementation of System Start-Up. Submit the plan 90 (OCONUS) days prior to System Start-Up. The Contractor are responsible for implementing System Start-Up in coordination with ongoing operations.

1.3.6 Performance Testing Plan

Submit a detailed written plan prepared by the Contractor for implementation of Performance Testing. Submit the plan 90 (OCONUS) days prior to System Start-Up. The Contractor are responsible for implementing the Performance Testing Plan in coordination with ongoing operations.

1.3.7 Fuel Provisioning Plan

Submit a detailed written Fuel Provisioning Plan prepared by the Contractor. Submit the plan 90 days prior to System Start-Up. The Fuel Provisioning Plan must include a timeline for the required receipt date(s), grade of fuel, and quantity of fuel on each receipt date required to execute the Start-up and the Performance Testing plans.

1.3.8 Certification (Ready for Start-Up and Performance Testing)

As a prerequisite to fuel system start-up, the Contractor must submit a certificate that certifies all work provided on the fuel system, except for touch-up field painting, has been inspected and approved by the specified approving authorities. Further, the Contractor must certify on this certificate that all specified checks and inspections have been successfully completed prior to start-up. Submit the plan 45 (OCONUS) notice prior to commencement of fuel system start-up. The Contractor must submit the Certificate of Completion to the Contracting Officer at least 7 calendar days prior to commencement of system start-up and performance testing. The Contracting Officer must then be responsible for scheduling the Government representatives and appropriate military command authority for participation in the inspection, performance testing, and final approval activities. Any contractual deficiencies observed must be corrected by the Contractor without cost to the Government.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Copies of API RP 1595; G

Existing System Cleanliness Confirmation Plan; G

System Start-Up Plan; G

Performance Testing Plan; G

Fuel Provisioning Plan; G

SD-06 Test Reports

Final Reports; G

Piping Flushing Checklist; G

SD-07 Certificates

Fuel Testing Laboratory Qualifications; G

Certification (Ready for Start-Up and Performance Testing); G

Contractor Start-Up Qualifications; G

Certificate of Calibration; G

Disposal Of Waste Materials

SD-11 Closeout Submittals

Certification of Entire System

1.5 CLOSEOUT SUBMITTALS

1.5.1 Final Reports

Contractor must prepare a final report that documents the execution of the approved start-up and performance testing plan. All items of work specified in the Start-up and Performance Testing plans must be carried out and reported in this report unless otherwise approved by the Contracting Officer. Include as a part of this report:

- a. Verification letters of approved fuel storage tank hydrostatic tests and the approved piping hydrostatic tests, as generated under other specification sections.
- b. Final settings of the valves and switches.
- c. A copy of the flow and pressure output graphs and excel data and charts on CDR media with an explanation of what the graph indicates and what the system is doing.
- d. A schematic flow diagram drawing of the resultant system showing major valves with the "normal" position of each manual valve noted, equipment, and the type, location, and setpoints of each safety relief valve and thermal relief valve.
- e. A certified pipeline inventory - a detailed list with sizes, lengths, quantity, and volumes must be provided for the systems in this project. Such systems include, but are not limited to: marine receipt, pipeline receipt, truck off-loading receipt, pumphouse, pump pad, truck loading, marine loading, transfer pipeline, product recovery, and other miscellaneous piping systems. Fuel system volume must be calculated using as constructed pipe lengths, internal diameters, fittings, and components. Totals must be provided for all items containing fuel with the exception of tanks which is covered by other specifications.

1.6 QUALITY ASSURANCE

1.6.1 Contractor Start-Up Qualifications

Submit the following data for approval:

- a. Certification stating that the Contractor has provided start-up services (including start-up, existing system cleanliness confirmation, and performance testing experience) for five similar systems in the last five years.
- b. Project names, locations, system description, and items provided at these installations. Include user point-of-contact and current telephone numbers.

1.6.2 Certification of Entire System

Prior to the acceptance of the newly constructed system by the Government, all installed mechanical and electrical system components must be inspected and approved by the Contracting Officer. Provide the Contracting Officer 60 days notice in order to schedule the Installation, DLA and Service Headquarters and/or their designated appointee for participation in the inspection, Performance Testing, and approval. Any

deficiencies observed must be corrected by the Contractor without cost to the Government.

1.6.3 Service Headquarters Definition

SME is defined as Service Headquarters Subject Matter Experts: Air Force - The Air Force Fuels Facilities Subject Matter Expert (AFCEC/COS); Army - Headquarters, U.S. Army Corps of Engineers; POL-MCX Facilities Proponent (CECW-EC) through the Army Petroleum Center (APC); Navy/Marine Corps - NAVFAC POL Facility Subject Matter Expert (NAVFAC EXWC, CI11). ER 1110-1-8167 Engineering and Design Petroleum, Oil, and Lubricants Mandatory Center of Expertise must be followed.

The Service Control Point (SCP) is defined as follows: Army - Army Petroleum Center; Air Force - Air Force Petroleum Office; Navy/Marine Corps - Naval Supply Systems Command - Energy.

1.7 DISPOSAL OF WASTE MATERIALS

The Contractor must be responsible for properly disposing of any sludge, debris, filtration elements, waste water, contact water, and waste fuel resulting from piping flushing and cleaning activities. Comply with all applicable Installation, local, and Host nation Regulations for hazardous waste disposal. Comply with the requirements of Ley 22/2011, Directiva 2008/98/CE, and Decreto 293/2009. Hire and provide the services of a firm accredited by Junta de Andalucia to perform sampling, testing and disposal of contaminated soils classified as waste (vertidos).

PART 2 PRODUCTS

2.1 GOVERNMENT-FURNISHED MATERIAL AND EQUIPMENT

The Government will furnish the following materials, equipment and services during the performance of the work under this section.

2.1.1 Fuel

Some fuel loss should be expected. Fuel that is lost during flushing and cleaning, equipment testing, and performance testing is viewed as a Start-up and Performance Testing expense and should be budgeted for in Contractor estimates. As Basis of Bid, the Contractor must allow for 200 gallons of fuel losses in the bid.

The Contractor is responsible for the disposal of any waste product.

The Contractor must establish a Department of Defense Activity Address Code and must reimburse the US Government for any US Government fuel lost resulting from poor workmanship or contamination.

The Government will provide fuel given that the requirements are submitted with sufficient lead time to allow the US Government to secure the grade and volume required and make the necessary transportation requirements. The Contractor must provide a copy of the Start-up, Fuel Provisioning, and Performance Testing plans a minimum of 90 days in advance of System Start-up. When Defense Working Capital Fund (DLA Owned) fuel is used, provide a copy of the Start-up, Fuel Provisioning, and Performance Testing plans to DLA Energy at the same time.

FLUSHING, CLEANING, EQUIPMENT TESTS and PERFORMANCE TESTING: The flushing

and cleaning and testing phases will be identified separately in the start-up and Performance Testing plans. Fuel used for flushing and cleaning, and testing will become contaminated, the fuel must be isolated and is considered off-specification until the quality can be verified. Fuel volumes required for flushing and cleaning and testing should be minimized to limit costs, so estimates should be made accordingly. The Contractor will be responsible for disposing of any waste product.

Upon satisfactory completion of the flushing and cleaning operations, the Government will supply the additional quantities of fuel required to complete the other work under this section. This larger volume of fuel will not be delivered to the system until the Contractor has satisfactorily completed all work and, in particular, the removal of preservatives and foreign matter from those portions coming in contact with the fuel valves, pumps, filter-separators and other such system components.

2.2 CONTRACTOR-FURNISHED MATERIAL AND EQUIPMENT

2.2.1 Contractor-furnished

Provide material, equipment and labor not specified to be Government-furnished and required for proper System Start-Up of the system. Equipment being provided must be calibrated and the certificate of calibration be submitted to the Contracting Officer prior to use. Equipment must include but not be limited to the following:

- a. Temporary strainers.
- b. Pipe spools.
- c. Flow meters.
- d. Pressure gauges.
- e. Vacuum Trucks for removing water and fuel from pits, drains, and other systems.
- f. Fuel Bowser for removing water and fuel from pits, drains, and other systems.
- g. Barge(s) or Ship(s) to provide the fuel required for system start-up activities
- h. Over-the-road (OTR) tank trucks for flushing, cleaning and draining of the system.
- i. Temporary Tankage for flushing, cleaning and draining of the system.
- j. The Contractor must be responsible for providing the electrical power from a source identified by the Government to the testing locations.
- k. Temporary filtration/strainers.
- l. Pigging equipment and services per Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT) including pig launching and receiving barrels, kicker line, and receiver line.

2.2.2 Design Conditions

Use temporary flushing lines and equipment that are equal in strength, stability, and materials to the associated permanent components. However, spools may be carbon steel.

PART 3 EXECUTION

3.1 SEQUENCE OF EVENTS

The following events take place in the Start-Up and Performance Testing of a system and their definitions:

- a. System Start-Up Phase: The System Start-Up phase encompasses all of the events described below up to the Performance Tests Phase.
- b. Preliminary Requirements: Confirms completion of the system before fuel is received.
- c. Preparations for Flushing: Activities that take place before initial receipt of fuel.
- d. Pneumatic Testing: Test for leaks in a piping system using air under pressure as the test medium. For requirements see Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- e. Pipe Pigging - Pneumatic: Use of pneumatically propelled foam pigs to confirm that lines are clear and free of debris and obstructions. For requirements see Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- f. Initial Fuel Receipt: The point in time at which fuel is introduced into the system with all that implies for safety, spill prevention and control, and inventory control.
- g. Hydrostatic testing - Fuel: Use of fuel to test for leaks in a piping system using liquid under pressure as the test medium. For requirements see Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- h. Soak Testing: Perform "Soak Testing" of the piping system using fuel, as per Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- i. Pipe Pigging - Cleaning with Fuel: Use of fuel propelled pigs to clean the line of gross contamination. Perform this as part of the work in Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- j. Pipe Pigging - Verification with Fuel: Use of fuel propelled pigs to inspect the condition of the pipeline. For requirements see Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).
- k. Flushing: In this context, refers to removing gross contamination (free water and solids) from a fuel system using fuel.
- l. Control Valve Adjustment: Control valve adjustment so that equipment can be safely tested and adjusted.
- m. Performance Tests Phase: Government witnessed demonstration of the proper operation of the individual pieces of equipment and the system or systems as a whole.

3.2 PRELIMINARY REQUIREMENTS

All activities listed in paragraph PART 3 EXECUTION must be performed sequentially in the order they are presented. Prior to any on-site system start-up activities, the Contractor must ensure that all requirements of the paragraph SAFETY are satisfied. Project must be substantially complete, except for touch-up field painting, and Contractor's work area must be free of debris, trash and obstacles. Perform the following activities prior to receipt of fuel:

3.2.1 Safety

Prior to any on-site system start-up activities, the following safety procedures must be accomplished in all fueling areas to be started up and performance tested under this specification section: testing/operation of emergency showers and eyewash stations, placement of Contractor-provided portable eyewash units within 31 meters or 10 seconds from the fueling point, verification of proper grounding throughout system, coordination with Government Fire and Safety Office and Fuels personnel, placement of Contractor-provided spill pads and containment booms, placement of Contractor-provided fire extinguishers capable of extinguishing a fuel fire. Ensure that all radios/devices at all Class I, Division 1 areas are intrinsically safe.

3.2.2 Electrical Preparations

Prior to energizing the electrical system components, verify that short-circuit links have been removed from current transformer and that secondary circuits have been connected. Confirm that all tests required for fire detection and suppression systems have been performed and accepted. Verify all electrical transmitter connections and ensure proper calibration. Verify all electrical system components meet the classification and division as required by the design. Verify correct rotation of all motors prior to testing. Verify paddle type flow switches by physically actuating vanes and checking outputs. Conduit explosion-proof sealoffs must be poured after initial electrical checks but before fuel receipt.

3.2.3 Piping System

Ensure that all piping weld integrity, leak detection systems, and coating inspections have been performed per the applicable specifications. Provide statement that required testing has been completed in the Final Reports. Evacuate all accumulated water from piping low point drains, valve cavities, and equipment drains. Verify all bolted connections are tightness tested to required torque using a calibrated torque wrench. Verify that all pressure gauges are properly calibrated, located and installed. Ensure that piping's cathodic protection system is tested, calibrated and operational. Ensure that pipe marking and identification are provided as specified. Ensure that piping system thermal relief provisions are installed as designed. Verify the correct installation of piping expansion loops and supports.

3.2.3.1 Pier Piping Systems

For pier delivery/receipt systems or other over-water piping installations, ensure compliance with the Contractor's previously approved spill control plans.

3.2.4 Existing System Cleanliness Confirmation

Contractor must conduct tests to confirm the cleanliness of the existing system before connecting it to the new system or an extension, upgrade or repair of the existing system. Confirm that the fuel in the existing system meets the intra-governmental receipt limits defined in MIL-STD-3004-1. Sampling must be performed by the Government and testing must be done by the Service or a DoD regional fuels testing laboratory.

3.3 PREPARATIONS FOR FLUSHING

Upon completion of the system to the satisfaction of the Contracting Officer and the Service Headquarters and/or their designated appointee, make the following preparations for flushing the system.

3.3.1 Protection of System Components

The following items must be removed from the system prior to start of flushing operations and, where applicable, replaced with spools of pipe, diameter equal to the item removed.

- a. Control valves. This also includes control valve tubing exterior to the valve such as that which runs from tank level control valves up to tank float pilots.
- b. Sensors which are exposed to the fluid such as pressure gauges and thermometers.

After flushing, the above items must be reinstalled in the system and the spool sections turned over to the Contracting Officer.

3.3.2 Strainers

Ensure strainers are clean before flushing. Any damaged permanent or temporary strainers must be replaced by the Contractor at no additional cost to the Government.

3.3.3 Water Draw-off

Remove any accumulated water from piping, and storage tanks' sumps and bottoms. Drain water and return fuel via filtration to storage tank. Repeat process until all water is removed. Dispose of petroleum contaminated water in accordance with Installation, local, and Host Nation regulations.

3.4 INITIAL FUEL RECEIPT INTO STORAGE TANK

3.4.1 General

Utilize temporary tankage for initial fuel receipt to isolate contaminated fuel. Initial receipt of fuel must be done by gravity if possible. The Contractor must station personnel throughout piping system at high point vents to bleed air. All flanges and system components will be periodically inspected for leaks during filling procedures.

3.4.2 Storage Tanks

Receipt flow rate into an empty storage tank must not exceed 1 m/s (FPS), as measured in the main receipt piping, until outlet of tank fill tube is

submerged.

3.4.3 Components

Differential pressure across strainers must be continuously monitored. Any time a strainer differential pressure reaches 138 kPa, it must be cleaned.

3.4.4 Fuel Quality

Fuel used during initial receipt must be considered contaminated and must be positively isolated, with blind flanges or closed, padlocked manual valves, from any active transfer, or aircraft fueling or tank truck loading operations. Fuel isolation must continue until all flushing and cleaning is completed. A method of checking fuel quality that is acceptable to the Contracting Officer must be provided at the beginning and end of any new piping section.

3.4.5 Fuel Receipt

3.4.5.1 Fuel Receipt from Pier

Start-up personnel must meet with Government personnel in charge of existing fuel storage to discuss fuel transfer procedures. Topics must include: methods of communication to start/stop pumps; flow rate and head characteristics of transfer pumps; methods of restricting initial receipt flow rate; methods of straining and filtering initial receipt fuel; accommodating multiple pump starts resulting from required strainer and filter cleaning operations; required quantity of fuel to be transferred. Contractor must provide a written summary of pipeline receipt procedures to the Contracting Officer.

3.4.6 Initial Low Point Flush

Perform an initial low point flush operation by flushing each low point drain through a portable basket strainer for 10 seconds at a system pressure of 207 kPa. Repeat flush until basket strainer collects no additional debris.

3.4.7 Piping Hydrostatic Tests

Hydrostatically test the piping system with fuel as required by Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT).

3.4.8 Piping Soak Tests

Upon completion of hydrostatic testing with fuel, perform "Soak Testing" of the piping systems per API RP 1595. Duration of the test must be a minimum of 4 days and maximum of 7 days.

3.5 FLUSHING

Flushing procedures remove gross particulate and water contaminants and must precede cleaning procedures. All piping including but not limited to receipt piping, issue piping, supply and return lines to the storage tanks, thermal relief piping, high point vents, and low point drains must be flushed with fuel. Fuel used for flushing shall be same type and grade as intended service of piping system.

In the event flushing identifies the presence of clay or soil like material in the system, Contractor must immediately notify Contracting Officer for direction.

3.5.1 Flushing Requirements

Begin flushing of fuel system piping at low flow rates. Slowly increase flushing flow rate to specified velocity. Flush for a minimum of 30 minutes. Sampling must be performed by the Government and testing must be done by the Service or a DoD regional fuels testing laboratory. Fuel must be free of gross contamination and visible free water to the satisfaction of the Contracting Officer.

In the event flushing identifies the presence of clay or soil like material in the system, Contractor must immediately notify Contracting Officer for direction.

3.5.2 Fueling System Piping

During flushing procedure periodically bleed air through high point vent and drain water through low point drains.

3.5.2.1 Receipt Pipeline

Flushing of the receipt pipeline must occur during the initial receipt/filling operations. Samples of the incoming fuel must be taken at the point of connection with the upstream and downstream systems at the start and the end of the flushing and periodically between the two. These samples must be tested by the designated government agency and turned over to the Contracting Officer.

3.5.2.2 Pier Piping

Pier piping should be hydrostatically tested per the requirements of Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT). The Contractor will be allowed to use Government-furnished hoses. Perform flushing with product at 3.5 m/s for 30 minutes.

3.5.3 Piping Flushing Checklist

The Contractor must generate a comprehensive matrix of all new piping sections in the system. Matrix must serve as an Owner's piping inventory and a checklist for all Contractor-provided flushing operations. Column entries must include pipe section name, location, diameter, approximate length, flushing fuel velocity and volume achieved and acceptable results of sampling.

3.6 PERFORMANCE TESTS

Performance testing must occur after the Contractor has performed the Equipment Tests. Performance testing must demonstrate to the satisfaction of the Contracting Officer and Service Headquarters and/or their designated appointee these portions of the fueling system are working as specified. Performance testing must consist of repeating the Equipment Tests (indicated in previous paragraphs) and operating the fueling system during actual operations in the presence of Government Witnesses. The maximum rated capacity of the system must be demonstrated. The Contractor must notify the Contracting Officer 45 calendar days in advance of the test to permit arrangement for the use of Government-furnished items.

During the time period of performance testing, no construction activities will be allowed on the project site. The project site must be considered an operational (fuel) zone (verses a construction zone) during this performance testing period. Personnel, dressed for fuel's operation, will be present to witness testing and participate in Contractor provided training.

3.6.1 Final Performance Test

A final performance test must consist of demonstrating the operation of the system for the purpose for which it is intended:

- a. loading marine vessels
- b. off-loading marine vessels

3.6.2 Satisfactory Performance

In the event a portion of the system or any piece of equipment fails to meet the test, make the necessary repairs or adjustments and repeat the Performance Test until satisfactory performance is obtained. The determination of satisfactory performance must be made by the Contracting Officer and the Service Headquarters and/or their designated appointee.

3.6.3 Performance Testing Plan

Contractor must edit the example Performance Testing plan and submit for approval. An example Performance Testing plan can be found at the end of this Section as Attachment 2 - Performance Testing Plan. Submit plan a minimum of 60 days prior to performance testing.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 52 23.15

POL SERVICE PIPING WELDING

11/18

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 DEFINITIONS
- 1.3 SUBMITTALS
- 1.4 GENERAL REQUIREMENTS
- 1.5 PERFORMANCE
- 1.6 QUALIFICATIONS
 - 1.6.1 Welding Operations
 - 1.6.2 Welding Procedure Specification and Procedure Qualification Records
 - 1.6.3 Welder and Welding Operator Performance
 - 1.6.3.1 Certification
 - 1.6.3.2 Identification
 - 1.6.3.3 Renewal of Qualification
 - 1.6.4 Test Reports
 - 1.6.5 Inspection and NDE Personnel
 - 1.6.5.1 Inspector Certification
 - 1.6.5.2 NDE Personnel
- 1.7 DELIVERY, STORAGE, AND HANDLING
 - 1.7.1 Material Control
 - 1.7.1.1 Damaged Containers
 - 1.7.1.2 Partial Issues
 - 1.7.2 Damaged Materials
- 1.8 SYMBOLS
- 1.9 SAFETY

PART 2 PRODUCTS

- 2.1 WELDING MATERIALS

PART 3 EXECUTION

- 3.1 WELDING OPERATIONS
 - 3.1.1 Base Metal Preparation
 - 3.1.2 Weld Joint Fit-Up
 - 3.1.3 Butt Weld Joint Spacing
 - 3.1.4 Preheat and Interpass Temperatures
 - 3.1.5 Production Welding Instructions
 - 3.1.6 Postweld Heat Treatment
- 3.2 EXAMINATIONS, INSPECTIONS AND TESTS
 - 3.2.1 Visual Inspection
 - 3.2.2 NDE Testing Frequency
 - 3.2.3 NDE Testing
 - 3.2.4 Special Requirements for PAUT Testing
 - 3.2.4.1 Essential Variables

- 3.2.4.2 Non-Essential Variables
 - 3.2.5 Inspection and Tests by the Government
 - 3.3 ACCEPTANCE STANDARDS
 - 3.3.1 Visual
 - 3.3.2 Magnetic Particle Examination
 - 3.3.3 Liquid Penetrant Examination
 - 3.4 CORRECTIONS AND REPAIRS
 - 3.4.1 Defect Removal
 - 3.4.1.1 Methods of Defect Removal
 - 3.4.1.2 Rewelding
 - 3.4.1.3 Peening or Caulking
 - 3.5 MAINTAINING CLEANLINESS OF PIPING
 - 3.6 COMMISSIONING
- End of Section Table of Contents --

SECTION 33 52 23.15

POL SERVICE PIPING WELDING
11/18

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API Std 650 (2013; Errata 1 2013; Addendum 1 2014; Errata 2 2014; Addendum 2 2016; Addendum 3 2018) Welded Tanks for Oil Storage

AMERICAN SOCIETY FOR NONDESTRUCTIVE TESTING (ASNT)

ASNT SNT-TC-1A (2020) Recommended Practice for Personnel Qualification and Certification in Nondestructive Testing

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B31.3 (2020) Process Piping

ASME BPVC SEC V (2017) BPVC Section V-Nondestructive Examination

AMERICAN WELDING SOCIETY (AWS)

AWS A2.4 (2012) Standard Symbols for Welding, Brazing and Nondestructive Examination

AWS A3.0M/A3.0 (2020) Standard Welding Terms and Definitions

AWS A5.1/A5.1M (2012) Specification for Carbon Steel Electrodes for Shielded Metal Arc Welding

AWS A5.3/A5.3M (1999; R 2007) Specification for Aluminum and Aluminum-Alloy Electrodes for Shielded Metal Arc Welding

AWS A5.4/A5.4M (2012) Specification for Stainless Steel Electrodes for Shielded Metal Arc Welding

AWS A5.9/A5.9M (2017) Welding Consumables-Wire Electrodes, Strip Electrodes, Wires, and Rods for Arc Welding of Stainless and Heat Resisting Steels- Classification

AWS A5.10/A5.10M (2021) Welding Consumables - Wire Electrodes, Wires and Rods for Welding of

Aluminum and Aluminum-Alloys -
Classification

AWS A5.18/A5.18M	(2021) Specification for Carbon Steel Electrodes and Rods for Gas Shielded Arc Welding
AWS A5.22/A5.22M	(2012) Specification for Stainless Steel Flux Cored and Metal Cored Welding Electrodes and Rods
AWS A5.32/A5.32M	(2021) Welding Consumables-Gases and Gas Mixtures for Fusion Welding and Allied Processes
AWS C5.5/C5.5M	(2003) Recommended Practices for Gas Tungsten Arc Welding
AWS D1.1/D1.1M	(2020; Errata 1 2021) Structural Welding Code - Steel
AWS D10.4	(1986; R 2000) Recommended Practices for Welding Austenitic Chromium-Nickel Stainless Steel Piping and Tubing
AWS D10.7/D10.7M	(2008) Guide for the Gas Shielded Arc Welding of Aluminum and Aluminum Alloy Pipe
AWS D10.10/D10.10M	(2021) Recommended Practices for Local Heating of Welds in Piping and Tubing
AWS D10.11M/D10.11	(2007) Guide for Root Pass Welding of Pipe Without Backing
AWS D10.12M/D10.12	(2000) Guideline for Welding Mild Steel Pipe
AWS QC1	(2016) Specification for AWS Certification of Welding Inspectors
AWS WHB-4.9	(2010) Welding Handbook, Volume 4 - Materials and Applications Part 1
AWS Z49.1	(2021) Safety in Welding and Cutting and Allied Processes

ASTM INTERNATIONAL (ASTM)

ASTM E2700	(2014) Standard Practice for Contact Ultrasonic Testing of Welds Using Phased Arrays
------------	--

1.2 DEFINITIONS

Definitions must be in accordance with AWS A3.0M/A3.0 except as follows:

- a. Weld slag is defined as the crystalline residue remaining on the weld surface following a weld procedure which uses flux as a shielding method.

- b. POL service piping consists of piping and components used for petroleum, oil and lubricants (POL) under pressure or gravity force including modifications to existing hydrant fueling systems.

1.3 SUBMITTALS

Government approval is required for submittals with a "G" classification. Submittals not having a "G" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Welding Procedure Specifications (WPS); G

Welder Performance Qualification (WPQ); G

Procedure Qualification Record (PQR); G

Welding Operations; G

SD-02 Shop Drawings

POL Service Piping; G

SD-06 Test Reports

Welding Reports

Examinations, Inspections and Tests

SD-07 Certificates

Qualifications; G

1.4 GENERAL REQUIREMENTS

This section covers the welding of Petroleum, Oil and Lubricant (POL) Service systems. Deviations from applicable codes, approved procedures, and approved detail drawings will not be permitted without prior written approval by the Contracting Officer. Materials or components with welds made offsite will not be accepted if the welding does not conform to the requirements of this specification, unless otherwise specified. Procedures must be developed by the Contractor for welding all metals included in the work. Welding must not be started until welding procedures, welders, and welding operators have been qualified. Qualification testing must be performed by an approved testing laboratory, or by the Contractor if approved by the Contracting Officer. Costs of such testing must be borne by the Contractor. The Contracting Officer must be notified at least one week in advance of the time and place of the tests. If the Contracting Officer elects to witness the tests, the qualification tests must be performed at or near the worksite. The Contractor must maintain current records of the test results obtained in the welding procedure, welding operator, welder performance qualifications, and nondestructive examination (NDE) procedures readily available at the site for examination by the Contracting Officer. The procedures for making transition welds between different materials or between plates or pipes of different wall thicknesses must be qualified.

Unless otherwise specified, the choice of welding process must be the responsibility of the Contractor. It should be noted that ASME B31.3 incorporates by reference, other requirements of ASME Section V, Section IX and specific AWS requirements.

1.5 PERFORMANCE

The Contractor will be responsible for the quality of all joint preparation, welding, and examination. All materials used in the welding operations must be clearly identified and recorded. The inspection and testing defined in this specification are minimum requirements. Additional inspection and testing must be the responsibility of the Contractor when he deems it necessary to achieve the quality required.

1.6 QUALIFICATIONS

Welding procedures, welders, and welding operators previously qualified by test may be accepted for the work without requalification, provided that all of the following conditions are fulfilled:

- a. Copies of the welding procedure specifications (WPS), the procedure qualification record (PQR) record, and the welder performance qualification (WPQ) are submitted and approved in accordance with paragraph SUBMITTALS.
- b. Testing was performed by an approved testing laboratory or approved technical consultant or by the Contractor's approved quality assurance organization.
- c. The welding procedures, welders, and welding operators were qualified in accordance with ASME B31.3 and base materials, filler materials, electrodes, equipment, and processes conformed to the applicable requirements of this specification.
- d. The requirements of paragraph RENEWAL OF QUALIFICATION below are met and records showing name of employer and period of employment using the process for which qualified are submitted as evidence of conformance.

1.6.1 Welding Operations

The Contractor must provide a description of how the critical welding operations will be accomplished. Provide the welding procedures to be used for each operation, the sequence of welding to minimize heat distortion, sequence of welding piping sections both in the trench and outside, machine welding if used, and multiple welders on same pipe weld. Submit detailed procedures which define methods of compliance to contract drawings and specifications.

1.6.2 Welding Procedure Specification and Procedure Qualification Records

The Contractor must record in detail and must qualify the Welding Procedure Specifications for every proposed welding procedure. Qualification for each welding procedure must conform to the requirements of ASME B31.3 and to this specification. The welding procedures must specify back purge gas requirements, end preparation for butt welds including cleaning, alignment, and root openings. Preheat, interpass temperature control, and postheat treatment of welds must be as required by approved welding procedures, unless otherwise indicated or specified.

Copies of the welding procedure specifications and weld procedure qualification record results for each type of welding required must be submitted in accordance with paragraph SUBMITTALS. Approval of any procedure does not relieve the Contractor of the sole responsibility for producing acceptable welds. Welding procedures must be identified individually and must be referenced on the POL service piping shop drawings. Submit detail drawings showing location, length, and type of welds; and indicating preweld and postweld heat treatment and NDE as required. The drawings must show the welding procedure specification (WPS) to be used at each weld location.

1.6.3 Welder and Welding Operator Performance

Each welder and welding operator assigned to work must be qualified in accordance with ASME B31.3.

1.6.3.1 Certification

Before assigning welders or welding operators to the work, the Contractor must provide the Contracting Officer with their names together with certification that each individual is performance-qualified as specified. The certification must state the type of welding and positions for which each is qualified, the code and welding procedure specification under which each is qualified, date qualified, and the firm and individual certifying the qualification tests. The Contractor must provide a summary table showing all welders and the WPS with which they are qualified to weld.

1.6.3.2 Identification

Each particular weld must be identified with the personal number, letter, or symbol assigned to each welder or welding operator. To identify welds, written records indicating the location of welds made by each welder or welding operator must be submitted, and each welder or welding operator must apply the personal mark adjacent to the welds using a rubber stamp or felt-tipped marker with permanent, weatherproof ink or other methods approved by the Contracting Officer that do not deform the metal. Identification by die stamps or electric etchers will not be allowed.

1.6.3.3 Renewal of Qualification

Requalification of a welder or welding operator must be required under any of the following conditions:

- a. When a welder or welding operator has not used the specific welding procedure for a period of 3 months; the period may be extended to 6 months if the welder or welding operator has been employed on another welding procedure.
- b. When a welder or welding operator has not welded with any procedure during a period of 3 months, all the personal qualifications must be considered expired, including any extension by virtue of a. above.
- c. There is specific reason to question the person's ability to make welds that will meet the requirements of the specifications.
- d. The welder or welding operator was qualified by an employer, other than those firms performing work under this contract, and a qualification test has not been taken within the preceding 12 months.

- e. Renewal of qualification for a specific welding procedure under conditions a., b., and d., above, needs to be made on only a single test joint or pipe of a thickness, position, or material required by the welding procedure specifications to reestablish the welder's or welding operator's qualification for the previous qualification.

1.6.4 Test Reports

Test reports must consist of the following.

- a. Records made by the AWS certified inspector for all duties performed per paragraph 4.2 of AWS QC1.
- b. All NDE (radiograph, PAUT) reports with unique weld ID for each weld tested.
- c. "Weld Maps." These maps/drawings correlate the shop drawings submitted to the NDE reports. The NDE report that shows a weld number as acceptable is correlated with weld number on the drawings.
- d. Provide the location of each weld, what procedure was used, which welder did the weld, the results of the visual test, and the results of the NDE.

1.6.5 Inspection and NDE Personnel

Contractor must provide a commercially independent organization for all weld examinations. All inspection and NDE personnel must be qualified in accordance with the following requirements. The contractor must submit the qualifications of all the testing personnel that will perform all field tests for review by the Contracting Officer. The qualifications of all personnel on the job site that will perform welding inspections and NDE must be submitted for approval. All inspectors and NDE personnel must have a minimum of one year experience inspecting the piping material being used and five years in military or commercial aircraft hydrant fueling systems or truck fueling systems, petroleum refineries, power generating plants, or chemical process plants. In addition, all PAUT personnel must meet ASNT Level III requirements.

1.6.5.1 Inspector Certification

Visual welding inspectors must be qualified in accordance with AWS QC1.

1.6.5.2 NDE Personnel

NDE personnel must be certified Level II in accordance with ASME Section V, ASME B31.3 and ASNT SNT-TC-1A for each NDE procedure he is required to use, and a written procedure for the control and administration of NDE personnel training, examination, and certification must be established. The procedures must be based on appropriate specific and general guidelines of training and experience recommended by ASNT SNT-TC-1A. Should the NDE examiner also be a welder, that individual is disqualified from examining their own work. Examiners performing PAUT must be qualified in accordance with ASME Section 5, Article 4, Mandatory Appendix VII.

1.7 DELIVERY, STORAGE, AND HANDLING

All filler metals, electrodes, and other welding materials must be delivered to the site in manufacturers' original packages and stored in a dry space until used. Packages must be properly labeled and designed to give maximum protection from moisture and to insure safe handling.

1.7.1 Material Control

Materials must be stored in a controlled access and clean, dry area that is weathertight and is maintained at a temperature recommended by the manufacturer. The materials must not be in contact with the floor and must be stored on wooden pallets or cribbing.

1.7.1.1 Damaged Containers

Low-hydrogen steel electrodes must be stored in their sealed shipping container. If the seal is damaged during shipment or storage, and the damage is not immediately detected, the covered electrodes in that container must be rebaked in accordance with the manufacturer's instructions prior to issuance or must be discarded. If a container is damaged in storage and the damage is witnessed, the electrodes from that container must be immediately placed in a storage oven. The storage oven temperature must be as recommended by the manufacturer or the welding material specification.

1.7.1.2 Partial Issues

When a container of covered electrodes is opened and only a portion of the content is issued, the remaining portion must, within the limits established by AWS D1.1/D1.1M be placed in a storage oven.

1.7.2 Damaged Materials

Materials which are damaged must be discarded. Covered electrodes which are oil or water-soaked, dirty, or on which the flux has separated from the wire must be discarded.

1.8 SYMBOLS

Symbols must be in accordance with AWS A2.4.

1.9 SAFETY

Safety precautions must conform to AWS Z49.1.

PART 2 PRODUCTS

2.1 WELDING MATERIALS

Welding materials for carbon steel, stainless steel and aluminum must comply with AWS WHB-4.9. Welding equipment, electrodes, welding wire, shielding and backing gas, and fluxes must be capable of producing satisfactory welds when used by a qualified welder or welding operator using qualified welding procedures. All field girth root pass welds must be made with non-covered electrodes or welding wire. All root passes must be made with shielding and backing gas. External welds on the pipe such as attaching pipe supports may be made with covered electrodes or welding wire. Electrodes, welding wire and fluxes must be in accordance with

Table 1. Welding materials for aluminum and aluminum alloy must comply with AWS D10.7/D10.7M.

TABLE I				
AWS	Process	Alloy	Consumable	Use
			(Examples) (1)	
AWS A5.1/A5.1M	SMAW	Low Carbon	E7018	Fill
AWS A5.4/A5.4M	SMAW	Stainless	E308L, E309L	Fill
AWS A5.3/A5.3M	SMAW	Aluminum		Fill
AWS A5.9/A5.9M	GTAW	Stainless	ER308L, ER309L	Root and Fill
AWS A5.10/A5.10M	GTAW	Aluminum		Root and Fill
AWS A5.18/A5.18M	GTAW	Low Carbon	E70S-3, E70S-6	Root and Fill
AWS A5.22/A5.22M	GTAW	Stainless	E308LT1-1	Root (Backing and Shielding Gas Required)
AWS A5.32/A5.32M	GTAW	All		Shielding Gas
Note(1): The consumable material designations shown are examples only and are not intended to limit the Contractor's selection of consumable materials.				

PART 3 EXECUTION

3.1 WELDING OPERATIONS

Welding must be performed in accordance with qualified procedures using qualified welders and welding operators. Welding must not be done when the quality of the completed weld could be impaired by the prevailing working or weather conditions. The Contracting Officer must determine when weather or working conditions are unsuitable for welding. Welding of hangers, supports, and plates to structural members must conform to Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS.

Welding must be performed in accordance with ASME B31.3 and the applicable portions of AWS D10.4, AWS D10.7/D10.7M, AWS D10.10/D10.10M, AWS D10.11M/D10.11, AWS D10.12M/D10.12, and AWS C5.5/C5.5M.

All joints unless indicated otherwise, in carbon steel, aluminum and stainless steel piping systems must be welded. Unless otherwise approved, all girth welds must be complete penetration groove welds made in accordance with qualified welding procedures. The root pass on stainless steel, aluminum, and carbon steel pipe must be by the GTAW process.

- a. Weld Preparation must comply with the requirements of ASME B31.3 and the qualified Welding Procedure Specification. The use of "rice paper" as purge blocks is not permitted. Contractor must submit alternate method for approval.

- b. Backing Rings. The use of backing rings for making or repairing welds will not be permitted.

3.1.1 Base Metal Preparation

Oxy-fuel cutting must not be used on austenitic stainless steel or nonferrous materials.

Mechanical grinding of thermal cut ends must be used to remove the heat affected area but should be limited to maximum 1/8 inch.

3.1.2 Weld Joint Fit-Up

Parts that are to be joined by welding must be fitted, aligned, and retained in position during the welding operation by the use of bars, jacks, clamps, or other mechanical fixtures. End welds must be properly aligned prior to welding in accordance with Chapter V of ASME B31.3. All socket-welded joints must be properly fitted with gaps between the pipe and the bottom of the socket in accordance with ASME B31.3. Contractor must implement a program to ensure gaps are properly measured and documented. Welded temporary attachments must not be used except when it is impractical to use mechanical fixtures. When temporary attachments are used, they must be the same material as the base metal, and must be completely removed by grinding or thermal cutting after the welding operation is completed. If thermal cutting is used, the attachment must be cut to not less than 6 mm from the member and the balance removed by grinding. After the temporary attachment has been removed, the area must be visually examined.

3.1.3 Butt Weld Joint Spacing

Butt weld joints must be spaced a minimum of 150 mm apart. Measurement must be taken from the toe of the first weld to the toe of the second weld. The measurement must be taken at the closest point between the welds when visually inspected. If spacing is not possible due to existing conditions, a reduction in spacing to not less than 50 mm may be made.

3.1.4 Preheat and Interpass Temperatures

Preheat temperatures must meet the requirements specified by ASME B31.3. However, in no case will the preheat be below 10 degrees C for ferritic steel or austenitic stainless steel, or 0 degrees C for nonferrous alloys. The maximum interpass temperatures must not exceed 149 degrees C for austenitic stainless steels, nickel alloys, and copper alloys; and 260 degrees C for carbon steels. Preheat techniques must be such as to ensure that the full thickness of the weld joint preparation and adjacent base material, at least 75 mm in all directions, is at the specified temperature. Preheating by induction or resistance methods is preferred. When flame heating is used, only a neutral flame must be employed. Oxy-fuel heating must not be used on austenitic stainless steel; however, air-fuel heating is acceptable if controlled to ensure that the surface temperature does not exceed 66 degrees C. Interpass temperatures must be checked on the surface of the component within 25 mm of the weld groove and at the starting location of the next weld pass, and for a distance of about 150 mm ahead of the weld, but not on the area to be welded.

3.1.5 Production Welding Instructions

- a. Welding must not be done when the ambient temperature is lower than minus 18 degrees C.
- b. Welding is not permitted on surfaces that are wet or covered with ice, when snow or rain is falling on the surfaces to be welded, or during periods of high winds, unless the welders and the work are properly protected.
- c. Gases for purging and shielding must be welding grade and must have a dew point of minus 40 degrees C or lower.
- d. Back purges are required for austenitic stainless steels and nonferrous alloys welded from one side and must be set up such that the flow of gas from the inlet to the outlet orifice passes across the area to be welded. The oxygen content of the gas exiting from the purge vent must be less than 2 percent prior to welding. The flow rate must be that required by the approved weld procedure specification.
- e. The purge on groove welds must be maintained for at least two passes or 5 mm whichever is greater.
- f. Removable purge dam materials must be made of expandable or flexible plugs, such as Plexiglas, plywood (which must be dry when used). Wood dams must be kiln-dried quality. Purge dams must not be made of polyvinyl alcohol.
- g. Any welding process which requires the use of external gas shielding must not be done in a draft or wind unless the weld area is protected by a shelter. This shelter must be of material and shape appropriate to reduce wind velocity in the vicinity of the weld to a maximum of 8 km/hour.
- h. Tack welds to be incorporated in the final welds must have their ends tapered by grinding or welding technique. Tack welds that are cracked or defective must be removed and the groove must be retacked prior to welding. Temporary tack welds must be removed, the surface ground smooth, and visually inspected. For low-alloy and hardenable high-alloy steels, the area must be magnetic particle examination inspected.
- i. Grinding of completed welds is to be performed only to the extent required for NDE, including any inservice examination, and to provide weld reinforcement within the requirements of ASME B31.3. If the surface of the weld requires grinding, reducing the weld or base material below the minimum required thickness must be avoided. Minimum weld external reinforcement must be flush between external surfaces.
- j. Each qualified welder must be assigned an identification symbol. All welds must be permanently marked with the symbol of the individual who made the weld.
- k. Direct welded connection of carbon steel and stainless steel must not be made. See Section 33 52 40 FUEL SYSTEMS PIPING (NON-HYDRANT) for method of connection.

3.1.6 Postweld Heat Treatment

- a. When required postweld heat treatment must be performed in accordance with ASME B31.3. Temperatures for local postweld heat treatment must be measured continuously by thermocouples in contact with the weldment.
- b. Postweld heat treatment of low-alloy steels, when required, must be performed immediately upon completion of welding and prior to the temperature of the weld falling below the preheat temperature. However, postweld heat treatment may be postponed after the completion of the weld, if, immediately after the weld is completed, it is maintained at a minimum temperature of 149 degrees C or the preheat temperature, whichever is greater, for 2-hours per 25 mm of weld thickness.

3.2 EXAMINATIONS, INSPECTIONS AND TESTS

Weld inspection and NDE must be performed by the Contractor to detect surface and internal discontinuities in completed welds. The services of a qualified commercial inspection or testing laboratory or technical consultant meeting the requirements of paragraph INSPECTION AND NDE PERSONNEL, approved by the Contracting Officer, must be employed by the Contractor. All completed welds must be visually inspected in accordance with the visual inspection requirements of ASME B31.3 and AWS D1.1/D1.1M. Radiographic Liquid penetrant Magnetic particle or PAUT examination must be required as indicated below. When in-process weld quality control is required for tie-in welds, it must be performed in accordance with ASME B31.3. When inspection and testing indicates disqualifying defects in a weld joint, the weld must be repaired by a qualified welder in accordance with paragraph CORRECTIONS AND REPAIRS. The Contractor must submit weld inspection and NDE field testing reports to the Contracting Officer. In all cases where PAUT is used, all welds must be recorded and the results reviewed by an ASNT Level III certified inspector and the requirements in this section.

The person performing the weld inspection must perform the following:

- a. Verify that the base materials and consumable welding materials conform to the specifications and that welding filler metals used are as specified for each base material.
- b. Verify that the welding equipment to be used for the work is appropriate for use with the welding procedure specification and has the capability to meet the applicable requirements of the welding procedure.
- c. Verify that only approved or qualified welding procedures are used for the work.
- d. Verify that the edge preparation or joint geometry meet the requirements of the welding procedure and drawings.
- e. Verify that the specified filler metals are used and that filler metals are maintained in proper condition, per requirements, or as recommended by the manufacturer.
- f. Verify that the technique and performance of each welder, welding operator, and tack welder are as specified.

- g. Verify that the work conforms to requirements of the applicable standards, drawings, or other documents.
- h. Verify that the work inspected is identified and documented in accordance with specified requirements.
- i. Prepare clear and concise reports and verify that records of the results of examinations are maintained.
- j. Verify the approved WPS pre-heat and post heat procedures are being used.

Welders found making defective welds must be removed from the work or must be required to be requalified in accordance with ASME B31.3.

3.2.1 Visual Inspection

Weld joints must be inspected visually as follows:

- a. Before welding - for compliance with requirements for joint preparation, alignment and fit-up, and cleanliness in accordance with ASME B31.3.
- b. During welding - for cracks and conformance to the approved welding procedure only when in-process weld quality control is required by ASME B31.3.
- c. After welding - for cracks, contour and finish, bead reinforcement, undercutting, overlap, weld slag on the interior of the pipe and size of welds in accordance with ASME B31.3 and AWS D1.1/D1.1M. Visual examination of the interior of the pipe may be performed by any of the remote means allowed by ASME BPVC SEC V, visual inspection. Visual examination of the weld must be performed prior to any other NDE examinations as required by this specification.

3.2.2 NDE Testing Frequency

All pipe field welds, including high point vent pipe tees, insert butt welded weld-o-lets, and low point drain pipe, must be examined by radiographic methods to determine conformance to the paragraph ACCEPTANCE STANDARDS. All socket welds and sock-o-lets or weld-o-lets to pipe welds must be examined with either magnetic particle or liquid penetrant methods, in addition to the visual examinations. The services of a qualified commercial or testing laboratory approved by the Contracting Officer must be employed by the Contractor for testing of piping welds. Costs of testing, including retesting of repaired welds, must be borne by the Contractor.

- a. Provide 100 percent radiographic testing or PAUT for all underground piping and hydrant pump discharge piping.
- b. Provide select radiographic testing or PAUT in accordance with ASME B31.3 for all aboveground piping. The inspection must include an examination of welds made by each welding operator or welder. If the testing reveals that any welds fail to meet minimum quality requirements, an additional percent of the welds in that same group must be inspected in accordance with ASME B31.3. If all of the additional welds inspected meet the quality requirements, the entire group of welds represented must be accepted and the defective welds

must be repaired. If any of the additional welds inspected also fail to meet the quality requirements, that entire group of welds must be rejected. The rejected welds must be removed and rewelded, or the rejected welds must be 100 percent inspected and all defective weld areas removed and rewelded.

3.2.3 NDE Testing

NDE must be as required by ASME B31.3 and in accordance with written procedures. Procedures for radiographic liquid penetrant magnetic particle ultrasonic or PAUT tests and methods must conform to ASME BPVC SEC V. Only Radiography or PAUT are acceptable test methods for butt welded joints. Refer to ASTM E2700 for PAUT. The approved procedure must be demonstrated to the satisfaction of the Contracting Officer. In addition to the information required in ASME BPVC SEC V, the written procedures must include the timing of the NDE in relation to the welding operations and safety precautions.

The services of a commercially independent qualified testing agency approved by the Contracting Officer must be employed by the Contractor for testing of piping welds. Costs of testing, including retesting of repaired welds, must be borne by the Contractor. Weld ripples or surface irregularities that might mask or be confused with the radiographic image of any objectionable defect must be removed by grinding, or other suitable mechanical means. The weld surface must be merged smoothly with the base metal surface.

3.2.4 Special Requirements for PAUT Testing

Phased array inspection must meet the review requirements of API Std 650 Twelfth Edition, Annex U, Section U5 and the Inspection requirements of ASME Section V 2017.

Calibration standards must meet the requirements of ASME Section V 2017 figure T-434.3-1, T-434.3-2.

The final data package shall be reviewed by a UT Level-III individual qualified in accordance with their employer's written practice. ASNT SNT-TC-1A or CP-189 shall be used as a guideline. Only Level-II or Level-III personnel shall perform UT examinations, analyze the data, or interpret the result. Alternatively, the review may be achieved by arranging for a data acquisition and initial interpretation by a Level-II individual, and a final interpretation and evaluation shall be performed by a Level-III individual qualified. The review shall include the following:

3.2.4.1 Essential Variables

Essential variables are listed below. Changes in any one of them beyond that allowed in the specification must require procedure requalification.

- a. Weld configurations to be examined, including thickness dimensions and base material product form (pipe, plate)
- b. The surfaces from which the examination shall be performed
Technique(s) Angle(s) and mode(s) of wave propagation in the material
- c. Search unit type(s), frequency(ies), and element size(s)/shape(s)
element size and number, and pitch and gap dimensions

- d. Focal range (identify plane, depth, or sound path)
- e. Virtual aperture size (i.e., number of elements, effective height, and element width)
- f. Angular range used (i.e., 40 deg. to 50 deg., 50 deg. to 70 deg.)
- g. Angle incremental change (i.e., 12 deg , 1 deg.)
- h. Element incremental change (i.e., 1,2)
- i. Additional S-scan, E-scan requirements Range of element numbers used (i.e., 1-126, 10-50)
- j. Scan plan
- k. Weld axis reference point marking
- l. Rastering angle(s)
- m. Aperture start and stop element numbers
- n. Aperture incremental change(s) (number of elements stepped)
- o. Additional S-scan requirements: Sweep angular range(s)
- p. Angular sweep increment (incremental angle change, deg)
- q. Aperture element numbers (first and last)
- r. Calibration calibration block(s) and technique(s)
- s. Scanning technique (automated vs. semi-automated)
- t. Scanner adhering and guiding mechanism
- u. Method for discriminating geometric from flaw indications
- v. Method for sizing indications
- w. Computer enhanced data acquisition, when used Scan overlap
- x. Computer software revision
- y. Personnel performance requirements, Personnel qualification requirements
- z. Surface condition (examination surface, calibration block)

3.2.4.2 Non-Essential Variables

- a. Couplant: brand name or type
- b. Post-examination cleaning technique
- c. Automatic alarm and recording equipment, when applicable Records, including minimum calibration data to be recorded (e.g., instrument settings)

3.2.5 Inspection and Tests by the Government

The Government may perform inspection and supplemental nondestructive or destructive tests as deemed necessary. The cost of supplemental NDE will be borne by the Government. The correction and repair of defects and the reexamination of weld repairs must be performed by the Contractor at no additional cost to the Government. Inspection and tests will be performed as required for visual inspection and NDE, except that destructive tests may be required also. When destructive tests are ordered by the Contracting Officer and performed by the Contractor and the specimens or other supplemental examinations indicate that the materials and workmanship do not conform to the contract requirements, the cost of the tests, corrections, and repairs must be borne by the Contractor. When the specimens or other supplemental examinations of destructive tests indicate that materials or workmanship do conform to the specification requirements, the cost of the tests and repairs will be borne by the Government. When destructive tests are made, repairs must be made by qualified welders or welding operators using welding procedures which will develop the full strength of the members cut. Welding must be subject to inspection and tests in the mill, shop, and field. When materials or workmanship do not conform to the specification requirements, the work may be rejected at any time before final acceptance of the system containing the weldment.

3.3 ACCEPTANCE STANDARDS

Acceptance standards must be in accordance with ASME B31.3 paragraph 341.3.2, Chapter VI in addition to the following specified items.

Interpretation of test results and limitations on imperfections in welds must comply with the requirements of 100 percent radiography, per ASME B31.3, paragraph 341.3.2, Chapter VI. For hydrant systems and stainless steel systems the evaluation must be based on severe cyclic conditions in addition to the following-specified items.

3.3.1 Visual

The following indications are unacceptable:

Weld Slag on the interior of the pipe.

3.3.2 Magnetic Particle Examination

The following relevant indications are unacceptable:

- a. Any linear indications.
- b. Rounded indications with dimensions greater than 5 mm.
- c. Four or more rounded indications in a line separated by 2 mm or less edge-to-edge.
- d. Ten or more rounded indications in any 3870 square mm of surface with the major dimension of this area not to exceed 150 mm with the area taken in the most unfavorable location relative to the indications being evaluated.

3.3.3 Liquid Penetrant Examination

Indications with major dimensions greater than 2 mm must be considered relevant. The following relevant indications are unacceptable:

- a. Any cracks or linear indications.
- b. Rounded indications with dimensions greater than 5 mm.
- c. Four or more rounded indications in a line separated by 2 mm or less edge-to-edge.
- d. Ten or more rounded indications in any 3870 square mm of surface with the major dimension of this area not to exceed 150 mm with the area taken in the most unfavorable location relative to the indications being evaluated.

3.4 CORRECTIONS AND REPAIRS

Disqualifying defects must be removed and repaired as specified in ASME B31.3 unless otherwise specified. Disqualifying defects discovered between weld passes must be repaired before additional weld material is deposited. After defect removal is complete and before rewelding, the area must be examined by the same test method which first revealed the defect to ensure that the defect has been eliminated. After rewelding, the repaired area must be reexamined by the same test method originally used for that area. Any indication of a defect must be regarded as a defect unless reevaluation by NDE or by surface conditioning shows that no disqualifying defects are present.

3.4.1 Defect Removal

Defective or unsound weld joints must be corrected by removing and replacing the entire weld joint, or for the following defects corrections must be made as follows:

- a. Excessive Convexity and Overlap: Reduce by removal of excess metal.
- b. Excessive Concavity of Weld, Undersized Welds, Undercutting: Clean and deposit additional weld metal.
- c. Excessive Weld Porosity, Inclusions, Lack of Fusion, Incomplete Penetration: Remove defective portions and reweld.
- d. Cracks or liner indications in Weld or Base Metal: Remove crack throughout its length, including sound weld metal for a distance of twice the thickness of the base metal or two inches, whichever is less, beyond each end of the crack, followed by the required rewelding. Complete removal must be confirmed by magnetic particle inspection for carbon steel or liquid penetrant inspection for stainless steel. Inspection procedures must comply with the requirements of ASME B31.3.
- e. Poor Fit-Up: Cut apart improperly fitted parts, and reweld.

3.4.1.1 Methods of Defect Removal

The removal of weld metal or portions of the base metal must be done preferably by chipping, grinding, sawing, machining, or other mechanical

means. Defects also may be removed by thermal cutting techniques. If thermal cutting techniques are used, the cut surfaces must be cleaned and smoothed by mechanical means to remove the heat affected zone. In addition, a maximum of 1/8-inch of metal must be removed by mechanical means from the cut surfaces of stainless steel.

Wherever a defect is removed, and repair by welding is not required, the affected area must be blended into the surrounding surface eliminating sharp notches, crevices, or corners.

3.4.1.2 Rewelding

Repair welds must be made using an electrode or filler wire smaller than that used in making the original weld. Rewelding must be done using qualified welding procedures. The surface must be cleaned before rewelding. Repair welds must meet the requirements of this specification.

3.4.1.3 Peening or Caulking

The use of force (peening) or foreign materials to mask, fill in, seal, or disguise any welding defects must not be permitted.

3.5 MAINTAINING CLEANLINESS OF PIPING

The Contractor must keep the interior and ends of all new piping affected by the Contractor's operations thoroughly cleaned of foreign matter and water before and after being installed. Piping systems must be kept clean during installation by means of plugs or other approved methods. When work is not in progress, open ends of piping and fittings must be closed so that no water or other foreign substance will enter the pipes or fittings. Piping must be inspected before placing into position. The interior of each length of pipe must be cleaned after welding; A swab, with a leather or canvas belt disc to fit the inside diameter of pipe, must be pulled through each length of pipe after welding in place. It must be the Contractor's responsibility for insuring that the interior of the piping is free of foreign matter including weld slag when it is connected into the system.

3.6 COMMISSIONING

For commissioning of POL service piping systems see Section 33 08 55 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT).

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 52 40

FUEL SYSTEMS PIPING (NON-HYDRANT)

11/18, CHG 2: 11/20

PART 1 GENERAL

- 1.1 SUMMARY
- 1.2 REFERENCES
- 1.3 ADMINISTRATIVE REQUIREMENTS
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Contractor Qualifications
 - 1.5.2 Work Plan
 - 1.5.3 Pigging Plan
 - 1.5.4 Hydrotesting Plan
 - 1.5.5 Design Data
 - 1.5.5.1 Pipeline Inventory
 - 1.5.6 Delivery, Storage, and Handling

PART 2 PRODUCTS

- 2.1 ELECTRICAL WORK
 - 2.1.1 General
 - 2.1.2 Grounding and Bonding
- 2.2 MATERIALS
 - 2.2.1 Types of Fuel
 - 2.2.2 Carbon Steel Piping
 - 2.2.3 Stainless Steel Piping
 - 2.2.4 External Protective Coatings for Aboveground Piping
 - 2.2.5 External Protective Coatings for Buried Steel Piping
 - 2.2.5.1 Carbon Steel Piping
 - 2.2.6 Fittings
 - 2.2.6.1 General
 - 2.2.6.2 Carbon Steel Fittings
 - 2.2.6.3 Stainless Steel Fittings
 - 2.2.7 Insulating Flange Kits (Electrically Isolating)
 - 2.2.8 Bolts, Nuts and Washers
 - 2.2.9 Flange Gaskets, Non-Metallic, Non-Electrically Isolating
 - 2.2.9.1 Nitrile Butadiene (Buna-N)
 - 2.2.9.2 Acrylonitrile Butadiene Rubber (NBR)
 - 2.2.9.3 Polytetrafluoroethylene (PTFE)
 - 2.2.9.4 Fluoro Rubber FKM
 - 2.2.9.5 Fluoroelastomer FPM
- 2.3 MANUAL VALVES
 - 2.3.1 Ball Valves
 - 2.3.1.1 Materials
 - 2.3.2 Plug (Double Block and Bleed) Valves
 - 2.3.2.1 General
 - 2.3.2.2 Valve Operation
 - 2.3.2.3 Relief Valves

- 2.3.2.4 Bleed Valves
- 2.3.2.5 Electric Valve Actuator
- 2.4 THERMAL RELIEF VALVE
 - 2.4.1 Valve Material
 - 2.4.2 Thermal Relief Valve (ASME Type)
 - 2.4.3 Thermal Relief Valve (Balanced Type)
- 2.5 PIPING ACCESSORIES
 - 2.5.1 Mechanically Adjustable Segmented Elastomeric Seal
 - 2.5.2 Pipe Sleeves
 - 2.5.3 Pressure Gauge
 - 2.5.4 Differential Pressure Gauge
 - 2.5.5 Pipe Supports
 - 2.5.5.1 General
 - 2.5.5.2 Adjustable Pipe Supports
 - 2.5.5.3 Low Friction Supports
 - 2.5.5.4 U-bolt Half Round Supports
 - 2.5.5.5 Concrete and Grout
 - 2.5.6 Sample Connections
- 2.6 PIGGING SYSTEM COMPONENTS
 - 2.6.1 Maintenance Pig Launchers and Receivers
 - 2.6.2 Smart Pig Launchers and Receivers
 - 2.6.3 Launcher and Receiver Closure Door
 - 2.6.4 Signaler
- 2.7 MISCELLANEOUS ACCESSORIES
 - 2.7.1 Concrete Anchor Bolts
 - 2.7.2 Coatings for Bolts, Studs, Nuts, and Washers
 - 2.7.3 Polytetrafluoroethylene (PTFE) Tape
 - 2.7.4 Pipe Sleeves
 - 2.7.5 Buried Utility Tape
 - 2.7.6 Pipeline Markers
- 2.8 FINISHES
 - 2.8.1 Factory Coating
 - 2.8.1.1 Valves
 - 2.8.1.2 Equipment and Components
 - 2.8.2 Field Painting

PART 3 EXECUTION

- 3.1 GENERAL
- 3.2 VERIFICATION OF DIMENSIONS
- 3.3 CLEANING OF PIPING
- 3.4 TRENCHING AND BACKFILLING
- 3.5 PIPING LAYOUT REQUIREMENTS
 - 3.5.1 Pipe Fabrication
 - 3.5.2 Interferences and Measurements
 - 3.5.3 Space and Access
 - 3.5.4 Location
 - 3.5.5 Pipe Supports
 - 3.5.6 Structural Support
 - 3.5.7 Grade
 - 3.5.8 Size Changes
 - 3.5.9 Direction Changes
 - 3.5.10 Threaded End Connections
 - 3.5.11 Existing Pipe Systems
 - 3.5.12 Bolted Connections
 - 3.5.13 Flanges and Unions
 - 3.5.14 Manual Valves
 - 3.5.15 Air Vents
 - 3.5.16 Drains

- 3.5.17 Pipe Sleeves
- 3.6 STRUCTURAL ATTACHMENTS
- 3.7 WELDING
 - 3.7.1 General
- 3.8 INSTALLATION
 - 3.8.1 Precautions
 - 3.8.2 Protective Coatings for Buried Piping Including Stainless Steel Piping
 - 3.8.2.1 Application of Coating System
 - 3.8.2.2 Inspection and Testing
 - 3.8.2.3 Damage Repair
- 3.9 INTERIOR EPOXY COATING
- 3.10 INSTALLATION OF UNDERGROUND PIPE
 - 3.10.1 Pipe Assembly
 - 3.10.2 Warning Tapes in Earth Trenches
 - 3.10.3 Clearances
 - 3.10.4 Protective Coating
 - 3.10.5 Pipeline Markers
- 3.11 SYSTEM COMMISSIONING
- 3.12 TESTING
 - 3.12.1 Before Backfilling
 - 3.12.1.1 Exterior Coating Holiday Test
 - 3.12.2 Pneumatic Test
 - 3.12.2.1 Pneumatic Test Procedure
 - 3.12.3 Hydrostatic Test
 - 3.12.4 Soak Testing
 - 3.12.5 Performance Testing
- 3.13 PIPE PIGGING - CLEANING
 - 3.13.1 General
 - 3.13.2 Use of Fuel in Cleaning Pigging
 - 3.13.3 Cleaning Pig Run
 - 3.13.4 Wire Brush Pig Run
- 3.14 PIPE PIGGING VERIFICATION
 - 3.14.1 Geometry/Ultrasonic Tool Reports
 - 3.14.2 Pipeline Internal Inspection Operations
 - 3.14.2.1 General
 - 3.14.2.2 Preparatory Work
 - 3.14.2.3 Pig Load And Launch
 - 3.14.2.4 Pipeline Operation During Pigging
 - 3.14.2.5 Brush and Gauging Survey
 - 3.14.2.6 Geometry/Ultrasonic Survey
 - 3.14.2.7 Pipe Wall Thickness Survey
 - 3.14.2.8 Lost Pig

-- End of Section Table of Contents --

SECTION 33 52 40

FUEL SYSTEMS PIPING (NON-HYDRANT)
11/18, CHG 2: 11/20

PART 1 GENERAL

1.1 SUMMARY

This section defines the requirements for pipe, piping components, and valves as related to an non-hydrant, non-service station, fuel distribution system. Such systems include, but are not limited to: marine receipt, pipeline receipt, truck off-loading receipt, pump house, pump pad, truck loading, marine loading, transfer pipeline, product recovery, and other miscellaneous piping systems. Provide the entire fuel distribution system as a complete and fully operational system. Size, select, construct, and install equipment and system components to operate together as a complete system. Substitutions of functions specified herein will not be acceptable. Coordinate the work of the system manufacturer's service personnel during construction, testing, calibration, and acceptance of the system. System components and piping specified herein must be designed to handle a working pressure of 1900 kPa at 38 deg C. Components specified herein must be compatible with the fuel to be handled. Components to be suitable for outside, unsheltered location, and to function normally in ambient temperatures.

1.2 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN PETROLEUM INSTITUTE (API)

API RP 540	(1999; R 2004) Electrical Installations in Petroleum Processing Plants
API RP 1110	(2013; R 2018) Recommended Practice for the Pressure Testing of Steel Pipelines for the Transportation of Gas, Petroleum Gas, Hazardous Liquids, Highly Volatile Liquids, or Carbon Dioxide
API RP 1595	(2012; R 2019; 2nd Ed) Design, Construction, Operation, Maintenance, and Inspection of Aviation Pre-Airfield Storage Terminals
API RP 2003	(2015; 8th Ed) Protection Against Ignitions Arising out of Static, Lightning, and Stray Currents
API RP 2009	(2002; R 2007; 7th Ed) Safe Welding, Cutting, and Hot Work Practices in Refineries, Gasoline Plants, and Petrochemical Plants

API RP 2200	(1999; R 2004) Electrical Installations in Petroleum Processing Plants
API STD 608	(2012) Metal Ball Valves - Flanged, Threaded, And Welding End
API Spec 5L	(2018; 46th Ed; ERTA 2018) Line Pipe
API Spec 6D	(June 2018, 4th Ed; Errata 1 July 2018; Errata 2 August 2018) Specification for Pipeline and Piping Valves
API Spec 6FA	(1999; R 2006; Errata 2006; Errata 2008; R 2011) Specification for Fire Test for Valves
API Std 607	(2016) Fire Test for Quarter-turn Valves and Valves Equipped with Non-metallic Seats

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1	(2003; R 2018) Unified Inch Screw Threads (UN and UNR Thread Form)
ASME B16.5	(2020) Pipe Flanges and Flanged Fittings NPS 1/2 Through NPS 24 Metric/Inch Standard
ASME B16.9	(2018) Factory-Made Wrought Buttwelding Fittings
ASME B16.11	(2016) Forged Fittings, Socket-Welding and Threaded
ASME B16.21	(2021) Nonmetallic Flat Gaskets for Pipe Flanges
ASME B18.2.1	(2012; Errata 2013) Square and Hex Bolts and Screws (Inch Series)
ASME B18.2.2	(2022) Nuts for General Applications: Machine Screw Nuts, and Hex, Square, Hex Flange, and Coupling Nuts (Inch Series)
ASME B31.3	(2020) Process Piping
ASME BPVC SEC VIII D1	(2019) BPVC Section VIII-Rules for Construction of Pressure Vessels Division 1
ASME B40.100	(2013) Pressure Gauges and Gauge Attachments

ASTM INTERNATIONAL (ASTM)

ASTM A36/A36M	(2019) Standard Specification for Carbon Structural Steel
ASTM A53/A53M	(2020) Standard Specification for Pipe, Steel, Black and Hot-Dipped, Zinc-Coated, Welded and Seamless

ASTM A105/A105M	(2021) Standard Specification for Carbon Steel Forgings for Piping Applications
ASTM A193/A193M	(2020) Standard Specification for Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service and Other Special Purpose Applications
ASTM A194/A194M	(2020a) Standard Specification for Carbon Steel, Alloy Steel, and Stainless Steel Nuts for Bolts for High-Pressure or High-Temperature Service, or Both
ASTM A234/A234M	(2019) Standard Specification for Piping Fittings of Wrought Carbon Steel and Alloy Steel for Moderate and High Temperature Service
ASTM A269/A269M	(2015; R 2019) Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service
ASTM A961/A961M	(2021) Standard Specification for Common Requirements for Steel Flanges, Forged Fittings, Valves, and Parts for Piping Applications
ASTM B696	(2000; R 2015) Standard Specification for Coatings of Cadmium Mechanically Deposited
ASTM B766	(1986; R 2015) Standard Specification for Electrodeposited Coatings of Cadmium
ASTM D229	(2019) Standard Test Methods for Rigid Sheet and Plate Materials Used for Electrical Insulation
ASTM D1418	(2010; R 2016) Standard Practice for Rubber and Rubber Lattices - Nomenclature
ASTM D3308	(2012; R 2017) Standard Specification for PTFE Resin Skived Tape
ASTM F336	(2002; R 2016) Standard Practice for Design and Construction of Nonmetallic Enveloped Gaskets for Corrosive Service
ASTM F436	(2011) Hardened Steel Washers

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

IEEE 142	(2007; Errata 2014) Recommended Practice for Grounding of Industrial and Commercial Power Systems - IEEE Green Book
IEEE 1100	(2005) Emerald Book IEEE Recommended Practice for Powering and Grounding Electronic Equipment

IEEE C62.41.2 (2002) Recommended Practice on
Characterization of Surges in Low-Voltage
(1000 V and Less) AC Power Circuits

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 1629 (2013) Rubber and Latexes - Nomenclature

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS
INDUSTRY (MSS)

MSS SP-58 (2018) Pipe Hangers and Supports -
Materials, Design and Manufacture,
Selection, Application, and Installation

NACE INTERNATIONAL (NACE)

NACE SP0188 (1999; R 2006) Discontinuity (Holiday)
Testing of New Protective Coatings on
Conductive Substrates

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 30 (2021; TIA 20-1; TIA 20-2) Flammable and
Combustible Liquids Code

NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; TIA
20-1; TIA 20-2; TIA 20-3; TIA 20-4)
National Electrical Code

NFPA 77 (2014) Recommended Practice on Static
Electricity

NFPA 407 (2022) Standard for Aircraft Fuel Servicing

NFPA 780 (2020) Standard for the Installation of
Lightning Protection Systems

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC SP 5/NACE No. 1 (2007) White Metal Blast Cleaning

SOCIETY OF AUTOMOTIVE ENGINEERS INTERNATIONAL (SAE)

SAE AMS3275 (2009; Rev C) Sheet, Acrylonitrile
Butadiene (NBR) Rubber and Non-Asbestos
Fiber Fuel and Oil Resistant

SAE J514 (2012) Hydraulic Tube Fittings

U.S. DEPARTMENT OF DEFENSE (DOD)

MIL-STD-161 (2005; Rev G; Notice 1 2010)
Identification Methods for Bulk Petroleum
Products Systems Including Hydrocarbon
Missile Fuels

MIL-STD-3004-1 (2018) Quality Assurance for Bulk Fuels,

Lubricants and Related Products

1.3 ADMINISTRATIVE REQUIREMENTS

Design conditions must be as specified. Refer to Section 01 78 23
OPERATION MAINTENANCE DATA.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" classification.
Submittals not having a "G" classification are for Contractor Quality
Control approval. Submit the following in accordance with Section 01 33 00
SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Work Plan; G

Pigging Plan; G

Hydrotesting Plan; G

SD-02 Shop Drawings

Grounding and Bonding

Pipe Supports; G

Pigging System Components; G

SD-03 Product Data

Carbon Steel Piping; G

Stainless Steel Piping; G

External Protective Coatings For Aboveground Piping; G

External Protective Coatings For Buried Steel Piping; G

Fittings; G

Carbon Steel Fittings; G

Insulating Flange Kits; G

Lightning Surge Arrester; G

Bolts, Nuts And Washers; G

Ball Valves; G

Plug (Double Block and Bleed) Valves; G

Electric Valve Actuator

Thermal Relief Valve; G

Mechanically Adjustable Segmented Elastomeric Seal; G

Pipe Sleeves; G

Sample Connections; G

Buried Utility Tape; G

Pipeline Markers; G

Flange Gaskets, Non-Metallic, Non-Electrically Isolating; G

SD-06 Test Reports

Exterior Coating Holiday Test

Pneumatic Test

Hydrostatic Test

Performance Testing

Pipe Pigging Verification

SD-07 Certificates

Contractor Qualifications; G

Pipeline Inventory; G

Survey Final Elevations; G

SD-10 Operation and Maintenance Data

Insulating Flange Kits; G

Lightning Surge Arrester; G

Ball Valves; G

Plug (Double Block and Bleed) Valves; G

Electric Valve Actuator; G

Thermal Relief Valve; G

Mechanically Adjustable Segmented Elastomeric Seal; G

Sample Connections; G

1.5 QUALITY ASSURANCE

1.5.1 Contractor Qualifications

Each installation Contractor must have successfully completed at least 3 projects of the similar scope and the same size or larger within the last 6 years. Each installation Contractor must demonstrate specific installation experience in regard to the specific system installation to be performed. Each installation Contractor must have taken, if applicable, manufacturer's training courses on the installation of piping

and must meet the licensing requirements in the state. Experience must include the erection of piping systems in compliance with the requirements of ASME B31.3, and NFPA 30. Submit a letter listing prior projects, the date of construction, a point of contact for each prior project, the scope of work of each prior project, and a detailed list of work performed. Provide in the letter evidence of prior manufacturer's training and state licensing.

1.5.2 Work Plan

Submit a comprehensive work plan that provides sufficient detail to demonstrate a thorough understanding of the project. Document that all components to be provided will function together and produce the result expected by the Government. Include any proposed dates for piping system shutdowns as well as the Contractor's ability to complete the work within the allotted shutdown periods. Show proposed dates and nature of piping system operations required of the Government. Include a list of manpower, spare piping and system component that will be on hand for each phase of the work. Describe, in detail, the means of:

- a. Coordinating work with Government and third parties.
- b. Preparing for safe piping repair work.
- c. Pneumatic pressure testing new piping sections.
- d. Hydrostatic pressure testing new piping sections.
- e. Interrupting or isolating an existing fuel service or system.
- f. Purging piping.
- g. Vapor monitoring.
- h. Preparations for containing and disposing of residual fuel.
- i. Cutting, sealing, and welding into existing piping systems.
- j. Welding tie-ins in place.
- k. Examining repair section tie-in welds.
- l. Collecting, storing and disposing of waste fuel generated during work.

1.5.3 Pigging Plan

The Contractor must submit a detailed written plan covering all aspects of the pipeline pigging operation, including anticipated pig runs, types of pigs, sequence of work, and retrieval/repair procedures. The Contractor must identify for each pigging evolution, the characteristics of the pig (type/purpose) and method/medium for propulsion.

1.5.4 Hydrotesting Plan

The Contractor must submit a detailed written plan covering all aspects of the pipeline hydrostatic testing operations, including procedures and sequencing of testing, segments of piping to be tested, fluid to be used during testing, equipment removal and isolation, hydrostatic testing pressures, safety protocols, etc.

1.5.5 Design Data

1.5.5.1 Pipeline Inventory

Fuel system volume must be calculated using as constructed pipe lengths, internal diameters, fittings, and components. Totals must be provided for all items containing fuel with the exception of tanks which is covered by other specifications. A certified pipeline inventory - a detailed list with sizes, lengths, quantity, and volumes must be provided for the systems in this project. Such systems include, but are not limited to: marine receipt, pipeline receipt, truck off-loading receipt, pump house, pump pad, truck loading, marine loading, transfer pipeline, product recovery, and other miscellaneous piping systems.

1.5.6 Delivery, Storage, and Handling

If fuel is used for hydrotesting or cleaning pigging, comply with all the requirements in Section 33 08 55 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT).

PART 2 PRODUCTS

2.1 ELECTRICAL WORK

2.1.1 General

Motors, manual or automatic motor control system components except where installed in motor control centers, and protective or signal devices required for the operation specified herein must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Any wiring required for the operation specified herein, but not shown on the electrical plans, must be provided under this section in accordance with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM.

2.1.2 Grounding and Bonding

Ground and bond as indicated on the drawings and in accordance with NFPA 70, NFPA 77, NFPA 407, NFPA 780, API RP 540, API RP 2003, IEEE 142, and IEEE 1100. Provide jumpers to overcome the insulating effects of gaskets, paints, or nonmetallic components.

2.2 MATERIALS

Pipe and fittings in contact with fuel must be stainless steel, interior epoxy coated carbon steel, or interior uncoated carbon steel as indicated on the drawings or as specified herein. No zinc coated metals, brass, bronze or other copper bearing alloys must be used in contact with the fuel. All carbon steel and stainless steel underground piping must have an exterior protective coating and must be cathodically protected in accordance with Section 26 42 17 IMPRESSED CURRENT CATHODIC PROTECTION (ICCP) SYSTEM. Cathodic protection for metal components that attach to a tank must be coordinated and compatible with the tank corrosion control system. Identification of piping must be in accordance with MIL-STD-161 unless specified otherwise. Material for manual valves must be as specified hereinafter. Do not use aluminum valves.

2.2.1 Types of Fuel

Components must be suitable for use with F-76 fuel.

2.2.2 Carbon Steel Piping

Subject each length of pipe to factory hydrostatic testing and ultrasonic testing in accordance with their respective pipe specification.

- a. Piping 300 mm and Larger: API Spec 5L Product Specification Level (PSL) 1, Grade B, seamless; or ASTM A53/A53M Grade B, seamless; all having a wall thickness of 9 mm.
- b. Piping 65 through 250 mm: Schedule 40, API Spec 5L Product Specification Level (PSL) 1, Grade B, seamless Grade B; or Schedule 40, Seamless, ASTM A53/A53M Grade B.
- c. Piping 50 mm and Smaller: Schedule 80, API Spec 5L Product Specification Level (PSL) 1, Grade B, seamless Grade B; or Seamless, Schedule 80 ASTM A53/A53M Grade B.

2.2.3 Stainless Steel Piping

- a. Stainless Steel Control Tubing: Seamless, fully annealed tubing conforming to ASTM A269/A269M, Grade TP316, Rockwell hardness B80 or less. Wall thickness for 13 mm tubing to be a minimum of 1.2 mm.

2.2.4 External Protective Coatings for Aboveground Piping

Provide exterior coating of aboveground piping and fittings, piping in pits, pipe supports, filter separators, and miscellaneous metal and system components in accordance with Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES. Color of finish coat must be white. Do not coat aboveground stainless steel or aluminum surfaces.

2.2.5 External Protective Coatings for Buried Steel Piping

2.2.5.1 Carbon Steel Piping

- a. New pipe and fittings must be factory coated in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEM.
- b. Field joints and repairs must be in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS.
- c. Field joints and repairs in tight spots (valve pits when heaters are too big) must be liquid epoxy in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS.
- d. Existing systems must match existing coating system and must be in accordance with Section 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS.

2.2.6 Fittings

2.2.6.1 General

Welding ells, caps, tees, reducers, must be of materials compatible for welding to the pipeline in which they are installed, and wall thickness, pressure and temperature ratings of the fittings must be not less than the

adjoining pipeline. Unless otherwise specified herein or required by the conditions of installation, all elbows must be the 1.5 diameter (D) type. Miter joints are not acceptable. Make odd angle offsets with pipe bends or elbows cut to the proper angle. Butt weld fittings must be factory-made wrought fittings manufactured by forging or shaping. Fabricated fittings will not be permitted. Welding branch fittings must be insert type suitable for radiographic inspections specified herein.

Make branch connections with butt-welded tees except where the branch is at least two pipe sizes smaller than the run, in which case the branch connection can be made with a forged or seamless branch outlet fitting. The branch outlet fitting must be designed in such a way that the connection can be radiographed. The branch outlet fittings may be a non-radiographicable if: the piping it is connected to is aboveground, the branch outlet size is 2.5 inches or less in diameter, and the branch outlet is located in contained pumphouses, contained truck offloads, contained truck fill stands, and other visibly contained areas equipped with containment curb.

2.2.6.2 Carbon Steel Fittings

- a. Fittings 65 mm and Larger: Butt weld, conforming to ASTM A234/A234M, grade WPB and ASME B16.9 of the same wall thickness as the adjoining pipe. All welds on buried piping must be radiographically examined throughout the entire length of each weld. Each fitting must be subjected to the Supplementary Requirements S3 and S4, Liquid Penetration examination and Magnetic-Particle Examination. Detectable flaws will not be accepted in the supplementary examinations. Fittings must be identified to relate them to their respective radiograph.
- b. Fittings 50 mm and Smaller: Forged, butt weld or socket welded (except flanges, see below). If specifically indicated on drawings, non-flange fittings may be threaded. Socket welded or threaded fittings must be Class 3000, conforming to ASTM A105/A105M, Grade 2 and ASME B16.11. Threaded fittings must only be used for above grade applications. Underground and in pits, low point drain pipe, and high point vent pipe must be butt welded.
- c. Flanges: Class 150 weld neck, butt weld, forged flanges conforming to ASTM A105/A105M, and ASME B16.5 except flanges that are to be connected to pumps must match the pump flanges rating. Threaded and slip-on flanges are not allowed. Flanges to be 2 mm raised face with modified spiral serrated gasket surface finish, except where required otherwise to match system components furnished. Match flange face to valves or system components furnished. Flange face must be machined to match valves or system components furnished. Use of spacing rings or gaskets discs are not allowed. Flanges must be subjected to the Supplementary Requirements S56, Liquid Penetrant Examination as outlined in ASTM A961/A961M. Detectable flaws will not be accepted. For flanges 50 mm and smaller located in contained pumphouses, contained truck offloads, contained truck fill stands, and other visibly contained areas the fitting may be forged (socket welded), Class 150, conforming to ASTM A105/A105M, Grade 2 and ASME B16.5. In pits, vaults, on thermal relief valve piping for pipeline routes, and other uncontained locations the flanges must be radiographicable, butt welded, weld neck type.
- d. Piggable System

(1) Provide barred tees on all branch outlets 50 mm and larger when within 6.1 meters of pig launcher or receiver barrel, including the barrels. Provide barred tees on all size outlets greater than 50 mm in size with any part of the outlet on the bottom half of the pipe. Provide barred tees on all branch connections equal to or greater than 50 percent of piggable line size.

(2) Use 3 D sweeps between pig launchers and receivers.

2.2.6.3 Stainless Steel Fittings

- a. Stainless Steel Tube Fittings. Flareless, 316 stainless steel fittings conforming to SAE J514.

2.2.7 Insulating Flange Kits (Electrically Isolating)

Provide ASTM D229 electrical insulating material of 1,000 ohms minimum resistance or 500 Volts per mil (VPM) minimum dielectric strength; material must be resistant to the effects of aviation and non-aviation hydrocarbon fuels. Provide full face insulating gaskets between flanges. Provide full surface 0.75 mm thick wall thickness, spiral-wound mylar insulating sleeves between the bolts and the holes in flanges; bolts may have reduced shanks of a diameter not less than the diameter at the root of threads. Provide 3 mm thick high-strength phenolic insulating washers next to flanges and provide flat circular stainless steel washers over insulating washers and under bolt heads and nuts. Provide bolts 12 mm longer than standard length to compensate for the thicker insulating gaskets and the washers under bolt heads and nuts. Above grade flanges separated by electrically insulating flange kits must be provided with weatherproof lightning surge arrester devices. The surge arrester must bolt across flanges separated by insulating gasket kits per detail on contract drawings. Provide with flange protector as described in this section. The arrester must have the following features:

- a. Weatherproof NEMA 4 enclosure.
- b. Bidirectional and bipolar protection.
- c. Constructed of solid state components, no lights, fuses or relays and used without required maintenance or replacement.
- d. Withstand unlimited number of surges at 50,000 Amperes.
- e. Maximum clamping voltage of 700 Volts based on a IEEE C62.41.2 8x20 microsecond wave form at 50,000 Amperes peak measured at the device terminals (zero lead length).
- f. A UL listed arrester for installation in Class 1, Division 1 and Division 2, Group D, hazardous areas.

Install the mounting bracket and leads on the flange side of the bolt insulating sleeve and washer, and size in accordance with this schedule:

Line Size	Bolt Size
50 mm	16 mm

Line Size	Bolt Size
65 mm	16 mm
80 mm	16 mm
100 mm	16 mm
150 mm	19 mm
200 mm	19 mm
250 mm	22 mm
300 mm	22 mm
350 mm	25 mm
400 mm	25 mm
Note: Make allowance for the 1 mm thickness of the insulating sleeve around the bolts when sizing the mounting lugs.	

2.2.8 Bolts, Nuts and Washers

- a. Bolts and nuts for pipe flanges, flanged fittings, valves and accessories must conform to ASME B18.2.1 and ASME B18.2.2, except as otherwise specified.
- b. Bolts must be of sufficient length to obtain full bearing on the nuts and must project no more than three full threads and no less than two full threads beyond the nuts with the bolts tightened to the required torque.
- c. Bolts must be regular hexagonal bolts conforming to ASME B18.2.1 with material conforming to ASTM A193/A193M, Class 2, Grade B8, stainless steel, when connections are made where a stainless steel flange is involved, and Grade B7, chromium molybdenum alloy, when only carbon steel flanges are involved. Bolts and nuts chosen must have sufficient strength to seat gasket types chosen. Bolts must be threaded in accordance with ASME B1.1, Class 2A fit, Coarse Thread Series, for sizes 25 mm and smaller and Eight-Pitch Thread Series for sizes larger than 25 mm.
- d. Nuts must conform to ASME B18.2.2, hexagonal, heavy series with material conforming to ASTM A194/A194M, Grade 8, stainless steel for stainless steel bolts, and Grade 7, chromium molybdenum alloy for chromium molybdenum alloy bolts. Nuts must be threaded in accordance with ASME B1.1, Class 2B fit, Coarse Thread Series for sizes 25 mm and smaller and Eight-Pitch Thread Series for sizes larger than 25 mm.
- e. Provide washers under bolt heads and nuts. Use chromium molybdenum alloy washers dimensioned to ASTM F436 flat circular for chromium molybdenum bolts. Stainless steel washer dimensioned similar to ASTM F436 flat circular, use material the same as the bolt.
- f. Use torque wrenches to tighten all flange bolts to the torque recommended by the gasket manufacturer. Tight in the pattern recommended by the gasket manufacturer. Use anti-seize compound on stainless steel bolts.

2.2.9 Flange Gaskets, Non-Metallic, Non-Electrically Isolating

ASME B16.21, composition ring, using a Nitrile Rubber such as Buna-N and NBR, polytetrafluoroethylene (PTFE), or a fluoro rubber such as FKM, FPM and Viton®. The gasket must be 3 mm thick. Gaskets must be resistant to the effects of aviation and non-aviation hydrocarbon fuels and manufactured of fire-resistant materials. Full-face gaskets must be used for flat-face flanged joints. Ring gaskets must be used for raised-face flanged joints. Gaskets must be of one piece factory cut. Select a gasket suitable for the working and test pressure of the fluid.

2.2.9.1 Nitrile Butadiene (Buna-N)

Provide Buna-N material that conforms to SAE AMS3275.

2.2.9.2 Acrylonitrile Butadiene Rubber (NBR)

Provide NBR material that conforms to SAE AMS3275.

2.2.9.3 Polytetrafluoroethylene (PTFE)

Provide PTFE material that conforms to ASTM F336.

2.2.9.4 Fluoro Rubber FKM

Provide FKM material that conforms to ASTM D1418.

2.2.9.5 Fluoroelastomer FPM

Provide FPM material that conforms to ISO 1629.

2.3 MANUAL VALVES

All portions of a valve coming in contact with fuel in stainless steel pipelines or epoxy lined carbon steel pipelines must be of noncorrosive material. Valves in stainless steel pipelines or epoxy lined carbon steel pipelines must be Type 304 or Type 316 stainless steel or carbon steel internally plated with chromium or nickel or internally electroless nickel plated. Valves in unlined carbon steel pipelines must have carbon steel body. Stem and trim must be stainless steel for all valves. Manually operated valves 150 mm and larger must be worm-gear operated and valves smaller than 150 mm must be lever operated or handwheel operated. Valves smaller than 50 mm must have lever-type handles. Valves installed more than 2.4 m above finished floor must have chain operators and a position indicators visible from ground level. Sprocket wheel for chain operator must be aluminum. Valves in the piggable line flow path between the pig launchers and the pig receivers, including the valves in the isolation valve pits must be full bore, piggable, double block and bleed type. The full bore piggable valves at the launcher and the receiver must be ball type. Valves must be true full bore with no projections extending into the flow path of the pig train.

2.3.1 Ball Valves

Ball valves must be fire tested and qualified in accordance with the requirements of API Std 607 and API STD 608. Seal material for the fire test must be graphite, seal material for the project must be as indicated below. Ball valves must be nonlubricated valves that operate from fully open to fully closed with 90-degree rotation of the ball.

Valves 2 inches and larger must conform to applicable construction and dimension requirements of API Spec 6D, ANSI Class 150 and must have flanged ends. Valves 50 mm and larger must conform to applicable construction and dimension requirements of API Spec 6D, ANSI Class 150 and must have flanged ends. The balls in valves 250 mm and larger full port and 300 mm and larger regular port and larger must have trunnion type support bearings. Except as otherwise specified or indicated, reduced port or full port valves may be provided at the Contractor's option. Balls must be solid, not hollow cavity.

2.3.1.1 Materials

Ball must be stainless steel. Ball valves must have tetrafluoroethylene (TFE) or fluoroelastetomer (FKM), commonly referred to as Viton seats, body seals and stem seals. Valves 100 mm and smaller must have a locking mechanism.

2.3.2 Plug (Double Block and Bleed) Valves

API Spec 6D, API Spec 6FA, ANSI Class 150, non-lubricated, resilient, double seated, trunnion mounted, tapered lift plug capable of two-way shutoff. Valve must have tapered plug of steel or ductile iron with chrome or nickel plating and plug supported on upper and lower trunnions. Sealing slips must be steel or ductile iron, with Viton seals which are held in place by dovetail connections. Valve design must permit sealing slips to be replaced from the bottom with the valve mounted in the piping. Valves must operate from fully open to fully closed by rotation of the handwheel to lift and turn the plug. Valves must have weatherproof operators with mechanical position indicators. Indicator shaft must be stainless steel. Minimum bore size must be not less than 65 percent of the internal cross sectional area of a pipe of the same nominal diameter unless bore height of plug equals the nominal pipe diameter and manufacturer can show equal or better flow characteristics of the reduced bore size design. Valves in the piggable line flow path between the pig launchers and the pig receivers, including the valves in the isolation valve pits, in fuel piping between the pig launchers and the pig receivers must be full bore, piggable. Valves must be true full bore with no projections extending into the flow path of the pig train. Full port plug valves in distribution piping must be provided with a 15 mm threaded body drain.

2.3.2.1 General

Valves in the operating tank suction and fill lines and the valves at the four valve manifold in the pump room in the tank fill lines must be provided with a factory-installed limit switch that is actuated by the valve closure. Tank fill line valve and four valve manifold limit switches must be provided with one double pole double throw contacts or four single pole, double throw contracts, two for open, two for closed. Tank suction line valve limit switches must be provided with one double pole double throw contacts or four single pole, double throw contacts, for closed, and one single pole double throw contact or two single pole, double throw contacts for open. All components must be watertight and U.L. listed for Class I, Division 1, Group D hazardous areas.

2.3.2.2 Valve Operation

Rotation of the handwheel toward open must lift the plug without wiping the seals and retract the sealing slips so that during rotation of the

plug clearance is maintained between the sealing slips and the valve body. Rotation of the handwheel toward closed must lower the plug after the sealing slips are aligned with the valve body and force the sealing slips against the valve body for positive closure. When valve is closed, the slips must form a secondary fire-safe metal-to-metal seat on both sides of the resilient seal. Plug valves located in Isolation Valve Pits or vaults must be provided with handwheel extensions.

2.3.2.3 Relief Valves

ANSI Class 150. Provide plug valves with automatic thermal relief valves to relieve the pressure build up in the internal body cavity when the plug valve is closed. Relief valves must open at 175 kPa differential pressure and must discharge to the throat of, and to the upstream side, of the plug valve.

2.3.2.4 Bleed Valves

ANSI Class 150, stainless steel body valve. Provide manually operated bleed valves that can be opened to verify that the plug valves are not leaking when in the closed position.

2.3.2.5 Electric Valve Actuator

The actuator, controls and accessories must be the responsibility of the valve-actuator supplier for sizing, assembly, certification, field-testing and any adjustments necessary to operate the valve as specified. The electric valve actuator must include as an integral unit the electric motor, actuator unit gearing, limit switch gearing, position limit switches, torque switches, drive bushing or stem nut, declutch lever, wiring terminals for power, remote control, indication connections and handwheel. The electrically actuated plug valve must be set to open and close completely in 30 to 60 seconds against a differential pressure of 2 MPa. The actuator settings of torque and limit contacts must be adjustable. The valve actuator must be suitable for mounting in a vertical or horizontal position and be rated for 30 starts per hour. The valve actuator must be capable of functioning in an ambient environment temperature.

- a. The electrical enclosure must be specifically approved by UL or Factory Mutual for installation in Class I, Division 1, Group D locations.
- b. The electric motor must be specifically designed for valve actuator service and must be totally enclosed, non-ventilated construction. The motor must be capable of complete operation at plus/minus 10 percent of specified voltage. Motor insulation must be a minimum NEMA Class F. The motor must be a removable subassembly to allow for motor or gear ratio changes as dictated by system operational requirements. The motor must be equipped with an embedded thermostat to protect against motor overload and also be equipped with space heaters. It must de-energize when encountering a jammed valve.
- c. The reversing starter, control transformer and local controls must be integral with the valve actuator and suitably housed to prevent breathing or condensation buildup. The electromechanical starter must be suitable for 30 starts per hour. The windings must have short circuit and overload protection. A transformer, if needed, must be provided to supply all internal circuits with 24 VDC or 110 VAC may be

used for remote controls.

- d. The actuator gearing must be totally enclosed in an oil-filled or grease-filled gearcase. Standard gear oil or grease must be used to lubricate the gearcase.
- e. The actuator must integrally contain local controls for Open, Close and Stop and a local/remote three position selector switch: Local Control Only, Off, and Remote Control plus Local Stop Only. A metallic handwheel must be provided for emergency operation. The handwheel drive must be mechanically independent of the motor drive. The remote control capability must be to open and close. Rim pull to operate valve manually must not exceed 28 kg.
- f. Position limit switches must be functional regardless of main power failure or manual operation. Four contacts must be provided with each selectable as normally open or normally closed. The contacts must be rated at 5A, 120 VAC, 30 VDC.
- g. Each valve actuator must be connected to a PLC supplied by "others."
- h. The actuator must have a local display of position even when power has been lost.
- i. The actuator must be supplied with a start-up kit comprising installation instruction, electrical wiring diagram and spare cover screws and seals.
- j. The actuator must be performance tested and a test certificate must be supplied at no extra charge. The test should simulate a typical valve load with current, voltage, and speed measured.

2.4 THERMAL RELIEF VALVE

2.4.1 Valve Material

Valves must have carbon steel bodies (stainless steel on stainless steel pipelines) and bonnets with stainless steel springs and trim. Valves must be Class 150 flanged end connections.

2.4.2 Thermal Relief Valve (ASME Type)

Thermal relief valves must be the fully enclosed, spring loaded, angle pattern, single port, hydraulically operated type with plain caps, and must be labeled in accordance with ASME BPVC SEC VIII D1. Valve stems must be fully guided between the closed and fully opened positions. The valves must be factory-set to open at pressures indicated on the drawings. Operating pressure must be adjustable by means of an enclosed adjusting screw. The valves must have a minimum capacity of 20 GPM at 10 percent overpressure. Valves must have a replaceable seat. Relief valves that do not relieve to a zone of atmospheric pressure or tank must be a balanced type relief valve.

2.4.3 Thermal Relief Valve (Balanced Type)

Thermal relief valves that do not relieve to a zone of atmospheric pressure or atmospheric tank must be a balanced type relief or regulator valve.

Thermal relief valves must be the fully enclosed, spring loaded, angle pattern, single port, fully balanced type (back pressure must not affect relief pressure) back pressure regulator/relief valve. Set valve at pressure indicated on drawings. Valve body must have 25 mm (minimum) raised face flange connections unless otherwise indicated. Orifice must have a minimum orifice size of 15 mm in diameter. Valve must have bubble-tight piston and seat design with stainless steel piston and Viton seat. Valve must be selected for the nominal flow condition of: pass a minimum of 18 liters per minute, at a differential pressure of 380 kilopascal, with a nominal set pressure of 345 kilopascal. Valve must be factory configured to open at required set pressure but must be field adjustable by means of an enclosed adjusting screw.

2.5 PIPING ACCESSORIES

2.5.1 Mechanically Adjustable Segmented Elastomeric Seal

Mechanically adjustable segmented elastomeric seals must be constructed of fuel resistant Buna-N elastomers and Type 316 stainless steel fasteners and hardware.

2.5.2 Pipe Sleeves

Pipe sleeves must be installed where indicated and at all points where the piping passes through concrete construction. Such sleeves must be of sufficient inside diameter to provide a minimum clear distance between the pipe and the sleeve of 13 mm. Sleeves through concrete pits or slabs must be standard weight carbon steel pipe with a protective coating. Each sleeve must extend through the respective pit wall or slab and must be provided with a wrap around Buna-N end seal (boot). (Viton when exposed to sunlight) and secured to the pipe sleeve and piping with adjustable stainless steel hose clamps. Sleeves where piping passes under roads or piping indicated to be double walled must be standard weight carbon steel pipe with a protective coating as previously specified. Alignment of the sleeve and piping must be such that the pipe is accurately centered within the sleeve by a nonconductive centering element. The sleeve must be securely anchored to prevent dislocation. Closure of space between the pipe and the pipe sleeve must be by means of a mechanically adjustable segmented elastomeric seal. The seal must be installed so as to be flush.

2.5.3 Pressure Gauge

Pressure gauges must conform to ASME B40.100 with metal cases and 100 mm diameter white dials. Gauges must be bottom connected, without back flanges. A pulsation dampener, adjustable to the degree of dampening required, must be provided for each gauge. Range of gauges must be as indicated. A ball valve must be provided for each pressure gauge. Gauges must be labeled with the calibration date.

2.5.4 Differential Pressure Gauge

The gauge must consist of a spring-supported, corrosion resistant piston moving inside a glass cylinder, with high pressure applied on top of the piston and low pressure applied below it. Under a differential pressure of 206.8 kPa, leakage past the piston must not exceed 120 drops per minute. The cylinder and flanges must be stainless steel with Viton O-ring seals. The high pressure inlet of the gauge must have a 10-micron pleated paper filter and the low pressure connection must have a fine mesh stainless steel strainer. The gauge must have an operating pressure of

2068 kPa. Differential pressure range of the gauge through approximately 75 mm of piston movement must be 0-2068 kPa with an accuracy of plus 34.5 kPa, calibrated linearly. High and low pressure connections must be 8 mm NPT female with a stainless steel bar stock valve at each connection. Construction of the gauge must be such that a 3-valve manifold is not necessary. If only one bar stock valve is closed, the gauge must not be damaged by up to 206.8 kPa differential pressure in either direction. A pressure gauge must be attached to the differential pressure gauge to indicate the high pressure and have a range of 2068 kPa.

2.5.5 Pipe Supports

2.5.5.1 General

Pipe supports must conform to MSS SP-58. Design pipe supports to meet the applicable requirements of ANSI/ASME B31.3 or ANSI/ASME B31.4. Provide hot-dip galvanized finish on rods, nuts, bolts, washers, and supports. Provide miscellaneous metal that conforms to ASTM A36/A36M, standard mill finished structural steel shapes, hot-dipped galvanized.

2.5.5.2 Adjustable Pipe Supports

Adjustable pipe supports must consist of a cast iron saddle and a threaded nipple connected to a carbon steel pipe by means of a special reducer conforming to MSS SP-58. The supports must be provided with PTFE insulation strips.

2.5.5.3 Low Friction Supports

Supports must have self-lubricating anti-friction bearing elements composed of 100 percent virgin tetrafluoroethylene polymer and reinforcing aggregates, prebonded to appropriate backing steel members. The coefficient of static friction between bearing elements must be 0.06 from initial installation for both vertical and horizontal loads and deformation must not exceed 51 micrometers under allowable static loads. Bonds between material and steel must be heat cured, high temperature epoxy. Design pipe support elements for the loads applied. Provide anti-friction material with a minimum of 2.3 mm thick. Provide hot-dipped galvanized steel supports. Provide supports that are factory designed and manufactured.

2.5.5.4 U-bolt Half Round Supports

Supports must have anti-friction bearing half-round in contact with the bottom of the pipe. Provide Polytetrafluoroethylene or like hydrophobic, anti-corrosive material half-round with a compressive strength of at least 69 mPa or greater as required. U-bolts must be installed in either a loose or limited guide configuration as indicated on the design drawings. Provide hot-dip galvanized u-bolts with seamless non-metallic low friction coating. U-bolt connection must be double nutted on the bottom and single bolted on top.

2.5.5.5 Concrete and Grout

Concrete and grout for anchors and supports must comply with Section 03 30 00 CAST-IN-PLACE CONCRETE.

2.5.6 Sample Connections

- a. Sample connections must be factory assembled units specifically designed for obtaining representative samples from fuel pipelines. Each connection must include a 6 mm sampling probe where the probe faces upstream, ball valve and 6 mm quick disconnect coupling with dust plug, all assembled into a unit that is suitable for installation in a pipe nipple. The sampling probe must extend not less than 25 mm into the fuel pipe. All materials in the sample connections must be stainless steel or aluminum.
- b. Furnish two sampling hose assemblies to the Contracting Officer at the project site. Each assembly must consist of a 1.8 m length of 6 mm clear plastic tubing with internal bonding/grounding wire. One end of the tubing will contain a male connector that actuates flow when inserted into the quick disconnect coupler. Each end of the bonding/grounding wire must be equipped with clips for attaching to the pipe and metal sample container.

2.6 PIGGING SYSTEM COMPONENTS

2.6.1 Maintenance Pig Launchers and Receivers

Construct of the same materials as the pipe, valves and fittings for a Class 150 system. The length of the straight barrel and the receiver line size section must be 1.5 meters each. Provide associated launcher kicker piping and receipt bypass piping, not less than 80 mm in size for 150 mm lines and 100 mm for up to 250 mm and 150 mm for up to 350 mm pipelines.

2.6.2 Smart Pig Launchers and Receivers

Construct of the same materials as the pipe, valves and fittings for a Class 150 system. The length of the straight barrel and the line size section must each be 4.5 meter. Provide associated launcher kicker piping and receipt bypass piping, not less than 80 mm in size for 150 mm lines and 100 mm for up to 250 mm and 150 mm for up to 350 mm pipelines.

2.6.3 Launcher and Receiver Closure Door

The closure must be hinged, swing bolted closure of the same material as the pipe and for a Class 150 system. Gasket must be nitrile or viton. Eye bolts must be pinned to lugs on the hub.

2.6.4 Signaler

The pig signaler must be mechanical flag type with manual reset, and be located on the pig launcher and the pig receiver. Material in contact with the fuel must be stainless steel. Units must be suitable for removal and installation under line pressure of 1.90 mPa. Signaler must be capable of withstanding line pressure of a Class 150 system.

2.7 MISCELLANEOUS ACCESSORIES

2.7.1 Concrete Anchor Bolts

Concrete anchors must conform to ASTM A307, Grade C, hot-dipped galvanized.

2.7.2 Coatings for Bolts, Studs, Nuts, and Washers

Carbon steel bolts, studs, nuts, and washers must be provided with a factory applied cadmium coating that conforms to ASTM B696 or ASTM B766.

2.7.3 Polytetrafluoroethylene (PTFE) Tape

Tape must conform to ASTM D3308.

2.7.4 Pipe Sleeves

Provided sleeves constructed of Coated Carbon steel piping, ductile iron, or cast-iron pipe conforming to ASTM A53/A53M, Standard weight.

2.7.5 Buried Utility Tape

Provide detectable aluminum foil plastic-backed tape or detectable magnetic plastic tape for warning and identification of buried piping. Tape must be detectable by an electronic detection instrument. Provide tape in minimum 75 mm width rolls, color coded for the utility involved, with warning identification imprinted in bold black letters continuously and repeatedly over entire tape length. Warning identification must be at least 25 mm high and must state as a minimum "BURIED JET FUEL PIPING BELOW". Provide permanent code and letter coloring that is unaffected by moisture and other substances contained in trench backfill material.

2.7.6 Pipeline Markers

Provide pipeline markers constructed of 150 mm diameter, one-half inch thick bronze disk with a 75 mm long bronze headed bolt welded to the back of the disk. Engrave the front of the disk with the words "UNDERGROUND FUEL LINE" in the case of one line and "UNDERGROUND FUEL LINES" in the case of multiple fuel lines.

2.8 FINISHES

2.8.1 Factory Coating

2.8.1.1 Valves

Valve surfaces must be blasted clean according to SSPC SP 5/NACE No. 1. Valve surfaces must be primed and coated in accordance with Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES.

2.8.1.2 Equipment and Components

Unless otherwise specified, provide equipment and components fabricated from ferrous metal with the manufacturer's standard factory finish. For equipment and component surfaces subject to temperatures above 50 degrees C, the factory coating must be appropriately designed for the temperature service.

2.8.2 Field Painting

Painting required for surfaces not otherwise specified must be field painted as specified in Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES. Do not paint aboveground stainless steel and aluminum surfaces. Do not coat equipment or components provided with a complete factory coating. Hot dip galvanized pipe support steel must not receive

the Zinc-Rich Epoxy Primer Coat only the Epoxy Intermediate Coat and Polyurethane Topcoat must be applied to these surfaces. Prior to any field painting, clean surfaces to remove dust, dirt, rust, oil, and grease.

PART 3 EXECUTION

3.1 GENERAL

Installation, workmanship, fabrication, assembly, erection, examination, inspection, and testing must be in accordance with ASME B31.3 and NFPA 30, except as modified herein. Safety rules as specified in NFPA 30 and NFPA 407 must be strictly observed. Never direct bury threaded connections, socket welded connections, unions, flanges, valves, air vents, or drains. Install all work so that parts requiring periodic inspection, operation, maintenance, and repair are readily accessible.

3.2 VERIFICATION OF DIMENSIONS

After becoming familiar with details of the work, verify dimensions in the field, and advise the Contracting Officer of any discrepancy before performing any work.

3.3 CLEANING OF PIPING

Keep the interior and ends of all new piping, affected by construction operations, thoroughly cleaned of foreign matter and water before and after being installed. Piping systems must be kept clean during installation by means of plugs or other approved methods. When work is not in progress, open ends of piping and fittings must be closed so that no water or other foreign substance will enter the pipes or fittings. Piping must be inspected before placing into position. The interior of each length of pipe must be cleaned after welding ensuring that the interior of the piping is free of foreign matter when it is connected into the system.

3.4 TRENCHING AND BACKFILLING

Trenching and backfilling must conform to Section 31 23 00.00 20 EXCAVATION AND FILL, and the following bedding and backfill requirements. The pipe must be laid in a bed of sand 150 mm deep, compacted in accordance with Section 31 23 00.00 20 EXCAVATION AND FILL. Sand must meet the requirements of Section 31 23 00.00 20 EXCAVATION AND FILL. The full length of each section of pipe without any protective covering must be excavated to permit installation of the protective covering. Pipe that has the grade or joint disturbed after laying, must be taken up and relaid. Pipe must not be laid in water or when the trench or weather conditions are unsuitable for such work. After testing and application of protective covering to joints, sand backfill must be placed and compacted around the pipe or protective coating to a depth of 300 mm above top of pipe. The remainder of the backfill must be the same as for other types of pipe.

3.5 PIPING LAYOUT REQUIREMENTS

3.5.1 Pipe Fabrication

Fabricate piping to measurements established on the project site and position into place without springing or forcing. Make provisions for absorbing expansion and contraction without undue stress in any part of

the system. The use of flexible connectors in permanently mounted pump suction and discharge lines as a method of compensating for piping misalignment is not acceptable.

3.5.2 Interferences and Measurements

Provide offsets, fittings, and accessories required to eliminate interferences and to match actual system components connection locations and arrangements. Verify measurements before commencing work. Submit discrepancies for clarification before proceeding with the installations to the Contracting Officer.

3.5.3 Space and Access

Keep piping, control tubing, which is not detailed close to structures and columns so as to take up a minimum amount of space. Ensure that access is provided for maintenance of system components, valves and gauges.

3.5.4 Location

Do not place unions in locations that will be inaccessible after the completion of the work. Place unions on each side of equipment.

3.5.5 Pipe Supports

Where provided hot dip galvanized pipe support steel must be surface prepped to SSPC SP 16, Brush-Off Blast Cleaning of Coated and Uncoated Galvanized Steel, Stainless Steels, and Non-Ferrous Metals. Once the steel is prepped, apply only the Epoxy Intermediate Coat and Polyurethane Topcoat per manufacturer's instructions.

Provide pipe supports with the maximum spacing as defined in Table 1 below, unless otherwise indicated. At the indicated locations. Provide additional pipe supports at concentrated piping loads (valves).

Table 1. Maximum Support Spacing									
Nominal Pipe Size (mm)	25 mm	40 mm	50 mm	80 mm	100 mm	150 mm	200 mm	250 mm	300 mm
Maximum Support Spacing (m)	2 m	2.75 m	3 m	3.5 m	4.25 m	5 m	5.75 m	6.50 m	7 m

Provide anchors where required to absorb or transmit thrust or eliminate vibration or pulsation. Provide supports near each change of direction. Select support components which do not restrict the movement of the pipe due to thermal expansion. Space supports uniformly and arrange symmetrically.

3.5.6 Structural Support

Provide supplementary or intermediate steel or other structural members as required for transmission of loads to members forming part of the

supporting structure. Piping must not be supported from other piping.

3.5.7 Grade

Where profiles of piping lines are shown on the drawings, grade the line uniformly between changes in slope or direction. Maintain gradient to within 6 mm over the entire length of pipe. When backfilling has been completed to the top of the pipe, the pipe must be surveyed at each joint, and logged by station number. Submit to the Contracting Officer for approval the survey final elevations before backfilling can continue.

3.5.8 Size Changes

Make changes in pipe size with reducing fittings. Do not use bushings. Make branch connections with butt-welded tees except where the branch is at least two pipe sizes smaller than the run, in which case the branch connection can be made with a forged or seamless branch outlet fitting. The branch outlet fitting must be designed in such a way that the connection can be radiographed. The branch outlet fittings may be a non-radiographicable if: the piping it is connected to is aboveground, the branch outlet size is 65 mm or less in diameter, and the branch outlet is located either in a pump house or on a system component pad equipped with containment curb.

3.5.9 Direction Changes

Make changes in direction of pipes with 1.5 D fittings. Where piping is to be piggable, make changes in direction with 3 D sweeps as indicated. Provide special fittings when required. Make odd-angle offsets with pipe bends or elbows cut to the proper angle. Make changes in direction with fittings, except that bending of pipe 100 mm and smaller will be permitted, provided a pipe bender is used and wide sweep bends are formed. Mitering or notching pipe or other similar construction to form elbows or tees will not be permitted. The centerline radius of bends must not be less than 6 diameters of the pipe if a pipe bender is used.

3.5.10 Threaded End Connections

Provide threaded end connections only on piping 50 mm in nominal size or smaller and only where indicated on the drawings. Provide threaded connections with PTFE tape or equivalent thread-joint compound applied to the male threads only. Not more than three threads must show after the joint is tightened.

3.5.11 Existing Pipe Systems

No interruptions or isolation of an existing fuel handling service or system may be performed unless the actions are appropriately documented in the approved work plan. Perform initial cutting of existing fuel pipe with a multiwheel pipe cutter, using a nonflammable lubricant. After cut is made, seal interior of piping with a gas barrier plug. Refer to API RP 2009 and API RP 2200. Purge interior of piping with carbon dioxide or nitrogen prior to performing any welding process.

3.5.12 Bolted Connections

For each bolted connection of stainless steel components (e.g., pipes, piping components, valves, and system components) use stainless steel bolts or studs, nuts, and washers. For each bolted connection of carbon

steel components, use carbon steel bolts or studs, nuts, and washers. Bolts project no more than three full threads and no less than two full threads beyond the nuts with the bolts tightened to the required torque. Prior to installing nuts, apply a compatible anti-seize compound to the male threads.

3.5.13 Flanges and Unions

Provide flanged joints in each line immediately preceding the connection to system components or material requiring maintenance such as pumps, general valves, control valves, strainers, and other similar items and as indicated. Assemble flanged joints square and tight with matched flanges, gaskets, and bolts.

3.5.14 Manual Valves

Install isolation plug or ball valves on each side of each system equipment, at the midpoint of looped mains, and at any other points indicated or required for draining, isolating, or sectionalizing purpose. Install valves with stems vertically up unless otherwise indicated. Provide individual supports and anchors for each valve.

3.5.15 Air Vents

Provide 40 mm air vents at all high points and where indicated to ensure adequate venting of the piping system.

3.5.16 Drains

Provide 50 mm drains at all low points and where indicated to ensure complete drainage of the piping. Drains must be schedule 120.

3.5.17 Pipe Sleeves

Provide a pipe sleeve around any pipe that penetrates a wall, floor, or slab. Do not install sleeves in structural members except where indicated or approved. Install pipe sleeves in masonry structures at the time of the masonry construction. Sleeves must be of such size as to provide a minimum of 12 mm all-around clearance between bare pipe and the sleeve. Align sleeve and piping such that the pipe is accurately centered within the sleeve by a nonconductive centering element. Securely anchor the sleeve to prevent dislocation. Closure of the space between the pipe and the pipe sleeve must be by means of a mechanically adjustable segmented elastomeric seal. The seal must be installed so as to be flush. For wall or floor penetrations, extend each sleeve through its respective wall or floor and cut flush with each surface. Seal around sleeves that penetrate through valve or fuel related pits with a Buna-N casing seal.

3.6 STRUCTURAL ATTACHMENTS

Provide attachment to building structure concrete and masonry by cast-in concrete inserts, built-in anchors, or masonry anchor devices. Apply inserts and anchors with a safety factor not less than 5. Do not attach supports to metal decking. Construct masonry anchors for overhead applications of ferrous materials only. Structural steel brackets required to support piping, headers, and system components, but not shown, must be provided under this section. Material used for support must be as specified under Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS.

3.7 WELDING

3.7.1 General

All joints, unless indicated otherwise, in carbon steel and stainless steel piping systems must be welded. Do not weld carbon steel to stainless steel. Welding of fuel pipe joints must comply with Section 33 52 23.15 POL SERVICE PIPING WELDING.

3.8 INSTALLATION

3.8.1 Precautions

Take special care to ensure that the protective coating on buried pipe is not damaged during installation and that the completed system is free of rocks, sand, dirt, water, weld slag, and foreign objects including construction debris. Take the following steps to ensure these conditions.

- a. Coated pipe must be handled only with canvas or nylon slings or padded clamps. Any coating damaged by improper handling or storage must be repaired as specified.
- b. Pipe brought to the site must be stored on blocks or horses at least 450 mm above the ground and adequately supported to prevent sagging. Padded blocks or horses must be used for coated pipe. The method and height of storing coated pipe must be in accordance with the coating manufacturer's instructions. Pipe ends must be protected and capped against weather at all times, except to accommodate immediate installation.
- c. Visual inspection must be made of the inside of each length of pipe to ensure that it is clear and clean prior to installation.
- d. The open ends of the pipe system must be closed at the end of each day's work or when work is not in progress by use of expansion plugs and must not be opened until the work is resumed.
- e. A swab, with a leather or canvas belt disc to fit the inside diameter of pipe, must be pulled through each length of pipe after welding in place.
- f. Obstruction remaining in the pipe after completion of the system must be removed at the expense of the Contractor.
- g. Plasma cutters and torches are not to be used to make penetrations in the pipe or to cut pipe.
- h. After installation and backfill of the piggable piping is complete and before fuel is put in the pipe, the pipe will be cleaned per the paragraphs CLEANING PIG RUN and WIRE BRUSH RUN in this Section.

3.8.2 Protective Coatings for Buried Piping Including Stainless Steel Piping

3.8.2.1 Application of Coating System

Application of coating system must be in accordance with 33 52 80 LIQUID FUELS PIPELINE COATING SYSTEMS. Application of coating system must be in accordance with 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES.

3.8.2.2 Inspection and Testing

The condition of factory field coated piping must be the responsibility of the Contractor and all damage to the protective covering during transit and handling must be repaired at no additional cost to the Government. All field coating must be subject to approval by the Contracting Officer. The entire pipe must be inspected as specified in sub-paragraph TESTING OF PROTECTIVE COATINGS under paragraph PROTECTIVE COATINGS FOR BURIED STEEL PIPING. The inspection for holidays must be performed just prior to lowering the pipe into the ditch and every precaution must be taken during lowering and backfilling to prevent damage to the protective covering.

3.8.2.3 Damage Repair

Damaged areas of coating must be repaired as specified in the preceding paragraph for fittings.

3.9 INTERIOR EPOXY COATING

When internally epoxy lined pipe is cut, the lining must be ground back from the end a minimum of 25 mm but not more than 38 mm.

3.10 INSTALLATION OF UNDERGROUND PIPE

Underground fuel pipelines must be pitched as shown on the drawings. Where not indicated they must be pitched a minimum of 50 mm per 30.5 m. 50 mm pipe size valved drain connections must be provided at all low points and 38 mm pipe size valved outlet vent connections must be provided at all high points. Vent and drain lines must terminate in male cam-type locking end connectors with matching female dust covers and installed in pits. The pipe must have cover as shown on the drawings. Drain lines must be installed at the slopes indicated.

3.10.1 Pipe Assembly

Pipe must be strung parallel and adjacent to or above a trench. The pipe must be supported on padded skids during welding and inspection of joints. Protective coating must be inspected and repaired prior to lowering the pipe into the trench. The pipe must be lowered using only canvas or nylon slings. The sling must be dug from underneath the pipe after placements and must not be pulled from underneath the pipe while in contact with it. Care must be taken to prevent damage to the pipe, welded joints or coating and any such damage must be repaired as directed by the Contracting Officer. Pressure testing of the pipe must be done after it has been placed in final position in the trench.

3.10.2 Warning Tapes in Earth Trenches

For the purpose of early warning and identification of buried pipes outside of building walls during future trenching or other excavation, continuous identification tapes must be provided in the trench. Provide metallic core or metallic-faced, acid- and alkali-resistant, polyethylene plastic warning tape manufactured for the purpose of early warning and identification of utilities buried below the tape. Tape must be at least 80 mm in width. Color of tape must be as standard with the manufacturer with respect to the type of utility buried below the tape. Tape must have lettering at least 25 mm high with warning and identification imprinted in bold black letters continuously over the entire tape length with not less

than the following identification on the tape: BURIED JET FUEL PIPING BELOW. Tape must be installed in accordance with the printed recommendations of the tape manufacturer, as modified herein. Tapes must be buried at a depth of 150 mm from the top of the subgrade or 300 mm below the top surface of earth. Provide permanent color and printing, unaffected by moisture or soil.

3.10.3 Clearances

Install pipe to be clear of contact with other pipes, pipe sleeves, casings, reinforcing steel, conduits, cables, or other metallic structures. Where pipes cross other pipes or structures with a separation of less than 150 mm, install an insulating separator. Protect the pipe from contact with a 300 mm square by 25 mm thick bituminous-impregnated canefiber board.

3.10.4 Protective Coating

When the protective coating on pipe is damaged, the Contracting Officer must be notified and must inspect the pipe before the coating is patched. If the damage to the pipe is deeper than 1.2 mm, the damage must be repaired by welding in accordance with paragraph WELDING. If the pipe is dented, out of round or damaged to the point that welding will not make it good as new, the length of pipe must be rejected.

3.10.5 Pipeline Markers

Provide above underground fuel piping spaced every 90 meters, at tees, and at changes in direction. For sections of underground piping less than 90 meters long, place at midpoint. Provide directly above pipe for single lines and between pipes where pipes run in pairs. Provide additional marker over each mid-line fitting connections for steel reinforced flexible pipe. Cast marker into 450 mm diameter, 300 mm thick concrete plug unless it is set in an area with concrete paving in which case it must be cast into the concrete paving.

3.11 SYSTEM COMMISSIONING

Conform to Section 33 08 55 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT).

3.12 TESTING

Piping must be tested by pneumatic and hydrostatic pressure. Testing must comply with applicable requirements of ASME B31.3, NFPA 30 and the requirements specified herein. Hydrostatic testing must be performed using fuel as the liquid. Pressure and hydrostatic testing must be performed only after welding inspection has been completed. Labor, materials, equipment, electricity, repairs, and retesting necessary for any of the tests required herein must be furnished by the Contractor. Provide provisions to prevent displacement of the piping during testing. Keep personnel clear of the piping during pneumatic testing. Only authorized personnel must be permitted in the area during pneumatic and hydrostatic testing. Isolate system components such as pumps, tanks, filter separators, and meters from the piping system during the testing. Do not exceed the pressure rating of any component in the piping system during the testing. Following satisfactory completion of each test, relieve the test pressure and seal the pipe immediately. Piping to be installed underground must not receive field applied exterior coatings at

the joints or be covered by backfill until the piping has passed the final pneumatic tests described herein.

3.12.1 Before Backfilling

Fuel piping must not be backfilled without completing the coating, pneumatic, weld radiograph and other piping examinations and approvals. Fuel piping in trench must not have exposed welds for longer than 30 days.

3.12.1.1 Exterior Coating Holiday Test

Following installation, test the exterior coating of direct buried piping for holidays using high-voltage spark testing in accordance with NACE SP0188. Repair holidays and retest to confirm holiday-free coating. Text must include all existing underground piping exposed for this project.

3.12.2 Pneumatic Test

Piping to be installed underground must not receive field applied protective covering at the joints or be covered by backfill until the piping has passed the pneumatic test described herein. To facilitate the tests, isolate various sections of the piping system and test each one separately. Where such sections terminate at flanged valve points, the line must be closed by means of blind flanges in lieu of relying on the valve. Furnish tapped flanges that can be attached to the end of the section of line being tested, and that will permit a direct connection between the piping and the air compressor or pressurizing pump. No taps in the permanent line will be permitted. Furnish all necessary equipment for testing; all gauges must be subject to testing and approval of the Contracting Officer. The air used for pneumatic testing must have a dew point of no more than 5 degrees C. Provide dehumidifying equipment on the suction or discharge side of the air compressor used to provide air for testing. Pressurizing pump must not exceed 4.7 L/s.

3.12.2.1 Pneumatic Test Procedure

Special safety measures, including the wearing of face mask, must be taken during testing under pressure. Only authorized personnel must be permitted in the area during testing. The pneumatic test pressure must be applied in increments. A preliminary 167 kPa test must be applied. Maintain the pressure while soapsuds or equivalent materials are applied to the exterior of the piping. While applying the soapsuds, visually inspect the entire run of piping, including the bottom surfaces, for leaks (bubble formations). If leaks are discovered, repair the leaks accordingly and retest. Repeat process until system is leak-free at 167 kPa. Unless otherwise directed by the Contracting Officer, all piping must then be tested at a pressure of 167 kPa for not less than 2-hours, during which time there must be no drop in pressure, only pressure rises with temperature. The pressure source must be disconnected during the final test period. Any leaks revealed by the test must be repaired and the test repeated.

3.12.3 Hydrostatic Test

Upon completion of pneumatic testing and after backfilling, hydrostatically test each underground piping system with fuel at in accordance with ASME B31.3 and API RP 1110, with no leakage or reduction in gauge pressure for four hours. Upon completion of pneumatic testing, hydrostatically test each aboveground piping system with fuel in

accordance with ASME B31.3 and API RP 1110, with no leakage or reduction in gauge pressure for four hours. Furnish electricity, instruments, connecting devices, and personnel for test. If fuel is used for testing, comply with all the requirements in Section 33 08 55 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). Upon completion of hydrostatic testing, perform "Soak Testing" of the piping systems per API RP 1595. Duration of the test must be a minimum of 4 days and maximum of 7 days. Defects in work must be corrected at the Contractor's expense, and the test repeated until the work is proven to be in compliance with the Contract requirements.

3.12.4 Soak Testing

Upon completion of hydrostatic testing with fuel, perform "Soak Testing" of the piping systems per API RP 1595. Duration of the test must be a minimum of 4 days and maximum of 7 days.

3.12.5 Performance Testing

After the fuel system testing is completed (including pneumatic and hydrostatic testing) the fuel system must be flushed, cleaned and performance tested as specified in Section 33 08 55 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). All control valves, both manual and automatic, must be checked for leaks (any area wetted with fuel) and proper operation and adjusted, repaired or replaced to correct any defects.

3.13 PIPE PIGGING - CLEANING

3.13.1 General

Track all pigs, using transmitter and receivers, at no less than 805 m increments, but no less than at four locations. The Contractor must prepare a contingency plan for retrieving a stuck pig and repairing any piping deformations. After pigging, plug valves must be flushed of all debris using the drain port at the bottom of the valve.

3.13.2 Use of Fuel in Cleaning Pigging

Cleaning pigging with fuel will take place after the initial receipt of fuel as per Section 33 08 55 FUEL DISTRIBUTION SYSTEM START-UP (NON-HYDRANT). Ensure that the fuel that is returned to the storage tanks during the pig runs is free of gross contamination and passes the color assessment method, and meets the requirements of MIL-STD-3004-1. Provide temporary storage tanks for the high particulate and dark color fuel that accumulates in front of and behind each pig. The contractor is responsible for cleaning the off-spec fuel in order to meet the requirements of MIL-STD-3004-1 or dispose of the off-spec fuel off-base.

3.13.3 Cleaning Pig Run

Initially, a proving pig run (foam density 32 kg/m³ should be performed to ensure the system is fully piggable. Upon completion of the successful proving pig run, the piping system must be cleaned with a standard cleaning pig. This will provide thorough cleaning of the interior of the piping system. Cleaning pig must be the bi-directional disk scraper style with steel body and replaceable polyurethane guiding and sealing disks, as well as gauge plates of 80 percent pipe diameter with 3 mm segmented aluminum fins. The pig body should include bypass nozzles and transmitter cavity. Propellant must be pressurized fuel using portable pumps. The

pig must be examined after the initial run for signs of possible pipe damage, interior slag or other adhered particles. Additional runs must be performed until the amount of collected sludge or debris is minimized, as determined by the System Supplier.

3.13.4 Wire Brush Pig Run

After the cleaning pig runs, the piping system must be cleaned with a wire brush style pig. This will remove weld slag and adhered particles from the system. Wire brush pig must be the bi-directional disk style or directional cup style with two circular steel wire brushes. The pig body should include bypass nozzles and transmitter cavity. Perform wire brush pig runs until the amount of collected weld slag or debris is minimized, as determined by the System Supplier.

3.14 PIPE PIGGING VERIFICATION

3.14.1 Geometry/Ultrasonic Tool Reports

After the system is installed and prior to performance testing, a field/preliminary report must be issued and a debrief given to Government personnel onsite on the condition of the piping system that was pigged. This must be comprised of raw data in the form of a PC download or equivalent which shows a continuous scan of each data unit output. Results of a preliminary interpretation of the data must be reported. These must include as a minimum all critical anomalies. A final report must include a description of the principle of operation, explanation of raw data, presentation of raw data, data to be clearly marked with distance traveled scale with classified anomaly location and all identifiable pipeline features, and all anomalies to be classified with locations in summary tabular form, pipe wall thickness survey, as well as the software necessary to read the data. Submittal must be in the form of digital media copied to a CD or DVD (flash drives are unacceptable).

3.14.2 Pipeline Internal Inspection Operations

3.14.2.1 General

The following pigs will be propelled through the pipeline with fuel in order to inspect the pipeline: 1.7 kg density foam swab, combination poly scraper-magnetic, stainless steel wire brush (plastic brush for internally lined piping), aluminum plate gauge, and geometry/ultrasonic tool. Tracking devices must be used on all pigs. At a minimum, the sequence of pig runs must be as follows: 1) foam swab for proving and cleaning, 2) wire brush for cleaning, 3) scraper-magnetic for cleaning, 4) aluminum plate gauge for gauging internal anomalies, 5) scraper-magnetic for cleaning, 6) wire brush for cleaning, 7) scraper-magnetic for cleaning, 8) foam swab for cleaning, (Note: the number of pig flights of each type of cleaning pigs must be determined by the amount and type of debris removed. The conclusion of the cleaning process must be when debris recovered is only that from the pigs themselves. This determination will be determined by the project's System Supplier and the Contracting Officer), 9) geometry/ultrasonic tool. The pipe wall must be continuously monitored on a real-time basis during the geometry/ultrasonic pig run. Anomalies such as patches, couplings, or flanges must also be identified, and the wall thickness given. The geometry/ultrasonic pig's technician will determine if additional runs are necessary. A permanent data set of internal inspection survey findings must be generated.

3.14.2.2 Preparatory Work

The Government will bring to the attention of the Contractor all statutes, rules and regulations relevant to the performance of the work on the site (on Government property) and will also provide the Contractor with a copy of its own site regulations (if any). Provide the pigging vendors with all-available pipeline records and drawings.

3.14.2.3 Pig Load And Launch

The pig must be loaded into the pig launcher by the Contractor. The method of loading and lodging the front pig cup into the launcher must not involve the use of uncontrolled mechanical force applied to the rear of the pig.

3.14.2.4 Pipeline Operation During Pigging

All pig runs must be made with the line packed with product. The system pumps will be used to propel the pig. The new pig traps will be used for pig launch and retrieval.

3.14.2.5 Brush and Gauging Survey

Run a brush pig at least as often as previously indicated. The brush pig must be designed and provided by the geometry/ultrasonic pig vendor. Additional runs may be required based upon the amount of debris found in the pipeline. The onsite geometry/ultrasonic pig vendor's personnel must determine if additional runs are required. Immediately following the brush pig run and immediately prior to the geometry/ultrasonic survey, run, as a minimum, a single batching pig fitted with a gauge plate equal to 90 percent of the pipeline normal inside diameter. The plate is to be a segmented aluminum disk of 3 mm thickness. The plate gauge pig must also include a tracker and tracking equipment. Track the pig assembly above ground during the operation.

3.14.2.6 Geometry/Ultrasonic Survey

After a satisfactory gauging pig run, the pipeline geometric defects must be determined by a geometry/ultrasonic tool. The geometry/ultrasonic tool must provide accuracy geometric anomaly detection, and bend radius measuring capability. The data obtained must be presented in a PC software format to allow user friendly analysis and presentation. The geometry/ultrasonic tool assembly must be capable of:

- a. Operating in hydrocarbon liquid environment, specifically diesel fuel, at a pressure of up to ANSI 300 rating.
- b. Traversing the pipeline with nominal wall thickness and possible bore restrictions down to 90 percent of nominal pipe inside diameter.
- c. Traversing the pipeline length at a speed of between 60 and 100 m/min when propelled by pumped diesel fuel. Pressure differential across pig not to exceed 34 kPa.
- d. Traversing through smooth pipe bends as small as 3D (3 pipe diameters) radius and single miter bends of up to 10 degrees change of direction.
- e. Include a tracker and tracking equipment. Track the pig assembly above ground during the operation. The battery life of the tracker

must not be less than 72-hours.

f. Manual loading into the new horizontal pig trap.

The geometry/ultrasonic tool assembly instrumentation performance must be capable of:

- a. Battery life to be minimum 18-hours at operating conditions.
- b. Principle of operation to be electronically stored geometry system.
- c. Geometry sensing to span full circumference and length of pipe, with associated distance measuring method.
- d. Geometry system must be capable of:
 - (1) positive location and identification of each geometric anomaly.
 - (2) positive location and identification of each bend.
 - (3) positive location and identification of distance marker reference points of either magnetic or electronic type placed on or above the pipe.
- e. Classification of geometric anomalies to be as minimum:
 - (1) discrimination between ovality and intrusion anomalies.
 - (2) mechanical damage such as mill defects, dents, internal gouges, and buckles.
 - (3) pipeline weld defects (such as excess weld penetration).
 - (4) geometric thickness anomalies. As a minimum, these must be reported in the following categories within the listed accuracy.
 - (a) magnitude of anomaly (plus/minus 25 mm)
 - (b) span of anomaly (plus/minus 25 mm)
 - (c) ovality (plus/minus 2.5 mm)
 - (d) span of ovality (plus/minus 25 mm)
 - (e) anomaly station (plus/minus 1:2,000)

3.14.2.7 Pipe Wall Thickness Survey

The geometry/ultrasonic tool must provide accuracy measurement of pipe wall thickness (plus/minus 0.25 mm). The data obtained must be presented in a PC software format to allow user friendly analysis and presentation.

3.14.2.8 Lost Pig

The Contractor is responsible for a lost pig, finding the pig, retrieval of the pig, and all repairs, radiographs to the pipeline system and the pig.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 52 80

LIQUID FUELS PIPELINE COATING SYSTEMS

08/22

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SYSTEM DESCRIPTION
- 1.3 DEFINITIONS
- 1.4 SUBMITTALS
- 1.5 QUALITY ASSURANCE
 - 1.5.1 Contract Errors, Omissions, and Other Discrepancies
 - 1.5.2 Corrective Action (CA)
 - 1.5.2.1 Corrective Action Procedures
 - 1.5.2.2 Corrective Action Request (CAR) Form
 - 1.5.2.3 Corrective Action Log
 - 1.5.3 Coatings Work Plan
 - 1.5.4 Design Data
 - 1.5.4.1 Containment System for Field Coating
 - 1.5.5 Test Reports
 - 1.5.5.1 Ferrous Metallic Abrasive Qualification Test Reports
 - 1.5.5.2 Non-Metallic Abrasive Qualification Test Reports
 - 1.5.5.3 Recycled Ferrous Metallic Abrasive Field Test Reports (Daily and Weekly)
 - 1.5.6 Qualifications
 - 1.5.6.1 Qualifications of Certified Industrial Hygienist (CIH)
 - 1.5.6.2 Qualifications of Pipe Coating Shop
 - 1.5.6.3 Qualifications of Certified Protective Coatings Specialist (PCS)
 - 1.5.6.4 Qualifications of Coatings Inspection Company
 - 1.5.6.5 Qualifications of Quality Assurance Coatings Inspector
 - 1.5.6.6 Qualifications of Coatings Contractors for Field Coating
 - 1.5.6.7 Qualifications of Individuals Performing Abrasive Blasting for Field Coating
 - 1.5.6.8 Qualifications of Individuals Performing Coating Application for Field Coating
 - 1.5.6.9 Qualifications of Testing Laboratory for Abrasive
 - 1.5.6.10 Ferrous Metallic Abrasive Certificate of Conformance
 - 1.5.6.11 Non-Metallic Abrasive Certificate of Conformance
 - 1.5.7 QA and QC Personnel
 - 1.5.7.1 QC Manager
 - 1.5.7.2 Protective Coatings Specialist (PCS)
 - 1.5.7.3 Quality Assurance Coatings Inspector
 - 1.5.7.4 Coatings Inspector for Shop Coating
 - 1.5.7.5 Coatings Contractor QC Coatings Inspector
 - 1.5.8 Pre-Application Meeting for Field Coating
- 1.6 PRODUCT DATA
 - 1.6.1 Coating System Instructions
- 1.7 DELIVERY AND STORAGE
- 1.8 COATING HAZARDS

- 1.9 JOB SITE REFERENCES - SHOP
- 1.10 JOB SITE REFERENCES - FIELD

PART 2 PRODUCTS

- 2.1 SHOP-APPLIED COATING FOR BURIED PIPING
 - 2.1.1 External Pipe Coating
- 2.2 FIELD-APPLIED EXTERNAL PIPE COATING
- 2.3 COATING FIELD SAMPLE COLLECTION AND SHIPPING KIT
- 2.4 ABRASIVE FIELD SAMPLE COLLECTION AND SHIPPING KIT
- 2.5 INSPECTION TEST KITS
 - 2.5.1 Test Kit for Measuring Chloride, Sulfate, and Nitrate Ions on Steel and Coated Surfaces
 - 2.5.2 Test Kit for Measuring Chlorides in Abrasives
 - 2.5.3 Test Kit for Identifying Amine Blush on Epoxy Surfaces
- 2.6 ABRASIVE
 - 2.6.1 Ferrous Abrasive
 - 2.6.1.1 New and Remanufactured Steel Grit
 - 2.6.1.2 Recycled Steel Grit
 - 2.6.2 Non-Metallic Abrasive

PART 3 EXECUTION

- 3.1 FIELD SAMPLE COLLECTION AND TESTING
 - 3.1.1 Coating Field Sample Collection
 - 3.1.2 Abrasive Field Sample Collection
 - 3.1.3 Abrasive Field Test Reports
- 3.2 LIGHTING
- 3.3 ENVIRONMENTAL CONDITIONS
 - 3.3.1 Automated Monitoring Requirements for Shop Applied Coating
 - 3.3.2 Containment for Field Coating
 - 3.3.3 Automated Monitoring Requirements for Field Coating
- 3.4 SHOP APPLIED COATING OF BURIED PIPING
 - 3.4.1 Surface Preparation for Shop Applied Coating
 - 3.4.1.1 Abrasive Blasting Equipment for Shop Applied Coating of Buried Piping
 - 3.4.1.2 Contamination Evaluation of Abrasive for Shop Applied Coating of Buried Piping
 - 3.4.1.3 Pre-Preparation Testing for Surface Contamination for Shop Applied Coating of Buried Piping
 - 3.4.1.3.1 Pre-Preparation Testing for Oil and Grease Contamination for Shop Applied Coating of Buried Piping
 - 3.4.1.3.2 Pre-Preparation Testing for Soluble Salts Contamination for Shop Applied Coating of Buried Piping
 - 3.4.1.4 Abrasive Blasting External Surfaces for Shop Applied Coating of Buried Piping
 - 3.4.1.5 Disposal of Used Abrasive for Shop Applied Coating of Buried Piping
 - 3.4.1.6 Pre-Application Testing for Surface Contamination for Shop Applied Coating of Buried Piping
 - 3.4.1.6.1 Pre-Application Testing for Oil and Grease Contamination for Shop Applied Coating of Buried Piping
 - 3.4.1.6.2 Pre-Application Testing for Surface Cleanliness for Shop Applied Coating of Buried Piping
 - 3.4.2 Mixing and Application of Shop Applied Coating of Buried Piping
 - 3.4.2.1 External Pipe Shop Applied Coating Application
 - 3.4.2.2 Holiday Testing and Shop Repair
- 3.5 FIELD EXTERIOR COATING OF ABOVEGROUND PIPING
- 3.6 FIELD REPAIRS TO EXTERNAL COATING OF BURIED PIPING

- 3.6.1 Surface Preparation for Field Coating of Buried Piping
 - 3.6.1.1 Abrasive Blasting Equipment for Field Coating of Buried Piping
 - 3.6.1.2 Contamination Evaluation of Abrasive for Field Coating of Buried Piping
 - 3.6.1.3 Surface Standard
 - 3.6.1.4 Pre-Preparation Testing for Surface Contamination for Field Coating of Buried Piping
 - 3.6.1.4.1 Pre-Preparation Testing for Oil and Grease Contamination for Field Coating of Buried Piping
 - 3.6.1.4.2 Pre-Preparation Testing for Soluble Salts Contamination for Field Coating of Buried Piping
 - 3.6.1.5 Abrasive Blasting for Field Coating of Buried Piping
 - 3.6.1.6 Disposal of Used Abrasive for Field Coating of Buried Piping
 - 3.6.1.7 Pre-Application Testing for Surface Contamination for Field Coating of Buried Piping
 - 3.6.1.7.1 Pre-Application Testing for Oil and Grease Contamination for Field Coating of Buried Piping
 - 3.6.1.7.2 Pre-Application Testing for Soluble Salts Contamination for Field Coating of Buried Piping
 - 3.6.1.7.3 Pre-Application Testing for Surface Cleanliness for Field Coating of Buried Piping
- 3.6.2 Final Inspection of Pipeline Prior to Burial
- 3.6.3 Final Cleanup
- 3.7 FIELD AND SHOP QUALITY CONTROL
 - 3.7.1 Inspection
 - 3.7.1.1 Inspection Requirements
 - 3.7.1.2 Inspection Report Forms
 - 3.7.1.3 Daily Inspection Reports
 - 3.7.1.4 Inspection Logbook
 - 3.7.1.5 Inspection Equipment
 - 3.7.2 Coatings Contractor QC Coatings Inspector's Responsibilities
 - 3.7.3 Quality Assurance Coatings Inspector's Responsibilities
- 3.8 PROJECT IDENTIFICATION
 - 3.8.1 Stencils

-- End of Section Table of Contents --

SECTION 33 52 80

LIQUID FUELS PIPELINE COATING SYSTEMS
08/22

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN WATER WORKS ASSOCIATION (AWWA)

- | | |
|-----------|---|
| AWWA C210 | (2015) Standard for Liquid Epoxy Coating Systems for the Interior and Exterior of Steel Water Pipelines |
| AWWA C213 | (2022) Fusion-Bonded Epoxy Coatings and Linings for Steel Water Pipe and Fittings |

ASTM INTERNATIONAL (ASTM)

- | | |
|------------|--|
| ASTM D3276 | (2015; E 2016) Standard Guide for Painting Inspectors (Metal Substrates) |
| ASTM D3335 | (1985a; R 2020) Low Concentrations of Lead, Cadmium, and Cobalt in Paint by Atomic Absorption Spectroscopy |
| ASTM D3718 | (1985a; R 2015) Low Concentrations of Chromium in Paint by Atomic Absorption Spectroscopy |
| ASTM D3925 | (2002; R 2015) Sampling Liquid Paints and Related Pigmented Coatings |
| ASTM D4285 | (1983; R 2018) Indicating Oil or Water in Compressed Air |
| ASTM D4417 | (2021) Standard Test Methods for Field Measurement of Surface Profile of Blast Cleaned Steel |

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

- | | |
|------------|--|
| ISO 8502-3 | (2017) Preparation of Steel Substrates Before Application of Paints and Related Products - Tests for the Assessment of Surface Cleanliness - Part 3: Assessment of Dust on Steel Surfaces Prepared for Painting (Pressure-Sensitive Tape Method) |
| ISO 9001 | (2015) Quality Management Systems- Requirements |

SOCIETY FOR PROTECTIVE COATINGS (SSPC)

SSPC AB 1	(2015; E 2017) Mineral and Slag Abrasives
SSPC AB 2	(2015; E 2016) Cleanliness of Recycled Ferrous Metallic Abrasive
SSPC AB 3	(2003; E 2004) Ferrous Metallic Abrasive
SSPC Guide 6	(2021) Guide for Containing Surface Preparation Debris Generated During Paint Removal Operations
SSPC Guide 12	(2023) Guide for Illumination of Industrial Painting Projects
SSPC PA 1	(2016) Shop, Field, and Maintenance Coating of Metals
SSPC PA 2	(2015; E 2018) Procedure for Determining Conformance to Dry Coating Thickness Requirements
SSPC QP 1	(2019) Standard Procedure for Evaluating the Qualifications of Industrial/Marine Painting Contractors (Field Application to Complex Industrial Steel Structures and Other Metal Components)
SSPC QP 3	(2010) Standard Procedure for Evaluating Qualifications of Shop Painting Applicators
SSPC QP 5	(2022) Standard Procedure for Evaluating the Qualifications of Coating and Lining Inspection Companies
SSPC QS 1	(2015) Standard Procedure for Evaluating a Contractor's Advanced Quality Management System
SSPC SP 1	(2015) Solvent Cleaning
SSPC SP 10/NACE No. 2	(2015) Near-White Blast Cleaning
SSPC VIS 1	(2002; E 2004) Guide and Reference Photographs for Steel Surfaces Prepared by Dry Abrasive Blast Cleaning

U.S. NATIONAL ARCHIVES AND RECORDS ADMINISTRATION (NARA)

29 CFR 1910.134	Respiratory Protection
29 CFR 1910.1000	Air Contaminants
29 CFR 1926.59	Hazard Communication

1.2 SYSTEM DESCRIPTION

This section specifies the requirements for exterior coating of

aboveground and buried carbon steel liquid fuel pipelines and exterior coating of aboveground and buried stainless steel liquid fuel pipelines. The exterior coating system for aboveground, carbon steel pipelines is specified with the same requirements as the exterior coating system for the exterior of fuel tanks, Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES. The exterior coating system for buried pipelines is a fusion bonded epoxy (FBE) system or a liquid epoxy system.

1.3 DEFINITIONS

Definitions are generally provided throughout this Section in the paragraphs where used and denoted by capital letters. The following definition is used throughout this Section:

- a. INDEPENDENT THIRD-PARTY - Impartial third-party not a part or affiliated with Contractor or subcontractor principal or subsidiary businesses, and not a materials supplier.

1.4 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-01 Preconstruction Submittals

Contract Errors, Omissions, and Other Discrepancies

Corrective Action Procedures

Corrective Action Request (CAR) Form

Coatings Work Plan

Inspection Report Forms

SD-05, Design Data

Containment System

SD-06 Test Reports

Daily Inspection Reports

Ferrous Metallic Abrasive Qualification Test Reports

Non-Metallic Abrasive Qualification Test Reports

Abrasive Field Test Reports

Recycled Ferrous Metallic Abrasive Field Test Reports (Daily and Weekly)

SD-07 Certificates

Qualifications of Certified Industrial Hygienist (CIH)

Qualifications of Pipe Coating Shop

Qualifications of Certified Protective Coatings Specialist (PCS)

Qualifications of Coatings Inspection Company

Qualifications of Quality Assurance Coatings Inspector

Qualifications of Coatings Contractors for Field Coating

Qualifications of Individuals Performing Abrasive Blasting for
Field Coating

Qualifications of Individuals Performing Coating Application for
Field Coating

Qualifications of Testing Laboratory for Abrasive

Ferrous Metallic Abrasive Certificate of Conformance

Non-Metallic Abrasive Certificate of Conformance

SD-08 Manufacturer's Instructions

Coating System Instructions

SD-11 Closeout Submittals

Disposal of Used Abrasive Shop Applied Coating

Disposal of Used Abrasive Field Applied Coating

Inspection Logbook

Corrective Action Log; G

1.5 QUALITY ASSURANCE

1.5.1 Contract Errors, Omissions, and Other Discrepancies

Submit all errors, omissions, and other discrepancies in contract documents to the Contracting Officer within 30 days of contract award for all work covered in this Section, other than the work that will not be uncovered until a later date. All such discrepancies must be addressed and resolved, and the Coatings Work Plan modified, prior to beginning the Initial and Follow-Up phases of work. Discrepancies that become apparent only after work is uncovered must be identified at the earliest discoverable time and submitted for resolution. Schedule time (float) must be built into the project schedule at those points where old work is to be uncovered or where access is not available during the first 30 days after award, to allow for resolution of contract discrepancies.

1.5.2 Corrective Action (CA)

CA must be included in the Contractor Quality Control Plan as outlined in Section 01 45 00.00 20 QUALITY CONTROL.

1.5.2.1 Corrective Action Procedures

Develop procedures for determining the root cause of each non-compliance,

developing a plan to eliminate the root cause so that the non-compliance does not recur, and following up to ensure that the root cause was eliminated.

1.5.2.2 Corrective Action Request (CAR) Form

Develop Corrective Action Request (CAR) forms for initiating CA and for tracking and documenting each step. The CAR should be included with the Corrective Action Procedures. A CAR must be initiated by either the Contractor or the Contracting Officer. The Protective Coatings Specialist (PCS) must approve each CAR at the root cause identification stage, the plan for elimination stage, and the close out stage after verification that the root cause has been eliminated.

1.5.2.3 Corrective Action Log

When a CAR is initiated, Contractor must take action to identify and eliminate the root cause of each non-compliance so as to prevent recurrence. These actions must apply to non-compliance in the work, and to non-compliance in the Quality Control (QC) System. Corrective actions must be appropriate to the effects of the non-compliance encountered. The corrective action must be documented in a report that is serialized and tracked in the Corrective Action Log until project completion and its acceptance by the Contracting Officer. All corrective action reports must be retained in project records. The Corrective Action Log, showing status of each CAR, must be submitted to the Contracting Officer monthly.

1.5.3 Coatings Work Plan

This work plan must be considered as part of the Contractor Quality Control Plan as outlined in Section 01 45 00.00 20 QUALITY CONTROL.

The Coatings Work Plan must be submitted and approved by the PCS prior to mobilization. The Coatings Work Plan must explain in detail all procedures including, but not limited to, all sequential processes, quality control for each process, quality assurance for each process, and safety considerations. Subsections must include at least the following:

- a. Purpose;
- b. Introduction;
- c. Safety, fire, and health information;
- d. Contractor and worker qualifications with certifications;
- e. Project management organization and documents;
- f. Timeline in a Gantt chart;
- g. Project document references;
- h. Reference to all applicable standards (e.g., AMPP, NACE, SSPC, ISO, and ASTM);
- i. Coatings manufacturer's supporting documentation;
- j. Descriptions and explanations of any exceptions from the coating manufacturer;

- k. Coating and blasting equipment, model names, and, if applicable, calibration dates;
- l. Containment design and/or details;
- m. Environmental testing;
- n. Material delivery, storage, and handling details;
- o. Surface preparation;
- p. Pre-application test panel validation for field-applied external coating as outlined in SURFACE STANDARD;
- q. Coating materials, mixing, application, recoat windows, and coating curing times, if applicable;
- r. Coating repairs and rework;
- s. Non-conformance;
- t. Spent material handling and effluent discharge containment and disposal;
- u. Inspection test plan (as outlined in FIELD INSPECTION, and including inspection hold points, both Quality Assurance and Coating Contractor QC Coatings Inspector's responsibilities, and daily documentation and delivery);
- v. Instruments and test kits;
- w. Warranty (in writing, signed by the Contractor and the coating manufacturer's representative);
- x. Demobilization;
- y. PCS and PM approval;

1.5.4 Design Data

1.5.4.1 Containment System for Field Coating

Submit complete design drawings and calculations for the scaffolding and containment system, including an analysis of the loads which will be added to the structure by the containment system and waste materials. A registered engineer must approve calculations and scaffold system design.

1.5.5 Test Reports

1.5.5.1 Ferrous Metallic Abrasive Qualification Test Reports

Submit qualification testing results for abrasive as required in paragraph 4 REQUIREMENTS of SSPC AB 3. Submit test results from an INDEPENDENT THIRD-PARTY laboratory of representative samples of each abrasive to be used on the jobsite. Samples must have been tested within the last three years. Note that this testing is for the purpose of pre-qualifying the abrasive.

1.5.5.2 Non-Metallic Abrasive Qualification Test Reports

Submit results for abrasive as required in paragraph 4 REQUIREMENTS of SSPC AB 1. Submit test results from an INDEPENDENT THIRD-PARTY laboratory of representative samples of each abrasive to be used on the jobsite. Samples must have been tested within the last three years. Note that this testing is for the purpose of pre-qualifying the abrasive.

1.5.5.3 Recycled Ferrous Metallic Abrasive Field Test Reports (Daily and Weekly)

Submit test results from an INDEPENDENT THIRD-PARTY laboratory of daily and weekly Quality Control testing required by SSPC AB 2, as modified in paragraph ABRASIVE.

1.5.6 Qualifications

The qualifications specified in this Section must be met throughout the duration of this contract. Personnel and corporate entities must provide proof of active certification to all qualifications specified in this Section. Work must not be provided by personnel and corporate entities unless all specified qualifications are met.

1.5.6.1 Qualifications of Certified Industrial Hygienist (CIH)

Submit name, address, telephone number, fax number, and e-mail address of the INDEPENDENT THIRD-PARTY CIH. Submit documentation that the hygienist is certified by the American Board of Industrial Hygiene in comprehensive practice, including certification number and date of certification/recertification. The CIH must remain certified during the entire project, and the Contracting Officer must be notified of any change in certification status within 10 days of the change. If a CIH's certification expires, the hygienist will not be allowed to perform any hygienist functions, and all hygienist work must stop, until the certification is reissued or another CIH is approved. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Provide evidence of experience with hazards involved in industrial coating application work.

1.5.6.2 Qualifications of Pipe Coating Shop

Each shop that applies coatings to pipe must be certified to SSPC QP 3 prior to contract award and must remain certified while accomplishing any surface preparation or coating application. If a shop's certification expires, the firm will not be allowed to perform any work until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in the shop's certification status within 10 days of the change. If a shop's qualification expires, the shop will not be allowed to perform any coatings related functions until the qualification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive qualification will not be considered and liquidated damages will apply. Notify the Contracting Officer of all scheduled and unannounced on-site audits from AMPP and furnish a copy of all audit reports.

1.5.6.3 Qualifications of Certified Protective Coatings Specialist (PCS)

Submit name, address, telephone number, fax number, and e-mail address of the INDEPENDENT THIRD-PARTY PCS. Submit documentation that the PCS is certified by the Association for Materials Protection and Performance (AMPP) (formerly SSPC: The Society for Protective Coatings (SSPC)), including certification number and date of certification/recertification. If the PCS is employed by the same coatings inspection company to which the Quality Assurance Coatings Inspector is employed, this does not violate the INDEPENDENT THIRD-PARTY requirements. The PCS must be an INDEPENDENT THIRD-PARTY to the pipe coating shop. The PCS must remain certified during the entire project, and the Contracting Officer must be notified of any change in certification status within 10 days of the change. If a PCS's certification expires, the PCS will not be allowed to perform any PCS functions, and all coatings work must stop, until the certification is reissued or another PCS is approved. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. The PCS must not be the designated Quality Assurance Coatings Inspector. The PCS's responsibilities are outlined in PROTECTIVE COATINGS SPECIALIST (PCS).

1.5.6.4 Qualifications of Coatings Inspection Company

Submit documentation that the selected quality assurance coatings inspection company is accredited by AMPP to the requirements of SSPC QP 5 prior to contract award. The coatings inspection company that is submitted and approved, must remain and cannot be changed through completion of the contract. The coatings inspection company must remain SSPC QP 5 certified for the duration of the coatings work. If a coatings inspection company's certification expires, the firm will not be allowed to perform any inspection functions, and all surface preparation and coating application work must stop, until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in the coatings inspection company certification status within 10 days of the change. Notify the Contracting Officer of all scheduled and unannounced on-site audits from AMPP and furnish a copy of all audit reports. The coatings inspection company must not engage in any activities that may conflict with their independence of judgment and integrity in relation to their inspection activities. In particular, they must not be engaged in the manufacture, supply, application, surface preparation, purchase, or maintenance of the applied coating in this project.

1.5.6.5 Qualifications of Quality Assurance Coatings Inspector

Submit documentation that each Quality Assurance Coatings Inspector is employed by the SSPC QP 5 company and is qualified to a minimum certification of NACE CIP Level II. Each inspector must remain employed by the coatings inspection company while performing any coatings inspection functions. The Quality Assurance Coatings Inspector's responsibilities are outlined in QUALITY ASSURANCE COATINGS INSPECTOR'S FIELD RESPONSIBILITIES. The roles of the Quality Assurance Coatings Inspector are in addition to, and distinct from, the role of the QC Coatings Inspector employed by the coatings Contractor and the Coatings Inspector for Shop Coating.

1.5.6.6 Qualifications of Coatings Contractors for Field Coating

All Contractors and Subcontractors that perform surface preparation or coatings application must be certified to SSPC QP 1 and SSPC QS 1 prior to contract award and must remain certified while accomplishing any surface preparation or coating application. If a Contractor's or Subcontractor's certification expires, the firm will not be allowed to perform any work until the certification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive certification will not be considered and liquidated damages will apply. Notify the Contracting Officer of any change in Contractor certification status within 10 days of the change. If a Contractor's qualification expires, the Contractor will not be allowed to perform any coatings related functions until the qualification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive qualification will not be considered and liquidated damages will apply. Notify the Contracting Officer of all scheduled and unannounced on-site audits from AMPP and furnish a copy of all audit reports.

For projects located outside the United States, Guam, and Puerto Rico, the certifications for the coatings Contractor (SSPC QP 1 and SSPC QS 1) can be substituted if the coatings Contractor meets all of the below requirements:

- a. ISO 9001 certified;
- b. Eight years of experience with industrial coatings;
- c. Evidence of recent work that has Contractor Performance Assessment Report System (CPARS) ratings, and/or other quality/performance ratings, that are equivalent to, or exceed, "Above Average";
- d. Evidence of an INDEPENDENT THIRD-PARTY audit from AMPP demonstrating equivalency to SSPC QP 1 and SSPC QS 1 within the last two years.

The coatings Contractors and coatings Subcontractors must be certified to ISO 9001 prior to contract award and must remain so certified for the duration of the project. Notify the Contracting Officer of any change in Contractor certification status within 10 days of the change. If a Contractor's qualification expires, the Contractor will not be allowed to perform any coatings related functions until the qualification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive qualification will not be considered and liquidated damages will apply. Notify the Contracting Officer of all scheduled and unannounced on-site inspections from the ISO certifying organization and furnish a copy of all inspection reports.

1.5.6.7 Qualifications of Individuals Performing Abrasive Blasting for Field Coating

Submit name, address, and telephone number of each person that will be performing abrasive blasting. All individuals performing abrasive blasting must be certified by AMPP to the SSPC C7 Dry Abrasive Blaster Qualification Program or CAS Coating Application Specialist Level 2 Certification Program (Interim Status), and must remain certified during the entire period of abrasive blasting. The Contracting Officer must be notified of any change in qualification status within 10 days of the change. If a blaster's qualification expires, the blaster will not be allowed to perform any blasting functions until the qualification is

reissued. Requests for extension of time for any delay to the completion of the project due to an inactive qualification will not be considered and liquidated damages will apply.

This requirement applies to all manual abrasive blasting performed in shop and field locations. This requirement does not apply to automated abrasive blasting performed in the shop.

1.5.6.8 Qualifications of Individuals Performing Coating Application for Field Coating

Submit name, address, telephone number, and evidence of certification of each person that will be performing coatings application by any method. All individuals performing coatings application must be certified by AMPP to the SSPC CAS Coating Application Specialist Level 2 Certification Program (Interim Status) or SSPC C12 Spray Application Program. Each applicator must remain certified during the entire period of coatings application, and the Contracting Officer must be notified of any change in qualification status within 10 days of the change. If an applicator's qualification expires, the applicator will not be allowed to perform any coatings application functions until the qualification is reissued. Requests for extension of time for any delay to the completion of the project due to an inactive qualification will not be considered and liquidated damages will apply.

1.5.6.9 Qualifications of Testing Laboratory for Abrasive

Submit name, address, telephone number, fax number, and e-mail address of the INDEPENDENT THIRD-PARTY laboratory or laboratories selected to perform testing of abrasive for compliance with this section. Submit documentation that the laboratory has experience in testing samples of abrasive for conformance with specifications and that the employees performing testing are qualified.

1.5.6.10 Ferrous Metallic Abrasive Certificate of Conformance

Provide manufacturer's certification of conformance that the materials are currently in conformance with SSPC AB 3 and as modified in this Section, and have been tested within the last three years.

1.5.6.11 Non-Metallic Abrasive Certificate of Conformance

Provide manufacturer's certification that the materials are currently in conformance with SSPC AB 1 and as modified in this Section, and have been tested within the last three years.

1.5.7 QA and QC Personnel

1.5.7.1 QC Manager

The QC Manager is as defined in Section 01 45 00.00 20 QUALITY CONTROL.

1.5.7.2 Protective Coatings Specialist (PCS)

The PCS must be considered a QC Specialist and must report to the QC Manager, as specified in Section 01 45 00.00 20 QUALITY CONTROL. The PCS must approve all submittals prior to submission to the QC Manager for approval or submission to the government for approval.

The PCS's responsibilities include, but are not limited to, the following:

- a. Obtain, review, and understand all project documentation including, but not limited to, this Section, project program, Coatings Work Plan, inspection and testing plan (ITP), and all submittals before the project starts, during the project, and all coatings related re-work;
- b. Attend all pre-job coatings meetings (in-person, phone, or virtually);
- c. Attend pre-final coatings walk-through (mandatory) and attend final coatings walk-through (as required).

1.5.7.3 Quality Assurance Coatings Inspector

The Quality Assurance Coatings Inspector must be considered a QC Specialist and must report to the QC Manager, as specified in Section 01 45 00.00 20 QUALITY CONTROL. The Quality Assurance Coatings Inspector must be present during all pre-preparation testing, surface preparation, coating application, initial cure of the coating system, during all coating repair work, and during completion activities. The Quality Assurance Coatings Inspector must provide complete documentation of conditions and occurrences on the job site, and be aware of conditions and occurrences that are potentially detrimental to the coating system. The responsibilities of the Quality Assurance Coatings Inspector are defined in QUALITY ASSURANCE COATING INSPECTOR'S RESPONSIBILITIES. These responsibilities are separate and distinct from the responsibilities of the Coatings Contractor QC Coatings Inspector and Coatings Inspector for Shop Coating.

1.5.7.4 Coatings Inspector for Shop Coating

The coatings inspector must be the shop Quality Manager or appropriate designee. The coatings inspector must be considered a QC Specialist and must report to the prime Contractor's QC Manager, as specified in Section 01 45 00.00 20 QUALITY CONTROL. The coatings inspector must be present during all pre-preparation testing, surface preparation, coating application, initial cure of the coating system, and during all coating repair work required of the shop. The coatings inspector must provide complete documentation of conditions and occurrences on the job site, and be aware of conditions and occurrences that are potentially detrimental to the coating system.

1.5.7.5 Coatings Contractor QC Coatings Inspector

The Coatings Contractor QC Coatings Inspector must stop non-compliant work. The responsibilities of the Coatings Contractor QC Coatings Inspector are defined in COATINGS CONTRACTOR QC COATINGS INSPECTOR'S RESPONSIBILITIES. These responsibilities are separate and distinct from the responsibilities of the Quality Assurance Coatings Inspector.

1.5.8 Pre-Application Meeting for Field Coating

After approval of submittals, but prior to the initiation of coatings work, Contractor representatives, including at a minimum, project superintendent, QC manager, paint foreman, Quality Assurance Coatings Inspector, and PCS must have a pre-application coating preparatory meeting. This meeting must be in addition to the pre-construction conference. Specific items addressed must include: corrective action requirements and procedures, coatings work plan, safety plan, coordination

with other Sections, inspection standards, inspection requirements and tools, test procedures, environmental control system, safety plan, and test logs. Notify Contracting Officer at least ten days prior to meeting.

1.6 PRODUCT DATA

1.6.1 Coating System Instructions

Submit manufacturer's printed instructions, including detailed mixing and application procedures, number and types of coats required, minimum and maximum application temperatures, and curing procedures of all coating materials. Include Safety Data Sheets (SDS) for materials to be used at the job site in accordance with 29 CFR 1926.59.

1.7 DELIVERY AND STORAGE

Ship, store, and handle materials in accordance with SSPC PA 1, applicable standards, and as modified in this Section. Maintain temperature in storage spaces between 5 and 24 degrees C, and air temperature more than 3 degrees C above the dew-point at all times. Inspect materials for damage and return non-compliant materials to manufacturer. Remove materials with expired shelf life from government property immediately and notify the Contracting Officer.

If materials are approaching shelf life expiration and an extension is desired, samples must be sent to the manufacturer, along with complete records of storage conditions, with a request for shelf life extension. If the manufacturer finds the samples and storage data suitable for shelf life extension, the manufacturer must issue an extension, referencing the product evaluation and the review of storage records. Products must not be extended longer than allowed in the product specification.

1.8 COATING HAZARDS

Ensure that employees are trained in all aspects of the safety plan. Specified coatings may have potential health hazards if ingested or improperly handled. The coating manufacturer's written safety precautions must be followed throughout mixing, application, and curing of the coatings. During cleaning, cleanup, surface preparation, and paint application phases, ensure that employees are protected from toxic and hazardous chemical agents which exceed concentrations in 29 CFR 1910.1000. Comply with respiratory protection requirements in 29 CFR 1910.134. The CIH must approve work procedures and personal protective equipment.

1.9 JOB SITE REFERENCES - SHOP

Make available to the Contracting Officer a copy of each standard to which the shop will be applying coatings under this Section.

1.10 JOB SITE REFERENCES - FIELD

Make available to the Contracting Officer a copy of each standard to which field coatings will be applied to under this Section, and an SSPC Certified Contractor Evaluation Form at the job site.

PART 2 PRODUCTS

2.1 SHOP-APPLIED COATING FOR BURIED PIPING

2.1.1 External Pipe Coating

Fusion Bonded Epoxy Coating: AWWA C213, minimum 375 microns, maximum 500 microns.

2.2 FIELD-APPLIED EXTERNAL PIPE COATING

All coating materials used to coat external girth welds or repair existing coating must be from the same manufacture as the shop applied coating. Use one or more of the following repair methods, as modified herein, to repair shop applied coatings and coat external girth welds:

- a. Liquid-Epoxy Coating System: AWWA C210

2.3 COATING FIELD SAMPLE COLLECTION AND SHIPPING KIT

Provide a kit for each sample to be collected. For liquid coatings, each kit must contain: a one liter can for the base of each liquid coatings material; one appropriately sized can for the activator of each liquid coatings material; dipping cups for each component to be sampled; a shipping box sized for the samples to be shipped; and packing material. Mark cans for the appropriate components (powder, base, or activator), including manufacturer's name, address, batch numbers, batch size shipped to the project site, and date of manufacture. Provide shipping documents, including either pre-paid shipping or a shipping number that can be used by the QC Manager to arrange pickup, addressed to the INDEPENDENT THIRD-PARTY coating testing laboratory.

2.4 ABRASIVE FIELD SAMPLE COLLECTION AND SHIPPING KIT

Provide a kit for each sample to be collected. Each kit must contain: one suitable plastic bag or container for each sample to be collected. Mark containers with manufacturer's name, address, batch number, batch size, and date of manufacture. Provide shipping documents, including either pre-paid shipping label or a shipping number that can be used by the QC Manager to arrange pickup, addressed to the approved coating testing laboratory.

2.5 INSPECTION TEST KITS

2.5.1 Test Kit for Measuring Chloride, Sulfate, and Nitrate Ions on Steel and Coated Surfaces

Provide test kits that meet the following requirements:

- a. Contains all materials, supplies, tools, and instructions for field testing and on-site quantitative evaluation of chloride, sulfate, and nitrate ions;
- b. Extract solution is acidic, factory pre-measured, pre-packaged, and of uniform concentration;
- c. Components and solutions are mercury free and environmentally friendly;
- d. Contains new materials and solutions for each test extraction;

- e. Contains an extraction test container (vessel, sleeve, cell) creates a sealed, encapsulated environment during salt ion extraction;
- f. Contains a test extract container is suitable for testing the following steel surfaces: horizontal (up/down configuration), vertical, flat, curved, smooth, pitted, and rough;
- g. All salt ion concentrations are directly measured in micrograms per square centimeter.

2.5.2 Test Kit for Measuring Chlorides in Abrasives

Provide test kits that meet all of the following requirements:

- a. Be a completely self-contained test kit with all materials, supplies, tools, and instructions to take tests and identify results;
- b. Use identifiable, consistent, factory pre-measured test extract solution;
- c. Provide for testing equal volumes of abrasive and test solution;
- d. Provide for taking direct measurements of the chloride ion in parts per million (PPM), without using conversion charts or tables;
- e. Provide all new components for extraction and titration for each test;
- f. Provide a factory sealed titration device for each test;
- g. Use the extract sampling container as the titration container.

2.5.3 Test Kit for Identifying Amine Blush on Epoxy Surfaces

Provide test kits that meet all of the following requirements:

- a. Be a completely self-contained field test kit with all materials, supplies, tools, and instructions to perform tests and indicate the presence of unreacted amines;
- b. Use an identifiable, consistent, uniform, pre-packaged, factory pre-measured indicating solution;
- c. Contains no mercury or lead and is environmentally friendly;
- d. Contains a solution of an unreacted amine for the purpose of "self checking" the indicator solution.

2.6 ABRASIVE

Use abrasive that is specifically selected to provide a sharp, angular profile to the specified depth. Abrasive must meet all requirements of this Section each time that it is placed in the blast pot. The referenced abrasive specifications have maximum limits for soluble salts contamination; however, this maximum level of contamination does not guarantee that contamination will not be transferred to the steel surface during abrasive blasting. Other factors, such as on-site handling and recycling, can allow contamination of abrasive that can be transferred to the steel surface. Contractors are cautioned to verify that the chosen

abrasive, along with work and storage processes, allow the final surface cleanliness requirements to be achieved. Successful testing of contamination in abrasive does not negate the final acceptance testing of steel surfaces.

Non-ferrous abrasives must be used for abrasive blasting stainless steel piping.

2.6.1 Ferrous Abrasive

2.6.1.1 New and Remanufactured Steel Grit

New and remanufactured steel grit abrasive must conform to the chemical and physical properties of SSPC AB 3 Class 1 (Steel) only; Class 2 (Iron) abrasive must not be used. Modify the requirements of SSPC AB 3 to substitute the requirement in paragraph 4.2.2 CONDUCTIVITY for one chloride test as measured using the test kit described in this Section (paragraph TEST KIT FOR MEASURING CHLORIDES IN ABRASIVES). The maximum allowable chloride content is 25 parts per million (PPM).

To develop a suitable work mix from new steel abrasive, a minimum of 200 to 400 recycles is required; therefore, it may be advantageous for a Contractor to use remanufactured steel grit or grit reclaimed from a previous project. Such grit must be traced to new grit conforming to SSPC AB 3 Class 1 and it meets all cleanliness requirements of SSPC AB 3 Class 1 when brought to the current jobsite. Submit one representative sample of this work mix to the INDEPENDENT THIRD-PARTY laboratory for testing, along with samples of new material. Acceptance and use of this work mix must not be used to justify any deviation from surface preparation requirements.

2.6.1.2 Recycled Steel Grit

Recycled steel grit abrasive media must conform to the chemical and physical properties of SSPC AB 2 except that:

- a. The maximum allowable chromium and cadmium content of the work mix must be 0.1 percent by weight when tested in accordance with ASTM D3718 for chromium and ASTM D3335 for cadmium. Modify the requirements of SSPC AB 2 to add requirement for one chromate test and one cadmium test for each "LEAD" test required.
- b. The maximum allowable chloride content is 25 parts per million (PPM) as measured with the test kit described in paragraph TEST KIT FOR MEASURING CHLORIDES IN ABRASIVES. Modify the requirements of SSPC AB 2 to substitute requirement for one chloride test for each "WATER SOLUBLE CONTAMINANTS" test.

2.6.2 Non-Metallic Abrasive

Non-metallic abrasive must be graded to the appropriate surface profile range and must conform to the chemical and physical properties of SSPC AB 1, Class A except that:

- a. The maximum allowable chromium and cadmium content of the work mix must be less than 0.1 percent by weight when tested in accordance with ASTM D3718 for chromium and ASTM D3335 for cadmium.
- b. Must contain less than 7 PPM chlorides when tested with the kit

provided in paragraph TEST KIT FOR MEASURING CHLORIDES IN ABRASIVES. Modify the requirements of SSPC AB 1 to substitute requirement for one chloride test for each "CONDUCTIVITY TEST" required in SSPC AB 1 (one random sample per 50 bags of abrasive or three random samples from each shipment, if abrasive is delivered in bulk).

PART 3 EXECUTION

Perform all work, rework, and repair in accordance with approved procedures in the Coatings Work Plan. The Coatings Work Plan must be submitted and approved by the PCS prior to mobilization in accordance with the paragraph entitled COATINGS WORK PLAN.

3.1 FIELD SAMPLE COLLECTION AND TESTING

Sample and test materials delivered to the jobsite as required in the subsequent subparagraphs. Notify Contracting Officer three days in advance of sampling. The QC Manager, and either the PCS or the Quality Assurance Coatings Inspector, must witness all sampling.

3.1.1 Coating Field Sample Collection

All coatings/resins applied in the shop and in the field require one sample to be collected from each coating type used. This sample must be collected and set aside for the duration of the project, and must be tested if unforeseen coatings issues arise or if testing is requested by the Contracting Officer.

Field applied coatings require a random field sample from each lot of coating material used on-site in accordance with ASTM D3925. Each random sample must be tested. Utilize sample collection kits as outlined in the paragraph entitled COATING FIELD SAMPLE COLLECTION AND SHIPPING KIT. For two component liquid coatings, obtain a one liter sample of each base material, and proportional samples of each activator based on mix ratio, by random selection from sealed containers from each lot. Prior to sampling, mix contents of each sealed container to ensure uniformity. As an alternative to collecting small samples from kits, entire kits may be randomly selected and shipped to the THIRD-PARTY INDEPENDENT laboratory, observing all requirements for witnessing and traceability. For purposes of quality conformance inspection, a lot is defined as that quantity of materials from a single, uniform batch produced and offered for delivery at one time. A batch is defined as that quantity of material processed by the manufacturer at one time and identified by number on the label. Identify samples by designated name, specification number, batch number, project contract number, sample date, intended use, and quantity involved. The QC manager will take possession of the packaged samples, contact the shipping company to arrange for pickup, and ship one complete sample of each material in question (including powder or base and activator) with all batch information to the INDEPENDENT THIRD-PARTY laboratory for testing as required in paragraph COATING FIELD TESTING REPORTS.

3.1.2 Abrasive Field Sample Collection

Utilize the sample collection kits as required in paragraph ABRASIVE FIELD SAMPLE COLLECTION AND SHIPPING KIT to obtain samples from each lot of abrasive delivered to site using the sampling techniques and schedule of one sample per every 50 bags for ferrous metallic abrasive, paragraph 4 REQUIREMENTS FOR RECYCLED WORK MIX ABRASIVES of SSPC AB 2 for recycled

ferrous metallic abrasives, or paragraph 5.3 SAMPLING FOR QUALITY CONTROL TESTS of SSPC AB 1 for non-metallic abrasives.

For purposes of quality conformance inspection, a lot must consist of all abrasive materials of the same type from a single, uniform batch produced and offered for delivery at one time. The addition of any substance to a batch must constitute a new lot. Identify samples by designated name, specification number, lot number, project contract number, sample date, intended use, and quantity involved. The QC manager will take possession of the packaged samples, contact the shipping company to arrange for pickup, and relinquish the samples only to the shipping representative for shipment to the approved INDEPENDENT THIRD-PARTY laboratory for testing required in paragraph ABRASIVE SAMPLE TEST REPORTS.

3.1.3 Abrasive Field Test Reports

Submit test results for each lot of abrasive delivered to the jobsite. Test samples of ferrous metallic abrasive to the requirements of paragraph 5.2 TEST PARAMETERS of SSPC AB 3, excluding paragraph 5.2.4 DURABILITY. Test samples of recycled ferrous metallic abrasives to the requirements of paragraph 4 REQUIREMENTS FOR RECYCLED WORK MIX ABRASIVES of SSPC AB 2. Test samples of non-metallic abrasive to the requirements of paragraph 5.3 SAMPLING FOR QUALITY CONTROL TESTS of SSPC AB 1. Reject entire lot represented by samples that fail one or more tests, select new lots, and test samples.

3.2 LIGHTING

Provide lighting for all work areas as prescribed in SSPC Guide 12.

3.3 ENVIRONMENTAL CONDITIONS

3.3.1 Automated Monitoring Requirements for Shop Applied Coating

Provide continuous monitoring of temperature, relative humidity, and dew point data at pertinent points in the shop, during surface preparation, coating application, and initial cure. This data does not suffice for documentation of conformity to surface conditions during application and cure of coating. Locate sensors to provide pertinent data for the surface preparation and coating application being performed, as well as the temperature extremes on the structure. Describe the location plan in the Coatings Work Plan. Monitor any heating, cooling, or dehumidification equipment used. Provide monitoring equipment to perform as follows:

- a. Data is collected in 15-minute increments, and available for download (on-site) in a standard database format. Contractor must collect these data and make available to the Contracting Officer and QC Manager;
- b. Monitoring equipment must have backup power such that data collection will be uninterrupted during the entire period of the dehumidification requirement;
- c. Monitoring equipment must have capability to measure interior and exterior dry bulb temperature (DB), relative humidity (RH), and dewpoint temperature (DP).

There is no requirement for connectivity of the monitoring system to control the dehumidification equipment; therefore, any combination of equipment having the required functionality will be accepted.

3.3.2 Containment for Field Coating

Design and provide a containment system for the capture, containment, collection, storage, and disposal of the waste materials generated by the work under this Section, to meet the requirements of SSPC Guide 6, Class 2. Vapor concentrations must be kept at or below 10 percent of Lower Explosive Limit (LEL) at all times. Containment may be designed as fixed containment for complete structure or portable containment for sections of structure; however, containment must remain in any one place from beginning of abrasive blasting through initial cure of coating. Waste materials covered by this paragraph must not include any material or residue from removal of coatings containing lead, chromium, cadmium, PCB, or any other hazardous material.

It is the Contractors responsibility to insure the feasibility and workability of the containment system. The Contractor must perform their operations and work schedule in a manner as to minimize leakage of the containment system. The containment system must be properly maintained and must not deviate from the approved drawings. If the containment system fails to function satisfactorily, the Contractor must suspend all operations, except those required to minimize adverse impact on the environment or government property. Operations must not resume until modifications have been made to correct the cause of the failure.

3.3.3 Automated Monitoring Requirements for Field Coating

Provide continuous monitoring of temperature, relative humidity, and dew point data at pertinent points on the structure, during surface preparation, coating application, and initial cure. This data does not suffice for documentation of conformity to surface conditions during application and cure of coating. Locate sensors to provide pertinent data for the surface preparation and coating application being performed, as well as the temperature extremes on the structure. Describe the location plan, including required moves, in the Coatings Work Plan. Monitor any heating, cooling, or dehumidification equipment used. Provide monitoring equipment to perform as follows:

- a. Data is collected in the field unit in 15-minute increments, and available for download (on-site) in a standard database format. Contractor must collect these data and make available to the Contracting Officer, Quality Assurance Coatings Inspector, and QC Manager;
- b. Monitoring equipment must have backup power such that data collection will be uninterrupted during the entire period of the dehumidification requirement;
- c. Monitoring equipment must have capability to measure surface temperatures at a minimum of four locations anywhere on a structure, regardless of the size of the structure;
- d. Monitoring equipment must have capability to measure interior and exterior dry bulb temperature (DB), relative humidity (RH), and dewpoint temperature (DP).

There is no requirement for connectivity of the monitoring system to control the dehumidification equipment; therefore, any combination of equipment having the required functionality will be accepted.

3.4 SHOP APPLIED COATING OF BURIED PIPING

3.4.1 Surface Preparation for Shop Applied Coating

Prepare exterior steel surfaces in accordance with AWWA C213.

3.4.1.1 Abrasive Blasting Equipment for Shop Applied Coating of Buried Piping

Confirm that air supply for abrasive blasting is free of oil and moisture when tested in accordance with ASTM D4285. Test air quality at each startup, but in no case, less often than every five operating hours.

3.4.1.2 Contamination Evaluation of Abrasive for Shop Applied Coating of Buried Piping

Test abrasive for salt contamination and oil contamination as required SSPC AB 1 for non-metallic abrasives, SSPC AB 2 for recycled ferrous abrasives, and SSPC AB 3 for ferrous metallic abrasives. Modify the schedule of testing to be daily, at startup, and every five operating hours thereafter.

3.4.1.3 Pre-Preparation Testing for Surface Contamination for Shop Applied Coating of Buried Piping

3.4.1.3.1 Pre-Preparation Testing for Oil and Grease Contamination for Shop Applied Coating of Buried Piping

Inspect all surfaces for oil or grease contamination using two or more of the following inspection techniques:

- 1) VISUAL INSPECTION - Observe surface for evidence of dirt or oil.
- 2) WATER BREAK TEST - Spray atomized mist of distilled water onto surface and observe for water beading. If water wets surface rather than beading up, surface can be considered free of oil or grease contamination. Beading of water (water forms droplets) is evidence of oil or grease contamination.
- 3) CLOTH RUB TEST - Rub a clean, white, lint-free, cotton cloth onto surface and observe for discoloration. To confirm oil or grease contamination in lightly stained areas, a non-staining solvent may be used to aid in oil or grease extraction. Any visible discoloration is evidence of oil or grease contamination.

Reject oil or grease contaminated surfaces, clean using a water based pH neutral degreaser in accordance with SSPC SP 1, and recheck for contamination until surfaces are free of oil and grease.

3.4.1.3.2 Pre-Preparation Testing for Soluble Salts Contamination for Shop Applied Coating of Buried Piping

Test each lot of pipe for soluble salt contamination on the surfaces using the test kit described in paragraph TEST KIT FOR MEASURING CHLORIDE, SULFATE, AND NITRATE IONS ON STEEL AND COATED SURFACES. For pipes with an outer diameter (OD) of less than 2.54 centimeters, one test per 6.09 meters is required. For pipes with an OD of more than 2.54 centimeters, three tests per 9.29 square meters must be taken, with a minimum of one test per

section of pipe. Label all test tubes and retain for test verification. Reject and wash surfaces of the entire lot if one or more readings greater than 3 micrograms per square centimeter of chlorides or 10 micrograms per square centimeter of sulfates or 5 micrograms per square centimeter of nitrates is measured.

Effective washing and removal of soluble salts will require removal of any barrier to the steel surface, including rust or mill scale. This procedure may necessitate combinations of wet abrasive blasting, high pressure water rinsing, and cleaning using a solution of water and soluble salt remover. The soluble salt remover must be acidic, biodegradable, non-toxic, non-corrosive, and after application, will not interfere with primer adhesion. Use potable water, or potable water modified with soluble salt remover, for all washing or wet abrasive blasting. Test methods and equipment used in this phase must be included in the Coatings Work Plan.

This phase is required because pre-preparation testing and washing are generally more advantageous than attempting to remove soluble salt contamination after abrasive blasting. Soluble salt testing is also required in paragraph PRE-APPLICATION TESTING FOR SOLUBLE SALTS CONTAMINATION as a final acceptance test of prepared surfaces after abrasive blasting. Successful completion of pre-preparation testing and washing does not negate the requirement for pre-application testing.

3.4.1.4 Abrasive Blasting External Surfaces for Shop Applied Coating of Buried Piping

For carbon steel, prepare surfaces in accordance with SSPC SP 10/NACE No. 2. The surface profile must be in accordance with AWWA C213 and measured in accordance with ASTM D4417, Method A and Method C. The appearance of the surface after blasting must have the appearance of a Sand or Grit comparator. A rounded profile shape or peened surface is not acceptable. Measure profile at a rate of three measurements per lot at the beginning of the day.

3.4.1.5 Disposal of Used Abrasive for Shop Applied Coating of Buried Piping

Provide Disposal of Used Abrasive Shop Applied Coating, in accordance with Local, EU, and Host Nation mandated regulations.

3.4.1.6 Pre-Application Testing for Surface Contamination for Shop Applied Coating of Buried Piping

3.4.1.6.1 Pre-Application Testing for Oil and Grease Contamination for Shop Applied Coating of Buried Piping

Ensure surfaces are free of contamination as described in paragraph PRE-PREPARATION TESTING FOR OIL AND GREASE CONTAMINATION FOR SHOP APPLIED COATING OF BURIED PIPING, except that only questionable areas need be checked for beading of water misted onto surface.

3.4.1.6.2 Pre-Application Testing for Surface Cleanliness for Shop Applied Coating of Buried Piping

Apply coatings to dust free surfaces. To test surfaces, use ISO 8502-3. Use a kit that is compliant with ISO 8502-3. A rating of 2 or better must be achieved for acceptance. If a test does not result in a rating of 2 or better, reject contaminated surfaces, clean by vacuum cleaning, and

retest. Test surfaces at rate of one test per joint or repair area where application will be performed. Provide two additional tests for each failed test or questionable test. Document test results in the Daily Inspection Report and attach tape to the Inspection Logbook.

Ferrous abrasives may become magnetized and difficult to remove from the steel substrate. If ferrous abrasives are used, additional visual inspection must be performed to ensure no surface contamination by the abrasive is present.

3.4.2 Mixing and Application of Shop Applied Coating of Buried Piping

3.4.2.1 External Pipe Shop Applied Coating Application

Apply coatings in accordance with SSPC PA 1 and as specified in AWWA C213. Apply coatings to surfaces that meet all stated surface preparation requirements.

For the application of AWWA C213, the DFT must be a minimum of 300 microns. The coating thickness must not exceed the manufacturers recommendations.

3.4.2.2 Holiday Testing and Shop Repair

Perform holiday testing on all shop-applied, externally coated areas and repair as outlined in AWWA C213.

3.5 FIELD EXTERIOR COATING OF ABOVEGROUND PIPING

Coat aboveground, carbon steel piping in accordance with Section 09 97 13.27 HIGH PERFORMANCE COATING FOR STEEL STRUCTURES.

3.6 FIELD REPAIRS TO EXTERNAL COATING OF BURIED PIPING

Field repairs must only be performed when repairing shop applied coatings in the field and coating external girth welds.

3.6.1 Surface Preparation for Field Coating of Buried Piping

Block or suspended pipeline at a height that will allow the blast nozzle to be perpendicular to the surface being blasted, and at the proper standoff distance, at all times.

3.6.1.1 Abrasive Blasting Equipment for Field Coating of Buried Piping

Use abrasive blasting equipment of conventional air, force-feed, or pressure type. Maintain a minimum pressure of 650 kPa at nozzle. Confirm that air supply for abrasive blasting is free of oil and moisture when tested in accordance with ASTM D4285. Test air quality at each startup, but in no case, less often than every five operating hours.

3.6.1.2 Contamination Evaluation of Abrasive for Field Coating of Buried Piping

Test abrasive for salt contamination and oil contamination as required SSPC AB 1 for non-ferrous abrasives, SSPC AB 2 for recycled ferrous abrasives, and SSPC AB 3 for ferrous abrasives. Modify the schedule of testing to be daily, at startup, and every five operating hours thereafter.

3.6.1.3 Surface Standard

Inspect surfaces to be coated, and select plate with similar properties and surface characteristics for use as a surface standard. Blast clean one or more 300 mm square steel panels in the manner specified in paragraph ABRASIVE BLASTING FOR FIELD COATING OF BURIED PIPING. Record blast nozzle type and size, air pressure at nozzle and compressor, distance of nozzle from panel, and angle of blast to establish procedures for blast cleaning. Measure surface profile in accordance with ASTM D4417, Method C. When the surface standard complies with all specified requirements, seal with a clearcoat protectant. Use the surface standard for comparison to abrasive blasted surfaces throughout the course of work.

3.6.1.4 Pre-Preparation Testing for Surface Contamination for Field Coating of Buried Piping

3.6.1.4.1 Pre-Preparation Testing for Oil and Grease Contamination for Field Coating of Buried Piping

Inspect all surfaces for oil or grease contamination using two or more of the following inspection techniques:

- 1) VISUAL INSPECTION - Observe surface for evidence of dirt or oil.
- 2) WATER BREAK TEST - Spray atomized mist of distilled water onto surface and observe for water beading. If water wets surface rather than beading up, surface can be considered free of oil or grease contamination. Beading of water (water forms droplets) is evidence of oil or grease contamination.
- 3) CLOTH RUB TEST - Rub a clean, white, lint-free, cotton cloth onto surface and observe for discoloration. To confirm oil or grease contamination in lightly stained areas, a non-staining solvent may be used to aid in oil or grease extraction. Any visible discoloration is evidence of oil or grease contamination.

Reject oil or grease contaminated surfaces, clean using a water based pH neutral degreaser in accordance with SSPC SP 1, and recheck for contamination until surfaces are free of oil and grease.

3.6.1.4.2 Pre-Preparation Testing for Soluble Salts Contamination for Field Coating of Buried Piping

Prior to abrasive blasting, test all exposed girth welds and areas to be recoated for soluble salt contamination using the test kit described in paragraph TEST KIT FOR MEASURING CHLORIDE, SULFATE, AND NITRATE IONS ON STEEL AND COATED SURFACES. Label all test tubes and retain for test verification. Reject and wash surfaces if one or more readings greater than 3 micrograms per square centimeter of chlorides or 10 micrograms per square centimeter of sulfates or 5 micrograms per square centimeter of nitrates is measured.

Effective washing and removal of soluble salts will require removal of any barrier to the steel surface, including rust. This procedure may necessitate combinations of wet abrasive blasting, high pressure water rinsing, and cleaning using a solution of water and soluble salts remover. The soluble salts remover must be acidic, biodegradable, non-toxic, non-corrosive, and after application, will not interfere with coating adhesion. Delays between testing and preparation, or testing and

coating application, may allow for the formation of new contamination. Use potable water, or potable water modified with soluble salt remover, for all washing or wet abrasive blasting. Test methods and equipment used in this phase must be included in the Coatings Work Plan.

This phase is required because pre-preparation testing and washing are generally more advantageous than attempting to remove soluble salt contamination after abrasive blasting. Soluble salt testing is also required in paragraph PRE-APPLICATION TESTING FOR SOLUBLE SALTS CONTAMINATION FOR FIELD COATING OF BURIED PIPING as a final acceptance test of prepared surfaces after abrasive blasting. Successful completion of pre-preparation testing and washing does not negate the requirement for pre-application testing.

3.6.1.5 Abrasive Blasting for Field Coating of Buried Piping

For carbon steel, prepare girth welds and repairs to bare steel to SSPC SP 10/NACE No. 2 immediately prior to coating application. Verify that prepared surfaces comply with SSPC VIS 1 and match the prepared test-panels from the paragraph SURFACE STANDARD at the time of coating application. The surface profile must be in accordance with the coating manufacturer's recommended range. Measure the surface profile in accordance with ASTM D4417, Method A and Method C. The appearance of the surface after blasting must have the appearance of a Sand or Grit comparator. A rounded profile shape or peened surface is not acceptable. For pipes with an outer diameter (OD) of less than 2.54 centimeters, one test per 6.09 meters is required. For pipes with an OD of more than 2.54 centimeters, three tests per 9.29 square meters must be taken, with a minimum of one test per section of pipe. When surfaces are re-blasted for any reason, retest profile as specified. Following abrasive blasting, remove dust and debris by vacuum cleaning. Dust and debris tend to collect at welds, plate overlaps, and surface irregularities. Do not attempt to wipe surface clean.

3.6.1.6 Disposal of Used Abrasive for Field Coating of Buried Piping

Provide Disposal of Used Abrasive Field Applied Coating off Government property in accordance with Local, EU, and Host Nation mandated regulations.

3.6.1.7 Pre-Application Testing for Surface Contamination for Field Coating of Buried Piping

3.6.1.7.1 Pre-Application Testing for Oil and Grease Contamination for Field Coating of Buried Piping

Ensure surfaces are free of contamination as described in paragraph PRE-PREPARATION TESTING FOR OIL AND GREASE CONTAMINATION FOR FIELD COATING OF BURIED PIPING, except that only questionable areas need to be checked for beading of water misted onto surface.

3.6.1.7.2 Pre-Application Testing for Soluble Salts Contamination for Field Coating of Buried Piping

Test surfaces for soluble salts using the test kit described in paragraph TEST KIT FOR MEASURING CHLORIDE, SULFATE, AND NITRATE IONS ON STEEL AND COATED SURFACES, test girth welds and areas to be recoated for soluble salts. Label all test tubes and retain for test verification. One or more readings greater than 3 micrograms per square centimeter of chlorides

or 10 micrograms per square centimeter of sulfates or 5 micrograms per square centimeter of nitrates is evidence of soluble salt contamination. Reject contaminated surfaces, wash as required in paragraph PRE-PREPARATION TESTING FOR SOLUBLE SALTS CONTAMINATION FOR FIELD COATING OF BURIED PIPING, allow to dry, and re-test, and re-wash until all required tests show allowable results. Re-blast tested areas using vacuum equipped blast equipment.

3.6.1.7.3 Pre-Application Testing for Surface Cleanliness for Field Coating of Buried Piping

Apply coatings to dust free surfaces. To test surfaces, use ISO 8502-3. Use a kit that is compliant with ISO 8502-3. A rating of 2 or better must be achieved for acceptance. If a test does not result in a rating of 2 or better, reject contaminated surfaces, clean by vacuum cleaning, and retest. Test surfaces at rate of one test per joint or repair area where application will be performed. Provide two additional tests for each failed test or questionable test. Document test results in the Daily Inspection Reports and attach tape to the Inspection Logbook.

Ferrous abrasives may become magnetized and difficult to remove from the steel substrate. If ferrous abrasives are used, additional visual inspection must be performed to ensure no surface contamination by the abrasive is present.

3.6.2 Final Inspection of Pipeline Prior to Burial

Verify that all surfaces of the pipeline are holiday-free at time of placement of backfill over pipe. Use holiday inspection requirements and acceptance criteria of the standards applicable to the coatings being tested.

3.6.3 Final Cleanup

Following completion of the work, remove debris, equipment, and materials from the site. Remove all foreign matter such as blast media, dust, dirt, debris, grease, and oils. Wipe all dry to handle coated surfaces with damp lint-free cloth. Remove temporary connections to Government- or Contractor-furnished water and electrical services. Restore existing facilities in and around the work areas to their original condition.

3.7 FIELD AND SHOP QUALITY CONTROL

For marking of surfaces, use chalk for marking bare steel and water based markers for marking coated surfaces. Remove marks prior to coating. Do not use any wax or grease based markers, or any other markers that leave a residue or stain.

3.7.1 Inspection

3.7.1.1 Inspection Requirements

- a. Perform inspections in accordance with ASTM D3276 and the approved Coatings Work Plan.
- b. Provide all tools and instruments required to perform the required testing, as well as any tools or instruments that the inspector considers necessary to perform the required inspections and tests. Document each inspection and test, including required hold points and

other required inspections and tests, as well as those inspections and tests deemed prudent from on-site evaluation to document a particular process or condition, as follows:

- (1) Location or area;
 - (2) Purpose (required or special);
 - (3) Method;
 - (4) Criteria for evaluation;
 - (5) Results;
 - (6) Determination of compliance;
 - (7) List of required rework;
 - (8) Observations.
- c. Collect and record environmental conditions as described in ASTM D3276 on a 24 hour basis, as follows:
- (1) During surface preparation, every hour, or when changes occur;
 - (2) During coating application and the first four days of initial cure, every hour, or when changes occur;
 - (3) Note location, time, and temperature of the highest and lowest surface temperatures each day;
 - (4) Use a non-contact thermometer to locate temperature extremes, then verify with contact thermometers.
- d. Data collected on environmental conditions in AUTOMATED MONITORING REQUIREMENTS may be used for overnight data; however, the data must be constantly verified as to location of sensors and validity of data with respect to the coating work being accomplished.
- e. Document all equipment used in inspections and testing, including manufacturer, model number, serial number, last calibration date and future calibration date, and results of on-site calibration performed. Work documented using data from equipment found to be out of calibration must be considered as non-compliant since last calibration or calibration check, as required.
- f. Document Contractors compliance with the Coatings Work Plan.

3.7.1.2 Inspection Report Forms

Develop project-specific report forms as required to report measurements, test results, and observations being complete and conforming to contract requirements. This includes all direct requirements of the contract documents and indirect requirements of referenced documents. Show acceptance criteria with each requirement and indication of conformity of each inspected item. The data may be in any format, but must be legible and presented so that entered data can be quickly compared to the appropriate requirement.

3.7.1.3 Daily Inspection Reports

Submit one copy of daily inspection report completed each day when performing work under this Section, to the Contracting Officer. Note all non-compliance issues, and all issues that were reported for rework in accordance with QC procedures of Section 01 45 00.00 20 QUALITY CONTROL. Each report must be signed by the Quality Assurance Coatings Inspector and the QC Manager. Submit report within 24 hours of date recorded on the report.

3.7.1.4 Inspection Logbook

A continuous record of all activity related to this Section must be maintained in an Inspection Logbook on a daily basis. The logbook must be hard or spiral-bound book or digital program with consecutively numbered pages, and must be used to record all information provided in the Daily Inspection Reports, as well as other pertinent observations and information. Submit the original Inspection Logbook to the Contracting Officer upon completion of the project and prior to final payment.

3.7.1.5 Inspection Equipment

All equipment must be in good condition, operational within its design range, and calibrated as required by the specified standard for use of each device.

3.7.2 Coatings Contractor QC Coatings Inspector's Responsibilities

The Coatings Contractor QC Coatings Inspector responsibilities include complete documentation of all daily inspection and production activities for the entire coatings project as outlined in the Coatings Work Plan, project program, and this Section. This includes, but is not limited to, the following:

- a. Attending and documenting the pre-job meeting and acquiring the project program, inspection and testing plan (ITP), schedule, and a list of who will receive the QC daily inspection reports;
- b. Performing a project site walk-through with the Quality Assurance Evaluator (QAE) or asset owner, Coatings Contractor QC, QC Manager, and Quality Assurance Coatings Inspector, inspecting at least the following:
 - (1) Asset(s) to be coated;
 - (2) Equipment and placement of equipment;
 - (3) Materials delivery and storage;
 - (4) Facility operational requirements during the project.
- c. Perform all daily and hold point inspections including, but not limited to, the following:
 - (1) Check equipment, including blotter test to verify compressed air cleanliness;
 - (2) Perform non-visible contaminants testing (in accordance with PRE-PREPARATION TESTING FOR SOLUBLE SALT CONTAMINATION FOR SHOP

APPLIED COATING OF BURIED PIPING, PRE-APPLICATION TESTING FOR SOLUBLE SALT CONTAMINATION FOR SHOP APPLIED COATING OF BURIED PIPING, PRE-PREPARATION TESTING FOR SOLUBLE SALT CONTAMINATION FOR FIELD COATING OF BURIED PIPING, and PRE-APPLICATION TESTING FOR SOLUBLE SALT CONTAMINATION FOR FIELD COATING OF BURIED PIPING);

- (3) Perform visible contaminants testing (in accordance with PRE-PREPARATION TESTING FOR OIL AND GREASE CONTAMINATION FOR SHOP APPLIED COATING OF BURIED PIPING, PRE-APPLICATION TESTING FOR OIL AND GREASE CONTAMINATION FOR SHOP APPLIED COATING OF BURIED PIPING, PRE-PREPARATION TESTING FOR OIL AND GREASE CONTAMINATION FOR FIELD COATING OF BURIED PIPING, and PRE-APPLICATION TESTING FOR OIL AND GREASE CONTAMINATION FOR FIELD COATING OF BURIED PIPING);
 - (4) Obtain environmental readings;
 - (5) Perform abrasive testing per SSPC AB 1, SSPC AB 2, or SSPC AB 3;
 - (6) Perform surface preparation monitoring and testing;
 - (7) Perform surface cleanliness testing;
 - (8) Perform dust quantity testing;
 - (9) Record materials storage documentation (record all coatings and abrasive materials information, batch numbers, segregation, and storage temperature);
 - (10) Witness all coatings materials mixing and record mix materials temperatures, with verification of time of coatings pot life;
 - (11) Verify, witness, and record application method;
 - (12) Perform random wet film thickness (WFT) readings;
 - (13) Perform inspection of coatings application;
 - (14) Obtain dry film thickness (DFT) readings per SSPC PA 2;
 - (15) Perform holiday testing in accordance with HOLIDAY TESTING AND SHOP REPAIR and HOLIDAY TESTING AND FIELD REPAIR;
 - (16) Label asset identification (label stickers);
 - (17) Write Correction Action Reports (CAR), if needed;
 - (18) Write Non-Conformance Reports (NCR), if needed.
- d. Writing a daily detailed summary of the work shift inspections, testing, and the day's events, including any meetings and prevalent conversations. The final daily report must include a project summary that must be part of the daily coatings inspection report.
 - e. The Coatings Contractor QC Coatings Inspector must stop all non-compliant work.

3.7.3 Quality Assurance Coatings Inspector's Responsibilities

The Quality Assurance Coatings Inspector's responsibilities include complete documentation of all on-site work associated with the coatings project. These responsibilities include, but are not limited to, the following:

- a. Attending and documenting the pre-job meeting and acquiring the project program, inspection and testing plan (ITP), schedule, and a list of who will receive the QC daily inspection reports;
- b. Performing a project site walk-through with the Quality Assurance Evaluator (QAE) or asset owner, prime Contractor, and coatings Contractor (QC Coatings Inspector and QC Manager), inspecting at least the following:
 - (1) Asset(s) to be coated;
 - (2) Equipment and placement of equipment;
 - (3) Materials delivery and storage;
 - (4) Facility operational requirements during the project.
- c. Verifying all daily and hold point inspections performed by the Coatings Contractor QC Coatings Inspector or QC Manager by performing mirror inspections including, but not limited to, the following:
 - (1) Verify equipment check, including blotter test to verify compressed air cleanliness;
 - (2) Verify visible contaminants testing;
 - (3) Take environmental readings;
 - (4) Perform surface preparation monitoring and testing;
 - (5) Perform surface cleanliness testing;
 - (6) Perform dust quantity test;
 - (7) Record materials storage documentation (record all coating and abrasive information, batch numbers, segregation, and storage temperature);
 - (8) Witness all coatings materials mixing and record mix materials temperatures, with verification of time of coatings pot life;
 - (9) Verify, witness, and record application method;
 - (10) Inspect coatings application;
 - (11) Perform dry film thickness (DFT) readings per SSPC PA 2;
 - (12) Inspect asset identification (label stickers);
 - (13) Write Correction Action Reports (CAR), if needed;
 - (14) Write Non-Conformance Reports (NCR), if needed.

- d. The following testing is witnessed by the Quality Assurance Coatings Inspector and performed by the Coatings Contractor QC Coatings Inspector or QC Manager:
- (1) Wet film thickness (WFT) readings by coatings applicator(s);
 - (2) Non-visible contaminants testing for chlorides, sulfates, and nitrates (CSN);
 - (3) Abrasive testing per SSPC AB 1, SSPC AB 2, or SSPC AB 3;
 - (4) Holiday testing.
- e. Writing a daily detailed summary of the work shift inspections, testing, and the day's events, including any meetings and prevalent conversations. The final daily report must include a project summary that will be part of the last daily coatings inspection report.

3.8 PROJECT IDENTIFICATION

At the completion of the work, affix pertinent coating data on structure at a location that is readily accessible and visible from the ground. Use either stencils or nameplates. The following list generally describes the pertinent coating data, but should be modified as required to describe the coating systems.

Date coated/accepted: _____/_____

Project Number: _____

Contractor: _____

Address: _____

Coating System

Manufacturer: _____

Surface Prep: SSPC SP ____ Profile: _____

Primer: _____ Thickness: _____

Intermediate: _____ Thickness: _____

Topcoat: _____ Thickness: _____

Total Thickness: _____

3.8.1 Stencils

Use stencils on piping 203 mm or larger. Use stencils with 3/4- to one-inch Helvetica style letters and acrylic stencil paint of contrasting color.

-- End of Section --

SECTION TABLE OF CONTENTS

DIVISION 33 - UTILITIES

SECTION 33 71 02

UNDERGROUND ELECTRICAL DISTRIBUTION

08/21

PART 1 GENERAL

- 1.1 REFERENCES
- 1.2 SYSTEM DESCRIPTION
- 1.3 RELATED REQUIREMENTS
- 1.4 DEFINITIONS
- 1.5 SUBMITTALS
- 1.6 QUALITY ASSURANCE
 - 1.6.1 Precast Underground Structures
 - 1.6.2 Certificate of Competency for Cable Splicer/Terminator
 - 1.6.3 Cable Installer Qualifications
 - 1.6.4 Directional Boring Certificate of Conformance
 - 1.6.5 Regulatory Requirements
 - 1.6.6 Standard Products
 - 1.6.6.1 Alternative Qualifications
 - 1.6.6.2 Material and Equipment Manufacturing Date

PART 2 PRODUCTS

- 2.1 CONDUIT, DUCTS, AND FITTINGS
 - 2.1.1 Rigid Metal Conduit
 - 2.1.1.1 Rigid Metallic Conduit, PVC Coated
 - 2.1.2 Intermediate Metal Conduit
 - 2.1.2.1 Intermediate Metal Conduit, PVC Coated
 - 2.1.3 Plastic Conduit for Direct Burial and Riser Applications
 - 2.1.4 Plastic Duct for Concrete Encasement
 - 2.1.5 Duct Sealant
 - 2.1.6 Fittings
 - 2.1.6.1 Metal Fittings
 - 2.1.6.2 PVC Conduit Fittings
 - 2.1.6.3 PVC Duct Fittings
 - 2.1.6.4 Outlet Boxes for Steel Conduit
- 2.2 LOW VOLTAGE INSULATED CONDUCTORS AND CABLES
 - 2.2.1 Conductor Types
 - 2.2.2 Conductor Material
 - 2.2.3 Jackets
 - 2.2.4 Direct Buried
 - 2.2.5 In Duct
 - 2.2.6 Cable Marking
- 2.3 LOW VOLTAGE WIRE CONNECTORS AND TERMINALS
- 2.4 LOW VOLTAGE SPLICES
 - 2.4.1 Heat Shrinkable Splice
 - 2.4.2 Cold Shrink Rubber Splice
- 2.5 TAPE
 - 2.5.1 Insulating Tape
 - 2.5.2 Buried Warning and Identification Tape

- 2.5.3 Fireproofing Tape
- 2.6 PULL ROPE
- 2.7 GROUNDING AND BONDING
 - 2.7.1 Driven Ground Rods
 - 2.7.2 Grounding Conductors
- 2.8 CAST-IN-PLACE CONCRETE
- 2.9 UNDERGROUND STRUCTURES
 - 2.9.1 Cast-In-Place Concrete Structures
 - 2.9.2 Precast Concrete Structures, Risers and Tops
 - 2.9.2.1 General
 - 2.9.2.2 Design for Precast Structures
 - 2.9.2.3 Construction
 - 2.9.2.4 Joints
- 2.10 CABLE SUPPORTS (RACKS, ARMS, AND INSULATORS)
 - 2.10.1 Cable Rack Stanchions
 - 2.10.2 Rack Arms
 - 2.10.3 Insulators
- 2.11 CABLE TAGS IN MANHOLES
 - 2.11.1 Polyethylene Cable Tags
- 2.12 PROTECTIVE DEVICES AND COORDINATION
- 2.13 SOURCE QUALITY CONTROL
 - 2.13.1 Arc-Proofing Test for Cable Fireproofing Tape

PART 3 EXECUTION

- 3.1 INSTALLATION
- 3.2 CABLE INSPECTION
- 3.3 CABLE INSTALLATION PLAN AND PROCEDURE
- 3.4 UNDERGROUND FEEDERS SUPPLYING BUILDINGS
- 3.5 UNDERGROUND STRUCTURE CONSTRUCTION
 - 3.5.1 Cast-In-Place Concrete Structures
 - 3.5.2 Precast Concrete Construction
 - 3.5.3 Pulling-In Irons
 - 3.5.4 Cable Racks, Arms and Insulators
 - 3.5.5 Field Painting
- 3.6 UNDERGROUND CONDUIT AND DUCT SYSTEMS
 - 3.6.1 Requirements
 - 3.6.2 Treatment
 - 3.6.3 Conduit Cleaning
 - 3.6.4 Jacking and Drilling Under Roads and Structures
 - 3.6.5 Galvanized Conduit Concrete Penetrations
 - 3.6.6 Multiple Conduits
 - 3.6.7 Conduit Plugs and Pull Rope
 - 3.6.8 Conduit and Duct Without Concrete Encasement
 - 3.6.8.1 Encasement Under Roads and Structures
 - 3.6.9 Duct Encased in Concrete
 - 3.6.9.1 Connections to Manholes
 - 3.6.9.2 Connections to Existing Underground Structures
 - 3.6.9.3 Connections to Existing Concrete Pads
 - 3.6.9.4 Connections to Existing Ducts
 - 3.6.9.5 Partially Completed Duct Banks
 - 3.6.9.6 Removal of Ducts
 - 3.6.10 Duct Sealing
- 3.7 CABLE PULLING
 - 3.7.1 Cable Lubricants
- 3.8 CABLES IN UNDERGROUND STRUCTURES
 - 3.8.1 Cable Tag Installation
- 3.9 CONDUCTORS INSTALLED IN PARALLEL
- 3.10 LOW VOLTAGE CABLE SPLICING AND TERMINATING

- 3.11 CABLE END CAPS
- 3.12 FIREPROOFING OF CABLES IN UNDERGROUND STRUCTURES
 - 3.12.1 Fireproofing Tape
 - 3.12.2 Tape-Wrap
- 3.13 GROUNDING SYSTEMS
 - 3.13.1 Grounding Electrodes
 - 3.13.2 Grounding Connections
 - 3.13.3 Grounding Conductors
 - 3.13.4 Ground Cable Crossing Expansion Joints
 - 3.13.5 Manhole Grounding
 - 3.13.6 Metal Splice Case Grounding
- 3.14 EXCAVATING, BACKFILLING, AND COMPACTING
 - 3.14.1 Reconditioning of Surfaces
 - 3.14.1.1 Unpaved Surfaces
 - 3.14.1.2 Paving Repairs
- 3.15 CAST-IN-PLACE CONCRETE
 - 3.15.1 Concrete Slabs (Pads) for Equipment
 - 3.15.2 Sealing
- 3.16 FIELD QUALITY CONTROL
 - 3.16.1 Performance of Field Acceptance Checks and Tests
 - 3.16.1.1 Low Voltage Cables, 600-Volt
 - 3.16.1.2 Grounding System
 - 3.16.2 Follow-Up Verification

-- End of Section Table of Contents --

SECTION 33 71 02

UNDERGROUND ELECTRICAL DISTRIBUTION
08/21

PART 1 GENERAL

1.1 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

AMERICAN ASSOCIATION OF STATE HIGHWAY AND TRANSPORTATION OFFICIALS
(AASHTO)

AASHTO HB-17 (2002; Errata 2003; Errata 2005, 17th Edition) Standard Specifications for Highway Bridges

AMERICAN CONCRETE INSTITUTE (ACI)

ACI 318M (2014; ERTA 2015) Building Code Requirements for Structural Concrete & Commentary

ACI SP-66 (2004) ACI Detailing Manual

ASTM INTERNATIONAL (ASTM)

ASTM B1 (2013) Standard Specification for Hard-Drawn Copper Wire

ASTM B3 (2013) Standard Specification for Soft or Annealed Copper Wire

ASTM B8 (2011; R 2017) Standard Specification for Concentric-Lay-Stranded Copper Conductors, Hard, Medium-Hard, or Soft

ASTM C309 (2019) Standard Specification for Liquid Membrane-Forming Compounds for Curing Concrete

ASTM C478M (2018) Standard Specification for Precast Reinforced Concrete Manhole Sections (Metric)

ASTM C857 (2016) Standard Practice for Minimum Structural Design Loading for Underground Precast Concrete Utility Structures

ASTM C990M (2009; R 2019) Standard Specification for Joints for Concrete Pipe, Manholes, and Precast Box Sections Using Preformed Flexible Joint Sealants (Metric)

INSTITUTE OF ELECTRICAL AND ELECTRONICS ENGINEERS (IEEE)

- IEEE 81 (2012) Guide for Measuring Earth Resistivity, Ground Impedance, and Earth Surface Potentials of a Ground System
- IEEE C2 (2017; Errata 1-2 2017; INT 1 2017) National Electrical Safety Code
- IEEE Stds Dictionary (2009) IEEE Standards Dictionary: Glossary of Terms & Definitions

INTERNATIONAL ELECTRICAL TESTING ASSOCIATION (NETA)

- NETA ATS (2021) Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

- ANSI C119.1 (2016) Electric Connectors - Sealed Insulated Underground Connector Systems Rated 600 Volts
- NEMA RN 1 (2005; R 2013) Polyvinyl-Chloride (PVC) Externally Coated Galvanized Rigid Steel Conduit and Intermediate Metal Conduit
- NEMA TC 2 (2020) Standard for Electrical Polyvinyl Chloride (PVC) Conduit
- NEMA TC 7 (2021) Smooth-Wall Coilable and Straight Electrical Polyethylene Conduit
- NEMA TC 9 (2020) Standard for Fittings for Polyvinyl Chloride (PVC) Plastic Utilities Duct for Underground Installation

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

- NFPA 70 (2020; ERTA 20-1 2020; ERTA 20-2 2020; ERTA 20-3 2020; TIA 20-1; TIA 20-2; TIA 20-3; TIA 20-4; TIA 20-5; TIA 20-6; TIA 20-7; TIA 20-8; TIA 20-9; TIA 20-10; TIA 20-11; TIA 20-12; TIA 20-13; TIA 20-14; TIA 20-15; TIA 20-16; ERTA 20-4 2022) National Electrical Code

TELECOMMUNICATIONS INDUSTRY ASSOCIATION (TIA)

- TIA-758 (2012b) Customer-Owned Outside Plant Telecommunications Infrastructure Standard

U.S. DEPARTMENT OF AGRICULTURE (USDA)

- RUS Bull 1751F-644 (2002) Underground Plant Construction

UNDERWRITERS LABORATORIES (UL)

UL 6	(2007; Reprint Sep 2019) UL Standard for Safety Electrical Rigid Metal Conduit-Steel
UL 44	(2018; Reprint May 2021) UL Standard for Safety Thermoset-Insulated Wires and Cables
UL 83	(2017; Reprint Mar 2020) UL Standard for Safety Thermoplastic-Insulated Wires and Cables
UL 94	(2013; Reprint Apr 2022) UL Standard for Safety Tests for Flammability of Plastic Materials for Parts in Devices and Appliances
UL 467	(2022) UL Standard for Safety Grounding and Bonding Equipment
UL 486A-486B	(2018; Reprint May 2021) UL Standard for Safety Wire Connectors
UL 510	(2020) UL Standard for Safety Polyvinyl Chloride, Polyethylene and Rubber Insulating Tape
UL 514A	(2013; Reprint Aug 2017) UL Standard for Safety Metallic Outlet Boxes
UL 514B	(2012; Reprint May 2020) Conduit, Tubing and Cable Fittings
UL 651	(2011; Reprint May 2022) UL Standard for Safety Schedule 40, 80, Type EB and A Rigid PVC Conduit and Fittings
UL 1242	(2006; Reprint Apr 2022) UL Standard for Safety Electrical Intermediate Metal Conduit -- Steel

1.2 SYSTEM DESCRIPTION

Items provided under this section must be specifically suitable for the following service conditions. Seismic details must conform to UFC 3-301-01, "Structural Engineering".

- a. Altitude 10 m.
- b. Ambient Temperature 55 degrees C.
- c. Frequency 60 Hz

1.3 RELATED REQUIREMENTS

Section 26 08 00 APPARATUS INSPECTION AND TESTING applies to this section, with the additions and modifications specified herein.

1.4 DEFINITIONS

- a. Unless otherwise specified or indicated, electrical and electronics terms used in these specifications, and on the drawings, are as defined in IEEE Std's Dictionary.
- b. In the text of this section, the words conduit and duct are used interchangeably and have the same meaning.
- c. In the text of this section, "medium voltage cable splices," and "medium voltage cable joints" are used interchangeably and have the same meaning.
- d. Underground structures subject to aircraft loading are indicated on the drawings.

1.5 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Precast Underground Structures; G

SD-03 Product Data

Precast Concrete Structures; G

Sealing Material

Pulling-In Irons

Cable Supports (racks, arms and insulators); G

Protective Devices and Coordination Study; G

Submit the study with protective device equipment submittals. No time extension or similar contract modifications will be granted for work arising out of the requirements for this study. Approval of protective devices proposed will be based on recommendations of this study. The Government will not be held responsible for any changes to equipment, device ratings, settings, or additional labor for installation of equipment or devices ordered or procured prior to approval of the study.

SD-06 Test Reports

Field Acceptance Checks and Tests; G

Arc-proofing Test for cable fireproofing tape; G

Cable Installation Plan and Procedure; G

Six copies of the information described below in 215.9 by 279.4 mm binders having a minimum of three rings from which material may readily be removed and replaced, including a separate section for

each cable pull. Separate sections by heavy plastic dividers with tabs, with all data sheets signed and dated by the person supervising the pull.

- a. Site layout drawing with cable pulls numerically identified.
- b. A list of equipment used, with calibration certifications. The manufacturer and quantity of lubricant used on pull.
- c. The cable manufacturer and type of cable.
- d. The dates of cable pulls, time of day, and ambient temperature.
- e. The length of cable pull and calculated cable pulling tensions.
- f. The actual cable pulling tensions encountered during pull.

SD-07 Certificates

Cable splicer/terminator; G

Cable Installer Qualifications; G

Directional Boring Certificate of Conformance; G

1.6 QUALITY ASSURANCE

1.6.1 Precast Underground Structures

Submittal required for each type used. Provide calculations and drawings for precast manholes and handholes bearing the seal of a registered professional engineer including:

- a. Material description (i.e., f'_c and F_y)
- b. Manufacturer's printed assembly and installation instructions
- c. Design calculations
- d. Reinforcing shop drawings in accordance with ACI SP-66
- e. Plans and elevations showing opening and pulling-in iron locations and details

1.6.2 Certificate of Competency for Cable Splicer/Terminator

The cable splicer/terminator must have a certification from the National Cable Splicing Certification Board (NCSCB) in the field of splicing and terminating shielded medium voltage (5 kV to 35 kV) power cable using pre-manufactured kits (pre-molded, heat-shrink, cold shrink). Submit "Proof of Certification" for approval, for the individuals that will be performing cable splicer and termination work, 30 days before splices or terminations are to be made.

Submit certification of the qualification of the cable splicer/terminator for approval, 30 days before splices or terminations are to be made in medium voltage (5 kV to 35 kV) cables. Include the training, and

experience of the individual on the specific type and classification of cable to be provided under this contract. Indicate that the individual has had three or more years recent experience splicing and terminating medium voltage cables. List a minimum of three splices/terminations that have been in operation for more than one year. In addition, the individual may be required to perform a dummy or practice splice/termination in the presence of the Contracting Officer, before being approved as a qualified cable splicer. If that additional requirement is imposed, the Contractor must provide short sections of the approved types of cables along with the approved type of splice/termination kit, and detailed manufacturer's instructions for the cable to be spliced. The Contracting Officer reserves the right to require additional proof of competency or to reject the individual and call for certification of an alternate cable splicer.

1.6.3 Cable Installer Qualifications

Provide at least one onsite person in a supervisory position with a documentable level of competency and experience to supervise all cable pulling operations. Provide a resume showing the cable installers' experience in the last three years, including a list of references complete with points of contact, addresses and telephone numbers. Cable installer must demonstrate experience with a minimum of three medium voltage cable installations. The Contracting Officer reserves the right to require additional proof of competency or to reject the individual and call for an alternate qualified cable installer.

1.6.4 Directional Boring Certificate of Conformance

Provide certification of compliance with the registered Professional Engineer's design requirements for each directional bore, including: HDPE conduit size and type, bend radius, elevation changes, vertical and horizontal path deviations, conductor size and type and any conductor derating due to depth of conduit. Record location and depth of all directional-bore installed HDPE conduits using Global Positioning System (GPS) recording means with "resource grade" accuracy.

1.6.5 Regulatory Requirements

In each of the publications referred to herein, consider the advisory provisions to be mandatory, as though the word, "must" had been substituted for "should" wherever it appears. Interpret references in these publications to the "authority having jurisdiction," or words of similar meaning, to mean the Contracting Officer. Equipment, materials, installation, and workmanship must be in accordance with the mandatory and advisory provisions of IEEE C2 and NFPA 70 unless more stringent requirements are specified or indicated.

1.6.6 Standard Products

Provide materials and equipment that are products of manufacturers regularly engaged in the production of such products which are of equal material, design and workmanship. Products must have been in satisfactory commercial or industrial use for 2 years prior to bid opening. The 2-year period must include applications of equipment and materials under similar circumstances and of similar size. The product must have been for sale on the commercial market through advertisements, manufacturers' catalogs, or brochures during the 2-year period. Where two or more items of the same class of equipment are required, these items must be products of a single

manufacturer; however, the component parts of the item need not be the products of the same manufacturer unless stated in this section.

1.6.6.1 Alternative Qualifications

Products having less than a 2-year field service record will be acceptable if a certified record of satisfactory field operation for not less than 6000 hours, exclusive of the manufacturers' factory or laboratory tests, is furnished.

1.6.6.2 Material and Equipment Manufacturing Date

Products manufactured more than 3 years prior to date of delivery to site are not acceptable, unless specified otherwise.

PART 2 PRODUCTS

2.1 CONDUIT, DUCTS, AND FITTINGS

2.1.1 Rigid Metal Conduit

UL 6.

2.1.1.1 Rigid Metallic Conduit, PVC Coated

NEMA RN 1, Type A40, except that hardness must be nominal 85 Shore A durometer, dielectric strength must be minimum 15.75 kV per mm at 60 Hz, and tensile strength must be minimum 25 MPa.

2.1.2 Intermediate Metal Conduit

UL 1242.

2.1.2.1 Intermediate Metal Conduit, PVC Coated

NEMA RN 1, Type A40, except that hardness must be nominal 85 Shore A durometer, dielectric strength must be minimum 15.75 kV per mm at 60 Hz, and tensile strength must be minimum 25 MPa.

2.1.3 Plastic Conduit for Direct Burial and Riser Applications

UL 651 and NEMA TC 2, EPC-40 EPC-80.

2.1.4 Plastic Duct for Concrete Encasement

Provide Type EPC-40 per UL 651 and NEMA TC 2, as indicated.

2.1.5 Duct Sealant

UL 94, Class HBF. Provide high-expansion urethane foam duct sealant that expands and hardens to form a closed, chemically and water resistant, rigid structure. Sealant must be compatible with common cable and wire jackets and capable of adhering to metals, plastics and concrete. Sealant must be capable of curing in temperature ranges of 2 degrees C to 35 degrees C. Cured sealant must withstand temperature ranges of -29 degrees C to 93 degrees C without loss of function.

2.1.6 Fittings

2.1.6.1 Metal Fittings

UL 514B.

2.1.6.2 PVC Conduit Fittings

UL 514B, UL 651.

2.1.6.3 PVC Duct Fittings

NEMA TC 9.

2.1.6.4 Outlet Boxes for Steel Conduit

Outlet boxes for use with rigid or flexible steel conduit must be cast-metal cadmium or zinc-coated if of ferrous metal with gasketed closures and must conform to UL 514A.

2.2 LOW VOLTAGE INSULATED CONDUCTORS AND CABLES

Insulated conductors must be rated 600 volts and conform to the requirements of NFPA 70, including listing requirements. Wires and cables manufactured more than 12 months prior to date of delivery to the site are not acceptable.

2.2.1 Conductor Types

Cable and duct sizes indicated are for copper conductors and THHN/THWN unless otherwise noted. Conductors No. 10 AWG and smaller must be solid. Conductors No. 8 AWG and larger must be stranded. All conductors must be copper.

2.2.2 Conductor Material

Unless specified or indicated otherwise or required by NFPA 70, wires in conduit, other than service entrance, must be 600-volt, Type THWN/THHN conforming to UL 83 or Type XHHW conforming to UL 44. Copper conductors must be annealed copper complying with ASTM B3 and ASTM B8.

2.2.3 Jackets

Provide multiconductor cables with an overall PVC outer jacket.

2.2.4 Direct Buried

Provide single-conductor and multi-conductor cables identified for direct burial.

2.2.5 In Duct

Cables must be single-conductor cable. Cables in factory-installed, coilable-plastic-duct assemblies must conform to NEMA TC 7.

2.2.6 Cable Marking

Insulated conductors must have the date of manufacture and other identification imprinted on the outer surface of each cable at regular

intervals throughout the cable length.

Identify each cable by means of a fiber, laminated plastic, or non-ferrous metal tags in each manhole, handhole, junction box, and each terminal. Each tag must contain the following information; cable type, conductor size, circuit number, circuit voltage, cable destination and phase identification.

Color code conductors. Provide conductor identification within each enclosure where a tap, splice, or termination is made. Conductor identification must be by color-coded insulated conductors, plastic-coated self-sticking printed markers, colored nylon cable ties and plates, heat shrink type sleeves, or colored electrical tape. Properly identify control circuit terminations. Color must be green for grounding conductors and white for neutrals; except where neutrals of more than one system are installed in same raceway or box, other neutrals may be white with a different colored (not green) stripe for each. Color of ungrounded conductors in different voltage systems are as follows:

- a. 208/120 volt, three-phase
 - (1) Phase A - black
 - (2) Phase B - red
 - (3) Phase C - blue
- b. 480/277 volt, three-phase
 - (1) Phase A - brown
 - (2) Phase B - orange
 - (3) Phase C - yellow
- c. 120/240 volt, single phase: Black and red
- d. On three-phase, four-wire delta system, high leg must be orange, as required by NFPA 70.

2.3 LOW VOLTAGE WIRE CONNECTORS AND TERMINALS

Provide a uniform compression over the entire conductor contact surface. Use solderless terminal lugs on stranded conductors.

- a. For use with copper conductors: UL 486A-486B.

2.4 LOW VOLTAGE SPLICES

Provide splices in conductors with a compression connector on the conductor and by insulating and waterproofing using one of the following methods which are suitable for continuous submersion in water and comply with ANSI C119.1.

2.4.1 Heat Shrinkable Splice

Provide heat shrinkable splice insulation by means of a thermoplastic adhesive sealant material applied in accordance with the manufacturer's written instructions.

2.4.2 Cold Shrink Rubber Splice

Provide a cold-shrink rubber splice which consists of EPDM rubber tube which has been factory stretched onto a spiraled core which is removed during splice installation. The installation must not require heat or flame, or any additional materials such as covering or adhesive. It must be designed for use with inline compression type connectors, or indoor, outdoor, direct-burial or submerged locations.

2.5 TAPE

2.5.1 Insulating Tape

UL 510, plastic insulating tape, capable of performing in a continuous temperature environment of 80 degrees C.

2.5.2 Buried Warning and Identification Tape

Provide detectable tape in accordance with Section 31 23 00.00 20 EXCAVATION AND FILL.

2.5.3 Fireproofing Tape

Provide tape composed of a flexible, conformable, unsupported intumescent elastomer. Tape must be not less than 0.762 mm thick, noncorrosive to cable sheath, self-extinguishing, noncombustible, adhesive-free, and must not deteriorate when subjected to oil, water, gases, salt water, sewage, and fungus.

2.6 PULL ROPE

Plastic or flat pull line (bull line) having a minimum tensile strength of 890 N.

2.7 GROUNDING AND BONDING

2.7.1 Driven Ground Rods

Provide copper-clad steel ground rods conforming to UL 467 not less than 19 mm in diameter by 3.1 m in length. Sectional type rods may be used for rods 6.2 m or longer.

2.7.2 Grounding Conductors

Stranded-bare copper conductors must conform to ASTM B8, Class B, soft-drawn unless otherwise indicated. Solid-bare copper conductors must conform to ASTM B1 for sizes No. 8 and smaller. Insulated conductors must be of the same material as phase conductors and green color-coded, except that conductors must be rated no more than 600 volts. Aluminum is not acceptable.

2.8 CAST-IN-PLACE CONCRETE

Provide concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE. In addition, provide concrete for encasement of underground ducts with 20 MPa minimum 28-day compressive strength. Concrete associated with electrical work for other than encasement of underground ducts must be 30 MPa minimum 28-day compressive strength unless specified

otherwise.

2.9 UNDERGROUND STRUCTURES

Provide precast concrete underground structures or standard type cast-in-place manhole types as indicated, conforming to ASTM C857 and ASTM C478M. Top, walls, and bottom must consist of reinforced concrete. Walls and bottom must be of monolithic concrete construction. Locate duct entrances and windows near the corners of structures to facilitate cable racking. Covers must fit the frames without undue play. Form steel and iron to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Install a pulling-in iron in the wall opposite each duct line entrance. Cable racks, including rack arms and insulators, must be adequate to accommodate the cable.

2.9.1 Cast-In-Place Concrete Structures

Concrete must conform to Section 03 30 00 CAST-IN-PLACE CONCRETE. Construct walls on a footing of cast-in-place concrete except that precast concrete base sections may be used for precast concrete manhole risers. Concrete block is not allowed in areas subject to aircraft loading.

2.9.2 Precast Concrete Structures, Risers and Tops

Precast concrete underground structures may be provided in lieu of cast-in-place subject to the requirements specified below. Precast units must be the product of a manufacturer regularly engaged in the manufacture of precast concrete products, including precast manholes.

2.9.2.1 General

Precast concrete structures must have the same accessories and facilities as required for cast-in-place structures. Likewise, precast structures must have plan area and clear heights not less than those of cast-in-place structures. Concrete materials and methods of construction must be the same as for cast-in-place concrete construction, as modified herein. Slope in floor may be omitted provided precast sections are poured in reinforced steel forms. Concrete for precast work must have a 28-day compressive strength of not less than 30 MPa. Structures may be precast to the design and details indicated for cast-in-place construction, precast monolithically and placed as a unit, or structures may be assembled sections, designed and produced by the manufacturer in accordance with the requirements specified. Structures must be identified with the manufacturer's name embedded in or otherwise permanently attached to an interior wall face.

2.9.2.2 Design for Precast Structures

ACI 318M. In the absence of detailed on-site soil information, design for the following soil parameters/site conditions:

- a. Angle of Internal Friction (ϕ) = 0.523 rad
- b. Unit Weight of Soil (Dry) = 1760 kg/m³, (Saturated)
= 2080 kg/m³

- c. Coefficient of Lateral Earth Pressure (K_a) = 0.33
- d. Ground Water Level = 915 mm below ground elevation
- e. Vertical design loads must include full dead, superimposed dead, and live loads including a 30 percent magnification factor for impact. Live loads must consider all types and magnitudes of vehicular (automotive, industrial, or aircraft) traffic to be encountered. The minimum design vertical load must be for H20 highway loading per AASHTO HB-17.
- f. Horizontal design loads must include full geostatic and hydrostatic pressures for the soil parameters, water table, and depth of installation to be encountered. Also, horizontal loads imposed by adjacent structure foundations, and horizontal load components of vertical design loads, including impact, must be considered, along with a pulling-in iron design load of 26,700 N.
- g. Each structural component must be designed for the load combination and positioning resulting in the maximum shear and moment for that particular component.
- h. Design must also consider the live loads induced in the handling, installation, and backfilling of the manholes. Provide lifting devices to ensure structural integrity during handling and installation.

2.9.2.3 Construction

Provide a uniform thickness for structure top, bottom, and wall not less than 150 mm. Thin-walled knock-out panels for designed or future duct bank entrances are not permitted. Provide quantity, size, and location of duct bank entrance windows as directed, and cast completely open by the precaster. Size of windows must exceed the nominal duct bank envelope dimensions by at least 305 mm vertically and horizontally to preclude in-field window modifications made necessary by duct bank misalignment. However, the sides of precast windows must be a minimum of 150 mm from the inside surface of adjacent walls, floors, or ceilings. Form the perimeter of precast window openings to have a keyed or inward flared surface to provide a positive interlock with the mating duct bank envelope. Provide welded wire fabric reinforcing through window openings for in-field cutting and flaring into duct bank envelopes. Provide additional reinforcing steel comprised of at least two No. 4 bars around window openings. Provide drain sumps a minimum of 305 mm in diameter and 100 mm deep for precast structures.

2.9.2.4 Joints

Provide tongue-and-groove joints on mating edges of precast components. Shiplap joints are not allowed. Design joints to firmly interlock adjoining components and to provide waterproof junctions and adequate shear transfer. Seal joints watertight using preformed plastic strip conforming to ASTM C990M. Install sealing material in strict accordance with the sealant manufacturer's printed instructions. Provide waterproofing at conduit/duct entrances into structures, and where access frame meets the top slab, provide continuous grout seal.

2.10 CABLE SUPPORTS (RACKS, ARMS, AND INSULATORS)

Zinc coat the metal portion of racks and arms after fabrication.

2.10.1 Cable Rack Stanchions

The wall bracket or stanchion must be 100 mm by approximately 38 mm by 4.76 mm channel steel, or 100 mm by approximately 25 mm glass-reinforced nylon with recessed bolt mounting holes, 1220 mm long (minimum) in manholes. Space slots for mounting cable rack arms at 200 mm intervals.

2.10.2 Rack Arms

Cable rack arms must be steel or malleable iron or glass reinforced nylon and must be of the removable type. Rack arm length must be a minimum of 200 mm and a maximum of 305 mm.

2.10.3 Insulators

Insulators for metal rack arms must be dry-process glazed porcelain. Insulators are not required for nylon arms.

2.11 CABLE TAGS IN MANHOLES

Provide polyethylene tags for each power cable located in manholes. Do not provide handwritten letters. The first position on the power cable tag denotes the voltage. The second through sixth positions on the tag identifies the circuit. The next to last position denotes the phase of the circuit and include the Greek "phi" symbol. The last position denotes the cable size. As an example, a tag could have the following designation: "11.5 NAS 1-8(Phase A)500," denoting that the tagged cable is on the 11.5kV system circuit number NAS 1-8, underground, Phase A, sized at 500 kcmil.

2.11.1 Polyethylene Cable Tags

Provide tags of polyethylene having an average tensile strength of 22.4 MPa; and that are 2 millimeter thick (minimum), non-corrosive non-conductive; resistive to acids, alkalis, organic solvents, and salt water; and distortion resistant to 77 degrees C. Provide 1.3 mm (minimum) thick black polyethylene tag holder. Provide a one-piece nylon, self-locking tie at each end of the cable tag, having a minimum loop tensile strength of 778.75 N and black block letters, numbers, and symbols 25 mm high on a yellow background. Letters, numbers, and symbols must not fall off or change positions regardless of the cable tags' orientation.

2.12 PROTECTIVE DEVICES AND COORDINATION

Provide protective devices and coordination as specified in Section 26 28 01.00 10 COORDINATED POWER SYSTEM PROTECTION.

2.13 SOURCE QUALITY CONTROL

2.13.1 Arc-Proofing Test for Cable Fireproofing Tape

Manufacturer must test one sample assembly consisting of a straight lead tube 305 mm long with a 65.5 mm outside diameter, and a 3.175 mm thick wall, and covered with one-half lap layer of arc and fireproofing tape per manufacturer's instructions. The arc and fireproofing tape must withstand

extreme temperature of a high-current fault arc 13,000 degrees K for 70 cycles as determined by using an argon directed plasma jet capable of constantly producing and maintaining an arc temperature of 13,000 degrees K. Temperature (13,000 degrees K) of the ignited arc between the cathode and anode must be obtained from a dc power source of 305 (plus or minus 5) amperes and 20 (plus or minus 1) volts. Direct the arc toward the sample assembly accurately positioned 5 (plus or minus 1) millimeters downstream in the plasma from the anode orifice by fixed flow rate of argon gas (0.18 g per second). Test each sample assembly at three unrelated points. Start time for tests must be taken from recorded peak current when the specimen is exposed to the full test temperature. Surface heat on the specimen prior to that time must be minimal. The end point is established when the plasma or conductive arc penetrates the protective tape and strikes the lead tube. Submittals for arc-proofing tape must indicate that the test has been performed and passed by the manufacturer.

PART 3 EXECUTION

3.1 INSTALLATION

Install equipment and devices in accordance with the manufacturer's published instructions and with the requirements and recommendations of NFPA 70 and IEEE C2 and CALPUC G.O.128 as applicable. In addition to these requirements, install telecommunications in accordance with TIA-758 and RUS Bull 1751F-644.

3.2 CABLE INSPECTION

Inspect each cable reel for correct storage positions, signs of physical damage, and broken end seals prior to installation. If end seal is broken, remove moisture from cable prior to installation in accordance with the cable manufacturer's recommendations.

3.3 CABLE INSTALLATION PLAN AND PROCEDURE

Obtain from the manufacturer an installation manual or set of instructions which addresses such aspects as cable construction, insulation type, cable diameter, bending radius, cable temperature limits for installation, lubricants, coefficient of friction, conduit cleaning, storage procedures, moisture seals, testing for and purging moisture, maximum allowable pulling tension, and maximum allowable sidewall bearing pressure. Perform pulling calculations and prepare a pulling plan and submit along with the manufacturer's instructions in accordance with SUBMITTALS. Install cable strictly in accordance with the cable manufacturer's recommendations and the approved installation plan.

Calculations and pulling plan must include:

- a. Site layout drawing with cable pulls identified in numeric order of expected pulling sequence and direction of cable pull.
- b. List of cable installation equipment.
- c. Lubricant manufacturer's application instructions.
- d. Procedure for resealing cable ends to prevent moisture from entering cable.
- e. Cable pulling tension calculations of all cable pulls.

- f. Cable percentage conduit fill.
- g. Cable sidewall bearing pressure.
- h. Cable minimum bend radius and minimum diameter of pulling wheels used.
- i. Cable jam ratio.
- j. Maximum allowable pulling tension on each different type and size of conductor.
- k. Maximum allowable pulling tension on pulling device.

3.4 UNDERGROUND FEEDERS SUPPLYING BUILDINGS

Terminate underground feeders supplying building at a point 1525 mm outside the building and projections thereof, except that conductors must be continuous to the terminating point indicated. Coordinate connections of the feeders to the service entrance equipment with Section 26 20 00 INTERIOR DISTRIBUTION SYSTEM. Provide PVC, Type EPC-40 conduit from the supply equipment to a point 1525 mm outside the building and projections thereof. Protect ends of underground conduit with plastic plugs until connections are made.

Encase the underground portion of the conduit in a concrete envelope and bury as specified for underground duct with concrete encasement.

3.5 UNDERGROUND STRUCTURE CONSTRUCTION

Provide standard type cast-in-place construction as specified herein and as indicated, or precast construction as specified herein. Horizontal concrete surfaces of floors must have a smooth trowel finish. Cure concrete by applying two coats of white pigmented membrane forming-curing compound in strict accordance with the manufacturer's printed instructions, except that precast concrete may be steam cured. Curing compound must conform to ASTM C309. Locate duct entrances and windows in the center of end walls (shorter) and near the corners of sidewalls (longer) to facilitate cable racking and splicing. Covers for underground structures must fit the frames without undue play. Form steel and iron to shape and size with sharp lines and angles. Castings must be free from warp and blow holes that may impair strength or appearance. Exposed metal must have a smooth finish and sharp lines and arises. Provide necessary lugs, rabbets, and brackets. Set pulling-in irons and other built-in items in place before depositing concrete. Manhole locations, as indicated, are approximate. Coordinate exact manhole locations with other utilities and finished grading and paving.

3.5.1 Cast-In-Place Concrete Structures

Construct walls on a footing of cast-in-place concrete except that precast concrete base sections may be used for precast concrete manhole risers.

3.5.2 Precast Concrete Construction

Set commercial precast structures on 150 mm of level, 90 percent compacted granular fill, 19 mm to 25 mm size, extending 305 mm beyond the structure on each side. Compact granular fill by a minimum of four passes with a plate type vibrator. Installation must additionally conform to the

manufacturer's instructions.

3.5.3 Pulling-In Irons

Provide steel bars bent as indicated, and cast in the walls and floors. Alternatively, pipe sleeves may be precast into the walls and floors where required to accept U-bolts or other types of pulling-in devices possessing the strengths and clearances stated herein. The final installation of pulling-in devices must be made permanent. Cover and seal exterior projections of thru-wall type pulling-in devices with an appropriate protective coating. In the floor, locate the irons a minimum of 150 mm from the edge of the sump, and in the walls, locate the irons within 150 mm of the projected center of the duct bank pattern or precast window in the opposite wall. However, the pulling-in iron must not be located within 150 mm of an adjacent interior surface, or duct or precast window located within the same wall as the iron. If a pulling-in iron cannot be located directly opposite the corresponding duct bank or precast window due to this clearance limitation, locate the iron directly above or below the projected center of the duct bank pattern or precast window the minimum distance required to preserve the 150 mm clearance previously stated. In the case of directly opposing precast windows, pulling-in irons consisting of a 915 mm length of No. 5 reinforcing bar, formed into a hairpin, may be cast-in-place within the precast windows simultaneously with the end of the corresponding duct bank envelope. Irons installed in this manner must be positioned directly in line with, or when not possible, directly above or below the projected center of the duct bank pattern entering the opposite wall, while maintaining a minimum clear distance of 75 mm from any edge of the cast-in-place duct bank envelope or any individual duct. Pulling-in irons must have a clear projection into the structure of approximately 100 mm and must be designed to withstand a minimum pulling-in load of 26,700 N. Hot-dip galvanize irons after fabrication.

3.5.4 Cable Racks, Arms and Insulators

Cable racks, arms and insulators must be sufficient to accommodate the cables. Space racks in power manholes not more than 915 mm apart, and provide each manhole wall with a minimum of two racks. Space racks in signal manholes not more than 420 mm apart with the end rack being no further than 305 mm from the adjacent wall. Methods of anchoring cable racks are as follows:

- a. Provide a 15 mm diameter by 125 mm long anchor bolt with 75 mm foot cast in structure wall with 50 mm protrusion of threaded portion of bolt into structure. Provide 15 mm steel square head nut on each anchor bolt. Coat threads of anchor bolts with suitable coating immediately prior to installing nuts.
- b. Provide concrete channel insert with a minimum load rating of 1192 kg per meter. Insert channel must be steel of the same length as "vertical rack channel;" and cast flush in structure wall. Provide 15 mm steel nuts in channel insert to receive 15 mm diameter by 75 mm long steel, square head anchor bolts.
- c. Provide concrete "spot insert" at each anchor bolt location, cast flush in structure wall. Each insert must have minimum 365 kg load rating. Provide 15 mm diameter by 75 mm long steel, square head anchor bolt at each anchor point. Coat threads of anchor bolts with suitable coating immediately prior to installing bolts.

3.5.5 Field Painting

Clean cast-iron frames and covers not buried in concrete or masonry of mortar, rust, grease, dirt and other deleterious materials, and coat with bituminous paint.

3.6 UNDERGROUND CONDUIT AND DUCT SYSTEMS

3.6.1 Requirements

Run conduit in straight lines except where a change of direction is necessary. Provide numbers and sizes of ducts as indicated. Provide a 4/0 AWG bare copper grounding conductor above medium-voltage distribution duct banks. Bond bare copper grounding conductor to ground rings (loops) in all manholes and to ground rings (loops) at all equipment slabs (pads). Route grounding conductor into manholes with the duct bank (sleeving is not required). Ducts must have a continuous slope downward toward underground structures and away from buildings, laid with a minimum slope of 75 mm per 30 m. Depending on the contour of the finished grade, the high-point may be at a terminal, a manhole, a handhole, or between manholes or handholes. Terminate all PVC conduit end points in utility holes, switching cabinets, transform handholes and buildings with end bells. The bell end of the conduits that enter manholes and handholes must be flush with the wall.

Perform changes in ductbank direction as follows:

- a. Short-radius manufactured 90-degree duct bends may be used only for pole or equipment risers, unless specifically indicated as acceptable.
- b. The minimum manufactured bend radius must be 450 mm for ducts of less than 80 mm diameter, and 900 mm for ducts 80 mm or greater in diameter.
- c. As an exception to the bend radius required above, provide field manufactured longsweep bends having a minimum radius of 7.6 m for a change of direction of more than 5 degrees, either horizontally or vertically, using a combination of curved and straight sections. Maximum manufactured curved sections allowed for use in field manufactured longsweep bend: 30 degrees.

3.6.2 Treatment

Keep ducts clean of concrete, dirt, or foreign substances during construction. Make field cuts requiring tapers with proper tools and match factory tapers. Use a coupling recommended by the duct manufacturer whenever an existing duct is connected to a duct of different material or shape. Store ducts to avoid warping and deterioration with ends sufficiently plugged to prevent entry of any water or solid substances. Thoroughly clean ducts before being laid. Store plastic ducts on a flat surface and protected from the direct rays of the sun.

3.6.3 Conduit Cleaning

As each conduit run is completed, for conduit sizes 75 mm and larger, draw a flexible testing mandrel approximately 305 mm long with a diameter less than the inside diameter of the conduit through the conduit. After which, draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs. For

conduit sizes less than 75 mm, draw a stiff bristle brush through until conduit is clear of particles of earth, sand and gravel; then immediately install conduit plugs.

3.6.4 Jacking and Drilling Under Roads and Structures

Conduits to be installed under existing paved areas which are not to be disturbed, and under roads and railroad tracks, must be zinc-coated, rigid steel, jacked into place. Where ducts are jacked under existing pavement, install rigid steel conduit because of its strength. To protect the corrosion-resistant conduit coating, predrilling or installing conduit inside a larger iron pipe sleeve (jack-and-sleeve) is required. For crossings of existing railroads and airfield pavements greater than 15 m in length, the predrilling method or the jack-and-sleeve method will be used. Separators or spacing blocks must be made of steel, concrete, plastic, or a combination of these materials placed not farther apart than 1.2 m on centers.

3.6.5 Galvanized Conduit Concrete Penetrations

Galvanized conduits which penetrate concrete (slabs, pavement, and walls) in wet locations must be PVC coated and extend from at least 50 mm within the concrete to the first coupling or fitting outside the concrete (minimum of 150 mm from penetration).

3.6.6 Multiple Conduits

Separate multiple conduits by a minimum distance of 75 mm, except that light and power conduits must be separated from control, signal, and telephone conduits by a minimum distance of 300 mm. Stagger the joints of the conduits by rows (horizontally) and layers (vertically) to strengthen the conduit assembly. Provide plastic duct spacers that interlock vertically and horizontally. Spacer assembly must consist of base spacers, intermediate spacers, ties, and locking device on top to provide a completely enclosed and locked-in conduit assembly. Install spacers per manufacturer's instructions, but provide a minimum of two spacer assemblies per 3050 mm of conduit assembly.

3.6.7 Conduit Plugs and Pull Rope

Provide new conduit indicated as being unused or empty with plugs on each end. Plugs must contain a weephole or screen to allow water drainage. Provide a plastic pull rope having 915 mm of slack at each end of unused or empty conduits.

3.6.8 Conduit and Duct Without Concrete Encasement

Depths to top of the conduit must be not less than 610 mm below finished grade. Provide not less than 75 mm clearance from the conduit to each side of the trench. Grade bottom of trench smooth; where rock, soft spots, or sharp-edged materials are encountered, excavate the bottom for an additional 75 mm, fill and tamp level with original bottom with sand or earth free from particles, that would be retained on a 6.25 mm sieve. The first 150 mm layer of backfill cover must be sand compacted as previously specified. The rest of the excavation must be backfilled and compacted in 75 to 150 mm layers. Provide color, type and depth of warning tape as specified in Section 31 23 00.00 20 EXCAVATION AND FILL.

3.6.8.1 Encasement Under Roads and Structures

Under roads, paved areas, and railroad tracks, install conduits in concrete encasement of rectangular cross-section providing a minimum of 75 mm concrete cover around ducts. Extend concrete encasement at least 1525 mm beyond the edges of paved areas and roads, and 3660 mm beyond the rails on each side of railroad tracks. Depths to top of the concrete envelope must be not less than 610 mm below finished grade.

3.6.9 Duct Encased in Concrete

Construct underground duct lines of individual conduits encased in concrete. Depths to top of the concrete envelope must be not less than 450 mm below finished grade, except under roads and pavement, concrete envelope must be not less than 610 mm below finished grade. Do not mix different kinds of conduit in any one duct bank. Concrete encasement surrounding the bank must be rectangular in cross-section and provide at least 75 mm of concrete cover for ducts. Separate conduits by a minimum concrete thickness of 75 mm. Before pouring concrete, anchor duct bank assemblies, prevent floating during concrete pouring by driving reinforcing rods adjacent to duct spacer assemblies and attaching the rods to the spacer assembly. Provide steel reinforcing in the concrete envelope as indicated. Provide color, type and depth of warning tape as specified in Section 31 23 00.00 20 EXCAVATION AND FILL.

3.6.9.1 Connections to Manholes

Duct bank envelopes connecting to underground structures must be flared to have enlarged cross-section at the manhole entrance to provide additional shear strength. Dimensions of the flared cross-section must be larger than the corresponding manhole opening dimensions by no less than 300 mm in each direction. Perimeter of the duct bank opening in the underground structure must be flared toward the inside or keyed to provide a positive interlock between the duct bank and the wall of the structure. Use vibrators when this portion of the encasement is poured to assure a seal between the envelope and the wall of the structure.

3.6.9.2 Connections to Existing Underground Structures

For duct bank connections to existing structures, break the structure wall out to the dimensions required and preserve steel in the structure wall. Cut steel and extend into the duct bank envelope. Chip the perimeter surface of the duct bank opening to form a key or flared surface, providing a positive connection with the duct bank envelope.

3.6.9.3 Connections to Existing Concrete Pads

For duct bank connections to concrete pads, break an opening in the pad out to the dimensions required and preserve steel in pad. Cut the steel and extend into the duct bank envelope. Chip out the opening in the pad to form a key for the duct bank envelope.

3.6.9.4 Connections to Existing Ducts

Where connections to existing duct banks are indicated, excavate the banks to the maximum depth necessary. Cut off the banks and remove loose concrete from the conduits before new concrete-encased ducts are installed. Provide a reinforced concrete collar, poured monolithically with the new duct bank, to take the shear at the joint of the duct banks.

Remove existing cables which constitute interference with the work. Abandon in place those no longer used ducts and cables which do not interfere with the work.

3.6.9.5 Partially Completed Duct Banks

During construction wherever a construction joint is necessary in a duct bank, prevent debris such as mud, and, and dirt from entering ducts by providing suitable conduit plugs. Fit concrete envelope of a partially completed duct bank with reinforcing steel extending a minimum of 610 mm back into the envelope and a minimum of 610 mm beyond the end of the envelope. Provide one No. 4 bar in each corner, 75 mm from the edge of the envelope. Secure corner bars with two No. 3 ties, spaced approximately 305 mm apart. Restrain reinforcing assembly from moving during concrete pouring.

3.6.9.6 Removal of Ducts

Where duct lines are removed from existing underground structures, close the openings to waterproof the structure. Chip out the wall opening to provide a key for the new section of wall.

3.6.10 Duct Sealing

Seal all electrical penetrations for radon mitigation, maintaining integrity of the vapor barrier, and to prevent infiltration of air, insects, and vermin.

3.7 CABLE PULLING

Test existing duct lines with a mandrel and thoroughly swab out to remove foreign material before pulling cables. Pull cables down grade with the feed-in point at the manhole or buildings of the highest elevation. Use flexible cable feeds to convey cables through manhole opening and into duct runs. Do not exceed the specified cable bending radii when installing cable under any conditions, including turnups into switches, transformers, switchgear, switchboards, and other enclosures. Cable with tape shield must have a bending radius not less than 12 times the overall diameter of the completed cable. If basket-grip type cable-pulling devices are used to pull cable in place, cut off the section of cable under the grip before splicing and terminating.

3.7.1 Cable Lubricants

Use lubricants that are specifically recommended by the cable manufacturer for assisting in pulling jacketed cables.

3.8 CABLES IN UNDERGROUND STRUCTURES

Do not install cables utilizing the shortest path between penetrations, but route along those walls providing the longest route and the maximum spare cable lengths. Form cables to closely parallel walls, not to interfere with duct entrances, and support on brackets and cable insulators. Support cable splices in underground structures by racks on each side of the splice. Locate splices to prevent cyclic bending in the spliced sheath. Install cables at middle and bottom of cable racks, leaving top space open for future cables, except as otherwise indicated for existing installations. Provide one spare three-insulator rack arm for each cable rack in each underground structure.

3.8.1 Cable Tag Installation

Install cable tags in each manhole as specified, including each splice. Tag wire and cable provided by this contract. Install cable tags over the fireproofing, if any, and locate the tags so that they are clearly visible without disturbing any cabling or wiring in the manholes.

3.9 CONDUCTORS INSTALLED IN PARALLEL

Group conductors such that each conduit of a parallel run contains one Phase A conductor, one Phase B conductor, one Phase C conductor, and one neutral conductor.

3.10 LOW VOLTAGE CABLE SPLICING AND TERMINATING

Make terminations and splices with materials and methods as indicated or specified herein and as designated by the written instructions of the manufacturer. Do not allow the cables to be moved until after the splicing material has completely set. Make splices in underground distribution systems only in accessible locations such as manholes, handholes, or aboveground termination pedestals.

3.11 CABLE END CAPS

Cable ends must be sealed at all times with coated heat shrinkable end caps. Cables ends must be sealed when the cable is delivered to the job site, while the cable is stored and during installation of the cable. The caps must remain in place until the cable is spliced or terminated. Sealing compounds and tape are not acceptable substitutes for heat shrinkable end caps. Cable which is not sealed in the specified manner at all times will be rejected.

3.12 FIREPROOFING OF CABLES IN UNDERGROUND STRUCTURES

Fireproof (arc proof) wire and cables which will carry current at 2200 volts or more in underground structures.

3.12.1 Fireproofing Tape

Tightly wrap strips of fireproofing tape around each cable spirally in half-lapped wrapping. Install tape in accordance with manufacturer's instructions.

3.12.2 Tape-Wrap

Tape-wrap metallic-sheathed or metallic armored cables without a nonmetallic protective covering over the sheath or armor prior to application of fireproofing. Wrap must be in the form of two tightly applied half-lapped layers of a pressure-sensitive 0.254 mm thick plastic tape, and must extend not less than 25 mm into the duct. Even out irregularities of the cable, such as at splices, with insulation putty before applying tape.

3.13 GROUNDING SYSTEMS

NFPA 70 and IEEE C2, except provide grounding systems with a resistance to solid earth ground not exceeding 10 ohms.

3.13.1 Grounding Electrodes

Provide cone pointed driven ground rods driven full depth plus 150 mm, installed to provide an earth ground of the appropriate value for the particular equipment being grounded. If the specified ground resistance is not met, provide an additional ground rod in accordance with the requirements of NFPA 70 (placed not less than 6 feet from the first rod). Should the resultant (combined) resistance exceed the specified resistance, measured not less than 48 hours after rainfall, notify the Contracting Officer immediately.

3.13.2 Grounding Connections

Make grounding connections which are buried or otherwise normally inaccessible, by exothermic weld or compression connector.

- a. Make exothermic welds strictly in accordance with the weld manufacturer's written recommendations. Welds which are "puffed up" or which show convex surfaces indicating improper cleaning are not acceptable. Mechanical connectors are not required at exothermic welds.
- b. Make compression connections using a hydraulic compression tool to provide the correct circumferential pressure. Tools and dies must be as recommended by the manufacturer. An embossing die code or other standard method must provide visible indication that a connector has been adequately compressed on the ground wire.

3.13.3 Grounding Conductors

Provide bare grounding conductors, except where installed in conduit with associated phase conductors. Ground cable sheaths, cable shields, conduit, and equipment with No. 6 AWG. Ground other noncurrent-carrying metal parts and equipment frames of metal-enclosed equipment. Ground metallic frames and covers of handholes and pull boxes with a braided, copper ground strap with equivalent ampacity of No. 6 AWG. Provide direct connections to the grounding conductor with 600 v insulated, full-size conductor for each grounded neutral of each feeder circuit, which is spliced within the manhole.

3.13.4 Ground Cable Crossing Expansion Joints

Protect ground cables crossing expansion joints or similar separations in structures and pavements by use of approved devices or methods of installation which provide the necessary slack in the cable across the joint to permit movement. Use stranded or other approved flexible copper cable across such separations.

3.13.5 Manhole Grounding

Loop a 4/0 AWG grounding conductor around the interior perimeter, approximately 305 mm above finished floor. Secure the conductor to the manhole walls at intervals not exceeding 914 mm. Connect the conductor to the manhole grounding electrode with 4/0 AWG conductor. Connect all incoming 4/0 grounding conductors to the ground loop adjacent to the point of entry into the manhole. Bond the ground loop to all cable shields, metal cable racks, and other metal equipment with a minimum 6 AWG conductor.

3.13.6 Metal Splice Case Grounding

Ground metal splice cases for medium-voltage direct-burial cable by connection to a driven ground rod located within 600 mm of each splice box using a grounding electrode conductor having a current-carrying capacity of at least 20 percent of the individual phase conductors in the associated splice box, but not less than No. 6 AWG.

3.14 EXCAVATING, BACKFILLING, AND COMPACTING

Provide in accordance with NFPA 70 and Section 31 23 00.00 20 EXCAVATION AND FILL.

3.14.1 Reconditioning of Surfaces

3.14.1.1 Unpaved Surfaces

Restore to their original elevation and condition unpaved surfaces disturbed during installation of duct. Preserve sod and topsoil removed during excavation and reinstall after backfilling is completed. Replace sod that is damaged by sod of quality equal to that removed. When the surface is disturbed in a newly seeded area, re-seed the restored surface with the same quantity and formula of seed as that used in the original seeding, and provide topsoiling, fertilizing, liming, seeding, sodding, sprigging, or mulching.

3.14.1.2 Paving Repairs

Where trenches, pits, or other excavations are made in existing roadways and other areas of pavement where surface treatment of any kind exists, restore such surface treatment or pavement the same thickness and in the same kind as previously existed, except as otherwise specified, and to match and tie into the adjacent and surrounding existing surfaces.

3.15 CAST-IN-PLACE CONCRETE

Provide concrete in accordance with Section 03 30 00 CAST-IN-PLACE CONCRETE.

3.15.1 Concrete Slabs (Pads) for Equipment

Unless otherwise indicated, the slab must be at least 200 mm thick, reinforced with a 152 mm by 152 mm - MW19 by MW19 (6 by 6 - W2.9 by W2.9) mesh, placed uniformly 100 mm from the top of the slab. Place slab on a 150 mm thick, well-compacted gravel base. Top of concrete slab must be approximately 100 mm above finished grade with gradual slope for drainage. Edges above grade must have 15 mm chamfer. Slab must be of adequate size to project at least 200 mm beyond the equipment.

Stub up conduits, with bushings, 50 mm into cable wells in the concrete pad. Coordinate dimensions of cable wells with transformer cable training areas.

3.15.2 Sealing

When the installation is complete, seal all conduit and other entries into the equipment enclosure with an approved sealing compound. Seals must be of sufficient strength and durability to protect all energized live parts of the equipment from rodents, insects, or other foreign matter.

3.16 FIELD QUALITY CONTROL

3.16.1 Performance of Field Acceptance Checks and Tests

Perform in accordance with the manufacturer's recommendations, and include the following visual and mechanical inspections and electrical tests, performed in accordance with NETA ATS.

3.16.1.1 Low Voltage Cables, 600-Volt

Perform tests after installation of cable, splices and terminations and before terminating to equipment or splicing to existing circuits.

a. Visual and Mechanical Inspection

- (1) Inspect exposed cable sections for physical damage.
- (2) Verify that cable is supplied and connected in accordance with contract plans and specifications.
- (3) Verify tightness of accessible bolted electrical connections.
- (4) Inspect compression-applied connectors for correct cable match and indentation.
- (5) Visually inspect jacket and insulation condition.
- (6) Inspect for proper phase identification and arrangement.

b. Electrical Tests

- (1) Perform insulation resistance tests on wiring No. 6 AWG and larger diameter using instrument which applies voltage of approximately 1000 volts dc for one minute.
- (2) Perform continuity tests to insure correct cable connection.

3.16.1.2 Grounding System

a. Visual and mechanical inspection

Inspect ground system for compliance with contract plans and specifications.

b. Electrical tests

Perform ground-impedance measurements utilizing the fall-of-potential method in accordance with IEEE 81. On systems consisting of interconnected ground rods, perform tests after interconnections are complete. On systems consisting of a single ground rod perform tests before any wire is connected. Take measurements in normally dry weather, not less than 48 hours after rainfall. Use a portable ground resistance tester in accordance with manufacturer's instructions to test each ground or group of grounds. The instrument must be equipped with a meter reading directly in ohms or fractions thereof to indicate the ground value of the ground rod or grounding systems under test. Provide site diagram indicating location of test probes with associated distances, and provide a plot of resistance vs. distance.

3.16.2 Follow-Up Verification

Upon completion of acceptance checks and tests, show by demonstration in service that circuits and devices are in good operating condition and properly performing the intended function. As an exception to requirements stated elsewhere in the contract, the Contracting Officer must be given 5 working days advance notice of the dates and times of checking and testing.

-- End of Section --